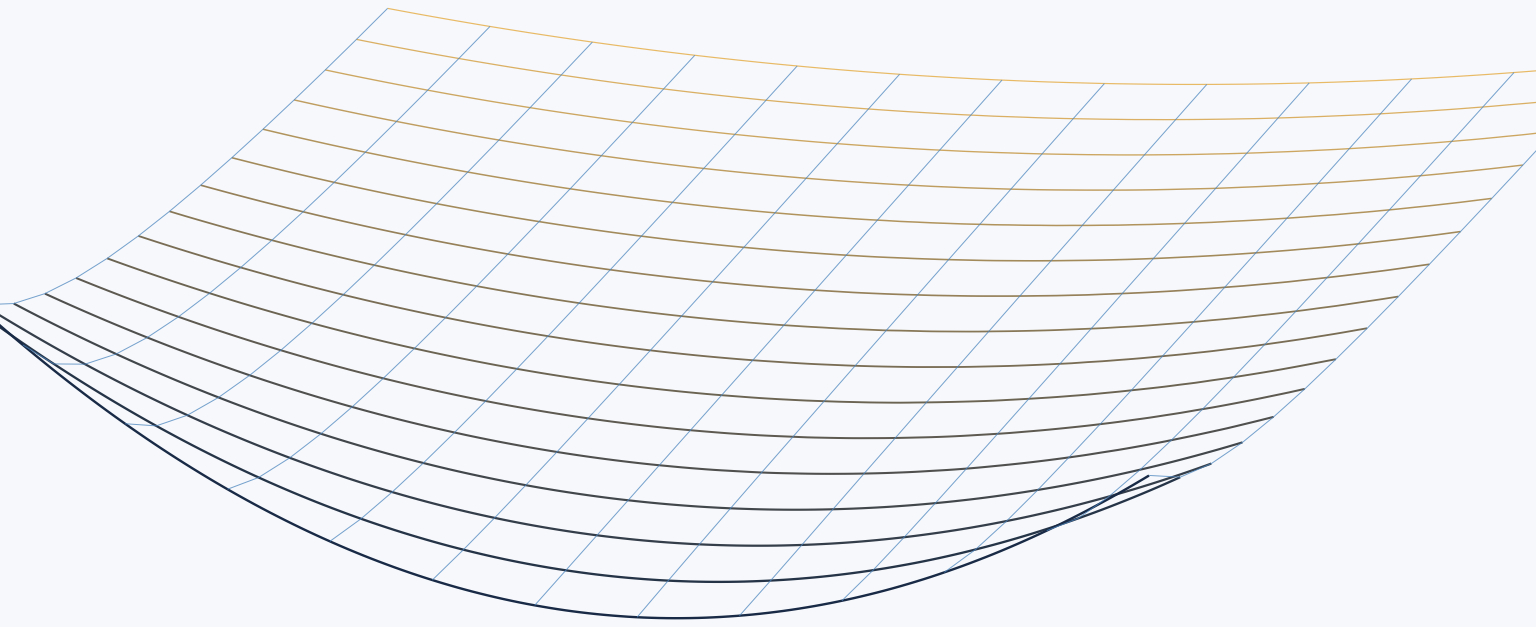
DRAFT EDITION

# DERIVATIVES

## CONTRACTS, MODELS, AND MARKETS

## SOLUTIONS TO PRACTICE QUESTIONS

**WULIN SUO**



# Derivatives: Contracts, Models, and Markets

Solutions to Practice Questions

Wulin Suo

Draft — October 29, 2026

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**Derivatives: Contracts, Models, and Markets**
**Solutions to Practice Questions**
Wulin Suo

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This volume accompanies *Derivatives: Contracts, Models, and Markets* and answers the Practice Questions of all forty chapters. It does not answer the Further Questions, which are the set an instructor assigns for grading. Its cross-references (to sections, equations, examples, and figures) are references into that book, and it is of little use without it.

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Typeset by the author in LaTeX. The cover is drawn in TeX from the same formula as the book's, the volatility surface of Chapter 19, with the fill removed and the lattice that generates it left visible.

Draft of October 29, 2026.

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# How to Use This Companion

This volume answers every *Practice Question* in *Derivatives: Contracts, Models, and Markets*. It does not answer the *Further Questions*, which are the set an instructor assigns for grading.

Each solution restates its question, so you can work through this volume without the book open, and then answers it in full: the reasoning, the arithmetic carried through, and, at the end, the point the question was testing. *Read the last part even when you got the number right*: several questions are constructed so that a correct answer arrived at by the wrong route is indistinguishable from a correct answer, and the closing note is where that is said.

Solutions are numbered as the questions are, *chapter.number*, so Solution 19.6 answers Question 19.6 of Chapter 19. Cross-references to sections, equations, examples, and figures are references into the book.

*A worked answer is not the only answer.* Where a question asks for an explanation rather than a number, what follows is one good response and not a template; a different argument that reaches the same place is worth as much. Where a question asks for a number, the number is not negotiable, and the arithmetic here has been recomputed independently of the chapter that poses the question.

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† indicates an elective chapter or section; the preface explains what that means.


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# Chapter 1
# Introduction: Derivatives and Their Markets

---

**1.1** *Explain why a forward contract has zero value at inception but an option does not, and what this implies about when money changes hands in each.*

A forward is *symmetric*. Both parties are obliged to trade, gains to one are losses to the other, and the delivery price is chosen precisely so that the two obligations have equal present value. At that price neither side would pay the other to enter, so the contract is worth zero to both. Nothing about this says the contract is riskless — only that the risk is shared on terms neither side would pay to alter.

An option is *asymmetric*. The holder exercises only when exercise pays, so the payoff is never negative to them and is strictly positive in some states. No choice of strike can make that worth zero, because the holder is never obliged to do anything. A contract with a nonnegative payoff and no corresponding obligation must be bought, and the price is the premium.

The consequence for cash is direct. On a forward, nothing changes hands at inception and everything at maturity, so credit exposure starts at zero and accumulates as the market moves, which is what Chapters 8 and 9 are about. On an option, the buyer pays at inception and then owes nothing further, so they post no margin; the writer receives cash at inception and carries an obligation for the life of the contract, and posts margin throughout.

*What the question tests.* That "zero value" is a statement about symmetry, not about safety. A forward with zero value can still ruin you.

---

**1.2** *A company enters a \$50 million notional five-year swap paying 3.8% fixed and receiving floating. In the first year floating averages 5.1%. Compute the net payment and state who pays whom. What fraction of notional is it?*

Only the difference between the two rates is exchanged:

$$(5.1\% - 3.8\%) \times \$50{,}000{,}000 \;=\; 1.30\% \times \$50{,}000{,}000 \;=\; \$650{,}000.$$

Floating exceeded fixed, so the company, the fixed payer, *receives* \$650,000 from its counterparty. As a fraction of notional it is 1.30%.

Two things follow. First, the \$50 million never moves. It is a multiplier on a rate difference, and calling this "\$50 million of derivatives" describes the multiplier rather than anything at stake. Second, a real swap resets each period and pays on each period's own accrual, so the annual figure is the sum of the period payments; taking the average rate times notional, as the question does, is exact only when the accrual factors are equal. The approximation is harmless here and is the reason the answer is quoted to the nearest thousand rather than the nearest dollar.

*What the question tests.* The gap between a headline notional and the cash it generates (1.3% in the first year), which is Question 1.3 in miniature.

**1.3** *Distinguish notional amount, gross market value, and gross credit exposure. A newspaper reports that a bank "has $3 trillion of derivatives." What further question should a reader ask?*

**Notional amount** is the reference quantity used to compute payments. It is not exchanged, except in cross-currency swaps where the principals genuinely are. The Bank for International Settlements puts the over-the-counter total at roughly $845 trillion at the end of 2025.

**Gross market value** is what the outstanding contracts are worth: the sum of what every position with a positive value would be paid if everything settled today, before any offsetting. Roughly $23 trillion, under 3% of notional.

**Gross credit exposure** nets each pair of counterparties' offsetting positions against one another, as their master agreements permit. Roughly $3.4 trillion, under half a percent of the headline, and most of that remainder is collateralized under the mechanisms of Chapter 8.

The question a reader should ask is: three trillion of which measure? The three differ by a factor of about 250 across the market, so the sentence is consistent with a bank running an enormous book or a modest one. A second question follows immediately: before or after collateral? Post-netting exposure that is fully collateralized is a different proposition from the same number uncollateralized.

*What the question tests.* That the three numbers answer three different questions (how much business, what is it worth, what could be lost), and that a figure quoted without saying which is not a small imprecision but an uninformative one.

**1.4** *An investor obtains $4 million of index exposure through futures requiring 6% initial margin. The index falls 12%. Compute the loss, express it as a multiple of the initial margin, and state what the investor owes.*

Initial margin is $0.06 \times \$4{,}000{,}000 = \$240{,}000$. The loss is the full move on the full exposure,

$$0.12 \times \$4{,}000{,}000 = \$480{,}000,$$

which is $\$480{,}000/\$240{,}000 = 2.0$ times the initial margin.

The investor owes the whole $480,000. Margin is a performance bond, not a premium and not a cap: as the index fell, variation margin calls demanded the loss in cash, day by day. The $240,000 already posted absorbs half of it, and the remaining $240,000 has to be found: fresh money, on demand, against a position that has just lost. If it is not found the position is closed out at whatever the market is then offering.

Compare an option. A buyer who paid $240,000 in premium for the same exposure could lose $240,000 and not a dollar more, and would never receive a call. That difference, not the leverage itself, is what distinguishes the two instruments as ways to be wrong.

*What the question tests.* That leverage runs both ways and that the margin figure describes the deposit, not the risk. Section 1.10 and Example 1.5 set up the arithmetic; most of the episodes in Section 1.11 are this calculation with a larger multiple and a counterparty who could not meet the call.

**1.5** *Explain why a market consisting only of hedgers would not function, and identify what speculators are being paid for.*

Hedgers as a group are net *sellers* of risk, and a sale requires a buyer. For a market of hedgers alone to clear, every exposure would need an equal and opposite exposure held by someone who wanted to shed it at the same moment — the same commodity, grade, quantity, delivery point, and date. Wheat farmers are short the harvest together, in the same season, in

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the same direction; the millers who want the other side are fewer, want different quantities, and want them at different times. Requiring that double coincidence would leave most exposures unhedged and would make the price of a hedge depend on whether a natural counterparty happened to exist that week.

Speculators supply the willingness to hold an exposure no one else wants, and to hold it *now* rather than when a match appears. That is a service, and it has a price: it shows up as the gap between the forward price and the expected future spot price, which Sections 5.13 and 5.14 work out in detail. The hedger pays it, knowingly, in exchange for certainty.

*What the question tests.* That speculation is structurally necessary rather than merely tolerated, and that the risk premium is the price of a service rather than evidence of a defect in the market.

---

**1.6** *Classify each as hedging, speculation, or arbitrage, and state what additional information you would need in each case: (a) a copper producer selling copper futures; (b) a fund buying gold futures and selling gold forward at a higher price; (c) an importer buying dollars forward; (d) a bank buying protection on a borrower to whom it has lent.*

**(a) Hedging**, if the quantity and the delivery dates line up with planned output. Needed: the production schedule and any forward sales already made. Futures sold in excess of output are not a hedge but a short position in copper, exposed to a rising price. Even a genuine hedge carries a cash risk: the LME nickel episode of Chapters 2 and 40 began with a producer's hedge, and when the price more than tripled in two days the losses on it ran to billions of dollars before any of the physical output could offset them.

**(b) Arbitrage**, if the two legs reference the same metal, quantity, and date, and the locked spread survives the cost of holding the position. Needed: the financing cost, the margin on the futures leg (a cash requirement the forward leg does not have, and a timing mismatch rather than a valuation one), the delivery specification, and the counterparty risk on the forward. If any of these differ, the spread is payment for a risk rather than a riskless profit, and the trade is speculation labeled as arbitrage.

**(c) Hedging**, if there is a contracted dollar payable of matching size and date. Needed: the underlying commercial exposure. Without a payable this is a long dollar position, whatever the importer calls it.

**(d) Hedging**, if the notional, maturity, and reference obligation match the loan. Needed: whether the contract references the same legal entity and the same seniority as the loan (the basis risk Chapter 25 takes seriously), and whether the protection bought exceeds the loan. Protection in excess of the exposure is a short credit position that profits from the borrower's default.

*What the question tests.* That the label is a property of the whole balance sheet rather than of the trade. The same contract is a hedge for one party and speculation for another, which is why the categories blur in practice and why the useful question is always what else does this party hold?

---

**1.7** *Using Figure 1.1, explain why the writer of a call is in a different business from the buyer of one, and what the writer must be able to do to be in it safely.*

Figure 1.1 plots profit (the payoff net of the premium), and the call panel is the asymmetry made visible. The buyer's line is flat at minus the premium and then rises without limit: bounded loss, unbounded gain, and no obligation once the premium is paid. The writer's line is the mirror image: flat at plus the premium and then falling without limit. Their maximum gain

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is known on day one and their maximum loss is not.

These are different businesses. The buyer has taken a view, or bought insurance, and can walk away. The writer has *sold* insurance and cannot. The premium is not profit; it is payment received for an obligation whose cost becomes known later.

To be in that business safely the writer must be able to do three things. First, hedge dynamically: hold delta shares against the position and rebalance as the price moves, which converts an open-ended payoff into a stream of rebalancing costs (Chapter 18). Second, fund the collateral that the mark-to-market will demand along the way; the obligation is a cash requirement before it is an accounting loss, which is Solution 1.4 again in another form. Third, survive gaps: dynamic hedging works when prices move continuously and fails when they jump, so the writer needs either capital sized to the jump or an offsetting option that caps the tail.

*What the question tests.* That the kink in the profit diagram is a statement about who is obliged, and that writing options is a manufacturing business with an inventory to manage, not the mirror trade of buying them.

---

**1.8** *Explain what changes when the same economic trade is executed on an exchange rather than bilaterally. Identify which of those changes transfers risk to someone else and which merely changes when cash moves, and state one exposure that the exchange does not remove.*

Three things change, and they are not the same kind of change.

**Standardization.** The exchange fixes the contract size, the underlying, the delivery months, and the settlement terms; the parties choose only price and quantity. This transfers no risk at all. It is a restriction, and what it buys is liquidity: many participants trading one contract rather than a few trading each of many bespoke ones.

**Novation to a central counterparty.** The clearing house becomes buyer to every seller and seller to every buyer, so exposure to a particular institution becomes exposure to the CCP and, through its default waterfall, to its members collectively. This is the transfer. It does not destroy counterparty risk; it pools it, which is why Chapter 2 spends most of its length on what happens when the pool is called on.

**Daily settlement and margin.** The gain or loss that a bilateral forward would accumulate to maturity is instead paid in cash, day by day. This is the change in timing. The same total changes hands; it changes hands earlier and in installments, and that is exactly why it creates a funding requirement rather than a credit exposure — the subject of Chapter 8, and the mechanism behind the 2020 dash for cash.

What the exchange does not remove is *market risk*. A short futures position and a short forward on the same underlying lose the same amount when the price rises; the clearing house changes who is owed and when, never whether. The distinction also blurs in practice: most standardized over-the-counter trades are now centrally cleared, so the useful question about a trade is no longer which venue it was struck in but whether it is cleared and collateralized.

*What the question tests.* That "exchange-traded is safer" is three separate claims of unequal strength, and that separating them is what makes Chapters 2 and 8 legible.

---

**1.9** *Using Example 1.1, explain what the single agreement clause prevents an administrator from doing in a bankruptcy, and use it to account for the fall from $23 trillion of gross market value to $3.4 trillion of gross credit exposure.*

In Example 1.1, A's five trades with B mark at +40, −35, +20, −15, −8 million. A's gross positive exposure is $60 million and the net is $2 million in A's favor.

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The single agreement clause makes those five trades *one* contract. What it prevents is **cherry-picking**: without it, an administrator would enforce the three contracts on which A owes (35 + 15 + 8 = $58 million, payable in full) while disclaiming the two on which A is owed, leaving A an unsecured claim on $60 million that recovers $12 million. A's net payment, what it pays out less what it recovers, becomes

$$58 - 12 = \$46 \text{ million}$$

in place of a $2 million claim. The trades did not change. The document did.

The market-level numbers are that arithmetic aggregated. Gross market value, roughly $23 trillion, is the sum of positive replacement values before any offsetting, the analog of A's $60 million. Netting under master agreements collapses it to gross credit exposure of roughly $3.4 trillion, the analog of A's $2 million: a reduction of about six-sevenths. Collateral posted under the Credit Support Annex then covers most of what survives.

*What the question tests.* That the reduction from $23 trillion to $3.4 trillion is a legal fact rather than an economic one. It exists only where close-out netting is enforceable, which is why enforceability in a jurisdiction is a precondition for a derivatives market existing there, and why capital rules recognize netted exposures only where it holds.

---

**1.10** *Archegos, as this chapter describes it, held a position no counterparty could see in full. Identify the instrument that made it invisible, state what each prime broker could observe and what it could not, and explain what would have had to change (in the instrument, the reporting, or both) for the aggregate position to be visible to any one of them.*

The instrument was the **total return swap**. The prime broker bought the shares and passed their return to Archegos, so Archegos held the economics of a concentrated equity position while owning no shares at all.

Each prime broker could see its own book: the notional it had written, the stocks it referenced, the collateral it held, and the concentration of its own exposure. Judged on that, each dealer's position was defensible. What no dealer could see was the same stocks financed simultaneously at several competitors — therefore not Archegos's aggregate position, not its true leverage, and not how large the combined holding was relative to the daily volume of the underlying shares. Ownership disclosure attaches to owning shares, and Archegos owned none, so nothing was filed.

Two changes would each have helped, and they act at different points. Change the *instrument's treatment*: count a total return swap as beneficial ownership for disclosure purposes, and a position of that size surfaces publicly, visible to every dealer at once. Or change the *reporting*: require financed positions to be reported to a repository that lets a prime broker see a client's total across dealers before extending more. The first makes the position public; the second makes it visible only to those extending the credit, which is the narrower fix and the more politically achievable one.

*What the question tests.* The consequence of the definition this chapter opens with. A derivative separates exposure to a risk from ownership of the thing that carries it — so a disclosure regime built around ownership has a blind spot exactly the shape of the derivatives market. Nothing about Archegos required a complicated instrument or a valuation model: it was leverage that was individually reasonable at each dealer and collectively enormous, seen through a regime that was looking at the wrong thing.

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# Chapter 2

# Futures Markets, Clearing, and Central Counterparties

**2.1** *Explain, using an arbitrage argument in both directions, why the futures price converges to the spot price at delivery.*

Suppose at delivery the futures price stands *above* the spot. Sell the futures contract, buy the asset in the spot market, and deliver it immediately. The proceeds are the futures price, the cost is the spot price, and the difference is riskless profit taken in the same instant. Traders doing this push the futures price down and the spot price up until the gap closes.

Now suppose the futures price stands *below* the spot. Anyone who wants the asset buys the futures contract and takes delivery rather than paying the higher spot price; the long cannot choose the moment of delivery, which the short controls, so this side is enforced by buyers' preference rather than by a textbook arbitrage, but it is enforced all the same. This pushes the futures price up.

Neither side requires a view about where the price is going, and the first needs no holding period or capital beyond the moment, because at delivery the futures contract *is* the asset: the contract has run out of time in which anything could happen. Away from delivery the two prices differ by the cost of carry, and that difference is not an arbitrage because carrying the asset costs money; at delivery there is nothing left to carry.

*What the question tests.* That convergence is enforced rather than observed. Nothing makes the two prices meet except that someone gains if they do not, and the argument has to be run in both directions because a one-sided argument only bounds the price from one side.

**2.2** *An investor sells three silver futures contracts, each for 5,000 ounces, at \$28.00 per ounce. Initial margin is \$9,000 per contract and maintenance margin \$7,000 per contract. Construct the margin account over five days if the settlement prices are \$28.20, \$28.55, \$28.30, \$28.95, and \$28.70. On which days is there a margin call, for how much, and why is the account restored to the initial rather than the maintenance level?*

Three contracts of 5,000 ounces is 15,000 ounces, so each cent of price movement is \$150. The position is short, so it loses when silver rises. Initial margin is $3 \times \$9{,}000 = \$27{,}000$ and maintenance is $3 \times \$7{,}000 = \$21{,}000$.

| Day | Settle | Change | Gain/loss | Balance | Action |
|---|---|---|---|---|---|
| 0 | 28.00 | — | — | 27,000 | — |
| 1 | 28.20 | +0.20 | −3,000 | 24,000 | above maintenance |
| 2 | 28.55 | +0.35 | −5,250 | 18,750 | **call \$8,250** |
| 3 | 28.30 | −0.25 | +3,750 | 30,750 | — |
| 4 | 28.95 | +0.65 | −9,750 | 21,000 | exactly at maintenance |
| 5 | 28.70 | −0.25 | +3,750 | 24,750 | — |

On day 2 the balance of \$18,750 is below the \$21,000 maintenance level, and the call is for the amount that restores the *initial* margin: \$27,000 − \$18,750 = \$8,250. The account then opens day 3 at \$27,000.

On day 4 the balance falls to exactly \$21,000 — the maintenance level, not below it — so on the usual convention that a call is triggered by a breach there is no call, though the position is one adverse tick away from one. A trader reading only the rule would report nothing; a trader reading the account would already be arranging cash.

The account is restored to the initial rather than the maintenance level because maintenance is a trigger and initial is a target. Topping up only to the trigger would leave the account one tick from another call, and the clearing house would spend its day issuing them. The gap between the two levels is deliberately a buffer, sized so that calls are occasional rather than continuous.

*What the question tests.* That the two margin numbers do different jobs, and that the arithmetic of a short position runs opposite to intuition trained on long ones.

---

**2.3** *Six firms trade with one another bilaterally. How many bilateral exposures exist, how many remain after novation to a CCP, and what is the general result for n firms?*

Bilaterally, every pair of firms has an exposure, so with six firms there are

$$\binom{6}{2} = \frac{6 \times 5}{2} = 15 \text{ bilateral exposures.}$$

After novation each firm faces the clearing house and no one else, so there are **six**. In general $n$ firms have $n(n-1)/2$ bilateral relationships and $n$ after novation — quadratic growth replaced by linear.

Two things follow, and only one is usually said. The obvious one is that $n(n-1)/2$ grows fast: a hundred firms have 4,950 bilateral exposures and a hundred cleared ones. The less obvious one is what the reduction costs. Every one of those $n$ exposures is to the same counterparty, so the network's risk has not been removed but concentrated into a single node whose failure would be a systemic event. That is why Sections 2.8 through 2.10 exist and why they are longer than the section that introduces novation.

*What the question tests.* That the case for clearing is a statement about network topology, and that the same arithmetic which makes it compelling is what makes the clearing house systemically important.

---

**2.4** *A clearing member defaults, generating a close-out loss of \$1.4 billion. The defaulter's margin is \$600m, its default fund contribution \$120m, the CCP's skin in the game \$75m, and the surviving members' default fund \$2.5 billion. Work through the waterfall in order, state what percentage of the surviving members' fund is consumed, and explain why the CCP's own capital sits ahead of surviving members' contributions.*

The waterfall is applied strictly in order, each layer exhausted before the next is touched:

| Layer | Absorbs (\$m) | Loss remaining (\$m) |
|---|---|---|
| Close-out loss | — | 1,400 |
| Defaulter's initial margin | 600 | 800 |
| Defaulter's default fund contribution | 120 | 680 |
| CCP skin in the game | 75 | 605 |
| Surviving members' default fund | 605 | 0 |

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The surviving members' fund absorbs \$605m of its \$2.5 billion, or

$$\frac{605}{2{,}500} = \mathbf{24.2}\%.$$

The fund survives, and no assessments or recovery tools are reached.

The CCP's own capital sits ahead of the surviving members' contributions for a reason that is about incentives rather than about size — \$75m against \$2.5 billion could never be about size. The clearing house sets the margin model and decides who may become a member. If its own money were behind the members' money, it could compete for volume by setting margin loosely and let the mutualized fund absorb the consequence. Placing its capital in front means the first institution to pay for a bad margin model is the institution that chose it.

*What the question tests.* That the ordering of the waterfall is not arbitrary and not merely legal. Each layer is placed where it is so that the party who can control a risk is the party who pays for it first.

---

**2.5** *A \$500 million position has an initial margin requirement set at three standard deviations of a three-day move. Calculate the requirement at annualized volatilities of 12% and 30%, state how much additional cash must be found, and explain why a practice that is prudent for one clearing house may be destabilizing for the system.*

The three-day standard deviation of a \$500 million position at annualized volatility $\sigma$ is $500\sigma\sqrt{3/252}$, and $\sqrt{3/252} = 0.10911$.

| Volatility | Three-day s.d. (\$m) | Requirement at $3\sigma$ (\$m) |
|---|---|---|
| 12% | 6.55 | 19.64 |
| 30% | 16.37 | 49.10 |

The additional cash required is $49.10 - 19.64 = \$\mathbf{29.5}$ million, and the requirement scales exactly with volatility — two and a half times the volatility, two and a half times the margin — because the position, the horizon and the confidence level are all unchanged.

Why prudence at one clearing house can destabilize the system: the call arrives *because* volatility has risen, which is to say in the middle of the disturbance rather than before it. Every clearing house runs the same calculation on the same day, so demands for cash are perfectly correlated across the system and land on members who are simultaneously being called elsewhere. A member short of cash sells assets; selling raises volatility; higher volatility raises margin again. The individual calculation is unimpeachable and the aggregate of many such calculations is a feedback loop — which is Section 2.9's subject, and the reason margin models are now required to look through the cycle rather than only at it.

*What the question tests.* That procyclicality is not a mistake in the model. It is what a correct model does when everyone runs one.

---

**2.6** *Distinguish volume from open interest over three days: (i) two new participants open a 20-contract position between them; (ii) one of them closes 8 to a new entrant; (iii) both original parties close 12. Give both figures for each day.*

| Day | Event | Volume | Open interest |
|---|---|---|---|
| 1 | two new participants open 20 between them | 20 | 20 |
| 2 | one closes 8 to a new entrant | 8 | 20 |
| 3 | both original parties close 12 | 12 | 8 |

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Day 1 creates twenty contracts where none existed, so both figures are 20. Day 2 is the instructive one: eight contracts change hands, so volume is 8, but one party has simply been replaced by another. No contract was created and none was extinguished, so open interest does not move. Day 3 has both original parties closing against each other, which extinguishes twelve contracts: volume 12 and open interest down to 8, held by the day-2 entrant against the remaining short.

*What the question tests.* That volume counts transactions and open interest counts obligations, and that the second is the one that says how much risk the market is carrying. A day of enormous volume that leaves open interest unchanged is a day on which positions moved between hands; a day of modest volume that lifts open interest sharply is a day on which the market took on exposure.

---

**2.7** *A trader is long 200 September contracts and short 200 December, with outright margin of $1,500 per contract and a spread rate of $250 per pair. Compare the portfolio and contract-by-contract requirements, and explain what happens to the position if the spread relationship breaks down.*

Charged contract by contract, the position is 400 contracts:

$$400 \times \$1{,}500 = \$600{,}000.$$

Charged as a spread, it is 200 pairs:

$$200 \times \$250 = \$50{,}000,$$

one twelfth of the outright figure. The clearing house grants the difference because the two legs are highly correlated: a parallel move in the September and December prices produces offsetting gains and losses, so what the position is exposed to is not the level of the price but the *difference* between two deliveries, which historically moves far less.

What happens if that relationship breaks down is the whole risk of the position, and it arrives in two forms at once. The spread itself moves against the trader, which is the loss; and the clearing house withdraws the spread credit and re-margins the legs as outrights, which is the call. The requirement jumps from $50,000 to $600,000 — a twelvefold increase — at exactly the moment the position is losing money, so the trader must find twelve times the cash to hold a position that has just been shown to be riskier than the margin assumed.

*What the question tests.* That a margin offset is a statement about correlation, and that every such statement fails precisely when it matters. This is the March 2020 basis-trade unwind in miniature and the reason Section 2.9 treats margin as a market risk rather than an administrative detail.

---

**2.8** *Explain the difference between individual and omnibus segregation of client collateral, why a client might not know which it has, and why the chance of being ported is lowest exactly when it is most needed.*

Under **individual segregation** a client's collateral is held in an account of its own, identifiable as that client's. Under **omnibus segregation** clients are pooled in one account, separate from the member's own assets but not from each other. Omnibus is cheaper, and the saving is real; what it buys is a worse outcome in a default, because a shortfall caused by any client in the pool can reach the collateral of the others, who did nothing.

A client may not know which it has because the choice is made in the account documentation at onboarding, is priced as a fee difference rather than presented as a risk decision, and does not surface again in any routine report. The distinction has no effect whatsoever until the member fails, which is the one day it determines everything.

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Porting is the transfer of a client's positions and collateral from the failed member to a surviving one. The chance of it succeeding is lowest exactly when it is most needed, and for a reason that is structural rather than administrative. Porting requires another member willing to accept the positions *and* the collateral supporting them. A member fails in a stressed market; in a stressed market the surviving members are managing their own calls and are least willing to take on someone else's book; and the positions on offer are precisely those the market has just moved against. The willingness to accept a client is lowest when the need to place one is highest, so the mechanism is most likely to work in the circumstances where it is least required.

*What the question tests.* That protections which hold in calm conditions and fail in stress are not protections, and that the client's own paperwork is where this one is decided.

---

**2.9** *Explain what a corner or squeeze is and how position limits are meant to prevent it. Why did the nickel episode produce similar market effects without being a manipulation, and what does the cancellation of executed trades imply about finality?*

A **corner** is a dominant long futures position held together with control of the deliverable supply; a **squeeze** is what follows when shorts must buy from that holder to close or deliver, at whatever price is asked. Position limits cap how much of a contract any one participant may hold, so that no single party can reach the size at which the deliverable market becomes theirs.

The nickel episode produced the same market effect without the same cause. There was no attempt to control supply and no manipulation: the short position was a hedge, held by a producer against its own output, and the price rose because shorts were scrambling to cover, not because the shorts were being held to ransom by a cornering party. The mechanism was a short position that could not be covered, arrived at honestly, and it demonstrates that the harm position limits target can be produced by ordinary hedging at sufficient size.

The cancellation of executed trades is the part with consequences beyond that market. Clearing rests on the premise that a matched trade is final — it is what allows novation, netting, and margin to mean anything, because each depends on knowing what the book contains. If executed trades can be undone after the fact, then no participant can be certain a completed hedge is a hedge, and the value of every position becomes contingent on a later decision by the venue. The immediate loss fell on those whose profitable trades were canceled; the lasting cost is to the assumption of finality that the rest of the apparatus is built on.

*What the question tests.* That a rule can be well designed for the case it was written for and silent about a case that looks identical from the outside, and that finality is not a technicality but the premise clearing depends on.

---

**2.10** *Futures markets are defended on the grounds that they produce price discovery. Explain what this means, who benefits, why it is an argument for tolerating speculation that serves no hedging purpose, and under what conditions the argument fails.*

**Price discovery** is the production, by many participants trading on their own information, of a continuously updated and publicly visible forward curve. No individual could produce it: each knows a little about their own supply, demand, or funding, and the price is the aggregation of all of it into a single number per delivery date that anyone may read without paying for it.

Who benefits is wider than who trades. A farmer who never opens an account can read the September price and decide what to plant; a utility can see whether forward power justifies a plant; a central bank can read expected policy from the prices of short-rate futures across successive delivery months. The information is a public good produced as a by-product of private speculation, and consumed mostly by people who contribute nothing to it.

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That is the argument for tolerating speculation with no hedging purpose. The speculator is not required to be socially useful in intent. Their orders carry information into the price and their willingness to take the other side is what lets a hedger transact at all, and both effects are unaffected by whether they own any of the underlying.

The argument fails under conditions worth naming precisely. It fails when trading is not informed but mechanical — positions taken because an index rebalances or a rule fires, which add volume and no information. It fails when the price reflects the positioning rather than the information, as in a squeeze, where the curve stops describing supply and demand and starts describing who owes what to whom. And it fails when liquidity is present in calm markets and absent in disturbed ones, since a price that is discoverable only when no one urgently needs it is not doing the job claimed for it.

*What the question tests.* That the standard defense of speculation is consequentialist rather than moral, and that a consequentialist defense carries its own falsification conditions — which is why the question asks for them.

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# Chapter 3
# Hedging Strategies Using Futures

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**3.1** *Set out the argument that a firm should not hedge because its shareholders can diversify, and give two reasons the argument may fail.*

The argument runs: a shareholder who dislikes exposure to the oil price can sell oil shares, hold a diversified portfolio, or hedge on their own account. They can do it more cheaply than the firm, because they need no treasury function, and more precisely, because only they know how much oil exposure the rest of their portfolio already carries. A firm that hedges is therefore spending money to supply something its owners can obtain for nothing, and in the process it removes exposure some of them may have bought the shares to get. In a world of frictionless markets and fully informed shareholders the argument is correct.

It fails for two reasons that the frictions supply.

**Financial distress is costly, and the cost is not linear.** A firm's value does not fall smoothly with a bad price outcome. Below a threshold it loses customers who doubt it will be there to honor warranties, suppliers tighten terms, lenders reprice, and management spends its time on refinancing. None of that is diversifiable by a shareholder, because it is destroyed value rather than transferred value. Hedging removes the left tail where those costs live, and that is worth paying for even to a perfectly diversified owner.

**The firm knows its exposure and the shareholder does not.** Only the firm knows its hedged position, its input mix, its contracted volumes, and the timing of its commitments. A shareholder replicating the hedge would be guessing at all of it. There is also a planning argument: a firm that knows its input cost can commit to investment and to prices for its own customers, which a firm facing an unknown cost cannot.

*What the question tests.* That the case for corporate hedging is not "risk is bad" but a specific claim about frictions, and that a reader who cannot state the argument against hedging cannot tell when hedging is actually warranted.

---

**3.2** *Define the basis, show that a short hedger who lifts a hedge before delivery receives $F_1 + b_2$, and explain what this implies about where the residual risk lies.*

The **basis** is the spot price minus the futures price:

$$b = S - F.$$

A short hedger holds the asset and is short futures. At $t_1$ they place the hedge with the spot at $S_1$ and the futures at $F_1$; at $t_2$ they sell the asset and close the futures. They receive $S_2$ for the asset and $F_1 - F_2$ on the futures leg, so the effective price is

$$S_2 + (F_1 - F_2) = F_1 + (S_2 - F_2) = F_1 + b_2.$$

The decomposition is the whole point. $F_1$ is known when the hedge is placed; $b_2$ is not. The hedge has therefore converted an exposure to the level of the price into an exposure to the basis at the moment of lifting — and nothing else.

What that implies about the residual risk needs care. It is *not* that the hedge eliminates risk; it is that the risk left over is the behavior of one price relative to another, at one date. Whether that is a good trade depends on how variable the basis is compared with the spot, which is why a hedge in a market whose basis is erratic (an unusual grade, a remote delivery point, a commodity that cannot be stored) can be worse than no hedge at all.

*What the question tests.* That "the hedge works" means the basis is stable, not that the price is stable, and that the identity is what tells you where to look.

---

**3.3** *A hedge is placed when the spot price is \$44 and the futures price is \$41.50, and lifted when the spot price is \$47 and the futures price is \$45.20. Compute the effective price for a short hedger, state whether the basis strengthened or weakened, and explain why a June exposure is usually better hedged with a July contract than a June one.*

The bases at the two dates are

$$b_1 = 44.00 - 41.50 = 2.50, \qquad b_2 = 47.00 - 45.20 = 1.80.$$

The short hedger's effective price comes out the same by either route:

$$S_2 + (F_1 - F_2) = 47.00 + (41.50 - 45.20) = \$\mathbf{43.30},$$

$$F_1 + b_2 = 41.50 + 1.80 = \$43.30. \quad \checkmark$$

The basis fell from 2.50 to 1.80, which is a **weakening** of the basis, and it cost the short hedger 0.70 a unit against the \$44.00 spot price prevailing when the hedge was placed. The sign follows directly from $F_1 + b_2$: a short hedger is long the basis, so a weakening basis hurts them and a strengthening one helps.

On contract choice, a June exposure is normally hedged with the July contract rather than the June one, for three reasons that compound. Futures prices behave erratically during the delivery month, as the contract converges and the residual open interest thins. A short hedger holding a contract into its delivery month may be required to deliver, which for a hedger with a different grade or location is an expensive accident. And liquidity has by then migrated to the next contract, so lifting the hedge is done into a thinner market. The general rule is to hold a contract that expires *after* the exposure and to lift it before delivery, accepting the basis risk that creates.

*What the question tests.* That the choice of contract month is a risk decision rather than a clerical one, and that the price of avoiding delivery-month trouble is exactly the basis risk of question 2.

---

**3.4** *A company will buy 800,000 units of a commodity in three months. The standard deviation of the change in its price over three months is \$0.65, that of the futures price \$0.81, and the correlation 0.72. Each contract covers 50,000 units. Compute the minimum-variance hedge ratio and the number of contracts, state what proportion of the variance remains, and explain why that ratio is the slope of a regression.*

The minimum-variance hedge ratio is

$$h^* = \rho \, \frac{\sigma_S}{\sigma_F} = 0.72 \times \frac{0.65}{0.81} = \mathbf{0.5778}.$$

The exposure is 800,000 units and each contract covers 50,000, so the unhedged position is 16 contract-equivalents and the hedge is

$$0.5778 \times 16 = 9.24 \longrightarrow \mathbf{9 \text{ contracts}},$$

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bought, not sold: the company is a future *buyer* of the commodity, so it is short the price and hedges long.

The proportion of variance remaining is

$$1 - \rho^2 = 1 - 0.72^2 = 1 - 0.5184 = \mathbf{48.2}\%,$$

so the optimal hedge removes only about half the variance. That is a property of the correlation alone, not of the hedge ratio, and not improvable by trading more contracts. Hedging more than $h^*$ increases variance rather than reducing it.

$h^*$ is the slope of the regression of $\Delta S$ on $\Delta F$ because the slope is

$$\frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \frac{\rho\, \sigma_S \sigma_F}{\sigma_F^2} = \rho\, \frac{\sigma_S}{\sigma_F},$$

which is the same expression. That is not a coincidence but the content of the result: minimizing the variance of $\Delta S - h\,\Delta F$ over $h$ is least squares, and the answer is the least-squares slope. It also explains why $1 - \rho^2$ is what is left: it is the regression's unexplained fraction.

*What the question tests.* That the hedge ratio and the regression slope are the same object, and that $\rho$ sets a ceiling on what any hedge in that contract can achieve.

---

**3.5** *A $12 million portfolio has a beta of 0.9. Index futures are worth $50 per point and the index stands at 4,000. How many contracts hedge it fully, and how many reduce beta to 0.4?*

One index point is worth \$50, and the index stands at 4,000, so one contract covers

$$4{,}000 \times 50 = \$200{,}000.$$

To hedge fully, short

$$\beta \times \frac{V}{\text{contract value}} = 0.9 \times \frac{12{,}000{,}000}{200{,}000} = 0.9 \times 60 = \mathbf{54\ contracts}.$$

To move beta from 0.9 to 0.4, short the difference:

$$(0.9 - 0.4) \times 60 = 0.5 \times 60 = \mathbf{30\ contracts}.$$

The general form is $(\beta - \beta^*)V/(\text{contract value})$, with a positive answer meaning a short position and a negative one meaning a long. It is worth noticing that the manager does not sell a single share to achieve either result: the portfolio is untouched and its beta is changed by an overlay. That is the practical reason index futures exist, and it is Section 3.6's point that adjusting exposure and trading the underlying are separable decisions.

*What the question tests.* That beta hedging is a scaling of the same contract count, and that the second part is the first with a target rather than zero.

---

**3.6** *An analyst estimates a hedge ratio of 0.46 from daily data and 0.64 from monthly data for a hedge held one month. State which should be used, explain why the two differ in this direction, and explain what goes wrong if the regression is run on levels rather than changes.*

**Use 0.64.** The estimation interval should match the hedging horizon: the hedge is held for a month, so what matters is how spot and futures move together over a month. A ratio fitted to daily changes answers a different question.

The two differ in this direction (short-horizon estimates lower) for a reason that recurs whenever prices are sampled finely. Over a day, the two series do not move together cleanly: they trade in different venues at different moments, quotes go stale, one market closes before the other, and a move in the futures may be reflected in the cash market only the following

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morning. Each of those puts noise into $\Delta F$ that is unrelated to $\Delta S$, and noise in the regressor *attenuates* the slope toward zero. Lengthening the interval lets the lagged responses complete inside the observation, so the measured relationship approaches the economic one. Estimating on daily data and hedging for a month therefore systematically underhedges.

Running the regression on levels rather than changes goes wrong more seriously. Spot and futures prices both wander without a fixed mean, and two such series will appear strongly related whether or not they have anything to do with each other, because both are trending. The regression reports a high $R^2$ that measures shared drift rather than shared movement; the residuals are heavily autocorrelated, so the standard errors are meaningless; and the slope is not the hedge ratio, because a hedge is a statement about *changes* — the hedger does not care where the price is, only how far it moves between placing the hedge and lifting it.

*What the question tests.* That the hedge ratio is an empirical estimate with the failure modes of an estimate, and that both failures here push in a direction a hedger would not notice: one leaves them underhedged, the other leaves them confident.

---

**3.7** *Give three reasons a firm might deliberately hedge less than the variance-minimizing amount, and explain what layered hedging achieves that a single transaction does not.*

Section 3.7 gives three reasons, each defensible.

**The exposure itself is uncertain.** The ratio is applied to an *expected* quantity. A producer that hedges all of its expected output and then produces less holds the surplus as a naked short futures position, which loses if prices rise — and a shortfall shared across the industry tends to raise them. An overhedge is a speculative position, which is the failure the firm was trying to avoid, and hedging a fraction leaves room for the quantity to come in low. The same logic applies to the ratio: $h^*$ is a regression estimate with a standard error, and hedging below it accepts a little more price risk in exchange for less exposure to the ratio being wrong.

**Shareholders may want some exposure.** An investor who buys a copper miner is buying copper. A fully hedged miner has removed the exposure its owners chose to hold, which is the diversification argument of question 1 turned around: owners who can manage the exposure themselves may prefer the firm to leave it with them.

**The hedge consumes cash.** Futures require variation margin, and the hedge loses cash precisely when the spot position is gaining, which is a timing mismatch rather than an economic loss. A firm with limited liquidity may rationally carry less hedge than the variance calculation recommends, because the full hedge could fail on funding before it pays off. That is Chapter 8's problem, and Metallgesellschaft's.

**Layered hedging**, placing the hedge in tranches over time rather than at one moment, achieves something a single transaction cannot: it removes the timing decision. A single hedge locks one price, chosen on one day, and its quality is judged against every price that followed; that is a forecast presented as a policy, and it exposes whoever chose the day. Layering delivers the average of several dates, which is worse than the best single day and better than the worst, and converts an unrepeatable judgment into a rule the treasury can follow and defend. It also spreads the margin requirement across time instead of concentrating it.

*What the question tests.* That the variance-minimizing hedge answers a precisely posed question which is often not the firm's question, and that the governance value of a rule is a real consideration rather than a soft one.

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**3.8** *A manager hedges an equity portfolio with index futures using the beta-adjusted contract count. Show that if the index moves by a proportion m, the position required changes by the factor* $(1+\beta m)/(1+m)$*. Explain why a portfolio with a beta of one never needs rebalancing on account of a market move alone, state the condition on* $\beta$ *and m under which the hedge must be* increased*, and evaluate the factor for a portfolio with a beta of 0.8 after the index rises 12%.*

The count is $N = \beta V_A / V_F$. A move of $m$ in the index revalues the portfolio to $V_A(1+\beta m)$, by the definition of beta, and one contract to $V_F(1+m)$, since a contract is worth the index times a fixed multiplier. So

$$\frac{N_{\text{new}}}{N_{\text{old}}} = \frac{\beta V_A(1+\beta m)/\left[V_F(1+m)\right]}{\beta V_A/V_F} = \frac{1+\beta m}{1+m}.$$

*Beta of one.* The factor becomes $(1+m)/(1+m) = 1$ for every $m$. The exposure and the contract are revalued in exactly the same proportion, so their ratio, which is all the hedge depends on, does not move. What creates the need to rebalance is not the market move but the amount by which $\beta$ departs from one.

*When the hedge must be increased.* The factor exceeds one when $1+\beta m > 1+m$, that is when $m(\beta-1) > 0$: a high-beta portfolio in a rising market, or a low-beta portfolio in a falling one. In both the exposure has grown relative to the value of one contract. The opposite pairings call for a reduction.

*The numerical case.* With $\beta = 0.8$ and $m = +0.12$,

$$\frac{1+0.8\times 0.12}{1.12} = 0.9786$$

so the required position falls by about 2.1%. Here $m > 0$ while $\beta < 1$, so $m(\beta-1) < 0$ and the condition above correctly predicts a reduction. On a 200-contract hedge that is four contracts, which is why a band rather than a daily adjustment is the sensible policy.

*What the question tests.* Whether the reader sees that a beta hedge drifts only through $\beta - 1$, and can therefore judge how often rebalancing is actually needed instead of assuming it must be continuous.

**3.9** *A bank hedges a portfolio of corporate bonds with Treasury futures. Explain what basis risk remains, why a historical regression over a calm period would understate it, and what happens to both legs in a credit sell-off.*

Treasury futures hedge the *risk-free* component of a corporate bond's yield. What remains is the **credit spread**: the difference between the corporate yield and the Treasury yield, plus the smaller mismatches of duration, convexity, and the behavior of the cheapest-to-deliver bond (the particular Treasury the futures price tracks, as Chapter 6 explains). The hedge is a bet that the spread is stable.

A historical regression over a calm period understates the risk for a structural reason rather than a statistical one. In calm markets corporate and Treasury yields move together, driven by the same rate expectations, so the measured correlation is high and the residual spread variance is small. The regression is an accurate description of the calm regime and says nothing about the other one, because the other one is not in the sample. The estimate is not noisy; it is answering a question about a period the hedge will eventually leave.

In a credit sell-off both legs lose, and that is the point of the question. The corporate bonds fall as spreads widen. Simultaneously, money moves into government bonds (the flight to quality), so Treasury prices *rise* and the short futures position loses too. The correlation the hedge relied on does not merely weaken; it inverts. The hedger is left with a loss on the asset

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and a loss on the hedge, at the same moment, and margin is owed in cash on the futures leg while the bond loss is unrealized.

*What the question tests.* That basis risk is not a small residual but the whole of what a cross hedge retains, and that estimating it from a benign sample produces a number whose failure mode is correlated with everything else going wrong.

---

**3.10** *Using the roll table of Example 3.6, recompute the effective price if the market had been in backwardation, with each roll closing at a price 0.40* above *the next contract, and state who gains in each case.*

In the example the market is in contango and each roll closes a contract and opens a more expensive one, gaining the short hedger +0.60, +0.50, +0.40, or \$1.50 in total. Reverse the curve. If each roll now closes at a price 0.40 *above* the contract being opened, each roll earns the short

$$-0.40 \quad \text{instead of a gain,}$$

and with three rolls the roll contribution is

$$3 \times (-0.40) = -\$1.20.$$

Holding the rest of the example fixed (the hedge placed at 80.00 and a final basis of $73.90 - 74.00 = -0.10$), the effective price becomes

$$80.00 - 1.20 - 0.10 = \$\mathbf{78.70},$$

against \$81.40 in the contango case: a swing of \$2.70 a barrel, or \$2.7 million on the million-barrel hedge, produced entirely by the shape of the curve and not by the level of the price.

Who gains is determined by the direction of the position, not by the merits of the hedge:

| | **Contango** | **Backwardation** |
|---|---|---|
| Short hedger (producer) | gains on each roll | loses on each roll |
| Long hedger (consumer) | loses on each roll | gains on each roll |

A rolling hedge is therefore two positions: a hedge against the price, and an unavoidable exposure to the shape of the curve, which no one chose and which compounds with every roll. Metallgesellschaft is the extreme case — long futures rolled in a market that moved into contango, so the roll losses accumulated while the offsetting supply contracts stayed unrealized.

*What the question tests.* That the roll is not an administrative step but a recurring trade, and that its sign depends on the curve rather than on whether the hedge is otherwise well constructed.

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# Chapter 4
# Interest Rates in the Risk-Free-Rate Era

**4.1** *Explain the two distinct defects in LIBOR's construction that led to its replacement, and state which of the two would have justified replacement even if no manipulation had ever occurred.*

The two defects are of different kinds, and only one of them is about dishonesty.

*It was a submission, not a transaction.* Panel banks were asked at what rate they *could* borrow, not at what rate they had. The answer was a judgment, unverifiable against any trade, and a judgment that its author had reasons to shade toward a level that flattered the bank's perceived creditworthiness, or that suited a derivatives position benchmarked to the result. That is the manipulation story, and it produced the fines.

*The market it described had largely ceased to exist.* Unsecured interbank term lending shrank after 2008 and never recovered. By the end, submissions for some currencies and tenors rested on almost no underlying transactions at all, so the benchmark was an expert estimate of the price in a market where hardly anyone traded.

The second would have justified replacement even if every submission had been honest. A benchmark referenced by hundreds of trillions of dollars of contracts cannot rest on a market that has disappeared, however scrupulous the panel: there is nothing to estimate, and no amount of good faith supplies the missing transactions. The manipulation made the reform urgent; the emptiness made it necessary.

*What the question tests.* That the case against LIBOR is usually told as a scandal and is more accurately a structural one, which is why its replacements are all built on observed overnight transactions.

**4.2** *An investment pays 12% per annum with quarterly compounding. Give the equivalent continuously compounded and annually compounded rates, and give the semiannual equivalent of a rate quoted as 5% continuously compounded.*

With $m$ compoundings a year at rate $R_m$, and a continuously compounded rate $R_c$,

$$R_c = m \ln\!\left(1 + \frac{R_m}{m}\right), \qquad R_m = m\left(e^{R_c/m} - 1\right).$$

For 12% with quarterly compounding:

$$R_c = 4\ln\!\left(1 + \frac{0.12}{4}\right) = 4\ln 1.03 = \mathbf{11.8235}\%,$$

$$R_1 = (1.03)^4 - 1 = \mathbf{12.5509}\%.$$

For 5% continuously compounded, the semiannual equivalent:

$$R_2 = 2\left(e^{0.05/2} - 1\right) = 2 \times (1.025315 - 1) = \mathbf{5.0630}\%.$$

Notice the ordering: 11.8235% continuous, 12% quarterly, 12.5509% annual all describe the *same* investment. The more frequent the compounding, the lower the quoted rate needed to deliver a given return, so a rate is meaningless until its frequency is attached, and comparing two quoted rates without converting them first is the most common arithmetic error in the subject.

*What the question tests.* Fluency in both directions of the conversion, which every later chapter assumes without comment.

---

**4.3** *Explain why SOFR can be described as risk-free even though it rose from 2.43% to 5.25% in a single day in September 2019, and state what kind of risk that episode illustrates.*

**Risk-free** in this context means free of *credit* risk, not free of variability. SOFR is the rate on overnight loans secured by Treasury collateral: the lender holds government securities against the cash, so if the borrower fails the collateral is sold. There is essentially no expectation of loss, which is what the term denotes.

What moved in September 2019 was not credit but the *supply of cash*. A corporate tax date and a large Treasury settlement drained reserves from the system on the same morning; dealers needing to finance inventory found fewer lenders; and the rate cleared far higher because cash was scarce, touching 5.25% against a prior 2.43%. Every one of those loans remained fully collateralized, and none of them defaulted.

The episode therefore illustrates **liquidity** or funding risk: the price of cash today rather than the probability of repayment tomorrow. A rate can be free of credit risk and still be extremely volatile, because the two are answers to different questions, and a reader who treats "risk-free" as "stable" will misread every overnight benchmark in the book.

*What the question tests.* That the replacement benchmarks removed credit risk from the reference rate and did not remove, and were never intended to remove, the volatility that comes from funding conditions.

---

**4.4** *An interest period runs Monday to Monday, seven calendar days. The overnight rates on Monday through Friday are 3.95%, 3.96%, 3.94%, 3.97%, and 3.96%. Using (4.1), calculate the compounded rate on an actual/360 basis and compare it with the day-weighted simple average.*

Compounding in arrears on actual/360 accrues each day's rate over the days it covers, with Friday's rate carrying the weekend:

$$\left[\prod_i \left(1 + r_i \frac{n_i}{360}\right) - 1\right] \times \frac{360}{7}.$$

| Day | Rate | Days | Factor |
|---|---|---|---|
| Monday | 3.95% | 1 | 1.00010972 |
| Tuesday | 3.96% | 1 | 1.00011000 |
| Wednesday | 3.94% | 1 | 1.00010944 |
| Thursday | 3.97% | 1 | 1.00011028 |
| Friday | 3.96% | 3 | 1.00033000 |
| **Product** | | **7** | **1.00076966** |

$$(1.00076966 - 1) \times \frac{360}{7} = \mathbf{3.9583}\%.$$

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The day-weighted simple average is

$$\frac{3.95 + 3.96 + 3.94 + 3.97 + 3 \times 3.96}{7} = \frac{27.70}{7} = \mathbf{3.9571}\%,$$

so compounding adds **0.11 basis points** over the week.

The size of the gap is the useful part of the answer. It is negligible here because the period is short and the rates are low, and it is *not* negligible over a year or at high rates, since the difference grows roughly with the square of the rate level and in proportion to the term. The convention matters for a second reason as well: Friday's rate carrying three days means a policy move or a year-end distortion landing on a Friday is weighted three times as heavily as one landing on a Tuesday.

*What the question tests.* That the compounded rate and the average rate are different objects, and that knowing when the difference is immaterial is as useful as being able to compute it.

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**4.5** *Explain the difference between a lookback with an observation shift and one without, and why the market generally prefers the shifted convention.*

Both conventions exist because a payment computed from rates observed up to the end of a period cannot be settled at the end of that period: no one can pay an amount that is only known that morning.

A **lookback** observes each day's rate a fixed number of business days before the day it applies to, so the rate for the final day of the accrual is known several days early. The accrual *weights*, however, still come from the interest period itself.

A lookback **with an observation shift** moves the weights as well: each observed rate is applied with the day count of the day on which it was *observed*, not the day to which it would otherwise apply.

The market prefers the shifted version for two reasons. The valuation reason is the one the chapter gives: without the shift each rate is weighted by days that belong to a different date, a weighting mismatch, so the compounded rate cannot be reproduced from discount factors and is not exactly the rate the curve prices; with the shift the observation period is a genuine compounding period. The practical reason is reproducibility. Without the shift, the rate and its weight come from different calendars, so the compounded amount cannot be reproduced from the published index alone. The calculation agent must rebuild the accrual by hand, and two parties doing so can differ. With the shift, the whole period's interest is a ratio of two published index values, so both sides compute the same number from the same public data and there is nothing to reconcile.

*What the question tests.* That the conventions of this market were chosen for operational reproducibility, not elegance, and that a convention which cannot be checked against a public index will be disputed sooner or later.

---

**4.6** *Explain why it mattered that the ISDA fallback spread adjustments were fixed permanently on a single date rather than recalculated periodically.*

The fallback spread adjustment converts a LIBOR-referencing contract into one referencing a risk-free rate plus a fixed increment, and the increment is meant to capture the average credit and term premium LIBOR carried.

Fixing it permanently on one date, from a five-year historical median, mattered because a floating adjustment would have been an open position. Had the spread been recalculated periodically, every legacy contract would have contained an unknown future adjustment, and its

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value would depend on a number no one could hedge, so the transition itself would have created a new risk in hundreds of trillions of dollars of contracts written before anyone contemplated it.

Fixing it also removes the incentive to influence it. A recalculated spread is something a large enough participant might try to move, at exactly the moment positions were most sensitive to it, which is the defect LIBOR was being replaced for.

The cost of fixing is that the number is right on average and wrong on any particular day: a contract falling due when actual bank credit spreads are far from the historical median transfers value from one side to the other. That was accepted deliberately, on the view that a known arbitrary number is easier to live with than an unknown moving one.

*What the question tests.* That the transition's design problem was distributional (somebody had to bear the difference) and was solved by making it certain rather than fair.

---

**4.7** *Annual-paying OIS rates are 3.50%, 3.70%, and 3.85% for one, two, and three years. Bootstrap the zero rates and discount factors, calculate the one-year forward rates beginning in one and two years, and explain why an upward-sloping zero curve implies a forward curve that rises more steeply.*

For annual-paying OIS, the par rate $s_n$ satisfies $s_n \sum_{i=1}^{n} \mathrm{DF}_i + \mathrm{DF}_n = 1$, solved outward:

| Maturity | Par rate | Discount factor | Zero rate |
|---|---|---|---|
| 1y | 3.50% | 0.966184 | 3.5000% |
| 2y | 3.70% | 0.929847 | 3.7037% |
| 3y | 3.85% | 0.892636 | 3.8584% |

The one-year forwards follow from the ratio of adjacent discount factors:

$$f_{1\times 2} = \frac{0.966184}{0.929847} - 1 = \mathbf{3.9078}\%, \qquad f_{2\times 3} = \frac{0.929847}{0.892636} - 1 = \mathbf{4.1687}\%.$$

Why an upward-sloping zero curve forces a forward curve that rises more steeply: the $n$-year zero rate is an average of the one-year forwards out to $n$. If an average is increasing as more terms are added, each new term must exceed the running average, or it would pull the average down. So the forwards sit above the zero rates and climb faster. Here the zero rates rise 3.50 to 3.86 while the forwards run 3.50, 3.91, 4.17.

The practical consequence is that the forward curve exaggerates the slope of the zero curve, so a modest upward tilt in observed rates implies forwards that look like a strong prediction of rising rates, a reading question 11 returns to.

*What the question tests.* That bootstrapping is a sequence of one-unknown problems, and that the forward-above-average relation is arithmetic rather than an economic claim.

---

**4.8** *A five-year bond pays a 6% annual coupon and yields 5%. Calculate its price, duration, and modified duration, estimate the price change for a 25 basis point increase in yield, and compare with the exact repricing.*

Discounting the 6% annual coupons and the redemption at 5%:

$$P = \sum_{t=1}^{5} \frac{6}{1.05^t} + \frac{100}{1.05^5} = \mathbf{104.3295}.$$

The bond trades above par because its coupon exceeds its yield.

Macaulay duration is the cash-flow-weighted average time:

$$D = \frac{1}{P} \sum_{t} t \, \frac{C_t}{1.05^t} = \frac{467.1614}{104.3295} = \mathbf{4.47775} \text{ years},$$

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shorter than the five-year maturity because the coupons return value earlier. Modified duration adjusts for the compounding frequency:

$$D^* = \frac{D}{1+y} = \frac{4.47775}{1.05} = \mathbf{4.2645}.$$

The estimated price change for a 25 basis point rise:

$$\Delta P \approx -D^* P\,\Delta y = -4.2645 \times 104.3295 \times 0.0025 = \mathbf{-1.1123}.$$

Repricing exactly at 5.25% gives 103.2248, a change of $\mathbf{-1.1047}$. The duration estimate overstates the loss by 0.0076, about 0.7% of the move.

The direction of the error is not accidental. Price is a convex function of yield, and duration is the tangent to it, so a straight line drawn at the current yield lies *below* the curve on both sides: it overstates the fall when yields rise and understates the gain when they fall. The gap is convexity, it is in the holder's favor for an ordinary bond, and it grows with the square of the yield move: negligible at 25 basis points, material at 200.

*What the question tests.* That duration is a first derivative and inherits a first derivative's limitations, and that the sign of its error is predictable.

---

**4.9** *A forward rate agreement fixes a rate on a notional for a future period. Explain how the forward rate is extracted from two zero rates, why the contract settles at the start of the period rather than at its end, and what that settlement convention requires of the discounting.*

A forward rate agreement fixes the rate on a notional for a period running from $T_1$ to $T_2$. The rate is not chosen; it is implied by the two zero rates already observable today, because borrowing to $T_2$ must cost the same as borrowing to $T_1$ and rolling. An FRA quotes a simple rate over the accrual $\tau = T_2 - T_1$, so the forward $F$ is extracted from the ratio of the two discount factors, or equivalently from the two annually compounded zero rates:

$$1 + F\tau = \frac{\mathrm{DF}_1}{\mathrm{DF}_2} = \frac{(1+z_2)^{T_2}}{(1+z_1)^{T_1}},$$

so $F$ is whatever makes the two routes cost the same. If it were not, one route would be cheaper and the difference could be taken without risk. The annually compounded forward $F_a$ of (4.8) is a different number unless $\tau$ is one year, and using it as the FRA rate misstates the contract.

The contract settles at $T_1$, the *start* of the period it covers, rather than at $T_2$. The reason is that by $T_1$ everything is known: the reference rate has fixed, so the payment can be computed, and there is no reason to wait a further period to exchange a determined amount; waiting would only add counterparty exposure for no purpose.

Settling early requires the payment to be *discounted*. The economic gain belongs at $T_2$, so paying it at $T_1$ means paying its present value. With the contract rate $R_K$ set equal to the forward $F$ at inception and $R$ the rate that fixes at $T_1$, the party paying fixed receives

$$\text{settlement} = \frac{(R - R_K)\,\tau\,L}{1 + R\,\tau},$$

discounted at the rate that actually fixed, not at the contract rate. The detail matters because the discounting is what makes early settlement equivalent to waiting, and a contract that settled the undiscounted amount at $T_1$ would overpay whichever side the fixing favored.

*What the question tests.* That the FRA rate is an arbitrage consequence rather than a forecast, and that the settlement convention carries an adjustment which is easy to omit and changes who gains.

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**4.10** *Explain why a single discount curve was adequate before 2008 and is not now. State which curve discounts a collateralized swap and which forecasts its floating payments, and explain why the two being different is a statement about collateral rather than about the credit of the counterparties.*

Before 2008 the difference between an unsecured interbank rate and a collateralized one was a few basis points, and the market treated one curve as serving both purposes: forecasting the floating payments of a swap and discounting them. The approximation was good enough that the error sat within the bid–ask spread.

The crisis widened those spreads to hundreds of basis points, and they never returned to the few basis points of before. Once the gap is material, using one curve for both jobs prices the same swap two ways depending on which curve is chosen, and the difference is no longer inside anyone's spread.

For a collateralized swap the two curves are:

*Discounting*: the overnight rate, because that is the rate paid on the collateral. A fully collateralized position is funded by the collateral it receives, so the correct discount rate is the one the collateral earns.

*Forecasting*: the curve of the index the floating leg actually references, which is whatever the contract names.

The essential point is what the difference is a statement *about*. It is not that the counterparty might default: the trade is collateralized, and the discounting question would be identical if the counterparty were riskless. The discount rate is determined by the collateral agreement (what may be posted, in what currency, and what it earns), so two otherwise identical swaps under different credit support annexes (CSAs) have different values. That is why Chapter 8 treats the CSA as a pricing document rather than a legal one.

*What the question tests.* That multi-curve pricing follows from collateralization rather than from credit, which is the distinction most short explanations blur.

---

**4.11** *State the theories of the term structure the chapter sets out, and give what each predicts about the shape of the curve when short rates are expected to stay flat. Explain which prediction the historical record supports, and what that implies for reading a forward rate as a forecast.*

The chapter sets out three.

*Expectations.* Forward rates are the market's expected future short rates, so with short rates expected to stay flat the curve is flat.

*Liquidity preference.* Lenders prefer to lend short and borrowers prefer to borrow long, so long rates carry a premium for the mismatch. With flat expected short rates the curve still slopes *upward*.

*Market segmentation.* Maturities are separate markets with their own supply and demand, so the curve takes whatever shape those flows produce and no general prediction follows.

The historical record supports liquidity preference: yield curves are upward sloping far more often than not, and much more often than any plausible sequence of expectations about future short rates would produce. Something is being paid for term, persistently, independent of the direction rates are expected to move.

What that implies for reading forwards is the practical content. A forward rate is not a forecast. It is a forecast plus a premium, and question 7 showed the arithmetic already exaggerates the slope. So an upward-sloping curve routinely implies forwards that appear to predict sharply rising rates when the market expects nothing of the kind: the steepness is partly the premium and partly the algebra. Treating forwards as predictions makes the curve look perpetually wrong; recognizing the premium explains why carry trades (holding longer bonds

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funded at short rates) earn on average, and Chapter 28 returns to it as the market price of risk.

*What the question tests.* That the three theories are distinguished by what they predict when expectations are neutral, and that the answer determines whether a forward rate can be read as information.

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# Chapter 5
# Determination of Forward and Futures Prices

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**5.1** *A non-dividend-paying stock trades at \$72 and the four-month risk-free rate is 3.6%. Compute the four-month forward price, and describe the arbitrage if the forward is quoted at \$74, giving the profit per share.*

$$F_0 = S_0 e^{rT} = 72\, e^{0.036\times 4/12} = 72 \times 1.012072 = \$\mathbf{72.87}.$$

At a quoted forward of \$74 the contract is dear, so sell it and carry the stock: borrow \$72, buy the share, and short the forward. At delivery, repay $72e^{0.036/3} = \$72.87$ and receive \$74, leaving

$$74.00 - 72.87 = \$\mathbf{1.13} \text{ per share},$$

locked at inception, requiring no capital, and independent of where the stock finishes. The trade is riskless because the share bought today is the share delivered later — there is nothing left uncertain.

*What the question tests.* That the forward price is a carry calculation rather than a forecast, and that quoting one away from it names the size of the riskless profit.

---

**5.2** *A stock will pay a \$1.50 dividend in two months and \$1.50 in five months. It trades at \$64 and the six-month rate is 4%, flat. Compute the six-month forward price.*

Discrete dividends are removed at present value before carrying:

$$I = 1.50e^{-0.04\times 2/12} + 1.50e^{-0.04\times 5/12} = 1.4900 + 1.4752 = \$2.9652,$$

$$F_0 = (S_0 - I)e^{rT} = (64 - 2.9652)e^{0.04\times 0.5} = 61.0348 \times 1.020201 = \$\mathbf{62.27}.$$

The forward is *below* spot even though the interest rate is positive, because \$2.97 of dividends outweighs \$1.23 of interest. The forward buyer does not receive the dividends, so they must not pay for them.

*What the question tests.* That income on the asset enters as a present value subtracted from spot, not as a yield, when the payments are discrete and dated.

---

**5.3** *An index stands at 3,800 with a dividend yield of 2.1% and a six-month rate of 4.0%. Compute the six-month futures price, state what it would be if the dividend yield exceeded the interest rate and why, and explain why that yield is a forecast rather than a known quantity.*

$$F_0 = S_0 e^{(r-q)T} = 3800\, e^{(0.040-0.021)\times 0.5} = 3800 \times 1.009545 = \mathbf{3836.3}.$$

If the dividend yield exceeded the rate, $r - q < 0$ and the futures price would sit *below* the index. There is nothing paradoxical in it: holding the index earns dividends and costs financing,

and when the dividends are worth more the net cost of carry is negative, so the futures price must be below the spot price to leave no arbitrage. It is not a forecast of a falling market.

The yield is a forecast in a way the interest rate is not. $r$ is contracted; $q$ is the dividends the index constituents will actually declare over the next six months, which no one knows. Hundreds of companies each decide, and a few large cuts move the number. So the futures price rests on one observable and one estimate, and the discrepancy between the estimate and the outcome is a real risk carried by anyone arbitraging the basis.

*What the question tests.* That the sign of $r - q$ has no predictive content, and that the carry formula's two inputs are not equally solid.

---

**5.4** *An investor is short a forward on a non-dividend-paying stock with a delivery price of \$95 and five months remaining. The stock trades at \$88 and the rate is 3.5%. Value the position.*

The value of a long forward with delivery price $K$ is

$$f = (F_0 - K)e^{-rT} = S_0 - Ke^{-rT}.$$

With $S_0 = 88$, $K = 95$, $r = 3.5\%$, $T = 5/12$:

$$f = 88 - 95e^{-0.035\times 5/12} = 88 - 95 \times 0.985522 = 88 - 93.625 = -\$5.625.$$

The investor is *short*, so their position is worth

$$+\$\mathbf{5.62} \text{ per share.}$$

The short is ahead because the stock has fallen well below a delivery price of 95: they have contracted to sell at 95 an asset now worth 88, and the present value of that advantage is \$5.62 rather than the full \$7 because the sale happens in five months.

*What the question tests.* That valuing a forward after inception is a discounting exercise on the difference between the delivery price and the current forward, and that the sign convention is the only place to go wrong.

---

**5.5** *The spot rate is 1.08 dollars per euro, the one-year dollar rate 4.2% and the euro rate 2.6%. Compute the one-year forward rate, state which currency is at a forward discount and why, and explain why a hedged carry trade earns nothing while an unhedged one is a different proposition entirely.*

$$F_0 = S_0 e^{(r_d - r_f)T} = 1.08\, e^{(0.042-0.026)} = 1.08 \times 1.016129 = \mathbf{1.0974} \text{ dollars per euro.}$$

The euro buys more dollars forward than spot, so the dollar is at a forward discount — which is the currency with the higher interest rate. That is covered interest parity: a currency that pays more interest must buy less forward than spot, or borrowing in the cheap currency and lending in the dear one would be a riskless profit.

Which is exactly why a hedged carry trade earns nothing. Borrow euros at 2.6%, convert, lend dollars at 4.2%, and sell the dollar proceeds forward: the forward rate is set precisely so the 1.6% interest advantage is given back on the currency leg. The arithmetic closes to zero, and it must, or the trade would be an arbitrage.

The unhedged trade is a different proposition because the forward is removed. The investor keeps the 1.6% and takes the spot rate in a year as it comes. That is not arbitrage but a bet that the dollar will not fall by more than the interest differential — and the historical record is that it often does not, which is why the carry trade persists, and that it occasionally falls by far more all at once, which is why it periodically destroys the people doing it.

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*What the question tests.* That parity kills the hedged trade by construction, and that the unhedged version's return is a risk premium rather than a riskless profit.

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**5.6** *Copper trades at \$4.20 per pound with a one-year futures price of \$4.05, a rate of 4.3%, and storage costs of 1.5%. Compute the implied convenience yield and interpret it. Repeat the carry calculation for silver at \$29 per ounce with storage of 0.4% and a rate of 4.3%, and explain why the two cases require different arguments.*

For a commodity, $F_0 = S_0 e^{(r+u-y)T}$, so

$$y = r + u - \frac{1}{T}\ln\frac{F_0}{S_0} = 0.043 + 0.015 - \ln\frac{4.05}{4.20} = 0.058 + 0.036368 = \mathbf{9.44}\%.$$

The futures price is below spot, so the market is in backwardation, and the convenience yield is what makes the carry equation balance. Interpreted: holders of physical copper are being compensated at about 9.4% a year for holding metal rather than paper — which is what a market with tight inventories, immediate industrial demand, and a real cost to running out looks like. It is a residual, not an observation: it is defined as whatever reconciles the two prices, so it absorbs any error in $r$ or $u$ as well.

Silver with no convenience yield:

$$F_0 = 29\, e^{(0.043+0.004)} = 29 \times 1.048122 = \$\mathbf{30.40}.$$

The two differ because the metals are held for different reasons. Silver is largely an investment asset, held in vaults in quantities far exceeding industrial need, so no one is short of it and full carry holds. Copper is consumed: a smelter that runs out stops, and the value of having metal on hand is real. Convenience yield is a statement about whether the commodity is held to use or held to own.

*What the question tests.* That the same formula produces contango for one metal and backwardation for another, and that the difference is physical rather than financial.

---

**5.7** *Explain the difference between the two uses of the word "contango" in this chapter, and why it is wrong to say that a market in contango expects prices to rise.*

The chapter uses the word for two different things.

**Contango as a shape.** The futures curve slopes upward: distant contracts cost more than near ones. This is a description of observable prices and nothing more.

**Contango as a theory.** The older usage, from Keynes and Hicks, refers to the futures price sitting above the expected future spot price — a statement about a risk premium, involving a quantity no one can see.

Saying a market in contango expects prices to rise conflates them. The upward slope is explained by carry: financing, storage, and insurance make a barrel deliverable in a year cost more than one deliverable now, whatever anyone expects. A curve can slope upward while every participant forecasts a falling price, and routinely does. Expectations enter only through the second sense, and that one cannot be read off the curve at all, because the expected future spot price is not quoted anywhere.

*What the question tests.* That the shape of the curve is a cost calculation, and that reading it as a forecast is the single most common error in commodity markets — one the roll-return arithmetic of question 11 punishes directly.

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**5.8** *A stock trades at \$120 with a three-month rate of 4% and a borrowing fee of 5% a year. Compute the no-arbitrage band for the three-month forward price, and explain why a quote inside the band is not a mispricing.*

The upper bound comes from cash-and-carry: borrow, buy the stock, sell the forward.

$$F \le S_0 e^{rT} = 120\, e^{0.04\times 0.25} = \$\mathbf{121.21}.$$

The lower bound comes from the reverse trade: short the stock, invest the proceeds, buy the forward. The short seller must borrow the shares and pay the 5% fee $b$, so the return to the strategy is $r - b$:

$$F \ge S_0 e^{(r-b)T} = 120\, e^{(0.04-0.05)\times 0.25} = \$\mathbf{119.70}.$$

The no-arbitrage band is therefore \$119.70 to \$121.21, about \$1.51 wide, and a quote anywhere inside it is not a mispricing. Inside the band the cash-and-carry earns less than its financing costs, and the reverse trade earns less than the borrow fee it must pay, so neither covers its costs. The band is not a measurement error or a market imperfection to be corrected: it is the price of the stock loan, made visible in the forward market.

The narrower point is that a hard-to-borrow stock has a wide band, so the forward can sit well below full carry with no arbitrage available to anyone who must borrow the shares. A holder who owns them outright and is free to sell still has one—selling spot, investing, and buying the forward needs no stock loan at all—so the band is reached only when the register is held by people who cannot do that, which is the usual condition of a heavily shorted stock. Chapter 10 finds the same effect in the option chain, where it appears as puts trading rich to calls.

*What the question tests.* That arbitrage bounds are two-sided and asymmetric once shorting has a cost, and that the width of the band is itself a quoted price.

---

**5.9** *A bond has a cash price of 101.20 and can be delivered into a futures contract in 60 days for an invoice amount of 101.90. Compute the implied repo rate on a 365-day basis. If the bond can be financed at 4.10%, should the cash-and-carry trade be done?*

The implied repo rate is the return to buying the bond and delivering it:

$$\frac{101.90 - 101.20}{101.20} \times \frac{365}{60} = 0.0069170 \times 6.0833 = \mathbf{4.21}\%.$$

Financing is available at 4.10%, which is below the 4.21% the trade earns, so **yes, the cash-and-carry should be done**: buy the bond, finance it in repo, short the futures contract, deliver in 60 days, and keep the 11 basis point spread on the position for two months.

Two qualifications keep this honest. The spread is thin — 11 basis points annualized is under two basis points of actual return over the 60 days — so it is a trade for someone with cheap financing and no transaction costs, which means a dealer rather than an investor. And the calculation assumes this bond is the one delivered; if it ceases to be cheapest to deliver, the short is holding the wrong bond, which is the delivery option Chapter 6 prices.

*What the question tests.* That the implied repo rate is a comparison against a financing cost rather than a number meaningful on its own.

---

**5.10** *Derive the value of a long forward contract by comparing two portfolios. Explain why the underlying asset price does not appear in the result, and what this permits operationally.*

Compare two portfolios, as the chapter does.

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**Portfolio A**: the existing long forward, with delivery price $K$, worth $f$.

**Portfolio B**: a long forward entered today for the same delivery date at the current forward price $F_0$, which by definition is worth zero.

At $T$ both portfolios take delivery of the same asset, worth $S_T$. A pays $K$ for it and B pays $F_0$, so A ends ahead of B by exactly $F_0 - K$ in every state. That difference is certain, so it is discounted at the risk-free rate:

$$f - 0 = (F_0 - K)e^{-rT} \quad \Longrightarrow \quad f = (F_0 - K)e^{-rT}.$$

The asset price does not appear because both portfolios hold the same asset at $T$: $S_T$ enters both and cancels, leaving only the gap between two delivery prices. The value depends only on how far today's forward price has moved from $K$. (For a non-dividend-paying stock $F_0 = S_0 e^{rT}$, and the same value reads $S_0 - Ke^{-rT}$, the form question 4 uses.) Nor does the asset enter as a *forecast* at any stage — no distribution, drift, or volatility is used. The argument never asks where the asset will go, only that whatever it does, the two portfolios do it together.

Operationally that permits a great deal. A dealer can quote and value forwards on anything with a spot price and a financing rate, with no view and no model, and can hedge the quote by holding the asset. It also means two dealers with opposite opinions must quote the same forward price — the reason this market is competitive on financing rather than on forecasting.

*What the question tests.* That replication removes expectations from the answer, which is the template every valuation result in the book follows.

---

**5.11** *A commodity trades at \$60 spot with the one-month futures price at \$60.75, and an investor holds continuous long exposure by rolling monthly. If the spot price is unchanged after a year, what is the return? Repeat for a one-month futures price of \$59.40, and identify who receives what the futures investor pays.*

With the spot unchanged, each contract bought at 60.75 converges to 60 by expiration, so each month loses 0.75 on a 60 investment. Rolling twelve times:

$$\left(\frac{60}{60.75}\right)^{12} - 1 = \mathbf{-13.9}\%.$$

The investor's view was right — the price did not fall — and the position lost 14%. That is roll return, and it is not a fee or a friction: each month the investor buys a claim priced above the spot and watches it converge.

In backwardation the sign reverses. Buying at 59.40 each month:

$$\left(\frac{60}{59.40}\right)^{12} - 1 = \mathbf{+12.8}\%,$$

earned with the spot again unchanged.

What the futures investor pays in contango goes to the short side of each contract, and ultimately to whoever holds the physical commodity and carries it: a holder who buys at spot, stores, and sells the deferred contract above spot is paid exactly for financing and storage. That is the cost of carry of Section 5.11, not a risk premium. In backwardation the flow reverses and the long collects, because holders of physical stock value having it now and the futures price sits below spot. Whether the long also earns a premium over the *expected* spot price is the separate question of Section 5.13.

*What the question tests.* That a commodity index's return has a component unrelated to the commodity price, large enough to dominate it, and that the sign depends on the curve rather than on the market's direction.

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**5.12** *Silver trades at \$29 per ounce, and storage and insurance cost \$0.12 per ounce per quarter, payable at the start of each quarter. The interest rate is 4.3% for all maturities. Compute the nine-month futures price. If the nine-month futures contract trades at \$30.40, describe the arbitrage and give the profit per ounce.*

Storage is stated as an amount, so it enters as a present value $U$, exactly as known income does but with the sign reversed. Three payments fall within the nine months, at 0, 3 and 6 months; the payment due at nine months buys a quarter after delivery and does not belong to this contract. Their present value is

$$U = 0.12\left(1 + e^{-0.043\times0.25} + e^{-0.043\times0.5}\right) = 0.12\times(1+0.989308+0.978729) = 0.12\times2.968037 = \$0.3562,$$

and from $F_0 = (S_0 + U)e^{rT}$

$$F_0 = (29 + 0.3562)\, e^{0.043\times0.75} = 29.3562 \times 1.032776 = \$\mathbf{30.32}.$$

At \$30.40 the futures price is above this, so the cash-and-carry applies: borrow \$29.3562 today, buy an ounce of silver for \$29, set aside \$0.3562 to meet the three storage bills as they fall due, and short the nine-month futures contract. At delivery the silver is delivered for \$30.40 and the loan is repaid with $29.3562 \times 1.032776 = \$30.32$, leaving

$$30.40 - 30.32 = \$\mathbf{0.08} \text{ per ounce},$$

with no capital committed and no exposure to the silver price. The calculation treats the futures price as a forward price, as the chapter does; over nine months on a metal the difference is negligible.

The step that is easy to get wrong is the borrowing. The trade has to finance the storage as well as the metal, and a trader who borrows only \$29 and pays the fees out of pocket has committed capital and understated the cost. Counting the payment at nine months as well would raise $U$ to about \$0.47 and make the futures look less rich than it is.

*What the question tests.* That a storage cost stated as an amount is discounted payment by payment and added to the spot price, and that the cash-and-carry must finance the storage as well as the asset.

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# Chapter 6
# Interest Rate Futures

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**6.1** *A Treasury bond is quoted at* `104-20` *with accrued interest of \$0.85 per \$100 of face. Express the quoted price as a decimal and give the cash price.*

Treasury bonds are quoted in thirty-seconds, so `104-20` means $104 + 20/32$:

$$104 + 0.62500 = \mathbf{104.625}.$$

The cash (dirty) price adds accrued interest:

$$104.625 + 0.85 = \mathbf{105.475} \text{ per \$100 of face,}$$

or \$105,475 on a \$100,000 bond.

The distinction is not cosmetic. The quoted price is what is negotiated, and it moves only with yield; the cash price is what is paid, and it also grows steadily between coupon dates as accrual builds. Quoting clean is what lets a trader see the price change that reflects the market rather than the calendar.

*What the question tests.* The thirty-seconds convention and the clean/dirty distinction, both of which every subsequent calculation in the chapter assumes.

---

**6.2** *A trader buys 150 three-month SOFR futures at 96.10 and the contract settles at 96.34. Calculate the profit, state what happened to the implied rate, and explain why the contract is quoted as* $100 - R$ *in the first place.*

Each contract is \$1 million of notional for three months, so one basis point is worth $1{,}000{,}000 \times 0.0001 \times 0.25 = \$25$. The price rose $96.34 - 96.10 = 0.24$, which is 24 basis points:

$$150 \times 24 \times \$25 = \$\mathbf{90{,}000} \text{ profit.}$$

The implied rate *fell*, from $100 - 96.10 = 3.90\%$ to $100 - 96.34 = 3.66\%$. The trader was long the contract and long futures gain when the price rises, which for this contract means when rates fall.

The $100 - R$ convention exists so the contract behaves like a price rather than a yield. A bond gains when rates fall; quoting the futures price as $100 - R$ makes it gain when rates fall too, so a long position is a long-duration position and the sign conventions of the rest of a fixed-income book carry over without translation. Quoting the rate directly would invert every intuition and every hedge ratio's sign.

*What the question tests.* That the tick value follows from notional and accrual, and that the quotation convention is chosen to make positions read the same way as cash bonds.

---

**6.3** *Explain why the rate sensitivity of a three-month SOFR futures contract declines through its reference period, why this did not happen with Eurodollar futures, and what it implies for a hedger.*

A three-month SOFR futures contract settles against the average of overnight rates across

its reference quarter. On the first day of that quarter every rate in the average is still unknown, so the whole of it is exposed to rate changes. By the midpoint, half the rates have already fixed and are frozen into the settlement; only the remainder can still move. So the contract's sensitivity declines roughly linearly through its reference period, reaching zero on the last day, when everything is determined.

Eurodollar futures did not behave this way because they settled against a single LIBOR fixing on the last trading day, at the start of the three months that rate covered. Nothing was determined until that fixing, so the contract carried full sensitivity right up to expiration and then lost it all at once.

For a hedger this matters in two ways. The hedge ratio is not constant: a position hedged with a contract in its reference period is progressively less hedged as the quarter runs, and needs topping up. And the decline is predictable rather than random (it is a property of the contract, known in advance), so it can be scheduled rather than discovered.

*What the question tests.* That the shift from a terminal fixing to an average changed the risk profile of the instrument, not merely the arithmetic of its settlement.

---

**6.4** *A bond pays a 6% coupon semiannually and has exactly 10 years to maturity after rounding. Calculate its conversion factor at the 6% notional coupon rate and explain the result. Then use equation (6.5) to calculate the conversion factor of a 4% coupon bond with 7 years and 3 months remaining after rounding.*

The conversion factor is the bond's price per dollar of face when it is priced at the notional coupon rate: 6% in the US contract, compounded semiannually, so 3% per half-year.

*The first bond: a 6% coupon and exactly 10 years.* Its coupon *equals* the notional rate, so discounted at that rate it prices exactly at par, and

$$\text{CF} = \mathbf{1.0000} \text{ (exactly).}$$

Equation (6.5) confirms it: $c = 0.06$, $n = 10$ and $z = 0$, so $v = 0$ and $\kappa = 1.03^{-20}$, and

$$\text{CF} = 0.03 + \kappa + \frac{0.06}{0.06}(1 - \kappa) - 0.03 = 0.03 + \kappa + 1 - \kappa - 0.03 = 1.$$

The $\kappa$ cancels, and that is the point: when the coupon equals the notional rate, the conversion factor is 1 whatever the maturity. This is also what the notional rate is for: it marks the line at which no rescaling is applied.

*The second bond: a 4% coupon with 7 years and 3 months remaining.* Here $c = 0.04$, $n = 7$ and $z = 3$, so $v = 3$, and because $z < 7$, $\kappa = 1.03^{-14} = 0.661118$. Substituting into equation (6.5),

$$\left[\frac{c}{2} + \kappa + \frac{c}{0.06}(1 - \kappa)\right] = 0.02 + 0.661118 + \frac{0.04}{0.06} \times 0.338882 = 0.907039,$$

$$\text{CF} = \frac{0.907039}{1.03^{0.5}} - \frac{0.04}{2} \cdot \frac{6 - 3}{6} = 0.985329 \times 0.907039 - 0.01 = \mathbf{0.8837}.$$

The factor is well below 1 because the 4% coupon is far below the 6% standard, and the contract scales the invoice down accordingly. The departure is also much larger than the first bond's, which needed no adjustment at all, and that is where the next point comes from: the conversion factor rescales different bonds by very different amounts, and whether the rescaling is right depends on how far the market yield sits from 6%. Question 6.10 develops this.

*What the question tests.* Using the formula, and recognizing the identity that a coupon equal to the notional rate gives a conversion factor of exactly 1, which says what the notional rate is, and anticipates why setting it far from the market causes trouble.

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**6.5** *The futures price is 108.75 and three bonds are deliverable, quoted at 96.20, 101.85, and 118.40, with conversion factors 0.8825, 0.9350, and 1.0880. Identify the cheapest-to-deliver bond.*

The cheapest to deliver minimizes the gross basis, quoted price minus futures price times conversion factor:

| Quoted | CF | Futures × CF | Gross basis |
|---|---|---|---|
| 96.20 | 0.8825 | 95.9719 | 0.2281 |
| 101.85 | 0.9350 | 101.6813 | 0.1688 |
| 118.40 | 1.0880 | 118.3200 | **0.0800** |

The **third bond**, quoted at 118.40 with a conversion factor of 1.0880, is cheapest to deliver: it costs the short the least, net of what the contract pays for it.

Note that the cheapest bond is not the one with the lowest price. The conversion factor rescales all three onto comparable terms, so what matters is the residual after that rescaling, which is why the answer requires the calculation rather than an inspection.

*What the question tests.* That "cheapest" is a statement about the gap between what a bond costs and what the contract pays for it, not about its absolute price.

**6.6** *Distinguish gross basis from net basis. A deliverable bond has a 3.0% coupon, is quoted at 94.60, can be financed at 4.40% for the 60 days to delivery, and has a gross basis of 0.0410. Calculate carry and the net basis, and comment on the sign.*

**Gross basis** is the raw difference, quoted bond price minus futures price times conversion factor. **Net basis** subtracts the carry earned while holding the bond to delivery, and so measures what the position actually costs after financing, which is the value of the short's delivery options.

Carry over the 60 days is coupon income less financing cost, with repo at actual/360 as in Example 6.6:

$$\text{coupon} = 3.00 \times \frac{60}{365} = 0.4932,$$

$$\text{financing} = 94.60 \times 0.044 \times \frac{60}{360} = 0.6937,$$

$$\text{carry} = 0.4932 - 0.6937 = \mathbf{-0.2006}.$$

$$\text{net basis} = \text{gross basis} - \text{carry} = 0.0410 - (-0.2006) = \mathbf{0.2416}.$$

The carry is *negative* because the bond's 3.0% coupon is below the 4.40% cost of financing it: the holder pays more to carry the bond than the bond pays them. That was unusual through the low-rate decade and became ordinary once short rates rose above the coupons of bonds issued earlier, a regime change that flipped the sign of the carry term across much of the deliverable basket.

Because carry is negative, the net basis exceeds the gross basis rather than falling below it, and the short's delivery options are worth more than the raw gap suggests.

*What the question tests.* That carry can be either sign, and that the sign depends on the relation between the coupon and the financing rate rather than on anything about the futures contract.

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**6.7** *A portfolio worth \$25 million has a duration of 5.4 years, ten-year note futures are priced at 109-00, and the CTD has a duration of 8.1 years at contract maturity. How many contracts should be shorted to immunize it? What position instead moves a \$40 million portfolio of duration 4.2 years to a target of 7.5 years, with futures at 111-00 and a CTD duration of 8.0?*

The contract count for a duration hedge is equation (6.11),

$$N^* = \frac{P\,(D_P - D_T)}{F\,D_F},$$

with the target $D_T = 0$ for immunization.

**Immunizing the \$25 million portfolio.** Futures at 109-00 give a contract value of $109.00 \times 1{,}000 = \$109{,}000$:

$$N^* = \frac{25{,}000{,}000 \times 5.4}{109{,}000 \times 8.1} = \frac{135{,}000{,}000}{882{,}900} = 152.9 \longrightarrow \textbf{short 153 contracts}.$$

**Moving the \$40 million portfolio from 4.2 to 7.5 years.** The hedge targets the difference in duration, and futures at 111-00 give a contract value of \$111,000:

$$N^* = \frac{40{,}000{,}000 \times (4.2 - 7.5)}{111{,}000 \times 8.0} = \frac{-132{,}000{,}000}{888{,}000} = -148.6 \longrightarrow \textbf{long 149 contracts}.$$

The sign is the whole difference between the two. Reducing duration to zero means selling futures; *increasing* duration means buying them, and the same formula produces both: a negative $N^*$ is a long position.

*What the question tests.* That immunization is the special case of a duration target, and that the CTD's duration — not the portfolio's — sets the denominator.

---

**6.8** *Describe the cash–futures basis trade, and state where the profit comes from and where the leverage comes from.*

The trade buys the cheapest-to-deliver bond, sells the corresponding futures contract against it, and finances the bond in the repo market. Held to delivery, the bond is handed over into the contract and the position unwinds itself.

**Where the profit comes from.** The gap between the bond's price and what the futures contract pays for it: the net basis, after carry. Asset-manager demand for duration in futures form tends to leave futures rich relative to cash, and the trade harvests that richness as the two converge at delivery. It is a few basis points.

**Where the leverage comes from.** The repo financing. The bond is bought with borrowed money and posted as collateral for that borrowing, so the trader's own capital is only the haircut, one or two percent. A few basis points of convergence on a position funded at a 1–2% haircut is a return on equity of fifty to a hundred times the raw spread.

The two are not separate facts. The profit is small precisely because the trade is nearly riskless in price terms, and it is worth doing only at leverage that makes the funding leg the dominant risk — which is what March 2020 demonstrated, and what the Treasury clearing rules of Section 2.11 are addressed to.

*What the question tests.* That the trade's return and its fragility have the same source, so describing one without the other misstates it.

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**6.9** *A ten-year note futures contract settles at 112-08. The delivered bond has a conversion factor of 0.9125 and accrued interest of $0.60 per $100. Calculate the invoice amount for one $100,000 contract.*

The invoice amount is the futures settlement price times the conversion factor, plus accrued interest, scaled to the contract:

$$\left(112 + \frac{8}{32}\right) = 112.25,$$

$$112.25 \times 0.9125 = 102.428125 \text{ per \$100 of face},$$

$$(102.428125 + 0.60) \times 1{,}000 = \$\mathbf{103{,}028.13}.$$

The multiplication by 1,000 converts a per-$100 quote to a $100,000 contract. The conversion factor is what makes a bond with a coupon other than the notional 6% deliverable on comparable terms; the accrued interest is added because the buyer receives a bond carrying it.

*What the question tests.* The mechanics of the invoice price, and that the conversion factor multiplies the futures price rather than adjusting the bond's own quote.

---

**6.10** *Explain why conversion factors do not make all deliverable bonds exactly equivalent, and state which bonds tend to be cheapest to deliver when yields are below 6% and why.*

Conversion factors are computed by pricing each deliverable bond to yield exactly 6%, the contract's notional coupon. That makes the bonds equivalent at a 6% yield and nowhere else. As soon as the market yield differs, the factors are wrong in a systematic direction, because bonds of different coupon and maturity have different durations and so respond differently to the move away from 6%.

When yields are below 6%, **high-coupon, short-maturity bonds tend to be cheapest to deliver**. Those are the bonds with the lowest duration. Falling yields lift every bond's price, but they lift the high-duration bonds most and the low-duration bonds least, while the conversion factor, fixed and computed once at 6%, does not adjust. The price of the short, high-coupon bond therefore rises least relative to what the contract pays for it (the futures price times its conversion factor), and it becomes the cheapest to hand over. Above 6% the ordering reverses, and long, low-coupon bonds become cheapest.

*What the question tests.* That the conversion factor is an approximation anchored at one yield, and that knowing the direction of its error tells you which bond the contract is really tracking — which is what makes the delivery option worth something.

---

**6.11** *Explain why steady demand from asset managers for duration in futures form tends to leave Treasury futures rich relative to the cash bond, and connect this to Section 6.5.*

An asset manager wanting duration has two routes: buy bonds, or buy futures and hold cash. The futures route requires margin rather than the full purchase price, leaves the cash available, and can be executed in large amounts in minutes. For a manager adjusting duration against a benchmark, a decision taken repeatedly and reversed often, that is decisively more convenient.

The consequence is a persistent, large, and fairly price-insensitive stream of *buying* interest in the futures contract. The buyers are not comparing the futures price with the cash bond and declining if it looks a basis point rich; they are implementing a duration decision. Steady one-directional demand that does not withdraw on price leaves the contract trading rich relative to the bond it tracks.

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That richness is exactly what the basis trade of Section 6.5 harvests. The basis trader takes the other side, selling the futures the asset managers are buying and holding the cash bond against it, and is paid the richness as the two converge at delivery.

So the two sections describe one arrangement from opposite ends. Asset managers pay a small premium for the convenience of duration in futures form; levered basis traders supply that convenience and collect the premium. The market-structure fact explains the price, and the price explains the trade — which is why the chapter presents them together.

*What the question tests.* Whether a persistent price anomaly can be traced to who is trading and why, rather than treated as a puzzle.

---

**6.12** *A dealer has received fixed on a $300 million swap with quarterly floating periods. How many three-month SOFR futures should it sell against each accrual period, ignoring discounting, and give two reasons the true hedge ratio differs.*

Each three-month SOFR futures contract covers $1 million of notional for one quarter, so against a $300 million accrual period the naive hedge is

$$\frac{300{,}000{,}000}{1{,}000{,}000} = \mathbf{300} \text{ contracts}$$

sold per period. The dealer has received fixed, so it is exposed to rising rates and sells the contract, which falls in price as rates rise.

Two reasons the true ratio differs.

**Discounting.** The swap's cash flow for a given accrual period is received at the end of that period and is worth its present value today, while the futures position settles in cash daily at face. Hedging the undiscounted amount over-hedges, and the correction is a factor of roughly the discount factor to that payment date: material for the far periods of a long swap and negligible for the near ones.

**Convexity.** A futures contract settles daily and a forward rate agreement does not, and the difference is worth something: futures gains are received early when rates have risen and financing is dear. That makes the futures rate exceed the forward rate by the convexity adjustment of Chapter 30, which grows with the square of maturity, so the hedge ratio drifts further from 1:1 the further out the accrual period sits.

*What the question tests.* That the notional-matching answer is the starting point rather than the answer, and that both corrections push in a knowable direction.

---

**6.13** *Drawing on Sections 3.9 and 3.10, explain what a stack-and-roll hedge is and the two costs it incurs, and why a hedge of this kind can fail even when the underlying exposure behaves exactly as expected.*

A **stack-and-roll** hedge places the entire long-dated exposure in the nearest liquid contract (stacking it) and then rolls that stack forward each time the contract approaches expiration, because the distant contracts a matched hedge would need are not liquid enough to trade in large amounts.

The two costs are:

**The bid–ask spread.** Every roll closes one contract and opens another, crossing the spread on both, and a multi-year hedge rolls many times.

**The roll spread.** More importantly, each roll happens at whatever spread between the two contracts prevails on the day, which is not known in advance. In contango a long hedger buys each successive contract dearer than the one closed, and pays the difference at every roll, for the life of the hedge. Question 5.11 quantifies the same mechanism.

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Beyond these two costs, the stack creates a **curve exposure**. A stacked hedge is not the exposure it is hedging. The underlying is spread over years; the hedge sits entirely at the front. The position is therefore long the front and short the back of the curve — a calendar spread no one chose — and it profits or loses on changes in the shape of the curve independently of the level.

The hedge can fail even when the underlying behaves exactly as expected because its two legs settle on different schedules. The futures leg pays or receives cash daily; the physical exposure it protects delivers value only later, at delivery or over years of production. A hedge that is correct in value can therefore be fatal in cash: margin is owed now against a gain that is real but not yet realized. That is Metallgesellschaft, and it is the mechanism Chapter 8 develops in full.

*What the question tests.* That a hedge which is right about the price can still be wrong about the timing, and that the stack construction creates a second exposure, to the curve, that the hedger never intended to take.

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# Chapter 7
# Swaps and the Multi-Curve Framework

**7.1** *State the comparative advantage argument for swaps, explain the flaw in it, and give the better explanation for the size of the swap market.*

**The argument.** Company A borrows fixed at 5.0% or floating at RFR + 0.3%; Company B, weaker, borrows fixed at 6.4% or floating at RFR + 1.0%. A's advantage is 140 basis points in fixed and only 70 in floating, so A has a comparative advantage in fixed and B in floating. Each borrows where it is relatively stronger, they swap, and the 70 basis point difference in differences is shared.

**The flaw.** The fixed quotes are commitments for the full term; the floating quotes are not. B's RFR + 1.0% is rolled periodically and will be repriced if B deteriorates; the lender can decline to renew. So B does not really face RFR + 1.0% for five years; it faces the RFR plus whatever it can get each period. The apparent 70 basis points is largely the credit option the floating lender retains, and after the swap B is left paying fixed on a stream whose floating leg it may not be able to keep funding. The advantage was not free; it was compensation for a risk that the comparison hid.

**The better explanation.** The swap market is enormous because it separates the interest rate exposure from the borrowing. A firm can borrow where it has access (its relationship banks, its home currency, its natural investor base) and then set the rate profile independently, at any tenor, and change it later without renegotiating the loan. That is a genuine and permanent service, and it does not depend on any arbitrage between two borrowers' quotes.

*What the question tests.* That the textbook motivation for swaps is wrong, and that the correct one is the same separation of exposure from ownership the book opens with.

**7.2** *A two-year annual swap on a notional of \$50 million pays fixed at 3.50% and receives floating. Discount factors are 0.9709 and 0.9426; forward rates are 3.00% and 3.30%. Value the swap to the fixed payer and find the par swap rate.*

The floating leg pays each period's forward rate on the notional, discounted:

$$V_{\text{flt}} = 50{,}000{,}000\,(0.0300 \times 0.9709 + 0.0330 \times 0.9426) = \$3{,}011{,}640.$$

The fixed leg pays 3.50%:

$$V_{\text{fix}} = 50{,}000{,}000 \times 0.0350 \times (0.9709 + 0.9426) = \$3{,}348{,}625.$$

To the fixed payer, who receives floating and pays fixed:

$$V = 3{,}011{,}640 - 3{,}348{,}625 = -\$\mathbf{336{,}985}.$$

The annuity is $A = 0.9709 + 0.9426 = 1.9135$, so the par swap rate (the fixed rate making

the swap worth zero) is

$$S = \frac{V_{\text{flt}}}{L\,A} = \frac{3{,}011{,}640}{50{,}000{,}000 \times 1.9135} = \mathbf{3.1478}\%.$$

The payer is out of the money because 3.50% is well above the 3.15% the market implies, and the loss is that 35 basis point gap valued over the annuity: $0.003522 \times 50\text{m} \times 1.9135 \approx \$337{,}000$.

*What the question tests.* That a swap is valued leg by leg and that the par rate is the annuity-weighted average of the forwards, the same relation that reappears in Chapter 25 as the index spread.

---

**7.3** *Show that when the floating index and the discount curve are the same, the floating leg is worth $L(1 - \text{DF}_n)$, and state precisely the condition under which this fails.*

Take the floating leg as a sequence of payments $L\tau_i F_i$ at each date, where $F_i$ is the forward rate for accrual period $i$ and $\tau_i$ its length. If the index being forecast is the same curve used for discounting, then by definition, with $\text{DF}_0 = 1$,

$$F_i = \frac{1}{\tau_i}\left(\frac{\text{DF}_{i-1}}{\text{DF}_i} - 1\right),$$

so each discounted payment is

$$L\tau_i F_i\,\text{DF}_i = L\left(\text{DF}_{i-1} - \text{DF}_i\right),$$

and the sum telescopes:

$$V_{\text{flt}} = L\sum_{i=1}^{n}\left(\text{DF}_{i-1} - \text{DF}_i\right) = L\left(1 - \text{DF}_n\right).$$

The interpretation is that a floating leg is worth lending the notional today and getting it back at maturity. No forecasting is required.

**The condition under which it fails** is precisely that the forecast curve and the discount curve differ. Question 2 is an instance: there, $L(1 - \text{DF}_2) = 50\text{m}(1 - 0.9426) = \$2{,}870{,}000$, while the actual floating leg is worth \$3,011,640. The \$141,640 gap is the basis between the two curves, and it is a material amount rather than rounding.

So the identity is not a general truth about floating legs; it is a single-curve result, and the multi-curve world of Section 7.5 is exactly the world in which it stops holding.

*What the question tests.* That a familiar shortcut carries an assumption that the post-2008 market no longer satisfies, and that the question's own data violate it.

---

**7.4** *Explain why the correct discount rate for a fully collateralized derivative is the rate paid on its collateral, and state what follows about two otherwise identical swaps collateralized in different currencies.*

A fully collateralized derivative is funded by its own collateral. When the trade moves in your favor you receive collateral; you hold that cash and it earns whatever the CSA specifies. When it moves against you, you post, and the collateral you posted earns that same CSA rate, paid to you by the receiver. The position therefore finances itself at the collateral rate, and a cash flow should be discounted at the rate at which it can actually be financed, so the discount curve is the collateral rate curve, not the counterparty's borrowing rate and not any government curve.

The credit of the counterparty does not enter, because the collateral removes it. That is the point worth holding onto: this is a funding argument, not a credit argument.

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What follows for two otherwise identical swaps collateralized in different currencies is that they are not the same instrument and do not have the same value. One is discounted at the overnight rate of one currency, the other at another, and those curves differ. The same trade, same counterparty, same cash flows, is worth more under one CSA than another, so the collateral agreement is a pricing document, and a dealer quoting without knowing the CSA is not quoting a price. Where a CSA allows a choice among currencies, the poster holds an option over which curve applies, which is the subject of Question 7.15.

*What the question tests.* That discounting is determined by funding rather than by credit, and that the CSA is therefore part of the instrument's definition.

---

**7.5** *Repeat Question 7.2 assuming the floating index has forward rates 15 basis points above those given, with discount factors unchanged, and verify that the change in value equals the basis times the annuity.*

Raising each forward by 15 basis points, with discount factors unchanged:

$$V_{\text{flt}} = 50{,}000{,}000\,(0.0315 \times 0.9709 + 0.0345 \times 0.9426) = \$3{,}155{,}152,$$

so the swap's value to the fixed payer becomes

$$3{,}155{,}152 - 3{,}348{,}625 = -\$\mathbf{193{,}473},$$

an improvement of

$$-193{,}473 - (-336{,}985) = +\$\mathbf{143{,}512}.$$

The check: the basis multiplied by the annuity and the notional,

$$0.0015 \times 1.9135 \times 50{,}000{,}000 = \$143{,}512, \quad \checkmark$$

matching to the dollar.

The identity is worth seeing rather than merely verifying. A parallel shift in the forecast curve, with discounting held fixed, changes the floating leg by exactly $L \times \Delta \times A$: the annuity is the sensitivity of the swap to a parallel move in the forecast index. That is why the annuity is the natural unit in which rate risk is quoted, and why Chapter 29 uses it as a numeraire.

*What the question tests.* That the annuity converts a rate change into a value change, and that separating forecasting from discounting makes the effect of each visible on its own.

---

**7.6** *Describe the order in which curves must be built in a multi-curve framework, explain why the discount curve must come first, and explain why a single currency needs more than one forecast curve.*

**The order.** Build the discount curve first, from overnight indexed swaps. Then build each forecast curve (in euros, the three-month EURIBOR curve, the six-month EURIBOR curve, and so on) using the already-fixed discount curve to price the instruments that reference them.

**Why discounting must come first.** Every instrument used to calibrate a forecast curve is quoted as a present value: a swap on three-month EURIBOR is at par, which is a statement about discounted cash flows. To infer the forecast rates from that quote, the discounting must already be known. Inferring both from the index swaps alone leaves the problem underdetermined, since many combinations of forecasting and discounting reproduce the same par quotes, so the discount curve has to be pinned down by instruments of its own, which the OIS market supplies because its own floating leg references the discount rate itself. (In practice the curves are then solved jointly, as Section 7.6 notes, but it is the OIS quotes that fix the discounting.)

**Why one currency needs several forecast curves.** The euro is the live case: euro swaps still trade against both three-month and six-month EURIBOR, which are different rates, and the spread between them is not zero and not constant. Before 2008 it was small enough to ignore;

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since then a three-month and a six-month curve must be built separately, and the 3s6s basis swap, one of the basis swaps of Question 7, is the quoted instrument that prices the difference. Each tenor the market quotes needs its own curve, all discounted on the single collateral curve. In dollars and sterling the IBOR tenors are gone, but the point survives: term SOFR and the legacy fallback rates are separate forecast curves on the same discount curve.

*What the question tests.* That the multi-curve framework is a sequence with a required order, and that the order follows from what each quoted instrument actually pins down.

---

**7.7** *Explain what a basis swap is, give two distinct reasons the market needs them, and explain why notional principals are exchanged in a cross-currency swap but not in a single-currency one.*

A **basis swap** exchanges two floating legs (three-month against six-month EURIBOR in euros, or one currency's floating against another's) with a spread on one side to make the exchange fair at inception.

**Two reasons the market needs them.** First, tenor mismatch: a bank whose assets reprice quarterly and whose liabilities reprice semiannually holds a real exposure to the spread between the two indices, and the basis swap is the only instrument that isolates it. Second, price discovery: the basis swap quote *is* the market's price for that spread, and it is the input that pins down the forecast curves of Question 6 wherever no outright swap on an index trades. Without it, the curves could not be separated there.

**Why cross-currency swaps exchange notionals.** In a single-currency swap the two notionals are identical amounts of the same money, so exchanging them would be handing over a sum and receiving it straight back. The two payments cancel and are dropped. In a cross-currency swap they are different currencies, and the whole point is to convert a borrowing in one into an exposure in the other, which requires the principal to actually change hands at the start and to be returned at maturity. The exchange is not a convention; it is the substance of the trade, and it is why a cross-currency swap carries far larger counterparty exposure than a single-currency one of the same size.

*What the question tests.* That the notional exchange follows from what the instrument is for, and that basis swaps are simultaneously a hedging tool and the market's pricing mechanism for a curve that could not otherwise be built.

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**7.8** *The eight-year swap rate is 3.60% and the eight-year government bond yields 3.72%. Compute the swap spread, describe the trade it appears to offer, calculate its present value on a \$250 million position using an eight-year annuity of 6.85, and explain both what prevents the trade from being done and why a negative spread does* not *imply that banks are better credits than the government.*

The swap spread is the swap rate minus the government yield:

$$3.60\% - 3.72\% = -\mathbf{12} \text{ basis points.}$$

**The apparent trade.** Buy the eight-year government bond at 3.72%, pay fixed on an eight-year swap at 3.60%, and finance the bond in repo. The interest rate exposure of the two legs offsets, leaving 12 basis points a year on a position that appears to be a hedged arbitrage against the government's own credit. Its present value on \$250 million:

$$250{,}000{,}000 \times 0.0012 \times 6.85 = \$\mathbf{2{,}055{,}000}.$$

**What prevents it.** The bond must be financed and held, and that consumes balance sheet. Since the leverage ratio, balance sheet has a capital cost that is charged regardless of how riskless the position is; the ratio is deliberately insensitive to risk. Twelve basis points does not

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cover the capital a dealer must hold against a $250 million bond position, so the institutions that could put the trade on decline to. The spread is not an unclaimed arbitrage; it is the price of the balance sheet needed to close it.

So a negative swap spread is not a mispricing but a measurement: it measures what intermediation costs after the post-crisis capital rules. It says the swap, which consumes almost no balance sheet, is cheaper than the bond, which does.

*Why it does not imply that banks are better credits than the government.* Because a swap rate today contains *no bank credit at all*. Its floating leg references SOFR, a secured overnight rate on Treasury repo, so the rate the fixed leg is set against carries no bank's credit. That is what changed. LIBOR-era swaps were collateralized too; the bank credit entered through the index, since LIBOR was the rate at which banks said they could borrow unsecured, and a swap rate set against it carried a bank credit component. The swap spread could then be read as a credit comparison. Collateral, as Question 4 showed, settles a separate matter: it fixes the discount curve, not the index. The credit reading has gone out of date while the name has stayed.

*What the question tests.* That an apparent arbitrage against the safest credit available persists because the constraint binding is regulatory rather than economic, and that recognizing which constraint binds is the useful skill.

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**7.9** *Explain why a swap creates two-sided credit exposure and why the exposure is far smaller than the notional. For the swap of Question 7.2, state the exposure of each party.*

A swap creates **two-sided** exposure because its value can be positive to either party. Rates rise and the fixed payer is in the money, so it is exposed to the receiver defaulting; rates fall and the position reverses. Unlike a loan, where one party is always the creditor, a swap's creditor is determined by the market and changes over the contract's life.

The exposure is far smaller than the notional because the notional is never exchanged. What is at risk is the replacement cost: the difference between the contracted fixed rate and the rate at which the position could be replaced today, valued over the remaining annuity. That is a difference of rates on a notional, not the notional itself, which is why gross market value across the market runs at a few percent of notional, as Chapter 1 established.

For the swap of Question 2, the value to the fixed payer is −$336,985. So:

$$\text{fixed payer's exposure} = \$0, \qquad \text{floating payer's exposure} = \$\mathbf{336{,}985}.$$

The receiver is in the money and therefore the one carrying credit risk; the payer carries none at present, and would begin to if rates rose enough to move the swap through zero. On a $50 million notional, the exposure is 0.67% of it.

*What the question tests.* That exposure is a mark-to-market quantity with a sign that can flip, and that the headline notional overstates it by two orders of magnitude.

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**7.10** *Explain what close-out netting achieves, what would happen without it, and why its legal enforceability is a jurisdiction-by-jurisdiction question.*

**What it achieves.** Close-out netting makes all trades under a master agreement a single contract, so that on default they are terminated together, valued, and reduced to one net amount owed one way. A portfolio of hundreds of trades with offsetting values collapses to a single number.

**Without it.** The defaulter's administrator would cherry-pick: enforce the contracts on which the surviving party owes money, demanding payment in full, and disclaim those on

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which the estate owes, leaving the survivor an unsecured claim. Chapter 1's Example 1.1 works the arithmetic: a $2 million net claim becomes a $46 million net payment. Exposure would be measured on gross positive value rather than net, and the capital required to run a derivatives book would rise by roughly the ratio between the two, which across the market is nearly seven to one.

**Why enforceability is jurisdiction by jurisdiction.** Netting works only if a court will honor it, and insolvency law is national. Many jurisdictions have passed statutes explicitly protecting close-out netting; some have not; and in some the answer differs by counterparty type: a bank, an insurer, a municipality and a sovereign may be treated differently within one legal system. So the question is not whether the contract says the trades net, but whether the relevant court will agree when it matters, which is why firms obtain and maintain legal opinions per jurisdiction and per counterparty type, and why capital rules permit netted exposure only where such an opinion exists.

*What the question tests.* That the single largest risk reducer in the derivatives market is a legal fact rather than a financial one, and that it has to be established separately in every country the book trades in.

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# Chapter 8
# Collateral, Margin, and Funding

**8.1** *Explain the difference between a one-way and a two-way CSA, why a one-way CSA makes an apparently mid-market trade loss-making for the dealer, and why the rate paid on collateral is the most economically important term in the document.*

A **two-way CSA** obliges both parties to post as the trade moves: whichever party the trade has moved against delivers collateral. A **one-way CSA** obliges only one (in practice the dealer), while the client posts nothing however far the trade moves in the dealer's favor.

That makes an apparently mid-market trade loss-making for the dealer because of the hedge. The dealer offsets the client trade with a trade that is itself collateralized. When the client trade moves in the dealer's favor, the hedge moves against it, so the dealer posts collateral on the hedge and receives nothing from the client. It therefore funds that collateral out of its own balance sheet, at its own cost of funds, for the life of the trade. The funding is real, continuous, and asymmetric, and its expected cost, the funding valuation adjustment of Chapter 9, has to be earned back somewhere. A dealer quoting the mid-market level on a one-way CSA is quoting a price that does not cover its own cost of carrying the trade.

**Why the collateral rate is the most important term.** It sets the discount curve. Section 7.4 established that a collateralized derivative is discounted at the rate its collateral earns, so that one clause determines the value of every cash flow in the contract. A CSA paying an overnight rate and one paying overnight minus a spread produce different prices for identical trades, and the difference compounds over the life. Everything else in the document (thresholds, minimum transfer amounts, the timing of calls) adjusts the mechanics; the collateral rate changes the number.

*What the question tests.* That the CSA is priced, not merely negotiated, and that the dealer's willingness to accept a one-way agreement is itself a price concession.

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**8.2** *A fund posts $40 million of variation margin and $25 million of initial margin, and its funding costs 60 basis points more than posted margin earns. Give the annual economic cost of each, and explain why they differ along the dimensions of what each covers, direction, and reusability.*

Posted margin earns its own return (interest on cash, the yield on bonds), so what each posting costs to carry is the 60 basis point spread over that return:

$$\text{VM: } \$40\text{m} \times 0.0060 = \$240{,}000 \text{ a year}, \qquad \text{IM: } \$25\text{m} \times 0.0060 = \$150{,}000 \text{ a year}.$$

Those are the funding carries, and they are computed the same way. The economic costs differ: $150,000 a year for the initial margin and nothing for the variation margin. The three dimensions show why.

**What each covers.** Variation margin settles a loss that has already happened: it transfers value the poster no longer has. Initial margin covers a loss that has *not* happened: the potential future move over the close-out period. The first is a settlement, the second is a deposit against a contingency.

**Direction.** VM flows one way at a time, netted (whichever party the trade has moved against pays), and a position that moves back returns it. IM is posted by both parties simultaneously and stays posted for as long as the trade exists, regardless of which way it moves.

**Reusability.** VM is transferred outright or held with a right of use, so the receiver may use it. Within the system, VM funding therefore largely nets out: someone is funding it, but someone else is holding usable cash. IM is segregated with a third-party custodian precisely so it *cannot* be used, which is question 3's subject.

So the \$150,000 is the deadweight and the \$240,000 is not a cost of the trade at all: it funds a loss the fund had already suffered and would bear whether or not collateral moved, and the cash sits usable in the dealer's hands. IM removes cash from the system for the life of the trade; VM moves it from one party to another. That asymmetry is why the industry fought over IM thresholds and accepted VM without much argument.

*What the question tests.* That two numbers computed the same way are economically different, and that reusability is the dimension that decides which.

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**8.3** *Explain why initial margin is posted gross by both parties to a third-party custodian rather than netted between them.*

Netting initial margin between the two parties would defeat its purpose. Each party owes the other a similar amount, so netting two equal IM obligations would leave neither side holding any protection at all. IM exists to cover the loss that arises *after* a counterparty has already defaulted: the move in the position during the days it takes to close out or replace it. At that moment the defaulter is, by construction, not performing. Any IM it was supposed to be holding on the survivor's behalf, or any offset against what it owed, is worth nothing.

Posting gross means each party's protection is genuinely there when needed: the survivor holds, or can call from a custodian, an amount that does not depend on the defaulter's cooperation. Netting would leave each side's protection contingent on the other side being solvent, which is the one thing that is certainly untrue in the scenario the margin is for.

The third-party custodian is the second half of the same argument. Collateral held by the counterparty itself is an unsecured claim on that counterparty presented as protection, recoverable only through the insolvency, at the same time and on the same terms as everything else. Segregating it with a custodian puts it outside the defaulter's estate, so it can be returned or applied without waiting for an administrator.

The cost is exactly the benefit. The collateral is immobilized: neither party can use it, and the system as a whole holds more cash idle than a netted arrangement would require. That is the price of protection that survives the event it was bought for.

*What the question tests.* That IM's design follows from asking what will be true at the moment it is needed, rather than what is efficient beforehand.

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**8.4** *A portfolio has sensitivities of \$50,000 per basis point in a bucket with a risk weight of 55 bp and \$90,000 per basis point in a bucket with a risk weight of 45 bp, with a prescribed correlation of 0.6. Calculate the initial margin, compare it with the undiversified sum, and explain why the result leads firms to concentrate their trades with one counterparty.*

The margin for two correlated buckets is

$$\mathrm{IM} = \sqrt{K_1^2 + K_2^2 + 2\rho K_1 K_2},$$

with each bucket's charge the sensitivity times its risk weight:

$$K_1 = 50{,}000 \times 55 = \$2{,}750{,}000, \qquad K_2 = 90{,}000 \times 45 = \$4{,}050{,}000.$$

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$$\text{IM} = \sqrt{2.75^2 + 4.05^2 + 2 \times 0.6 \times 2.75 \times 4.05}\ \text{million} = \$\mathbf{6{,}109{,}828}.$$

The undiversified sum is 2,750,000 + 4,050,000 = \$6,800,000, so the correlation saves

$$6{,}800{,}000 - 6{,}109{,}828 = \$690{,}172, \quad \mathbf{10.1}\%.$$

The benefit is modest because $\rho = 0.6$ is high: the two buckets mostly move together, so there is little genuine offset to recognize. At $\rho = 0$ the same charges would give $\sqrt{2.75^2 + 4.05^2} = \$4.90$m, a 28% saving; at $\rho = 1$ the formula collapses to the plain sum and the saving vanishes.

**Why this leads firms to concentrate their trades with one counterparty.** The offset is recognized only within one master agreement and one margined portfolio. Split the same two positions across two counterparties and each computes IM on its own portfolio, the offset is lost, and the firm posts \$6.8 million rather than \$6.1 million. Spreading counterparties costs money and concentrating them saves it. The margin rules therefore reward concentration, which is exactly what the counterparty risk management of Chapter 27 tries to limit.

The structure is worth noting: the model grants diversification through a prescribed correlation matrix rather than an estimated one. That makes the result reproducible by both parties (essential when the number is exchanged daily and disputes are expensive), at the cost of being wrong whenever realized correlations differ from the prescription, which is precisely when it matters.

*What the question tests.* That margin models aggregate by a correlation rule, and that the size of the diversification benefit is governed by $\rho$ rather than by the number of buckets; and how the portfolio on which margin is computed shapes the choice of counterparties.

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**8.5** *Explain why a CSA permitting several collateral currencies gives the poster an option, and why that option is an option over the discount curve.*

A CSA permitting several eligible currencies lets the poster choose which to deliver, and the choice is free and repeatable: collateral can be substituted. A rational poster delivers whichever currency is cheapest for it to fund at that moment, and switches when the ranking changes. That is an option, held by the poster and written by the receiver, and it was given away in the documentation without anyone pricing it.

It is an option over the discount curve because of Section 7.4's result: the discount rate is the rate paid on the collateral actually held. If dollars are posted, the trade discounts on the dollar overnight curve; if euros, on the euro curve. So the poster's choice determines the curve, and the poster will choose the one that makes the trade cheapest for itself, which is the one that makes it least valuable to the receiver.

The correct valuation is therefore not on any single curve but on the cheapest to deliver across the eligible set, evaluated at every date, because the poster can switch at any time. That is the same cheapest-to-deliver structure as Chapter 6's Treasury basket, and it has the same consequence: the receiver must value assuming the worst admissible choice, and the option's value moves with the cross-currency basis.

*What the question tests.* That optionality can be created by a clause about operational eligibility, and that recognizing it requires connecting the collateral schedule to the discounting result of the previous chapter.

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**8.6** *A pension fund must post £300 million of cash margin, which earns SONIA at 4.0% once posted. It holds bonds yielding 3.8% and can repo them at 4.3% with a 3% haircut. How many bonds must it deliver, and what is the annual cost of the transformation?*

With a 3% haircut, delivering bonds of market value $B$ raises $0.97B$ of cash, so raising £300 million requires

$$B = \frac{300}{0.97} = \text{£}\mathbf{309.3}\text{ million of bonds.}$$

The bond yield is a distraction. The fund earns 3.8% on the bonds whether or not it repos them, so the yield is the same on both sides of the comparison and drops out. What the transformation adds is a repo payment on the cash borrowed and a SONIA receipt on the same cash once it is posted as margin, so the annual drag is the spread between those two:

$$(4.3\% - 4.0\%) \times \text{£}300\text{m} = \text{£}\mathbf{0.9}\text{ million a year.}$$

Netting the repo rate against the bond yield instead, as if the bonds earned only because they were pledged, would count the yield the fund already had as a benefit of the trade.

Two features of this are worth drawing out. The haircut immobilizes an extra £9.3 million of bonds that raise no cash at all; they are encumbered purely as overcollateralization, and they earn their coupon but cannot be sold or pledged again. And the 30 basis point spread is the cost of holding the wrong kind of asset: the fund owns bonds and owes cash, and the transformation between them is not free. Nor is the spread the main risk: the repo is short-dated and must be rolled, and a haircut that widens or a dealer that declines at the next roll costs far more than the spread.

The wider point is that this cost is invisible in any measure of the fund's market risk. Its duration is unchanged, its exposure is unchanged, its hedges are unchanged, and it is paying £0.9 million a year for the privilege of meeting margin calls in the form they are demanded.

*What the question tests.* That collateral transformation has a price set by the repo rate against what the posted cash earns, not against the bond yield, and that a fund can be perfectly hedged and still lose money every year.

**8.7** *An airline expects to buy 500 million gallons of jet fuel evenly over the next two years and hedges by paying a fixed $2.80 a gallon in swaps that are margined daily. Jet fuel then falls to $2.10 a gallon. Calculate the variation margin the airline must pay and the offsetting saving on its fuel bill, and state when each becomes cash. Explain why the airline can be perfectly hedged against price risk and still fail.*

**The margin call.** The airline pays fixed and receives the floating price, so a fall of $0.70 a gallon is a loss on the swaps of

$$500\text{m gallons} \times (\$2.80 - \$2.10) = \$\mathbf{350}\text{ million,}$$

and because the swaps are margined daily, it is paid in cash **now**.

**The offsetting saving.** The fuel the airline has yet to buy is cheaper by the same $0.70 a gallon, so its fuel bill falls by the same **$350 million**. But that saving arrives only as the fuel is bought: about $175 million a year over the next two years, a little at every refueling.

**Why it can fail while hedged.** The two positions offset in *value*, as the hedge intends: on a mark-to-market basis the airline has neither gained nor lost. They do not offset in *cash*. The swaps settle daily; the hedged item, future fuel purchases, produces its offsetting cash only when the fuel is bought. A fall in prices that leaves the airline's net value unchanged therefore demands $350 million today against savings spread over two years. If the airline cannot raise that sum, the swaps are closed out, which converts an offsetting position into a realized loss:

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the hedge is gone and the exposure is not. The airline fails while being right about everything except its cash.

The quantity that would have warned it is the gross cash outflow the hedge can demand before the offsetting cash arrives, which is what the stress tests of Section 8.10 measure. Metallgesellschaft, the 2022 European utilities and the liability-driven investment funds of 2022 are the same failure.

*What the question tests.* That price risk and funding risk are different quantities, and that hedging perfectly against the first can create the second.

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**8.8** *A fund must cover a $250 million exposure with bonds subject to a 5% haircut. How much collateral must it deliver? Recalculate at a 12% haircut, and explain why the two other things happening at the same moment make the position worse than the comparison suggests. Now suppose the $250 million is initial margin under the uncleared margin rules, posted in euro-denominated government bonds with more than five years to maturity against trades that settle in dollars. Using the standardized schedule (4% on the bonds plus the 8% currency add-on), how much must the fund deliver, and how much of that is due to the currency?*

With a haircut $h$, delivering collateral of market value $C$ covers $C(1-h)$ of exposure, so covering $250 million requires

$$h = 5\%: \quad C = \frac{250}{0.95} = \$\mathbf{263.2}\text{m}, \qquad h = 12\%: \quad C = \frac{250}{0.88} = \$\mathbf{284.1}\text{m},$$

an additional **$20.9 million** of bonds to support an unchanged exposure.

The comparison understates the position because the haircut does not widen on its own. It widens in stress, and two other things are happening in the same moment.

**The exposure has grown.** Haircuts widen when markets move, and the move that widened them has also moved the derivative, so the $250 million being collateralized is not $250 million any more. The fund is covering a larger number at a worse rate.

**The collateral is worth less.** The bonds being delivered have fallen in price, so the same nominal holding delivers less market value, and more of it must be found to reach the required amount.

The three effects (the wider haircut and these two) arrive together because they have one cause. Each is manageable alone; arriving together, on the same morning, they are the mechanism by which a solvent fund becomes a forced seller, while every other fund holding the same collateral is selling it into the same market.

**Posted as initial margin in another currency.** The standardized schedule charges the bonds 4% and, because they are in euros against trades that settle in dollars, the 8% currency add-on as well, for a combined haircut of 12%:

$$C = \frac{250}{1 - 0.04 - 0.08} = \frac{250}{0.88} = \$\mathbf{284.1}\text{m}.$$

The same bonds in dollars would need $250/0.96 = $260.4 million, so the currency accounts for **$23.7 million**. The add-on alone takes a posting made in a calm market to the level the stressed 12% haircut produced above. It applies here because this is initial margin posted in bonds; cash variation margin in a major currency is exempt. A fund that holds its bonds in one currency and trades in another pays it on every posting.

*What the question tests.* That a haircut comparison holding everything else fixed is the wrong comparison, because nothing else stays fixed when haircuts move; and that the currency of the collateral carries a haircut of its own.

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**8.9** *A swap has a five-day 99% risk of $20 million. Compare the collateral immobilized if it is cleared with the amount immobilized bilaterally under UMR, stating your assumptions. Why were the rules calibrated this way, and what is the cost of that design?*

**Assumptions.** A cleared trade's initial margin is set to a five-day close-out at 99%, which is the $20 million given. Uncleared margin rules require a ten-day close-out at the same confidence, and require both parties to post gross; other trades in the relationship have already used up its €50 million threshold, so the whole amount is posted. On the cleared side only the client's margin is counted: the dealer's leg offsets against the rest of its cleared book and adds little to its own IM, as in Example 8.5.

Scaling the horizon by the square root of time:

$$\mathrm{IM}_{\text{bilateral}} = 20 \times \sqrt{\frac{10}{5}} = 20 \times 1.414 = \$28.3\text{m each side},$$

$$\text{total immobilized} = 2 \times 28.3 = \$\mathbf{56.6}\text{ million},$$

against **$20 million** cleared, roughly **2.8 times** as much collateral locked away for the same economic risk. If the dealer's leg found no offset and it posted $20 million too, the comparison would be $40 million against $56.6 million, and only the margin period's factor of 1.41 would remain.

**Why calibrated this way.** Deliberately, to make clearing cheaper than staying bilateral. The post-crisis objective was to move standardized derivatives into central counterparties, and rather than mandate it for everything, the rules priced the alternative so that clearing won on cost. The longer close-out period is defensible on its own terms (a bilateral position genuinely takes longer to replace than a cleared one), but the gap's size reflects policy as much as risk.

**The cost of that design.** It worked, and what it bought is the concentration Chapter 2 examines. Risk that was diffused across a network of bilateral relationships now sits in a small number of clearing houses whose margin models are themselves a systemic variable, and whose failure is not survivable. The rules did not remove the risk; they chose where to put it, and the choice was made on the view that a measurable, mutualized, well-capitalized concentration is preferable to an opaque dispersion.

The cost the chapter names falls on the other side of the line: genuinely bespoke hedges, which cannot be cleared, now carry a collateral cost that may exceed the value of the tailoring, so some risk-reducing hedges are no longer done.

*What the question tests.* That a margin rule's calibration can be an instrument of policy, and that the policy's success and its principal danger are the same fact.

**8.10** *Explain what rehypothecation is, why it is economically efficient, and why initial margin may not be rehypothecated while variation margin may.*

**Rehypothecation** is the reuse of collateral received: a party that takes in securities or cash against an exposure pledges the same assets onward to support its own borrowing or its own margin obligations.

**Why it is efficient.** A given pool of high-quality assets can support several obligations in sequence rather than sitting idle against one. The market needs less collateral in total to support the same volume of trading, financing costs fall, and the assets are held by whoever values them most. Restricting reuse raises the demand for collateral and its price.

**Why IM may not be reused and VM may.** The distinction follows from what each is for, exactly as in question 2. Variation margin settles a loss that has occurred: it is transferred outright or held with a right of use, so reusing it is simply using money the receiver is entitled to

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use. Initial margin covers a loss that has not occurred and is held *against the possibility* of the poster defaulting. If the receiver has pledged it onward, then at the moment the poster defaults (or, worse, at the moment the *receiver* defaults and the poster wants its margin back) the assets are somewhere else in a chain, and recovering them requires the solvency of every link.

So the prohibition on reusing IM is the same argument as segregation: protection that depends on a chain of counterparties being solvent is not protection against a counterparty failing. The efficiency is real and was given up knowingly.

*What the question tests.* That the reuse rules are not arbitrary, and that the line between IM and VM is drawn by whether the collateral is settling something or securing something.

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**8.11** *Give two examples of wrong-way risk in the choice of collateral, and explain the common feature that makes them dangerous.*

**Two examples follow**, both from Section 8.7. A dealer takes the government bonds of a bank's own sovereign as collateral against its exposure to that bank. And a dealer takes energy-sector securities as collateral against its exposure to an energy trading firm. The extreme case, a counterparty posting its own securities, is the same hazard in its plainest form, which is why it is prohibited outright.

**The common feature.** In both, the collateral's value falls at the moment the counterparty defaults, because the same event drives both. Collateral is supposed to be worth what it is worth *independently* of the counterparty; the whole construction assumes the two are unrelated. When they are correlated, the protection thins exactly as the need for it arrives, and the measured coverage before the event tells you nothing about the coverage during it. The collateral is not wrong on average; it is wrong in the only state where it is called on.

*What the question tests.* That correlation between collateral and counterparty destroys the premise of collateralization.

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**8.12** *A dealer calls a counterparty for \$60 million of variation margin. The counterparty pays \$35 million and disputes the rest, and the gap is outstanding for 20 days, which the dealer funds at 4.8%. Calculate the funding cost of the dispute. The dispute is the third on the netting set in two quarters to outlast the margin period of risk, so the period doubles from ten days to twenty for the next two quarters. If the netting set's potential future exposure add-on is \$150 million and capital costs 40 basis points a year on the exposure, what does the doubling cost? Explain why a margin dispute is a funding event that clusters at the worst possible time.*

**The funding cost.** The dealer is short the disputed \$25 million for 20 days:

$$\$25\text{m} \times 0.048 \times \frac{20}{360} = \$\mathbf{66{,}667}.$$

**The cost of the doubling.** The potential future exposure add-on scales with the square root of the margin period of risk through the maturity factor (Section 27.3); replacement cost is unaffected. Doubling the period therefore raises the add-on to

$$\$150\text{m} \times \sqrt{2} = \$212.1\text{m},$$

an extra \$62.1 million of exposure for two quarters. At 40 basis points a year,

$$\$62.1\text{m} \times 0.0040 \times 0.5 \approx \$\mathbf{124{,}000},$$

nearly twice the funding cost of the dispute itself, and incurred whichever party's valuation was right.

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**Why a dispute is a funding event.** A dispute means one party believes it is owed collateral and has not received it. Whatever the merits, the party that made the call must fund the shortfall itself while the disagreement is resolved, so the dispute converts a valuation disagreement into an immediate cash requirement.

**Why disputes cluster at the worst time.** Disputes arise when valuations are hardest: when markets have moved sharply, when prices are stale or unobservable, and when models disagree. That is the same moment every other margin call is large, funding is tight, and counterparties are least willing to concede. Disputes are not independent events: they are generated by volatility, and so is everything else straining the firm's liquidity that day.

*What the question tests.* That the visible funding cost of a dispute is the smaller part of it, and that the operational machinery around margin has its own failure mode, timed to coincide with every other strain.

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**8.13** *A CSA gives a counterparty a threshold of \$5 million and requires it to post an independent amount of \$2 million. The minimum transfer amount is \$500,000, and calls are rounded up to the next \$100,000. The dealer's exposure is \$14.23 million and it holds collateral worth \$8.0 million after haircuts. Calculate the margin call. The next day the exposure is \$14.6 million: is there a further call? How much of the exposure is then uncollateralized, and why?*

**The call.** The credit support amount is exposure plus independent amount minus threshold:

$$14.23 + 2 - 5 = \$11.23\text{m}.$$

The dealer holds \$8.0 million, so the shortfall is $11.23 - 8.0 = \$3.23$ million. That exceeds the \$500,000 minimum transfer amount, and rounding up to the next \$100,000 makes the call **\$3.3 million**. Once it is met the dealer holds \$11.3 million.

**The next day.** The credit support amount is $14.6 + 2 - 5 = \$11.6$ million, \$0.3 million more than the dealer holds. That is below the minimum transfer amount, so **no call is made**.

**What is left uncollateralized.** The exposure is \$14.6 million against \$11.3 million of collateral, so **\$3.3 million** is unsecured. Three terms account for it: the \$5 million threshold, less the \$2 million independent amount that offsets part of it, plus the \$0.3 million shortfall that the minimum transfer amount lets stand:

$$5 - 2 + 0.3 = \$3.3\text{m}.$$

Rounding works the other way: it made the first day's call \$70,000 larger than the shortfall, which the second day's increase in exposure then absorbed. The threshold is an unsecured credit line by design and the dealer prices it as one (Chapter 9); the minimum transfer amount adds a small, fluctuating unsecured slice on top.

*What the question tests.* That a margin call is a mechanical consequence of the CSA's terms, and that even a fully performing CSA leaves the receiver exposed by the threshold net of the independent amount, plus whatever the minimum transfer amount allows to accumulate.

---

**8.14** *A dealer posts \$600 million of cash collateral under a CSA permitting dollars, euros, or sterling. Dollar collateral earns SOFR, the cross-currency basis against the dollar is −20 basis points for euros and −8 for sterling, and the portfolio annuity is 6. Which currency does the dealer post, and what is the approximate value of that choice relative to posting dollars? In what respect does this estimate understate the value of the CSA's optionality, and in what respect does it overstate it?*

**Which currency.** As in Example 8.6, a currency's collateral, swapped back into dollars, earns SOFR minus its basis against the dollar: SOFR plus 20 basis points for euros, SOFR plus

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8 for sterling, and SOFR flat for dollars. The dealer posts **euros**, the currency that earns most and is therefore cheapest to fund.

**The value.** Posting euros rather than dollars raises the effective discount rate on the dealer's liability by 20 basis points, worth roughly

$$\$600\text{m} \times 0.0020 \times 6 = \$\mathbf{7.2}\text{ million}.$$

Sterling, the second choice, would be worth $\$600\text{m} \times 0.0008 \times 6 = \$2.88$ million.

**Why it understates.** The estimate prices only the currency that is cheapest today. The CSA lets the dealer switch whenever the ranking changes, and that option to switch has value of its own, which grows as the bases become more volatile and less closely correlated.

**Why it overstates.** Using the annuity assumes the $600 million balance stays constant over the portfolio's life. A swap book amortizes, so the collateral balance, and with it the value of the choice, shrinks over time.

*What the question tests.* That the collateral currency choice can be valued from the basis and the annuity, and which way the simple estimate errs on each side.

---

**8.15** *A dealer's portfolio with one counterparty holds an interest rate swap whose SIMM margin on its own is $8 million and an equity swap whose margin on its own is $5 million, and the two positions have tended to move in opposite directions. What initial margin does SIMM require on the two together? Replace the equity swap with an FX forward whose margin on its own is $5 million. SIMM prescribes a correlation of about 0.3 between the interest rate and FX margins, whatever the positions' signs: what is the margin now? Explain the difference.*

**Interest rate swap and equity swap.** Interest rates and equity sit in different SIMM product classes (rates and FX, and equity), and SIMM adds the margins of its product classes with no offset at all:

$$\text{IM} = 8 + 5 = \$\mathbf{13}\text{ million}.$$

That the two positions have tended to move in opposite directions makes no difference: SIMM recognizes no diversification across product classes, whatever the history shows.

**Interest rate swap and FX forward.** Interest rates and FX are different risk classes within the same product class, and their margins are aggregated with a correlation. The correlation applies to the two margins, which are both positive, so the directions of the positions do not enter:

$$\text{IM} = \sqrt{8^2 + 5^2 + 2 \times 0.3 \times 8 \times 5} = \sqrt{113} = \$\mathbf{10.63}\text{ million},$$

$2.37 million, or 18%, below the sum.

**The difference.** Within a product class SIMM recognizes partial offset between risk classes; across product classes it recognizes none. A dealer whose book with a counterparty spans several product classes therefore posts the sum of their margins, however well the positions hedge one another. Splitting such a book across counterparties costs nothing extra in SIMM terms, while splitting the rates and FX trades of one product class does.

*What the question tests.* That SIMM's diversification stops at the product class boundary, so a correlation the dealer can observe is not one the model will credit.

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# Chapter 9
# XVA: Valuation Adjustments in Practice

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**9.1** *Explain why counterparty exposure is* $\max(V(t), 0)$ *rather than* $|V(t)|$*, and why this makes counterparty risk an option.*

If a counterparty defaults when the trade is worth $V(t) > 0$ to the dealer, the dealer loses that amount: it is owed the money and will recover only a fraction. If the counterparty defaults when $V(t) < 0$, the dealer owes *it*, and the obligation does not disappear: the administrator will collect it in full. So the loss on default is

$$\max(V(t), 0),$$

never $|V(t)|$. The negative states contribute nothing, not because they are harmless but because they are not losses at all.

That truncation at zero is precisely a call payoff on the trade's own value, with strike zero, exercised at a random time. Counterparty risk is therefore an option, and it inherits everything options do: its value rises with the volatility of $V$, it depends on the whole distribution rather than the mean, and it cannot be priced from the expected value of $V$ alone; CVA needs $\mathbb{E}[\max(V, 0)]$, the expected exposure, not $\mathbb{E}[V]$.

Two consequences follow immediately. A trade whose expected value is zero still carries positive counterparty risk, because $\mathbb{E}[\max(V, 0)] > 0$ whenever $V$ can move. And hedging CVA requires hedging an option, which is why an XVA desk trades gamma and vega, not merely credit, and why question 12's loss is possible on a fully spread-hedged book.

*What the question tests.* That the asymmetry in the definition is what makes the subject hard, and that recognizing the option is what connects this chapter to the rest of the book.

---

**9.2** *Explain why an interest rate swap's expected exposure peaks partway through its life while a cross-currency swap's rises to maturity, and state which carries more counterparty risk and why.*

An interest rate swap's exposure is driven by two effects pulling opposite ways. Rates diffuse, so the range of possible values widens with the square root of time. But payments run off, so the remaining annuity shrinks toward zero at maturity. Their product rises, peaks, and falls (the hump of Figure 9.1), with the maximum roughly a third of the way through.

A cross-currency swap has the same diffusion but not the same run-off, because the principals are exchanged at maturity. The largest single cash flow sits at the very end, and its value moves with the exchange rate right up to the final day. There is no amortization to offset the diffusion, so the profile rises monotonically and is at its highest when the contract terminates.

The cross-currency swap carries far more counterparty risk, for two reinforcing reasons. Its exposure peaks at the point of maximum accumulated uncertainty rather than partway along;

and the quantity at risk is a principal amount rather than a stream of rate differences, so it is several times larger. This is Chapter 7's point that notionals are exchanged because the exchange is the substance of the trade. Figure 9.1's schematic curve does not show the second reason at full size: as drawn it gives a CVA only about 9% above the interest rate swap's, where a principal exposed to ordinary currency volatility would put the profile well above it.

*What the question tests.* That the shape of an exposure profile follows from the contract's cash-flow structure, and that a single structural feature, the final principal exchange, changes both the shape and the scale.

---

**9.3** *Explain why exposure is computed on the netting set rather than trade by trade, and why a dealer might quote a better price for a trade that offsets its existing position with a client.*

Exposure is computed on the netting set because that is what would actually be lost. Under a master agreement with close-out netting, a default terminates every trade with that counterparty and reduces them to one net amount, so a trade worth +\$10m alongside one worth −\$9m produces an exposure of \$1m, not \$10m. Measuring trade by trade would ignore the legal fact that makes the portfolio a single claim, and would overstate the risk by roughly the ratio Chapter 1 puts at nearly seven to one across the market.

It follows that a new trade's contribution to exposure depends on what is already there. A trade that offsets the dealer's existing position with that client reduces the netting set's exposure, and therefore reduces the CVA the dealer must charge. The incremental CVA can be negative (the trade makes the portfolio safer), and a dealer pricing correctly passes some of that back, quoting inside its usual spread.

This is not a favor but arithmetic, and it has a consequence worth noticing: the same trade has different prices at different dealers, depending on each one's existing book, and a client can sometimes get a better quote from a dealer whose existing trades with it point the other way. Counterparty risk makes derivatives pricing portfolio-dependent, which nothing in the earlier chapters would lead one to expect.

*What the question tests.* That netting turns pricing into a portfolio problem, and that the pricing consequence is observable in dealers' quotes.

---

**9.4** *A three-year uncollateralized swap has expected exposures of \$1.2m, \$1.5m, and \$0.9m at mid-year points. The counterparty's CDS trades at 180 basis points, recovery is 40%, and discount factors are 0.97, 0.93, and 0.89. Compute the CVA, first as the cost of insuring the exposure and then as loss given default times the default rate the spread implies, and confirm the two agree.*

**As the cost of insurance.** The credit triangle puts the default rate the spread implies at $\lambda = 0.018/0.60 = 3\%$ a year. Protection on each year's expected exposure costs the 180 basis point premium, paid only while the counterparty survives, so the cost of cover for year $i$ is $0.018\, e^{-0.03(i-1/2)}$: 1.773%, 1.721% and 1.670%. The contributions are 1.2m×1.773%×0.97 = \$20,640, 1.5m × 1.721% × 0.93 = \$24,005 and 0.9m × 1.670% × 0.89 = \$13,376, a CVA of \$58,021.

**As loss given default times the implied default rate.** The same CVA is the loss given default applied to the expected exposure at each date, weighted by the probability of defaulting then and discounted:

$$\text{CVA} = (1-R)\sum_i \text{EE}_i \left[Q(t_{i-1}) - Q(t_i)\right] \text{DF}_i, \qquad Q(t) = e^{-\lambda t}.$$

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| Year | EE ($m) | Marginal PD | DF | Contribution ($) |
|---|---|---|---|---|
| 1 | 1.2 | 0.029554 | 0.97 | 34,401 |
| 2 | 1.5 | 0.028681 | 0.93 | 40,010 |
| 3 | 0.9 | 0.027833 | 0.89 | 22,295 |
| | | | | **96,706** |

$$\text{CVA} = 0.60 \times 96{,}706 = \$\mathbf{58{,}024}.$$

The two readings agree to within $3, the gap being the mid-year approximation of the survival weight. They must agree, because a CDS is priced so that its premiums and its expected payouts balance.

Two details follow. The marginal default probabilities decline across the years (2.955%, 2.868%, 2.783%) because defaulting in year three requires surviving the first two, and the survival probability is falling. And the second year contributes most despite a lower default probability than the first, because the exposure is highest there: CVA is a product, and the year that dominates is the one where exposure and default probability are jointly largest, not where either is largest alone.

*What the question tests.* That the calculation is a sum over dates of three factors, that insuring the exposure and bearing its expected loss cost the same, and that the shape of the exposure profile matters as much as the credit.

---

**9.5** *Explain why CVA is computed from the counterparty's CDS spread rather than from its historical default rate, and what would happen to a dealer that used the historical rate.*

CVA is a price (it is charged to the client and it must be hedged), so it has to be what the hedge costs. The instrument that hedges CVA is a credit default swap on the counterparty, and its spread is the price of that protection: it embeds a premium for bearing a risk that pays off in bad times, and compensation for the seller's funding and capital, as well as the expected loss. A CVA built from the historical default rate would be a forecast of the loss, inconsistent with the cost of the hedge.

The gap is large, not marginal. Section 24.9 measures it: most of a credit spread is not expected loss (the risk premium and liquidity components dominate for investment-grade issuers), so the default rate a spread implies has run at several times the historical one.

A dealer using the historical rate would therefore systematically undercharge for counterparty risk. It would win business, because its quotes would beat competitors pricing correctly, and the business it won would be exactly the business whose risk it had underpriced. When it then tried to hedge in the CDS market, the protection would cost more than it had collected, and the difference would be a loss recognized immediately. Adverse selection would do the rest: the counterparties most eager to trade at those prices are the ones the market thinks are riskiest.

*What the question tests.* That the choice of input follows from what the number is for, and that "the real probability" is the wrong answer when the number is a price rather than a forecast.

---

**9.6** *State the argument for DVA and the objection to it, explain why regulators removed DVA from capital, and state what that leaves banks managing.*

**The argument for DVA.** CVA is symmetric. If a dealer charges a client for the possibility that the client defaults, then by the same logic the client should charge the dealer for the possibility that the *dealer* defaults, and from the dealer's side that possibility is an asset, because

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its own default would extinguish liabilities at a fraction of their value. Accounting standards agree: fair value means exit value, and a liability's exit value reflects the obligor's own credit.

**The objection.** DVA is a gain that can be realized mainly by defaulting, and only in part through the funding benefit of Section 9.4. A bank whose credit deteriorates books a profit, and the worse its condition the larger the profit, which is perverse on its face. It cannot be hedged in any straightforward way, because hedging it means selling protection on oneself. And it is not available to the shareholders whose accounts record it: in the default that monetizes it, they receive nothing.

**Why regulators removed it from capital.** Capital exists to absorb losses in going-concern conditions. Allowing a bank to count as capital a gain that materializes only in its own failure would let deteriorating banks report improving capital ratios; the 2011 gains showed how large the effect could be. So DVA is excluded from regulatory capital while remaining in accounting fair value.

**What that leaves banks managing.** A permanent gap between the accounting number and the regulatory one: DVA volatility runs through reported earnings and is stripped out of capital, so the two measures move differently on the same event. Managing it means explaining, every quarter, a profit that regulators do not recognize and shareholders cannot spend.

*What the question tests.* That the symmetry argument is logically sound and practically unusable, and that the resolution was to split the answer between two frameworks rather than to settle it.

---

**9.7** *Set out the Hull–White argument against FVA and the practitioner response. In what sense was the debate settled and in what sense was it not? Identify where FVA and DVA arguably double-count, and state what a firm would have to believe to include both.*

**The Hull–White argument.** FVA should not exist. A derivative's value is determined by its cash flows and their risk, and a dealer's own funding cost is a property of the dealer, not of the trade. Two dealers with different funding costs must still quote the same value, or one of them is mispricing. Charging a client for the dealer's funding spread confuses the cost of running the firm with the value of the contract and, worse, double-counts, because a funding spread compensates lenders for default risk that DVA already recognizes.

**The practitioner response.** The argument holds in a world where the dealer can borrow and lend at the risk-free rate in unlimited size. No dealer can. An uncollateralized trade in the dealer's favor must be funded until it settles, at the dealer's actual cost of money, and that cash outflow is real, contractual, and appears in the accounts. A model that says the cost is zero disagrees with the treasury statement.

**Settled and unsettled.** It was settled in practice: essentially every dealer computes FVA, auditors accept it, and the initial adoptions ran to billions of dollars of one-off charges. In theory it was settled later, and by splitting the difference: Andersen, Duffie, and Song (2019) showed that funding an uncollateralized position transfers value from the bank's shareholders to its creditors, so FVA is a real cost to the shareholders who decide whether to trade and belongs in the price, while Hull and White are right that it is not part of the instrument's value to the bank's stakeholders taken together. What remains unsettled is the overlap with DVA, below, which each firm still resolves its own way, so that two dealers can quote different prices on the same trade.

**Where they double-count.** DVA values the dealer's own default from the liability side; the funding spread the dealer pays exists largely *because* the dealer might default. Both therefore monetize the same credit. To include both at full value, a firm would have to believe its funding spread is compensating lenders for something other than its own default risk (liquidity premia,

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balance sheet scarcity, market segmentation) and that the two components are separable and non-overlapping.

*What the question tests.* That the debate was settled in practice by accounting and in theory by asking who bears the cost, that the overlap with DVA is still open, and that the reader can say which part of each side is right.

---

**9.8** *Using the trade of Example 9.4, recompute MVA at a funding spread of 90 basis points and average initial margin of $2.5 million, compare it with the CVA of Example 9.2, and explain why KVA is nonetheless often the largest adjustment on a long-dated uncollateralized trade.*

MVA is the funding cost of initial margin over the life of the trade, approximately the funding spread multiplied by the average margin and by the discount-weighted duration:

$$\text{MVA} \approx s_F \times \overline{\text{IM}} \times \text{duration}.$$

With Example 9.4's duration of 4.53 years, a 90 basis point spread and $2.5 million of average initial margin give

$$\text{MVA} = 0.0090 \times \$2.5\text{m} \times 4.53 = \$\mathbf{102{,}000},$$

slightly more than the $94,000 CVA of Example 9.2, and three times the $34,000 of Example 9.4 itself, a change that comes entirely from the dealer's funding spread and the margin model, with the counterparty unchanged. Twice the example's margin is what a stressed look-back period in the margin model, or a ten-day close-out on a longer trade, can produce.

That comparison is the point of the question: MVA is charged on a counterparty that may be entirely safe, because initial margin is posted regardless of credit quality. A trade with a pristine counterparty can carry negligible CVA and substantial MVA.

KVA is nonetheless often the largest adjustment on a long-dated uncollateralized trade, because it is the cost of the regulatory capital the trade requires, carried for its whole life at the firm's cost of equity rather than its cost of debt. Equity is expensive (a hurdle of ten percent or so against a funding spread of under one percent), and long-dated uncollateralized exposure attracts capital under several charges at once: counterparty risk, the CVA capital charge of question 11, and the leverage ratio. The adjustment that dominates is the one priced off equity rather than off credit or funding.

*What the question tests.* That the XVA family is a set of separately caused charges, and that which one dominates depends on the trade's collateral and tenor rather than on the counterparty.

---

**9.9** *A portfolio's value at the peak of its exposure profile is normally distributed about zero with a standard deviation of $9 million. Compute the expected exposure and the 95% potential future exposure, and explain which is used for pricing, which for credit limits, and why.*

For $V \sim N(0, \sigma^2)$ with $\sigma = \$9$ million:

$$\text{EE} = \mathbb{E}[\max(V, 0)] = \frac{\sigma}{\sqrt{2\pi}} = 9 \times 0.39894 = \$\mathbf{3.59} \text{ million},$$

$$\text{PFE}_{95\%} = 1.645\,\sigma = \$\mathbf{14.81} \text{ million}.$$

**Expected exposure is used for pricing.** CVA is the cost of insuring the exposure, and the hedge is sized to its average, so the quantity that enters it is the mean of the truncated distribution. Using a percentile would overstate the expected cost by a factor of four here.

**Potential future exposure is used for credit limits.** A limit is not a price; it answers "how bad could this plausibly get?" which is a question about the tail. A credit officer sizing how much business to do with a counterparty cares about the adverse case, not the average one.

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The two differ by more than four times on the same position, and using either for the other's purpose fails in a predictable direction: pricing off PFE makes the dealer uncompetitive on every trade, and setting limits off EE lets exposures accumulate that the firm could not absorb. They are different questions about the same distribution, and the distribution is asymmetric precisely because of question 1's truncation.

*What the question tests.* That the choice of statistic follows from the decision it supports, and that the gap between mean and percentile is large enough to matter.

---

**9.10** *A client requests a $3 million threshold in its CSA. Using a CDS spread of 150 basis points, a funding spread of 70 basis points, an average uncollateralized exposure of $1 million, and a discount-weighted duration of 4.0, estimate the cost of the concession. How would the funding cost change if the client were instead a sovereign that insisted on a one-way CSA, under which only the dealer posts collateral?*

A threshold of $3 million means the client posts nothing until the exposure exceeds that amount, so the dealer carries an uncollateralized exposure of up to $3 million continuously. That exposure costs the dealer twice.

**Credit.** It must insure the uncollateralized amount at the client's CDS spread, as in Example 9.5:

$$s\,\overline{\mathrm{EE}} \times \text{duration} = 0.0150 \times \$1\text{m} \times 4.0 = \$60{,}000.$$

**Funding.** It must fund that exposure until it settles, at its own spread:

$$s_F\,\overline{\mathrm{EE}} \times \text{duration} = 0.0070 \times \$1\text{m} \times 4.0 = \$28{,}000.$$

$$\text{Total cost of the concession} \approx \$\mathbf{88{,}000}.$$

Two observations follow. The funding component is about a third of the total, so a dealer pricing only the credit cost recovers barely two-thirds of what the concession costs, which is the practical case for FVA that question 7 discusses in theory. And the client asked for an operational convenience, not a price concession; it is unlikely to have been told that its threshold has a number attached, and the number is what the dealer must recover in the spread.

**A sovereign on a one-way CSA.** The funding cost rises on both sides of the trade. The sovereign posts nothing at all, so the dealer funds the whole positive exposure rather than the $1 million that the threshold left uncovered; at the same 70 basis points that is the cost of an uncollateralized trade. And when the trade moves against the dealer it must post collateral to the sovereign while its own hedge returns margin to it, so the funding benefit that an uncollateralized trade would have earned on negative exposure (Section 9.4) disappears. FVA becomes all cost, which is why sovereigns and supranationals that insist on one-way terms pay the largest funding charges in the market.

*What the question tests.* That every clause in a CSA has a computable price, and that the reader can put one on a term that looks purely administrative.

---

**9.11** *Distinguish CVA the valuation adjustment from the CVA capital charge, explain why the latter exists, and give one action that reduces both and one that hedges the price while leaving most of the charge in place.*

**CVA the valuation adjustment** is a price: the market cost of insuring the counterparty's default, deducted from the value of the derivative and charged to the client. It is an accounting number, it flows through profit and loss, and it moves with credit spreads and exposure.

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**The CVA capital charge** is regulatory capital held against the *volatility* of that adjustment: against the risk that CVA moves against the bank, not against the default itself, which is capitalized separately.

The latter exists because of what happened in 2008. Roughly two-thirds of counterparty-related losses came not from counterparties actually failing but from mark-to-market losses on CVA as credit spreads widened. Banks were capitalized against default and not against the revaluation, so the losses that arrived were in the category no one held capital for.

**Reduces both:** novate the trade to a clearing house, which eliminates both, or buy single-name CDS protection on the counterparty, which hedges the price and is recognized by the charge, nearly in full under SA-CVA and up to three-quarters of the charge under BA-CVA. Collateralizing the trade also lowers both, though by different amounts: the price falls by what the firm's own exposure model says, the charge by what the standardized formulas allow.

**Hedges the price but leaves most of the charge:** hedge with index or proxy protection. Most counterparties have no traded CDS, so the protection a desk can actually buy is on an index or on a similar issuer. It offsets moves in the price, but the capital rules recognize index protection only partly and proxy protection less still (Section 27.7). A hedge that works economically can leave most of the capital charge in place, and the instrument a desk can buy is the one the framework recognizes least.

*What the question tests.* That the two CVAs answer different questions, and that the actions available to a desk affect them differently, which is why the XVA desk's hedging decisions are made against two scorecards at once.

---

**9.12** *An XVA desk reports a large loss in a quarter during which it was fully hedged on credit spreads. Give three distinct explanations consistent with the hedging having been done competently.*

Three explanations follow, all consistent with competent hedging.

**The exposure moved, not the spread.** CVA is expected exposure multiplied by default probability. A desk hedged on credit spreads has neutralized the second factor and not the first. If rates, currencies or equities moved sharply, the expected exposure on the underlying portfolio rises, and CVA rises with it, while the credit hedge, correctly sized for the old exposure, does nothing. This is CVA's first-order sensitivity to the market factors driving the exposure, which a credit hedge does not touch; where rates and spreads move together, the cross-gamma between them adds to it, and the two are a large and persistent source of unexplained XVA P&L.

**The hedge was a proxy.** Many counterparties have no traded CDS. The desk hedges with an index or a proxy issuer, which tracks in normal conditions and decouples in stress, which is when the adjustment moves. A proxy hedge that was competently chosen can still leave a large basis when idiosyncratic and systemic credit diverge.

**FVA and collateral effects moved independently.** The funding spread the bank pays, the cross-currency basis, and the collateral currency choice valued in Example 8.6 all feed the valuation adjustments and none of them is a credit spread. A quarter in which the bank's own funding widened, or the basis moved, produces a loss on the XVA book with credit spreads flat.

The common thread is that "XVA" is not one exposure. A desk hedged perfectly along one dimension is unhedged along the others, and the adjustments are products of terms drawn from different markets, which is why an XVA desk requires a full trading infrastructure rather than a credit book.

*What the question tests.* Whether the reader understands CVA as a multiplicative object with several independent inputs, rather than as a credit position.

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# Chapter 10
# Mechanics of Options Markets

**10.1** *An investor buys 5 call contracts at a quoted premium of \$2.85. What is the total cost, and what is the maximum loss?*

A US equity option contract covers 100 shares, and the premium is quoted per share:

$$5 \times 100 \times \$2.85 = \$\mathbf{1{,}425}.$$

The maximum loss is the same **\$1,425**. A purchased option confers a right and no obligation, so once the premium is paid the holder can never be asked for another cent, whatever the stock does and however long the position is held.

That is worth stating plainly because it is the single feature distinguishing a long option from every other leveraged position in the book. The futures position of Example 1.5 obtained comparable exposure for a comparable outlay and could demand unlimited further cash; this cannot. Chapter 3's producer choosing between selling futures and buying puts is choosing between exactly these two profiles, and Chapter 8's argument about funding is the reason the choice is not obvious.

*What the question tests.* The contract multiplier, and that maximum loss for a buyer is known at inception rather than discovered later.

---

**10.2** *Using Table 10.2, compute the proportional bid–ask spread for the 45-strike call and the 55-strike call. Explain why they differ and what it implies for strategy selection.*

From Table 10.2:

| Strike | Bid | Ask | Spread | Mid | Proportional |
|---|---|---|---|---|---|
| 45.00 | 4.35 | 4.50 | 0.15 | 4.425 | **3.4%** |
| 55.00 | 0.28 | 0.33 | 0.05 | 0.305 | **16.4%** |

The far out-of-the-money option has the *smaller* absolute spread (five cents against fifteen) and a proportional spread nearly five times as wide.

They differ because a market maker's spread reflects the cost of hedging and holding the position, which does not shrink in proportion to the option's price. The 55-strike call is cheap because it is unlikely to pay, not because it is cheap to make a market in. It still has to be hedged and held, and the quote must cover the tick size and the dealer's fixed costs, none of which shrinks with the premium.

The implication for strategy selection is direct and often ignored. A structure with legs in far out-of-the-money options pays a proportional cost several times that of an at-the-money trade, and the more legs it has the worse it gets. A strategy with three or four option legs crosses three or four such spreads, so a strategy whose theoretical edge is a few percent of premium can be entirely consumed at execution. Section 12.14 works this through; the chain shows why it happens.

*What the question tests.* That cheapness and cost are different things, and that the proportional spread is the one that decides whether a strategy survives its own execution.

---

**10.3** *A stock trades at \$74 and pays no dividend. A trader writes 8 naked call contracts struck at \$85 for a premium of \$1.10. Compute the margin requirement from both formulas and state which binds.*

Eight contracts on 100 shares is 800 shares. The stock is at \$74 and the strike is \$85, so the options are \$11 out of the money; the premium collected is $800 \times \$1.10 = \$880$ and the underlying value is $800 \times \$74 = \$59{,}200$.

Formula (10.1):

$$0.20 \times 59{,}200 - (85 - 74) \times 800 + 880 = 11{,}840 - 8{,}800 + 880 = \$3{,}920.$$

Formula (10.2):

$$0.10 \times 59{,}200 + 880 = 5{,}920 + 880 = \$6{,}800.$$

The requirement is the greater, so **\$6,800 binds**: the second formula wins by a wide margin.

This is exactly the case the floor was written for. The options are far out of the money, so the deduction in the first formula is large and would permit the writer to carry an unlimited-loss position on \$3,920 of collateral. The floor ignores moneyness entirely and charges against the size of the underlying, which is the quantity that actually determines how much can be lost.

*What the question tests.* That the two formulas bind in different regimes, and that recognizing which one applies requires computing both rather than reasoning about the option.

---

**10.4** *A company declares a 2-for-1 stock split. State how the contract terms of an outstanding option are adjusted under each of the two conventions described in Section 10.2, and verify that the aggregate strike cost is preserved in both.*

A 2-for-1 split doubles the share count and halves the price, leaving the holder's economic position unchanged. Option contracts are adjusted so that holders are likewise unaffected, and the chapter gives two conventions.

**Adjust the contract terms.** The strike is halved and the number of shares per contract doubled: one contract on 100 shares at \$60 becomes one contract on 200 shares at \$30.

$$\text{aggregate strike cost} = 200 \times \$30 = \$6{,}000.$$

**Adjust the contract count.** The holder receives two contracts in place of one, each on 100 shares, with the strike halved to \$30.

$$\text{aggregate strike cost} = 2 \times 100 \times \$30 = \$6{,}000.$$

Before the split, the position's aggregate strike cost was $100 \times \$60 = \$6{,}000$, so both conventions preserve it exactly, and both leave the holder controlling the same fraction of the company at the same total outlay.

The conventions differ operationally rather than economically: one produces a non-standard contract on 200 shares that must be tracked separately, the other produces two standard contracts that trade normally. Which is used depends on whether the split ratio leaves a whole number of contracts: a 3-for-2 split cannot be handled by the second method and must adjust the terms.

*What the question tests.* That the adjustment rule is designed to hold one quantity fixed, and that identifying which quantity makes both conventions obvious.

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**10.5** *Explain why an option buyer posts no margin while a writer must, and relate the answer to Figure 10.1.*

The buyer of an option has paid in full at inception and holds a right with no obligation. There is no state of the world in which they owe anything further, so there is nothing for margin to secure. Their maximum loss is the premium, and it has already been paid.

The writer is in the opposite position. Having received the premium, they hold an obligation that can grow large (without limit for a naked call), and they have posted nothing against it. Margin exists to make that obligation credible, and it is recalculated as the position moves.

Figure 10.1 makes the asymmetry visible: each long panel is bounded below at minus the premium, and each short panel is its mirror image, capped above at the premium and falling to a large loss: without limit for the call, to the strike less the premium for the put. The margin rule follows the open-ended side. Margin is required of whoever can lose more than they have already paid, which is a statement about the shape of the payoff rather than about who is more likely to be right.

The same asymmetry explains the funding argument that recurs from Chapter 3 onward: a purchased option never generates a margin call; a written option or a futures position can.

*What the question tests.* That the margin convention is derived from the profit diagram rather than imposed on it.

---

**10.6** *A trader is short 50 call contracts struck at $30 and the stock closes at $30.00 on expiration Friday. Describe the trader's position on Saturday morning and explain what could have been done on Friday to avoid it.*

The stock closed exactly at the strike, so the calls are at the money and their intrinsic value is zero. Whether they are exercised is not the trader's decision: each holder chooses independently, and some will exercise, whether to establish a position or on an expectation about Monday's opening. Exercise by exception does not settle it, since the calls are not a cent in the money.

So on Saturday morning the trader learns they have been assigned on *some unknown fraction* of the 50 contracts. If 30 are assigned, they are short 3,000 shares at $30, with the market closed and no way to hedge until Monday. The weekend is spent holding an unhedged equity position whose size they did not choose, exposed to any news over two days. This is pin risk, and the uncertainty is not about the price but about the quantity.

The remedy on Friday was to close the position. Buying back the 50 contracts, at whatever small price the market asked, converts an unknown assignment into a known cost. A trader who reasons that the options are worthless and not worth paying to close has priced the option correctly and ignored the risk of the *assignment*, which is a different thing and does not go to zero as the option does.

The same mechanism drives the same-day expiration effects of Section 10.12 and the dealer hedging of Section 18.12: near the strike at expiration, delta jumps between 0 and 1 over a few cents, so the risk is in the quantity of stock owed rather than in the option's price.

*What the question tests.* That an option worth nothing can still create a position worth a great deal, and that the remedy costs money and is worth paying.

---

**10.7** *An option expiring in three hours and one expiring in three months are both at the money. Compare their intrinsic value, time value, and sensitivity to a $1 move in the underlying.*

Both are at the money, so **both have zero intrinsic value**: intrinsic value depends only on

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where the stock is relative to the strike, not on time remaining.

**Time value** is where they part. The three-month option has substantial time value, because the stock has three months in which to move and the option captures the upside without the downside. The three-hour option has very little: there is almost no time left for anything to happen, so almost nothing to pay for. Time value decays roughly with $\sqrt{T}$, so the final hours are where it disappears fastest.

**Sensitivity to a $1 move** is where the surprise lies. At the money both start with a delta near one-half, so the first cents of a move affect them alike; what differs is how fast that changes. The three-hour option's delta is close to a step function: a dollar in the money and it will almost certainly be exercised, a dollar out and it will almost certainly expire worthless. On a $50 stock at 30% volatility, a $1 rise takes its delta to about 0.94 and a $1 fall to about 0.06, while the three-month option's delta moves only to about 0.58 or 0.48. Its gamma is enormous. Measured against the premium the contrast is starker: the three-hour option, worth about $0.26, gains about $0.76 on the rise and loses $0.24, nearly all of it, on the fall, while the three-month option, worth about $2.99, moves by about 17–19% either way.

The cheap option is the volatile one. That combination of almost no premium and extreme sensitivity is what makes very short-dated options attractive to buyers and dangerous to writers, and it is the mechanism behind the 0DTE material in Section 10.12 and Chapter 23.

*What the question tests.* Whether the reader separates the option's price from its sensitivity, which move in opposite directions as expiration approaches.

---

**10.8** *Distinguish a warrant from a listed call option on the same stock at the same strike. Which is worth more, and why?*

A **listed call** is created by the market: it is written by another participant, cleared centrally, and its exercise transfers existing shares from the writer to the holder. The company is not a party and its share count is unaffected.

A **warrant** is issued by the company itself. Exercise causes the company to issue *new* shares in exchange for the strike, which increases the share count and dilutes existing holders.

The listed call is worth more, and the reason is the dilution. When a warrant is exercised the new shares are sold below market value, so the value of every share falls slightly, including the value of the share the warrant holder is receiving. The payoff is therefore not $\max(S_T - K, 0)$ but that quantity scaled by a dilution factor of roughly $n/(n+m)$, where $m$ new shares are issued against $n$ existing. Two instruments with identical strikes and maturities pay different amounts, and the warrant pays less.

Two secondary differences reinforce it. A warrant carries the issuer's credit risk, since the company must deliver, whereas a listed option is guaranteed by the clearing house of Chapter 2. And warrants are typically far less liquid, so the proportional spreads of question 2 apply with force.

*What the question tests.* That two instruments with the same payoff formula can differ in value because one of them changes the quantity the formula is written on.

---

**10.9** *An index option, a currency option, and an option on a futures contract differ in what changes hands on exercise. State what each delivers, and explain why exercising an option on a futures contract leaves the holder with a position to manage rather than an asset to keep.*

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| | Delivers on exercise | Holder ends with |
|---|---|---|
| Index option | cash, against a computed settlement value | cash, position closed |
| Currency option | a fixed amount of foreign currency (or cash) | an asset, position closed |
| Futures option | a futures position plus cash of the last settlement price less $K$ | an open, margined position |

The index option settles in cash because delivering the basket is impractical, and settles against a value computed from the constituents' opening or closing prices, which the holder cannot trade at (Section 10.3).

Exercising a futures call does not deliver a commodity or a security; it delivers a *long futures position* at the most recent settlement price, together with that settlement price less $K$ in cash, so the holder collects the intrinsic value and is left holding a futures contract. They now have an obligation rather than an asset: the position is margined, settles daily, and can demand cash tomorrow.

That is precisely why these contracts dominate exchange-traded options on commodities and interest rates. The holder never has to fund the underlying (no barrels, no bonds), and the resulting position is one the margin system already supports. What they must manage afterward is a futures position with all of Chapter 2's mechanics attached, which a holder expecting to have "bought oil" will not have planned for.

*What the question tests.* That settlement mechanics are part of the instrument, and that one of the three leaves the holder with something that keeps moving.

---

**10.10** *Dealers quote many options in volatility rather than in price. Explain what the convention buys them, state what else must be agreed before a volatility quote determines a cash amount, and explain why such a quote stays meaningful while the underlying moves and a price quote does not.*

**What the convention buys.** A price quote goes stale the instant the underlying moves: a call quoted at \$2.85 is a different proposition if the stock ticks up ten cents while the phone call is in progress. A volatility quote does not: volatility is the one input that does not move mechanically with the spot, so the quote remains valid while the market moves and the two parties agree the price at the moment of execution. It also lets a dealer quote a whole surface consistently and compare options across strikes and maturities on a single scale, which prices cannot do.

**What else must be agreed.** A volatility is not a cash amount until the model and its other inputs are fixed: the pricing formula (Black–Scholes–Merton on spot, or Black's model on the forward), the reference spot or forward, the interest rate, the dividend or borrowing assumption, and the exact time to expiration. Two desks with the same volatility and different forwards compute different premiums. In practice this is handled by agreeing the forward alongside the volatility, and by both sides computing with the same convention, which is why Chapter 11's implied forward is the first number a desk calculates.

**Why it stays meaningful.** The quote is a parameter of the distribution rather than a point on it. The underlying moving changes where the option sits on the surface; it need not change the surface. The dealer has quoted the model input, leaving the model to translate it into a price at whatever spot prevails.

*What the question tests.* That quoting in volatility is a solution to a practical problem about stale quotes, and that it works only because a further set of conventions is agreed and usually left unstated.

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**10.11** *An investor wants six months of downside protection on a $5 million S&P 500 portfolio, with the index at 5,250 and SPY at $525. Compare buying index puts (SPX) with buying SPY puts: the number of contracts, settlement and exercise, what each leaves the investor holding at expiration, and US tax treatment.*

Take the portfolio to track the index, so the hedge is sized on notional.

**Number of contracts.** An SPX contract carries the $100 multiplier, and an SPY contract covers 100 shares at one-tenth of the index level:

$$\frac{5{,}000{,}000}{5{,}250 \times 100} = \frac{5{,}000{,}000}{525{,}000} = 9.52, \qquad \frac{5{,}000{,}000}{525 \times 100} = \frac{5{,}000{,}000}{52{,}500} = 95.24.$$

SPX forces a choice between 9 contracts, which cover $4,725,000 and leave 5.5% of the portfolio unhedged, and 10, which cover $5,250,000 and overhedge by 5%. SPY sizes the hedge almost exactly: 95 contracts cover $4,987,500, within 0.25%. The coarseness is the price of a contract worth half a million dollars; Cboe's Mini-SPX (XSP), one-tenth the size and otherwise like SPX, removes it.

**Settlement and exercise.** SPX puts are European and settle in cash, ($K$ − settlement value) × $100 a contract. A standard monthly SPX expiration settles in the morning against a value computed from the constituents' opening prices, which the investor cannot trade at (Section 10.3); the weekly and daily series settle against the afternoon close. SPY puts are American and settle by delivery: exercising one means selling 100 SPY shares at the strike, and an in-the-money put is exercised against the closing price of the expiration day. Early exercise is the holder's right, so as a buyer the investor faces no early-assignment risk in either contract; that risk sits with the writers, and it is why SPY puts carry a small premium for the American feature (Chapter 11).

**What each leaves at expiration.** If the market has fallen, the SPX puts pay cash and the hedge is finished: the portfolio is intact and the account holds the payoff. The SPY puts, if still held, are exercised automatically (Section 10.9) and require the investor to deliver 9,500 SPY shares it does not own, since the portfolio holds the constituent stocks rather than the fund. The result is a short position of 9,500 SPY shares against a long stock portfolio, which has to be bought back on the next trading day. The investor avoids this only by closing the puts with a sale before expiration, which an SPX buyer need not remember to do.

**Tax.** SPX options are Section 1256 contracts: every gain or loss is taxed 60% at the long-term and 40% at the short-term rate, however long the puts were held, and open positions are marked to market at year end, so a six-month hedge that spans December 31 is taxed on its value at that date. SPY options are ordinary equity options: a gain on puts held six months is entirely short-term. At top federal rates of 20% and 37% (before the 3.8% net investment income tax), the Section 1256 blend is

$$0.60 \times 20\% + 0.40 \times 37\% = 12.0\% + 14.8\% = 26.8\%,$$

against 37% on the SPY gain.

For a taxable investor protecting a $5 million stock portfolio the choice usually favors SPX: cash settlement fits a portfolio that does not hold the fund, there is no delivery to manage, and the tax rate on a gain is about ten points lower. SPY's advantages are its sizing, and its use in an account where tax does not matter.

*What the question tests.* That the same exposure in two wrappers differs in contract size, settlement, the position left at expiration, and tax, and that each difference comes from a specification in Section 10.3 rather than from the payoff.

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**10.12** *Name three distinct institution types that buy options and state, for each, the exposure it is buying or removing. For one of them, explain how its demand shows up in the prices the rest of the market sees.*

**Asset managers** buy index puts to remove downside exposure without selling holdings, reducing tail risk while keeping the portfolio, its dividends, and its tax position intact. They are buying insurance against a fall.

**Companies** buy currency options to remove exchange-rate exposure on contracted foreign receipts, and prefer them to forwards when the underlying transaction is uncertain, such as a contract not yet won. They are removing an exposure that may not exist, which a forward cannot hedge without creating one.

**Hedge funds and volatility traders** buy options to acquire exposure to volatility itself rather than to direction: a straddle removes the directional view and leaves a position in how far the underlying moves. They are buying an exposure they cannot get any other way.

**How the demand shows up in prices.** Asset-manager demand for index puts is persistent, one-directional, and concentrated at strikes below the current index level. Dealers are the natural sellers and must be paid to hold the resulting short-put position, so the implied volatility of low-strike index options sits systematically above that of high-strike ones. That is the equity skew of Chapter 19: not a statement about the distribution of returns so much as the price of the insurance one identifiable group persistently buys. Section 19.10 measures what they pay for it as the volatility risk premium.

*What the question tests.* A theme of this book applied to options: knowing who is on each side and why, and being able to trace a persistent feature of the price surface back to the demand that produces it.

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# Chapter 11
# Properties of Options and Arbitrage Bounds

**11.1** *Explain why both a call and a put increase in value with volatility, and why a forward contract's value does not depend on volatility at all.*

An option holder keeps the upside and is protected on the downside: the payoff is $\max(S_T - K, 0)$ for a call and $\max(K - S_T, 0)$ for a put, and both are truncated at zero. Widening the distribution of $S_T$ therefore adds to the states that pay and adds nothing to the states that do not, because those already pay zero. Volatility increases the good tail without deepening the bad one, and the truncation is what makes the effect one-directional. It applies to puts for the same reason with the tails exchanged — which is why more volatility raises both, despite their being opposite bets on direction.

A forward has no truncation. Its payoff is $S_T - K$, linear, unbounded in both directions, so a wider distribution adds symmetric weight to gains and losses and they cancel in expectation. Its value depends on the *mean* of the distribution, which is pinned by the carry argument of Chapter 5, and not on the spread around it.

The distinction is the one that organizes the rest of the book: linear payoffs are priced by static replication and carry, and volatility never enters; kinked payoffs need dynamic replication, and it always does.

*What the question tests.* That the kink, rather than any statement about risk preference, is what makes volatility a price input.

**11.2** *A non-dividend-paying stock trades at \$45. A six-month European call struck at \$42 trades at \$3.10 and the rate is 6%. Is the lower bound violated? If so, construct the arbitrage and compute the profit.*

The lower bound for a European call on a non-dividend stock is

$$c \geq S_0 - Ke^{-rT} = 45 - 42e^{-0.06\times 0.5} = 45 - 40.7587 = \$4.2413.$$

The call trades at \$3.10, so the bound is violated by **\$1.14**.

**The arbitrage.** Buy the call and short the stock, investing the proceeds:

| | |
|---|---|
| Buy the call | −3.10 |
| Short the stock | +45.00 |
| Invest \$41.90 at 6% for six months | −41.90 |
| Net today | **0.00** |

The investment grows to $41.90e^{0.03} = \$43.18$. At expiration:

*If $S_T > 42$:* exercise the call, pay \$42 for the stock, return it to close the short. Left with $43.18 - 42.00 = \$1.18$.

*If* $S_T \le 42$: let the call lapse, buy the stock in the market for $S_T$ and return it. Left with $43.18 - S_T \ge \$1.18$.

So the profit is **at least \$1.18 per share** in every state, with no capital committed, and more if the stock falls. The \$1.18 is the \$1.14 violation carried forward at the risk-free rate.

*What the question tests.* That a bound violation is converted into a trade by buying the cheap side and shorting the replication, and that the profit must be checked in both states rather than asserted.

---

**11.3** *A stock trades at \$28 with no dividends. Four-month European options struck at \$27 are quoted: call \$2.60, put \$1.10, rate 5%. Test parity and, if it fails, state the trade and the profit.*

Put–call parity requires $c + Ke^{-rT} = p + S_0$:

$$\text{left} = 2.60 + 27e^{-0.05\times 4/12} = 2.60 + 26.5537 = \$29.1537,$$

$$\text{right} = 1.10 + 28.00 = \$29.1000.$$

Parity **fails by \$0.0537**: the call side is expensive relative to the stock-and-put side.

**The trade.** Sell the expensive side and buy the cheap one, a conversion (Example 11.3):

| | |
|---|---|
| Sell the call | +2.60 |
| Buy the put | −1.10 |
| Buy the stock | −28.00 |
| Borrow $27e^{-rT}$ | +26.5537 |
| Net today | +0.0537 |

At expiration the loan repayment is \$27 and the position closes at exactly zero in both states: if $S_T > 27$ the call is exercised against you and the stock is delivered for \$27; if $S_T \le 27$ you exercise the put and sell the stock for \$27. Either way \$27 arrives and \$27 is repaid.

The profit is therefore the **\$0.0537 taken at inception** (\$5.37 per contract), and nothing further is at risk. Whether it survives the four bid–ask spreads it must cross is a separate question, and at this size it almost certainly does not.

*What the question tests.* That a parity violation is banked at the start rather than earned at expiration, and that the size of the violation must be compared against execution costs before calling it an arbitrage.

---

**11.4** *Using put–call parity, express a synthetic long put in terms of a call, the stock, and cash. Verify the payoff in both states at expiration.*

Rearranging $c + Ke^{-rT} = p + S_0$ for the put:

$$p = c - S_0 + Ke^{-rT}.$$

So a **synthetic long put** is: long one call, short one share, and long cash of $Ke^{-rT}$, cash that grows to exactly $K$ by expiration.

Verifying the payoff at $T$, when the cash is worth $K$:

| | $S_T \le K$ | $S_T > K$ |
|---|---|---|
| Long call | 0 | $S_T - K$ |
| Short stock | $-S_T$ | $-S_T$ |
| Cash | $K$ | $K$ |
| **Total** | $\mathbf{K - S_T}$ | **0** |

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which is $\max(K - S_T, 0)$ in both states: a long put exactly.

The construction is not merely algebraic. It says that anyone who can trade the call, the stock, and cash does not need a put to be quoted, which is why parity is a no-arbitrage condition rather than an empirical regularity: if the put traded away from this, buying the cheaper of the two and selling the more expensive would lock in the difference.

*What the question tests.* That parity is a recipe as well as an identity, and that checking a payoff state by state is how a claimed replication is verified.

---

**11.5** *Explain why an American call on a non-dividend-paying stock should never be exercised early, giving both the bounds argument and the economic argument. Which of the two still works when $r = 0$?*

**The bounds argument.** An American call is worth at least the European one, which by the lower bound (11.1) is worth at least $S_0 - Ke^{-rT}$. Exercising yields only $S_0 - K$. Since $Ke^{-rT} < K$ whenever $r > 0$,

$$\underbrace{S_0 - Ke^{-rT}}_{\text{worth if sold}} > \underbrace{S_0 - K}_{\text{worth if exercised}},$$

so selling always beats exercising. It follows that early exercise is never optimal and $C = c$.

**The economic argument.** Exercising early does two harmful things. It pays $K$ now instead of at $T$, forfeiting interest of $K - Ke^{-rT}$. And it discards insurance: the holder who exercises owns the stock and takes the full loss if it falls, while the holder who waits keeps a floor at zero.

**At $r = 0$ the second argument survives and the first does not.** With no interest, $Ke^{-rT} = K$ and the bounds inequality becomes an equality — selling and exercising look identical, and the bounds argument is silent. But the insurance is still there: waiting preserves the downside protection, exercising throws it away for nothing. So the conclusion holds, and only the economic argument establishes it.

*What the question tests.* That two proofs of one result can have different domains, and that knowing which assumption each rests on is what tells you when a result will generalize (here, to the dividend case of question 7, where both arguments have to be revisited).

---

**11.6** *A stock has fallen to \$0.25 and an American put struck at \$40 has three months remaining. Should it be exercised? Give the numerical comparison with $r = 4\%$.*

Exercising immediately yields

$$K - S = 40 - 0.25 = \$39.75.$$

The most that waiting can ever produce is $K$ itself, since the stock cannot fall below zero, and received at expiration that is worth

$$Ke^{-rT} = 40e^{-0.04\times 0.25} = 40 \times 0.990050 = \$39.60.$$

$$39.75 > 39.60, \quad \text{so } \textbf{exercise now}.$$

Even the best case for waiting to expiration, the stock falling all the way to zero, is worth less today than the exercise proceeds. Waiting a shorter time pays only if the stock reaches zero within about eight weeks, since $40\left(1 - e^{-0.04\tau}\right) = 0.25$ gives $\tau = 0.157$ years, which no realistic model allows. This is the chapter's argument for the stock at five cents, with different numbers: three months' interest on the strike at 4% is worth about 40 cents, while waiting can add at most the remaining 25 cents of the stock's fall.

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The comparison works only this deep in the money. Setting $K - S = Ke^{-rT}$ gives the price below which it settles the question,

$$K\left(1 - e^{-rT}\right) = 40 \times 0.009950 = \$0.40.$$

Below forty cents, exercise beats waiting to expiration with no model at all, and beats any shorter wait unless the stock collapses almost at once. Above it the comparison is silent: it does *not* say to wait, because the put's value today is below that best case $Ke^{-rT}$ and exercising may still beat it. The true critical price $S^*$, which depends on volatility and takes a model to find, lies well above forty cents at any realistic volatility.

*What the question tests.* Whether the reader applies the comparison and knows what it proves: when exercising beats the best possible outcome of waiting, the decision is settled; when it does not, the comparison has not decided anything.

---

**11.7** *A stock at \$80 goes ex-dividend tomorrow, paying \$0.90. An investor holds American calls struck at \$75 with three months remaining; the rate is 4%. Apply condition* (11.13) *and state what conclusion is and is not justified.*

The condition compares the dividend captured against the interest forgone by paying the strike early:

$$D > K\left(1 - e^{-r(T-t)}\right).$$

Here

$$K\left(1 - e^{-r(T-t)}\right) = 75\left(1 - e^{-0.04\times 0.25}\right) = 75 \times 0.009950 = \$0.746,$$

against a dividend of \$0.90. Since $0.90 > 0.746$, **the condition is satisfied**.

**What is justified.** Early exercise is not ruled out. The dividend is large enough to outweigh the interest cost of paying \$75 three months early, so the first two effects together favor exercising just before the ex-dividend date.

**What is not justified.** That exercise *is* optimal. The condition is necessary, not sufficient: it compares only the dividend against the interest and ignores the third effect, the insurance the holder gives up. Exercising converts the option into stock, and the holder then bears the full downside for the remaining three months. With the stock at \$80 and the strike at \$75 the option retains real protective value, and that value can exceed the \$0.154 by which the dividend beats the interest.

The correct test is to compare the exercise proceeds against the option's market value, not against a rearranged inequality. The condition identifies candidates for exercise, and the valuation decides.

*What the question tests.* That a necessary condition is routinely quoted as if it were sufficient, and that naming the omitted term is what distinguishes understanding the result from applying it.

---

**11.8** *Construct a box spread from options struck at \$40 and \$50 expiring in six months, list the four legs, and state the payoff. If the box costs \$9.70, what rate is implied?*

A box combines a synthetic long at the lower strike with a synthetic short at the higher one:

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| Leg | Strike |
|---|---|
| Buy call | 40 |
| Sell put | 40 |
| Sell call | 50 |
| Buy put | 50 |

The 40 pair is a synthetic long forward at 40; the 50 pair is a synthetic short forward at 50. The stock exposure cancels exactly, and what remains pays

$$K_2 - K_1 = \$10$$

at expiration, **regardless of where the stock finishes**. The box is a zero-coupon bond assembled from four options.

At a cost of \$9.70 for \$10 in six months, the implied continuously compounded rate is

$$r = \frac{1}{0.5}\ln\left(\frac{10}{9.70}\right) = 2 \times 0.030459 = \mathbf{6.09}\%,$$

or 6.19% quoted as a simple semiannual rate.

An investor buying the box is lending at 6.09% on a centrally cleared position requiring no credit assessment of any counterparty; selling it borrows at the same rate. That is why institutions use boxes as a financing instrument competing with repo — and why the American-exercise trap of the chapter's case study is so costly, since the whole construction assumes the four legs cannot be exercised early.

*What the question tests.* That the four legs collapse to a bond, and that the implied rate is read from the discount rather than from any option formula.

---

**11.9** *State the three relations that constrain call prices at different strikes on the same underlying and maturity, and for each identify the position that profits if it is violated.*

Write $c(K)$ for the call price at strike $K$, with $K_1 < K_2 < K_3$ equally spaced.

**Monotonicity.** $c(K_1) \geq c(K_2)$: a lower strike confers the right to buy more cheaply. If violated, buy the low-strike call and sell the high-strike one: a bull call spread established for a credit, which can never pay out negatively.

**The spread bound.** $c(K_1) - c(K_2) \leq (K_2 - K_1)e^{-rT}$: the spread cannot be worth more than the maximum it can pay. If violated, sell the spread and invest the proceeds, which grow to more than the largest possible payout.

**Convexity.** $c(K_1) - 2c(K_2) + c(K_3) \geq 0$: the middle option cannot be too expensive relative to its neighbors. If violated, buy the butterfly (long the outer two calls, short two of the middle one) for a credit, and it pays nonnegatively in every state.

**Of the three, convexity is the one worth checking most often.** Monotonicity and the spread bound are violated only by gross errors and are usually caught by the quoting system itself. Convexity fails subtly, because it involves three strikes and a second difference, and a quote can drift into violation through ordinary repricing of one strike without anything looking wrong. It is also the relation that matters: Chapter 19 shows that convexity in the strike *is* the statement that the implied density is nonnegative, so a violation is not merely an arbitrage but an announcement that the surface implies negative probabilities.

*What the question tests.* That the three constraints are ordered by subtlety, and that the most fragile one is the one carrying the distributional meaning.

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**11.10** *Using the mid prices of Table 10.2 at every strike, with $r = 5\%$, compute the implied forward (11.12) and the dividend it implies, $S_0 - F_0 e^{-rT}$. At which strikes does the reading drift, in which direction, and why? Which strike would you quote?*

With 45 days to expiration, $e^{rT} = e^{0.05\times 45/365} = 1.006183$ and $e^{-rT} = 0.993855$. At each strike the implied forward is $F_0 = K + (C - P)e^{rT}$ and the implied dividend is $S_0 - F_0 e^{-rT} = S_0 - Ke^{-rT} - (C - P)$, with $S_0 = \$48.30$. At the 52.50 strike, for instance,

$$F_0 = 52.50 + (0.71 - 5.65) \times 1.006183 = 52.50 - 4.971 = \$47.529,$$

$$S_0 - F_0 e^{-rT} = 48.30 - 47.529 \times 0.993855 = 48.30 - 47.237 = \$1.063.$$

The whole chain:

| Strike | Call mid | Put mid | $C - P$ | Implied $F_0$ | Implied $D$ |
|---|---|---|---|---|---|
| 42.50 | 6.650 | 1.585 | 5.065 | 47.596 | 0.996 |
| 45.00 | 4.425 | 1.850 | 2.575 | 47.591 | 1.002 |
| 47.50 | 2.670 | 2.575 | 0.095 | 47.596 | 0.997 |
| 50.00 | 1.475 | 3.875 | −2.400 | 47.585 | 1.007 |
| 52.50 | 0.710 | 5.650 | −4.940 | 47.529 | 1.063 |
| 55.00 | 0.305 | 7.750 | −7.445 | 47.509 | 1.083 |

**Where the reading drifts.** From 42.50 to 50 the implied dividend stays within a cent of \$1.00, a forward of about \$47.59. At 52.50 and 55 it drifts: the forward falls by 6 to 8 cents and the implied dividend rises to \$1.06 and \$1.08.

**Why.** The options are American, and at 52.50 and 55 the puts are deep in the money, with intrinsic values of $52.50 - 48.30 = \$4.20$ and $55 - 48.30 = \$6.70$. Section 11.9 showed that such a put may be worth exercising early, so its price carries an early-exercise premium that the European put lacks. That premium raises $P$, lowers $C - P$ below the European $c - p$, and the formula, which assumes $C - P = c - p$, reads the shortfall as a lower forward and a larger dividend. Nothing about the stock's dividend changes across strikes; what changes is how far $C - P$ sits from the European value inside the band of Section 11.11, and the deep in-the-money put pushes it toward the band's lower edge. The deep in-the-money 42.50 call could in principle carry the mirror-image premium, since a \$1.00 dividend exceeds the interest on its strike and condition (11.13) holds, which would push the reading the other way; its 0.996 shows little of it.

**Which strike to quote.** The 47.50 strike, nearest the money: both options are liquid and neither is deep enough in the money to carry much early-exercise premium. Its reading of \$47.596 agrees with the 42.50 and 45 strikes to half a cent, which is what lets Example 11.4 treat the \$1.00 as good to ten cents or so. A desk would usually average the near-the-money strikes and discard the deep in-the-money puts, for exactly the reason the last two rows show.

*What the question tests.* That the implied forward is a property of the stock and should not depend on the strike, so a reading that drifts with the strike is measuring the options' American feature rather than the dividend.

**11.11** *State the parity bounds for American options, explain why the European equality becomes a band, identify which side of the band binds when the put is deep in the money and why, and state how the band changes when the stock pays dividends of present value $D$.*

For American options on a non-dividend-paying stock the equality becomes

$$S_0 - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT}.$$

**Why the equality becomes a band.** The European proof works by holding two portfolios to expiration and observing they coincide. With American options either side may be exercised

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before expiration, at a moment the arbitrageur does not control, so the portfolios cannot be held to a common horizon. What survives is a pair of one-sided arguments: each bound is established by a trade that remains profitable whenever the counterparty exercises, and neither pins the value exactly.

**Which side binds for a deep in-the-money put.** The *left*. A deep in-the-money American put is a candidate for early exercise (question 6 sets the condition), so $P$ is worth more than its European counterpart, which pushes $C - P$ down toward its lower limit $S_0 - K$. In the limit where the put should be exercised immediately, $P = K - S_0$ and $C \approx 0$, so $C - P \approx S_0 - K$: the left bound is all but attained.

The right bound, $S_0 - Ke^{-rT}$, is the European $c - p$. Since $C = c$ by question 5, $C - P$ comes close to it when the put's early-exercise premium is negligible, which is when the put is far out of the money.

**With dividends.** Only the left edge moves, down by $D$: $S_0 - D - K \le C - P \le S_0 - Ke^{-rT}$, equation (11.15). The share in the lower-bound portfolio collects the dividends, while the right-hand argument holds with or without them, so the band widens from $K\left(1 - e^{-rT}\right)$ to $D + K\left(1 - e^{-rT}\right)$.

*What the question tests.* That the two bounds correspond to the two options' early-exercise possibilities, so knowing which one is live tells you which inequality is tight.

---

**11.12** *A chain shows puts trading 40 cents above parity on a \$50 stock with six months to expiration. Using Example 10.4, state the borrow fee that would account for it, and explain why observing a violation of this size is not the same as being able to profit from it.*

A borrow fee $f$ makes the replicating short position cost $S_0(1 - e^{-fT})$ over the option's life, and that cost shows up as puts trading above parity. Setting this equal to the observed 40 cents on a \$50 stock over six months:

$$50\left(1 - e^{-0.5f}\right) = 0.40 \implies f = -2\ln(0.992) = \mathbf{1.61}\%.$$

Example 10.4's 3% fee opens a 74-cent band on the same stock and horizon, so 40 cents is consistent with a fee of roughly half that — a stock that is somewhat hard to borrow but not extremely so.

**Why observing it is not the same as profiting from it.** The trade that would close the gap is: sell the overpriced put, buy the call, short the stock. That short is the whole problem. To establish it the arbitrageur must borrow the shares and pay the very fee that created the discrepancy, so the 40 cents is consumed by the cost of capturing it. Worse, the fee is not fixed: a stock that is hard to borrow today can become harder, and the lender may recall the shares entirely, forcing the position closed at the least convenient moment.

So the violation is not a mispricing but a quotation. The option chain is reporting the price of borrowing the stock, in a market where that price is otherwise opaque, and reading it that way turns an apparent riskless profit into a useful piece of information.

*What the question tests.* That the arbitrage-free band widens with a friction, and that the friction is precisely what makes the trade inside it unavailable.

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# Chapter 12
# Trading Strategies Involving Options

**12.1** *Show, using put–call parity, that a covered call has the same profit profile as a written put. State one respect in which the two positions nevertheless differ in practice.*

A covered call is long stock plus a written call. Rearranging parity,

$$S_0 - c = Ke^{-rT} - p,$$

so the covered call equals cash of $Ke^{-rT}$ plus a *written put*: a cash-secured put, not a naked one. Both positions collect a premium, both keep it if the stock rises, and both absorb the full fall if it drops: the profit profiles coincide exactly. Strictly the identity holds for European options; listed options on single stocks are American, and that is where the first difference comes from.

**One respect in which they differ in practice.** Early assignment comes at different moments. The covered call's short call is most likely to be exercised just before an ex-dividend date, when the dividend exceeds its remaining time value, and the shares are delivered; the short put is exercised once it is deep in the money, and the writer buys the shares with the cash set aside.

Other answers are also defensible. The covered call crosses two bid–ask spreads rather than one. Dividends arrive as cash to the shareholder but are embedded in the put's price. The tax treatment differs. And the naked form of the written put, unlike either covered form, is levered: brokers, who set options-approval levels, commonly approve covered calls and cash-secured puts at the same lowest level and require a higher one, and the margin of Section 10.8, for the naked put.

*What the question tests.* That identical payoffs can differ in every practical dimension, and that the marketing of the covered call as conservative describes its funding rather than its risk.

**12.2** *Using Table 10.2, construct a bear put spread from the 47.50 and 45.00 puts at the quoted bid and ask. Compute the debit, the maximum profit, the maximum loss, and the breakeven.*

A bear put spread buys the higher strike and sells the lower, paid at the quoted sides: buy the 47.50 put at its \$2.63 ask, sell the 45.00 put at its \$1.81 bid.

$$\text{debit} = 2.63 - 1.81 = \$\mathbf{0.82} \text{ per share, } \$82 \text{ per spread.}$$

| | |
|---|---|
| Maximum profit, at $S_T \le 45$ | $(47.50 - 45.00) - 0.82 = \$\mathbf{1.68}$ |
| Maximum loss, at $S_T \ge 47.50$ | $\$\mathbf{0.82}$ |
| Breakeven | $47.50 - 0.82 = \$\mathbf{46.68}$ |

Read what the position actually requires. The stock is at \$48.30, so it must fall **3.4%** in 45 days merely to break even, and **6.8%** to earn the maximum. The reward-to-risk ratio is 1.68 to

0.82, or 2.05 to 1: the trade can make twice what it risks, but only after a substantial move. At mid prices the debit would be 2.575 − 1.85 = \$0.725, so the quotes add \$0.095, 13%.

That is not an argument against it; it is the price of a bearish position with a capped loss. But a trader who describes it as a cheap way to be bearish has not looked at the breakeven.

*What the question tests.* That executing at the bid and ask rather than the mid changes the arithmetic materially, and that the breakeven expressed as a percentage move is the number that says what the trade needs.

---

**12.3** *Using Table 10.2, price the 47.50/50/52.50 call butterfly at the quoted bid and ask and at mid prices. Compute what execution costs as a percentage of the mid value, both breakevens at the quotes, and where the profitable range lies relative to the \$48.30 stock price.*

The butterfly buys the 47.50 and 52.50 calls and sells two 50 calls. At the quotes the calls bought are paid at the ask and the calls sold go at the bid:

$$\text{cost} = 2.72 - 2 \times 1.44 + 0.74 = 3.46 - 2.88 = \$\mathbf{0.58}.$$

At mid prices,

$$\text{value} = 2.67 - 2 \times 1.475 + 0.71 = 3.38 - 2.95 = \$\mathbf{0.43}.$$

Execution costs 0.58 − 0.43 = \$0.15, which is **35%** of the mid value (26% of the price paid), close to the 38% that Example 12.3 lost on the 45/47.50/50 butterfly.

The payoff is zero outside [47.50, 52.50] and peaks at \$2.50 at \$50, so the maximum profit is 2.50 − 0.58 = \$1.92 against a maximum loss of \$0.58. The breakevens are

$$\text{lower} = 47.50 + 0.58 = \$\mathbf{48.08}, \qquad \text{upper} = 52.50 - 0.58 = \$\mathbf{51.92}.$$

The profitable range runs from **0.5%** below \$48.30 to **7.5%** above it, a band \$3.84 wide, or 8.0% of the stock price.

The range is centered on the middle strike, 3.5% above spot, not on the current price. This butterfly is therefore not a bet that the stock goes nowhere: it makes little if the stock stays where it is, loses if it falls by more than half a percent, and pays most if the stock drifts up to about \$50 and stops there. Choosing the middle strike is choosing the target.

*What the question tests.* That the bid–ask spread takes a similar share of any butterfly's value at entry, and that the breakevens, expressed relative to spot, show what view the chosen strikes actually express.

---

**12.4** *Explain why a strangle that costs 36% less than a straddle can require a larger move to break even. What does this imply about "cheap" option strategies generally?*

A straddle buys the at-the-money call and put; a strangle buys options that are out of the money on both sides. The strangle is cheaper because both its legs are further from the money and therefore worth less (36% less, here).

But the breakeven is not the premium; it is the strike plus the premium. The straddle's upper breakeven is $K$ + premium, and the strangle's is $K_{\text{call}}$ + premium, with $K_{\text{call}}$ already above the current price. The strangle saves on premium and pays it back in strike distance, and here the distance costs more than the premium saved: the strangle's breakevens, \$41.60 and \$53.40, lie outside the straddle's \$42.15 and \$52.85, so the stock must travel further before the position makes money.

What the strangle actually changes is the shape of the outcome, not the hurdle. It loses less if the stock does not move, because less was spent; once the stock is beyond both pairs of

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breakevens the straddle earns \$0.55 more, because its payoff starts rising dollar for dollar from closer points. It is a different distribution of the same bet, not a cheaper version of it.

**The general implication.** A strategy described as cheaper is usually one whose payoff has been moved further away, and the saving is the price of the distance. The comparison that matters is not the premium alone but the breakeven, and strategies should be ranked by what they require the underlying to do. That is the discipline the whole chapter is organized around.

*What the question tests.* That "cheap" and "likely to pay" are different properties, and that the premium comparison alone systematically flatters the cheaper structure.

---

**12.5** *A five-year principal-protected note is offered when the five-year rate is 3% and a five-year at-the-money call costs 22% of notional. Compute the participation rate with no fee and with a 1.5% fee.*

The note must return the principal in five years, so the issuer sets aside its present value and spends what is left on the option.

$$\text{PV of principal} = e^{-0.03\times 5} = 0.8607, \qquad \text{option budget} = 1 - 0.8607 = 0.1393.$$

At an at-the-money call costing 22% of notional, the participation rate is

$$\frac{0.1393}{0.22} = \mathbf{63.3}\%.$$

With a 1.5% fee taken from the same budget:

$$\frac{0.1393 - 0.015}{0.22} = \mathbf{56.5}\%.$$

Two observations follow. A 1.5% fee costs the investor 6.8 percentage points of participation, because it is taken from the option budget rather than from the notional: the fee is 1.5% of the note and 10.8% of the only part of it that does anything. And the participation is highly sensitive to the interest rate: at a 1% rate the budget would be 4.9% and the participation 22%, which is why principal-protected notes became scarce during the decade of low interest rates.

*What the question tests.* That the participation rate is a residual, and that fees and rates act on it with far more leverage than their headline size suggests (the machinery Chapter 35 uses to evaluate structured notes generally).

---

**12.6** *Identify what each of the following is a bet on, using the four things to bet on named in the chapter's introduction: (a) a calendar spread; (b) a risk reversal; (c) a condor; (d) a covered call.*

Using the four categories (direction, magnitude, time, and skew):

**(a) Calendar spread.** Principally *time*, with a secondary bet on magnitude. It profits from the near option decaying faster than the far one, and it wants the stock to stay near the strike while that happens. Through its two maturities it also carries a view on the term structure of implied volatility, which Table 12.2 records as "term."

**(b) Risk reversal.** *Direction* and *skew* together. Long the call and short the put is a bullish position, and its cost is determined by the relative implied volatilities of the two strikes, so it also expresses a view that the skew is too steep.

**(c) Condor.** *Magnitude*, in the negative direction. It pays if the stock stays within a band, so it is short movement: a bet that realized volatility comes in below what the market has priced.

**(d) Covered call.** *Direction* (mildly bullish, since the stock is held) combined with short *magnitude* (a sharp fall is borne in full and a sharp rise is given up above the strike). By question 1 it is a written put, a bet that the market does not fall far.

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The point of the taxonomy is that most positions bet on two or three of these simultaneously, and traders lose money on positions whose *directional* view was correct because a different category went against them: the risk reversal that was right about direction and wrong about skew, or the calendar that was right about the stock and wrong about when.

*What the question tests.* Whether the reader can decompose a structure into what it is actually exposed to, rather than classifying it by name.

---

**12.7** *Explain why a portfolio containing butterflies, condors, and calendar spreads is not diversified.*

All three structures are short a sharp move—the calendar too, although Table 12.2 lists it as "term" rather than short volatility. A butterfly and a condor pay when the stock stays inside a band and lose when it leaves; a calendar spread profits from decay and is damaged by a large move in either direction. Every one of them is a position that wants the underlying to sit still.

So a portfolio of all three is not diversified: it is the same bet in three notations. The positions differ in the width of the band and in which expiration they sit on, but their exposures point the same way, and they lose together in the same event, a large move, which is precisely the event a portfolio is supposed to be diversified against.

The formal statement is the one Chapter 18 supplies. Each of these has negative gamma, and gammas add: a book of such positions has the sum of their gammas and none of the cancellation that genuine diversification would produce. The relation $\Theta \approx -\frac{1}{2}\Gamma\sigma^2 S^2$ then says the position collects theta every quiet day and pays for it in one move: a return profile that looks like skill for a long time.

Diversification requires exposures that respond differently to the same event, and counting positions is not a test of that. The test is to shock the underlying and see whether they move together, which these do.

*What the question tests.* That structural variety and risk variety are different things, and that the Greeks are what distinguish them.

---

**12.8** *A trader establishes a four-legged condor by executing each leg separately. Describe two distinct ways this can cost more than expected, and state how each is avoided.*

**Crossing four spreads.** Each leg is executed at the bid or the ask rather than the mid, so a four-legged structure pays the half-spread four times. On the chain of Table 10.2 the cheapest options carry proportional spreads as wide as 16%, and a condor's value at mid prices, often a few tenths of a dollar, can be entirely consumed before the position exists. It is avoided by executing the condor as a single package, quoted and filled as one instrument, so the market maker prices the net risk rather than four separate ones.

**Legging risk.** Executing sequentially means holding an incomplete position while the remaining legs are being filled. Between the first and last fill the trader has a directional exposure they never intended, and if the underlying moves the remaining legs are no longer available at the prices assumed. The structure that ends up on the book is not the one that was priced. It is avoided the same way, by trading the package; or, if legging is unavoidable, by filling the hardest leg to execute first and completing the rest in offsetting pairs (each vertical spread as one order), so the partial position carries little delta. Filling the largest-delta leg first does the opposite: it maximizes the exposure carried while the rest are filled.

The two are distinct: the first is a cost that is certain and calculable in advance, the second is a risk that may cost nothing or a great deal. Both are reasons the multi-leg structures of this chapter are traded as packages by anyone doing them in large amounts.

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*What the question tests.* That execution is part of a strategy's design, and that the chapter's profit diagrams assume an execution that must be arranged.

---

**12.9** *A calendar spread sells the near-dated option and buys the longer-dated one at the same strike. Explain why this is neither a directional trade nor straightforwardly a volatility trade, identify the two effects that determine its profit, and state the move in the underlying that hurts it most.*

It is not a directional trade because both legs are struck at the same price, so a move in the underlying affects them similarly: their deltas largely offset. It is not straightforwardly a volatility trade either, because the position gains when the far option's implied volatility rises and loses when the near one's does. To a parallel rise in implied volatility it is net long, because the far option is the more sensitive to volatility (roughly by the square root of the ratio of the two maturities); beyond that, what it holds is a view on the difference between two volatilities at two maturities (the term structure).

**Two effects determine the profit.** Time decay: the near option loses time value faster than the far one, since value grows roughly with $\sqrt{T}$ and decay per day is steepest at the end. That works for the position while the stock stays near the strike. And the volatility term structure: the position gains if long-dated implied volatility rises relative to short-dated, and loses if the curve flattens or inverts.

**The move that hurts most** is a large move in the underlying, in either direction. Both options end far from the strike, where neither has much time value and the difference between them collapses. The position is worth most when the stock sits at the strike as the near option expires, and its value falls away on both sides, so it is short magnitude even though it is not a straightforward volatility position.

*What the question tests.* That a spread across maturities isolates a different quantity from a spread across strikes, and that a position can be short movement without being simply short volatility.

---

**12.10** *A risk reversal buys an out-of-the-money call and sells an out-of-the-money put. Explain why its cost is a direct reading of the skew rather than of the level of volatility, and state what a risk reversal that costs nothing to enter tells you about the two implied volatilities.*

Buying an out-of-the-money call and selling an out-of-the-money put trades one implied volatility against another. The level of volatility largely cancels: raise every implied volatility on the surface by a point and both legs gain value, the long and the short roughly offsetting. What does not cancel is the *difference* between the two strikes' implied volatilities, and that difference is the skew.

So the price of a risk reversal is a direct quotation of skew, which is why the currency market quotes the risk reversal itself, in volatility points, as one of its standard instruments (Chapter 16).

**A zero-cost risk reversal** tells you the two implied volatilities are such that the call and put have equal value, not that they are equal. Unless the surface is flat, the two strikes that achieve this sit at different distances from the forward, and the difference in those distances is the skew expressed in strikes. In an equity market, where the put's implied volatility exceeds the call's, the put sold must be further from the money than the call bought. Seen from the other side, a zero-cost collar must buy its put further out than the call it sells: the buyer of protection pays the skew, and gives up more upside for a given floor than a flat surface would require.

*What the question tests.* That a structure can isolate a second-order feature of the surface, and that reading its cost tells you about the shape rather than the level.

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**12.11** *A founder holds a large single-stock position and wants to reduce the risk without selling. Describe the collar that achieves this, explain in whose favor the skew works when the collar is on an equity, and state what the shareholder remains exposed to that a sale would have removed.*

**The collar.** Hold the shares, buy an out-of-the-money put to floor the position, and sell an out-of-the-money call to pay for it. Struck so the premiums offset, it costs nothing at inception and confines the outcome to a band: protected below the put strike, participating between the strikes, and capped above the call strike.

**Whose favor the skew works in.** It works against the founder. On an equity the put's implied volatility exceeds the call's (the protection demand of Section 10.11, priced in Chapter 19), so the put being bought is expensive and the call being sold is cheap in volatility terms. A zero-cost collar must therefore set its put further from the forward than its call: for a given floor the cap comes closer, and for a given cap the floor drops lower. The skew works in favor of whoever supplies the protection (the dealer who sells the put and buys the call), and it is the price of the persistent demand from holders like this one. Example 12.5 shows the asymmetry, and Example 12.6 isolates the skew's share of it on index quotes, where the forward sits close to spot.

**What the shareholder remains exposed to.** First, the stretch above the floor: the fall from today's price to the put strike (6.8% in Example 12.5) is borne in full, where a sale would have fixed today's price. Second, the horizon: the collar expires, and the holding is then exposed at whatever terms the next collar can be struck, which after a fall are set around a lower price. Third, everything that comes with still owning the shares: the concentration on the founder's balance sheet, the shares encumbered as collateral for the written call and unavailable to pledge elsewhere, and the disclosure that large holders' hedges typically require. A sale ends all of that. A collar is a hedge, not an exit, and it leaves the holder still holding.

*What the question tests.* That the skew works against the buyer of protection, and that the real cost of a collar is not in the payoff diagram.

**12.12** *A manager hedges an equity portfolio with index puts rather than index futures. State the beta-adjusted contract count, explain what shape the option hedge buys that the futures hedge does not, and explain why the floor it sets on the portfolio is only approximate even when the beta is correct.*

**The contract count** is the same beta-adjusted calculation as for futures:

$$N = \beta \times \frac{V_P}{\text{index level} \times \text{multiplier}},$$

rounded to whole contracts, with puts bought rather than futures sold.

**What the option hedge buys.** It buys a floor rather than a lock. Index futures fix the portfolio's value and remove the upside with it; index puts remove the downside below the strike and leave the upside intact, in exchange for a premium paid at the start. The manager is buying a *shape* (asymmetry), and the premium is its price. It is the producer's choice between short futures and long puts, which this chapter draws for a commodity, transferred to a portfolio.

**Why the floor is only approximate even with the right beta.**

First, the floor is set on the index, not on the portfolio. Beta describes how the portfolio moves with the index on average; the residual, idiosyncratic part of its return is unhedged by construction. A portfolio can fall further than beta implies, or fall while the index does not, and the puts pay only on the index. The more concentrated or style-specific the portfolio, the larger this residual, and it is precisely the part a manager was paid to take.

Second, the arithmetic drifts. The beta scaling ignores interest and dividends, which Section 16.2's strike corrects for. The contract count must be rounded (104.8 to 105 in

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Example 12.6), so the puts pay at a slightly different rate from the portfolio's beta-scaled loss and the floor moves a little as the fall deepens; and the premium comes on top, so the net floor sits below the one the strike implies, 6.7% rather than 5.5% in that example.

In practice the premise fails as well: beta is estimated from past returns and tends to rise in sell-offs, so a hedge sized on a normal-market beta is undersized in the event it was bought for.

*What the question tests.* That the contract count is the easy half, and that the floor is a statement about the index which beta translates to the portfolio only approximately.

---

**12.13** *A producer can hedge by selling futures or by buying puts. Compare the two on cost at inception, retained upside, and cash demands over the life of the hedge, and identify which consideration the textbook comparison usually omits.*

| | **Short futures** | **Long puts** |
|---|---|---|
| Cost at inception | none (margin is a deposit) | the premium, paid up front |
| Retained upside | none: the price is locked | all above the strike, less the premium |
| Cash over the life | variation margin, unbounded | none after the premium |

The futures hedge fixes the price. It costs nothing to enter beyond a returnable deposit, and it gives up the entire upside: if the commodity doubles, the producer sells at the price it locked. The put fixes a *floor*. It costs a premium that is gone whatever happens, and it leaves the upside intact above the strike.

The comparison textbooks usually make stops there, and the omission is the third row. The futures hedge's cash demands arrive when the hedge is working. A producer short futures loses on the futures leg exactly when the commodity price rises, which is when its unsold output is gaining value that has not yet been realized. The margin is due in cash, daily; the offsetting gain arrives at delivery, months later. The bought put has the opposite profile: the entire cost is known and paid at the start, and no further cash is ever demanded, because the holder of an option has no obligation to margin (unless the option is margined futures-style, Section 16.5).

So the choice is not simply "free but no upside" against "costly but keeps upside." It is between a known financing cost and an unknown, adversely timed one, and a firm whose liquidity is thin may rationally pay a premium to avoid margin calls it could not meet. That is Section 3.10's point and the mechanism behind both Metallgesellschaft and the 2022 European utilities.

*What the question tests.* Whether the reader compares instruments on economics alone or also on the timing of the cash, which is the distinction the rest of the book keeps returning to.

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# Chapter 13
# Binomial Trees

**13.1** *A stock at \$40 will move to \$44 or \$36 in six months. The rate is 5%. Value a European call struck at \$40 by constructing the riskless portfolio, then verify the answer using (13.2).*

**The riskless portfolio.** Hold $\Delta$ shares and short one call. At expiration the portfolio is worth $44\Delta - 4$ in the up state and $36\Delta - 0$ in the down state. Setting them equal,

$$44\Delta - 4 = 36\Delta \implies \Delta = \frac{4-0}{44-36} = \mathbf{0.5}.$$

The portfolio is then worth $36 \times 0.5 = \$18$ with certainty, so today it must be worth $18e^{-0.05\times 0.5} = \$17.5556$. Since it consists of half a share less one call,

$$40 \times 0.5 - c = 17.5556 \implies c = 20 - 17.5556 = \$\mathbf{2.44}.$$

**Verification by the risk-neutral formula.** With $u = 1.10$, $d = 0.90$,

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{1.025315 - 0.90}{0.20} = 0.626576,$$

$$c = e^{-r\Delta t}\,[p \times 4 + (1-p) \times 0] = 0.975310 \times 2.506303 = \$\mathbf{2.44}. \quad \checkmark$$

The two routes agree because they are the same argument written differently: the hedge ratio and the risk-neutral probability both come from the requirement that the portfolio's two outcomes coincide.

*What the question tests.* That the formula is a shortcut for the construction, not an alternative to it, which is what question 11 develops.

---

**13.2** *In the previous question, compute the risk-neutral probability and verify that the expected stock price under it equals $S_0 e^{rT}$.*

From question 1, $p = 0.626576$. The expected stock price under it:

$$puS_0 + (1-p)dS_0 = 0.626576 \times 44 + 0.373424 \times 36 = \$\mathbf{41.0126},$$

and

$$S_0 e^{rT} = 40e^{0.05\times 0.5} = 40 \times 1.025315 = \$\mathbf{41.0126}. \quad \checkmark$$

They agree exactly, and by construction: $p$ was defined as the probability making $e^{-r\Delta t}\mathbb{E}[S_T] = S_0$, so the identity is the definition rearranged rather than a discovery.

The content is in what it licenses. Under $p$, every asset (the stock, the bond, and any derivative on them) earns the risk-free rate in expectation. That is what makes the discounted expectation a valid price: the measure has been chosen so that discounting at $r$ is correct for everything at once, which no real-world probability does.

Note also that $p = 0.6266$ falls short of the real-world probability one would expect for a stock with a positive risk premium: a stock expected to earn more than the risk-free rate must have a higher real-world chance of the up state than the risk-neutral measure assigns.

The risk-neutral probability is not a belief about the world; it is the weight that makes the arbitrage-free price come out right.

*What the question tests.* That the martingale property is definitional, and that recognizing it is the step from a two-state trick to the general framework of Chapter 28.

---

**13.3** *Explain why the real-world probability of an up move does not appear in the option price. Would the answer change if it were 0.9 rather than 0.5?*

The real-world probability does not appear because the option was priced by *replication*, and the replication works state by state. The portfolio of $\Delta$ shares and a short call was constructed to have the same value in the up state and the down state; a portfolio whose value is identical in every state has a value that does not depend on how likely the states are. Only the two prices, 44 and 36, and the two payoffs enter.

**No, the answer would not change if the probability were 0.9.** The call would still be worth \$2.44.

That looks wrong on first encounter and is worth stating carefully: it does not mean the market ignores the probability. It means the probability is *already in the stock price*. If everyone believed a 90% chance of the up move, the stock would not be trading at \$40; it would be higher, and the option would be worth more as a consequence. Holding $S_0 = 40$ fixed while changing the probability describes a market that has not repriced the stock, and in that market the option's value is determined by the stock's, whatever anyone believes.

*The option is priced relative to the stock, not absolutely.* That is the single idea the rest of the book is built on, and it is why no expected return, no risk premium, and no forecast appears in any formula that follows.

*What the question tests.* Whether the reader can explain the absence of the probability without saying that beliefs do not matter.

---

**13.4** *A stock at \$100 has $u = 1.15$, $d = 0.87$, and two six-month steps, with $r = 6\%$. Value a European put struck at \$100.*

With $u = 1.15$, $d = 0.87$, $\Delta t = 0.5$, $r = 6\%$:

$$p = \frac{e^{0.03} - 0.87}{1.15 - 0.87} = \frac{1.030455 - 0.87}{0.28} = 0.573052.$$

Terminal nodes and put payoffs:

| Node | Stock | Payoff |
|---|---|---|
| $uu$ | 132.25 | 0.00 |
| $ud$ | 100.05 | 0.00 |
| $dd$ | 75.69 | 24.31 |

Rolling back one step:

$$f_u = e^{-0.03}\,[p \times 0 + (1-p) \times 0] = 0,$$

$$f_d = e^{-0.03}\,[p \times 0 + (1-p) \times 24.31] = 0.970446 \times 10.3791 = 10.0724.$$

And back to today:

$$f = e^{-0.03}\,[p \times 0 + (1-p) \times 10.0724] = \$\mathbf{4.17}.$$

Note that the middle node lands at 100.05 rather than exactly 100 ($ud = 1.0005$ rather than 1) because $u$ and $d$ were given rather than derived as $u = 1/d$. The put is a whisker out of the money there and pays nothing, which is what makes the value rest entirely on the $dd$ path.

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*What the question tests.* Backward induction on a two-step tree, and attention to whether the middle node actually recombines at the strike.

---

**13.5** *Repeat the previous question for an American put and compute the early-exercise premium. At which nodes is exercise optimal?*

The American put is valued by the same backward induction with an exercise test at every node.

**Upper node** ($S = 115$): continuation 0, exercise $\max(100 - 115, 0) = 0$. Value 0; exercise irrelevant.

**Lower node** ($S = 87$): continuation 10.0724, exercise $100 - 87 = \mathbf{13.00}$. Exercise is optimal, and the node takes 13.00 rather than 10.07.

**Today** ($S = 100$): continuation

$$e^{-0.03}\,[p \times 0 + (1 - p) \times 13.00] = 0.970446 \times 5.550325 = \$5.3863,$$

against immediate exercise of 0. So

$$P = \$\mathbf{5.39}, \qquad \text{early-exercise premium} = 5.3863 - 4.1733 = \$\mathbf{1.21}.$$

Exercise is optimal at exactly one node (the down node after one step), and the premium of \$1.21 is about 29% of the European value, which is large because the tree is coarse and that single node carries the whole of the option's value.

The reason exercise pays there is question 11.6's: with the stock at 87 and the strike at 100, receiving \$13 now beats waiting for a payoff whose present value is \$10.07: waiting can add little to a put already this deep in the money, while the interest on \$100 received six months early is certain.

*What the question tests.* That the American value is the European calculation with a maximum taken at each node, and that identifying *where* exercise binds is what explains the size of the premium.

---

**13.6** *A stock has volatility 25% a year. Compute u, d, and p for a tree with three-month steps when $r = 3\%$. Verify that $d < e^{r\Delta t} < u$.*

With $\sigma = 25\%$ and $\Delta t = 0.25$:

$$u = e^{\sigma\sqrt{\Delta t}} = e^{0.25 \times 0.5} = e^{0.125} = \mathbf{1.1331},$$

$$d = \frac{1}{u} = \mathbf{0.8825},$$

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{1.007528 - 0.8825}{0.2506} = \mathbf{0.4988}.$$

The no-arbitrage condition:

$$0.8825 \;<\; e^{r\Delta t} = 1.0075 \;<\; 1.1331. \quad \checkmark$$

The condition must hold, or the tree admits an arbitrage. If $e^{r\Delta t} \geq u$, the riskless bond would beat the stock in *both* states, so no one would hold the stock; if $e^{r\Delta t} \leq d$, the stock would beat the bond in both states and no one would hold the bond. Either way one asset dominates the other and there is an arbitrage. Equivalently, the condition is exactly what makes $0 < p < 1$, so a violation shows up as a "probability" outside the unit interval, which is how it is usually noticed in practice.

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That $p$ is almost exactly one half here is not a coincidence. As Example 13.7 shows, $p \approx 1/2 + \left[(r - \sigma^2/2)/(2\sigma)\right]\sqrt{\Delta t}$, and here $(0.03 - 0.03125)/0.50 \times 0.5 = -0.00125$: the drift over a quarter is small relative to a 12.5% move, so the up and down states are nearly equally weighted.

*What the question tests.* That the Cox–Ross–Rubinstein parameters follow from matching volatility, and that the ordering condition is the no-arbitrage requirement in disguise.

---

**13.7** *Explain why the binomial value oscillates as n increases, and why the existence of a limit matters for the argument of Section 13.1.*

**Why it oscillates.** The tree's terminal nodes form a discrete grid of prices, and the option's payoff has a kink at the strike. Whether a node lands exactly on the strike, just above it, or just below changes how the kink is resolved, and as $n$ increases the grid shifts relative to $K$ in a way that alternates, typically with the parity of $n$, since an even number of steps places a node at $S_0$ and an odd number straddles it. The error therefore does not decline smoothly; it zigzags, its amplitude shrinking roughly as $1/n$. Whether it also changes sign depends on the strike: at the money the even and odd values fall on opposite sides of the limit, while in Figure 13.6, with $K = 52$ against $S_0 = 50$, both are still above it at $n = 50$.

**Why the limit matters.** Section 13.1's argument establishes that a *one-period* model with two states admits a unique arbitrage-free price. That is a statement about a model, not about the world, and its relevance depends on the model approximating something real. The convergence result is what supplies that: as the steps shorten, the tree's terminal distribution approaches the lognormal, and its price approaches the Black–Scholes–Merton value. Without a limit, the one-step argument would be a curiosity about a two-state world; with it, the two-state construction is a discretization of a continuous market, and the hedging argument survives the passage to the limit.

The practical corollary is Chapter 20's: because the error oscillates rather than decaying monotonically, no single value of $n$ can be trusted on its own, and a control variate removes far more error than simply increasing $n$.

*What the question tests.* That the oscillation has a structural cause worth knowing, and that convergence is what connects a toy model to the rest of the book.

---

**13.8** *Show that for a futures contract $p = (1 - d)/(u - d)$, and explain in words why the interest rate disappears.*

For a futures contract the underlying requires no investment: entering a futures position costs nothing, so a portfolio holding $\Delta$ futures and a short derivative has value equal to the derivative alone.

As before, a position of $\Delta$ futures and a short option is riskless when

$$\Delta(Fu - F) - f_u = \Delta(Fd - F) - f_d,$$

that is, when $\Delta = (f_u - f_d)/(Fu - Fd)$. Because the futures position costs nothing to enter, the portfolio is worth $-f$ today, so its certain outcome must equal $-fe^{r\Delta t}$. Substituting $\Delta$ into the up-state side and solving for $f$ gives $f = e^{-r\Delta t}\,[pf_u + (1 - p)f_d]$ with

$$p = \frac{1 - d}{u - d},$$

which is the usual expression with $e^{r\Delta t}$ replaced by 1.

**Why the interest rate disappears.** The risk-neutral drift of an asset is the cost of carrying it. A stock must drift at $r$ because holding it ties up $S_0$ of capital that could have earned $r$. A

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futures position ties up nothing (it is an agreement, not a purchase), so there is no carry to compensate, and the futures price is a martingale under the risk-neutral measure rather than growing at $r$. Setting the drift to zero is what turns $e^{r\Delta t}$ into 1.

The same substitution reappears in Chapter 16 as $q = r$ in Black's model, and Section 28.7 shows why the futures price, unlike the forward price, is a martingale under the risk-neutral measure itself, with no change of numeraire.

*What the question tests.* That the risk-neutral drift is a statement about funding, and that a zero-cost instrument therefore has zero drift.

---

**13.9** *A stock at \$50 can move to \$56, \$50, or \$44 over one period, with $r = 4\%$ and $\Delta t = 0.5$. Show that a call struck at \$52 cannot be replicated, and find the range of arbitrage-free prices by imposing positive state prices.*

**Why replication fails.** A portfolio of $\Delta$ shares and $B$ in cash produces three equations, one per state, in two unknowns:

$$56\Delta + Be^{0.02} = 4, \qquad 50\Delta + Be^{0.02} = 0, \qquad 44\Delta + Be^{0.02} = 0.$$

The last two give $\Delta = 0$ and $B = 0$, which contradicts the first. No combination of the stock and the bond reproduces the call's payoff, so the option is not replicable and no single price is enforced by arbitrage.

**The range of arbitrage-free prices.** Any state prices $(\psi_1, \psi_2, \psi_3)$ that price the bond and the stock correctly are admissible:

$$\psi_1 + \psi_2 + \psi_3 = e^{-0.02} = 0.980199,$$

$$56\psi_1 + 50\psi_2 + 44\psi_3 = 50,$$

with every $\psi_i > 0$. The call's price is $4\psi_1$, so the range is found by pushing $\psi_1$ to its limits.

Subtracting $50\times$ the first equation from the second eliminates $\psi_2$:

$$6(\psi_1 - \psi_3) = 50\left(1 - e^{-0.02}\right) \implies \psi_1 - \psi_3 = 0.165011.$$

Positivity then bounds $\psi_1$ from both sides: $\psi_3 > 0$ forces $\psi_1 > 0.165011$, and $\psi_2 > 0$ forces $\psi_1 < (e^{-0.02} + 0.165011)/2 = 0.572605$. Since the call pays 4 only in the top state, $c = 4\psi_1$ and

$$\mathbf{0.6600 < c < 2.2904}.$$

Each bound is approached as one state price goes to zero, leaving a two-state model: the lower when $\psi_3 \to 0$, the upper when $\psi_2 \to 0$. Both limiting vectors reprice the stock at exactly 50 and the bond at $e^{-0.02}$, which is the check that they are admissible.

*What the question tests.* That incompleteness is what produces a price *range*, and that the range is found by imposing positivity rather than by any pricing formula.

---

**13.10** *Explain why a one-period model with two states can be completed by the stock and the bond, and what fails when a third state is added. State what completeness has to do with the option having a single arbitrage-free price rather than a range of them.*

A one-period model with two states has two unknown state prices, and two traded assets (the stock and the bond) give two equations. The system is square and, provided the stock's two payoffs differ, has a unique solution. Every payoff is then a combination of the two assets, so any derivative can be manufactured and its cost is its price.

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Adding a third state adds a third unknown and no new equation. The system is underdetermined: infinitely many positive state-price vectors reprice the stock and the bond, and each assigns a different value to a payoff that distinguishes the three states. Question 9 exhibits exactly this, and the arbitrage-free prices form an interval rather than a point.

**What completeness has to do with a single price.** An arbitrage argument can only pin down a price by showing that a portfolio reproduces the payoff: if it does, the two must cost the same, or there is a riskless profit. In a complete market such a portfolio exists for every payoff, so every derivative has one price. In an incomplete market it does not, and the only constraint on the price is that it create no arbitrage: any value inside the range is consistent with no arbitrage, and choosing among them requires something arbitrage cannot supply: a preference, a utility, or a model assumption.

This is why the number of sources of randomness matters more than the number of assets. A market with two states and ten assets is still complete; one with three states and two assets is not, and Chapter 34's barrier pricing is where the consequence becomes expensive.

*What the question tests.* That completeness is a counting condition, and that it is precisely the condition under which "the" price exists.

---

**13.11** *Explain why pricing an option and hedging it are one calculation in a binomial tree rather than two. State where the hedge ratio appears in the pricing arithmetic, and what that implies about a model that prices well but hedges badly.*

The two are one calculation because the hedge ratio is computed *first* and the price is derived from it. In question 1, $\Delta = 0.5$ was found by requiring that the portfolio's two outcomes coincide, and only then was the price obtained by discounting the resulting certain value. The price is the cost of the hedge, so it cannot be produced without producing the hedge along the way.

The hedge ratio appears in the pricing arithmetic as

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d},$$

the quantity that makes the portfolio riskless. It is the same object that reappears in the limit as $\partial f / \partial S$, the delta of Chapter 18. In the risk-neutral formulation it is hidden, because discounting the expectation gives the price directly, but it is still there: the formula is only valid because the hedge exists.

**What that implies about a model that prices well but hedges badly.** That it is not pricing well; it is matching quoted prices. A model calibrated to reproduce today's option prices while producing wrong deltas has fitted the outputs without capturing the mechanism, and since the price is supposed to be the cost of running the hedge, a wrong hedge means the price is wrong for the purpose it exists to serve. The dealer who trades on such a model books a profit at inception and discovers the true cost over the life of the position.

Chapter 21 makes this concrete: local volatility fits every vanilla price exactly and produces demonstrably wrong hedges and forward smiles.

*What the question tests.* That calibration and correctness are different properties, and that the binomial construction shows why by putting the hedge before the price.

---

**13.12** *State four assumptions the binomial tree makes, and for each name a market condition that violates it. Identify which of the four the convergence result of Section 13.9 is most sensitive to, and why.*

**Four assumptions follow, each with a violation.**

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*The stock moves to one of two prices.* Violated by any market that gaps: an earnings announcement, a takeover, a currency devaluation. Reality has a continuum of possible prices, and jumps.

*Volatility is known and constant.* Violated by every market in the book: Chapter 19's smile is the statement that the market does not believe it, and Chapter 21 replaces it.

*Trading is possible at every node, at no cost, so the hedge can be rebalanced at each step.* Violated by transaction costs, by bid–ask spreads, and by markets that close. Chapter 18's rebalancing table prices the consequence.

*Borrowing and lending take place at one riskless rate.* Violated by funding spreads (Section 5.15 and Chapter 8), by the stock borrow fees a short hedge pays (Section 11.12), and by the multi-curve reality of Chapter 7.

**Which the convergence result is most sensitive to.** The *two-state* assumption, and specifically the absence of jumps, matters most. Convergence to the Black–Scholes–Merton value works because the binomial distribution of many small moves tends to a lognormal one, which is a central-limit argument: it requires each step's contribution to be small and the steps to be independent. A jump violates the first condition outright, so no refinement of $\Delta t$ recovers it: shrinking the time step makes the tree finer without ever producing a large move. The other three assumptions degrade the model's realism at any $n$; this one breaks the limit itself.

*What the question tests.* That the result does not depend equally on every assumption, and that distinguishing an approximation that improves with $n$ from one that does not is what tells you when the tree can be trusted.

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# Chapter 14
# Wiener Processes and Itô's Lemma

**14.1** *A variable's change over one year has standard deviation 5. What is the standard deviation of its change over three years, and over one month? State the property you used.*

Variances of independent increments add, so standard deviations scale with the square root of time:

$$\text{3 years:}\quad 5\sqrt{3} = \mathbf{8.66}, \qquad \text{1 month:}\quad 5\sqrt{1/12} = \mathbf{1.44}.$$

The property used is that the process has **independent increments**: the change over one interval is independent of the change over any disjoint interval. Independence (zero correlation is enough) makes the variances additive; it is that additivity, not anything about the normal distribution, that produces the $\sqrt{T}$ rule.

The rule is worth distrusting as well as knowing. Every application of $\sqrt{T}$ in this book (volatility scaling, margin horizons, the square-root-of-time rule in Chapter 38) rests on the same independence assumption, and it fails whenever moves are serially correlated. Daily price limits, as on Chinese exchanges, give the sharpest case: three consecutive limit-down days are perfectly correlated, so the three-day loss is three times the one-day loss, $\sqrt{3} \approx 1.7$ times what $\sqrt{T}$ scaling gives.

*What the question tests.* That the scaling law is a consequence of independence rather than of normality, which is what tells you when it breaks.

---

**14.2** *A variable follows a generalized Wiener process with drift 3 per year and variance rate 16 per year, starting at 25. Give its distribution after two years and compute the probability that it is negative.*

A generalized Wiener process $dx = a\,dt + b\,dW$ over time $T$ gives

$$x_T \sim N\left(x_0 + aT,\ b^2T\right).$$

With $x_0 = 25$, $a = 3$, $b^2 = 16$ so $b = 4$, and $T = 2$:

$$x_2 \sim N(25 + 6,\ 16 \times 2) = \mathbf{N(31,\ 32)},$$

a mean of 31 and a standard deviation of $\sqrt{32} = 5.657$.

The probability of being negative:

$$\Pr(x_2 < 0) = N\left(\frac{0 - 31}{5.657}\right) = N(-5.480) \approx \mathbf{2 \times 10^{-8}},$$

about two chances in a hundred million.

The number is small, and the fact that it is not zero is the point of the question. A generalized Wiener process is normally distributed and therefore assigns positive probability to every real value, including negative ones. For a variable that cannot go negative (a stock price, a volatility, a default intensity), that is a defect of the model rather than a remote possibility, and it is exactly

why question 3 rejects this process for a stock price and why the book uses geometric Brownian motion instead.

*What the question tests.* The distribution of a Wiener increment, and whether the reader notices that a probability of $10^{-8}$ is still a modeling failure when the event is impossible.

---

**14.3** *Explain why $dS = \mu\,dt + \sigma\,dW$ is unsuitable as a model for a stock price, giving two distinct reasons.*

**First, it permits negative prices.** As question 2 shows, a generalized Wiener process is normally distributed and can take any real value. A stock is a limited-liability claim and cannot be worth less than zero, so a model that assigns positive probability to $S_T < 0$ is describing something else.

**Second, the absolute drift and volatility are wrong in scale.** The process $dS = \mu\,dt + \sigma\,dW$ says a stock moves by $\sigma$ in absolute terms regardless of its price: a \$5 stock and a \$500 stock would have the same dollar volatility. What is stable in real markets is the *proportional* move: returns, not price changes, are comparable across stocks and across time. A model in absolute terms would need a different $\sigma$ for every price level, which is another way of saying it has the wrong variable.

Both defects are cured by modeling the return rather than the price:

$$\frac{dS}{S} = \mu\,dt + \sigma\,dW,$$

which keeps $S$ positive (its solution is $S_0$ times an exponential, Section 14.9) and makes $\mu$ and $\sigma$ per-year rates comparable across assets.

*What the question tests.* That geometric Brownian motion is chosen for two specific reasons, and that a reader who can name them will also recognize where the choice is wrong: interest rates and spreads, which can be negative, and are modeled arithmetically in Chapters 29 and 31 for exactly that reason.

---

**14.4** *A stock at \$80 has $\mu = 12\%$ and $\sigma = 25\%$. Give the distribution of the proportional change over two weeks, and compare the sizes of the drift and noise terms.*

Over $\Delta t = 2/52 = 0.03846$ years, the proportional change is

$$\frac{\Delta S}{S} \sim N\left(\mu\,\Delta t,\ \sigma^2\Delta t\right) = N\left(0.004615,\ 0.049029^2\right),$$

so a mean of $\mathbf{0.46}\%$ and a standard deviation of $\mathbf{4.90}\%$. On \$80 that is an expected gain of 37 cents against a typical move of \$3.92.

**Comparing the two terms** is the point of the question. The drift scales with $\Delta t$ and the noise with $\sqrt{\Delta t}$, so over short horizons the noise dominates completely: here by a factor of

$$\frac{0.049029}{0.004615} = 10.6.$$

Over a day the ratio would be about 33; over a year it falls to about 2.

This is why estimating $\mu$ from historical data is hopeless while estimating $\sigma$ is routine. Pricing does not need $\mu$ for a different reason: the hedge of Section 14.8 removes it.

*What the question tests.* The differing time-scaling of drift and diffusion, and the consequence that follows for what can be measured and what can be ignored.

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**14.5** *Using Itô's lemma, find the process followed by $G = S^2$ when $S$ follows (14.4).*

With $G = S^2$, the derivatives are

$$\frac{\partial G}{\partial S} = 2S, \qquad \frac{\partial^2 G}{\partial S^2} = 2, \qquad \frac{\partial G}{\partial t} = 0.$$

Itô's lemma gives

$$dG = \left(\frac{\partial G}{\partial S}\mu S + \frac{\partial G}{\partial t} + \frac{1}{2}\frac{\partial^2 G}{\partial S^2}\sigma^2 S^2\right) dt + \frac{\partial G}{\partial S}\sigma S\, dW,$$

so

$$dG = \left(2S \cdot \mu S + \frac{1}{2} \times 2\sigma^2 S^2\right) dt + 2S\sigma S\, dW = \left(2\mu + \sigma^2\right) G\, dt + 2\sigma G\, dW.$$

So $G = S^2$ is itself a geometric Brownian motion, with drift $2\mu + \sigma^2$ and volatility $2\sigma$.

Two features stand out. The volatility simply doubles, as one would guess from the exponent. The drift does *not* simply double: it gains an extra $\sigma^2$, which is entirely the Itô term. Ordinary calculus applied to $G = S^2$ would have produced $2\mu$ and missed it; and the error grows with volatility, so it is largest exactly where it matters.

*What the question tests.* Mechanical fluency with the lemma, and whether the reader can point to which part of the answer the ordinary chain rule would have gotten wrong.

---

**14.6** *A stock at \$60 has $\mu = 14\%$ and $\sigma = 35\%$. Compute the expected price and the median price after two years, and explain why they differ.*

Under geometric Brownian motion, $\ln S_T$ is normal with mean $\ln S_0 + (\mu - \sigma^2/2)T$ and standard deviation $\sigma\sqrt{T}$. So

$$\mathbb{E}[S_T] = S_0 e^{\mu T} = 60e^{0.28} = \$\mathbf{79.39},$$

$$\text{median}(S_T) = S_0 e^{(\mu-\sigma^2/2)T} = 60e^{(0.14-0.06125)\times 2} = 60e^{0.1575} = \$\mathbf{70.23}.$$

They differ by more than \$9 (13% of the median), and the reason is that the lognormal distribution is right-skewed. The price can rise without limit and can fall by at most 100%, so the upper tail is long and the lower one is bounded. A few very large outcomes pull the *mean* well above the *typical* outcome, and the median is the better guide to what will probably happen.

The gap is exactly the $\sigma^2/2$ correction, and it grows with volatility: at $\sigma = 35\%$ over two years it is 13%, and at $\sigma = 60\%$ it would be over 40%. The more volatile the stock, the further the average outcome lies above the usual one, which is question 10's subject, and the reason a track record's arithmetic average return is not the return its typical investor experienced.

*What the question tests.* That mean and median diverge under lognormality by a known amount, and that the reader can say which one answers "what will probably happen."

---

**14.7** *For the stock in the previous question, find the range within which the price after two years lies with 95% probability.*

A 95% range for $\ln S_T$ is its mean plus and minus 1.96 standard deviations:

$$\ln S_T \sim N\left(\ln 60 + 0.1575,\ (0.35\sqrt{2})^2\right), \qquad \sigma\sqrt{T} = 0.4950.$$

Exponentiating the two ends:

$$60\, e^{0.1575 - 1.96\times 0.4950} = \$\mathbf{26.62}, \qquad 60\, e^{0.1575 + 1.96\times 0.4950} = \$\mathbf{185.30}.$$

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So with 95% probability the stock finishes between roughly \$27 and \$185.

The asymmetry is the striking part. The interval runs from 56% below the current price to 209% above it, and the median sits at \$70.23, much closer to the lower end than the upper. That is the lognormal's shape made concrete: the interval is symmetric in logs and violently asymmetric in prices.

It is also a useful corrective to intuitions about volatility. A 35% annual volatility sounds moderate; over two years it makes a range of nearly seven to one the ordinary 95% outcome, which is why long-dated options are so much more expensive than short-dated ones and why forecasts of long-horizon prices carry so little information.

*What the question tests.* That the range is built in log space and transformed, and that the resulting asymmetry is a feature rather than an arithmetic slip.

---

**14.8** *Explain in words why the term $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2$ appears in Itô's lemma but not in the ordinary chain rule.*

The ordinary chain rule computes the change in $f(S)$ from a change in $S$ using a first-order expansion, and discards second-order terms because $(\Delta S)^2$ is negligible compared with $\Delta S$ when $\Delta S$ is small.

For a variable driven by a Wiener process that reasoning fails. Over a short interval $\Delta t$ the standard deviation of $\Delta S$ is of order $\sqrt{\Delta t}$, not $\Delta t$; that is the $\sqrt{T}$ scaling of question 1. Its square is therefore of order $\Delta t$, the same order as the time term $f_t\,\Delta t$ the chain rule keeps, so it does not vanish faster than the terms that make up the drift; it survives into the limit and appears as $\frac{1}{2}f_{SS}\sigma^2 S^2$, writing $f_{SS}$ for $\partial^2 f/\partial S^2$.

In words: a Brownian path is so jagged that over any interval, however short, the variable moves much further than its drift would suggest. A function with curvature therefore picks up a systematic contribution from that jaggedness, in a direction set by the sign of $f_{SS}$ (upward for a convex function, downward for a concave one), even when the underlying has no drift at all.

That systematic contribution is convexity, and it is the same quantity that appears as gamma in Chapter 18, as the convexity adjustment in Chapter 30, and as the $\sigma^2/2$ gap in question 6. Itô's lemma is the general statement of it.

*What the question tests.* That the extra term is a consequence of $\Delta S \sim \sqrt{\Delta t}$, and that the reader can connect it to the several places the book meets it again under other names.

---

**14.9** *State the Markov property and explain why it is an assumption about information rather than about volatility. Explain what weak-form market efficiency has to do with it, and give one observed feature of returns that sits uneasily with it.*

The **Markov property** says the distribution of the future depends only on the present value, not on the path that reached it: given $S_t$, the history $S_u$ for $u < t$ carries no additional information about $S_T$.

It is an assumption about *information*, not volatility, because it constrains what is knowable rather than how much things move. A Markov process can be extremely volatile or barely move at all; what the property forbids is that yesterday's path tells you something about tomorrow that today's price does not already reflect.

**Weak-form market efficiency** is the economic reason for it: past prices contain no information useful for predicting future prices. If they did, the pattern would be traded on and removed. So the Markov assumption is the mathematical form of a claim about competition among traders, not a claim about the physics of prices, which is why it is plausible for liquid markets and less so elsewhere.

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**One feature that sits uneasily with it: volatility clustering.** Large moves are followed by large moves. Knowing that yesterday's return was $-6\%$ genuinely does change the distribution of tomorrow's, even holding today's price fixed, so the price alone is not a sufficient statistic. This is what Chapter 38's volatility forecasts model, and what stochastic volatility in Chapter 21 restores by adding a second state variable, recovering the Markov property in a larger space.

*What the question tests.* That the assumption is economic in origin, and that the reader can name a well-documented violation and say how the book responds to it.

---

**14.10** *A stock has expected return $\mu$. Explain why the expected continuously compounded return per year is $\mu - \sigma^2/2$ rather than $\mu$, and state which of the two a manager quoting the arithmetic average of past annual returns is reporting.*

Over a period $T$ the continuously compounded return is $\ln(S_T/S_0)$, and from question 6's distribution its expectation is

$$\mathbb{E}\left[\ln \frac{S_T}{S_0}\right] = \left(\mu - \frac{\sigma^2}{2}\right) T.$$

The $-\sigma^2/2$ arises because the logarithm is concave. By question 8's reasoning, applying a concave function to a diffusing variable produces a systematic downward contribution: the average of the logs is below the log of the average, by exactly half the variance. For example, a series of proportional gains and losses of the same size loses money (up 10% then down 10% leaves 99), and the shortfall is the volatility drag.

**Which a manager is reporting.** The arithmetic mean of annual returns estimates $\mu$, not $\mu - \sigma^2/2$. So a manager quoting that average is reporting the larger number, and it exceeds what an investor who held throughout earned over the period. The gap is $\sigma^2/2$: for a fund with 30% volatility, 4.5 percentage points a year. (The standardized average annual total return a US mutual fund must publish is the compounded figure, so the warning applies to track records summarized by their arithmetic average.)

Two funds reporting the same arithmetic average return, one with 10% volatility and one with 40%, would be expected to deliver compound returns differing by 7.5 points a year; and the volatile one is the one whose advertised number flatters it most.

*What the question tests.* That the two means answer different questions, and that the difference is large enough to change how a track record should be read.

---

**14.11** *Explain why Itô's lemma delivers a hedge and not merely a change-of-variable formula. State which term in the lemma the hedge removes, which term survives it, and why the surviving term is what the option is paid for.*

Itô's lemma gives the process for a derivative $f(S, t)$:

$$df = \left(f_S \mu S + f_t + \frac{1}{2} f_{SS} \sigma^2 S^2\right) dt + f_S \, \sigma S \, dW.$$

The crucial feature is that the *same* $dW$ drives both $S$ and $f$, and it enters $f$ multiplied by the known quantity $f_S$. So a portfolio short one derivative and long $f_S$ shares has

$$-df + f_S \, dS = -\left(f_t + \frac{1}{2} f_{SS} \sigma^2 S^2\right) dt,$$

with the $dW$ terms canceling exactly. That cancellation is the hedge, and it is available only because the lemma expresses the derivative's randomness as a multiple of the underlying's, rather than as some new source of noise.

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**The term removed** is $f_S \sigma S\, dW$, the first-order exposure, delta. **What survives** is $\frac{1}{2} f_{SS} \sigma^2 S^2$, the convexity term, together with time decay $f_t$.

The surviving term is what the option is paid for. A delta-hedged long option is riskless only in the limit of continuous rebalancing; over any real interval it gains from the underlying's moves, at a rate set by $f_{SS}$ and $\sigma^2$, and loses time decay. The writer holds the opposite position, and the premium compensates for the gamma the writer must manage: Chapter 18's subject, and, ignoring interest, the relation $\Theta \approx -\frac{1}{2}\Gamma\sigma^2 S^2$ written out.

*What the question tests.* That the lemma's structure (one shared noise term, entering linearly) is what makes replication possible, and that the reader can identify what is left over and why it is not a residual but the point.

---

**14.12** *State the risk-neutral process for a stock price. Explain what changes and what does not when moving to it from the real-world process, and why the volatility is untouched while the drift is not.*

The risk-neutral process replaces the expected return with the risk-free rate:

$$dS = rS\, dt + \sigma S\, dW^{\mathbb{Q}}.$$

**What changes.** The drift, from $\mu$ to $r$. Under this measure every asset earns the risk-free rate in expectation, so discounting any payoff at $r$ is correct, which is what makes $f = e^{-rT}\mathbb{E}^{\mathbb{Q}}[f_T]$ a valid pricing formula.

**What does not.** The volatility, and the set of possible paths. $\sigma$ is identical under both measures.

**Why.** Girsanov's theorem says a change of measure adds a drift to a Wiener process and does nothing else. Intuitively, the two measures must agree on which events are possible (they are equivalent, assigning zero probability to the same sets), and volatility is a statement about the *roughness* of a path, which can be read off any single realization by summing squared increments. Reweighting which paths are likely cannot change how jagged each one is; it can only change how much they are expected to drift.

That asymmetry is why the framework is usable. The drift, which the data cannot pin down, is replaced and so does not matter; the volatility, which the data can measure (question 4), is kept unchanged.

*What the question tests.* That the measure change is drift-only, and that the reader can say why that particular asymmetry is what makes risk-neutral pricing practical rather than merely formal.

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# Chapter 15
# The Black–Scholes–Merton Model

**15.1** *A non-dividend stock trades at \$30. Value a three-month European call struck at \$29 with $r = 5\%$ and $\sigma = 25\%$. Then value the put and verify put–call parity.*

With $S_0 = 30$, $K = 29$, $r = 5\%$, $\sigma = 25\%$, $T = 0.25$:

$$d_1 = \frac{\ln(30/29) + (0.05 + 0.03125) \times 0.25}{0.25\sqrt{0.25}} = 0.4337, \qquad d_2 = d_1 - 0.125 = 0.3087.$$

$$c = 30\,N(d_1) - 29e^{-0.0125}N(d_2) = 30 \times 0.667751 - 28.63976 \times 0.621230 = \$\mathbf{2.2407}.$$

$$p = 29e^{-0.0125}N(-d_2) - 30\,N(-d_1) = 28.63976 \times 0.378770 - 30 \times 0.332249 = \$\mathbf{0.8804}.$$

Carried to five decimals, parity holds:

$$c + Ke^{-rT} = 2.24067 + 28.63976 = 30.88043, \qquad p + S_0 = 0.88043 + 30 = 30.88043. \quad \checkmark$$

The identity holds exactly, and must: the Black–Scholes–Merton formulas for the call and the put are derived from the same risk-neutral expectation, so parity is built into them rather than being an additional property to verify. Computing both and checking is nonetheless the standard way to catch an arithmetic slip, since an error in $d_1$ or $d_2$ will almost always break it.

*What the question tests.* Mechanical fluency with the formula, and the recognition that parity is a consistency check the model satisfies by construction.

---

**15.2** *Explain why the stock's expected return does not appear in (15.1), and state which chapter established the same result in a discrete setting.*

A portfolio short one derivative and long $\partial f/\partial S$ shares has, by Itô's lemma, no $dW$ term: the two random components cancel exactly, and $\mu$, which enters both drifts through the same $\frac{\partial f}{\partial S}\mu S$, cancels with them. Being riskless over the instant, the portfolio must earn $r$, and imposing that gives the equation.

The economic content is that the derivative is priced *relative* to the stock. Whatever the market believes about the stock's prospects is already reflected in $S_0$, which is an input; the option's value follows from the cost of manufacturing its payoff out of the stock, and that cost does not depend on where anyone expects the stock to go.

**Chapter 13 established the same result in a discrete setting.** There the real-world probability of an up move never appeared in the binomial price, and Section 13.2 makes the point explicitly: the true probability might be 0.7 or 0.3, and with $S_0$ held fixed the option is worth \$0.545 either way. The continuous argument is the same replication carried to the limit.

*What the question tests.* That the disappearance of $\mu$ is a consequence of hedging rather than an assumption of risk neutrality, and that the reader can point to where the same thing happened two chapters earlier.

**15.3** *In Example 15.1, state the risk-neutral probability that the option is exercised and explain why this is not the probability an investor should use when forecasting.*

The risk-neutral probability that a call is exercised is $N(d_2)$, the probability, under $\mathbb{Q}$, that $S_T > K$. For the option of Example 15.1 that probability is

$$N(d_2) = N(0.3460) = \mathbf{0.6353},$$

about a 64% chance.

**Why an investor should not use it as a forecast.** Because $\mathbb{Q}$ is not the real-world measure. Under $\mathbb{Q}$ the stock drifts at the risk-free rate; in reality it is expected to drift at $\mu$, which for equities is higher. The real-world probability of exercise uses $\mu$ in place of $r$ in $d_2$, and for a stock with any equity risk premium it is *larger* than the risk-neutral figure.

The two answer different questions. $N(d_2)$ is a pricing weight: the number that makes the discounted expectation equal the cost of the hedge. The real-world probability is a statement about what will happen, and it depends on $\mu$, which is unobservable and irrelevant to the price.

An investor who reads 64% as "the option will probably finish in the money" has imported a pricing artifact into a forecast. The error is systematic and in one direction: risk-neutral probabilities of good outcomes understate real ones for assets carrying a positive risk premium, which is the same reason risk-neutral default probabilities in Chapter 24 overstate historical ones.

*What the question tests.* That $N(d_2)$ is a probability under a measure chosen for convenience, and that using it as a forecast is a specific and predictable mistake.

**15.4** *Compute the historical volatility from the last six observations of Table 15.1 and compare it with the estimate from all ten. What does the difference tell you about the reliability of short samples?*

Annualizing the sample standard deviation of daily log returns with 252 trading days:

$$\text{all ten observations: } \mathbf{20.4}\%, \qquad \text{last six: } \mathbf{19.4}\%.$$

The two differ by about one percentage point, which looks reassuringly close—and that impression is what the question is testing. The standard error of a volatility estimate from $n$ observations is roughly $\sigma/\sqrt{2n}$:

$$n = 10: \quad \frac{20.4}{\sqrt{20}} = 4.6\%, \qquad n = 6: \quad \frac{19.4}{\sqrt{12}} = 5.6\%.$$

So each estimate carries a standard error about five times the difference between them. The agreement is not evidence that either is right: since the six are a subset of the ten, it is partly mechanical, and shows only that they share data.

The chapter's own sensitivity table makes the consequence concrete: a two-standard-error band of about ±10 points around 24% runs from 14% to 34% volatility, which prices the call of Example 15.1 anywhere from \$4.83 to \$7.64, about 20% either side of its value.

That is why the working minimum is 90 to 180 observations, and why implied volatility, which requires no historical sample at all, displaced historical estimation for pricing.

*What the question tests.* That closeness of two estimates says nothing about their accuracy, and that the reader computes the standard error rather than eyeing the gap.

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**15.5** *A call on a \$60 stock struck at \$60 with six months to run and $r = 4\%$ trades at \$5.20. Estimate the implied volatility to the nearest half percent by trial and error.*

Trial and error on the Black–Scholes–Merton call value with $S_0 = K = 60$, $r = 4\%$, $T = 0.5$:

| $\sigma$ | **Call value** |
|---|---|
| 25.0% | 4.80 |
| 27.0% | 5.14 |
| 27.5% | 5.22 |
| 30.0% | 5.63 |

The market price of \$5.20 is bracketed between 27% and 27.5%, and interpolating gives 27.4%. To the nearest half percent the implied volatility is

$$\boxed{27.5\%}$$

(the exact root is 27.38%).

Two remarks follow. The call value is close to linear in $\sigma$ over this range (the sensitivity is vega, and for an at-the-money option it is nearly constant), which is what makes simple interpolation work so well and why the search converges in two or three trials.

And the inversion has a unique answer because the call price is increasing in volatility, as Section 11.1 argued without a model, and strictly so in the formula, whose vega is positive for every $\sigma > 0$. Without that monotonicity, "the" implied volatility would not be well defined.

*What the question tests.* That implied volatility is obtained numerically because the formula cannot be inverted analytically, and that the reader knows why the root is unique.

**15.6** *An index at 2,000 with a 2% dividend yield has three-month calls struck at 2,050 trading at 42. With $r = 3\%$, find the implied volatility.*

An index option uses the dividend-yield form, replacing $S_0$ with $S_0 e^{-qT}$:

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}.$$

With $S_0 = 2000$, $K = 2050$, $q = 2\%$, $r = 3\%$, $T = 0.25$ and a market price of 42, solving for $\sigma$ gives

$$\sigma = \mathbf{15.4}\%.$$

The option is 2.5% out of the money with three months to run, so its entire value is time value; at a 15.4% volatility the index's quarterly standard deviation is about 7.7%, comfortably enough for the strike to be reached.

Note how the carry enters. With $r - q = 1\%$, the three-month forward is $2000e^{0.0025} = 2005$, so the option is 2.24% out of the money against the *forward* rather than 2.5% against spot. The distinction is immaterial here but not always: Chapter 16's question 9 is the general statement, and a desk that inverts against spot instead of the forward will report a skew that is partly an artifact of its own carry assumption.

*What the question tests.* The dividend-yield adaptation, and whether the reader inverts against the right reference.

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**15.7** *A stock at \$40 pays a \$0.60 dividend in three months. Value a five-month European call struck at \$40 with $r = 5\%$ and $\sigma = 28\%$.*

A known discrete dividend is removed at present value before applying the formula:

$$S_0^* = 40 - 0.60e^{-0.05\times 0.25} = 40 - 0.5925 = \$39.4075.$$

Then with $K = 40$, $r = 5\%$, $\sigma = 28\%$, $T = 5/12$:

$$c = \$\mathbf{2.9468}.$$

The reasoning is Chapter 5's carried into option pricing. The option holder does not receive the dividend, so the part of today's share price that represents the imminent payment is not available to them. Stripping it out leaves the quantity the option is effectively written on, and treating that residual $S_0^*$ as geometric Brownian motion is an assumption, not a derivation.

That assumption is where the method is approximate. The true process is the share price, which jumps down by the dividend on the ex-date; modeling the dividend-stripped price as lognormal is a convenient choice, not a fact. Applying the same idea in a tree (Chapter 20's displaced tree) adds early exercise but keeps the assumption. Modeling $S$ itself requires the non-recombining tree of Chapter 20, and Appendix A's volatility adjustment (TN15.9) reconciles the two.

*What the question tests.* The mechanics of subtracting a dividend's present value from the stock price, and awareness that it is an approximation rather than an application of the model.

**15.8** *Explain why the European formula prices an American call on a non-dividend stock exactly, but not an American put.*

**The call.** Chapter 11 established that an American call on a non-dividend-paying stock should never be exercised early, because $S_0 - Ke^{-rT} > S_0 - K$ for any $r > 0$: selling always beats exercising. If the early-exercise right is never used, it is worthless, so the American call has exactly the value of the European one and the European formula prices it exactly. Nothing is being approximated.

**The put.** The same argument reverses. Exercising a put brings forward the *receipt* of $K$, which earns interest, so for a sufficiently deep in-the-money put immediate exercise dominates waiting (Chapter 11's question 6 is such a case). The right is therefore genuinely valuable, and a formula that ignores it must undervalue the option.

The deeper reason the formula cannot be repaired is that the American put's value depends on the *exercise boundary* $S^*(t)$, the critical price as a function of time remaining. That boundary is not known in advance: it is part of the solution, determined simultaneously with the value. The problem is a free-boundary problem, and free-boundary problems have no closed-form solution in general.

That is why most dealers' pricing systems use a lattice, Chapter 13's tree made efficient in Chapter 20, or the finite-difference grid of Chapter 20. The tree takes the maximum of continuing and exercising at every node and so discovers the boundary rather than assuming it.

*What the question tests.* That the difference is not a matter of degree but of mathematical type, and that naming the free boundary explains why numerical methods are required rather than merely convenient.

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**15.9** *Verify the limiting behavior of* (15.2) *as* $\sigma \to \infty$ *and confirm that it reproduces the upper bound of Section 11.2. Do the same as* $\sigma \to 0$.

**As** $\sigma \to \infty$**.** Then $d_1 = \frac{\ln(S_0/K)+(r+\sigma^2/2)T}{\sigma\sqrt{T}} \to +\infty$ and $d_2 = d_1 - \sigma\sqrt{T} \to -\infty$, so $N(d_1) \to 1$ and $N(d_2) \to 0$:

$$c \to S_0 \times 1 - Ke^{-rT} \times 0 = S_0.$$

That is precisely the upper bound of Section 11.2: a call can never be worth more than the stock, since it confers at most the right to own it. Infinite volatility sends almost every path toward zero while a vanishing few carry the stock's whole expected value, so the strike is almost never paid and the option converges to the asset itself.

**As** $\sigma \to 0$**.** The stock becomes deterministic and grows at $r$, so $S_T = S_0 e^{rT}$ with certainty. If $S_0 e^{rT} > K$ the option finishes in the money; $d_1$ and $d_2$ both $\to +\infty$ and

$$c \to S_0 - Ke^{-rT},$$

the discounted certain payoff, which is the lower bound of Section 11.2. If $S_0 e^{rT} < K$ both arguments go to $-\infty$ and $c \to 0$.

So the formula reproduces both bounds as limiting cases, which is the check that it is consistent with the model-free results derived four chapters earlier from arbitrage alone. A formula violating them would be wrong regardless of its derivation.

*What the question tests.* That limiting behavior is how a closed-form result is sanity-checked, and that both bounds emerge from the same expression.

---

**15.10** *Show that a forward contract satisfies* (15.1)*, and explain what this demonstrates about the relationship between Chapter 5 and this chapter.*

A forward contract with delivery price $K$ has value

$$f = S - Ke^{-r(T-t)}.$$

Its derivatives are:

$$\frac{\partial f}{\partial S} = 1, \qquad \frac{\partial^2 f}{\partial S^2} = 0, \qquad \frac{\partial f}{\partial t} = -rKe^{-r(T-t)}.$$

Substituting them into the equation gives:

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = -rKe^{-r(T-t)} + rS + 0 = r\left(S - Ke^{-r(T-t)}\right) = rf. \quad \checkmark$$

The equation is satisfied identically, for every $\sigma$.

**What this demonstrates.** That Chapter 5 and this chapter are not two theories but one. The forward's value was derived there by a static replication argument (two portfolios with identical payoffs), with no stochastic calculus and no mention of volatility. Here it emerges as a solution of the same partial differential equation that produces the Black–Scholes–Merton formula, under the boundary condition $f(S,T) = S - K$.

The role of $\sigma$ is the interesting part: it multiplies $f_{SS}$, which is zero, so the volatility term contributes nothing. A linear payoff has no convexity, so its value cannot depend on volatility, which is Chapter 11's question 1 restated as a property of the equation. Every instrument in the book divides on this line, and the equation shows exactly where the division comes from.

*What the question tests.* That the PDE governs all derivatives, and that the linear ones are the special case where its second-order term switches off.

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**15.11** *Using Table 15.3, state by what factor the implied volatilities differ across strikes and explain in one sentence what the model requires them to be.*

From the table, implied volatility falls monotonically as the strike rises:

$$\frac{56.9\%}{33.2\%} = \mathbf{1.71},$$

so the implied volatilities differ by a factor of about **1.7** across the chain, from 56.9% at the 42.50 strike to 33.2% at 55.00, on the same stock, at the same moment, for the same expiration. The calls are American on a stock paying $1.00, so part of each figure, most at the lowest strikes, is early-exercise premium folded into the implied volatility (how much depends on the ex-dividend date, which the chain does not give), but not enough to close the gap.

The model requires them to be identical. The Black–Scholes–Merton model assumes one constant volatility for the underlying process; the strike is a feature of the contract, not of the stock, so every option on that stock must be priced with the same $\sigma$. A surface that varies by 70% across strikes is the market announcing that it does not believe the model it is quoting in.

The reconciliation (how a wrong model remains the market's universal quoting convention) is Chapter 19's subject. In brief: implied volatility is used as a *transformation* of price rather than as a parameter estimate, a monotone map that puts option prices on a comparable scale. No one using it believes the process has that volatility, any more than a bond trader quoting a yield believes the curve is flat.

*What the question tests.* That the smile is a contradiction of the chapter's own central assumption, stated plainly, and that the reader can say what the model demands before learning how the market lives with its violation.

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# Chapter 16
# Options on Indices, Currencies, and Futures

**16.1** *An index stands at 800 with a dividend yield of 3%. Value a four-month European call struck at 820 with $r = 4\%$ and $\sigma = 22\%$.*

With $S_0 = 800$, $K = 820$, $q = 3\%$, $r = 4\%$, $\sigma = 22\%$, $T = 1/3$:

$$d_1 = \frac{\ln(800/820) + (0.04 - 0.03 + 0.0242) \times 1/3}{0.22\sqrt{1/3}} = -0.1047,$$

$$d_2 = d_1 - 0.22\sqrt{1/3} = -0.2317,$$

$$c = 800e^{-0.01}N(-0.1047) - 820e^{-0.0133}N(-0.2317) = \mathbf{32.56}.$$

The dividend yield enters by discounting the index rather than the strike, which is Section 5.4's carry argument applied to option pricing: the option holder does not receive the index's dividends over the option's life, so the quantity the option is written on is the index net of them.

Note that $r - q = 1\%$, so the four-month forward is $800e^{0.0033} = 802.7$ and the option is 2.2% out of the money against the forward rather than 2.5% against spot, the distinction Example 16.2 draws.

*What the question tests.* The dividend-yield substitution and the arithmetic of an index option, which the rest of the chapter generalizes.

**16.2** *Explain why $q = r$ for a futures contract, and state which chapters established this in discrete and continuous time.*

A futures contract costs nothing to enter. Buying the underlying asset ties up capital that would otherwise earn $r$, and the risk-neutral drift compensates for exactly that cost; a futures position ties up nothing, so there is no carry to compensate and the futures price drifts at zero under $Q$.

Setting the drift to zero in the general formula $S_0e^{(r-q)T}$ requires $q = r$: the asset behaves like one whose dividend yield exactly offsets the financing cost. That substitution turns the Black–Scholes–Merton formula into Black's model, with the futures price replacing the spot and $e^{-rT}$ discounting the whole payoff.

**Chapter 13 established it in discrete time**: Section 13.10 shows that for a futures contract the risk-neutral probability is $p = (1 - d)/(u - d)$, the usual expression with $e^{r\Delta t}$ replaced by 1, and for the same reason: no capital is committed, so no growth is required.

**Chapter 14 established it in continuous time**: Section 14.7 applies Itô's lemma to the forward price $F = Se^{r(T-t)}$ and finds a drift of $\mu - r$, which is zero once risk-neutral valuation sets $\mu = r$. Chapter 28 states it in measure terms: the futures price is a martingale under the risk-neutral measure, the forward price under the $T$-forward measure.

*What the question tests.* That $q = r$ is a statement about funding rather than about dividends, and that the reader can locate the same result in both the discrete and continuous treatments.

---

**16.3** *Spot GBP/USD is 1.2600, US rates are 4%, UK rates 4.5%, and six-month volatility is 9%. Value a call on sterling struck at 1.2800, and express the premium both in dollars per pound and as a percentage of the sterling notional.*

A currency option treats the foreign interest rate as a dividend yield: sterling held earns the UK rate, which the option holder forgoes. So $q = r_f = 4.5\%$ and $r = r_d = 4\%$, with $S_0 = 1.2600$, $K = 1.2800$, $\sigma = 9\%$, $T = 0.5$:

$$c = 1.2600e^{-0.0225}N(d_1) - 1.2800e^{-0.02}N(d_2) = \mathbf{0.021505}$$

dollars per pound.

As a percentage of the sterling notional:

$$\frac{0.021505}{1.2600} = \mathbf{1.71}\%.$$

Both quotations are used and they are not interchangeable. *Dollars per pound* is the premium a dollar-based buyer pays per unit of sterling bought, and it is what settles. *Percentage of notional* is how the option is compared with others: it is scale-free, so a desk can see at a glance whether 1.71% is expensive for a six-month call struck 1.8% out of the money against the forward.

The FX market's habit of quoting premiums in several conventions at once (percentage of foreign notional, percentage of domestic, pips) is a recurring source of settlement error, and the reason a confirmation states the convention explicitly.

*What the question tests.* The Garman–Kohlhagen substitution, and awareness that the premium's units are part of the quote.

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**16.4** *A dealer quotes three-month USD/JPY as ATM 9.50, $RR_{25} = +1.20$, $BF_{25} = 0.30$. Compute the two 25-delta wing volatilities and state which currency the market is paying more to protect against.*

The market quotes the smile through three numbers. The risk reversal is the difference between the two 25-delta wings and the butterfly is their average above the at-the-money level:

$$\sigma_{25\mathrm{C}} = \mathrm{ATM} + \mathrm{BF} + \frac{1}{2}\mathrm{RR} = 9.50 + 0.30 + 0.60 = \mathbf{10.40}\%,$$

$$\sigma_{25\mathrm{P}} = \mathrm{ATM} + \mathrm{BF} - \frac{1}{2}\mathrm{RR} = 9.50 + 0.30 - 0.60 = \mathbf{9.20}\%.$$

The risk reversal is +1.20, meaning *calls on the dollar* (equivalently puts on the yen) carry the higher implied volatility. In USD/JPY quoted as yen per dollar, a call on the base currency is a bet the dollar strengthens against the yen.

So the market is paying more to protect against **a stronger dollar / weaker yen**. That is the less common configuration for this pair. The yen is a funding currency and carry positions are short yen, so the crowded risk is a yen rally, and Japanese institutions holding foreign assets add persistent *hedging* demand for yen calls; both push the risk reversal negative. A positive reading says that, for now, demand for protection against a weaker yen outweighs both. Which side dominates flips with the regime, and the sign of the risk reversal is the market's answer on the day.

*What the question tests.* The FX quoting convention, and whether the reader can translate a positive risk reversal into a statement about which currency direction is being insured.

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**16.5** *Using Black's model, value a six-month call on a futures price of 60 struck at 58, with $r = 3\%$ and $\sigma = 30\%$.*

Black's model discounts the whole payoff and uses the futures price in place of the spot:

$$c = e^{-rT}\left[F_0 N(d_1) - K N(d_2)\right], \qquad d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}.$$

With $F_0 = 60$, $K = 58$, $r = 3\%$, $\sigma = 30\%$, $T = 0.5$:

$$d_1 = \frac{\ln(60/58) + (0.30)^2 \times 0.5/2}{0.30\sqrt{0.5}} = 0.2659, \qquad d_2 = 0.0537,$$

$$c = e^{-0.015}\left[60\, N(0.2659) - 58\, N(0.0537)\right] = \$\mathbf{5.957}.$$

The structural difference from the Black–Scholes–Merton formula is worth stating: there is no $S_0$ anywhere, and no carry term inside $d_1$. The forward already embeds the cost of carry, so the formula needs only the forward, the strike, the volatility of the forward, and a discount factor. That is why Black's model generalizes so far beyond futures: to caps, swaptions, and anything else quoted off a forward, as Chapter 29 develops.

*What the question tests.* That Black's model is the same formula with the forward as its input, and that the absence of a carry term inside $d_1$ is the reason it travels.

**16.6** *Show that at $K = F_0$ Black's model gives $c = p$, and identify the result from Chapter 11 that this reproduces.*

At $K = F_0$ the logarithm vanishes, so

$$d_1 = \frac{\sigma\sqrt{T}}{2}, \qquad d_2 = -\frac{\sigma\sqrt{T}}{2} = -d_1.$$

Then

$$c = e^{-rT} F_0\left[N(d_1) - N(-d_1)\right],$$
$$p = e^{-rT} F_0\left[N(d_1) - N(-d_1)\right],$$

using $d_2 = -d_1$ in both formulas, so $c = p$ identically.

**The Chapter 11 result this reproduces** is put–call parity. Parity says

$$c - p = e^{-rT}(F_0 - K),$$

which is zero exactly when $K = F_0$. So the equality of the at-the-money-forward call and put is not a special property of Black's model: it is parity evaluated at the forward, and it holds model-free.

That is why the at-the-money-forward straddle is the usual reference point. Its value depends on volatility alone, with the directional component removed by construction, so it is the cleanest instrument for quoting and trading volatility. Chapter 19's surface is anchored there.

*What the question tests.* That a formula's special case can be recognized as a model-free result, and that the reader sees why this particular strike is the one most volatility markets quote from.

**16.7** *Explain the difference between a quanto and a composite option, and state which one an investor wanting foreign equity exposure without currency risk should buy.*

Both instruments give a domestic investor exposure to a foreign asset, and they differ in what happens to the currency.

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**A composite option** is written on the foreign asset *converted at the prevailing exchange rate*: its payoff is $\max(V_T S_T - K, 0)$ in domestic currency, so the investor is exposed to the asset and the currency together, and to their correlation.

**A quanto** pays the foreign asset's return *in domestic currency at a fixed rate*: the payoff is $\max(V_T - K, 0)$ multiplied by a predetermined exchange rate, so the currency is eliminated entirely.

**An investor wanting foreign equity exposure without currency risk should buy the quanto.** The exchange rate is fixed in the contract, so the return is the foreign index's return, delivered at home, with no FX exposure at any point.

The price of that convenience is the quanto adjustment of question 8: the dealer hedging the position must hold a currency exposure that changes with the index, and the cost of that dynamic hedge depends on the correlation between the two. So the currency risk has not disappeared: it has been sold to the dealer, and priced on a correlation assumption.

*What the question tests.* That the two structures differ in a single clause with large consequences, and that removing a risk from a client means pricing it for a dealer.

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**16.8** *A foreign index at 2,500 has $q_f = 1.5\%$, $r_f = 2\%$, $\sigma_V = 22\%$, $\sigma_S = 9\%$, and $\rho = -0.25$. Compute the one-year quanto forward and state whether the adjustment raises or lowers it.*

The quanto forward adjusts the foreign risk-neutral drift by the covariance between the asset and the exchange rate:

$$F_0^{\text{quanto}} = V_0\, e^{(r_f - q_f - \rho\,\sigma_V\,\sigma_S)T}.$$

With $V_0 = 2500$, $r_f = 2\%$, $q_f = 1.5\%$, $\sigma_V = 22\%$, $\sigma_S = 9\%$, $\rho = -0.25$, $T = 1$:

$$\rho\,\sigma_V\,\sigma_S = (-0.25) \times 0.22 \times 0.09 = -0.004950,$$

$$F_0^{\text{quanto}} = 2500\, e^{(0.02 - 0.015 + 0.004950)} = 2500\, e^{0.009950} = \mathbf{2525.0}.$$

The adjustment **raises** the forward, because the correlation is negative and enters with a minus sign. Without it the forward would be $2500e^{0.005} = 2512.5$, so the quanto correction is worth 12.5 index points, about 0.5%.

The intuition: the dealer hedging a quanto holds the foreign asset, but owes its price movement in domestic currency at a fixed rate. If the currency tends to weaken as the asset rises (which is what negative $\rho$ means, with the exchange rate quoted so that a fall is a weakening), the hedge's gains arrive when the currency is cheap and its losses when the currency is dear, so measured in domestic currency the hedge systematically lags the index it is meant to replicate. The higher forward is what pays for that shortfall.

The adjustment rests entirely on $\rho$, a parameter no one can observe directly: it is estimated from history, it is unstable, and it is why quanto positions carry correlation risk that no vanilla hedge removes.

*What the question tests.* The formula, the sign, and the recognition that the adjustment's input is the least reliable quantity in the calculation.

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**16.9** *Every formula in this chapter takes a forward as its real input. Using Table 16.3, state the rule that governs how the forward is obtained for an equity index, a currency, and a commodity, and explain why a desk that computes the forward from spot and a carry rate can be wrong even when its rates are right.*

The rule is the same in all three markets: take the forward from the market that quotes it, and compute it only when no such market exists.

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For an **equity index**, the forward is read from index futures where they trade, and otherwise computed from spot and expected dividends. For a **currency**, it comes from the FX forward market directly, which is deeper and more reliable than the interest rate differential. For a **commodity**, the forward is the futures price itself, quoted continuously on an exchange.

**Why a desk computing from spot and a carry rate can be wrong with correct rates.** The carry calculation omits everything that is not an interest rate. For an equity, it omits the dividend forecast (Section 11.6 shows the option market's own forward embeds a dividend expectation that may differ from the declared calendar) and the cost of borrowing the stock, which Example 10.4 prices (Section 11.12). For a currency, it omits the cross-currency basis, the persistent failure of covered interest parity since 2008 that Section 5.7 documents. Each of these sits in the traded forward and in none of the inputs to the carry formula.

So the traded forward is not an approximation to the computed one; it is the correct number, and the computed one is the approximation.

*What the question tests.* That the forward is an observable to be sourced rather than a quantity to be derived, and that the reader can name what the derivation leaves out.

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**16.10** *Using Table 16.4, state which convention applies to an equity index option, a currency option, and an option on a futures contract. Explain what goes wrong when a convention is carried across from the market it belongs to, and give one case where two conventions would both be defensible.*

**Which convention applies.**

An *equity index option* uses the dividend-yield form: $S_0 e^{-qT}$ in place of the spot, with $q$ the index's continuous dividend yield.

A *currency option* uses the same form with $q = r_f$, the foreign interest rate (Garman–Kohlhagen).

An *option on a futures contract* uses Black's model, with $q = r$ so that the futures price is a martingale, and the entire payoff discounted at $r$.

**What goes wrong when a convention is carried across.** The carry term inside $d_1$ is then wrong, so both the value and every Greek are wrong, and the error grows with maturity. The most common instance is applying the spot-based equity formula to a futures option, which double-counts the carry (the futures price already embeds it) and, by growing the futures price at $r$ a second time, produces a call value that is too high whenever rates are positive. The error is invisible on a short-dated at-the-money option, where the carry term is small, and becomes material at long maturities.

**A case where two conventions are both defensible.** An option on an index futures contract. It can be priced with Black's model on the futures price, or with the dividend-yield formula on the index spot with $q$ the index yield; if the futures price is exactly $S_0 e^{(r-q)T}$ the two agree identically. They diverge when the futures price trades away from fair value, which Chapter 5 shows it can do within the band that arbitrage costs leave open, and the choice is then about which market's price to trust.

*What the question tests.* That the three formulas are one formula with three carry substitutions, and that knowing which to use requires knowing what the underlying quote already contains.

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# Chapter 17

# Employee Stock Options and Equity Compensation

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**17.1** *Explain why Section 11.8's theorem on early exercise does not apply to an employee option, being precise about which assumption fails.*

The theorem of Section 11.8 (that an American call on a non-dividend-paying stock should never be exercised early) rests on the holder having a better alternative to exercising: sell the option. Selling realizes intrinsic value *plus* time value; exercising realizes intrinsic value alone, so exercise is dominated.

The assumption that fails is **transferability**. An employee option cannot be sold, so the comparison the proof relies on is unavailable. The employee's choice is not "exercise or sell" but "exercise or continue to hold an undiversified, unhedgeable claim on their own employer," and the second is genuinely unattractive to someone whose salary is already exposed to the same firm.

Two further assumptions fail alongside it, and both push the same way. The holder cannot *short* the stock to lock in a gain without exercising (company policy sees to that, and for officers and directors so does Section 16 of the Exchange Act), so hedging is not a substitute for selling either. And the option is *forfeited on departure* if unvested and typically expires within 90 days of departure if vested, which imposes an exercise date the contract never mentions.

The consequence is that early exercise is not irrational here; it is the rational response of a risk-averse, undiversified holder who cannot trade. That is why the valuation methods of Section 17.3 replace *optimal* exercise with a *behavioral* rule, and why the resulting number is not a market price.

*What the question tests.* That a no-arbitrage theorem is only as good as its assumptions, and that the employee-option setting breaks the one that matters.

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**17.2** *A grant has a ten-year term but an expected life of five years. Explain why using the expected life in Black–Scholes is an approximation rather than a correction, and whether its error has a definite sign.*

*Why it is an approximation.* Expected life is not an input to the option: it is an *output* of the exercise behavior, and it depends on the stock price path. Black–Scholes with a five-year term prices a *European* option that certainly lives five years. The real grant is a claim that dies at a *random* time, and one that is systematically correlated with the stock: exercise comes early when the stock has risen through the employee's threshold, and late when it has not. Replacing a random, stock-dependent stopping time with its unconditional mean is a substitution no amount of care in estimating the mean can repair.

*The direction.* It depends on what the expected life includes. Folded into a life that counts pre-vesting forfeiture, the shortcut errs **high**, because it prices a forfeited option, which pays nothing, as though it lived the average life: Example 17.1's lattice ($M = 2.5$, exit 3%) is worth

**14.22** per option granted with an expected life of **7.6 years**, and Black–Scholes at 7.6 years returns **14.84**, 4.4% high. On the standards' basis, conditional on vesting, the sign is not fixed: the same lattice is worth **15.56** with an expected life of **8.2 years**, and Black–Scholes at that life returns **15.36**, 1.3% low. Jensen's inequality pushes the shortcut up, since value is concave in maturity; exercise at the multiple, which captures the gain on exactly the paths that rose, pushes the lattice up, and here it wins.

A last point of practice: under ASC 718 the *forfeiture* of unvested awards is handled by adjusting the *number* of awards expensed, not the fair value of each, so the expected life used for fair value should capture post-vesting exercise and departure, not turnover before vesting. Folding pre-vesting turnover into the term as well, while the award count is also reduced for it, counts it twice. (Example 17.5 works instead with the expected cost per option granted, which carries pre-vesting turnover once, in the value, and so makes no separate count adjustment.)

*What the question tests.* That the expected-term shortcut changes the contract rather than correcting the model, and that whether its bias has a sign depends on what the expected life includes.

---

**17.3** *Using Example 17.1, state which input the reported expense is most sensitive to, and explain why that is awkward for an auditor.*

The most sensitive input, within the example, is the exit (turnover) rate. Moving it from 0% to 6% moves the value from 16.10 to 12.59, a swing of 3.51, or 25% of the grant. The exercise multiple $M$, over its plausible range of 2.0 to 3.0, moves the value only from 13.54 to 14.52, a swing of 0.98. Turnover matters roughly three and a half times as much.

*Why that is awkward for an auditor.* Every other Black–Scholes input can be challenged against something outside the company. The rate comes from the Treasury curve; volatility can be benchmarked against listed peers, and against the company's own options if any trade; the stock price is observable. The exit rate cannot. It is an assertion about how many of *these* employees will leave over the *next* ten years, supported at best by the company's own historical turnover, data the company produces, over a period that need not resemble the future, for a workforce that need not resemble the one holding the grant.

Worse, the bias runs one way. A higher assumed exit rate lowers the reported expense, so the assumption management would prefer is the one hardest to refute. The auditor is reduced to testing that the number is *consistent with history* rather than that it is *right*, and history was the wrong sample to begin with.

The headline of Example 17.1 is the deeper version of the same point: the methods ASC 718 allows span 11.552 to 15.56 on the standards' basis (a factor of 1.35, or $4.0 million per million options that vest), and any of them would survive audit. The largest single choice is not an input at all but the method, and there is no market price to appeal to, because the instrument by construction cannot be traded.

*One caveat applies.* The example holds volatility fixed at 30%. Across companies, from 20% to 45%, volatility moves the lattice value from 11.48 to 17.87, a wider swing than turnover produces. For one company, the range an auditor would accept is narrower: say 25% to 35%, which moves the value from 12.86 to 15.51, a swing of 2.65 against turnover's 3.51. That is the sense in which the exit rate drives the number more than volatility does, and it is also much the harder of the two to corroborate.

*What the question tests.* That an unauditable input driving a material number is a governance problem, not a modeling one.

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**17.4** *A company with 20 million shares issues 5 million warrants, and the equivalent call is worth \$8.00. What is each warrant worth? Explain why an outstanding, disclosed employee grant instead requires no dilution adjustment, and state which of a covered and a corporate warrant needs the factor in* (17.1).

*The warrant value.* By (17.1), with $N = 20$ million shares and $M = 5$ million warrants, each warrant is $N/(N+M)$ of the equivalent call:

$$\frac{20}{20+5} \times \$8.00 = 0.80 \times \$8.00 = \$\mathbf{6.40}$$

The 20% reduction is the dilution: the exercise gain is spread over 25 million shares rather than 20 million.

*Why a disclosed employee grant needs no such adjustment.* The factor corrects for dilution the market has *not yet* priced. A freshly announced warrant issue changes the claim structure on announcement day. An employee grant that has been outstanding and disclosed (in the share-count footnote, in the diluted-EPS reconciliation) is already in the market's information set, so the observed share price already reflects the shares those options will create. Applying the factor as well would deduct the same dilution twice.

The rule is therefore about *timing of disclosure*, not about who holds the instrument. The dilution adjustment belongs at the moment new claims are created and the market learns of them; after that it lives in the share price.

*Which warrant needs the factor.* A **corporate warrant**: one the company itself issues, so that exercise creates new shares and the company receives $MK$. A **covered warrant** is written by a bank or dealer against shares it already owns or will buy in the market; exercise transfers existing shares and the share count is unchanged, so there is no dilution and (17.1) does not apply. A covered warrant is priced as a plain call.

*What the question tests.* That $N/(N+M)$ prices *new share creation*, and that neither the word "warrant" nor the identity of the holder is what triggers it.

---

**17.5** *A grant of 200,000 options has a grant-date fair value of \$12 each and vests 25% a year over four years. Compute the first-year expense under straight-line recognition, and compare it with graded recognition.*

Total grant-date fair value is $200{,}000 \times \$12 = \$2{,}400{,}000$, and each of the four annual tranches carries \$600,000 of it.

**Straight-line recognition** spreads the total evenly over the four-year requisite service period:

$$\text{Year-1 expense} = \frac{\$2{,}400{,}000}{4} = \$\mathbf{600{,}000}$$

**Graded (accelerated) recognition** treats each tranche as a separate award and expenses it over *its own* vesting period. In year 1 all four tranches are being earned, but each over a different horizon:

| Tranche | Fair value | Vesting | Year-1 expense |
|---|---|---|---|
| 1 | \$600,000 | 1 year | \$600,000 |
| 2 | \$600,000 | 2 years | \$300,000 |
| 3 | \$600,000 | 3 years | \$200,000 |
| 4 | \$600,000 | 4 years | \$150,000 |
| **Total** | \$2,400,000 | | **\$1,250,000** |

$$\$600{,}000\,(1 + 1/2 + 1/3 + 1/4) = \$600{,}000 \times \frac{25}{12} = \$\mathbf{1{,}250{,}000}$$

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*The comparison.* Graded recognition books **52%** of the total cost in the first year against straight-line's 25%, more than twice as much. The *total* is identical at $2.4 million; only the timing differs. ASC 718 permits either for a service-condition award; IFRS 2 requires the graded pattern. So two companies with identical grants can report first-year compensation expense differing by $650,000, and neither is misstating anything.

Note the simplification the question builds in. Example 17.5 values each tranche separately and finds them *unequal*: the expected cost per option granted falls from 14.52 to 14.05 as the cliff lengthens, because a longer cliff means more exposure to forfeiture before vesting, while the fair value conditional on vesting, which is what the standards measure, rises from 14.96 to 15.85. Assuming $12 for all four suppresses a spread of a few percent. It does not change the shape of the comparison.

*What the question tests.* That the recognition pattern is a policy choice with no effect on the total, and that graded front-loads it heavily.

---

**17.6** *Explain why one RSU is worth two to three options, why the ratio would be smaller for a more volatile company, and what happens to the expense of a performance award with a market and with a non-market condition if the target is missed.*

*Why two to three.* An RSU is a prepaid forward on the share (a zero-strike call), not an option: it has no strike, so its grant-date fair value is the share price less the present value of dividends forgone during vesting and, on the same basis as the option, less the chance of forfeiture before vesting. In Example 17.7's company, with the option's three-year cliff and 3% exit rate, that is $40e^{-(0.01+0.03)\times 3} = 35.48$ against an option whose expected cost per option granted is 14.22, so

$$\frac{35.48}{14.22} = \mathbf{2.50} \text{ options per RSU}$$

Leaving the forfeiture discount off the RSU alone would overstate the ratio, since the option's value already carries it. Measured instead conditional on vesting, as ASC 718 and IFRS 2 measure fair value, both values rise by the same factor $e^{0.03\times 3}$ and the ratio is unchanged. It sits between 2 and 3 because a long-dated at-the-money option is worth about a third of the stock (the option holder has given up the entire lower half of the distribution), while the RSU keeps nearly all of it.

*The volatility comparison.* An RSU's value does not depend on volatility at all (a prepaid forward is linear in the share price), while an option's value rises with it. So the *option* in the denominator grows and the ratio **falls**:

| $\sigma$ | **Option** | **RSU** | **Options per RSU** |
|---|---|---|---|
| 20% | 11.48 | 35.48 | 3.09 |
| 30% | 14.22 | 35.48 | 2.50 |
| 45% | 17.87 | 35.48 | 1.99 |
| 60% | 20.89 | 35.48 | 1.70 |

(Lattice values on Example 17.1's grant, with only $\sigma$ varied.) A volatile startup therefore exchanges options for RSUs at 2:1 or less, and a stable utility at nearer 3:1.

*Performance awards when the target is missed.* The two conditions are treated in opposite ways, and the distinction is central to ASC 718:

- **Market condition** (total shareholder return, a share-price hurdle). The probability of missing is built into the grant-date fair value by the valuation model itself: the award is priced as the exotic it is. So the expense is recognized in full over the service period and, provided the employee completes that service, is **never reversed**, even if the hurdle is

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missed and nothing vests. The company reports compensation cost for shares no one received. (An employee who leaves before the service period ends forfeits the award, and that cost is reversed.)

- **Non-market condition** (an EPS or revenue target, or continued service). No model prices these, so they are handled through the *count*: the company estimates how many units will vest and trues the estimate up each period. If the target is missed, the estimate goes to zero and all previously recognized expense is **reversed**.

The reason for the asymmetry is that a market condition is a *payoff* already valued at grant, while a non-market condition is a *quantity* not yet known.

*What the question tests.* Which grant features enter the price and which enter the count, and that volatility moves an option's value but not an RSU's.

---

**17.7** *An employee leaves a private company holding 50,000 ISOs struck at $1.00 with a 409A value of $6.00. Compute the cash needed to exercise and the AMT income created.*

*Cash to exercise.* The strike is paid in cash regardless of the company's valuation:

$$50{,}000 \times \$1.00 = \$\mathbf{50{,}000}$$

*AMT income.* An incentive stock option creates no *regular* taxable income on exercise (that is the ISO's advantage), but the bargain element is an adjustment item for the alternative minimum tax:

$$50{,}000 \times (\$6.00 - \$1.00) = \$\mathbf{250{,}000}$$

At a 28% AMT rate that is up to $70,000 of tax, though the exemption and its phase-out mean the actual amount depends on the rest of the return.

*The position this leaves.* The employee spends $50,000 of cash, may owe another $70,000 in April, and receives in exchange shares of a private company that **cannot be sold** to pay either bill. The $6.00 is a Section 409A appraisal (a valuation prepared for tax compliance, not a price anyone has paid), so the $300,000 of stock is worth $300,000 only in the sense that the appraiser said so.

If the company later fails, the shares go to zero, and the AMT paid on a gain that never materialized is recovered only slowly through the AMT credit, against future regular tax the employee may never owe enough of. Employees were bankrupted this way after 2000, holding tax bills on unrealized gains from stock that had become worthless.

The 90-day post-departure exercise window is what forces the decision: leave, and the choice is to write the check or abandon the grant. Some companies have extended the window to seven or ten years to defuse exactly this, at the cost of the options losing ISO status after 90 days and converting to non-qualified.

*Two limits on the ISO.* An ISO's strike may not be below the stock's value at grant, so the stock these options cover was worth at most $50,000 then, inside the $100,000 of stock on which ISOs may first become exercisable in any one year; above that the excess is non-qualified. And the company's side mirrors the employee's: an ISO exercise gives it no tax deduction unless the shares are sold before the holding periods end, whereas a non-qualified exercise here would have given it a $250,000 deduction.

*What the question tests.* That the ISO's tax advantage is a deferral with a contingent liability attached, and that illiquidity is what turns it into a trap.

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**17.8** *Explain how backdating was detected, and why the Sarbanes–Oxley two-day reporting rule stopped it where prohibition had not.*

*How it was detected.* It was detected statistically, not from any confession. Backdated grants were dated to local *minima* of the share price, so the price path around the grant date had a characteristic shape: an abnormal decline before the stated grant date and an abnormal rise after it. Lie (2005) showed that average returns around grant dates were far too favorable to have arisen by chance (the probability that so many grants landed at the month's low is vanishingly small), and Heron and Lie extended the finding across thousands of grants, estimating in 2009 that 29.2% of the firms they study manipulated grants to top executives at some point between 1996 and 2005. By March 2007 the research firm Glass Lewis counted 257 companies with internal reviews or SEC or Justice Department investigations, the SEC staff put its own count at about 140, dozens restated earnings, and several chief executives resigned or were prosecuted.

The pattern is a signature of the *whole population* of grants; no single grant at a low price is evidence of anything.

*Why the two-day rule stopped it.* Backdating was already illegal: it produced false accounting (an option granted in the money requires expense recognition), false tax filings, and false proxy disclosure. Prohibition did not stop it, because detection required someone to compare the stated grant date against a decision that left no independent trace. Under the pre-2002 regime, grants could be reported as late as 45 days after the company's fiscal year end. That window was the whole opportunity: with up to a year of hindsight, choose the date whose price is lowest, and no external record contradicts you.

Sarbanes–Oxley Section 403 cut the reporting deadline to **two business days**. The company must now name the grant date before it can know which date would have been favorable. The filing is made before the favorable date can be known, so backdating now requires filing, within two days, a document false on its face, a far more exposed act than a late filing that merely happens to be flattering.

The general lesson, and the reason this belongs in a derivatives book, is that the fix was not a stronger prohibition but a change to *when the record is made*. Where a valuation depends on a date, controlling the date controls the value, and the only reliable control is contemporaneous documentation. The same principle governs trade timestamping and the fixing of benchmark rates in Chapter 4.

*What the question tests.* That a disclosure timing rule can succeed where a substantive prohibition failed, and why.

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**17.9** *A vested option is worth \$18.40 with the stock at \$60 and a strike of \$45. State the time value, explain what an employee gives up by exercising, and use Example 17.4 to explain why "I believe in the company" is an argument against exercising rather than for it.*

*Time value.* Intrinsic value is \$60 − \$45 = \$15.00, so

$$\text{time value} = \$18.40 - \$15.00 = \$\mathbf{3.40}$$

or 22.7% of the intrinsic value.

*What exercising gives up.* It gives up two things, and only the first is usually noticed. One is the **\$3.40 of time value**, destroyed outright: exercise converts an \$18.40 asset into a \$15.00 one, and part of what is lost is the interest on the \$45 strike, which holding leaves invested elsewhere until expiration. The other, for an option taxed at exercise such as a non-qualified option, is the **tax deferral**: exercising makes the spread taxable now, where holding defers the same tax.

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Against these stands the reason to exercise, *diversification*, which brings its own tax and cash consequences.

*Why "I believe in the company" is an argument against exercising.* Because belief in the company is a reason to **keep the exposure**, and the option *is* the exposure, with leverage and with no cash outlay. Example 17.4 sets out the arithmetic. On 10,000 options struck at \$20 with the stock at \$50, exercise-and-hold costs \$200,000 in strike and \$120,000 in tax paid years early, so \$320,000 of cash buys \$500,000 of stock, exposure the employee already had for nothing. It also destroys \$34,826 of time value.

So exercise-and-hold is usually dominated. Someone who genuinely believes should *hold the option*; someone who wants cash or diversification should *exercise and sell*. The one action that neither belief justifies is the one that is most common. "I want to start the capital gains clock" is a real but different argument: prepaying the ordinary-income tax converts later appreciation into a capital gain, taxed at a lower rate, and that saving is worth having only if it exceeds the time value, the deferral and the cost of staying concentrated, which on most grants it does not.

The genuine tension is between the first two rows (hold, or exercise and sell), and they must be compared after tax on both sides. At a constant 40% rate the options are worth $0.6 \times \$334{,}826 = \$200{,}896$ after the tax due on eventual exercise, against \$180,000 from exercising and selling, so diversifying costs about \$20,900, the after-tax time value. That is a question about concentration risk, not about belief.

*What the question tests.* That time value is the price of an option's optionality, and that exercise-and-hold pays cash to give it away.

---

**17.10** *A company has 40 million shares, 4 million options struck at \$25, and an average share price of \$40. Compute the incremental shares under the treasury stock method ignoring unrecognized cost, explain why the method understates economic dilution, and identify the type of company for which the understatement is largest.*

*Incremental shares.* Under the treasury stock method the company is assumed to use the exercise proceeds to buy back shares at the average market price:

$$\begin{aligned} \text{Proceeds} &= 4{,}000{,}000 \times \$25 = \$100{,}000{,}000 \\ \text{Shares repurchased} &= \$100{,}000{,}000 / \$40 = 2{,}500{,}000 \\ \text{Incremental shares} &= 4{,}000{,}000 - 2{,}500{,}000 = \mathbf{1{,}500{,}000} \end{aligned}$$

Diluted share count is 40 + 1.5 = 41.5 million against a basic 40 million, or 3.75% dilution. Equivalently, the incremental shares are $M(1 - K/S) = 4(1 - 25/40) = 1.5$ million.

*Why it understates economic dilution.* The method counts only *intrinsic* value. An option struck at \$25 with the stock at \$40 is worth more than \$15 to its holder (it carries time value), and every dollar of that time value is value transferred from shareholders that the diluted share count does not register. The extreme case makes it plain: an **at-the-money** option contributes $M(1 - K/S) = 0$ incremental shares, so a company can grant ten million at-the-money options and report no dilution at all in the year of grant, though it has just handed over tens of millions of dollars of value. An **out-of-the-money** option is likewise excluded entirely, though it is plainly worth something.

The method is a rule about *share counts* applied to an instrument whose value is not a share count. It answers "how many shares would be created net of a buyback?" when the question shareholders care about is "how much value has been transferred to the option holders?"

*The unrecognized-cost refinement.* ASC 260 adds unrecognized compensation cost to the assumed proceeds, since the company also receives the remaining employee service. If

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$10 million of the grant's cost is still unamortized, proceeds become $110 million, shares repurchased 2.75 million, and incremental shares fall to **1.25 million**. For a critic, the adjustment moves the *wrong* way: it makes reported dilution smaller still. It is defensible on its own terms, since the service is real consideration, but it widens the gap between the reported figure and the economic one.

*Where the understatement is largest.* For companies that grant heavily at or out of the money and whose stock is volatile: typically growth and technology companies. The two causes compound: at-the-money grants contribute no incremental shares, and high volatility makes time value the largest share of those options' value, so the part the method ignores is largest. A stable company granting deep-in-the-money options reports dilution close to the economic figure; a volatile company paying most of its compensation in at-the-money options can report dilution near zero while its economic dilution is material.

*What the question tests.* That diluted EPS is an accounting convention rather than a measure of value transfer, and that at-the-money grants are exactly where it fails.

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**17.11** *Using Example 17.6, state the incremental expense of the repricing and of the value-for-value exchange, and explain why the exchange is cheaper to report and weaker as an incentive.*

Example 17.6 gives the incremental expense: $4,029,120 for the repricing and zero for the exchange, which delivers 225,025 options worth $3,297,207 against the repricing's 500,000 worth $7,326,327.

A repricing is a modification, so the company recognizes the *increase* in fair value: $14.6527 − $6.5944 = $8.0583 per option, times 500,000, is **$4.03 million**, on top of the original grant's cost, which continues to be recognized. The value-for-value exchange hands over 0.450 new options per old one, chosen precisely so that fair value is unchanged, and the incremental expense is **zero**.

*Why the exchange is cheaper to report.* Because the ratio is *engineered* to make it so: the board sets new fair value equal to old, which is the accounting definition of no modification cost, and nothing about the economics requires that number.

*Why it is the weaker incentive.* Incentive intensity is the *delta* of the grant, not its value. Using the deltas of Example 17.6 (0.452 underwater, 0.779 at the money), the exchange leaves the employee with 175,297 shares of exposure against the 225,931 they surrendered, **22% less**. The transaction that costs nothing to report has *reduced* the employee's stake in the share price.

The mechanism is worth stating because it is counterintuitive. A deeply out-of-the-money long-dated option has a low *value* but a surprisingly high *delta*: 0.452 here, nearly half a share, because six years and 45% volatility give it a real chance of finishing in the money. Equating values therefore does not equate deltas: it trades a large number of options with high delta per dollar for a small number with low delta per dollar.

So the two transactions rank in opposite orders on the two criteria. The repricing costs $4.03 million to report and raises exposure to 389,507 shares; the exchange costs nothing and cuts exposure to 175,297. Boards and proxy advisers choose on the reported number, which is the one criterion that has nothing to do with why the grant exists.

*What the question tests.* That value-neutral is not delta-neutral, and that the accounting-preferred repair is the incentive-destroying one.

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**17.12** *A company grants 100,000 cash-settled SARs, fully vested at grant, worth $8 each; a year later they are worth $13. Explain how the expense differs from an equity-settled option grant of the same value, and what the pay-versus-performance table of Item 402(v) would show for the year.*

*Cash-settled.* A cash-settled SAR is a liability, remeasured to fair value at every reporting date. At grant the company recognizes $100{,}000 \times 8$, or $800,000. A year later the liability is $100{,}000 \times 13$, or $1,300,000, so the year's expense is the $500,000 increase. Total expense equals the cash the award eventually pays, and it keeps moving with the share price until the award settles.

*Equity-settled.* An option grant of the same value is expensed at its grant-date fair value of $800,000 and never remeasured. With the same rise in the share price, the second year's expense is zero. The $500,000 gain accrues to the holder and is borne by shareholders as dilution, not through the income statement.

*Pay versus performance.* "Compensation actually paid" remeasures awards at fair value at each year end, so it includes the $500,000 rise for either type of award. For the SAR the table agrees with the expense. For the option it shows $500,000 that the income statement never records, which is the remeasurement the expense never makes.

*What the question tests.* That the accounting follows the form of settlement, not the economics: the two awards have the same value and the same exposure to the share price, and the liability classification is what puts the price's path into the income statement.

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# Chapter 18
# The Greek Letters

**18.1** *A dealer writes 50,000 calls with delta 0.62. How many shares must be held to be delta-neutral, and what happens to that number as the stock rises?*

The dealer is *short* the calls, so the position's delta is $-50{,}000 \times 0.62 = -31{,}000$. Neutrality requires holding

**31,000** shares

long, which cancels it.

*What happens as the stock rises.* Delta rises (it is $N(d_1)$, and $N(\cdot)$ is increasing), so the required hedge **grows**. If the stock moves up enough to take delta from 0.62 to 0.70, the hedge must go from 31,000 to 35,000 shares, and the dealer buys 4,000 more. In the limit, a deeply in-the-money call has delta 1 and the hedge is 50,000 shares; deeply out of the money it is zero and the hedge is unwound entirely.

Two consequences follow, and both recur through the chapter:

*The hedge is dynamic, not static.* A delta hedge is neutral only for the *next* small move. The number 31,000 has a shelf life, and the rate at which it goes stale is gamma (Section 18.3). This is the whole reason gamma is a quantity a desk tracks.

*The rebalancing rule is buy-high, sell-low.* A short-call dealer buys as the stock rises and sells as it falls. That is a systematic loss on every round trip, and it is the cash form of the premium the dealer collected. Question 11 pursues what it does to the market.

*What the question tests.* The sign convention on a written option, and that delta changes continuously rather than being a hedge ratio fixed at inception.

**18.2** *Explain why neither a naked nor a covered position is a hedge, using the two outcomes in Section 18.1.*

Section 18.1's two columns settle it. The dealer has written 100,000 calls struck at \$64 with the stock at \$63, collecting \$319,438:

| Stock at expiration | Naked | Covered |
|---|---|---|
| \$54 | +\$319,438 | −\$580,562 |
| \$74 | −\$680,562 | +\$419,438 |

Neither position is a hedge, because neither holds up in both rows: each wins in one outcome and loses in the other. A hedge is a position whose value is insensitive to the underlying; each of these is a directional bet, merely pointing in opposite directions. The naked dealer is short the stock in effect and profits if it falls; the covered dealer is long 100,000 shares and profits if it rises. The outcome in each case is decided entirely by where the stock ends up, which is precisely the variable the dealer has no view on and no business having a view on.

The magnitudes make the point sharper. Each strategy can lose roughly *twice* the premium collected: \$680,562 and \$580,562 against \$319,438 received. So the dealer is being paid \$319,438 to accept a risk of nearly \$700,000 on a single trade whose direction is a coin flip. A

market maker who intermediates thousands of such trades would be a leveraged directional fund, not a market maker, and would be out of business in a bad month.

What is actually needed is a position that holds up in both rows, whichever way the stock moves: hold a quantity of stock that makes the portfolio insensitive to the stock's movement. Naked holds zero shares, covered holds 100,000; neither number was chosen with reference to the option's sensitivity. The correct quantity is $N(d_1) = 0.5199$ per option, or 51,995 shares, between the two extremes and continuously adjusted. Naked and covered are the two endpoints of a range whose *interior* contains the hedge.

*What the question tests.* That a hedge is defined by insensitivity, not by owning the underlying, and that a covered position is a directional bet presented as prudence.

---

**18.3** *A delta-neutral portfolio has gamma −4,000. Estimate the profit or loss if the underlying moves by $3, and state whether the direction of the move matters.*

The delta-neutral portfolio's change in value is dominated by the gamma term:

$$\Delta\Pi \approx \frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2} \times (-4{,}000) \times (3)^2 = -\$\mathbf{18{,}000}$$

*The direction does not matter.* The move enters squared, so a $3 fall produces the same −$18,000 as a $3 rise. That is the defining property of a gamma position: it is a bet on the *size* of the move, not its sign. Delta neutrality has removed the first-order exposure to direction and left a second-order exposure to magnitude.

Being short gamma means losing on movement in either direction. The mechanism is the rebalancing rule of question 1: a short-gamma book buys after the stock has risen and sells after it has fallen, so every round trip is a realized loss.

*What compensates.* Nothing in this calculation does, which is why it must not be read as a forecast of $18,000 of loss. By (18.8), a delta-neutral book with $\Gamma < 0$ has $\Theta > 0$, so the same position earns time decay every day. The fuller expression is

$$\Delta\Pi \approx \Theta\,\Delta t + \frac{1}{2}\Gamma(\Delta S)^2$$

and the trade is profitable if the stock moves *less* than the theta is paying for. The $3 move is a loss only relative to the move the option's implied volatility had priced in. Short gamma is a short position in *realized* volatility, collected as theta and paid out through rebalancing.

*One caveat attaches to the approximation.* The expansion holds gamma constant over the move. For a $3 move on a stock in the sixties that is fine to the precision quoted; for a large move it is not, since gamma itself changes (its rate of change, the third derivative, is called "speed"), which is the first failure Section 18.10 describes.

*What the question tests.* That gamma P&L is quadratic and therefore sign-indifferent, and that it is only half of a trade whose other half is theta.

---

**18.4** *Using (18.8), find the theta of a delta-neutral portfolio worth $500,000 with gamma −2,000, when $S = 80$, $\sigma = 25\%$, and $r = 4\%$.*

For a delta-neutral portfolio, (18.8) reduces the Black–Scholes–Merton equation to

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi$$

so

$$\Theta = r\Pi - \frac{1}{2}\sigma^2 S^2 \Gamma$$

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Substituting $r = 0.04$, $\Pi = 500{,}000$, $\sigma = 0.25$, $S = 80$, $\Gamma = -2{,}000$:

$$\frac{1}{2}\sigma^2 S^2 \Gamma = 0.5 \times 0.0625 \times 6{,}400 \times (-2{,}000) = -400{,}000$$
$$r\Pi = 0.04 \times 500{,}000 = 20{,}000$$
$$\Theta = 20{,}000 - (-400{,}000) = +\$\mathbf{420{,}000} \text{ per year}$$

or about \$1,151 per calendar day (\$1,667 per trading day).

Read the sign. Theta is **positive**: the book *earns* \$1,151 a day simply from the passage of time. That is the required counterpart of $\Gamma < 0$, and the identity fixes the exchange rate between them. The trader chooses neither independently.

Check it against question 3: a short-gamma book earns theta daily and loses $\frac{1}{2}\Gamma(\Delta S)^2$ on movement, and this one also pays $r\Pi = \$20{,}000$ a year to finance its \$500,000 value. Net of that it earns $\Theta - r\Pi = \$400{,}000$ a year, or \$1,587 per trading day, so the daily break-even move solves

$$\frac{1}{2} \times 2{,}000 \times (\Delta S)^2 = 1{,}587 \quad \Longrightarrow \quad \Delta S = \$1.26$$

which is 1.57% of \$80, and exactly the stock's daily standard deviation at 25% volatility on the trading-day clock Section 18.4 uses, $80 \times 0.25/\sqrt{252} = \$1.26$. So the book breaks even when the stock moves by its implied volatility, as Section 18.4 found for the dealer's option.

That identity is the content of (18.8): theta is exactly the rent charged for gamma, priced at the implied volatility. A short-gamma book breaks even if the stock realizes its implied volatility, profits if it realizes less, loses if it realizes more. Nothing is earned for free.

*What the question tests.* Substitution into the theta–gamma identity, and reading what the resulting sign means for the book.

---

**18.5** *Explain why gamma cannot be hedged with the underlying asset but delta can.*

*Because the underlying has no curvature.* A position of $n$ shares is worth $nS$, which is *linear* in $S$:

$$\frac{\partial(nS)}{\partial S} = n, \qquad \frac{\partial^2(nS)}{\partial S^2} = 0$$

Its delta is the constant $n$ and its gamma is identically zero. Adding stock to a book shifts delta by any amount you like and leaves gamma exactly where it was. The same applies to a forward or a futures contract: both are linear in the underlying, which is why neither can supply gamma either.

Delta, by contrast, *can* be hedged with stock precisely because stock has a delta of 1 per share and nothing else. It is the ideal instrument for the job: it moves the one exposure you want to move and disturbs none of the others.

So the asymmetry is a statement about the instrument, not about the risk. Gamma is curvature, and only an instrument that is itself curved in $S$—another option—can supply it. This is why Example 18.6 insists on the *order* of operations: hedge gamma first with options, then delta with stock. Done the other way, buying stock to fix delta leaves gamma untouched, and the options then bought to fix gamma reintroduce delta, so the first step has to be redone. Stock cannot disturb what the options set, but options always disturb what the stock set.

The same argument extends one step further. Vega, like gamma, cannot be hedged with stock. And since every option carries *both* gamma and vega, one option cannot set them independently, which is why Example 18.7 needs two options of different maturities, exploiting the fact that short-dated and long-dated options deliver gamma and vega in different proportions.

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*What the question tests.* That hedgeable exposures are determined by the derivatives an instrument possesses, and that linearity is the reason stock is both perfect for delta and useless for everything else.

---

**18.6** *An option has vega 18. By how much does its value change if implied volatility rises from 22% to 24%? On a position of 20,000 options, what is the total?*

Vega is $\partial f/\partial\sigma$ with $\sigma$ in decimal form, so a vega of 18 means a one-percentage-point rise in volatility raises the option by $18 \times 0.01 = \$0.18$. For a two-point rise:

$$18 \times 0.02 = \$\mathbf{0.36} \text{ per option}$$

and across the position

$$20{,}000 \times \$0.36 = \$\mathbf{7{,}200}$$

This is the convention Section 18.5 uses for the dealer's option, whose $\mathcal{V} = 14.769$ raises each option by \$0.148 per point and \$14,769 across 100,000 of them.

*Two cautions are worth attaching to the arithmetic.*

The estimate is *linear* and volatility moves are not small. Vega is itself a function of $\sigma$ (the sensitivity of vega to volatility is volga, also called vomma), so a two-point move is at the edge of where the first-order number is reliable. In a stress event where implied volatility moves ten points, the linear estimate is worthless, which is why desks revalue rather than extrapolate the Greeks.

And the number *presumes a parallel shift*. A book of 20,000 options is almost never 20,000 identical options; a real book spans strikes and maturities, and implied volatility does not move by two points at all of them. Aggregate vega therefore answers a question no market ever asks. Desks decompose it by maturity bucket, and by the smile's level, skew, and curvature, which is Chapter 19's subject and the reason Section 18.8 distinguishes what a desk *computes* from what it actually *hedges*.

*What the question tests.* The units in which vega is quoted, and that scaling it to a position is where the practical limits of a first-order sensitivity start to show.

---

**18.7** *Explain why a book can be short gamma and long vega simultaneously, and which maturities contribute each.*

*Because gamma and vega have opposite term structures.* Both are largest near the money, but they scale with maturity in opposite directions:

$$\Gamma = \frac{N'(d_1)}{S\sigma\sqrt{T}} \propto \frac{1}{\sqrt{T}}, \qquad \mathcal{V} = S\sqrt{T}\,N'(d_1) \propto \sqrt{T}$$

A short-dated option is nearly all gamma and almost no vega; a long-dated option is nearly all vega and little gamma. So a book that is **short short-dated options** and **long long-dated options** is short gamma and long vega at once, with no inconsistency whatever.

Example 18.7 puts numbers on it. At the same strike:

| | Gamma | Vega | Gamma per unit vega |
|---|---|---|---|
| 8 weeks | 0.07321 | 9.835 | 0.00744 |
| 40 weeks | 0.03232 | 21.711 | 0.00149 |

The short-dated option delivers **five times** as much gamma per unit of vega. That ratio is what makes the two options non-redundant, and it is why two maturities suffice to set gamma and vega independently while one never can.

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*Which maturities contribute each.* Gamma comes from the front: the weeklies and the one-to-three-month strip, where the desk's rebalancing P&L is generated day to day. Vega comes from the back: six months and beyond, where the position is a view on the level of implied volatility rather than on realized movement.

*Why the combination is common rather than exotic.* It is the natural residue of a dealer's flow. Clients buy short-dated options for event and hedging demand, leaving the desk short gamma; the desk finances and offsets the vega by buying longer-dated volatility. The result is a book that loses on a sharp move tomorrow but profits if the whole volatility surface lifts: two exposures to "volatility" that can move in opposite directions, and a good reason not to speak of a book's volatility exposure as one number.

*What the question tests.* That gamma and vega are separate exposures with separate term structures, despite both being labeled "volatility risk."

---

**18.8** *A trader holds a portfolio of butterflies and calendar spreads and describes it as diversified. Using (18.8), explain why it is not.*

*The two structures share the exposure that matters.* At the money, a long butterfly is **short gamma** (the two short middle-strike options dominate the wings near the center), and a long calendar spread is also **short gamma**, because the near-dated option it sells carries far more gamma than the far-dated one it buys ($\Gamma \propto 1/\sqrt{T}$).

By (18.8), a delta-neutral book has

$$\Theta = r\Pi - \frac{1}{2}\sigma^2 S^2 \Gamma$$

so $\Gamma < 0$ forces $\Theta > 0$ for both. Each position earns time decay every day and pays it back through rebalancing whenever the stock moves. Both make money if the underlying sits still and lose if it moves, in either direction, since the gamma term is quadratic.

So the trader holds *one* trade in several forms: a short position in **realized volatility**. Adding a second short-gamma structure does not diversify the book; it doubles the size of the position. Nothing hedges anything, and the correlation between the P&L of the two legs is high on any day the stock moves. The book looks diversified only in the sense that it has several tickets and several strikes. It is an appearance of variety in *instruments* concealing a concentration in *risk factors*, which is the general diagnostic and the reason a desk aggregates by Greek rather than by position.

*Where they genuinely do differ.* In *vega*, and the trader could defend the word on that ground alone. A long butterfly is short vega: it wants the surface to fall as well as the stock to sit still. A long calendar is *long* vega, since the far-dated option it buys carries more vega than the near-dated one it sells (question 7's term structure). So the combination is short gamma twice over while being closer to vega-neutral.

That is a real offset, and it sharpens rather than rescues the point: the book is hedged against the exposure the trader did not name and doubly exposed to the one they did. A short-gamma book of any construction is short a jump, and a jump is what puts desks out of business.

*What the question tests.* That structures are not risk factors, and that the theta–gamma identity is what exposes two positions as one.

---

**18.9** *Using Table 18.7, state which rebalancing frequency minimizes the mean cost plus one standard deviation, and explain why the mean cost rises with the number of rebalances while the standard deviation falls. Explain also why the formula value of 3.1944 is not attainable at any frequency.*

*Which frequency wins.* From Table 18.7, $n =$ **100** rebalances minimizes mean-plus-one-

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standard-deviation at **3.7242**:

| $n$ | Mean | Std. dev. | Mean + s.d. |
|---|---|---|---|
| 5 | 3.2628 | 1.1924 | 4.4552 |
| 20 | 3.3158 | 0.6212 | 3.9370 |
| 100 | 3.4266 | 0.2976 | **3.7242** |
| 250 | 3.5457 | 0.2232 | 3.7689 |

The criterion is genuinely interior (5 and 250 are both worse), because the two columns move in opposite directions and the optimum is where their sum turns.

*Why the mean rises with n.* Every rebalance is a trade, and every trade pays the proportional cost $k = 0.10\%$. Rebalancing more often means trading more often, so expected cost climbs monotonically: 3.2628 at $n = 5$ to 3.5457 at $n = 250$. Transaction costs are the price of precision.

*Why the standard deviation falls.* Hedging error is the accumulated $\frac{1}{2}\Gamma(\Delta S)^2$ over intervals during which the hedge is stale. Halving the interval halves the variance of $\Delta S$ within it, so the dispersion of total hedging cost falls as roughly $1/\sqrt{n}$: 1.1924 at $n = 5$ falls to 0.6212 at $n = 20$, a factor of 1.92 against the 2.00 the law predicts. The fit loosens further out (0.2232 at $n = 250$ is 5.3 times smaller rather than the $\sqrt{50} = 7.1$ of the pure law), because part of the dispersion comes from the transaction-cost term, which does not shrink with $n$ at all.

So the trade-off is **cost against precision**, and the optimum depends on which is being priced. On mean alone the answer is to rebalance as rarely as possible; on variance alone, as often as possible. A dealer running thousands of positions weights variance lightly and hedges loosely; a desk with one large position and a tight risk limit hedges tightly. There is no universal frequency, and a desk that quotes one has assumed a risk aversion it has not stated.

*Why the formula value is not reached.* The dashed line in Figure 18.8 sits at **3.1944**, the Black–Scholes–Merton price and the cost of replication under *continuous*, *costless* rebalancing. Every entry in the table exceeds it, because of two frictions:

- *Discreteness.* Between rebalances the hedge is stale, so any one replication misses. This widens the dispersion of the cost but does not raise its mean (without costs, Table 18.3 finds the mean equal to the formula value, to within sampling error, at every frequency), and it shrinks with $n$.

- *Transaction costs.* These are what lift the mean, and they grow with $n$.

The two cannot be driven to zero together. Continuous rebalancing would eliminate the first and make the second *infinite*, since the paths of Brownian motion have unbounded variation: a continuously rebalanced hedge trades an infinite amount. The formula's world is one where trading is free, and in that world alone is the cost 3.1944.

The Black–Scholes–Merton price is therefore a **lower bound** on the cost of manufacturing the option, not an achievable cost. The gap at the optimum ($0.23 per option on the mean, $0.53 on the mean plus one standard deviation) is what a dealer must charge above theoretical value to break even, and it is one reason the bid–ask spread on options exists at all.

*What the question tests.* That the rebalancing decision is a bias–variance trade-off with an interior optimum, and that the model price is unattainable by construction rather than by inefficiency.

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**18.10** *A pension fund holds an index portfolio and wants the payoff of portfolio-plus-put without paying a premium. State the futures position it must run and how that position changes as the market falls, and give two respects in which the result differs from buying the put outright.*

*The futures position.* The fund must *manufacture* the put by trading: the synthetic option of Section 18.11. On an index with yield $q$ a put has delta $e^{-qT}[N(d_1) - 1]$, which is negative, so the fund runs a **short** index position worth

$$e^{-qT}\,[1 - N(d_1)] \times \text{portfolio value}$$

taken in futures through (18.13), which scales it by $e^{-(r-q)(T^*-t)}$ for futures maturing at $T^*$, and divided by the contract size to give a number of contracts, adjusted continuously.

*How it changes as the market falls.* The put moves toward the money, so its delta falls toward $-1$ and the required short grows: the fund **sells more futures as the market falls**, and buys them back as it rises. At the extremes the position is fully short (the portfolio is completely neutralized, replicating a deep-in-the-money put) or flat (the market has risen and the protection is unneeded).

Note the sign. This is the mirror of the dealer in question 1 (a short-call dealer buys into strength; the insurer sells into weakness), but both trade *with* the move, the way a short-gamma hedger must, and both are momentum-following. The insurer is manufacturing the long gamma of the put it did not buy, and selling into declines is what manufacturing it requires (Section 18.12).

*Two respects in which the result differs from a purchased put.*

*1. The premium is not avoided, only paid in a different currency.* A purchased put costs a known amount today. The synthetic costs an *unknown* amount over the option's life, paid out as the sell-low, buy-high losses of the rebalancing rule. This is question 9's table, and the cost varies with the path. On a choppy market with no net move, the synthetic can cost far more than the option; on a smooth trend, less. So the fund has not removed the premium, it has replaced a fixed cost with a random one, and taken on the volatility risk the option writer would have borne. If realized volatility comes in above the level assumed when the hedge was designed, the shortfall is the fund's.

*2. It needs a liquid market, and it consumes liquidity.* Delta hedging replicates only if the fund can trade as the price moves. A purchased put pays $\max(K - S_T, 0)$ however the market gets there: gap, halt, or overnight. The synthetic must *trade through* the fall to build the short, selling futures at the moment everyone running the same program is selling. If the market gaps or the bid disappears, the trades happen at prices far below where the model wanted them, and the insurance simply is not in place.

October 1987 is the case, and it belongs in the answer. Portfolio insurance was sold on the strength of the replication argument, the market fell 20% in a day, futures traded at a large discount to fair value, and the strategies could not execute. The protection was worth a fraction of the put it was meant to replicate—and, worse, the selling required by every insurer at once contributed to the fall itself, which is question 11's feedback loop. A purchased put has a *counterparty* who bears exactly these risks; the synthetic has only a trading desk and a model.

*A third difference is worth a line.* A purchased put is a contract with a strike and an expiration. The synthetic has neither in any binding sense: the fund can change the parameters, stop hedging, or be stopped out. That flexibility is sometimes an advantage and always a temptation, and discipline is not a property a portfolio can be assumed to have in a crisis.

*What the question tests.* That synthetic insurance converts a known premium into an unknown path cost plus gap risk, and that neither is disclosed by the replication argument.

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**18.11** *A dealer who is short gamma must buy the underlying as it rises and sell as it falls. State whether that damps or amplifies the move, and explain why pinning near expiration (Example 10.6) and retail call buying (Section 10.12) push in opposite directions even though both are dealer delta-hedging.*

*It amplifies.* Buying as the price rises adds demand to a rising market and selling as it falls adds supply to a falling one. The hedging flow is *momentum-following*, pushing price further in whichever direction it has already moved. A large short-gamma position across a market is a mechanical accelerant, and it needs no one to intend it: it is the rebalancing rule of question 1, executed by risk managers doing their jobs.

The magnitude is the second derivative in action. If dealers are short $50 million of gamma per point, a one-point move forces $50 million of same-direction trading, which moves the price again, which forces more. Nothing in the mechanism is self-limiting except the size of the position and the depth of the market.

Long gamma is the opposite, and this is the key to the comparison. A long-gamma book sells into strength and buys into weakness, supplying liquidity and **damping** the move. Whether option hedging stabilizes or destabilizes a market therefore depends entirely on which side of the gamma dealers are on, a fact about positioning, not about options.

*Why pinning and retail call buying push opposite ways.*

Pinning (Example 10.6) is the **long-gamma** case. Near expiration, gamma at a heavily traded strike becomes enormous, and dealers who are net long those options must sell as the stock rises through the strike and buy as it falls back. That flow pushes the stock *toward* the strike from both sides. Prices cluster at round strikes on expiration Friday for exactly this reason.

Retail call buying (Section 10.12) is the **short-gamma** case. Retail buys calls; someone must sell them, and that someone is a dealer, who is left short gamma. Now the dealer buys as the stock rises, and since the flow is concentrated in short-dated, out-of-the-money calls whose gamma explodes as they approach the money, the required buying accelerates precisely as the stock climbs. That is the mechanism behind the episodes of 2020–21, where heavy call buying in individual stocks produced sharp upward moves that the underlying's fundamentals did not explain, and equally sharp reversals when the flow stopped.

So the two are the same equation with opposite signs on $\Gamma$: one holds the price at a strike, the other pushes it away. And note the asymmetry in *when* each bites. Pinning operates near expiration at the money, over hours. The short-gamma case operates as options travel *into* the money, over days. Both are strongest at short maturities, because that is where gamma lives.

*What the question tests.* That hedging flow is a real order flow with a sign, and that the sign is set by dealer positioning rather than by anything about options as instruments.

**18.12** *An exchange-traded product offering inverse exposure to VIX futures must buy those futures when VIX rises. Explain why its rebalancing rule has that sign, why that rule trades the way a short-gamma hedger does, and what distinguishes this case from a dealer whose hedging incidentally moves the market.*

*Why the rule has that sign.* An inverse product promises $-1\times$ the daily return of its index, so its target position is short futures with notional equal to its net asset value (NAV). When VIX futures rise, the short loses and NAV *falls*, so the target short notional falls too, and shrinking a short means **buying**. The fund therefore buys the futures that have just risen. When VIX falls, NAV rises, the target short grows, and the fund sells into a falling market.

That is the buy-high, sell-low rule again, and the arithmetic behind it is unavoidable: any constant-leverage product must rebalance in the direction of the move, with a leveraged long doing it in the same direction for the mirror reason.

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*Why it trades like a short-gamma hedger.* The fund holds no option, but its rule makes it trade the way a short-gamma dealer's hedge trades: it buys after the index has risen and sells after it has fallen. Rebalancing in the direction of the move is exactly the cash-flow signature of question 1's short-call dealer, and the P&L of the rebalancing behaves accordingly: the product bleeds in choppy markets (each round trip a realized loss) and is destroyed by a large move. The pattern is produced by the rebalancing rule itself, which is why the fund trades like a short-gamma book without being short an option.

*What distinguishes it from a dealer's short-gamma book.* Four things distinguish it, and the first matters most:

1. **No premium was collected.** A dealer short gamma was *paid* for it and holds a positive theta: by (18.8), the rebalancing losses are financed by time decay, and the trade is profitable if realized volatility comes in below implied. The exchange-traded product (ETP) has no such offset. Its rebalancing losses are pure cost, compensated only by the roll-down its short position earns as VIX futures, normally priced above spot VIX, drift down toward it, which is a different bet entirely. Where the dealer's hedging flow serves a *position in volatility with a carry*, the ETP's is an *artifact of its own prospectus*.

2. **The trading is mandatory, timed, and public.** A dealer chooses when and whether to hedge; the fund must rebalance to the close, and its rule is disclosed. Anyone can compute the required trade in advance and trade ahead of it, so the flow moves the very price it is computed from. On February 5, 2018, the inverse VIX products needed to buy so many futures into the close that they moved the index they were tracking; VIX futures roughly doubled, and one product, XIV, lost more than 90% of its value overnight and was terminated. That is the feedback loop of question 11 with a published timetable.

3. **The exposure can be destroyed but not closed out.** The dealer's short gamma runs off at expiration. The ETP holds a rolling position with no expiration, so its obligation to rebalance never matures; instead, the prospectus provides that a large enough one-day move triggers termination: a discrete, total loss for holders rather than a hedge that has gone wrong.

4. **Whose risk it is.** A dealer's hedge protects the dealer's capital. The ETP's rebalancing losses fall on retail holders who bought a ticker and are unlikely to have read the compounding disclosure, and the product is often held for weeks despite promising only a *daily* return.

*The general lesson.* Hedging flow does not require an option. Any rule that mechanically scales exposure to a price trades as a delta hedge does, and the rule sets the direction. Leveraged and inverse ETPs, portfolio insurance (question 10), stop-loss orders, margin-driven liquidation, and variable-annuity hedging programs all trade with the move, the way a short-gamma dealer's hedge does, whether the rule is manufacturing protection, as portfolio insurance is, or simply holding a leverage ratio constant. The question a risk manager should ask of any strategy is not "does it hold options?" but "in which direction is it forced to trade when the market moves, and how large is the forced trade?"

*What the question tests.* That the direction of forced trading is a property of a *rebalancing rule*, not of an instrument, and that the missing premium is what separates a hedged dealer from a product that is short volatility for free.

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**18.13** *For a European put with the same inputs as the dealer's call ($S_0 = 63$, $K = 64$, $T = 18/52$, $r = 4\%$, $\sigma = 22\%$), find its value by put–call parity, then its delta and the elasticity $\Omega$ of Example 18.1. Interpret the sign of each, and compare the put's leverage with the call's.*

*The put's value.* Put–call parity gives

$$p = c - S_0 + Ke^{-rT} = 3.19438 - 63 + 64e^{-0.04\times 18/52} = 3.19438 - 63 + 63.11995 = 3.31433$$

*Delta.* By (18.3), $\Delta_{\text{put}} = N(d_1) - 1 = 0.5199 - 1 = -0.4801$. The sign says the put loses value as the stock rises: a \$1 rise lowers it by about \$0.48. Repricing at \$64 gives 2.8585, a fall of 0.4558, and the gap of 0.0242 is the same as the call's, because a put and a call on the same terms have the same gamma. Holding the put is in effect a short position in 0.48 shares, and a dealer who writes it hedges by selling 0.48 shares short per option.

*Elasticity.*

$$\Omega_{\text{put}} = \frac{S\,\Delta_{\text{put}}}{p} = \frac{63 \times (-0.4801)}{3.3143} = -9.13$$

The elasticity is negative because the put's value and the stock move in opposite directions. A 1% rise in the stock lowers the put by about 9.1% (8.83% exactly at \$63.63), and a 1% fall raises it by about 9.1% (9.42% exactly at \$62.37). The put's volatility is about $|\Omega|\sigma = 9.13 \times 22\% \approx 201\%$.

*Against the call.* In magnitude the put's leverage, 9.13, is a little below the call's 10.25, for two reasons. The put is worth more (3.3143 against 3.1944): by parity $c - p = S_0 - Ke^{-rT} = -0.12$, because the strike's present value, 63.12, exceeds the stock price. And its delta is smaller in magnitude (0.4801 against 0.5199), because $N(d_1)$ is just above one half. A larger denominator and a smaller numerator both lower $|\Omega|$.

The ranking is not a property of puts. The pattern of Example 18.1 holds for puts with the sign reversed: a put struck at 54, out of the money, has $\Omega = -15.92$, and one struck at 74, in the money, has $\Omega = -5.13$. Out-of-the-money puts carry the most leverage on a fall for the same reason out-of-the-money calls carry it on a rise.

*What the question tests.* That the put's delta and elasticity follow from the call's through parity and $\Delta_{\text{put}} = N(d_1) - 1$, that a negative elasticity is leverage on a fall, and that the size of the leverage is set by moneyness, not by whether the option is a call or a put.

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# Chapter 19
# Volatility Smiles and Surfaces

**19.1** *Explain why equity options show a skew while currency options show a roughly symmetric smile, giving the structural reason in each case.*

The two shapes reflect two different asymmetries, and in each case the asymmetry is a fact about the *underlying*, not about the options.

*Equities: a skew, because the underlying's risk is one-sided.* Three structural reasons point down:

- *Leverage.* A firm's equity is a residual claim on assets financed partly by debt. When the share price falls, the debt-to-equity ratio rises, so the *same* asset volatility translates into a higher *equity* volatility. Volatility and price are therefore negatively correlated by the structure of the balance sheet, before anyone trades anything.
- *Crash risk.* Equity markets fall much faster than they rise. A 20% decline in a day has happened; a 20% rise in a day has not. The distribution has a fat left tail and a thinner right tail, which by Section 19.2 is what a downward-sloping skew encodes in the risk-neutral density.
- *One-directional demand.* Institutions with long equity portfolios buy puts and sell calls: protective puts, covered calls, collars. The flow is systematically long low-strike volatility and short high-strike volatility, and dealers charge for absorbing it.

*Currencies: a smile, because there is no "down."* A currency pair is a *ratio*. A fall in EUR/USD is a rise in USD/EUR, and the two are the same market viewed from two desks. There is no leverage effect, because neither currency is a residual claim, and no asymmetry of catastrophe: a devaluation is a crash for one holder and a windfall for the other. Both tails are therefore fat relative to the lognormal (jumps happen in both directions, from central-bank action, pegs breaking, and interventions), but neither tail is systematically fatter, so the curve is a symmetric **smile** rather than a skew.

The formal version of the argument is worth stating. Under the Black–Scholes–Merton model, if $S$ is lognormal then $1/S$ is lognormal too, and the volatilities are equal. So any *consistent* implied surface must satisfy $\sigma_{\text{EUR/USD}}(K) = \sigma_{\text{USD/EUR}}(1/K)$: the smile seen from one side is the mirror of the smile seen from the other. The inversion identity is exact; what it implies is that a skew in one quotation is the opposite skew in the other, so nothing structural favors either sign.

One qualification keeps the answer honest: real currency smiles are not perfectly symmetric. A pair with a large interest-rate differential, or an emerging-market currency with a peg that can only break one way, does show a skew, and it points toward the direction in which the peg would break. What the argument establishes is that currencies have no *structural* reason for a skew, not that they never have one. The risk reversal, the quoted difference between the

25-delta call and put volatilities, is precisely the instrument that prices this residual asymmetry, and it is quoted separately because that asymmetry is usually small and occasionally large.

*What the question tests.* That the smile's shape is inherited from the underlying's distribution, and that the ratio structure of a currency pair is what rules out an equity skew.

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**19.2** *Using the three-month and six-month rows of Table 19.3, compute the slope between strikes 90 and 110 per 10% of spot and per standard deviation of standardized moneyness (use each row's at-the-money volatility), and explain why the two disagree about whether the skew flattens.*

*The slopes.* The 90 and 110 strikes are 20% of spot apart, so the slope per 10% of spot is half the fall in volatility. In standard deviations the same gap is $[\ln 1.1 - \ln 0.9]/(\sigma_{\text{ATM}}\sqrt{T}) = 0.2007/(\sigma_{\text{ATM}}\sqrt{T})$:

| Maturity | $\sigma(90)$ | $\sigma(110)$ | Per 10% of spot | $\sigma_{\text{ATM}}$ | Gap in s.d. | Per s.d. |
|---|---|---|---|---|---|---|
| 3 months | 21.9% | 16.0% | 2.95 | 18.5% | 2.17 | 2.72 |
| 6 months | 21.7% | 17.2% | 2.25 | 19.3% | 1.47 | 3.06 |

*Why they disagree.* Per 10% of spot the skew flattens by almost a quarter from three months to six. Per standard deviation it *steepens* by 12%. The same 20% of spot is 2.17 standard deviations at three months and only 1.47 at six, because $\sigma\sqrt{T}$ is larger; a given skew in the standardized coordinate $z = \ln(K/F)/(\sigma\sqrt{T})$ is therefore spread over more dollars of strike at the longer maturity and looks flatter there. Nothing about crash aversion has changed, only the yardstick. The text's one-month and two-year rows show the same thing at greater range: 3.75 falling to 1.15 per 10% of spot, 1.89 rising to 3.37 per standard deviation. Real surfaces flatten more slowly than $1/\sqrt{T}$ (here $2.95/2.25 = 1.31$ against $\sqrt{2} = 1.41$), so in $z$ their skew slowly steepens.

*Two practical consequences follow.* Skew risk sits in short maturities even though a desk's *vega* sits in long ones (question 10). And a smile fitted in strike space, one maturity at a time, has nothing tying the maturities together, which is why the standard parameterizations are written in log-moneyness with maturity entering through total variance (SVI and SSVI, Section 19.5).

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**19.3** *State the Breeden–Litzenberger result and explain, without algebra, why the second derivative of the call price with respect to strike is a probability density.*

*The result.* Breeden–Litzenberger (1978): the risk-neutral density of the terminal asset price is the second derivative of the call price with respect to strike, compounded forward,

$$g(K) = e^{rT}\,\frac{\partial^2 C}{\partial K^2}$$

so a continuum of European call prices at one maturity determines the entire risk-neutral distribution at that maturity. No model is required, only prices.

*Why, without algebra.* Build the answer out of the two derivatives in turn.

*The first derivative is a digital.* A call spread (long $K$, short $K + h$) pays \$0 below $K$, rises linearly to \$$h$ at $K + h$, and pays \$$h$ above. Divide by $h$ and shrink $h$: the payoff becomes an indicator, paying \$1 if $S_T > K$ and nothing otherwise. Its price is therefore the discounted probability of finishing above $K$:

$$-\frac{\partial C}{\partial K} = e^{-rT}\Pr(S_T > K)$$

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The minus sign is just the fact that calls get cheaper as the strike rises. So the *slope* of the call curve in strike is a cumulative probability.

*The second derivative is a butterfly.* Differentiate the cumulative distribution and you get the density, which in instrument terms means differencing the call spread again. Long $K - h$, short two at $K$, long $K + h$ is a **butterfly**: it pays nothing except in a narrow band around $K$, where it pays a small triangular amount peaking at $\$h$. Divide by $h^2$ and shrink $h$, and the payoff becomes a spike of unit mass at $K$. Its price is the discounted probability of landing *at* $K$, which is the density, discounted.

So the answer is really one sentence: a butterfly is a bet that the asset finishes in a narrow band, and a bet on a narrow band, priced per unit of width and compounded forward, is a probability density. The second derivative is not an analogy for a density; it is the price of the instrument that pays off exactly when the density says it should.

*Two properties fall out immediately.* A butterfly can never have a negative payoff, so its price cannot be negative, which is the convexity condition of Section 11.3, and now reads as "the density must be nonnegative." And a call spread's price lies in $[0, he^{-rT}]$, which, compounded forward and divided by $h$, reads as "a probability must lie in $[0, 1]$." The static arbitrage restrictions on an option chain *are* the axioms of probability, expressed in the prices of traded instruments. Example 19.2 carries this out on six real quotes and Example 19.3 shows a chain that violates them.

*The one caveat.* The density recovered is *risk-neutral*, not physical. It prices states, and states in which investors are desperate for money are priced above their objective probability. Example 19.1's fat left tail is therefore partly a statement about the probability of a crash and partly a statement about how much a dollar delivered in a crash is worth, and no amount of option data separates them.

*What the question tests.* That differentiating in strike is a portfolio operation, and that the resulting probabilities are prices.

---

**19.4** *Using the table in Example 19.1, state by what factor the market's probability of a 30% decline exceeds the lognormal's, and explain what feature of the smile produces it.*

*The factor.* From Example 19.1's table,

$$\frac{\Pr(S_T < 70) \text{ implied by the smile}}{\Pr(S_T < 70) \text{ lognormal}} = \frac{6.9\%}{4.6\%} = \mathbf{1.5}$$

The market assigns 50% more probability to a 30% decline than a constant 20% volatility would, and the factor grows further out: at a 40% decline it is 3.0% against 0.7%, about four times. In the other direction the comparison reverses: $\Pr(S_T > 130)$ is 6.0% implied against 7.9% lognormal, so the right tail is *thinner* than lognormal by about a quarter.

*The feature of the smile that produces it.* The raised left wing: implied volatility at low strikes well above the at-the-money level. On the example's smile it runs from 20.0% at the money to 27.3% at a strike of 70. By question 3's argument, the probability of finishing below a strike is the slope of the put price there, compounded forward, and along a smile that slope has two parts, $\Pr(S_T < K) = N(-d_2) + e^{rT}\mathcal{V}\, d\sigma/dK$. At 70 the level alone, $N(-d_2)$ at 27.3%, gives 12.1%; the wing's slope is negative, so the second term takes back 5.2 points and leaves the table's 6.9%. The high level fattens the tail and the steepness only moderates it. Expensive out-of-the-money puts and a fat left tail are the same fact stated twice.

The right tail follows from the same curve read the other way. Implied volatility falls below 20% at high strikes (18.7% at 110, 17.1% at 130), so out-of-the-money calls are *cheaper* than a constant-volatility model would make them. At 130 the level alone gives $\Pr(S_T > 130) = 5.3\%$,

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and the smile's slope adds back 0.8 points, giving the table's 6.0%, still below the lognormal's 7.9%.

This is the chapter's central point, stated without mystery. A skew is not a pricing anomaly and not evidence that the Black–Scholes–Merton formula is being "used wrong." It is what a left-skewed, leptokurtic distribution looks like after being pushed through a formula that insists on reporting one number per strike. The formula is a quoting convention, and the convention distorts a single distribution into a curve of volatilities.

*The caveat the example insists on.* A tail probability at a quoted strike depends only on the smile's level and slope at that strike, so the factors of 1.5 at a 30% decline and four at a 40% decline are read off the smile itself. What extrapolation decides is how the mass below the lowest quoted strike is spread, and anything integrated over it: in this calculation the quadratic wing turns the density negative below a strike of about 6, so that integrated upward from there it totals 1.06 rather than 1.00, a 6% error manufactured entirely outside the observable range. Any statement about the shape of the tail beyond the last quoted strike is a statement about the extrapolation scheme, not about the market, and should be labeled as such. Chapter 38's tail-risk measures inherit this limitation directly.

*What the question tests.* Translating a wing's level and slope into tail probability, and knowing where the translation stops being supported by data.

---

**19.5** *Using Example 19.3, state the largest rise in implied volatility the 115/120 spread can carry, describe the arbitrage that appears beyond it, and explain why the permitted slope shrinks as strikes move further out of the money.*

*The largest rise the spread can carry.* With $\sigma(115) = 25.0\%$, the chain tolerates $\sigma(120)$ up to **29.27%**, a rise of **4.27 volatility points** across 5 of strike. At the chain's own volatilities the local ceiling is 4.51 points per 5 of strike at 115 ($\sigma = 25.0\%$) and 4.05 at 120 ($\sigma = 29.27\%$); the discrete bound of 4.27 falls between them, because the ceiling tightens as the strike rises across the interval. (Example 19.3's table, built on a different smile, gives 4.45 at 115.)

*The arbitrage beyond it.* The 115/120 call spread's price turns **negative**. Quoting 31.0% at 120 against 25.0% at 115 gives calls of 2.6563 and 3.0442, so

$$C(115) - C(120) = 2.6563 - 3.0442 = -\$0.3878$$

You are *paid* \$0.39 to enter a position whose payoff, $\min(\max(S_T - 115, 0), 5)$, is never negative and is sometimes \$5. Buy the 115 call, sell the 120 call, collect \$0.39, and wait: there is no state of the world in which you lose. It is an arbitrage in the strict sense: positive cash today, nonnegative cash later.

The equivalent statement is that the implied digital probability $\Pr(S_T > 117.5) \approx -0.0787$. A negative probability and a negative-cost nonnegative payoff are the same violation in two vocabularies, which is question 3's identification working in reverse.

*Why the permitted slope shrinks further out of the money.* Differentiating the Black–Scholes–Merton price along the smile,

$$\frac{dc}{dK} = -e^{-rT}N(d_2) + \mathcal{V}\frac{d\sigma}{dK}$$

so the call spread turns negative once

$$\frac{d\sigma}{dK} > \frac{e^{-rT}N(d_2)}{\mathcal{V}}$$

Both terms fall as the strike rises, but $N(d_2)$ falls faster than vega does, so the ratio, and with it the permitted slope, tightens:

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| Strike | $\sigma$ | Vega | Maximum steepening per 5 of strike |
|---|---|---|---|
| 110 | 22.5% | 25.77 | 5.34 volatility points |
| 115 | 24.0% | 22.83 | 4.45 volatility points |
| 120 | 26.0% | 20.19 | 3.86 volatility points |
| 125 | 29.0% | 18.60 | 3.49 volatility points |

The numerator and denominator can be watched separately. From 110 to 125, $N(d_2)$ runs 0.2795, 0.2061, 0.1582, 0.1319 (down to 47% of where it started), while vega runs 25.77, 22.83, 20.19, 18.60, retaining 72%. The probability of finishing in the money is dying much faster than the option's sensitivity to volatility, and their ratio is the ceiling.

The intuition is that far out of the money there is very little call value left to redistribute. $e^{-rT}N(d_2)$ is the rate at which price falls with strike for a *fixed* volatility, and vega is the leverage a volatility change has on that price. Deep in the wing, a small change in volatility moves the price by relatively more than the strike does, so a smaller volatility slope suffices to overwhelm the strike effect and reverse the sign of the spread.

*Why this matters more than it looks.* The violating quote was 31.0% against a bound of 29.27%, 1.7 volatility points past the boundary. Plotted on a smile that runs from 22% to 31%, that is invisible. The check is arithmetic, not visual, and it must be run on the *quotes* before any curve is fitted through them: a smooth interpolating curve will happily pass through arbitrageable points and hand the arbitrage on to every exotic priced off the surface. That is the practical content of Section 19.5: fitting a smooth curve is not fitting an arbitrage-free surface.

*What the question tests.* That no-arbitrage bounds the smile's *slope*, that the bound tightens in the wings, and that a violation can be far too small to see.

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**19.6** *An at-the-money option has vega 40 and Black–Scholes delta 0.54. Under sticky delta with $\partial\sigma/\partial S = 0.0015$, compute the corrected delta and the error in shares on a 50,000-option position, and state which convention gives the larger hedge for a call under an equity skew.*

*The corrected delta.* Under sticky delta the implied volatility attached to a fixed strike moves as spot moves, so the total derivative picks up a second term:

$$\Delta_{\text{total}} = \frac{\partial c}{\partial S} + \frac{\partial c}{\partial \sigma}\frac{\partial \sigma}{\partial S} = \Delta_{\text{BS}} + \mathcal{V}\frac{\partial \sigma}{\partial S}$$

$$= 0.54 + 40 \times 0.0015 = 0.54 + 0.06 = \mathbf{0.60}$$

*The error in shares.* On 50,000 options the two conventions differ by

$$(0.60 - 0.54) \times 50{,}000 = \mathbf{3{,}000} \text{ shares}$$

an 11% error in the hedge, enough to matter, and entirely invisible to anyone reading delta off a pricing screen without asking which convention produced it.

*Which convention gives the larger hedge for a call under an equity skew: sticky delta.* The sign is worth deriving rather than remembering. An equity skew slopes *down* in strike. Sticky delta says the smile is a function of moneyness $K/S$, so it travels with spot: if $S$ rises, a fixed strike $K$ becomes relatively *lower* in moneyness terms, and a lower point on a downward-sloping skew carries a *higher* volatility. Hence $\partial\sigma/\partial S > 0$ for a fixed strike (the +0.0015 given), and since vega is positive, the correction is positive. Sticky delta gives the larger delta.

Sticky strike is the other convention, and it gives exactly $\Delta_{\text{BS}} = 0.54$: each strike keeps its own volatility as spot moves, so $\partial\sigma/\partial S = 0$ and there is no correction at all. The Black–Scholes delta is therefore not convention-free: it is the sticky-strike delta, and it is only one of the choices.

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*The two are testable, and for an index neither is right.* They are empirical claims about how the surface moves, so a desk can check them against its own data. For equity indices the measured answer lies outside both: the implied volatility at a fixed strike tends to fall as the index rises, so the minimum-variance delta lies below the Black–Scholes delta for calls and puts alike (Hull and White 2017, as Section 18.10 reports). Sticky delta, with its positive correction, describes currencies better than equities. Most desks estimate $\partial\sigma/\partial S$ empirically rather than adopting either convention wholesale, and carry the uncertainty as a risk in its own right.

*And note what the correction is not.* It is not a smile "adjustment" to a Black–Scholes number. The Black–Scholes delta answers "how does the price move if spot moves and volatility does not," and in a market with a skew that is an assumption about how the surface moves, not a fact. The 3,000 shares are not an error term; they are what the sticky-delta assumption adds to the exposure.

*What the question tests.* That delta depends on an assumption about surface dynamics, that the assumption has a sign under an equity skew, and that the difference is large enough to show up in the hedge.

---

**19.7** *Explain why the volatility risk premium implies that a straddle buyer is not simply betting the market will move.*

*The premium.* Implied volatility exceeds subsequently realized volatility on average, by two to four points on equity indices, and by more at low strikes. That gap is the **volatility risk premium**, and it is a systematic feature rather than a persistent mispricing.

*What it does to the straddle buyer's bet.* A straddle costs the implied volatility and pays off the realized volatility, so its expected profit is

$$\mathbb{E}[\text{profit}] \;\approx\; k\Big(\underbrace{\sigma_{\text{realized}}}_{\text{what you collect}} - \underbrace{\sigma_{\text{implied}}}_{\text{what you paid}}\Big)$$

The buyer wins only if the market moves by *more than the option already priced in*. So the bet is not "will the market move": the market will certainly move, and the price of the straddle already reflects how much. The bet is that realized volatility will exceed implied volatility, and on average it does not.

Stated as a threshold: a straddle bought at 20% implied on an index that goes on to realize 18% loses money if delta-hedged, and on average if not, even though the market moved a great deal. Confusing "the market will be volatile" with "the market will be more volatile than priced" is the single most common error in the use of these instruments, and the volatility risk premium is what makes it expensive.

*Why the premium exists, since a systematic loss demands an explanation.* Selling a straddle is selling insurance. The payoff is negatively skewed (small gains most of the time, occasional very large losses), and, crucially, the large losses arrive precisely in market crashes, when the seller's other assets are also falling and a dollar is worth most. A short volatility position therefore pays worst when money is most valuable, and standard asset pricing says such a position must earn a premium. Buying volatility is buying insurance, and insurance has a negative expected return by construction. No one expects to profit on average from fire insurance.

So the buyer is not mispricing anything; they are paying for a hedge. The question is whether they want the hedge, and the answer may perfectly well be yes: a fund whose portfolio is destroyed by a crash may rationally pay a premium for a payoff that arrives then. What is not rational is buying the straddle as a directional bet on movement and expecting a positive expected return.

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*Two qualifications follow.* First, the premium is not constant: it is large after crashes, when implied volatility overshoots, and can go *negative* before an anticipated event whose realized move exceeds what was priced. Second, a naked straddle is not a clean volatility bet at all, since it acquires a directional exposure the moment spot moves; harvesting the premium properly requires delta hedging, which is Chapter 18's machinery, with the transaction costs that chapter's question 9 measures.

*What the question tests.* That the strike price of the bet is implied volatility, not zero, and that the systematic gap between implied and realized has an economic reason.

---

**19.8** *In Example 19.2 the bucket probabilities and the two tail probabilities sum to exactly 100%. Explain why this had to happen and what it would mean if it did not; then replace the \$50 call's midquote of 1.475 with 1.050, and say which butterfly turns negative and what arbitrage it offers.*

*It had to happen because it is an algebraic identity, not a test.* Write the six call prices as $C_0, \ldots, C_5$ at strikes \$2.50 apart. The four butterfly prices are $C_{i-1} - 2C_i + C_{i+1}$ for $i = 1, \ldots, 4$, and summing them the middle terms cancel:

$$\sum_{i=1}^{4} (C_{i-1} - 2C_i + C_{i+1}) = (C_0 - C_1) - (C_4 - C_5)$$

Every interior price appears with coefficient $1 - 2 + 1 = 0$. What survives is exactly the two outer call spreads. Each bucket probability is its butterfly price times $e^{rT}/h$, and each outer call spread, times the same factor, is the chapter's approximation to the digital at the midpoint of its strikes. So

$$\underbrace{73.3\%}_{\text{four buckets}} = \underbrace{89.6\%}_{\Pr(S_T>43.75)} - \underbrace{16.3\%}_{\Pr(S_T>53.75)}$$

and adding $\Pr(S_T < 43.75) = 10.4\%$ and $\Pr(S_T > 53.75) = 16.3\%$ to the buckets necessarily returns 100%. *Any* six numbers would do this. The three pieces look independent (different formulas, different quotes), but the middle quotes enter the total with coefficient zero, so they cannot affect it.

*What it would mean if it did not.* It would mean an arithmetic error and nothing else: not an arbitrage, a data problem or a market signal. The identity holds for arbitrageable chains too. To see this concretely, take the same six quotes and lower the \$50 call from 1.475 to 1.050. The four butterfly prices become 0.470, 0.135, 1.280 and −0.065. The last, centered at \$52.50 (long the 50 and 55 calls, short two 52.50 calls), is priced below zero although the position can never pay less than zero, so buying it collects \$0.065 per share and risks nothing; the density it implies at \$52.50 is −0.0105. Yet the four buckets still total 73.3% and the pieces still total exactly 100%, since the \$50 call cancels from the sum.

What carries information, then, is the parts, not the sum. Arbitrage freedom shows up as:

- every butterfly price **positive**: the density is nonnegative, which is Section 11.3's convexity condition;
- every digital in $[\mathbf{0}, \mathbf{1}]$: the cumulative distribution is a probability, which is monotonicity (the floor Example 19.3 breaches) plus the spread bound of Section 11.3 (the ceiling).

Example 19.2's chain passes both, which is the substantive result. The 100% is a consequence of the arithmetic, and reporting it as evidence would be reading a tautology as a test.

The distinction generalizes and is worth carrying beyond this chapter. Many "internal consistency checks" in derivatives systems are identities in disguise: they verify that the code

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implements the formula, which is worth knowing, but they say nothing about the data. A check that cannot fail on bad input is a check on the implementation. A check that *can* fail (positivity here, the parity relations of Chapter 11, the total-variance condition of Section 19.5) is a check on the market. Know which kind you have before you rely on it.

*What the question tests.* That telescoping forces the total, and the difference between a tautology and a test.

---

**19.9** *An index of 100 equally weighted stocks has component implied volatilities of 25% and an index implied volatility of 16%. Compute the implied correlation, and explain why index skew is steeper than single-stock skew.*

*The implied correlation.* For $n$ equally weighted components each of volatility $\sigma$ and pairwise correlation $\rho$,

$$\sigma_I^2 = \frac{1}{n}\sigma^2 + \left(1 - \frac{1}{n}\right)\rho\,\sigma^2$$

Substituting $\sigma_I = 16\%$, $\sigma = 25\%$, $n = 100$:

$$0.0256 = \frac{1}{100} \times 0.0625 + \frac{99}{100} \times 0.0625\,\rho$$
$$0.0256 = 0.000625 + 0.061875\,\rho$$
$$\rho = \frac{0.024975}{0.061875} = \mathbf{40.4}\%$$

With 100 stocks the idiosyncratic term is small, so the quick approximation $\rho \approx (\sigma_I/\sigma)^2 = (16/25)^2 = 41.0\%$ is within a percentage point and is the figure a desk quotes.

Note what this number is: a *market price*, not a statistic. It is the correlation the index and single-stock option markets jointly imply, and selling it against subsequently realized correlation is the **dispersion trade**: short index volatility, long the components. The trade is profitable when implied correlation exceeds realized correlation, which it typically does, for the same insurance reason as question 7.

*Why index skew is steeper than single-stock skew.* The index smile prices *two* things that rise together in a sell-off; the single-stock smile prices one.

*The idiosyncratic component diversifies away; the common one does not.* Each stock's volatility has a market part and a firm-specific part. In the index the firm-specific parts cancel (which is why 16% is well below 25%), leaving the index driven almost entirely by the common factor. So the index is a purer bet on the systematic risk that crash fear is about.

*And correlation is itself higher in the left tail.* In a market-wide decline, everything falls together: dispersion collapses and realized correlation rises toward one. So the index's downside volatility is elevated by *two* compounding effects, higher component volatility *and* higher correlation, while a single stock's downside volatility is elevated by the first alone. The index's left tail is therefore fatter relative to its own body than any component's is relative to its body, and a fatter left tail is a steeper skew.

The mirror image holds on the upside and reinforces the asymmetry: markets rise idiosyncratically, sector by sector, with correlations low. Diversification works on the way up and fails on the way down, and the index skew is the price of that failure.

*Restating it in the language of the trade:* implied correlation is not a constant across strikes. It is higher for low-strike index options than for at-the-money ones, and that *correlation skew* is precisely the excess of index skew over the skew the components would generate on their own. A dispersion book is therefore short a strike-dependent quantity, and is most exposed exactly when correlations spike.

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*What the question tests.* Inverting the index variance formula, and recognizing that the extra steepness of the index smile is a correlation phenomenon rather than a volatility one.

---

**19.10** *Explain why the volatility term structure inverts in a crisis, and why this makes total vega a poor summary of an option book's exposure.*

*Why the term structure inverts.* In calm markets the at-the-money term structure slopes *up*: near-dated volatility is low, and uncertainty accumulates with horizon. In a crisis it inverts, with short maturities trading far above long ones, for two reasons that reinforce each other.

*Volatility is mean-reverting, and the market knows it.* Implied volatility at maturity $T$ is roughly the average volatility expected over $[0, T]$. A shock raises volatility *now*, and a one-month option prices a month that is nearly all crisis. A two-year option prices a period in which the crisis is expected to have passed, so the shock is diluted across mostly normal months, and the long end barely moves. The long end is anchored by the long-run mean; the short end is not anchored at all.

*Hedging demand concentrates at the front.* In a crisis, people want protection *now*, and near-dated options are the cheapest way in premium terms to buy it. The flow lands entirely on the short end, and dealers who are already short gamma raise the price of exactly the maturities being demanded.

The numbers are dramatic. In March 2020 VIX, a one-month implied volatility, exceeded 80% while two-year implied volatility stayed far lower. In calm periods the same two points might read 13% and 20%: not merely a flattening but a reversal of sign.

*Why this makes total vega a poor summary.* Vega scales with $\sqrt{T}$,

$$\mathcal{V} = S\sqrt{T}\,N'(d_1)$$

so a two-year option carries roughly $\sqrt{24} \approx 4.9$ times the vega of a one-month option of the same size. Aggregate vega is therefore dominated by the maturities that move least. Adding a one-month vega to a two-year vega adds two numbers denominated in different things, since a "one point" move is a routine event at one month and an extraordinary one at two years.

Two failures follow directly:

*A vega-neutral book is not volatility-neutral.* A desk long two-year options worth \$1 million per volatility point and short one-month options of the same vega reports zero total vega. In a crisis the short end rises 40 points and the long end 5, so the book loses $40 \times \$1\text{m}$ and gains $5 \times \$1\text{m}$, a catastrophic loss on a position the risk report called flat. This is not hypothetical; it is the standard way a calendar-spread book is destroyed.

*The sign of the exposure can reverse with the regime.* The same book profits in a slow grind higher in volatility, where the long end leads. A summary that reports zero for a book whose profit can take either sign, depending on the shape of the move, is not a risk measure.

*What desks do instead.* Bucket vega by maturity (one month, three months, six months, one year, two years and beyond), and stress each bucket by an amount reflecting how much that bucket actually moves. The standard weighting scales the shock by $1/\sqrt{T}$ relative to a reference maturity, since that is the empirical pattern of term-structure moves and it matches the $\sqrt{T}$ in vega, giving a comparable quantity across buckets. Better still, apply historical crisis scenarios to the whole surface at once and revalue, which captures the inversion and the skew steepening together rather than assuming a shape.

The general lesson is Chapter 18's, one dimension further on: a Greek is a first-order sensitivity to a *parallel* move in one variable, and the volatility surface never moves in parallel. Summing sensitivities across a curve whose points move by different amounts answers a question no market ever asks.

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*What the question tests.* Mean reversion as the reason the long end is anchored, and that $\sqrt{T}$ scaling makes aggregate vega weight the surface in inverse proportion to how much it moves.

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# Chapter 20
# Numerical Procedures

**20.1** *A binomial tree gives 3.5724 at 100 steps and 3.5777 at 200. Estimate the exact value by Richardson extrapolation, state what you must check before trusting it, and explain why the binomial error alternates in sign with the parity of n when $S_0 = K$.*

*The extrapolation.* If the error behaves like $C/n$, then $V(2n) - V(n) = -C/(2n)$, and adding that difference once more removes the leading term:

$$V_{\text{exact}} \approx 2V(2n) - V(n) = 2 \times 3.5777 - 3.5724 = \mathbf{3.5830}$$

Table 20.2's underlying values carry more digits (3.577688 and 3.572449) and give 3.582927 against a Black–Scholes–Merton value of 3.582934. The raw 200-step value erred by −0.0052; the extrapolated one errs by −0.000007, nearly three orders of magnitude better for no extra tree.

*What must be checked first: that n and 2n have the same parity.* Richardson assumes the error is a smooth function of $1/n$. The binomial error is not smooth (it oscillates), so the assumption holds only *along one branch* of the oscillation. Here 100 and 200 are both even, so both values sit on the same branch and the method works. Extrapolating from $n = 25$ to $n = 50$ instead would cross from the odd branch to the even one, and the doubling relation then subtracts errors of opposite sign rather than canceling errors of the same sign, so the answer is *worse than either input*: errors of +0.0407 and −0.0209 extrapolate to −0.0826, twice the larger of the two. A method that can degrade its own inputs is one to apply deliberately, not by default.

Two further checks belong with it. The doubling relation should be verified empirically: if $V(n)$, $V(2n)$, $V(4n)$ do not show the error halving, the $C/n$ premise is wrong and the extrapolation is unjustified. And for an *American* option the plain tree's error is not a smooth $O(1/n)$, because the exercise boundary is itself resolved only to the grid's accuracy, which is why the chapter extrapolates only after the control variate has smoothed the sequence.

*Why the error alternates in sign with the parity of n when $S_0 = K$.* In a Cox–Ross–Rubinstein tree $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$, so the terminal nodes sit at $S_0 u^{2j-n}$. The exponent $2j - n$ has the parity of $n$:

- $n$ **even.** The exponent can be zero, so a terminal node lands *exactly* on $S_0 = K$, precisely where the payoff has its kink.
- $n$ **odd.** The exponent is never zero, so no node lands on the strike; the two nearest straddle it symmetrically at $S_0 u^{\pm 1}$.

The tree replaces a continuous density with mass at nodes, and the error in pricing a kinked payoff depends on where the kink falls relative to the nodes. The two configurations are geometrically different, and they sit on opposite sides of the true value: the errors at $n = 100$ and $n = 101$ are −0.010485 and +0.010046, nearly equal in size and opposite in sign.

Section 20.3 exploits that symmetry through the choice of step count; the simplest use is to average $V(n)$ and $V(n+1)$, which cancels most of the error: at $n = 100$ the average errs by $-0.00022$, a factor of 48 better for twice the work. So the oscillation is a nuisance for Richardson and a gift for averaging: the same fact, used two ways.

*What the question tests.* That extrapolation is a claim about the *form* of the error, and that node placement relative to the strike is what sets that form.

---

**20.2** *A tree gives an American put value of 3.6914 and a European put value of 3.5724 at the same step count, and the Black–Scholes–Merton formula gives 3.5829 for the European. Compute the control-variate corrected American value, and explain why the correction helps enormously at odd step counts and only slightly at even ones.*

*The corrected value.* The control variate is the European option, whose exact value is known. Assume the tree makes the same error on both and subtract it:

$$V_{\text{Amer}}^{\text{corrected}} = V_{\text{Amer}}^{\text{tree}} + \left(V_{\text{Euro}}^{\text{BS}} - V_{\text{Euro}}^{\text{tree}}\right) = 3.6914 + (3.5829 - 3.5724) = \mathbf{3.7019}$$

*Whether that helped.* The step count here is $n = 100$, which is **even**: the figures 3.5724 and 3.6914 are the $n = 100$ values of Table 20.2 and Example 20.1. The true American value is 3.6970. So the raw tree erred by $-0.0057$ and the corrected value errs by $+0.0048$ (from the unrounded tree values): the error changes sign and shrinks by less than a fifth, which is to say almost not at all. The question's own numbers are the "only slightly" case.

*Why the parity matters so much.* The correction assumes the two errors are equal. They are, at odd step counts, and are not, at even ones:

| $n$ | European error | American error | Corrected error | Improvement |
|---|---|---|---|---|
| 99 | +0.010249 | +0.010352 | +0.000103 | 100× |
| 100 | −0.010485 | −0.005672 | +0.004813 | 1.2× |
| 199 | +0.005096 | +0.005163 | +0.000067 | 77× |
| 200 | −0.005246 | −0.002827 | +0.002419 | 1.2× |

At $n = 99$ the two errors agree to about 1%, so subtracting one from the other leaves almost nothing, and the corrected value is right to about one unit in the fourth decimal place. At $n = 100$ the American error is only *half* the European one, so subtracting the European error **overcorrects**, pushing the answer past the true value to the other side.

*Why the errors diverge at even n.* Question 1's node placement is the reason. At even $n$ a terminal node lands exactly on the strike, and that node-on-the-kink effect is the dominant source of the *European* error. The American value carries that terminal-node error too, but also a second one, from locating the *early-exercise boundary* that runs through the interior of the tree, and at even $n$ the two partly offset, which is why the American error is only half the European one. The European option has no exercise boundary, so its error cannot cancel the second component, and the difference of the two errors is not zero. At odd $n$ the strike falls between nodes, and the error from straddling it (as large as the even-$n$ error, but the same for both options) is what the two genuinely share, so it cancels.

The practical rule is therefore the opposite of the natural instinct: with a control variate, use an **odd** number of steps. Without one, use an even number, or average $n$ and $n+1$. And the general warning is worth extracting: a control variate is only as good as the correlation between the two errors, and here that correlation is a function of a grid choice no one would think to record. An error-correction technique that silently depends on the parity of a step count is exactly the kind of thing that works on the test cases and fails on live trades.

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*What the question tests.* The mechanics of the control variate, and that its benefit is governed by error *similarity* rather than by the size of either error.

---

**20.3** *State two things a trinomial tree buys over a binomial tree and one thing it does not, and name the finite difference scheme that is a trinomial tree in another notation.*

*A trinomial tree buys two things.*

*1. A free parameter, and with it the placement of nodes.* In (20.2) the space step $\Delta x$ and time step $\Delta t$ are chosen *independently*, whereas the binomial tree ties them together through $u = e^{\sigma\sqrt{\Delta t}}$. That freedom lets a node be placed exactly on a feature of the payoff, a barrier or a strike. It matters enormously: a barrier option priced on a tree whose nodes *straddle* the barrier converges badly, because the barrier is effectively relocated to the nearest node, and the option being priced is not the option that was traded. Chapter 34 returns to this.

*2. Room for a varying volatility.* A local-volatility model needs a different $\sigma$ at different nodes. In a binomial tree that changes $u$ and $d$ node by node and destroys recombination; the trinomial absorbs the variation in the three *probabilities* while leaving the geometry alone.

*One thing it does not buy: faster convergence.* Both schemes are $O(1/n)$, and numerically they are almost indistinguishable. On the European put of Table 20.2, the trinomial errors at $n = 50, 100, 200$ are $-0.0207$, $-0.0103$, $-0.0052$ against binomial errors of $-0.0209$, $-0.0105$, $-0.0052$, agreeing to within 0.0002. Anyone reaching for a trinomial tree expecting accuracy for free will not get it. The extra branch buys *flexibility*, not *order*.

*The finite difference scheme it is.* The **explicit** scheme ($\theta = 0$ in Section 20.12). At $\theta = 0$ each new grid value is an explicit weighted average of three neighbors, and on the log grid those weights are the trinomial probabilities of (20.2) to first order in $\Delta t$. This is not an analogy: in the limit it is the same arithmetic in two vocabularies.

The identification is useful at once, because each vocabulary knows something the other does not. Numerical analysis tells us the explicit scheme is only *conditionally* stable, requiring $\sigma^2\Delta t \lesssim \Delta x^2$ on the log grid (on the price grid of Section 20.12, $\Delta t \lesssim 1/(\sigma^2 M^2)$), so a trinomial tree whose $\Delta x$ is too small for its $\Delta t$ can produce probabilities outside $[0, 1]$ and explode, which question 10 pursues. Conversely, the tree vocabulary makes early exercise obvious: compare with the intrinsic value at each node. Neither insight is natural in the other language.

*What the question tests.* That the trinomial's advantage is geometric freedom rather than accuracy, and that trees and finite differences are the same method twice named.

---

**20.4** *Explain why a known cash dividend destroys the recombination of a binomial tree, and describe the standard remedy.*

*Why recombination fails.* A binomial tree recombines because up-then-down returns to the starting price: $S_0ud = S_0$, since $d = 1/u$. The moves are *multiplicative*, and multiplication commutes. A known **cash** dividend $D$ is *subtractive*, and subtraction does not commute with multiplication:

$$(S_0u - D)d = S_0 - Dd \qquad \text{but} \qquad (S_0d - D)u = S_0 - Du$$

and since $u \neq d$ these differ. The up-then-down node and the down-then-up node no longer coincide.

The cost compounds, and that is the whole point. With recombination, step $i$ has $i + 1$ nodes and the total work grows like $n^2$. Without it, every node standing at the ex-date starts a tree of its own: a single dividend halfway through a 100-step tree turns 101 terminal nodes into $51 \times 51 = 2{,}601$, and each further ex-date multiplies the count again. With quarterly dividends

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on a long-dated option it is not a question of a slower calculation but of an impossible one. And a *proportional* dividend causes no such problem, because it is multiplicative and commutes, which locates the difficulty precisely in the cash amount.

*The standard remedy.* Split the stock price into two parts:

$$S = \underbrace{S^*}_{\text{uncertain}} + \underbrace{\text{PV of dividends to be paid over the option's life}}_{\text{certain}}$$

Build the tree on $S^*$, whose starting value is $S_0$ minus the present value of the dividends. That component is purely multiplicative, so the tree recombines normally. Then add the present value of the remaining dividends back at each node when computing the actual stock price used for exercise decisions and payoffs.

The economics is that the announced dividend is not risky (it is a known cash payment), so it should not have volatility applied to it. Stripping it out and treating only the residual as lognormal is a more accurate description of the security as well as a computational convenience.

*This changes two things that are easy to miss.* The volatility must be recalibrated, because the $\sigma$ observed on $S$ is not the $\sigma$ of the smaller quantity $S^*$, and using the observed figure on the reduced price understates the absolute uncertainty. And the ex-dividend dates matter for American calls: early exercise is optimal only just before an ex-date (Section 11.10), and the tree must have a time step landing there.

*What the numbers look like when it is done wrong.* The chapter's example gives **4.6098** for the correct treatment, against **3.6970** if the dividend is ignored entirely and **4.6385** if it is subtracted undiscounted. Ignoring it is a 20% error: a different option. Subtracting it undiscounted is only 0.6% off, small enough to survive a casual comparison and wrong all the same, which makes it the more dangerous of the two mistakes.

*What the question tests.* That the obstacle is the failure of subtraction to commute with multiplication, and that the remedy is a decomposition into a risky and a riskless part.

---

**20.5** *A Monte Carlo estimate has a standard error of 0.04 at 10,000 paths. How many paths are needed for a standard error of 0.005?*

Monte Carlo's standard error falls with the square root of the number of paths, by (20.4):

$$\text{SE} = \frac{s}{\sqrt{M}}$$

so the required factor in $M$ is the *square* of the required factor in SE. To go from 0.04 to 0.005 is a factor of 8, so

$$M = 10{,}000 \times 8^2 = 10{,}000 \times 64 = \mathbf{640{,}000} \text{ paths}$$

Equivalently, $s = 0.04\sqrt{10{,}000} = 4.0$, and $M = (4.0/0.005)^2 = 640{,}000$.

The law that answer illustrates is the chapter's central complaint about Monte Carlo. Each additional decimal place of accuracy costs a **hundredfold** increase in paths. To reach an SE of 0.0005 from here would take 64 million paths; 0.00005 would take 6.4 billion. Brute force runs out long before the accuracy a trading desk wants.

Two consequences follow, and both shape how the method is used in practice.

*Variance reduction is not an optimization; it is the only thing that works.* Section 20.8's techniques attack $s$ rather than $M$, and they attack it far more effectively than more paths ever could. A control variate achieving a variance reduction of 583, as the geometric Asian does in question 7, is worth 583 times as many paths, turning 100,000 paths into the equivalent of 58 million. No amount of hardware competes with that.

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*The compensating virtue is dimension.* The $1/\sqrt{M}$ rate is slow, but it does not depend on the number of state variables. A grid with $m$ points along each of $d$ state variables has $m^d$ nodes, so a tree's work grows exponentially in dimension and is unusable beyond about three; Monte Carlo's convergence rate for a twenty-asset basket is the same $1/\sqrt{M}$ as for one asset. That is the whole reason to tolerate the slow rate, and it is why Monte Carlo is the only feasible method for the CVA calculations of Chapter 9 and the multi-asset structures of Chapter 34.

*A caution on reporting.* The standard error measures *simulation* noise only: how much the answer would move with a different set of random numbers. It says nothing about model error, discretization error from the time step, or the correctness of the parameters. An estimate quoted as $12.34 \pm 0.005$ is precise, and precision is not accuracy. Question 6's discretization bias, in particular, does not shrink with $M$ at all.

*What the question tests.* The square-root law, and knowing what the resulting error bar does and does not cover.

---

**20.6** *Explain why the exact lognormal simulation step is preferable to an Euler step, giving both reasons.*

*The two steps.* For geometric Brownian motion, the exact solution of the stochastic differential equation can be sampled directly:

$$S_{t+\Delta t} = S_t \exp\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon\right]$$

against the Euler discretization of the equation itself:

$$S_{t+\Delta t} = S_t\left(1 + r\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon\right)$$

*Reason 1: the exact step has no discretization error.* It is not an approximation to the process; it *is* the process, sampled at a point. Whatever $\Delta t$ is chosen, the distribution of $S_{t+\Delta t}$ given $S_t$ is exactly right, so a European option depending only on the terminal value can be priced in one step of length $T$. The Euler scheme has an error of order $\Delta t$ that survives no matter how many paths are run: it is a *bias*, not noise, and the confidence interval of question 5 does not see it. A million paths of a biased scheme converge tightly on the wrong number, which is the worse of the two failure modes because it looks like success.

*Reason 2: the exact step cannot produce a negative price.* The exponential is positive for every $\varepsilon$, so the simulated price stays positive by construction. The Euler step is a *linear* function of a normal variable and is therefore negative whenever

$$\varepsilon < -\frac{1 + r\Delta t}{\sigma\sqrt{\Delta t}}$$

which is a real possibility for large $\Delta t$ or large $\sigma$: at $\sigma = 40\%$ with $\Delta t = 1$ year and $r = 5\%$ it happens about once in 230 paths. A negative stock price is not merely inelegant. It is outside the state space, it makes a subsequent $\ln S$ undefined, and a practitioner who patches it by flooring at zero has introduced a second, undocumented bias on top of the first.

The chapter attaches a qualification. The exact step exists only because geometric Brownian motion happens to have a closed-form solution. Most processes worth simulating do not: a general local volatility $\sigma(S, t)$, the Heston variance process, or a jump-augmented model must be discretized, and the choice is then between Euler, a predictor–corrector where the drift depends on the state (Section 20.7), and schemes designed for the particular process; the Heston variance has its own, because a naive Euler step drives it negative. So the rule is: use the exact step whenever the process permits, and when it does not, treat the time step as a convergence parameter to be tested rather than a setting to be chosen once.

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*And note where the exact step stops helping.* A path-dependent payoff (an Asian average, a barrier monitored continuously) needs the path, not just the endpoint. The exact step still gives exactly right values *at the sampled dates*, but the payoff depends on what happened between them, and no step size removes that error. A barrier sampled monthly is a monthly barrier, whatever the sampling scheme, which is why Chapter 34 needs a continuity correction.

*What the question tests.* The distinction between simulation noise and discretization bias, and that positivity of the state variable is a property of the scheme rather than of the model.

---

**20.7** *A control variate has correlation 0.95 with the target. By what factor is the variance reduced? Compare with the 0.99914 achieved by the geometric Asian control variate and explain why the improvement is so much larger than the correlations differ.*

*The variance reduction.* With a control variate of correlation $\rho$ and an optimal coefficient, the residual variance is $(1-\rho^2)$ times the original, so the reduction factor is

$$\frac{1}{1-\rho^2}$$

At $\rho = 0.95$:

$$\frac{1}{1-0.9025} = \frac{1}{0.0975} = \mathbf{10.3}\times$$

worth ten times the paths, and a factor of $\sqrt{10.3} = 3.2$ on the standard error. At $\rho = 0.99914$ the formula gives $1/(1-0.99828) = 582\times$, and the chapter's measured figure for the geometric Asian control variate is **583**, equivalent to 58 million plain paths from 100,000. The two differ only by rounding: a reduction of 583 corresponds to $\rho = 0.999142$, and the question quotes $\rho$ to five decimals.

*Why the improvement is so much larger than the correlations differ.* Because the factor depends on $1-\rho^2$, the *residual*, and not on $\rho$ itself. The correlations differ by 5%; the residuals differ by a factor of 57:

| $\rho$ | $\rho^2$ | $1-\rho^2$ | **Reduction** |
|---|---|---|---|
| 0.90 | 0.8100 | 0.1900 | 5.3× |
| 0.95 | 0.9025 | 0.0975 | 10.3× |
| 0.99 | 0.9801 | 0.0199 | 50× |
| 0.999 | 0.998001 | 0.001999 | 500× |
| 0.99914 | 0.998281 | 0.001719 | 582× |

The function has a pole at $\rho = 1$, so all the interesting behavior is crowded into the last fraction of a percent. Moving from 0.95 to 0.99 gains a factor of 5; moving from 0.99 to 0.999 gains another 10. The right way to read a control variate's correlation is therefore to look at the digits after the nines, not at the number itself. The figures 0.95 and 0.99914 sound similar and are not remotely comparable.

*Why the geometric Asian is so extraordinarily good.* The two payoffs are not merely correlated; they are nearly the same random variable. Both are functions of the *same* set of sampled prices along the *same* path, differing only in whether the average is arithmetic or geometric. Since the geometric mean of a set of positive numbers is close to the arithmetic mean unless they are widely dispersed, and prices sampled at monthly intervals over a year are not widely dispersed, the difference between the two payoffs is tiny and almost deterministic. The control variate is therefore subtracting off nearly the entire payoff and simulating only the small residual, and it has an exact closed-form price, because a geometric average of lognormals is itself lognormal.

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That is the general design principle: choose a control variate that is the same instrument with one feature removed, not merely a related one. A European option is a fair control for an American; the geometric Asian is an outstanding one for the arithmetic Asian; and the difference between fair and outstanding is two orders of magnitude of computing time.

*One practical note.* The optimal coefficient $\beta = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$ is estimated from the same paths, which introduces a small bias. With the path counts in question this is negligible, but estimating $\beta$ on a separate pilot run removes it if a strictly unbiased estimate is needed.

*What the question tests.* That the reduction factor is $1/(1-\rho^2)$ and is therefore governed by the residual, and what makes one control variate far better than another.

---

**20.8** *Explain why bumping the spot price and revaluing gives a poor delta if the two simulations use independent random numbers.*

*The problem.* A bumped delta is a difference of two estimates:

$$\Delta \approx \frac{\hat{V}(S_0 + h) - \hat{V}(S_0 - h)}{2h}$$

If the two simulations use *independent* random numbers, each carries its own independent noise, with standard error $\eta$, and the difference carries

$$\mathrm{SE}\big[\hat{V}(S_0 + h) - \hat{V}(S_0 - h)\big] = \eta\sqrt{2}$$

Dividing by $2h$:

$$\mathrm{SE}[\Delta] \approx \frac{\eta\sqrt{2}}{2h} = \frac{\eta}{h\sqrt{2}}$$

The noise is **divided by** $h$, so it *diverges* as the bump shrinks. Meanwhile the true signal, the difference in value, shrinks *proportionally* to $h$ and is swamped.

The trap is that there is no good choice of $h$. Making $h$ small controls the finite-difference bias, which is $O(h^2)$ for a central difference, but multiplies the noise by $1/h$. Making $h$ large controls the noise but prices a different option. A concrete case: with $\eta = 0.01$ and $S_0 = 50$, a bump of $h = 0.50$ gives $\mathrm{SE}[\Delta] = 0.014$ against a delta of perhaps 0.5, or 3% noise. Halving the bump to 0.25 doubles it to 0.028, or 6%. The estimate gets *worse* the more carefully the derivative is approximated, which is the opposite of how numerical differentiation is supposed to behave.

*The fix: common random numbers.* Use the *same* draws $\varepsilon_i$ for both runs. With the exact lognormal step of question 6, the two paths are then related exactly:

$$S_T(S_0 + h) = \frac{S_0 + h}{S_0}\, S_T(S_0)$$

Every path in the bumped run is a scaled copy of the corresponding path in the base run, so the two payoffs are almost the same number and their difference is nearly *deterministic*. The noise in the difference is $O(h)$ rather than $O(1)$, the $h$ in the denominator cancels it, and $\mathrm{SE}[\Delta]$ becomes essentially independent of the bump size. Small bumps become safe, and the bias can be driven down without penalty, for a payoff that is continuous in $S_T$. A digital is the exception: there the bumped difference is nonzero only on the few paths that cross the strike, its variance grows like $1/h$ even with common random numbers, and the choice of $h$ is back. Typical variance reductions on the Greek are a factor of 100 or more, obtained by reusing the first run's random numbers, often a one-line change.

*Two extensions exist.* The same argument applies to *every* Greek and to scenario analysis generally: any quantity computed as a difference between two simulations must share random

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numbers, or the answer is noise. And for a smooth payoff, the pathwise derivative method differentiates the payoff analytically along each path, eliminating the bump entirely: no $h$, and therefore no bias. It fails where the payoff is discontinuous, as for a digital, where the likelihood-ratio method or a smoothed payoff is needed instead. Adjoint algorithmic differentiation (Section 20.10 and Example 20.8) applies the pathwise idea to every input at once: one backward pass delivers every sensitivity, at a fixed multiple of the cost of one price however many inputs there are.

*What the question tests.* That the difference of two noisy estimates is dominated by noise when the estimates are independent, and that correlating them is the standard and nearly free repair.

---

**20.9** *State the direction of the Longstaff–Schwartz bias, explain why it goes that way, and state when the method is nonetheless the right choice.*

*The direction: Longstaff–Schwartz is biased **downward**.* The estimate is a lower bound on the true American value, in expectation.

*Why.* The method estimates the continuation value by regressing realized discounted payoffs on functions of the current state, using the cross-section of in-the-money paths. The regression is imperfect (a finite basis, a finite sample), so the resulting exercise rule is *not* the optimal one. It sometimes exercises when it should hold, and holds when it should exercise.

But the true American value is the value under the *optimal* stopping rule, which is a maximum over all admissible rules:

$$V = \sup_{\tau} \mathbb{E}[e^{-r\tau}\,\text{payoff}(S_\tau)]$$

Any *particular* rule, including a bad one, attains at most this supremum. A suboptimal exercise policy earns less, so the value it delivers is below the true value. The bias is not a defect of the implementation; it is a structural property of evaluating a supremum by proposing a candidate.

*Knowing the sign has two useful consequences.*

The result is a **lower bound**, which is genuinely useful: one can state with confidence that the option is worth *at least* the computed figure. Pairing it with an upper bound from the dual (Andersen–Broadie) formulation brackets the answer, and the width of the bracket measures how good the exercise rule is, a diagnostic the method does not otherwise provide.

And a *second*, opposing bias enters if one is careless. Using the same paths both to estimate the regression and to value the option lets the rule use information from the paths it is applied to: a foresight bias that pushes *upward*. With many paths this is small; with few it is not, and it can mask the downward bias entirely. The clean practice is to estimate the coefficients on one set of paths and evaluate on an independent set, which restores the guaranteed lower bound.

*When it is nonetheless the right choice.* **When a tree cannot be built**, which in practice means when the state is high-dimensional or path-dependent:

- *Several underlyings.* A Bermudan option on a basket of five assets is a five-dimensional lattice problem, which is out of reach; Monte Carlo does not care about dimension (question 5).
- *Path dependence.* An American Asian option, or a callable structure whose payoff depends on the path so far, has a state variable a tree cannot carry.
- *Complex dynamics.* Stochastic volatility with jumps adds state variables that compound the same problem.

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- *Bermudan interest-rate products.* The callable swaps and Bermudan swaptions of a multi-factor model (Chapter 32) are where the method is needed most.

*And when it is the wrong choice.* For a **one-dimensional American option**, such as a plain American put, a tree beats it decisively: faster, more accurate, unbiased in the limit, and with no regression basis to choose. Reaching for Longstaff–Schwartz there is using the harder tool because it is the more interesting one. Section 20.13 puts the rule plainly: a tree or finite differences for one state variable with early exercise, and Longstaff–Schwartz only where early exercise and high dimension come together.

*What the question tests.* That the bias follows from the definition of an optimal stopping value, and that the method is a fallback for problems trees cannot represent rather than a general-purpose improvement.

---

**20.10** *Explain why the explicit finite difference scheme becomes unstable when the space grid is refined, and state how the time grid must respond.*

*The mechanism.* At $\theta = 0$ the scheme computes each new value as an explicit weighted average of three neighbors, the weights $a_j$, $b_j$ and $c_j$ of Section 20.12; on the log grid they approximate the trinomial probabilities of (20.2). They are probabilities (all nonnegative and, apart from discounting, summing to one) only if the middle weight $b_j = 1 - \Delta t\,(r + \sigma^2 j^2)$ stays nonnegative at every node. At the top of the grid, $j = M - 1$, that requires

$$\Delta t \lesssim \frac{1}{\sigma^2 M^2}$$

for $M$ space steps. The grid spacing has canceled out of the coefficients, so what matters is the number of nodes: the diffusion coefficient at node $j$ is $\sigma^2 j^2$, and refining the grid adds nodes, so the largest coefficient grows like $M^2$. Past the threshold the middle "probability" goes **negative**.

A negative weight means the scheme *amplifies* rather than averages. Any perturbation present (rounding error at the last bit is enough) is multiplied by a factor greater than one at every time step. Over a thousand steps even a factor of 1.01 becomes $2 \times 10^4$, and the factors past the threshold are far larger, so rounding error at the sixteenth decimal is carried to astronomical size. The growth is geometric, so the failure is not gradual.

*What it looks like.* Example 20.11 prices the American put at 3.6766 on 100 space and 1,000 time steps, at a number beyond $10^{40}$ on 200 and 1,000, and at 3.6918 on 200 and 4,000. Doubling the space resolution while leaving the time grid alone destroyed the calculation. The middle result is not slightly wrong; it is dozens of orders of magnitude wrong, and it came from an action that in every other numerical method makes the answer better. Refining a grid is not always an improvement, and a scheme that punishes refinement is one whose stability condition must be known before it is used.

*How the time grid must respond.* $\Delta t$ must fall like $1/M^2$: **doubling the space resolution requires four times as many time steps**, and the total work therefore grows like $M^3$. Since the space error is of order $\Delta S^2$, one more decimal place needs about $\sqrt{10} \approx 3.2$ times the space steps, ten times the time steps, and some thirty times the work. The third row confirms it: quadrupling the time steps to 4,000 restores a sensible answer at $M = 200$.

*Why the implicit and Crank–Nicolson schemes exist.* Both are **unconditionally stable**: $\Delta t$ and $\Delta S$ can be chosen independently on accuracy grounds alone, with no stability coupling. The price is that each time step requires solving a tridiagonal system rather than reading off an average, a Thomas-algorithm solve that is $O(M)$ per step, so the per-step cost is comparable and the freedom to choose $\Delta t$ costs almost nothing. Crank–Nicolson is additionally second-order

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accurate in time against the other two schemes' first order, for a European payoff; early exercise imposed after each solve gives some of that back, as Example 20.12 shows.

One caveat applies to Crank–Nicolson, since "unconditionally stable" is often misread. Stability is not the same as absence of oscillation. Crank–Nicolson damps high-frequency error slowly, so a discontinuous payoff (a digital, or the kink of a vanilla struck on a grid point) produces spurious ripples near the discontinuity that persist for many steps. The standard repair is Rannacher time-stepping: two or four fully implicit steps at the start to damp the node-to-node oscillation, then Crank–Nicolson thereafter.

*What the question tests.* That the explicit scheme's weights must remain probabilities, that the resulting condition couples the two grids quadratically, and why anyone tolerates a tridiagonal solve.

---

**20.11** *Explain why a relative total shareholder return award cannot be valued as expected units times expected share price, and state the direction and rough size of the error.*

*Why the naive product fails.* A relative-TSR award pays a *number of shares* that depends on performance, settled in *shares whose price* depends on the same performance. Both factors are random and they move together, so

$$\mathbb{E}[\text{units} \times S_T] = \mathbb{E}[\text{units}]\,\mathbb{E}[S_T] + \text{Cov}(\text{units}, S_T)$$

and the covariance term is exactly what the naive product discards. It is not a small correction; it is the structure of the instrument. Units are earned on precisely the paths where the company has beaten the index, which are the paths on which the share price is high. The award pays the most shares when each share is worth the most, and the fewest when each is worth least. The two multiply.

*The direction: the naive calculation **understates** the value.* The covariance is positive, because the same outperformance drives both factors.

*The size.* Example 20.7 simulates 400,000 correlated pairs for a three-year award with $S_0 = 50$, company volatility 35%, index volatility 20%, and correlation 0.60. The award is worth **60.25** per target unit; the naive product of expected units and expected share price gives **41.87**. The naive figure understates the value by

$$\frac{60.25 - 41.87}{60.25} = \mathbf{30}\%$$

or equivalently the correct value is $60.25/41.87 - 1 = 44\%$ higher. That is not a refinement: a compensation committee told the grant costs \$41.87 per unit when it costs \$60.25 has been given a number that misses almost a third of the cost.

Note that the award is worth more than the share price itself (60.25 against $S_0 = 50$), which looks impossible until the payoff structure is read. The scale runs to a **cap of 200%** of target on 30% outperformance, so the upside carries two shares per unit while the downside floors at zero shares. The award is closer to a leveraged call spread on relative performance than to a share, and valuing it as "about one share, times a probability" misses both the leverage and the correlation.

*Why this must be simulated rather than solved.* A market condition is priced into the grant-date fair value and never trued up (Section 17.7 and question 6 of that chapter), so the number has to be right on day one; there is no later correction. The payoff depends on the *joint* distribution of two correlated assets through a piecewise-linear, capped and floored function of their return difference, which has no convenient closed form. The machinery is Section 20.9's: factor the correlation matrix, generate $\varepsilon_1 = z_1$ and $\varepsilon_2 = \rho z_1 + \sqrt{1-\rho^2}\, z_2$, evolve both assets, apply the scale, and average. A pair of correlated assets is the smallest problem for which

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Monte Carlo is the natural method for a payoff of this shape, which is why the chapter puts the example here.

*The general lesson.* Never value a quantity-times-price payoff by multiplying expectations. The same error appears throughout derivatives: in quanto adjustments, where the payoff is denominated in a currency correlated with the underlying; in wrong-way risk (Chapter 27), where exposure is largest exactly when the counterparty is most likely to default; and in any structure where notional and price are driven by a common factor. Whenever the amount and the price share a driver, the covariance term is part of the price, and the sign of the error is the sign of the covariance.

*What the question tests.* That $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ is a pricing fact rather than a statistical footnote, and that the correlation here is positive by construction of the award.

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# Chapter 21

# Volatility Modeling: Local, Stochastic, Jump, and Rough

**21.1** *Explain why the denominator of Dupire's formula (21.2) is proportional to the implied density, and what that implies about when a local-volatility model exists.*

*Why the denominator is the density.* Dupire's denominator is

$$\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}$$

and Section 19.2's Breeden–Litzenberger result identifies $\partial^2 C/\partial K^2$ as the risk-neutral density of $S_T$, up to the discount factor: $g(K) = e^{rT}\partial^2 C/\partial K^2$. So the denominator is $\frac{1}{2}K^2$ times the density the surface already implies. Nothing has been assumed; the density was in the call prices all along, and Dupire's formula merely divides by it.

Read the whole formula that way and it becomes intelligible rather than magical. The numerator measures how fast option value accrues with maturity (the calendar dimension of the surface), and the denominator measures how much probability mass sits at that strike. The ratio is the variance rate that must be running *at* $(K, T)$ to produce the observed accrual given the observed density. Local volatility is not a model fitted to the surface; it is a quotient of two things the surface contains.

*What it implies about existence.* A local-volatility model exists precisely when the implied density is **positive** everywhere and, as below, the numerator is not negative. The density admits three cases:

- *Density positive.* The quotient is well defined and positive, $\sigma_{\text{loc}}$ is a real function, and Dupire's theorem says it is the *unique* one reproducing every quoted price.
- *Density zero.* The quotient diverges. Local volatility is infinite there, in a region the surface says the asset reaches with zero probability but which the calendar dimension says must still generate value.
- *Density negative.* $\sigma_{\text{loc}}^2 < 0$, so $\sigma_{\text{loc}}$ is imaginary and **no diffusion exists** that reproduces the surface.

The third case is the important one, because it is not hypothetical. A negative implied density is exactly the butterfly arbitrage of Section 19.5; Example 19.3's wing, where a call spread was priced at −\$0.39, breaks monotonicity, and a call price that rises in the strike forces a negative density somewhere beyond it. The numerator must not be negative either, which is Section 19.5's calendar condition: total variance falling with maturity would also make $\sigma_{\text{loc}}^2$ negative. So the no-arbitrage conditions on an option chain and the existence conditions for a local-volatility model are the same. A surface admitting arbitrage is not merely one a trader

could exploit; it is one no diffusion can generate, and any calibration routine handed such a surface will either fail or silently return nonsense.

*The practical consequence* is the order of operations in a dealer's pricing system: arbitrage checks on the raw quotes come *first*, then the smoothing and interpolation, then Dupire. A smooth interpolating surface fitted through arbitrageable quotes will hand the calibrator a density that dips negative in the wings, and the failure will surface as an exotic price that is wrong for reasons no one can trace back to a quote.

*What the question tests.* That Dupire's denominator is Breeden–Litzenberger in disguise, and that model existence and market arbitrage-freedom are one condition seen twice.

---

**21.2** *Explain the "rule of two" using the averaging argument, and state whether a steep local-volatility surface is evidence of a calibration error.*

*The rule.* The local-volatility surface is roughly **twice as steep** in strike as the implied surface derived from the same prices. Example 21.1 measures it: at one year, implied volatility falls at $-0.296$ per unit of log-strike and local volatility at $-0.518$, a ratio of **1.75**.

*The averaging argument.* An implied volatility is not a volatility. It is the single constant that, inserted into the Black–Scholes–Merton formula, reproduces the price of an option struck at $K$. The price is generated by paths that travel through many different local volatilities on their way to $K$, so the implied number is an *average* of the local volatilities visited.

Averaging flattens. Consider an option struck well below the money. Its price is driven by paths that start at $S_0$ and travel down to $K$, sampling local volatilities all the way from $\sigma_{\text{loc}}(S_0)$ to the much higher $\sigma_{\text{loc}}(K)$. The implied volatility comes out somewhere in between, nearer the middle than the endpoint. The same happens at every strike, so the implied curve is a smoothed version of the local curve, and a smoothed curve is a flatter curve.

Run that backward and the rule falls out: to produce a given slope *in the average*, the underlying function must slope about **twice** as steeply. If the average of the values between $S_0$ and $K$ is to move by one unit as $K$ moves by one, and the average is roughly the midpoint, the endpoint must move by two. The factor is 2 in the leading-order approximation, near the money and at short maturity; the 1.75 measured here is lower because a one-year implied volatility also averages local volatilities at earlier dates, which are steeper (Example 21.2).

*Is a steep local-volatility surface evidence of a calibration error? No.* It is **arithmetic**, and the trap is a well-worn one. A local-volatility surface always looks alarming. It is about twice as steep as the implied surface near the money, and far in the wings it reaches levels that no one would quote and no trader believes. The instinct is to conclude the calibration has gone wrong and to damp the surface until it looks reasonable. That instinct produces a surface that no longer reproduces the vanilla prices, which is the one thing local volatility is *for*.

*Where a genuine error would show instead.* The diagnostics are the vanillas, not the appearance of the surface. Reprice the quoted options through the calibrated model: if they come back, the calibration is right however steep the surface looks. Genuine failures announce themselves differently: as a negative or exploding $\sigma_{\text{loc}}$ from a density that went non-positive (question 1), as instability under small changes in the input quotes, or as sensitivity to the interpolation scheme in regions where no options trade.

One thing the steepness does tell you is worth separating from the false alarm: the far wings of a local-volatility surface are extrapolation, not data. Their steepness is arithmetic *plus* whatever the extrapolation scheme invented, and Example 19.1's density integrating to 1.06 is the same warning stated in terms of the density. The rule of two explains why the surface is steep; it does not certify the wings.

*What the question tests.* That implied volatility is an average along a path, and the discipline

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not to "fix" a model that is behaving correctly.

---

**21.3** *Using Example 21.2, explain why a local-volatility model misprices a forward-starting option and in which direction.*

*The mechanism.* A local-volatility model has no freedom left once calibrated: $\sigma_{\text{loc}}(S,t)$ is a *fixed* function, so calibrating to today's surface fully determines what the model says about every future date. In particular it determines the **forward smile**, the smile that will prevail at some future date $t_1$ for options expiring at $t_2$, and that prediction is testable.

Example 21.2 extracts it. Measuring both skews ±5% around the forward:

| Maturity | Implied skew | Local-vol skew |
|---|---|---|
| 3 months | −0.479 | −0.981 |
| 1 year | −0.296 | −0.518 |
| 3 years | −0.148 | −0.231 |

The local-volatility function's own skew decays by a factor of more than **four** between three months and three years. A one-year option *starting* in three years is priced from the local volatilities between years three and four, whose skew is less than a quarter of the three-month one, and the model predicts a three-year-forward smile far flatter than today's spot smile.

*The market disagrees.* Forward-starting options and cliquets trade with a skew comparable to today's. The smile is roughly **stationary in moneyness**: it travels with spot and with time rather than decaying.

*The direction: local volatility marks a product that gains when forward skew steepens too **cheap**.* A forward-starting option's value comes from the smile prevailing at the start date. Predicting a flat forward smile when the market delivers a steep one undervalues an out-of-the-money forward-starting put or a forward-starting risk reversal (long the put, short the call), which gain from that skew, and overvalues the mirror image, an out-of-the-money forward-starting call, which loses from it. A cliquet (a strip of forward-starting options) compounds the error across every reset.

*Why this is the deeper failure of the model, not a fixable one.* The error is not in the fit. The fit is exact by construction; every vanilla reprices. The error is in *what the fit implies*, and no recalibration touches it, because the flattening follows from the model's structure rather than from its parameters. Chapter 34 returns to this with cliquets, where the mispricing is the whole trade.

The consequence for risk management is sharper than the consequence for pricing. Two products with *identical* vanilla hedges can have quite different forward-smile exposure, and local volatility prices and hedges both as though the exposure were the same. A desk running the model therefore cannot see the risk on its own reports, which is the argument for the local-stochastic construction of Section 21.9, whose mixing parameter makes exactly this exposure visible as a dial.

*What the question tests.* That an exact fit constrains dynamics as well as prices, and that the forward smile is where a local-volatility model's dynamics are wrong.

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**21.4** *A Heston fit returns $\rho = -0.05$. What shape of smile does this imply, and in which market would you expect to see it?*

*The shape: an almost perfectly **symmetric smile**.* In Heston, the correlation $\rho$ is what tilts the smile, and it is the parameter that does so most directly. Table 21.2 isolates it (skews at $h = 0.10$): at $\rho = 0$ the skew is 0.000 (symmetric to three decimals) against −0.282 in the base

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case at $\rho = -0.7$ and $-0.398$ at $\rho = -0.9$. A fit returning $\rho = -0.05$ is reporting a surface with essentially no tilt, and whatever smile it has is *curvature*, supplied by $\sigma_v$: both wings lifted, roughly equally, above a minimum near the money.

*Why $\rho$ does that.* Negative correlation means volatility rises as the price falls, so downward paths are also high-volatility paths, and the terminal distribution acquires a fat left tail and a thin right one. That is Section 19.9's leverage explanation written as a parameter. At $\rho \approx 0$ the two processes are unlinked, up-moves and down-moves carry the same volatility, and the distribution is symmetric in log-price.

*Where you would expect to see it: a **currency** market.* Section 19.1's structural argument gives the reason, and it is not a coincidence of estimation. A currency pair is a *ratio*: a fall in EUR/USD is a rise in USD/EUR, and the two are one market seen from two desks. There is no leverage effect, because neither currency is a residual claim on a levered balance sheet, and no asymmetry of catastrophe, because a devaluation is a crash for one holder and a windfall for the other. The identity $\sigma_{\text{EUR/USD}}(K) = \sigma_{\text{USD/EUR}}(1/K)$ means any skew in one quotation is the opposite skew in the other, so a skew must favor one currency; for a major pair with no structural reason to favor either, the smile comes out roughly symmetric. A near-zero fitted $\rho$ on a major currency pair is the model recovering that structure from the quotes.

*Two qualifications keep the answer honest.* A major pair is the clean case; an emerging-market currency with a peg that can break in only one direction *does* show a skew, and the fitted $\rho$ will pick it up, pointing in the direction the peg would break. And $\rho = -0.05$ on an **equity index** would be a red flag rather than a result: equity surfaces are unambiguously skewed, so a near-zero $\rho$ there means the optimizer has landed in a bad local minimum, or the objective is weighted so heavily toward at-the-money quotes (Example 21.6) that the wings, which carry the skew information, never entered the fit.

*One structural caution follows.* $\rho$ in Heston is the *instantaneous* correlation between price and variance moves. A change of measure leaves it unchanged, so it can in principle be estimated from returns against changes in a variance proxy and compared with the fit, though the proxy's noise biases such estimates toward zero. A fitted value far from the historical one is therefore a warning, not a fact about the market; reading a fitted parameter as an economic quantity is the usual mistake, and question 5 pursues where that goes wrong.

*What the question tests.* That $\rho$ is the main source of skew, and that the shape it implies identifies an asset class.

---

**21.5** *Explain why raising $\sigma_v$ changes the at-the-money level as well as the curvature, and what this implies about interpreting a single fitted parameter.*

*The effect on the level.* Raising $\sigma_v$ lowers the at-the-money implied volatility substantially, even with the variance level $v_0 = \theta = 0.04$ (nominally 20%) held fixed:

| | ATM (%) | Skew | Curvature |
|---|---|---|---|
| $\sigma_v = 0.2$ | 19.44 | −0.112 | −0.01 |
| $\sigma_v = 0.6$ (base) | 16.94 | −0.282 | 0.36 |
| $\sigma_v = 1.0$ | 14.45 | −0.340 | 1.40 |

(Skews at $h = 0.10$, as in Table 21.2.) That is five volatility points of level, from a parameter described as "vol-of-vol."

*Why the level moves.* Implied volatility is a *volatility*, but the model's state variable is a *variance*. An at-the-money option's price is approximately governed by $\mathbb{E}\left[\sqrt{\bar{V}}\right]$, where $\bar{V}$ is the average variance realized over the option's life. The square root is **concave**, so by Jensen's

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inequality

$$\mathbb{E}\left[\sqrt{\bar{V}}\right] < \sqrt{\mathbb{E}[\bar{V}]} = 20\%$$

and the gap widens as $\mathrm{Var}(\bar{V})$ grows, which is what raising $\sigma_v$ does. Uncertainty about variance therefore *lowers* the at-the-money volatility even though it leaves expected variance untouched. The effect is present at $\rho = 0$ too: the base case with $\rho$ set to zero still gives 17.91%, two points below 20%, so this is genuinely a convexity effect and not a by-product of the correlation.

What this implies about interpreting a fitted parameter: the parameters are not orthogonal, so no single one can be read as "the" something. Moving $\sigma_v$ from 0.2 to 1.0 changes all three summary statistics at once: level 19.44 → 14.45, skew −0.112 → −0.340, curvature −0.01 → 1.40. It is described as the curvature parameter because curvature is what it moves *most*, not because it moves curvature *only*.

The consequence for calibration is the practical one. Because several parameters move the same features, the objective function has **flat directions**: a rise in $\sigma_v$ can be substantially offset by a rise in $v_0$, or by a change in $\kappa$, with very little deterioration in fit. So a fit reporting $\sigma_v = 0.9$ today and 0.5 tomorrow may be describing the same surface twice, and a trader who reads the change as "the market's uncertainty about volatility has collapsed" has read an optimizer's wandering as a market move. A fitted parameter is a coordinate in a fit, not a measurement of the world, which is why Section 21.10 penalizes day-to-day parameter movement (question 11), and why the honest summary of a calibration is the surface it produces rather than the parameter vector that produced it.

The remedy, when two parameters are redundant in the data being fitted, is to fix one using data of a different kind: a time series rather than a smile. Section 29.9 applies exactly this to SABR, whose $\beta$ is fixed from the backbone rather than fitted alongside $\rho$.

*What the question tests.* Jensen's inequality as the reason variance-of-variance moves a volatility, and that parameter redundancy is a property of the objective rather than a defect of the optimizer.

---

**21.6** *State what jumps produce that a diffusion cannot, and state the corresponding defect in the term structure they generate.*

*What jumps produce that a diffusion cannot: a steep **short-dated** skew.* Over a short horizon a continuous path simply cannot travel far. A diffusion covers a distance of order $\sigma\sqrt{T}$, so at one month with $\sigma = 17\%$ that is about 5%, and a strike 15% out of the money lies more than three such moves away, a probability of about one in two thousand, making a 15%-out-of-the-money put worth essentially nothing. The market prices such puts at a substantial premium. Only a **discontinuity** can get there: a jump crosses the distance in no time at all, so its contribution to the option's value does not vanish with $T$.

Example 21.3 shows the resulting surface, under Merton's model with $\sigma = 17\%$, $\lambda = 0.5$ a year, mean jump −10% and jump volatility 13%:

| Strike | 1 month | 3 months | 1 year |
|---|---|---|---|
| 85 | 30.99% | 23.82% | 21.21% |
| 100 | 18.77% | 19.49% | 20.18% |
| 105 | 18.25% | 18.99% | 19.94% |
| 110 | 18.42% | 18.73% | 19.74% |

At one month the smile spans 12.7 volatility points; at one year, 1.5 points. That short-end steepness is the achievement.

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*The corresponding defect: the skew **dies too fast** with maturity.* The same table read along the rows shows the smile's range collapsing by a factor of more than eight from one month to one year. Measured properly as the at-the-money skew, the jump model's one-month-to-one-year ratio is **3.7**; the market's is about **2.7**. Jumps overshoot the short end.

The mechanism is diversification over time. A jump process with intensity $\lambda$ delivers, on average, $\lambda T$ jumps by time $T$. At one month that is 0.04 jumps, so the option is essentially pricing the chance of a single discrete event, and the distribution is violently non-normal. At one year it is 0.5 jumps, still far from normal, but the diffusion has had twelve times as long to spread, so a jump of fixed size is a smaller part of the distribution. The skewness the jumps add decays like $1/\sqrt{\lambda T}$, and the implied-volatility skew it produces falls off faster still, approaching $1/T$ at longer maturities ([TN21] works this out); between one month and one year its average rate of decay, about $T^{-0.53}$, is well above the market's $T^{-0.4}$.

*What the defect costs.* A jump model fitted to one-month options misprices one-year options, and fitted to one-year options misprices one-month options. The only repair is to let the jump parameters vary with maturity, which is to say to abandon the model as a description and keep it as an interpolation scheme, with a different $\lambda$ for every expiration and no claim that any of them means anything.

*And a second defect, sharper than the first.* Jumps break market completeness more fundamentally than stochastic volatility does. A jump of unknown size cannot be hedged by any position established before it arrives, so the risk is not removed by the stock and any finite set of options, whatever dynamic strategy is run in them. With stochastic volatility one can hedge with another option; with jumps, no perfect hedge exists. Section 13.4's argument said an incomplete market has no unique price; here there is no perfect hedge either, and whoever holds the jump risk must be paid for it. That payment is part of Section 19.10's volatility risk premium.

*The resolution, which is Section 21.7's subject.* Jumps decay too fast, classical stochastic volatility saturates and does not steepen at all, and the market sits between them at $T^{-0.4}$. Combining the two (jumps for the very short end, stochastic volatility for the rest) is Bates's model (Section 21.6), widely used for short-dated skew, but it lands between the two errors rather than producing the market's shape. Rough volatility (questions 7 and 8) produces the observed law from a single parameter instead.

*What the question tests.* That short-dated skew requires a discontinuity, and that the same discontinuity makes the skew decay too quickly once several jumps can accumulate.

---

**21.7** *Define the at-the-money skew $\psi(T)$, state the empirical power law it obeys, and explain why a classical stochastic-volatility model cannot reproduce a skew that keeps steepening as maturity shortens.*

*The definition.* Write $k = \log(K/F)$ for log-moneyness. The at-the-money skew is the magnitude of the surface's slope there:

$$\psi(T) = -\left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}$$

The minus sign makes $\psi$ *positive* on an equity surface, where implied volatility falls in the strike, so that a steepening skew is a rising $\psi$.

*The empirical law.* Across equity indices, from about a month to a few years,

$$\psi(T) \sim T^{-\alpha}, \qquad \alpha \approx 0.4$$

The skew keeps steepening as maturity shortens; whether it does so below a month is disputed (Section 21.8). Normalized to one year, the law predicts a one-month skew of $(1/12)^{-0.4} = 12^{0.4} = \mathbf{2.70}$ times the one-year skew, which is Figure 21.4's market curve.

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*Why a classical stochastic-volatility model cannot reproduce it.* Because in such a model **variance is a diffusion**, and a diffusion cannot move in no time.

Take Heston: $dv = \kappa(\theta - v)dt + \sigma_v\sqrt{v}\,dW_2$. Over a horizon $T$, the variance moves by an amount of order $\sigma_v\sqrt{T}$. The skew is produced by the correlation between price moves and *variance moves*, so it is proportional to how far the variance has traveled, which vanishes as $T \to 0$. Meanwhile the price's own dispersion also scales as $\sqrt{T}$, and the two $\sqrt{T}$ factors cancel in the skew. The result is that

$$\psi(T) \longrightarrow \text{a finite constant as } T \to 0$$

The asymmetry the correlation creates shrinks like $\sqrt{T}$, but so does the width of the distribution against which the skew, a slope per unit of log-moneyness, is measured. The two cancel, and the at-the-money slope tends to a constant, $\rho\sigma_v/(4\sqrt{v_0})$ to leading order: $-0.525$ in the base case, where the one-month slope is already $-0.53$.

Figure 21.4 measures the shortfall. Normalized to one year, Heston's skew at one month is **1.77** against the market's **2.70**: it reaches about two-thirds of the market's ratio and then flattens out, while the market curve is still climbing. The mismatch is not a calibration failure that better parameters would fix; $\kappa$ and $\sigma_v$ change where the saturation sets in, not the fact of it. The model's short-end behavior is structural.

*Why this mattered enough to motivate a new class of models.* The shortfall is not confined to a corner of the surface. Between a month and a year, where the power law is well documented, Heston delivers two-thirds of the market's steepening, and that range governs the pricing of short-dated barriers and digitals. Below a month, where 0DTE and weekly options trade (Chapter 23), the stakes would be highest if the law continued, which is the disputed point (Section 21.8). Either way a model that saturates misprices the short end systematically, in the same direction, every day.

Two repairs were available before rough volatility. *Jumps* produce a short-end skew, but they overshoot, with a ratio of 3.7 against the market's 2.7 (question 6). *Stochastic volatility* undershoots, saturating at 1.77. Bolting them together, as Bates does, brings the short end close (2.92 against 2.70) but inherits the jump model's long end, and its eight parameters place the fit without producing the law. Rough volatility gets the exponent right from one parameter, which is question 8.

*What the question tests.* That the short-maturity limit of the skew is a structural property of the volatility process, and that "diffusive variance" and "a short-dated skew that keeps steepening" are incompatible.

---

**21.8** *State the relationship between the Hurst parameter and the skew term structure, and give the value of $H$ that reproduces the observed exponent.*

*The relationship.* A model whose log-volatility is driven by a fractional Brownian motion with Hurst parameter $H$ generates an at-the-money skew obeying

$$\psi(T) \sim T^{H-1/2}$$

The exponent is $H - 1/2$, so the Hurst parameter and the observed power law are the same fact.

*The value that reproduces the observation.* Setting the model exponent equal to the empirical one,

$$H - \frac{1}{2} = -0.4 \quad \Longrightarrow \quad \mathbf{H = 0.1}$$

which is the order of magnitude Gatheral, Jaisson, and Rosenbaum (2018) estimated directly from high-frequency realized volatility. Figure 21.4's "market" curve *is* $\psi(T) \sim T^{H-1/2}$ at $H = 0.1$.

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Reading the formula at its endpoints shows why it is the right generalization rather than a curve-fit. At $H = 1/2$ the exponent is zero, so $\psi(T)$ is constant: the flat short-end behavior of the classical models, recovered in the short-maturity limit. $H = 1/2$ *is* ordinary Brownian motion, so the classical case is nested, and question 7's saturation is not a separate phenomenon but the $H = 1/2$ point of this family. Values below $1/2$ mean the path is **rougher** than Brownian (more jagged, with increments negatively correlated at every scale), and the rougher the volatility path, the faster the skew steepens as maturity shortens.

*Why one parameter doing this much was taken as evidence rather than fitting.* Two independent points deserve emphasis.

*The estimate is stable across markets.* Roughly the same $H \approx 0.1$ for equity indices, for bond and commodity futures, for individual stocks, and in both halves of the sample. A parameter that had been absorbing model error would not be so consistent across asset classes with quite different smiles.

*Roughly the same H appears on both sides of the measure change.* The value estimated from realized volatility (a time-series quantity, under the physical measure) roughly matches the value implied by option prices (a cross-sectional quantity, under the risk-neutral measure). There is no guarantee these agree. Roughness is a property of paths, which an equivalent change of measure cannot alter, but the risk-neutral value is not measured on paths: it is read off option prices through a model, and if the model were wrong the two need not coincide. That they roughly coincide is evidence about the *process* rather than about the quality of a fit, and it is what moved rough volatility from an observation into a modeling program. Both points are now questioned (Section 21.8): estimation error can make a smooth process look rough, and below a month the power law may not hold.

*The cost, since nothing is free.* Fractional Brownian motion with $H \neq 1/2$ is **not Markovian**: the drift of volatility depends on the entire path so far. There is no state variable to put on a tree, no partial differential equation to solve, and simulating the volatility exactly, jointly with the price, costs $O(n^2)$ per path in the number of steps after a Cholesky factorization, against $O(n)$ for a standard diffusion. The model was initially one no desk could use in daily pricing; it entered practice through fast hybrid simulation, neural-network calibration and Markovian approximations, which represent the kernel $(t - s)^{H-1/2}$ of Section 21.8 as a sum of a few ordinary mean-reverting factors and restore a state vector at the cost of exactness.

*What the question tests.* That $H - 1/2$ is the exponent, that $H = 0.1$ recovers $T^{-0.4}$, and that the classical models are the $H = 1/2$ special case rather than a rival family.

---

**21.9** *Explain what the leverage function in a local-stochastic volatility model does, and what the mixing parameter measures.*

*What the leverage function does.* In a local-stochastic volatility model,

$$dS = (r - q)S\,dt + L(S,t)\sqrt{v}\,S\,dW_1$$

with $v$ following a stochastic-volatility process. The function $L(S,t)$ is chosen so that the model **reproduces the vanilla surface exactly**. It is the correction factor that closes the gap between the smile the stochastic component happens to generate and the smile the market actually quotes.

The construction is the whole point, and it is best read as a division of labor. The stochastic component supplies *realistic dynamics*: a forward smile that does not collapse (question 3), a genuine vol-of-vol, a volatility risk the stock cannot hedge. But a stochastic-volatility model with five parameters cannot fit hundreds of quotes exactly, and the residual would show up as vanilla mispricings a desk cannot tolerate. The leverage function absorbs whatever is left over.

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So: stochastic volatility for the dynamics, local volatility for the fit, with neither asked to do the other's job.

Setting $L \equiv 1$ recovers pure stochastic volatility, which fits badly. Switching off the stochastic component recovers pure local volatility, which fits exactly and forecasts badly. The combination is what exotics desks run.

*What the mixing parameter measures.* It is a single dial that controls **how much of the smile is explained by the stochastic component and how much by the local component**. At zero, the model is pure local volatility; at one, it is as close to pure stochastic volatility as an exact fit permits. Every setting in between reproduces *the same vanilla prices* (the leverage function is recomputed to guarantee it) while implying different dynamics.

That is what makes the parameter valuable rather than a nuisance. It is the model risk, made into a dial. Example 21.7 showed two models agreeing on every vanilla and disagreeing by 49.5% on a reverse knock-out; ordinarily, quantifying that requires implementing a second model and hoping the comparison is meaningful. Here a desk prices the exotic at both extremes of the mixing parameter *within one model* and reads off the range. It has measured its exposure to the modeling choice without needing a second model, and the range is the number to report (question 10).

*The cost, which is real.* Calibrating $L$ is a fixed-point problem: the leverage function is defined in terms of the model's own density, and the density depends on the leverage function. Solving it requires a forward partial differential equation for the density, iterated to convergence, or a particle method in which a cloud of simulated paths estimates the conditional expectation as it goes. Either is far more expensive than a Heston calibration, and both are delicate where the density is thin: the far wings, where the leverage function is least constrained and most volatile.

*One thing the mixing parameter does not do.* It does not tell you which setting is *right*. Nothing in the vanilla market does, because the vanillas are matched at every setting; that is the construction's premise. Choosing it requires either a view on forward-smile dynamics or, better, a market price for an instrument that isolates them. Foreign exchange has one: Section 21.9's one-touch and double-no-touch options are liquid enough to calibrate the dial to. Equities do not, because the forward-starting options and cliquets that would serve trade over the counter, so there the setting is a judgment and the range is what gets reported.

*What the question tests.* That the leverage function buys the fit while the stochastic part buys the dynamics, and that a parameter which leaves all quoted prices unchanged is precisely the one that measures model risk.

---

**21.10** *Two models reproduce every quoted vanilla price and disagree on a barrier. Explain how this is possible without either committing an arbitrage, and state what a desk should report as a result.*

*How it is possible.* Each model is arbitrage-free internally, since every price it produces is an expectation under one martingale measure; each also **reproduces every traded price**. So no trade in quoted instruments exploits either, and the barrier, which is not quoted, is left free. Nothing in the vanilla market pins it down, so the two models are free to disagree about it, and no trade exists that profits from the disagreement. To exploit it you would have to buy the barrier at one price and sell it at the other, and there is no market on which to do so.

Put more precisely: the vanilla surface identifies the marginal distribution of $S_T$ at each maturity, and nothing more. A barrier's payoff depends on the *joint* law of the whole path: on the probability of finishing in a region *and* never having touched a level along the way. Infinitely many path laws share a given set of marginals, and the two models pick different ones. Within either model the barrier has a hedge (in Heston, with an option added) and a unique

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price; it is the vanilla data that leave the model unidentified, so each model's price is unique only given its own dynamics.

*Where the disagreement is large, and where it is not.* Example 21.7 prices barriers under Heston and under the local-volatility surface derived from Heston's own vanillas, using common random numbers so the difference is measured directly:

| Barrier | Heston | Local vol | Relative |
|---|---|---|---|
| Down-and-out call, 80 | 6.5286 | 6.5821 | +0.8% |
| Down-and-out call, 90 | 5.5388 | 5.6138 | +1.4% |
| Up-and-out call, 130 | 5.8138 | 5.5787 | −4.0% |
| Up-and-out call, 115 | 2.4071 | 1.7138 | −28.8% |
| Up-and-out call, 110 | 0.8402 | 0.4244 | **−49.5**% |

The contrast between the halves is the result. A down-and-out call struck at the money pays off well *above* the barrier, in a region the vanillas pin down tightly, so the models agree to about a percent. A reverse knock-out pays off only in the narrow band *between* the strike and the barrier—and no vanilla option isolates the joint event "finish in this band" and "never touch the top of it." Where the surface constrains least, the models diverge most, and at a barrier of 110 they differ by a factor of two.

The direction repays care, because a plausible argument gets it backward. One might reason that local volatility is low at high spot (12.0% at a spot of 110 a year out, against 16.2% at 100), so paths approach the upper barrier slowly, knock out less often, and the option should be worth *more*. It is worth *less*, and the premise is false: local-volatility paths knock out more often (44.8% survive against 52.7% in Example 21.5). Both models put the same probability on every terminal level, so what matters is survival *given* where a path ends. Of the paths finishing between 108 and 110, 39% have never touched 110 under Heston and 14% under local volatility; of those finishing between 100 and 102, 77% against 57%. Under Heston some paths are quiet and some wild, and the quiet ones reach the band without touching the barrier; local volatility reproduces the same terminal distribution with one volatility at each spot and date, so it has no quiet paths to offer.

*What a desk should report: a **range**, not a number.* Two conclusions follow directly. First, the vanilla market does not price this product, and no amount of recalibration will make it do so—so a single mark carries a false precision that the risk report will then propagate. Second, the instrument for producing the range is Section 21.9's mixing parameter: price the exotic at both extremes, and the spread sets the reserve held against model risk.

Three practical riders belong with that. The reserve should be deducted from the profit reported when the trade is booked, not merely noted beside it, or the desk books a profit that is an artifact of a modeling choice. The range should be reported *alongside* the mark to the risk committee, since a 0.8% range and a 49.5% range demand different governance and the mark alone does not distinguish them. And the exercise is worth doing only where the range is wide, which is why the table matters *before* the trade: knowing in advance which products are model-sensitive is what lets a desk price the reserve into the quote rather than discover it later.

*What the question tests.* That vanillas determine marginals and not path law, and that the honest output of a model-risk analysis is an interval with a governance rule attached.

---

**21.11** *Explain why a calibration objective is usually stated in implied volatility (price errors divided by vega) rather than in price, what a price-error objective would be blind to, and what a desk gains and gives up by penalizing day-to-day parameter movement.*

*Why implied volatility.* Because it measures every error in **volatility points**, the unit in

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which options are quoted and bid–ask spreads are set. A calibration minimizes a sum of squared errors, and the choice of what to square is a choice about which quotes matter. A price error divided by the option's vega is, to first order, its implied-volatility error, so a one-point miss counts the same on a low-vega wing option as on a high-vega at-the-money one, and errors are commensurable across the surface. Weighting volatility errors *by* vega does the opposite: vega is the conversion factor between the two, so it turns them back into price errors, which is the price objective again.

Quote quality is a separate problem. A surface contains quotes of wildly different reliability (the one-month at-the-money option trades continuously, the five-year 10-delta put twice a month), and an unweighted fit lets the least reliable quotes drive the parameters. Liquidity weights (inverse bid–ask width, or open interest) damp that, on top of the choice of units.

*What a price-error objective would be blind to:* ***the wings***. Squared price errors are dominated by at-the-money options, which have the largest vega, so a given volatility miss costs most there. Example 21.6's one-month 80 put is worth \$0.05 and contributes almost nothing to a squared-price objective even with its implied volatility 7.08 points too low. So a price-error fit tracks the at-the-money term structure closely and can be badly wrong in the wings without the objective function registering it at all. Example 21.6 sets the two side by side: dividing by vega cuts the one-month 80-strike miss from 7.08 to 4.01 volatility points, at a cost of 0.19 points at the money against 0.12.

Neither is right in general, and that is the more important half of the answer. The price-error fit is right for a desk whose book is at-the-money straddles. The volatility fit is right for a desk whose book is *in the wings*, such as the structured-product issuers of Section 19.8, whose exposure sits precisely where the first objective is blind. The calibration objective encodes an assumption about the book, and it should be chosen deliberately rather than inherited from whatever the software's default was. The deepest version of the rule is that the objective should target the risk the model is being used to hedge: a model that reproduces every quote and mishedges the one position on the book has failed.

*Penalizing day-to-day parameter movement.*

*What a desk gains.* Parameters that mean something over time. A model refitted daily to the whole surface produces parameters that jump, and question 5 explains why: the objective has flat directions, so the optimizer trades one parameter against another with almost no change in fit quality. The damage is not cosmetic. A book revalued with jumping parameters generates **phantom profit and loss**, and the desk cannot distinguish a genuine mark-to-market move from a calibration artifact, which corrupts P&L attribution, hedge ratios, and the risk reporting that Chapter 39's regulatory tests depend on. A regularization term penalizing distance from yesterday's parameters buys stability, and with it a Greek profile that does not lurch overnight.

*What it gives up.* It gives up fit quality, deliberately (today's surface is reproduced slightly less well), and, more seriously, *responsiveness*. The penalty is a prior that the market has not changed, and when the market genuinely has changed, that prior is wrong. A regularized calibration lags a real regime shift, so the model is stalest precisely when accuracy matters most: in the days after a volatility shock, when the surface has moved and the penalty is holding the parameters back. The parameter path is also *history-dependent*: today's fit depends on yesterday's, so two desks with identical current data and different histories mark the same book differently.

The practical compromise is to make the penalty weight a decision rather than a constant: light enough that a genuine move is tracked within a day or two, heavy enough that noise is filtered, and monitored by watching whether fit error is *trending* rather than merely elevated. A persistent, one-directional deterioration in fit means the penalty is fighting the market and should be relaxed.

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*What the question tests.* That a calibration objective is a statement about which errors are expensive, and that stabilizing parameters trades a known cost in fit for a hidden cost in responsiveness.

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# Chapter 22

# Volatility Products: VIX, Variance Swaps, and Dispersion

**22.1** *A variance swap is struck at 20% with \$100,000 of vega notional. Compute the variance notional and the payoff at 35% realized, compare it with what a linear-in-volatility contract would give, and explain what determines the gap between a variance and a volatility swap strike.*

*The variance notional and the payoff at 35% realized.* From (22.2), with $K_{\text{var}} = 20$ quoted in volatility points, $N_{\text{var}} = 100{,}000/(2 \times 20) = \$\mathbf{2{,}500}$ per variance point, so by (22.1)

$$2{,}500 \times \left(35^2 - 20^2\right) = 2{,}500 \times (1{,}225 - 400) = 2{,}500 \times 825 = \$\mathbf{2{,}062{,}500}$$

*Against a linear-in-volatility contract.* Realized volatility exceeded the strike by 15 points, so a volatility swap of the same vega notional would pay

$$15 \times \$100{,}000 = \$1{,}500{,}000$$

The variance swap pays **\$562,500 more**, 37.5% above the linear payoff. That excess is the convexity, and it grows quadratically: at 5 points over the strike the gap is \$62,500, at 15 points it is \$562,500, and at 20 points over it is \$1 million.

*What determines the gap between the two strikes.* Not the payoff at any particular outcome but the **vol-of-vol**, the dispersion of realized volatility itself. The variance swap strike is $\sqrt{E^{Q}[\sigma_R^2]}$ and the volatility swap strike is $E^{Q}[\sigma_R]$, and since the square root is concave, Jensen's inequality makes the first exceed the second whenever $\sigma_R$ is uncertain at all. If realized volatility were known in advance, the two contracts would be identical and the spread would be zero.

Example 22.2 sizes it, with realized volatility lognormal about a mean of 20%:

| Vol-of-vol | Variance swap | Volatility swap | Spread |
|---|---|---|---|
| 20% | 20.40% | 20.00% | 0.40 pts |
| 30% | 20.92% | 20.00% | 0.92 |
| 50% | 22.66% | 20.00% | 2.66 |
| 70% | 25.55% | 20.00% | 5.55 |

The spread is $\bar{\sigma}(e^{\nu^2/2} - 1)$, where $\bar{\sigma}$ is the mean of realized volatility and $\nu$ the standard deviation of its logarithm over the period, so it grows fast. At a realistic one-year index value of $\nu$ (40% to 60%), the variance swap strike sits one and a half to four volatility points above the volatility swap strike on the same underlying.

*Two consequences follow.* First, a variance swap strike is not a volatility forecast. Quoting 22.66% and calling it "the market's expected volatility" overstates the forecast by 2.66 points. This is not a separate overstatement from the skew premium of Section 19.3: at-the-money implied volatility sits close to the volatility swap strike, so the variance strike's excess over

it is largely this same convexity, carried on the surface by the smile's curvature, with the spot–volatility correlation adding its part. The overstatement relative to the market's expected volatility is the 2.66 points, not 2.66 plus the skew premium.

Second, the spread is itself a traded view. Buying a volatility swap against a variance swap on the same underlying is a clean position in vol-of-vol, and it is one of the few ways to trade the second moment of volatility without touching options on variance, which barely trade.

*And the reason the market chose the awkward contract.* A volatility swap is what people think in, and a variance swap is what trades—because variance is **replicable** from vanillas and volatility is not (question 2). Volatility swaps do exist, but they are priced off variance swaps with the Jensen adjustment above, which makes them a derivative of a derivative and leaves the dealer with hard-to-hedge vol-of-vol risk.

*What the question tests.* That the convexity gain is quadratic in the move, and that the variance–volatility strike spread is Jensen's inequality applied to a square root.

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**22.2** *Explain why delta-hedging a log contract earns realized variance regardless of the path, what feature of the log contract's gamma is responsible, and why the replication strip weights options by* $1/K^2$ *rather than equally.*

*Why a delta-hedged log contract earns realized variance, path independently.* Take any derivative $f$ held delta-hedged. Over a short interval the profit and loss of the hedged position is the gamma term of Section 18.4,

$$\Delta\Pi \approx \frac{1}{2} S^2 \Gamma \left[ \left( \frac{\Delta S}{S} \right)^2 - \sigma_{\text{impl}}^2 \, \Delta t \right]$$

so each day the position collects the squared return actually realized, less the variance that was priced in, weighted by the dollar gamma $S^2\Gamma$. Summing over the life of the trade gives total realized variance only if that weight is the *same every day*. For a normal option it is not: dollar gamma peaks at the strike and dies away from it, so the total is a gamma-weighted average of daily squared returns, a quantity that depends on where the underlying happened to be on the days it moved.

*The feature responsible: the log contract has constant dollar gamma.* For $f(S) = -2\ln(S/F)$,

$$\frac{\partial f}{\partial S} = -\frac{2}{S}, \qquad \frac{\partial^2 f}{\partial S^2} = \frac{2}{S^2}$$

and the contract's value has gamma $\Gamma = 2e^{-r(T-t)}/S^2$ (Section 22.3), so $S^2\Gamma = 2e^{-r(T-t)}$. The weight is the same at every price; the discount factor only states each day's gain in today's money, and carried to expiration at the risk-free rate that gain is worth exactly the squared return, which is the $e^{rT}$ in (22.5). So the sum of daily gamma P&L is exactly proportional to the sum of squared returns (realized variance), with no weighting and therefore no path dependence. The contract does not care *where* the volatility happened, only how much of it there was. That is a remarkable property and it is the entire reason the variance swap market exists: it converts a path-dependent hedging P&L into a clean linear claim on a single number.

Every other feature follows. Because the payoff is replicated by a fixed portfolio of vanillas held *statically* plus a short forward position (there is no bond, since $f(F) = 0$), no model and no volatility forecast is required, and only the stock position, of constant dollar size, is traded as the price moves; the strike is read off traded option prices. The variance swap is therefore one of the few genuinely model-free instruments in the book.

*Why the strip weights options by* $1/K^2$. There are two ways to see it, and both are worth having.

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*The algebraic route.* Section 12.8's decomposition writes any twice-differentiable payoff $g$ as a bond, a forward, and a strip of options with weight $g''(K)\,dK$ at strike $K$. For $g(S) = -2\ln(S/F)$,

$$g''(K) = \frac{2}{K^2}$$

so the weight *is* $2/K^2$. The $1/K^2$ is not chosen; it is the second derivative of the log, and the log was chosen because its dollar gamma is constant.

*The gamma route*, which explains why the two facts are one fact. Each option contributes one unit of gamma in total, spread over a band around its strike, so a strip holding $w(K)\,dK$ at each strike has gamma close to $w(S)$ at price $S$ and dollar gamma close to $S^2 w(S)$. That is *flat* across all price levels only if $w \propto 1/K^2$. Weighting the options **equally** would give dollar gamma rising with $S^2$ (and a $1/K$ weighting would still leave it rising linearly): it would over-collect variance at high prices and under-collect at low ones, and the payoff would depend on the path after all.

*The practical residue*, since exactness is an assumption. The argument requires a *continuous* path and a *continuum* of strikes. Jumps break the first: a gap of log size $x$ adds $x^2$ to realized variance while the replicating portfolio collects $2(e^x - 1 - x)$, which falls short on a downward gap, so the hedge under-performs precisely on the days that matter. Truncation breaks the second, which is question 3.

*What the question tests.* That constant dollar gamma is what removes path dependence, and that the $1/K^2$ weight is the second derivative of the log rather than a convention.

---

**22.3** *Using Example 22.1, state whether widening the strike range or narrowing the strike spacing is the more effective way to improve a replication at low volatility, and explain why the answer changes at high volatility.*

*At low volatility, narrowing the spacing is the more effective repair.* Example 22.1 settles it on a flat 20% surface, one year, where the answer must be exactly 20.00%:

| Strike range and spacing | Replicated volatility |
|---|---|
| 50–200, step 5 | 19.956% |
| 30–300, step 5 | 19.956% |
| 20–500, step 5 | 19.956% |
| 10–800, step 2 | **20.015%** |

Widening the range from 50–200 all the way to 20–500 changes the answer by **nothing**, not to three decimal places. Cutting the strike step from 5 to 2 brings the answer within 1.5 basis points, now on the high side.

*Why.* The two failures are different in kind, and only one of them is binding here. *Truncation* error is the value of the options outside the range that the strip omits. At 20% volatility a one-year put struck at 50 is worth a fraction of a cent, so it contributes nothing to the integral despite its enormous $1/K^2$ weight. The weight is large and the price is zero, and their product is what matters. Extending into that region adds nothing because there is nothing there.

*Quadrature* error is different: the strip is a discrete sum approximating a continuous integral, and here the approximation fails at one place only, the forward, 102.02, where the strip switches from puts to calls and the integrand has a kink; elsewhere the integrand is smooth and the sum is very accurate. No amount of widening touches it; only a finer mesh does. Where the grid sits relative to the forward sets the error's sign and size: on the 5-point grid it runs from about −5 to +10 basis points depending on that offset (−4 on Example 22.1's grid), largest with a strike exactly on the forward. That spread shrinks with the square of the spacing, so the 2-point grid's

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+1.5, from its strike at 102, is close to that grid's worst case of +1.6. It is related to the split that Section 22.6's correction term handles for VIX, but it is not the error that term removes: the correction accounts for anchoring at a listed strike, and the discretization error remains. So at low volatility the entire residual error is in the discretization, and the fix is spacing.

*Why the ordering reverses at high volatility.* Raise volatility to 60% and the one-year options beyond 50 and 200 are worth something real: the underlying now has a genuine chance of getting there. A put struck at 20, whose $1/K^2$ weight is 25 times that at strike 100, now multiplies a non-negligible price, and the omitted tail becomes a material part of the integral. Truncation error grows with volatility while quadrature error does not, so beyond some level the ranking flips and **widening the range** becomes the binding constraint.

*And this is the uncomfortable part.* Section 19.3's warning applies with full force: the strip stops where the quoting rule stops, not where the integral does. In a calm market the far strikes are irrelevant to the calculation. At high volatility they carry real value, and the strikes the integral needs run further out than any listed range, so whatever the rule leaves out is no longer worth nothing. The replication is therefore most accurate when accuracy does not matter and degrades when it does.

Three consequences follow for practice. VIX's own construction stops walking outward after two consecutive strikes with no bid, so the index's value depends on how thickly the wings happen to be quoted, and VIX can move on a day when no volatility changed. A dealer's hedge degrades in the same conditions and the same direction, which is one of the four gaps in Section 22.10. And any statement about the far tail of an implied distribution is a statement about the extrapolation, not about the market; Example 19.1's density integrating to 1.06 is this same fact in Chapter 19's terms.

*What the question tests.* That truncation and quadrature are distinct error sources with different cures, and that which one binds is a function of the volatility level.

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**22.4** *Six-month variance swaps are quoted at 25% and one-year variance swaps at 22%. Compute the forward volatility for the second half-year, explain why it lies below both spot strikes, and state the positions in the two swaps, in variance notional, that make up one unit of forward variance.*

*Total variance to each date.* A variance swap strike is an annualized volatility, so the variance it delivers to its maturity is $TK_{\text{var}}^2$:

$$0.5 \times 0.25^2 = 0.5 \times 0.0625 = 0.03125, \qquad 1 \times 0.22^2 = 0.0484$$

*The forward.* As in Example 22.9, the variance over the second half-year is the difference, annualized over the half-year it covers:

$$\sigma_{\text{fwd}}^2 = \frac{0.0484 - 0.03125}{1 - 0.5} = \frac{0.01715}{0.5} = 0.0343, \qquad \sigma_{\text{fwd}} = \mathbf{18.52}\%$$

The check is the telescoping identity: the two half-year variances average to the one-year, $(0.0625 + 0.0343)/2 = 0.0484$.

*Why it lies below both spot strikes.* The one-year variance is the average of the two half-years. The first half already carries 25%, above the one-year 22%, so the second half must sit below 22% for the average to come out at 22%. It is Example 22.9 inverted, and the forward rate on an inverted yield curve of Chapter 4. The averaging is in variance, not in volatility: the first half sits 0.0141 of variance above the one-year and the second half 0.0141 below, but in volatility that is 3 points above and **3.48 points** below, because the same variance buys more volatility points at a lower level.

*The replicating positions.* Total variance adds, so realized annualized variance over the second half-year is twice the one-year realized variance less the six-month: $\sigma_{\text{fwd},R}^2 =$

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$2\sigma^2_{1,R} - \sigma^2_{0.5,R}$. One unit of forward variance is therefore **long 2 units** of variance notional in the one-year swap and **short 1 unit** in the six-month swap. The fixed legs then combine to $2 \times 0.0484 - 0.0625 = 0.0343$, the forward strike, so the package costs nothing to enter. The weights are set by the maturities alone and would be the same whatever the two quotes were.

*What the question tests.* That a forward variance is read off total variance rather than volatility, and that it needs no model: two spot swaps, each replicable from options, span it.

---

**22.5** *Explain why variance swaps, single-stock ones above all, are now often written with caps, and what the cap does to the replication argument.*

*Why the caps appeared, and why they spread.* Because the payoff is **unbounded above and bounded below**; single stocks carried caps from the start, and 2008 collected on the difference for everyone else. Realized volatility on equity indices reached several times the strikes at which variance had been sold, and sellers (many of them dealers who treated the short variance as an offset to the options they were long) took losses far beyond anything the vega notional suggested.

The scale is what makes the point. Take a one-month swap struck at 15% with \$100,000 of vega notional and realized volatility of 80%, roughly the March 2020 experience for a swap struck just before the sell-off:

$$100{,}000 \times \frac{80^2 - 15^2}{2 \times 15} = 100{,}000 \times 205.83 = \$\mathbf{20.6}\text{ million}$$

A factor of **206** on a position whose stated vega notional was one hundred thousand dollars. No model error is involved—it is simply what the payoff formula does when realized volatility comes in at more than five times the strike, and the quadratic that gives the buyer a pleasant 37.5% overshoot on a 15-point move (question 1) gives the seller this on a large one.

The market's response was structural rather than a matter of pricing. Single-stock variance swaps, because a stock that can gap 40% on an earnings miss or a takeover bid has a much fatter tail than an index, had carried a cap from the start, conventionally **2.5 times the strike**; after 2008 they nearly stopped trading even so. Capped index contracts appeared too, though uncapped ones still trade. Capped at 2.5, the same trade loses

$$100{,}000 \times \frac{37.5^2 - 15^2}{2 \times 15} = \$3.9\text{ million}$$

a reduction by a factor of 5.2 and, more to the point, an exposure that is *known in advance* rather than unbounded.

*What the cap does to the replication: it destroys it.* A capped variance swap is

$$\text{variance swap} \;-\; \text{a call on realized variance struck at } (2.5K_{\text{var}})^2$$

The first term is statically replicable by the strip of Section 22.3. The second is not. A call on realized variance has a payoff that is a convex function of realized variance, and no static portfolio of vanilla options can produce it: vanilla payoffs depend on $S_T$ alone, and the log strip reaches realized variance only because Itô's lemma makes the variance enter linearly. Pricing it requires a view on the distribution of realized variance, which means a model of how forward variance moves, such as the rough-volatility models of Section 21.8, and of vol-of-vol.

*And that is the irony.* The variance swap's selling point was that it needed **no model**: strike read off traded vanillas, payoff replicated exactly, no forecast required. The uncapped payoff that this model-free construction made possible was precisely what destroyed its sellers—so the market added a cap, and the cap reintroduced a model. The instrument that was attractive

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because it needed no model acquired one as a direct consequence of the payoff its replication permitted.

*Two practical riders follow.* The dealer's hedge is now imperfect by construction, and the residual is short vol-of-vol, an exposure that is small in normal conditions and large exactly when the cap is approaching. And a capped and an uncapped swap at the same strike are *different contracts*: comparing quotes without checking the cap compares two different instruments, which is the same documentation trap as the mean-subtraction convention of Section 22.2.

*What the question tests.* That the seller's exposure is quadratic and one-sided, and that capping it converts a model-free instrument into a model-dependent one.

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**22.6** *Explain why VIX exceeds the at-the-money implied volatility of 30-day S&P options, and why a VIX futures contract is not a forward contract on VIX in the sense of Chapter 5.*

*Why VIX exceeds at-the-money implied volatility.* There is one quantity, the skew premium, and Jensen's inequality accounts for part of it.

*The skew.* VIX is a variance swap rate, computed by (22.7) from a strip of *out-of-the-money* options weighted by $\Delta K/K^2$. That weight rises sharply as the strike falls, so the calculation loads heavily on out-of-the-money puts, and on an equity index those are exactly the options carrying the highest implied volatilities (Section 19.1). The at-the-money option is one point on the surface; VIX is a $1/K^2$-weighted average across it, pulled upward by the expensive left wing. This is Section 19.3's point: a variance swap strike exceeds at-the-money implied volatility by roughly the amount the skew is worth, which Section 22.6 measures at 4.38 points on a representative skew and 2.43 points on the strikes that skew actually quotes (21.99% against a flat control of 19.56%, since truncation costs something on its own).

*Jensen, again.* VIX is the *square root* of a variance swap rate, so it exceeds the risk-neutral expected volatility by the vol-of-vol adjustment of question 1; at-the-money implied volatility sits close to that expectation, and on the surface the adjustment is the smile's curvature, which is part of what the skew premium above measures; the rest comes from the spot–volatility correlation that tilts the smile.

The consequence is a widely repeated error: VIX is routinely described as "the market's expectation of 30-day volatility." It is not. It is above that expectation by exactly the Jensen gap of question 1, since VIX squared *is* the expected variance; the skew premium over at-the-money volatility contains that gap and, through the spot–volatility correlation, exceeds it. A VIX of 20 does not mean the market expects 20% realized volatility, and the gap is not a mispricing—it is what the index is defined to measure; under the physical measure the volatility risk premium widens it further.

*Two further peculiarities of the published number*, which follow from the rule rather than the theory. The forward it is built around comes from put–call parity applied to the options themselves, (22.6), not from the index level, so the level VIX is constructed around need not be the level the index is quoted at. And the strip stops after two consecutive strikes with no bid, so VIX depends on how thickly the wings happen to be quoted and can move on a day when no volatility changed (question 3).

*Why a VIX futures contract is not a forward on VIX.* Because Chapter 5's cost-of-carry argument requires that the underlying can be **bought and held**, and VIX cannot be. There is no portfolio one can buy today, carry, and deliver as VIX at expiration: VIX is a statistic computed from option prices, not an asset. So the arbitrage that forces $F = Se^{(r-q)T}$ has no mechanism, and the futures price is under no obligation to bear any particular relation to the spot index.

Being marked to market daily, a VIX futures price is instead the risk-neutral *expectation* of its final settlement value, $E^{Q}[\mathrm{VIX}_T]$, where $\mathrm{VIX}_T$ is the Special Opening Quotation of

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Section 22.7: the index formula run on that morning's opening option prices. By Jensen once more, since VIX is a square root, that expectation sits *below* the square root of the forward variance swap rate. A VIX futures price is therefore a genuine expectation with no carry tether to spot, which is why it is free to trade far from spot.

And that is not a technicality; it is the whole shape of the market. With no carry relation, the term structure is set by mean reversion (Section 19.6) and by the risk premium: upward sloping when spot volatility is low, which is most of the time, and inverted in stress. Figure 22.3 shows both. A long VIX futures position therefore pays a roll cost that has nothing to do with financing and everything to do with the curve's slope (question 7), and the misconception that a VIX futures contract should track VIX is what makes investors buy these products as hedges and then wonder where the money went.

*What the question tests.* That VIX is a $1/K^2$-weighted variance rate rather than a volatility quote, and that a non-storable underlying removes the cost-of-carry link entirely.

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**22.7** *Using the curve of Figure 22.3, compute the monthly and annualized roll cost of a constant one-month long VIX futures position, and state who receives it.*

*The roll cost.* From the calm curve of Figure 22.3, the front month is 16.4 and the second month 17.2. A fund holding constant one-month exposure must roll: it buys the second month at 17.2, and one month later, if the curve has not moved, that contract has become the front month and is marked at 16.4.

$$\text{loss} = 17.2 - 16.4 = 0.8 \text{ points}, \qquad \frac{0.8}{17.2} = \mathbf{4.65}\% \text{ per month}$$

Compounded over a year,

$$(1 - 0.0465)^{12} - 1 = \mathbf{-43.5}\% \text{ per year}$$

with the VIX index unchanged throughout. Nothing has to happen for this to be paid; the loss is the slope of the curve, collected twelve times.

*Who receives it: the **seller** of VIX futures*, the short volatility position on the other side of the roll. In practice that is the systematic short-volatility funds, the inverse exchange-traded products of Section 22.8, and the dealers running long-vega books who sell the futures as a hedge and are net receivers of the carry.

Two consequences follow, one for each side of the trade.

*Long VIX futures products are structurally poor hedges to* hold. An investor who buys volatility exposure as portfolio insurance and holds it for a year pays something like 40% of the position for the privilege, and any payoff must beat that before the hedge has accomplished anything. This is the arithmetic behind the long-run charts of VXX, the largest long VIX futures note, and its relatives, which lose more than 99% of their value over a decade, far more than VIX itself ever falls; the loss is the roll, not the index. The decay is not a defect in the product; it is the product doing exactly what its prospectus says.

*And short VIX futures products earn that 40% as a carry.* It is the volatility risk premium of Section 19.10, collected through a different instrument. Section 22.8 and question 8 are about what happened to the people collecting it.

*The unifying point, and the one worth carrying away.* The premium and the roll are **the same thing seen from two sides**. Mean reversion alone would tilt the curve upward from a low spot level, but a futures price that merely forecast the index would cost a long holder nothing on average. The curve slopes up by *more* than that because sellers demand compensation for bearing a risk that is small most days and catastrophic occasionally, and that excess roll cost to the buyer *is* the compensation to the seller. Neither is a market inefficiency. A strategy

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marketed as "harvesting contango" is not exploiting anything—it is being paid to take a risk that occasionally arrives, and the payment looks like a riskless profit for precisely as long as the risk does not.

*And a caveat on the arithmetic.* The 43.5% assumes the curve stays put. It will not: the slope is steepest when volatility is low and inverts in stress, so the realized carry is front-loaded into calm periods and reverses violently in the periods that matter. Quoting an annualized roll cost from a static curve describes the typical month, not the year.

*What the question tests.* Computing a roll yield from a futures curve, and recognizing that the buyer's cost and the seller's premium are one quantity, not two.

---

**22.8** *A $-1\times$ inverse VIX product has \$2bn of NAV. Compute the rebalancing trade required if futures rise 30% in a day, state its direction, and explain why the requirement grows as the product loses money.*

*The rebalancing trade.* From Example 22.6: a fund with net asset value $A$ maintaining $-1\times$ exposure is short futures with notional $A$. If futures return $R$ over the day, its NAV becomes $A(1-R)$ so the new *target* short is $A(1-R)$, while the existing short has grown to $A(1+R)$ of exposure. The required trade is

$$A(1+R) - A(1-R) = 2AR$$

With $A = \$2\text{bn}$ and $R = +30\%$:

$$2 \times \$2\text{bn} \times 0.30 = \$\mathbf{1.2}\text{ billion}$$

and the NAV afterward is $\$2\text{bn} \times 0.70 = \$1.4\text{bn}$.

*The direction: it must **buy**.* The fund purchases \$1.2 billion of the very futures that have just risen 30%, into a market already moving against it. The rebalance is **destabilizing**, and it is Section 18.12's short-gamma feedback in a different form—such an exposure with no option behind it, manufactured entirely by a rule in a prospectus.

*Why the requirement grows as the product loses money.* Measure the trade against what the fund has *left*, which is what determines whether it can execute:

$$\frac{2AR}{A(1-R)} = \frac{2R}{1-R}$$

| Move | NAV remaining | Buy required, % of remaining NAV |
|---|---|---|
| +30% | 70% | 86% |
| +50% | 50% | 200% |
| +80% | 20% | 800% |
| +96% | 4% | 4,800% |

The ratio **diverges** as $R \to 1$. The numerator (the notional to be bought) rises with the move, while the denominator (the capital left to support it) falls with the same move. Two quantities moving in opposite directions produce a requirement that explodes, and there is no level at which the mechanism self-limits.

Example 22.6 puts it at the industry's scale. With roughly \$3 billion in inverse VIX products, a 96% move leaves \$0.12bn of capital and demands \$5.76 billion of buying, nearly twice the products' original size, due in the fifteen minutes before the futures close, when the market normally trades a small part of its daily volume. The rebalance was not a large order in a deep market; it was a large order in a thin one, announced in advance.

That is exactly what happened on February 5, 2018. The products' required purchase and the products' remaining capital had crossed. XIV's indicative value collapsed in the after-hours

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session; the note was redeemed early under the clause in its prospectus, and holders received \$5.99 against more than \$100 a few days earlier.

Three features distinguish this from a dealer's short gamma, and they are why it is worse. The dealer *collected a premium* and holds positive theta; the fund collected nothing but the roll yield of question 7, which is a different bet entirely. The dealer chooses when to hedge; the fund must trade *to the close*, on a published rule, so the flow is predictable and can be traded ahead of. And the dealer's exposure runs off at expiration, while the fund's rolling position never matures, so the convexity is permanent, and the only terminal event available is liquidation.

*The general lesson.* Convexity does not require an option. Any rule that mechanically scales exposure to a price produces gamma, and the sign is set by the direction of the forced trade. The question to ask of a strategy is not "does it hold options?" but "in which direction must it trade when the market moves, and how large is that trade relative to what it will have left?"

*What the question tests.* The $2AR$ rebalance, and that the danger is measured by the trade as a fraction of surviving capital rather than in absolute dollars.

---

**22.9** *Explain why a dispersion trade weighted $(K_I/K_c)^2$ makes exactly zero when realized correlation equals implied correlation, whatever the components' common volatility, and why it loses so much more in a crash than it earns in a normal quarter.*

*Why the weighting makes the trade exactly flat at implied correlation.* Take $n$ equally weighted components of common volatility $\sigma_c$ and correlation $\rho$. For large $n$, Section 19.7's relation is $\sigma_I^2 \approx \rho\,\sigma_c^2$, and applying it to the quoted strikes defines the implied correlation:

$$K_I^2 = \rho_{\text{impl}}\,K_c^2$$

Now short one unit of index variance notional and hold $w = (K_I/K_c)^2$ units of component variance. The profit is

$$\Pi = -\left(\sigma_{I,R}^2 - K_I^2\right) + w\left(\sigma_{c,R}^2 - K_c^2\right)$$

Substituting $\sigma_{I,R}^2 = \rho_R\,\sigma_{c,R}^2$ and $w = K_I^2/K_c^2 = \rho_{\text{impl}}$:

$$\begin{aligned}\Pi &= -\rho_R\,\sigma_{c,R}^2 + \rho_{\text{impl}}K_c^2 + \rho_{\text{impl}}\,\sigma_{c,R}^2 - \rho_{\text{impl}}K_c^2 \\ &= \boxed{\sigma_{c,R}^2\left(\rho_{\text{impl}} - \rho_R\right)}\end{aligned}$$

The $K_c^2$ terms cancel *identically*, which is what the weighting was chosen to do. So the position pays zero whenever $\rho_R = \rho_{\text{impl}}$, and note that it does so **whatever realized volatility turns out to be**, which is what "whatever the components' common volatility" in the question means. The trade is neutral to a parallel move in volatility and exposed to correlation alone.

That is exactly Section 22.11's construction. With 50 stocks struck at 30% and the index at 18%, the implied correlation is 34.7%: the exact $n = 50$ figure, $(0.0324-0.09/50)/(0.09\times0.98)$, against the large-$n$ approximation $(0.18/0.30)^2 = 36.0\%$, which is separately the component weight $w$. With the exact $n$ formula the identity is $\Pi = (1 - 1/n)\,\sigma_{c,R}^2(\rho_{\text{impl}} - \rho_R)$, still zero at $\rho_{\text{impl}}$ whatever the component volatility. Figure 22.5 shows the P&L crossing zero at 34.7% and linear in correlation on either side.

*Why it loses so much more in a crash than it earns in a normal quarter.* Read the boxed result again: the P&L is a *product* of two terms, and in a crash **both** move against the seller at once.

*Correlation rises.* In a market-wide decline everything falls together, dispersion collapses, and realized correlation runs toward one. So $\rho_R - \rho_{\text{impl}}$ goes from a modest negative (the

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premium being collected) to a large positive. Selling correlation at 34.7% and realizing 70% is 35 points of correlation against the seller, where the example's good quarter was 10 points in its favor.

*And the multiplier explodes.* The correlation error is scaled by $\sigma^2_{c,R}$, realized *component variance*, which in a crash is several times its normal level. A quiet quarter collects a small correlation edge times a modest variance; a crash pays a large correlation error times a huge one. The two effects compound **multiplicatively**, so the loss is not twice a typical gain but many times it: eight times in Example 22.8.

Figure 22.5's two curves show precisely this. At unchanged component volatility of 30%, realized correlation of 70% costs 8.65 index volatility points. Raise component volatility to 45%, which is what accompanies a correlation of 70%, and the same correlation costs **19.46** points. The second curve is not a different scenario from the first; it is the same scenario with the variance multiplier restored.

So dispersion is a short insurance position, with the payoff profile that implies: a steady premium in ordinary quarters, and a loss in the rare state that is far larger than the sum of the premiums collected before it. The trade is not a correlation *arbitrage*. Implied correlation sits above realized correlation on average for the same reason implied volatility sits above realized (Section 19.10): someone must be paid to bear the state in which diversification stops working, and the dispersion seller is that someone.

*One practical rider follows.* The clean algebra above requires variance swaps on the single stocks, and since 2008 those trade only capped and thinly (question 12). A desk running dispersion with straddles instead does not get the exact cancellation, only an approximation that degrades as spot moves, which is question 12's subject.

*What the question tests.* That the $(K_I/K_c)^2$ weight is what isolates correlation, and that the crash loss is the product of two simultaneous moves rather than either one alone.

---

**22.10** *Implied volatility is 22% and realized comes in at 18%. Compute what a short variance swap earns in volatility points, and explain why a short delta-hedged straddle need not earn the same and why it typically earns less.*

*What the short variance swap earns.* Per unit of vega notional, the payoff to the short is

$$\frac{K^2_{\text{var}} - \sigma^2_R}{2K_{\text{var}}} = \frac{22^2 - 18^2}{2 \times 22} = \frac{484 - 324}{44} = \mathbf{3.64} \text{ volatility points}$$

Note that it is not 4. Realized volatility undershot the strike by four points, but the short collects only 3.64 of them, 91% of the naive figure. The convexity of Section 22.4 works *against* the short in both directions: it shrinks the gain on a favorable move and amplifies the loss on an adverse one. Had realized come in four points *above* at 26%, the short would lose $(676 - 484)/44 = 4.36$ points. Down four, earn 3.64; up four, lose 4.36. The elevated strike is what the short is paid for accepting that asymmetry.

*Why a short delta-hedged straddle typically earns less.* It need not: on a path that stays near the strike its gamma is high when the moves occur and it can collect more. Four reasons account for the usual shortfall, in descending order of importance.

*1. Non-constant dollar gamma, hence path dependence.* This is the essential difference, and question 2 supplies it. A delta-hedged position collects $\frac{1}{2}S^2\Gamma\left[(\Delta S/S)^2 - \sigma^2_{\text{impl}}\Delta t\right]$ each day. The log contract's dollar gamma is a *constant* 2, so the variance swap collects every day's squared return at equal weight and delivers exactly $\sigma^2_R - K^2_{\text{var}}$. A straddle's dollar gamma peaks at the strike and dies away from it, so what it collects is a *gamma-weighted* average of daily squared returns. If the index drifts away from the strike, the straddle stops earning almost

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entirely: the position can be right about volatility and collect a fraction of it, purely because the market moved somewhere the position has no gamma. The straddle is a bet on volatility *near the strike*; the variance swap is a bet on volatility.

*2. It sells at a lower price.* A straddle is struck at the money, so it sells the at-the-money implied volatility. The variance swap strike sits **above** that, by the skew premium of question 6, because the replication loads on expensive out-of-the-money puts. The two positions are short volatility at different prices, and the variance swap seller has the better one. Where 22% is the variance strike, the at-the-money implied the straddle sells is lower by the skew premium, so part of the 3.64 points is premium the straddle seller never sees.

*3. Transaction costs.* The variance swap is a single trade with a static hedge. The straddle must be delta-hedged continuously, and Section 18.9's table showed the cost of that rebalancing to be a real deduction: even at the frequency minimizing mean cost plus one standard deviation, the mean cost of hedging ran 0.23 per option above the formula value in a controlled simulation. Every rebalance is paid for, and the frequency that minimizes total cost still costs something.

*4. Vega decay.* A straddle's vega shrinks as it approaches expiration and as it moves away from the money, so the position's size in volatility terms is not constant even if nothing is traded. Maintaining a target short-volatility exposure requires re-striking, which is more trades and more cost.

*The one respect in which the straddle is better*, and it is not small: its gamma is bounded and decays away from its strike, so its loss on a large move grows roughly **linearly**, while the variance swap's constant dollar gamma makes its loss quadratic at every level. That is why the dealers who *sell* single-stock volatility write variance swaps only with a cap. The dispersion desk, which is long that volatility, moved to straddles for a different reason (question 12).

*What the question tests.* That the convexity is symmetric in form and asymmetric in effect, and that the variance swap's advantage over a straddle is constant dollar gamma bought with a worse tail.

---

**22.11** *A special dividend of 8% is paid on a stock during the life of a variance swap that does not adjust for dividends. Explain what happens to measured realized variance and who gains.*

*What happens to measured variance.* Realized variance is computed from **price** returns, and a price drops on the ex-dividend date. An 8% special dividend produces a one-day log return of

$$\ln(0.92) = -0.0834$$

which is not volatility—it is a scheduled, known transfer of value from the share price to the shareholder's cash account. But the contract does not know that, and the return enters the sum of squares like any other.

The contribution is

$$\frac{252}{n}\,(\ln 0.92)^2$$

For a one-year swap ($n = 252$) that is 0.00695 of variance. Against a base realized volatility of 20% (a variance of 0.0400), the measured figure becomes

$$\sqrt{0.0400 + 0.00695} = \mathbf{21.67}\%$$

So a single dividend adds **1.67 volatility points** of variance that never happened.

And the shorter the contract, the worse it is, because the annualization factor $252/n$ scales the single squared return up. On a three-month swap ($n = 63$) the same dividend adds 0.0278 of variance, and 20% becomes

$$\sqrt{0.0400 + 0.0278} = \mathbf{26.04}\%$$

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**Six volatility points** from one day. The added variance is inversely proportional to the contract's length, so a dividend landing inside a short-dated swap can dominate the entire payoff.

*Who gains: the **long**, the buyer of variance*, at the seller's expense. The buyer receives $N_{\text{var}}(\sigma_R^2 - K_{\text{var}}^2)$ and the measured $\sigma_R^2$ has been inflated by something with no volatility content. On a one-year swap struck at 20% with \$100,000 of vega notional, the transfer is

$$\frac{100{,}000}{2 \times 20}\left(21.67^2 - 20^2\right) = 2{,}500 \times 69.6 \approx \$174{,}000$$

of pure documentation risk: a payment the strike never priced.

*Why this is a live exposure and not a footnote.* Many contracts, especially single-stock ones, adjust for extraordinary dividends, adding back the dividend actually paid; index contracts usually do not. Terms differ, which is the reason to read the confirmation rather than the term sheet. A special dividend announced before the trade is known to both sides and is priced into the strike, but not by the strip alone: the strip prices the diffusive variance, and the dealer must add the dividend's own squared return, $(\ln 0.92)^2/T$. The transfer arises from one declared afterward.

*The general lesson*, which Section 22.2 states and this question makes concrete: realized variance is defined, not observed. The contract specifies the observation times, the return convention, the annualization factor, whether the mean is subtracted, and how corporate actions are handled. Two swaps with the same strike and different conventions are different contracts. This one belongs with the three other gaps of Section 22.10 (jumps, single-stock convexity, and the strip drifting out of range) as the things that separate an exact replication from a dealer's actual P&L.

*What the question tests.* That a price return and a total return are not the same thing, that the distortion scales with $252/n$, and that the gain accrues to whichever side the documentation happens to favor.

---

**22.12** *Explain why dispersion desks often use straddles rather than variance swaps on single stocks, and what they give up by doing so.*

*Why they switched.* Because single-stock variance swaps, capped from the start, nearly stopped trading after 2008 and have stayed thin, and the reason is specific to single stocks: **convexity applied to an underlying that gaps**.

Question 5's arithmetic is the general problem: the seller's loss is quadratic in realized volatility and unbounded above. On an index that is dangerous. On a single stock it is unmanageable, because a stock can move 40% in one day on an earnings miss or a takeover bid, and a single 40% fall contributes $(\ln 0.6)^2 = 0.261$ to a sum whose whole annual value at 35% volatility would be 0.1225. One day is worth two years. Index moves of that size are rare and diversified across constituents; single-stock moves of that size are a normal feature of the earnings calendar. The distribution the seller is writing against has a far fatter tail than an index's, and 2008 established that the premium had never been sized for it.

So sellers write single-stock variance only with a cap, and dispersion desks often use **straddles or vanilla options** on the single stocks, with variance swaps on the index.

*What they give up.*

*1. Constant dollar gamma, and with it path independence.* This is the loss that matters, and it is exactly the property the instrument existed to provide. The log contract's dollar gamma is a constant 2 (question 2), so a variance swap delivers realized variance regardless of where the stock was when it moved. A straddle's dollar gamma peaks at the strike and dies away from it, so the desk now captures a *gamma-weighted* average of squared returns. As Section 22.10 puts

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it, the substitution reintroduces exactly the path dependence the instrument was designed to remove.

*2. The clean correlation exposure.* Question 9's cancellation (P&L = $\sigma^2_{c,R}(\rho_{\text{impl}} - \rho_R)$, with the $K_c^2$ terms vanishing identically) depends on both legs being linear in realized variance. Replace the component leg with straddles and the cancellation becomes an approximation that degrades as the single stocks move away from their strikes. The position acquires a residual volatility exposure that was not intended and is not constant, so a trade sold to a risk committee as a correlation position is now a correlation position plus a drifting volatility position.

*3. Operational cost.* Each stock must be delta-hedged individually and re-struck as it moves, against a single trade with a static hedge. On a fifty-stock dispersion book that is fifty hedging programs, fifty sets of transaction costs (question 10), and fifty opportunities for the hedge to be stale. Some of the premium the trade is designed to collect is spent maintaining it.

*4. Basis between the legs.* The index leg is still a variance swap, linear in realized variance up to its cap, while the component legs are straddles. The two legs now respond differently to the same market, so the hedge is imperfect in a way that has nothing to do with correlation, and the residual is largest in exactly the fast markets where the correlation view is being tested.

*What they gain, since the trade was not irrational.* A component leg that trades freely and keeps its crash payoff. The desk is *long* single-stock volatility, so the quadratic tail was never its own problem: it was the problem of the dealers who had to sell that volatility, and they now sell it only capped. A capped swap leaves the long desk short a call on variance, which gives away the payoff on the large single-stock moves the component leg is long. A straddle's seller faces a loss that grows linearly with the move rather than quadratically with accumulated variance, so straddles are still offered, at prices that carry no charge for a quadratic tail. The desk has exchanged four measurable inefficiencies for a leg it can actually trade, which is the correct trade even though every line of the comparison except the last favors the old instrument.

*The broader pattern is worth naming.* A market found an elegant, model-free instrument; discovered that the elegance depended on a payoff no one could bound; and either capped it or retreated to a clumsier product whose risks it could see. Single-stock variance survives only *with a cap*, and some index contracts carry one too, which is the same retreat in a milder form. Neither move was a failure of the theory—the replication is still exact under its assumptions—but the assumptions and the tail were the same object viewed from two ends.

*What the question tests.* That the single-stock tail is what nearly killed the instrument, and that substituting straddles restores the path dependence variance swaps were invented to eliminate.

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# Chapter 23

# Short-Dated Options, Dealer Positioning, and Market Feedback

**23.1** *Explain why 0DTE contract volume overstates the amount of risk being transferred.*

Because a contract is a unit of *counting*, not a unit of *risk*, and at zero maturity the two diverge violently.

*1. A 0DTE contract carries a fraction of the risk of a longer-dated one.* From Table 23.1, an at-the-money call on the index at 5000 is worth 110.47 with thirty days to run and 3.71 with fifteen minutes left. Selling one of each is recorded as two contracts; the premium at risk differs by a factor of **thirty**. Vega shrinks by about the same factor (both scale with $\sqrt{T}$; the ratio is 28), while gamma moves the other way, 28 times *larger* on the 15-minute option. A volume figure that adds the two treats them as equals in risk, when one carries a twenty-eighth of the vega and twenty-eight times the hedging.

*2. The same risk is counted many times over.* A 0DTE position is opened and closed within hours, and each leg prints. A trader who buys at 10:00 and sells at 11:30 generates two contracts of volume and ends the day flat, having transferred risk for ninety minutes and none overnight. Longer-dated positions are opened and held, so volume and risk transfer stay closer together. Intraday round-tripping is the norm here, not the exception, which is why 0DTE volume can exceed half of SPX option volume.

*3. Spread structures multiply the count.* Systematic premium sellers trade defined-risk verticals and iron condors, so a single risk decision prints two or four contracts. The counting convention rewards the structures that use the most legs.

*4. Published open interest cannot see most of it.* Open interest is reported at the start of the day, so a position opened and closed within the session never appears in it, and the comparison that circulates sets a day's *flow* of 0DTE contracts against a *stock* of longer-dated ones.

*Why this matters beyond pedantry.* The headline "0DTE is now half of SPX option volume" has been used to argue that a large and growing share of equity risk has migrated to instruments that expire in hours. The better measures do not support the claim on the same scale: measured in premium, in vega, or in net dealer gamma, 0DTE is a much smaller share of the market than the contract count suggests, and Section 23.12 records that the aggregate net dealer gamma attributable to 0DTE appears small relative to the longer-dated book.

*What the honest measures are.* Premium traded, aggregate vega, aggregate gamma, and net dealer positioning, though question 8 shows the last of these is not observable either. None is as easy to publish as a contract count, which is most of why the contract count is what gets published.

*And one thing the volume figure does measure correctly.* Trading *activity*, and therefore the exchange's and market maker's revenue, the operational load on clearing, and the sheer number of decisions being made. Those are real, and a market structure question about capacity and message traffic is properly asked in contracts. It is the risk question that the contract count

answers badly.

*What the question tests.* That a count is not an exposure, and which measures to substitute.

---

**23.2** *State how at-the-money gamma behaves as $T \to 0$, give the ratio between one-hour and thirty-day gamma from Table 23.1, and explain why the narrowing of the gamma band matters more than the growth in gamma itself.*

*How at-the-money gamma behaves.* It **diverges like** $1/\sqrt{T}$. Section 18.3's formula makes it immediate:

$$\Gamma = \frac{N'(d_1)}{S\sigma\sqrt{T}}$$

At the money $N'(d_1)$ tends to a constant, so gamma grows without bound as $T \to 0$. There is no limiting value; the at-the-money option's curvature becomes arbitrarily large.

*The ratio.* From Table 23.1, one-hour gamma is 0.021528 against 0.001534 at thirty days:

$$\frac{0.021528}{0.001534} = \mathbf{14.0}$$

and the gamma term gives the same figure, 26.91 against 1.92, though the exact P&L does not (17.55 against 1.89), because a 1% move at one hour leaves the band. The square-root law confirms it: measuring time in *trading* hours, thirty days is $30 \times 6.5 = 195$ hours, and $\sqrt{195} = 13.96$. The table is $1/\sqrt{T}$ working almost exactly; the small remainder is $N'(d_1)$ at thirty days sitting just below its at-the-money limit.

*Why the narrowing of the band matters more than the growth.* Because gamma is only *somewhere*. Integrated across index levels, an option's gamma always totals one, since its delta runs from 0 to 1; as expiration approaches, that same total is compressed into an ever-narrower range of spot. At thirty days an option retains at least half its at-the-money gamma five or six percent from the strike; at one hour it is concentrated within a few tenths of a percent, and outside that band it is essentially zero. The delta surface is collapsing into a **step function**, and gamma is the height of the step divided by its width.

Three consequences follow, and two are later questions in this set.

*Hedging demand becomes a step function of the index level, not a smooth function of it.* A dealer's required trade for a given move depends discontinuously on where the index happens to be relative to the strike. The same 0.5% move produces billions of dollars of hedging for Example 23.6's 20,000-contract book if it starts at the strike, and comparatively little if it starts a percent away and moves further off. A risk system computing a sensitivity at the current spot describes the market's behavior only for as long as spot stays where it is.

*The stabilizing force saturates.* Example 23.6 measures it: a long-gamma dealer book sells $4.1 billion into a 0.5% move and only $876 million more over the next half percent, because delta is already 0.912 and can rise only to 0.996, so there is little left to sell. Large gamma inside a narrow band is a powerful restoring force for small moves and no force at all for large ones (question 7).

*And the exposure can vanish between one print and the next.* A move that carries the index out of the band takes the whole book from maximum gamma to none, so the market's character changes abruptly rather than gradually. That is question 12's dangerous configuration.

*The contrast worth holding onto.* A large but *broad* gamma, a monthly book, produces a hedging flow that is proportional, continuous, and predictable from a single sensitivity number. A large and *narrow* gamma produces a flow that is enormous, local, and discontinuous, and no single number summarizes it. The growth in gamma changes the size of the flow; the narrowing changes its *character*, and character is what determines whether a market breaks.

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*What the question tests.* The $1/\sqrt{T}$ divergence and its arithmetic, and the distinction between how much gamma there is and how it is distributed across spot.

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**23.3** *An at-the-money 0DTE call costs 0.38% of spot. State the move required to break even at expiration, compare it with a one-standard-deviation daily move, and explain why the bid–ask spread is a much larger percentage cost late in the day than early.*

*The break-even move.* An at-the-money call costing 0.38% of spot must finish 0.38% in the money to return the premium, since at expiration it is worth $\max(S_T - K, 0)$ and $K = S_0$:

$$\text{index must rise } \mathbf{0.38}\%$$

On the index at 5000 that is 19 points, matching the 19.10 premium of Example 23.1.

*Against a one-standard-deviation daily move.* At 15% implied volatility the one-day standard deviation is

$$\frac{15\%}{\sqrt{252}} = 0.945\% \qquad \text{or 47 index points}$$

so the break-even is

$$\frac{0.38}{0.945} = \mathbf{0.40} \text{ standard deviations}$$

That is not a lottery ticket, and the distinction matters. A hurdle of 0.4 standard deviations is cleared about 34% of the time, so the buyer is not facing long odds in the way the retail framing suggests. The option is priced without obvious error: its expected return is negative, set mainly by the volatility risk premium of Section 19.10, a modest, systematic edge against the holder of the same kind as buying any option.

*What makes it feel different is the shape, not the edge.* Example 23.1 shows the decay with the index unmoved: 100% of premium at the open, 78% at four hours, 39% at one hour, 19% at fifteen minutes. **Sixty-one percent of the premium is gone by the last hour if nothing happens.** The position is either right quickly or worthless, so a modest negative expected return is delivered as a rapid sequence of complete wins and complete losses rather than a slow loss, a design human beings find far more engaging, and one that is fair to describe as a feature of the product's appeal rather than an accident of it.

*Why the bid–ask spread is a much larger percentage cost late in the day.* Because the *spread* is quoted in ticks and does not shrink, while the *premium* collapses. A one-tick spread of ten cents (SPX's minimum tick for an option priced at \$3 or more) against a premium of 19.10 at the open is

$$\frac{0.10}{19.10} = 0.52\% \text{ of premium}$$

The same dime against 3.71 with fifteen minutes left is

$$\frac{0.10}{3.71} = 2.7\%$$

five times as expensive for the identical contract, purely because the denominator has fallen.

Two things aggravate this. The tick size is a *floor*: a market maker cannot quote a spread narrower than the tick even where competition would justify it, so the relative cost is bounded below by the exchange's rules rather than by liquidity. And **the volume is heaviest late**, so the traders paying the highest percentage cost are the majority of the traders, not a careless minority. Round-tripping a position in the final hour costs the whole spread: 1.3% of premium with an hour left and 2.7% with fifteen minutes on Example 23.1's figures, and more on the cheaper out-of-the-money strikes, so execution cost is not small beside the theoretical disadvantage.

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*What the question tests.* Converting a premium into a break-even and into standard deviations, and that percentage transaction cost is driven by the collapse of the denominator.

---

**23.4** *Define charm, explain why it is negligible for a monthly option and dominant for a 0DTE option, and use Example 23.2 to state three properties that distinguish charm hedging from gamma hedging.*

*The definition.* **Charm** is the rate at which delta changes with the passage of time,

$$\text{charm} = \frac{\partial \Delta}{\partial t} = \frac{\partial^2 f}{\partial S \, \partial t}$$

where $f$ is the option's value, sometimes called delta decay. It is the sensitivity that answers "how much will my hedge need to change if nothing happens?"

*Why it is negligible for a monthly option.* With thirty days to run, an at-the-money option's delta is 0.5323 (Table 23.1) and it drifts by a fraction of a percent per day. The whole delta surface is smooth in $S$ and changes shape slowly, so a day's passage barely moves the required hedge. Section 18.6 mentions charm and moves on, correctly, because for a monthly book it is a rounding error against the gamma flow.

*Why it is dominant for a 0DTE option.* As expiration approaches, an out-of-the-money option's delta must go to *zero* and an in-the-money option's to *one*. The delta surface is collapsing into a step function, and—this is the point—**that collapse happens on the clock, not on the price**. With the index perfectly still, a delta of 0.30 becomes 0.03 by the last half hour and zero at the close. For a monthly option there is no such deadline; for a 0DTE option the deadline is hours away and the entire surface must reach its terminal shape by the close.

*The size, from Example 23.2.* Take a call struck 0.5% above spot (half a daily standard deviation, an ordinary 0DTE strike), with the index pinned at 5000 all day:

| Time left | Delta | Charm, per hour |
|---|---|---|
| 6.5 hours | 0.3041 | −0.0146 |
| 2 hours | 0.1728 | −0.0614 |
| 1 hour | 0.0902 | −0.1099 |
| 0.5 hours | 0.0288 | −0.1254 |

A dealer short 10,000 of these contracts hedges by holding $1.52 billion of index at the open and $451 million with an hour left, so it must *sell* roughly $1.07 billion of index by the last hour (all $1.52 billion by the close), and the index has not moved at all.

*Three properties distinguish charm hedging from gamma hedging.*

*1. It is one-directional.* Gamma hedging buys on the way up and sells on the way down, so a round trip in the index leaves the hedge where it started and the flow *nets out*. Charm hedging runs **one way all day**, its direction fixed by the clock rather than by the price. There is no offsetting flow later, because time does not go back.

*2. It is predictable.* The only input is the passage of time. Anyone who knows the open interest by strike can compute the required flow in advance, to the hour, before the session starts. Gamma hedging requires knowing what the index will do; charm hedging requires only a calendar. A predictable flow is one that can be traded ahead of, which changes who bears its cost.

*3. It accelerates.* The size of charm rises from 0.015 to 0.125 per hour, more than **eightfold**, so the flow does not run evenly through the day but is fastest in the final hour. That is also when 0DTE volume is heaviest and when the day's other closing trades are competing for the same buyers and sellers, which is question 10's point about depth against volume.

*What is and is not in doubt.* Whether this flow stabilizes or destabilizes depends on the sign of dealer positioning, which is not observable (question 8). What is not in doubt is that a large,

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predictable, clock-driven flow now exists in the final hour of the US equity session, and that ten years ago it was a fraction of its size.

*What the question tests.* That charm is the clock's effect on delta, that it scales like the collapse of the delta surface rather than like time to expiration, and that a one-directional predictable flow is a different market-structure object from a mean-reverting one.

---

**23.5** *The overnight period carries 25% of daily variance. Compute the implied volatility a 0DTE option will show in a market whose close-to-close volatility is 16%, explain why this is not a term-structure effect, and state which contract would hedge an announcement made after the close.*

*The implied volatility shown.* A 0DTE option bought at the open expires at the close: it covers the *session* and not the night, yet it is quoted with a full trading day as its time to expiration. If the overnight period carries a fraction $\omega$ of daily variance, the session carries $1 - \omega$, and the implied volatility that reproduces the option's price is

$$\sigma_{\text{session}} = \sigma_{\text{daily}}\sqrt{1-\omega}$$

With $\omega = 25\%$ and a close-to-close volatility of 16%:

$$16\% \times \sqrt{0.75} = 16\% \times 0.8660 = \mathbf{13.86}\%$$

The option prints about **13.9%** against a market whose volatility is 16%, a shortfall of 2.14 points, or 13%.

*Why this is not a term-structure effect.* A term structure is a statement about *expectations*: that volatility over the next month is expected to differ from volatility over the next year, because volatility mean-reverts (Section 19.6). Nothing of the kind is happening here. The market's volatility is 16% at every horizon in this example. What has changed is the **measurement window**: the contract's life excludes a quarter of the day's variance, while the pricing formula divides by a $T$ that includes it. The discrepancy is a mismatch between the clock the option runs on and the clock the model is told about: an artifact of the quoting convention, not a feature of the market.

The distinction has teeth because the two are routinely mixed. For a US index, where most news arrives during the session, $\omega$ is typically 15–25%, so on the trading-day clock 0DTE implied volatilities print roughly a tenth below the 30-day implied volatility for purely mechanical reasons (a system that counts calendar minutes makes the opposite error). This can be misread as evidence that short-dated options are *cheap*. They are not. And plotting the 0DTE point on the same axes as the 30-day and one-year points produces a term structure with a **spurious kink at the front**: a real phenomenon (mean reversion) contaminated by a measurement artifact, with no way to separate them after the fact.

*The fix, where one is needed*, is to price in *business time*: measure $T$ in units of expected variance rather than calendar time, so that the overnight period is either included at its proper weight or excluded along with the variance it carries. Desks that trade the front of the surface do this; systems that publish surfaces for general consumption often do not.

*Which contract hedges an announcement made after the close:* ***tomorrow's*** *expiration, not today's.* Today's 0DTE option expires at the close, so it has settled and paid before the announcement is made. It cannot hedge the gap because the gap happens after the contract has ceased to exist. An institution using daily options to hedge event risk must own **the expiration that covers the event**, which for an after-the-close announcement means the following day's contract, whose life spans the overnight period.

*And note what that does to the price.* Tomorrow's contract includes the overnight variance *and* the event, so it will print a *higher* implied volatility than today's, partly for the mechanical

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reason above running in reverse, and partly because it genuinely contains the risk. The two effects are again confounded, and a hedger comparing the two quotes and concluding that the event is expensively priced may be reading the measurement window.

*What the question tests.* That $\sqrt{1-\omega}$ scaling is a convention artifact rather than an expectation, and that hedging an event requires owning the contract whose life contains it.

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**23.6** *Using the $T^{-0.4}$ law, compute how much steeper a four-hour smile is than a 30-day smile, and explain why quoting a 15-minute option in implied volatility terms conveys less information than quoting its price.*

*The computation.* With the at-the-money skew following $\psi(T) \sim T^{-0.4}$, the ratio of skews at two maturities is

$$\frac{\psi(T_1)}{\psi(T_2)} = \left(\frac{T_2}{T_1}\right)^{0.4}$$

Measuring $T$ in trading days of 6.5 hours (the convention Table 23.1 uses), four hours is $4/6.5 = 0.615$ days, so

$$\left(\frac{30 \times 6.5}{4}\right)^{0.4} = (48.75)^{0.4} = \mathbf{4.73}$$

The four-hour smile is roughly **4.7 times as steep** as the 30-day smile. In the units of Section 23.6's table, where the 30-day smile is 2.5 volatility points per 1% of moneyness, the four-hour smile is about **11.8 points per 1%**, sitting, as it should, between the one-day figure of 9.7 and the one-hour figure of 20.6.

*Why quoting a 15-minute option in implied volatility terms conveys less information than quoting its price.* Because at that maturity the map from price to implied volatility becomes extraordinarily steep, so implied volatility is a **badly conditioned coordinate**: tiny, economically meaningless differences in price map to enormous differences in the quoted number.

Three ways to see it all come to the same fact.

*Vega has collapsed.* Implied volatility is defined by inverting the price through vega, and a 15-minute option's vega is nearly zero. Dividing a price difference by a vanishing vega amplifies it without limit. A one-tick disagreement between two market makers is about two-fifths of a volatility point on the at-the-money option worth 3.71 (a ten-cent tick), but on a put far enough below spot to be priced at a dime it is about two points (a five-cent tick), so two quotes describing essentially the same price describe quite different volatilities.

*The tick size sets a resolution floor.* Prices are quoted on a discrete grid. Near expiration the grid is coarse relative to the option's value, so the set of achievable implied volatilities is a sparse, widely spaced set. The quoted number carries spurious precision: it has more digits than the underlying price grid can support.

*The skew swamps the level.* At about 36 volatility points per 1% of moneyness at fifteen minutes (20.6 at one hour), a put one percent out of the money quotes tens of points away from the at-the-money level. So the implied volatility of a 0DTE option tells you mostly *which strike it is*, and only secondarily anything about the market's view of volatility, which is the opposite of what the coordinate was introduced to do. Sections 15.7 and 19.12 described implied volatility as a language for quoting prices; at these maturities the language distorts.

*The price, by contrast, is the thing.* It is what is paid, it is quoted on the exchange's own grid with no transformation, and it needs no model to interpret. Section 23.6 draws the conclusion directly: quoting in volatility terms conveys less information than quoting the price.

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*And the steepness is not a quoting artifact.* It is evidence about the process. A path with smoothly varying volatility cannot travel far in fifteen minutes, so a classical diffusion gives an out-of-the-money 0DTE option essentially no value—the market gives it real value. Whatever generates this smile is not a classical diffusion with smoothly varying volatility, and the 0DTE market is the sharpest available evidence for the jumps and roughness of Chapter 21. It is also, uncomfortably, exactly where the retail flow goes: the out-of-the-money strikes a directional intraday buyer naturally reaches for are the ones whose implied volatilities are most extreme.

*What the question tests.* Applying the power law across maturities in consistent time units, and recognizing when a change of coordinates destroys information rather than organizing it.

---

**23.7** *Explain why the pinning force of Example 23.6 saturates, and why a market that damps small moves but not large ones may be worse than one that damps neither.*

*Why the force saturates.* Because delta is bounded. A dealer long 20,000 call contracts at a strike hedges by selling as the index rises through it, and the amount to sell is set by how much delta has *changed*, but delta can only travel from about 0.50 to 1.00. Once it arrives, there is nothing left to sell.

Example 23.6 measures the exhaustion, with one hour to expiration:

| Index move | Delta goes from | Dealer sells |
|---|---|---|
| +0.10% | 0.502 → 0.609 | \$1,063m |
| +0.50% | 0.502 → 0.912 | \$4,114m |
| +1.00% | 0.502 → 0.996 | \$4,990m |

A tenth of a percent generates a *billion dollars* of stabilizing supply: genuine pinning, and the reason an index often trades in a narrow range near a strike with large open interest. But the first half percent generates \$4.1 billion and the second half percent only \$876 million more. The stabilizer is strongest for the moves that did not need stabilizing and absent for the ones that did.

Contrast a 30-day book, which sells \$382 million on a 0.5% move and \$760 million on a 1% move, smoothly and roughly *in proportion*. The monthly book's response is linear in the move; the 0DTE book's is a saturating curve. That is question 2's narrow gamma band expressed as a supply schedule.

*Why a market that damps small moves but not large ones may be worse than one that damps neither.* The damage is informational, and it operates through what the price series stops telling people.

*Small fluctuations are how stress announces itself.* In an ordinary market, accumulating pressure (a build-up of positioning, a deteriorating bid, a widening dispersion) shows up first as elevated small-scale variability, well before anything breaks. Participants read it, size down, and hedge. Damping the small moves **removes the early warning** while leaving the underlying stress exactly where it was.

So the large move, when it comes, is more surprising than it should have been. Not larger, necessarily—but arriving out of a calm that was manufactured rather than real. Risk systems calibrated on the damped series report low realized volatility, so value-at-risk falls, position limits loosen, and leverage rises into a market whose measured tranquility is an artifact of the hedging flow. Chapter 38's procyclicality is this mechanism in the risk-management vocabulary.

*And the transition is discontinuous.* The market does not gradually become less stable as the move grows; it is heavily stabilized right up to the edge of the band and then unstabilized, because the dealers who were supplying liquidity have finished supplying it. The regime change happens at a spot level no one publishes (question 12).

A market that damps nothing is at least *honest*: its volatility is its volatility, the warning

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signs are visible, and participants size accordingly. A market that damps selectively delivers a false measurement, and false measurements propagate into every system that consumes them. That asymmetry is the defensible form of the worry in Section 23.14, and it is worth separating from the louder and less supported claim that 0DTE hedging amplifies moves outright; Section 23.12 finds the evidence for that mixed and the sample short.

*What the question tests.* That a bounded delta implies a bounded restoring force, and that suppressed volatility is a measurement problem before it is a stability problem.

---

**23.8** *A published gamma-exposure figure assumes dealers are long all calls and short all puts. State what is measured and what is assumed, and explain when the assumption is least defensible.*

*What is measured.* Only **open interest by strike** (the number of contracts outstanding at each strike and expiration, published by the exchange), together with each contract's gamma, which follows from the observable price, strike, maturity and implied volatility. Both are solid facts.

*What is assumed: the **sign**.* The construction assumes that dealers are long all calls and short all puts, aggregates the gammas with those signs, and reports a "gamma exposure" and a "flip level," the index level at which net gamma changes sign. But open interest records *that* a contract exists; it does not record *who is on which side*. Every contract has a buyer and a seller, and nothing in public data says which is the dealer.

Example 23.7 shows how much the assumption decides. Take 100,000 calls and 100,000 puts outstanding at one strike, with identical gamma per contract:

| Assumed split (calls long / puts short) | Net dealer gamma (thousands of contracts) |
|---|---|
| 100% / 0% | +200 |
| 60% / 40% | +40 |
| 50% / 50% | 0 |
| 40% / 60% | −40 |
| 100% / 100% | 0 |

The sign flips between the 60/40 and the 40/60 assumption, and nothing in public data determines which is right. The last row is the rule the question names: applied to equal call and put open interest it returns exactly zero, because the assumption is symmetric, not because dealers are flat. So a published gamma-exposure level should be read as a calculation of what dealer positioning would be if the assumed split were true, a considerably weaker statement than it usually receives, and one whose conclusion is contained in its premise.

*When the assumption is least defensible: when the flow is two-sided*, which is precisely the 0DTE case.

The assumption originates in a market that no longer describes this one. It is roughly right where investors sell calls against stock they hold and institutions buy puts for protection, leaving dealers long calls and short puts. But Section 23.7 argues that 0DTE flow is close to two-sided: retail directional buyers push one way and systematic premium sellers push the other (question 10), so the true split is near 50/50, exactly the point at which the naive rule is furthest from the truth and the computed sign is least informative. The assumption is worst where it matters most.

Two aggravating factors follow. In **single stocks** the assumption fails harder still, because a single large institutional trade can dominate a strike's open interest and there is no aggregate tendency to appeal to. And the assumption is at its worst in **unusual markets** (a heavy retail call episode, a large hedging program), which are the occasions on which someone bothers to look at the number.

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*And the flip level inherits the whole error.* It is the level at which the computed net gamma changes sign, so an error in the assumed split does not merely scale the number—it moves the level, or removes it entirely. A "gamma flip" that is quoted to the index point rests on a 50-versus-60 percent guess about who owns what.

*What this does not mean.* It does not mean dealer gamma is unimportant: Section 23.7's mechanism is real, and the hedging flows of questions 4 and 10 genuinely exist. It means the published *estimate* of its sign is a hypothesis, and that a trading decision resting on it rests on the hypothesis rather than on the data. The honest use is as a scenario ("if dealers are positioned this way, then"), with the conditional left visible.

*What the question tests.* Separating the observable (open interest, per-contract gamma) from the asserted (the side), and recognizing that an assumption is least reliable exactly where it is most consequential.

---

**23.9** *A 0DTE option shows 13.4% implied volatility with no event scheduled. A 2 p.m. announcement adds a move of 0.8% standard deviation. Compute the morning implied volatility and its value just after 2 p.m., with two hours left, and explain why the fall is not a fall in volatility.*

*The morning implied volatility: 18.5%.* Follow Example 23.4. With 15% close-to-close volatility and a fifth of the day's variance overnight, an ordinary session has a standard deviation of 0.845%, which is what the 13.4% measures. Variances add, so the announcement raises the session's standard deviation to

$$\sqrt{0.845\%^2 + 0.8\%^2} = \sqrt{0.714 + 0.640}\,\% = 1.164\%$$

and the implied volatility in proportion, $13.4\% \times 1.164/0.845 = \mathbf{18.5}\%$, a multiple of 1.38, between the 0.75% and 1.00% rows of the example's table.

*Just after 2 p.m.: back to 13.4%.* Two hours of the 6.5-hour session leave ordinary variance with a standard deviation of $0.845\% \times \sqrt{2/6.5} = 0.469\%$. Just before the announcement the event is still in the window, $\sqrt{0.469\%^2 + 0.8\%^2} = 0.927\%$, which annualizes over two hours to $13.4\% \times 0.927/0.469 = 26.5\%$. Just after it only the 0.469% remains, which annualizes to **13.4%**: a fall by a factor of 1.98 in two minutes, and of 5.1 points from the morning's 18.5%.

*Why the fall is not a fall in volatility.* The market's ordinary volatility is the same all day. An implied volatility is the variance left in the window, annualized, and in the morning the window held one known minute carrying nearly three times the variance of the rest of the afternoon. When the minute passes its variance leaves the window, and the quoted figure falls with it: the calendar, not a revision. On the business-time clock of Section 23.9 nothing has moved; the variance budget has been spent. A trader who bought at 1:59 for a rise in volatility misread the instrument; one who bought at 1:59 and sold at 2:01 bought the event and sold its aftermath.

*What the question tests.* That variances add within the session, so a scheduled event lifts the implied volatility until it happens and then leaves it.

---

**23.10** *Explain why retail flow makes dealers short gamma and systematic premium selling makes them long. A dealer book is net short $800m of gamma per 1%. Compute the hedging trade a 1.5% move requires, state its direction, and explain why the same flow can be negligible against daily volume and material against the depth at 3:55 p.m.*

*Why the two flows give dealers opposite gamma.* Retail traders are mostly option *buyers* (Example 23.1), so the dealer who sells to them is short options and **short gamma**: they buy as the index rises and sell as it falls, amplifying the move. The systematic sellers of Section 23.13 *sell* 0DTE spreads, so the dealer opposite them is long options and **long gamma**,

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trading against the move and damping it. The two roughly offset, which is the main reason Section 23.12 gives for the absence of an incident; but the net is a small difference of two large flows, unobservable (question 8), and a property of the flow rather than of the instrument, gone if the sellers withdraw (question 11). The book here is the case in which it comes out short.

*The hedging trade.* "Short $800m of gamma per 1%" means the book's delta changes by $800 million for each 1% move in the index, in the direction that requires the dealer to trade *with* the move. For a 1.5% move:

$$1.5 \times \$800\text{m} = \$\mathbf{1.2}\text{ billion}$$

*The direction:* ***buy*** *on a 1.5% rise* (and sell on a 1.5% fall). A short-gamma book's delta becomes more negative as the index rises, so restoring neutrality means buying into strength and selling into weakness, the momentum-following, destabilizing rule of Section 18.12.

Note the approximation. Constant gamma over the move is the assumption, and question 2 says it is a poor one at 0DTE maturities: the band is narrow, so a 1.5% move very likely carries the index *out* of the region where the gamma exists. If the book's gamma is concentrated near spot, the linear estimate **overstates** the trade, because much of the required hedging is completed within the first few tenths of a percent and then stops; if the move carries the index toward a concentration of strikes, it understates it. The $1.2 billion is the number a risk system reports; the realized flow is front-loaded and, for a book concentrated near spot, smaller.

*Why the same flow is negligible against volume and material against depth.* Because **volume and depth are different quantities**, and only one of them describes what happens when an order arrives.

*Against daily volume, $1.2 billion is nothing.* S&P 500 futures trade a few hundred billion dollars of notional in a session, so this is a fraction of a percent of the day's turnover. That is the comparison that reassures, and the one usually cited.

*Against depth, it is enormous.* Depth is what sits in the order book *now*, within a few ticks: tens of millions to about a hundred million dollars, not billions. Daily volume is that book being replenished thousands of times over the session. An order does not consume a day's volume; it consumes the book in front of it and then waits for replenishment. $1.2 billion is many multiples of the resting depth, so executing it moves the price, which, for a short-gamma book, generates more hedging demand, which is the feedback loop.

*And 3:55 p.m. is the worst moment for this.* Three effects coincide, and they are not independent:

- *The flow is all on one side.* Charm and gamma push the same way in the final hour, so the hedger meets only the orders resting on the other side; the close's heavy volume is no help to an order that must trade now.
- *Charm has concentrated the flow there.* Question 4's clock-driven hedging accelerates more than eightfold into the last hour, so the largest predictable flow of the day arrives in the same few minutes.
- *Everyone's deadline is the same.* 0DTE positions must be resolved by the close, so there is no option to wait for liquidity to return. A monthly book can hedge tomorrow; a 0DTE book cannot.

*The general point* is that "the market trades $X billion a day" is the wrong denominator for a question about market impact. The right one is depth over the horizon in which the trade must be completed, and for 0DTE hedging that horizon is minutes, imposed by the contract rather than chosen by the trader.

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*What the question tests.* Scaling a gamma figure to a move, and that liquidity is a rate-with-a-deadline rather than a daily total.

---

**23.11** *State the four reasons Section 23.12 gives for the absence of an incident, identify which is behavioral rather than structural, and explain why daily expiration bounds the size of the risk but not its speed.*

*The four reasons, from Section 23.12.*

*1. The flow is two-sided.* Retail buying leaves dealers short gamma and systematic premium selling leaves them long, so the net is small (question 10). This is the **main** reason and, Section 23.12 says explicitly, *the most fragile*.

*2. Expiration is a natural stop-loss.* A short 0DTE option cannot become a larger problem tomorrow, because tomorrow it does not exist. Compare the inverse VIX products of Section 22.8, whose short position was carried from day to day for years, or the snowball books of Chapter 35, whose knock-in exposure grew as the market fell over months.

*3. Margin is collected daily and the products are exchange-traded.* The 2018 episode ran through an exchange-traded note, XIV, whose prospectus allowed early termination; a listed option position is margined by the clearing house of Chapter 2 against a position that self-liquidates.

*4. The underlying is the deepest equity market in the world.* A billion dollars of S&P futures is absorbable in a way the same flow in a single stock, or in VIX futures, is not.

*Which is behavioral: the first.* Two-sidedness is a fact about *who happens to be trading*, not about how the contracts are constructed. The systematic premium sellers are under no obligation to be there. They will withdraw after a bad month, on a risk limit, or in a drawdown, and the withdrawal would be *caused* by deteriorating conditions, so the stabilizer disappears exactly when it is needed. The remaining three are structural: they follow from contract terms, from clearing arrangements, and from the size of the underlying market, and they hold whoever is trading.

That distinction is the point of the question. By the argument of Section 22.5's case study, a structural repair holds under stress and a behavioral regularity does not—which is why the reassuring summary of Section 23.12 rests on the weakest of the four legs.

*Why daily expiration bounds the size of the risk but not its speed.*

*It bounds the size* because it bounds accumulation. A position cannot be carried into a multi-day drawdown; it cannot double down; there is no equivalent of the inverse VIX products, whose required purchase grew to nearly twice their original size as they lost money (Section 22.8, question 8 of that chapter). Whatever is outstanding this afternoon is the whole of it, and by tomorrow morning it is gone. The exposure is a *flow* with a hard reset, not a *stock* that compounds.

*It does nothing about speed*, and speed is the other half of a stability problem. Everything in this chapter happens inside one session: gamma is fourteen times its monthly level and concentrated in a band of tenths of a percent (question 2), charm's flow accelerates more than eightfold into the final hour (question 4), and the flow peaks when it is all on one side and must trade against the depth available at that moment (question 10). A disorderly move that takes a full day to develop can be met by capital arriving from elsewhere. A disorderly move that develops in fifteen minutes cannot, because the capital cannot get there in fifteen minutes.

So the correct worry is not a slow-building crisis but a fast, bounded one—a disorderly half hour rather than a solvency event. That is a genuinely less alarming shape than the commentary usually implies, and it is also a shape that daily expiration does nothing whatever to prevent. The 2010 flash crash is the reference: it was over in thirty-six minutes, it accumulated nothing,

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and it required no leverage to compound overnight.

*What the question tests.* That structural safeguards outrank behavioral ones, and that bounding the magnitude of an exposure is a separate question from bounding the rate at which it can be transmitted.

---

**23.12** *Explain what happens to dealer hedging demand when a move takes the index outside the gamma band, and why this is the dangerous configuration.*

*What happens: dealer hedging demand falls off sharply, and soon to* ***zero****.* Question 2 established that the gamma band narrows as expiration approaches: at one hour, meaningful gamma survives only within a few tenths of a percent of the strike. By a percent away the delta has all but reached its terminal value, 0 or 1, so further moves in the index barely change it, and there is almost no hedge left to adjust.

Example 23.6 shows the exhaustion in progress: by a 1% move the delta has reached 0.996, and the last half-percent generated $876 million of hedging against $4.1 billion for the first half.

*Why this is the dangerous configuration.* There are four reasons, and the last is the one that makes it structural rather than merely unfortunate.

*1. The stabilizer vanishes exactly when it is needed.* A long-gamma dealer book supplies liquidity (selling into rallies, buying into declines), and it supplies most of it for the smallest moves. Once the index leaves the band that supply stops entirely. The market has been heavily cushioned against the moves that did not matter and is uncushioned for the one that does (question 7).

*2. The transition is discontinuous, not gradual.* The market does not become progressively less stable as the move grows. It is stabilized right up to the edge and then not, because the dealers who were supplying the liquidity have finished supplying it. A participant reading the morning's price action (quiet, mean-reverting, well-behaved) has observed a regime that can end without warning at a spot level no one publishes.

*3. If the book is short gamma, the reverse happens and it is worse.* A short-gamma dealer must trade *with* the move, so while the index is still inside the band the required flow is at its maximum and in the destabilizing direction (question 10). The narrow band means that flow is compressed into a small price range, and a hedge executed across half a percent is a very different order from the same amount spread over a day. Whether the configuration is stabilizing-then-absent or amplifying-then-absent depends on the sign of net dealer gamma, which question 8 says is not observable.

*4. And the exit is where liquidity is worst.* Leaving the band means the index has moved sharply, which is when depth is thinnest, when spreads widen, and when the market makers who provide the resting book step back. The hedging support disappears at the same moment the order book does, and the two are not independent events—the same move causes both.

*The general shape of the risk*, and the reason it belongs at the end of the chapter: this is not a mechanism that builds a large position over time and then unwinds it catastrophically, which is the 1987, 2008 and 2018 pattern. It is a mechanism in which a genuinely stabilizing force has a **boundary**, the boundary is invisible, and crossing it changes the market's character in minutes. Section 23.12's structural comforts (daily expiration, daily margin, a deep underlying) bound how large the resulting episode can be. None of them bounds how fast it happens, which is question 11's distinction, and speed is what makes a disorderly move disorderly.

*What the question tests.* That the gamma band has an edge, that hedging demand is a step function of spot rather than a smooth one, and that the danger lies in the discontinuity rather than in the magnitude of the exposure.

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# Chapter 24
# Credit Risk and Default Modeling

**24.1** *A borrower has a constant hazard rate of 4%. Compute the probability of default within five years and within ten, and explain why a 3% hazard rate does not mean a 30% probability over ten years.*

*The two probabilities.* Under a constant hazard rate $\lambda$, survival to $T$ is $Q(T) = e^{-\lambda T}$, so the probability of defaulting by $T$ is $1 - e^{-\lambda T}$. With $\lambda = 4\%$:

$$\mathbb{P}(\text{default within 5 years}) = 1 - e^{-0.04\times 5} = 1 - e^{-0.20} = \mathbf{18.13}\%$$

$$\mathbb{P}(\text{default within 10 years}) = 1 - e^{-0.04\times 10} = 1 - e^{-0.40} = \mathbf{32.97}\%$$

Note that doubling the horizon does not double the probability: 18.13% becomes 32.97%, not 36.26%. The shortfall is the point of the second half of the question.

*Why a 3% hazard rate is not a 30% ten-year probability.* Because a hazard rate is a **conditional** rate (the instantaneous probability of default *given survival so far*), and the population it applies to shrinks every year. The correct calculation is

$$1 - e^{-0.03\times 10} = 1 - e^{-0.30} = \mathbf{25.92}\%$$

against the naive 30%: an overstatement of 4.08 percentage points, or 16% in relative terms.

The mechanism is easiest to see one year at a time. In year one, 2.96% of the original cohort defaults (a continuous 3% rate is a 2.96% annual probability). In year two the same rate acts on the survivors, 97.0% of the original cohort, so 2.87% of it defaults. By year ten the annual contribution has fallen to 2.26%. Adding ten copies of 3% counts borrowers that were already gone.

*The same structure in three other places*, which is why it is worth internalizing rather than memorizing:

- *Compounding.* $\lambda$ is a continuously compounded rate, exactly as a continuously compounded interest rate is; $1 - e^{-\lambda T}$ is the credit analog of $e^{rT}$, and multiplying $\lambda$ by $T$ is the same error as treating a 3% rate as 30% growth over a decade.
- *The bound.* The linear formula gives probabilities above 1 for large $\lambda T$ (at a 20% hazard over ten years it returns 200%), while the exponential is automatically confined to $[0, 1)$. The exponential is not an approximation to the linear rule; the linear rule is a first-order approximation to it, and a bad one outside its range.
- *The small-$\lambda T$ regime.* For $\lambda T \ll 1$ the two agree: $\lambda = 0.5\%$ over one year gives 0.4988% against 0.5%, a difference in the fourth decimal. So the shortcut is harmless for high-grade issuers at short horizons and dangerous for distressed issuers or long horizons—which is when someone reaches for it.

*And a caution that questions 4 and 11 develop.* All of this assumes $\lambda$ is *constant*. It is not: Example 24.2 bootstraps a hazard curve that rises from 1.33% to 4.22% and then falls, and Example 24.6 shows why: an investment-grade issuer's forward default intensity rises with

horizon because default is reached through downgrade. So a flat 4% is a convenient fiction, and using a five-year average to price something that pays off in years three to five understates the risk (question 4).

*What the question tests.* That a hazard rate is conditional on survival, and that $1 - e^{-\lambda T}$ rather than $\lambda T$ is what a rate compounds into.

---

**24.2** *A CDS trades at 240 bp with assumed recovery of 40%. Estimate the hazard rate, and explain why a change in the discount curve alone barely moves a par CDS spread.*

*The hazard rate.* By the credit triangle (24.3), $s = \lambda(1 - R)$, so

$$\lambda = \frac{s}{1 - R} = \frac{0.0240}{1 - 0.40} = \frac{0.0240}{0.60} = \mathbf{4.00}\%$$

The conversion is a single division, which is why traders can do it mentally: at 40% recovery, divide the spread in basis points by 0.6, or equivalently multiply by $5/3$.

Note what the calculation did *not* require: no discount curve, no maturity, no payment schedule. That is not a simplification—it is exact, and the reason is the second half of the question.

*Why a change in the discount curve alone barely moves a par CDS spread.* Because the discount factor and the survival probability appear **identically on both legs** and cancel.

Over a short interval $dt$, conditional on survival to $t$:

- the *protection* leg pays $(1 - R)$ with probability $\lambda\, dt$, and the present value of that is $(1 - R)\,\lambda\, Q(t)\, P(0, t)\, dt$;
- the *premium* leg, paid continuously, receives $s\, dt$, worth $s\, Q(t)\, P(0, t)\, dt$.

With a flat hazard rate the par condition integrates both lines against the same weight $Q(t)P(0, t)$, and setting the integrals equal gives $s = \lambda(1 - R)$ with the weight dividing out. The par spread then does not depend on the discount curve at all, and depends on the hazard rate and the recovery only through the product $\lambda(1 - R)$. With a sloping hazard curve the par spread is $(1 - R)$ times a $Q(t)P(0, t)$-weighted average of $\lambda(t)$, so the discount curve enters only through the weights and barely moves it.

And for a continuously paid premium this is exact, not approximate, which is the point Example 24.1 makes about the textbook statement. Pricing a five-year CDS numerically:

| $\lambda$ | **Continuous premium** | **Quarterly, with accrual** | **Quarterly, no accrual** |
|---|---|---|---|
| 1% | 60.00 | 60.23 | 60.30 |
| 6% | 360.00 | 361.35 | 364.08 |
| 20% | 1200.00 | 1204.47 | 1235.17 |

(basis points). The first column is $\lambda(1 - R)$ exactly. A quarterly contract sits about 0.37% above it at every hazard rate (0.376%, 0.375% and 0.373% in the three rows), because a premium paid at the quarter end is discounted for half a quarter longer than one paid continuously, which is $r\Delta/2$ of the premium leg. The "approximation error" usually quoted is the third column against the second: it is the value of the **premium accrued** between the last payment date and the default date: 2.5% of the spread at $\lambda = 20\%$. That is a contract feature, not a mathematical error, and a contract that did not pay accrual would genuinely quote at the third column.

*The practical consequence, and it is not hypothetical.* CDS spreads did *not* move mechanically when the discounting convention changed in the reforms of Chapter 4, from LIBOR to OIS to SOFR. A swap rate would have moved; a par CDS spread did not, because the discount factor never entered it. What *did* move was the **upfront** payment on an off-market

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contract, since that is a present value and the risky annuity in it does depend on the curve. Question 2 of Chapter 25 is that calculation.

*What the question tests.* The one-line conversion, and that the cancellation making it exact is the same fact that makes a par spread insensitive to discounting.

---

**24.3** *Explain why the credit triangle is exact rather than approximate, and what the usually quoted error actually measures.*

*Why it is exact.* The usual presentation calls $s = \lambda(1 - R)$ an approximation valid for small $\lambda$. For a premium paid continuously it is not an approximation at all, and Section 24.2 derives it and Example 24.1 confirms it numerically.

The derivation is one line. Take the hazard rate flat at $\lambda$, as the triangle does. Over $[t, t + dt]$, conditional on survival to $t$ with survival probability $Q(t)$ and discount factor $P(0, t)$:

$$\underbrace{(1 - R)\,\lambda\,Q(t)P(0,t)\,dt}_{\text{protection leg pays}} \;=\; \underbrace{s\,Q(t)P(0,t)\,dt}_{\text{premium leg receives, paid continuously}}$$

Both sides carry $Q(t)P(0, t)\,dt$, so it cancels, leaving $s = \lambda(1 - R)$ pointwise, at every instant and for any discount curve, not merely in aggregate. Integrating over the contract's life changes nothing, because an identity that holds at every $t$ holds when summed.

The crucial step is that the premium is paid *continuously*. A real contract pays quarterly, and that departs from the identity in two separable ways. The survivors' premium is paid at the quarter end rather than through the quarter, so it is discounted for half a quarter longer; the premium leg is smaller by a factor $1 - r\Delta/2$ and the par spread higher by a factor $1/(1 - r\Delta/2)$, about $r\Delta/2$ to first order: 0.376% at $r = 3\%$ and $\Delta = 0.25$, whatever the hazard rate. And if the contract also omitted the premium accrued to the default date, a default midway through a quarter would deliver protection while the buyer had paid nothing for the period since the last payment, a second, larger gap that grows with $\lambda$.

*What the quoted error actually measures: the* ***value of the accrued premium***. It is the difference between a quarterly contract that pays accrued premium on default and one that does not:

| $\lambda$ | Continuous | Quarterly, with accrual | Quarterly, no accrual | Accrual gap |
|---|---|---|---|---|
| 1% | 60.00 | 60.23 | 60.30 | 0.1% |
| 6% | 360.00 | 361.35 | 364.08 | 0.8% |
| 20% | 1200.00 | 1204.47 | 1235.17 | 2.5% |

The gap grows with $\lambda$ for the obvious reason: accrual matters only when default occurs inside a premium period, and the probability of that rises with the hazard rate. At $\lambda = 1\%$ it is under a tenth of a basis point; at 20% it is 31 basis points. The 0.376% timing residual, by contrast, is nearly the same in every row.

So the standard statement is wrong in an instructive way. It attributes to mathematics an error that belongs to a contract term. Someone who "knows" the triangle is approximate will distrust it at high spreads (precisely where it is most useful, since distressed issuers are where the conversion between spread and hazard rate matters most) and will reach for a numerical solver that returns the same answer. Conversely, someone using it on a contract that genuinely omits accrual will be wrong by a knowable amount (2.5% at a 20% hazard rate) and will attribute the discrepancy to the approximation rather than to the term sheet.

*The general lesson*, which Chapter 25 develops at length: read the contract before blaming the model. A great many "model errors" in credit are documentation. The accrual here, the restructuring clause of Chapter 25's question 6, and the mean-subtraction convention of

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Section 22.2 are all the same species: a term that changes what is being priced, showing up as an unexplained residual in someone's calibration.

*The exactness buys two things.* Reading a spread as a hazard rate requires only a recovery assumption and no discount curve (question 2), which is why the conversion is done mentally. And with a sloping hazard *curve* the same cancellation makes the par spread $(1 - R)$ times a weighted average of the hazard rate, which is why the bootstrap of question 4 behaves like a zero-curve bootstrap, each spread an average and each bucket a forward.

*What the question tests.* The cancellation, and that the residual everyone quotes is a contract feature with a sign and a size.

---

**24.4** *Using Example 24.2, explain why applying the credit triangle to a five-year spread understates the hazard rate over years three to five.*

*The numbers.* Example 24.2's borrower quotes 170 bp at five years. Applying the credit triangle directly to that quote gives

$$\lambda = \frac{0.0170}{0.60} = 2.833\%$$

The bootstrapped hazard rate for the **3–5 year bucket** is **3.998%**, a factor of **1.41** larger.

*Why.* Because a spread is an *average* and a bucket hazard rate is a *forward*, and the relationship between them is exactly the relationship between a swap rate and a forward rate in Chapter 4.

The five-year spread compensates for default risk over the whole of years zero to five. The curve is upward sloping (80, 130, 170, 190, 200 bp), so the early years carry *less* risk than the five-year average and the later years carry *more*. Specifically the first year's hazard rate is 1.333% and the 1–3 year rate is 2.617%, both below the 2.833% average. Since the average is built from these low early rates together with the years three to five, the years three to five must lie *above* the average to bring it up to 2.833%. That is arithmetic, not modeling: whenever a curve slopes upward, the forward rate exceeds the spot rate at every point.

The whole table shows the divergence, which is largest in the 3–5 year bucket and narrows after it:

| Maturity | Spread | Bootstrapped $\lambda$ | Triangle guess |
|---|---|---|---|
| 1y | 80 | 1.333% (0–1y) | 1.333% |
| 3y | 130 | 2.617% (1–3y) | 2.167% |
| 5y | 170 | 3.998% (3–5y) | 2.833% |
| 7y | 190 | 4.222% (5–7y) | 3.167% |
| 10y | 200 | 3.885% (7–10y) | 3.333% |

The two agree exactly on the first bucket, where the average *is* the forward because there is nothing earlier to average with, and differ thereafter.

*Why it matters, stated as a rule.* Anyone using the five-year spread to price something that only pays off in years three to five underprices the risk by a factor of 1.41. That is not an edge case. It is a forward-starting CDS, the back end of a tranche's loss profile, the later exposure dates in a CVA calculation (Chapter 9), or a step-up structure whose protection begins after a lock-out. Each of these depends on the *forward* default intensity, and each is mispriced by using the spot spread.

*The economics behind the shape.* An upward-sloping spread curve is the norm for investment-grade issuers and reflects the same mean reversion Section 19.6 found in volatility: a currently healthy firm is more likely to deteriorate than to improve. Example 24.6 supplies the mechanism rather than the fitted curve: a BBB issuer alive in year four is disproportionately

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likely to be BB or B by then, facing a much higher hazard than the 0.28% it started with. Distressed issuers *invert*, for the same reason inverted volatility curves appear in stress: if the firm survives the next year, the danger has largely passed and the hazard falls.

*And note the hump.* The bootstrapped rate rises through the 5–7 year bucket at 4.222% and then *falls* to 3.885%. That is the concavity of the spread curve at work (10 bp more spread for three more years), and it is invisible in the spread curve, which rises monotonically throughout. A monotone spread curve can imply a non-monotone hazard curve, and only the bootstrap reveals it.

*What the question tests.* That spot-to-forward is the same transformation in credit as in rates, and that the error from ignoring it is large and one-directional.

---

**24.5** *A five-year CDS trades at 170 bp. State the implied hazard rate at 40% recovery and at 60%, and name one instrument the two assumptions would price differently.*

*The two implied hazard rates.* By the credit triangle, $\lambda = s/(1-R)$:

$$R = 40\%: \quad \lambda = \frac{0.0170}{0.60} = \mathbf{2.833}\%$$

$$R = 60\%: \quad \lambda = \frac{0.0170}{0.40} = \mathbf{4.250}\%$$

a ratio of exactly 1.50, and the implied five-year default probabilities are 13.21% and 19.14%.

*Why both are consistent with the same quote.* Because the CDS observes only the **product** $\lambda(1-R)$ and nothing more. Example 24.3 extends the table to $R = 20\%$ ($\lambda = 2.125\%$) and $R = 80\%$ ($\lambda = 8.500\%$), and all four price the CDS identically. So the recovery assumption is not *extracted* from the spread—it is **imposed** on it. The 40% convention is a coordinate choice, not a measurement.

*An instrument the two assumptions price differently: a* ***digital default swap***, which pays a fixed amount on default regardless of the realized recovery. Its value depends on $\lambda$ alone, so the 60% assumption prices it about **45% higher** than the 40% assumption on the identical market quote: a five-year default probability of 19.14% against 13.21%, a little less than the hazard rate's 50% excess because default probabilities compound rather than add. Across the full range of the example, the four rows price it over a range of more than three to one.

A **first-to-default basket** is less exposed than it looks. It pays $1-R$ on the first default, so to first order its value is $\sum_i \lambda_i(1-R)$, the product again; the split matters only through the concavity of the probability of at least one default, and on five independent borrowers at 170 bp the basket prices about 14% lower at 60% recovery than at 40%, where the digital prices 45% higher. Any **expected-loss calculation for capital or provisioning**, by contrast, needs the split, because regulation asks for probability of default and loss given default *separately* (question 13).

*Where the convention is safe and where it is not.* For pricing a CDS, or anything else whose value depends only on the product, the assumption is harmless; the damage is done when the extracted $\lambda$ is exported to other uses, as question 13 shows.

*One further reason to distrust the split remains, beyond identification.* Recovery is not a constant even in principle. It varies enormously by seniority (secured bank debt recovers far more than senior unsecured, which recovers far more than subordinated, which often recovers nothing), and it is **correlated with the default rate** (question 6). So the quantity being assumed fixed is one that moves systematically in the direction that hurts, and the practical convention of fixing $R$ by seniority and sector is defensible for pricing and indefensible for risk measurement.

*What the question tests.* That a single quote cannot identify two parameters, and knowing which instruments break the degeneracy.

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**24.6** *Explain why recoveries are lowest in the years when defaults are highest, and what this does to portfolio losses.*

*Why recoveries are lowest when defaults are highest.* Because a firm defaulting in a recession is **selling its assets into a market full of other distressed sellers**. Four channels reinforce one another:

*The natural buyers are impaired.* The parties best placed to value a defaulted firm's assets are its competitors, and in an industry-wide downturn they are conserving cash, cutting capital expenditure, and facing their own credit constraints. The bidder with the highest valuation is absent precisely when the asset comes to market.

*Supply spikes.* Defaults cluster, so many similar assets are liquidated at once: aircraft in an airline recession, drilling rigs when energy prices collapse, retail leases in a retail contraction. The price is set by a glut.

*And the assets themselves are worth less.* Recovery is realized in the same economy that caused the default. Commercial real estate, inventory, and receivables are all marked down by the conditions that pushed the borrower under, so both the numerator and the number of claimants move the wrong way.

Financing conditions add a fourth: distressed acquirers rely on credit, and credit is scarce in exactly these periods, so even a willing buyer may be unable to bid.

*What it does to portfolio losses: it makes them worse than a fixed-recovery model implies, and it does so **twice over**.* Portfolio loss is

$$L = \sum_i \mathbb{1}\{\text{default}_i\} \times (1 - R_i)$$

and a model fixing $R$ computes $\mathbb{E}[L] = \mathbb{E}[\text{defaults}] \times (1 - \bar{R})$. The true expectation carries a covariance term:

$$\mathbb{E}[L] = \mathbb{E}[\text{defaults}](1 - \bar{R}) + \text{Cov}\left(\text{defaults},\ 1 - R\right)$$

and that covariance is **positive**, because both factors rise together. This is the same $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ error as the relative-TSR award of Chapter 20's question 11, with the sign running against the holder rather than for them.

The effect on the *tail* is far larger than on the mean. In a bad year the number of defaults is high *and* each one recovers less: say 30 defaults at 25% recovery instead of the expected 10 at 45%, which is not three times the loss but more than four. A fixed-recovery model understates the tail twice: once through the count and once through the severity. Since capital is set by a tail quantile rather than by the mean, the understatement lands squarely where it does the most damage.

*Two practical consequences follow.* A **senior tranche** is hit hardest, because it suffers only in the states where both effects are operating (question 10's point that a senior tranche is a bet on the joint behavior rather than on the average). And **stochastic recovery** models, in which $R$ is driven by the same common factor as default, exist for exactly this reason and are standard in portfolio credit work; they are one of the mechanisms Chapter 26 examines.

*And it explains the survival of a convention the data contradict.* The 40% figure is an *unconditional average*. It is roughly right across the cycle and roughly wrong in every individual year: too high in recessions, too low in expansions. Averaging over the cycle is acceptable for pricing (question 5) and unacceptable for a stress scenario, whose entire purpose is to describe one particular bad year rather than the average of many.

*What the question tests.* That the correlation is economic rather than statistical, and that a positive covariance between frequency and severity compounds in the tail.

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**24.7** *Explain why equity is a call option on the firm and debt is short a put, what this implies about the shareholder–creditor conflict, and what the ratio of equity to asset volatility measures as the firm approaches default.*

*Equity is a call on the firm.* If the firm's assets are worth $V_T$ when debt of face value $D$ matures, shareholders receive

$$\max(V_T - D, 0)$$

They keep the surplus and walk away from the shortfall—limited liability *is* the option's floor at zero. So equity is a call struck at the face value of debt, and Chapter 15's formula prices it.

*Debt is short a put.* Debt holders receive the complement,

$$\min(V_T, D) = D - \max(D - V_T, 0)$$

which is a riskless bond of face $D$ *less* a put struck at $D$. **Lending is selling a put on the borrower's assets**, and the credit spread is the premium on that put. Figure 24.2 shows both payoffs on one axis, and they sum to $V_T$ as they must.

*What this implies about the shareholder–creditor conflict.* It makes it a matter of *derivatives*, not of governance. Equity is long an option, so equity holders **gain** from volatility. Debt is short an option, so debt holders **lose** from it. An increase in firm risk that leaves total firm value unchanged is a pure transfer from creditors to shareholders, because it raises the value of a call and the value of the put the creditors have written.

Three familiar corporate behaviors follow immediately, and none requires anyone to behave badly: *asset substitution*, swapping safe projects for risky ones of the same net present value; *levering up*, which raises the strike and moves the creditors' put toward the money; and the *gamble for resurrection*, in which a near-insolvent firm takes enormous risk because shareholders own an out-of-the-money option whose only route to value is volatility.

This is the sharpest available statement of the conflict flagged in Section 17.11. An executive paid in options is paid in a claim whose value rises with firm risk, and every increase in that risk is a transfer from creditors. Nothing in the payoff structure aligns the two, and no amount of governance language changes the derivative. Covenants exist precisely because the conflict cannot be resolved by incentives: they are contractual constraints imposed on a party whose optimum genuinely differs.

What the ratio of equity to asset volatility measures as the firm approaches default: **the leverage of the option**, which is its elasticity. Differentiating $E = f(V)$ and applying Itô,

$$\sigma_E = \frac{V}{E} N(d_1)\, \sigma_V \qquad \Longrightarrow \qquad \frac{\sigma_E}{\sigma_V} = \frac{V\, N(d_1)}{E}$$

which is the option's delta times the ratio of the underlying to the option's value, the option's elasticity. It is the **effective leverage** of the equity claim.

Its behavior is what matters. For a healthy firm, $E$ is large and $N(d_1)$ near 1, so the ratio is modest: equity volatility is a small multiple of asset volatility. As the firm deteriorates, $E$ collapses toward zero faster than $N(d_1)$ does, so the ratio **rises without bound**. A firm approaching default has equity volatility many times its asset volatility, which is why distressed equities trade with implied volatilities of 80% and 100% on assets that are far less volatile than that.

Two practical uses follow. It is one of the two simultaneous equations in Section 24.6's calibration, since $\sigma_V$ is unobservable and $\sigma_E$ is not. And it explains a puzzle in the equity smile: leverage rises mechanically as the price falls, so equity volatility rises as the price falls—Section 19.9's leverage effect, derived here rather than asserted.

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*What the question tests.* That the capital structure is an option decomposition, that the agency conflict is a consequence of the payoffs rather than of behavior, and that the volatility ratio is an elasticity that diverges at the default boundary.

---

**24.8** *Explain why Merton's credit spread goes to zero as maturity shortens, and name the chapter in which the same defect appeared in a different market.*

*Why the spread vanishes.* In Merton's model the firm's asset value follows a **continuous** diffusion, and default occurs only if $V_T < D$ at maturity. A firm whose assets exceed its debt today cannot default tomorrow, because a continuous path must travel through every intermediate level and there is not enough time to travel that far. The probability of covering a finite distance in a vanishing time falls to zero faster than any power of $T$, so the spread follows it down.

Figure 24.3's left edge measures the collapse, with assets 100, debt 80, asset volatility 25%:

| Maturity | Spread | Default probability |
|---|---|---|
| 1 week | 0.0000 bp | $6 \times 10^{-9}$% |
| 1 month | 2.2 bp | 0.09% |
| 3 months | 64.6 bp | 3.42% |
| 1 year | 200.5 bp | 16.66% |

A one-week default probability of $6 \times 10^{-9}$% is not a small number—it is a statement that the event is impossible, which is a different claim.

*Real short-dated credit does not behave this way.* One-year CDS on investment-grade issuers trades in the tens of basis points and does not collapse toward zero as maturity shortens. Short-dated spreads on stressed issuers can be enormous: the market's whole concern with a deteriorating credit is precisely whether it survives the next few weeks, and Credit Suisse in March 2023, whose one-year CDS spread ran well above its five-year spread, traded with a steeply inverted CDS curve (Chapter 25 tells the rest of that episode). A model asserting that near-term default is impossible has nothing to say about the situations credit markets care most about.

*The chapter in which the same defect appeared in a different market:* **Chapter 21**, Section 21.7. There, a diffusive volatility model could not produce the observed short-dated skew, because $\psi(T)$ approaches a finite constant as $T \to 0$ while the market's steepens like $T^{-0.4}$. Here, a diffusive *asset* process cannot produce short-dated credit spreads. Both failures have the identical cause: a continuous path cannot generate short-horizon tails, so any model built on one assigns near-zero probability to events that happen at short horizons and prices the corresponding claims at near zero.

That two literatures discovered the same limitation independently, in different decades and different asset classes, is the point worth carrying away. It is not a shortcoming of Merton or of Heston; it is a property of diffusions.

*And the repairs run parallel too.* In volatility the answers were jumps and roughness. In credit they are **jump-diffusion** models of asset value; **first-passage** models, in which default occurs the first time $V$ touches a barrier rather than only at maturity, which helps but does not fix the short end for a firm starting well above the barrier; and models with **incomplete information**, in which investors cannot observe $V$ exactly. This last is the most satisfying, because uncertainty about where the firm currently stands produces a nonzero short-dated spread from an otherwise continuous process, and it corresponds to something true about the world.

*One caveat attaches to the diagnosis.* Merton's short end is wrong, but the model is valuable

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elsewhere: it delivers the option decomposition of question 7, it makes distance to default an observable quantity, and the commercial EDF implementations built on it forecast defaults respectably at the one-year horizon. The failure is specific and located, not general.

*What the question tests.* That continuity is what kills the short end, and recognizing the same structural failure across two unrelated chapters.

---

**24.9** *Historical five-year default rates for BBB issuers are about 1.36% and the spread is about 130 bp. Compute the expected loss, state what fraction of the spread it accounts for, and explain why the ratio of risk-neutral to historical default probability is much larger for investment-grade than for single-B issuers.*

*Expected loss.* A five-year cumulative default probability of 1.36% corresponds to an annual hazard rate of

$$\lambda = -\frac{\ln(1 - 0.0136)}{5} = 0.274\% \text{ per year}$$

and at 40% recovery the expected annual loss is

$$0.274\% \times 0.60 = \mathbf{16.4} \text{ basis points}$$

(The cruder $1.36\%/5 \times 0.60 = 16.3$ bp gives the same answer to a fifth of a basis point, because $\lambda T$ is small, question 1's regime.)

*The fraction of the spread.*

$$\frac{16.4}{130} = \mathbf{12.6}\%$$

so **87% of the BBB spread is not expected loss**. That is the headline of Example 24.5 and the reason Section 9.2 insists on the distinction between risk-neutral and historical probabilities.

*Why the ratio of risk-neutral to historical default probability is much larger for investment-grade than for single-B issuers.* The table shows the pattern:

| Rating | Spread | Historical | Risk-neutral | Ratio |
|---|---|---|---|---|
| AAA | 40 bp | 0.34% | 3.28% | 9.6× |
| A | 80 bp | 0.39% | 6.45% | 16.5× |
| BBB | 130 bp | 1.36% | 10.27% | 7.5× |
| B | 550 bp | 15.60% | 36.77% | 2.4× |

An A issuer's risk-neutral default probability is about sixteen times its historical rate (AA's is 16.0), and a single-B issuer's under two and a half times. AAA is lower only because its 0.34% rests on very few issuers and exceeds AA's 0.28%, a small-sample artifact. Whatever the excess compensates for, it does not scale with default risk.

Three explanations are offered, none established, and they are not exclusive.

*1. The highest-grade defaults are mostly systematic.* On this account such an issuer defaults only in a catastrophe, a systemic event that impairs everything at once. Its default is therefore maximally correlated with the states in which a dollar is worth most, so it carries the largest risk premium per unit of probability. A single-B issuer defaults for idiosyncratic reasons most of the time (bad management, a failed product, a covenant breach), and idiosyncratic risk is diversifiable and commands no premium. The premium is compensation for *systematic* default risk, and the systematic share of default risk is highest exactly where total default risk is lowest.

*2. The non-credit components of the spread do not scale.* Illiquidity, the cost of balance sheet, and a premium for parameter uncertainty are roughly similar in absolute terms across ratings. Against 16.4 bp of expected loss that residual is a large multiple; against 550 bp it is a rounding error. A near-fixed additive component divided by a rapidly shrinking denominator produces exactly this pattern.

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*3. Parameter uncertainty is proportionally worst where the estimate is thinnest.* The 0.34% figure for AAA rests on a handful of defaults among few issuers, which is why it exceeds the AA figure. A protection seller is compensated for the possibility that the true rate is several times the estimate, and that uncertainty is a larger multiple of a small number.

*The practical consequence, and it is Section 9.2's.* Use risk-neutral probabilities to price and hedge, historical probabilities to forecast losses, and never mix them. An institution that provisions using risk-neutral probabilities will over-reserve by a factor of seven or more on investment-grade portfolios, and by about sixteen at AA and A. One that prices using historical probabilities will systematically sell protection too cheaply, and will do so worst on exactly the issuers that look safest.

*What the question tests.* Decomposing a spread into expected loss and residual, and the candidate explanations for why the ratio is largest where default risk is smallest.

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**24.10** *A portfolio of 100 borrowers has 5% default probability per borrower. Explain why raising correlation leaves expected defaults unchanged but raises both the probability of zero defaults and the probability of twenty or more, and why a senior tranche is a bet on correlation while an equity tranche benefits from it.*

*Why expected defaults are unchanged.* Expectation is linear, so

$$\mathbb{E}[\text{defaults}] = \sum_{i=1}^{100} \mathbb{P}(\text{borrower } i \text{ defaults}) = 100 \times 0.05 = 5$$

whatever the dependence between the borrowers. Correlation is a statement about *joint* behavior; it leaves every *marginal* untouched. In the one-factor Gaussian copula (24.10) this is explicit by construction: the threshold $N^{-1}(\pi)$ is set from each borrower's own market- implied default probability, and $\rho$ supplies only the dependence. Marginals from the market, dependence from an assumption: that separation is the entire appeal of a copula.

*Why both tails grow.* With the mean pinned, more dependence must move probability out of the middle, and it has nowhere to go but the ends. Example 24.7:

| $\rho$ | Expected | Std. dev. | P(no defaults) | P(20 or more) |
|---|---|---|---|---|
| 0.00 | 5.00 | 2.18 | 0.59% | 0.00% |
| 0.15 | 5.00 | 4.89 | 10.86% | 1.85% |
| 0.30 | 5.00 | 7.12 | 24.62% | 4.91% |
| 0.60 | 5.00 | 11.56 | 53.02% | 7.87% |

At $\rho = 0.6$ there is a **53% chance nothing at all defaults** and an **8% chance at least a fifth of the portfolio does**.

The mechanism is transparent in the copula. Conditional on the common factor $M$, the borrowers are independent with a common probability (24.12), and the portfolio's loss is then a tight binomial around $100\,p(M)$. All the dispersion therefore comes from the variation of $p(M)$ across draws of $M$. At $\rho = 0$, $p$ is the constant 5% and the only randomness is binomial sampling, which diversifies, giving a standard deviation of $\sqrt{100 \times 0.05 \times 0.95} = 2.18$. As $\rho$ rises, $p(M)$ swings: a good draw of the factor makes default nearly impossible for *every* borrower simultaneously, and a bad draw makes it likely for all of them. The portfolio stops being 100 independent bets and becomes closer to one bet repeated 100 times, at which point diversification does nothing.

*Why a senior tranche is a bet on correlation.* A senior tranche suffers only if losses exceed a large attachment point, which requires *many simultaneous* defaults. At low correlation that is essentially impossible: the probability of 20 or more defaults is 0.00% at $\rho = 0$, so the

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tranche's expected loss is negligible and it is genuinely safe. At high correlation it is 7.87%, so the tranche's expected loss is material. The portfolio's expected loss has not moved at all, so everything the senior tranche's value depends on is the correlation. It is a correlation instrument that looks like a credit instrument.

*Why an equity tranche benefits from correlation.* The equity tranche absorbs the *first* losses, so it is destroyed by the ordinary case of a few scattered defaults and survives only if *nothing* happens. Correlation raises $\mathbb{P}$(no defaults) from 0.59% to 53%, a ninety-fold increase in the one outcome under which the equity tranche pays in full. Its expected loss therefore *falls* as correlation rises, and it is **long** correlation where the senior tranche is short.

Two consequences follow, and Chapter 26 develops both. The tranches on one portfolio have *opposite* correlation exposures while sharing an identical pool, so a dealer holding the mezzanine (the tranche between them) and hedging with the senior and the equity is running a correlation book, not a credit book. And pricing any of them requires a number, $\rho$, that no observable market quoted before tranches themselves began to trade. The circularity in that sentence is Chapter 26's subject, and it is what implied correlation and base correlation are attempts to escape.

*One warning attaches to the model.* That tractability (one integral prices the whole portfolio), and not any evidence about how firms actually fail is why the Gaussian copula became standard. Its tail dependence is zero, meaning that in the limit extreme joint defaults are less likely than any credible account of contagion would suggest, which is one reason senior tranches lost far more in 2007–08 than the models implied.

*What the question tests.* That correlation redistributes without changing the mean, and that the two ends of the capital structure take opposite sides of the same parameter.

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**24.11** *Using Table 24.2, state the probability that a BBB issuer is still investment grade after five years and the probability it has fallen to speculative grade without defaulting, then explain why compounding a one-year default rate understates the five-year probability and give the factor from Example 24.6.*

*Where a BBB issuer stands after five years.* Raising Table 24.2 to the fifth power gives Example 24.6's distribution:

| | AAA | AA | A | BBB | BB | B | CCC | D |
|---|---|---|---|---|---|---|---|---|
| Probability (%) | 0.10 | 1.37 | 14.64 | 62.17 | 12.90 | 5.06 | 0.87 | 2.89 |

*Still investment grade* (AAA through BBB):

$$0.10 + 1.37 + 14.64 + 62.17 = \mathbf{78.3}\%$$

Fallen to speculative grade without defaulting (BB, B, CCC):

$$12.90 + 5.06 + 0.87 = \mathbf{18.8}\%$$

with the remaining 2.89% in default.

*Why compounding a one-year rate understates the five-year probability.* The one-year BBB default rate is 0.28%. Compounding it as though each year were an independent repetition gives

$$1 - (1 - 0.0028)^5 = 1.39\%$$

against the matrix's **2.89%**, a factor of **2.08**.

The naive calculation assumes the issuer is **still BBB** at the start of every year. It is not. The matrix says a fifth of surviving issuers have fallen to speculative grade by year five, and a BB issuer defaults at 1.52% a year and a B issuer at 6.45%, five and twenty-three times the BBB rate. So a BBB issuer alive in year four is disproportionately likely to be facing a much higher

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hazard than the 0.28% it started with. Default is reached through downgrade, and treating the starting rating's default rate as constant ignores the migration that does most of the work.

This is the same structure as question 4's hazard curve, arrived at from the other direction. The bootstrap of Example 24.2 *fitted* a forward default intensity that rises with horizon for an investment-grade issuer; the transition matrix *generates* one, and here the mechanism is visible rather than merely the fitted result. The upward-sloping spread curve, the rising bootstrapped hazard rate, and the factor of 2.08 are three views of one fact: a healthy firm has more room to deteriorate than to improve.

*Two uses for the matrix beyond default.* Migration is a risk even without default, because a downgrade widens spreads and triggers the contractual consequences of Section 24.10: rating triggers, collateral calls, forced sales by mandate- constrained holders. A portfolio marked to market bears migration risk whether or not anything defaults, and the 18.8% falling to speculative grade is a much larger number than the 2.89% defaulting. And the matrix is the natural input to a credit portfolio model, which Section 24.14 builds from it.

*One caution that limits all of the above.* The matrix is not stable. Transition probabilities are strongly cyclical (downgrades cluster in recessions), so a single average matrix describes no actual year. Using it for a five-year projection assumes the cycle averages out over that horizon, which it may not, and assumes transitions are Markov in the rating alone, which they are not: an issuer recently downgraded is more likely to be downgraded again than an issuer that has held its rating for a decade.

*What the question tests.* Reading a powered transition matrix, and that the Markov chain and the fitted hazard curve encode the same migration effect.

---

**24.12** *State two reasons a structural model cannot be applied to a sovereign borrower, and explain why the natural sellers of sovereign protection create wrong-way risk.*

*Two reasons a structural model cannot be applied.*

*1. There is no asset value and no debt threshold.* Merton's mechanism is that default happens when $V_T$ falls below $D$. A sovereign has no balance sheet of that kind. The relevant "asset" is the government's *capacity to tax* and its *willingness to use it*, a present value of future political decisions, not a quantity that trades or that anyone can mark. There is nothing to put on the $x$-axis of Figure 24.2, so the option decomposition of question 7 has no referent.

*2. Ability and willingness are separate, so default is a* ***choice***. A company defaults when it runs out of money. A country defaults when repaying costs more, politically, than not repaying—and it may have the money. Russia in 1998 defaulted on ruble debt it could in principle have monetized; Argentina in 2001 had given up that option under its currency board, and its case shows the other half of the point: even where the strain was real, the timing and the terms were political. A structural model is a model of *insolvency*, and sovereign default is frequently not insolvency. No threshold-crossing story captures a decision that turns on an election, a negotiation, or the relative pain of austerity.

*Two further differences are worth naming.* There is no court and no seniority: a defaulting sovereign cannot be liquidated, and the ranking of its obligations is whatever it chooses to respect, so "recovery" is the outcome of a negotiation settled by a mixture of law and pressure rather than a claim on identifiable assets. And local-currency debt is different again, since a government borrowing in its own currency can always print, making default there a policy choice with an inflationary alternative, which is why the same issuer carries different ratings for local- and foreign-currency obligations.

*Why the natural sellers create wrong-way risk.* The institutions that are often most willing to sell protection on a sovereign are **domiciled in that sovereign**: its own banks, insurers

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and pension funds. They sell for reasons that make sense individually: they know the credit best, they hold the bonds already, they are paid to take a risk they believe is overpriced, and regulation has often encouraged them to treat domestic government exposure as riskless.

But the correlation between the protection seller's solvency and the reference entity's default is high, and it runs the wrong way at several points:

- Domestic banks hold large portfolios of their own government's bonds, so a sovereign default impairs their assets directly.
- Their deposit base, their loan book and their funding all depend on the domestic economy, which a sovereign default wrecks.
- Their own credit rating is effectively capped by the sovereign's.
- And they may be recapitalized by—the sovereign.

So the protection is worth most at exactly the moment the seller cannot pay. This is Section 9.9's wrong-way risk in its purest form: not a statistical correlation between exposure and counterparty quality but a mechanical identity between them. A buyer of sovereign protection from a domestic bank has purchased something whose payout is conditional on a state in which the payer has failed—which is to say, has purchased very little, however the CVA calculation scores it.

The mechanism runs in a loop: the sovereign's weakness impairs the banks, the banks' weakness creates a contingent liability for the sovereign, and each repricing worsens the other. The euro-area crisis of 2010–12 is the case study, and it is why a buyer of sovereign protection should ask who the seller is and prefer one whose solvency does not hang on the reference entity.

*The lesson generalizes.* Wherever default is a legal event rather than an economic one, the contract's definition matters as much as the borrower's finances—and Chapter 25 is largely about what happens when those two come apart.

*What the question tests.* That structural models require both an asset process and an involuntary trigger, and that wrong-way risk here is structural rather than statistical.

---

**24.13** *Explain why fixing recovery at 40% is harmless for pricing a CDS and damaging for a capital calculation.*

*Why it is harmless for pricing.* Because a CDS's value depends only on the **product** $\lambda(1-R)$, and never on the two factors separately. Example 24.3's table makes the point: a 170 bp five-year quote is reproduced exactly by $(\lambda, R)$ of $(2.125\%, 20\%)$, $(2.833\%, 40\%)$, $(4.250\%, 60\%)$ or $(8.500\%, 80\%)$. A desk that fixes $R = 40\%$ and lets $\lambda$ absorb the residual prices the contract **correctly whatever** $R$ **truly is**, because the error in $R$ is exactly offset by the compensating error in the $\lambda$ extracted using it.

The convention is therefore not a belief about recovery. It is a coordinate choice, a normalization that fixes one degree of freedom in an under-determined system, and within its domain it is entirely safe. Marking, hedging, and unwinding a CDS book at 40% recovery is correct even if realized recoveries turn out to be 25%.

*Why it is damaging for a capital calculation.* Because regulation asks for probability of default and loss given default **separately**, and the moment the two are separated the cancellation stops working. Three distinct failures follow.

*1. The coordinate gets paired with something else.* Multiplying the CDS-implied $\lambda$ back by the same 60% loss given default is harmless: it returns the spread. But a risk system that

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takes the $\lambda$ extracted at 40% and applies its own loss-given-default estimate of, say, 75% has used two recoveries for one borrower, and a capital formula that is nonlinear in the default probability does not reproduce the spread even when the same 40% is used throughout. Either way the coordinate has been *exported* beyond the domain in which it was arbitrary.

2. *The error does not cancel, it compounds.* Suppose the true recovery is 20%. The extracted $\lambda$ at an assumed 40% is 2.833% against a true 2.125%, so probability of default is overstated by a third while loss given default is understated by a quarter. For a product depending on $\lambda(1-R)$ these offset exactly. For a *digital* default swap, or any calculation that uses $\lambda$ alone, the error is the full 33%—and it has a *sign*, so it does not average out across a portfolio.

3. *And the assumed constant is not even a constant.* Question 6 established that recovery is strongly correlated with the default rate. So the fixed 40% understates tail losses *twice over*: once because more borrowers default in a bad year, and once because each recovers less. Capital is set by a tail quantile rather than by a mean, so the understatement lands precisely where the calculation is meant to bite. An unconditional average is the wrong input to a conditional question.

*The general principle*, which is a direct analog of Section 21.10's warning: a calibration encodes an assumption about what the model is for. Fixing $R$ by seniority and sector and letting $\lambda$ absorb everything is defensible for pricing and indefensible for risk measurement, and the two uses live in the same system, drawing on the same parameter table, usually maintained by the same team. Nothing in the data announces which purpose is being served, which is why the error is so durable.

*Where the split has to come from instead.* Not from CDS quotes, which cannot supply it, but from external evidence: realized recovery studies by seniority and sector, the firm's own asset composition, secured-versus-unsecured structure, and jurisdiction, and, for a stress calculation, a recovery assumption conditioned on the stressed state rather than averaged over the cycle. Recovery rate swaps and digital default swaps would identify it directly, which is precisely why they exist, but they trade too thinly to calibrate against.

*What the question tests.* That an under-identified parameter is safe exactly as long as it stays inside the product it was normalized within, and that regulation's insistence on the split is what forces it out.

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**24.14** *A large, granular loan book has a one-year default probability of 2%, a loss given default of 45% and a copula correlation of 0.15. Compute the worst-case default rate at 99.9% and the credit VaR per \$100 million, and explain why the credit VaR would be zero if the loans defaulted independently.*

*The worst-case default rate.* From (24.15), with $N^{-1}(0.02) = -2.0537$, $N^{-1}(0.999) = 3.0902$ and $\sqrt{0.15} = 0.3873$:

$$\mathrm{WCDR}(0.999) = N\left(\frac{-2.0537 + 0.3873 \times 3.0902}{\sqrt{0.85}}\right) = N(-0.9294) = \mathbf{17.63}\%$$

In the worst year in a thousand, almost nine times the average fraction of the book defaults.

*The credit VaR.* Per \$100 million the worst-case loss is $100 \times 0.45 \times 0.1763 = \$7.93$ million and the expected loss is $100 \times 0.45 \times 0.02 = \$0.90$ million, so by (24.16)

$$\text{Credit VaR}(0.999) = 100 \times 0.45 \times (0.1763 - 0.02) = \$\mathbf{7.03} \text{ million}$$

about 7.8 times the expected loss. The expected loss is not part of it: it is charged in the loans' spread and held in provisions.

*Why independence would leave nothing.* At $\rho = 0$ the conditional default probability $p(M)$ of (24.12) equals $\pi$ whatever the common factor does, so every loan defaults independently

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with probability 2%. In a large book the law of large numbers then fixes the fraction that defaults at 2% in every year, the loss at its expected value, and the unexpected loss at zero; (24.15) returns $\pi$ at every confidence level. A finite book keeps some binomial noise, but its standard deviation falls with the square root of the number of loans, so for a granular book it is small. What a large book cannot diversify is the common factor, and correlation is the size of its exposure to it.

*What the question tests.* That credit VaR on a granular portfolio is a measure of correlation, not of default probability.

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**24.15** *Using Example 24.10, explain why the 99% credit VaR of a BBB bond is set by downgrade and its 99.9% credit VaR by default, and state what a model that ignores migration reports at each level.*

*Where the quantiles fall.* Read the BBB row of Table 24.2 from the bottom: default has probability 0.28%, CCC or worse 0.48%, and B or worse 1.38%. The first percentile is the state at which the cumulative probability first reaches 1%, which is B, so the 99% quantile of the bond's value is the B value, 93.54, and the 99% credit VaR is $106.97 - 93.54 = 13.44$ per \$100 face. The 0.1% quantile is reached already in default, because default alone carries 0.28% > 0.1%, so the 99.9% credit VaR is $106.97 - 40.00 = 66.97$.

*What a default-only model reports.* It values the bond at 107.52 unless it defaults, so its mean is $0.9972 \times 107.52 + 0.0028 \times 40 = 107.33$. At 99% its quantile is 107.52, because default is less likely than 1%; that is above the mean, and the model reports no credit VaR at all. At 99.9% its quantile is the default value, and it reports $107.33 - 40.00 = 67.33$, within half a dollar of the migration model.

*The general rule.* A quantile is set by whichever event first accumulates the tail probability. For a high-grade bond default is rarer than 1%, so a 99% measure lives entirely in downgrades, and a model without migration is blind there. Default is more likely than 0.1%, so a 99.9% measure lives in default and the migration model adds little. For a BB issuer, with a one-year default probability of 1.52%, default would set the 99% quantile as well.

*What the question tests.* That the confidence level decides which risk a credit measure sees, which is why a 97.5% expected shortfall treats credit as spread risk while the 99.9% default charges, in either book, treat it as default (Sections 39.3 and 39.10).

---

**24.16** *Explain why credit VaR subtracts the expected loss while Chapter 38's VaR is measured from zero, and why its horizon is a year rather than a few days.*

*The expected loss is already paid for.* A loan book's expected loss is predictable in aggregate and is a cost of the business, not a surprise. It is priced into the spread at origination and recognized in advance in provisions (IFRS 9's expected credit loss, CECL in the United States), so it reduces earnings before it is ever realized. Capital is for what provisions and margins do not cover, which is the loss in excess of the expected loss. Counting the expected loss again in capital would charge for it twice.

*A trading book's expected loss is negligible.* Over ten days the expected profit or loss of a trading position is tiny against its volatility, and nothing is provisioned for it, so measuring VaR from zero or from the mean gives almost the same number, and zero is the convention.

*Why a year.* A loan cannot be sold within days without a large discount, and often not at all, so the horizon over which the bank is exposed is the time it takes losses to emerge and capital to be rebuilt. Defaults and recoveries play out over months, ratings are reviewed roughly annually, and the data from which default and transition probabilities are estimated are annual. Ten days is the time to hedge or exit a liquid position; nothing comparable exists for most loans.

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*What the question tests.* That the two VaRs answer the same question about different balance sheets, and that each convention follows from what the position is and how it is accounted for.

---

**24.17** *A five-year bond with a 5% annual coupon trades at 97.00 against the flat 3% SOFR curve of Example 24.11. Find its Z-spread and the hazard rate it implies at 40% recovery by the credit triangle, and state the CDS–bond basis if the issuer's five-year CDS trades at 170 bp. Explain why a basis trader would not simply buy the bond and buy protection.*

*The Z-spread.* It is the $z$ that solves

$$5\left(e^{-(0.03+z)}+e^{-2(0.03+z)}+e^{-3(0.03+z)}+e^{-4(0.03+z)}\right)+105\,e^{-5(0.03+z)}=97.00,$$

found by trial: Example 24.11 showed that $z = 2.20\%$ gives 98.55, and the price falls by about 4.5 per percentage point of spread, so $z$ is near $2.20\% + 1.55/4.5 \approx 2.54\%$. Solving exactly gives $z =$ **255 bp**.

*The hazard rate.* By the credit triangle, $\lambda = 0.0255/0.60 =$ **4.25**%. Pricing the bond with a constant hazard rate and 40% recovery of face, as in Example 24.11, gives 4.23%, so the triangle is again close for a bond near par.

*The basis.* By (24.17), $170 - 255 =$ **−85 bp**. The bond's spread is 85 bp wider than the CDS spread for the same issuer's default risk.

*Why the trade is not a riskless profit.* Buying the bond and buying protection leaves no credit risk and earns the 85 bp only if the bond can be financed at the risk-free rate. It cannot. It is funded in repo with a haircut and a spread over SOFR, and the package uses balance sheet that carries a capital charge, so the trade earns the basis minus the funding cost minus the cost of capital. As Example 24.12 shows, a basis of this size can be fully absorbed by funding, and it widens precisely when funding is scarce, which is when a trader holding the package would be forced to unwind it. The cheapest-to-deliver option and the broader credit-event definition also push the CDS spread up relative to the bond, so part of the gap is not mispricing at all.

*What the question tests.* Reading a hazard rate from a bond as well as from a CDS, and seeing that the gap between them is a price of funding rather than a riskless profit.

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# Chapter 25
# Credit Derivatives and the CDS Market

**25.1** *Explain why a protection buyer need not own the reference obligation, giving one benefit and one problem this creates, and explain why gross CDS notional outstanding is not a measure of the risk transferred.*

*Why ownership is not required.* A CDS is a **bilateral contract between two parties about a third**, and the reference entity is not a party to it. The premium leg and the protection leg run between buyer and seller; the reference entity's role is only to supply the event that triggers the payout. Nothing in the structure requires the buyer to hold the debt, any more than a buyer of a put on an index must hold the index.

*One benefit: it makes the instrument tradeable, and therefore useful.* Requiring ownership would make credit protection an insurance contract: bespoke, illiquid, held to maturity. Without it, credit becomes a two-way market with a price. Three concrete consequences: a **short** position in credit becomes possible at all, where selling a bond one does not own requires a repo market that barely exists for corporate debt; a **price** for credit risk is discovered continuously rather than inferred from occasional bond trades; and a bank can hedge a loan *portfolio*, or a CVA desk hedge counterparty exposure (Chapter 9), without touching the underlying relationship.

*One problem: protection outstanding can exceed the debt outstanding.* That decoupling caused the failure of the original settlement mechanism. *Physical settlement* required the protection buyer to deliver a defaulted bond and receive par. When more protection existed than deliverable debt, buyers bid for the bonds, driving their price *up*—which reduced the payout on exactly the contracts that were supposed to be paying out. The instrument's own settlement mechanism destroyed its value at the moment of exercise, and question 4 is about the repair.

The deeper problem is the same decoupling seen as an incentive: a party with protection and no underlying exposure has a strictly one-sided interest in the credit event occurring. That is not hypothetical: engineered defaults, in which a borrower was paid to miss a payment so that a protection holder could collect, occurred often enough to prompt contract amendments in 2019.

*Why gross notional is not a measure of risk transferred.* Because it counts contracts, not exposures, and most contracts offset. A dealer that has bought $100 million of protection on an issuer and sold $100 million on the same issuer, maturity and coupon holds **$200 million of gross notional and no risk at all**.

That is not a corner case; it is the normal state of a dealer book, which intermediates rather than positions. Two mechanisms then make the gross figure still less meaningful. *Compression* services tear up offsetting trades in bulk (made possible by the Big Bang's standardization, question 13), removing enormous gross notional and zero net exposure. And *clearing* nets multilaterally rather than bilaterally, shrinking the reported number again without changing

anyone's economic position.

So the pre-crisis figure of tens of trillions of dollars, routinely quoted as though it were the amount at risk, was mostly the same exposures counted repeatedly. The right measure is **net notional** by reference entity, what would actually change hands on a credit event, which is a small fraction of gross and is now published by the trade repositories the reforms of Chapter 2 created. This is Section 1.6's argument, and the fact that gross notional was ever treated as a risk measure is the reason that section exists.

*What the question tests.* That the contract's independence from the debt is what makes it a market and also what broke its settlement, and that gross notional counts intermediation rather than exposure.

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**25.2** *A five-year CDS has a par spread of 250 bp and trades on a 100 bp coupon with a risky annuity of 4.19. Compute the upfront payment, and explain why the risky annuity falls as the spread widens.*

*The upfront payment.* A standardized contract pays a fixed coupon, so any difference between the market's par spread and that coupon is settled in cash at inception:

$$\text{Upfront} = (s - c)\,A = (250 - 100)\text{bp} \times 4.19 = 628.5 \text{ bp} = \mathbf{6.285}\% \text{ of notional}$$

or \$628,500 on \$10 million. The **protection buyer pays** it: the market price of protection is 250 bp but the contract only obliges them to pay 100 bp, so they must compensate the seller for the 150 bp shortfall over the contract's expected life. The risky annuity is precisely the factor converting a running spread into a present value.

This is the Big Bang convention of Section 25.3, and its purpose is fungibility: every contract on an issuer carries the same coupon and maturity, so two contracts can be netted and torn up against each other. The upfront absorbs everything that would otherwise make them different instruments, which is what made the compression of question 13 possible.

*Why the risky annuity falls as the spread widens.* The annuity is the expected present value of one basis point of premium:

$$A = \sum_i \Delta_i \, D(t_i) \, Q(t_i)$$

where $Q(t_i)$ is the probability of surviving to the payment date (the accrual-on-default term of (25.2), about half a percent of the total, is not in this sum but is in the 4.19). A wider spread implies a higher hazard rate (by the credit triangle, $\lambda = s/(1 - R)$), so the survival probabilities fall, and with them every term in the sum. The premium stream is shorter in expectation because default arrives sooner and the payments stop.

The effect is not small. At 250 bp and 40% recovery the hazard rate is 4.17%, so roughly a fifth of the notional is expected to have defaulted by year five and stopped paying. On a riskless annuity the five-year figure at 3% would be about 4.6; the risky annuity is 4.19.

The consequence is a concavity that works against the protection buyer, and it is the subject of question 3. Because the annuity shrinks as the spread widens, each additional basis point of widening is worth *less* than the last. Protection gains less than a linear duration calculation suggests, and the shortfall grows with the size of the move. A desk sizing a hedge on duration alone is under-hedged exactly when the hedge is working.

*And the annuity is where the discount curve enters, which the par spread did not.* Chapter 24's question 2 established that a *par* spread barely depends on discounting, because the discount factor cancels between the two legs, exactly so for a flat hazard rate. An *upfront* is a present value, so it does not cancel: a change in the discounting convention moves the upfront on an off-market contract while leaving the par spread almost untouched. That distinction is why the reforms of Chapter 4 changed CDS marks without changing CDS quotes.

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*What the question tests.* The upfront convention, and that the annuity is a survival-weighted duration whose shrinkage is what makes the instrument concave for the buyer.

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**25.3** *A desk bought five-year protection on the standard 100 bp coupon when the par spread was 150 bp, paying an upfront of 2.18% of notional. One year later the spread is 400 bp, and the risky annuity on the remaining four years at that level is 3.31. Compute the position's value and its gain since inception as percentages of notional, explain why marking against 150 bp or using the annuity at the original spread misstates them, and state in which direction the annuity error biases a hedge.*

*The value.* What the desk holds is a contract paying the 100 bp coupon; the 150 bp at which it traded was settled in the upfront on the trade date. By (25.6), with the annuity *at the current spread*,

$$\text{MTM} = (s_t - c)\, A_t = (400 - 100)\text{bp} \times 3.31 = 993 \text{ bp} = \mathbf{9.93}\% \text{ of notional},$$

which is what a dealer would pay to take the position over.

*The gain since inception* is the value less the upfront paid: $9.93\% - 2.18\% = \mathbf{7.75}\%$ of notional, before the coupons the desk has paid to carry the position. (The 2.18% is (150 − 100) bp times the five-year annuity at 150 bp, 4.3550, as in Example 25.2.)

*Why marking against 150 bp is wrong.* $(400 - 150)\text{ bp} \times 3.31 = 8.28\%$ is neither the value nor the gain. It treats the traded spread as though it were still being paid, when the contract pays 100 bp and the difference was exchanged as cash a year ago. It happens to land between the two correct numbers, which is what makes the error easy to miss.

*Why the original annuity overstates the value.* Both objects in (25.6) carry the subscript $t$. The annuity to use is the one describing the premium stream that will *actually* be exchanged from here, and the market now expects that stream to be shorter, because it expects default sooner. On the same four-year horizon, 3% discounting and 40% recovery, the annuity at 150 bp is about **3.58** against 3.31 at the current level; using it gives $300 \times 3.58 = 1{,}074$ bp, or 10.74%, an overstatement of **8%**. Example 25.3 works the same error on its own numbers and finds 7.22% against a correct 6.82%, an overstatement of 5.9%.

*The direction is systematic.* As a credit deteriorates the annuity *shortens*, so the naive calculation, which holds the annuity fixed, always overstates the protection buyer's gain, and by more the larger the move. The position is concave in the spread, and the concavity works against the protection buyer: the further the spread widens, the less each additional basis point is worth.

*Which direction it biases a hedge: toward being* ***under-hedged****, and precisely when the hedge is needed.* A desk that computes its credit sensitivity as "notional times spread duration" using the current annuity believes each basis point of widening delivers a fixed amount. It does not. As the spread widens the delivered amount per basis point shrinks, so the hedge that was correctly sized at 150 bp is too small at 400 bp, and too small by more at 800 bp. The hedge decays as the event it hedges approaches.

Two practical consequences follow. A protection buyer must **re-size dynamically**, adding notional as spreads widen. This is the credit market's version of a short-gamma rebalancing rule, with the same procyclical character: buying more protection into a widening market. And a hedge ratio computed from a linear spread duration is a first-order approximation to a concave relation, so risk systems that report only "CS01" (the change in value for a one-basis-point move in the spread) describe the position for small moves only, exactly as a delta describes an option for small moves only.

*What the question tests.* That the mark is taken against the coupon, not the traded spread, because the traded spread was settled upfront; that both terms in the mark are evaluated on the

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current curve; and that the annuity's shrinkage makes the position concave in a direction that flatters the naive calculation.

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**25.4** *Explain why physical settlement failed, what the auction replaced it with, and why an auction recovery is not the same as the eventual realized recovery.*

*Why physical settlement failed.* The original design had the protection buyer deliver a defaulted bond and receive par. It works only if there are **enough bonds**, and question 1 established that protection outstanding can exceed the debt outstanding, because a buyer need not own what they insure.

When it does, the mechanism turns on itself. Every protection buyer must obtain a deliverable obligation, so they bid for a fixed and small supply of defaulted bonds. The price is driven **up**, in some cases far above any plausible recovery, which *reduces* the payout on exactly the contracts that are supposed to be paying out, since the buyer's gain is par less the price they paid for the bond. A short squeeze in the deliverable, produced by the settlement process itself, at the moment of maximum stress. This happened, and it is why the mechanism changed.

*What the auction replaced it with.* Run under separate protocols before 2009 and written into the standard contract by the 2009 reforms, a **two-stage auction** determines a single recovery rate for the whole market, and all contracts settle in *cash* against it. The first stage collects dealer markets and physical settlement requests, establishing an initial price and a *net open interest*: the residual amount that genuinely needs to change hands after offsetting requests are matched. The second stage runs a limit-order auction to clear that interest, and the clearing price becomes the final recovery $R$ applied to every contract.

The design solves the squeeze directly. Because settlement is in cash against one number, no one needs to source a bond, so there is no bidding for scarce deliverables. And because the first stage nets offsetting requests, the amount actually auctioned is a small fraction of gross notional.

*Why an auction recovery is not the eventual realized recovery.* Because they are different quantities measured at different times by different processes.

The auction price is a market price on one day, set weeks after the credit event, by whoever chooses to participate. The realized recovery is the **cash actually distributed** at the end of a bankruptcy process that may run for years, discounted back. Nothing forces the first to equal the second, and empirically they diverge substantially in both directions.

Three specific reasons follow. The auction happens *early*, when the outcome of the restructuring is genuinely unknown, so it prices an expectation rather than an outcome. Participation is *limited*: the bidders are dealers and distressed funds, and a thin auction is a poor price discovery mechanism for an asset no one has valued yet. And the auction determines a single number for *all* deliverable obligations, though in reality different bonds of the same issuer recover differently by seniority and maturity, so it prices a basket at the cheapest-to-deliver level.

*The consequence for how recovery should be understood.* Chapter 24's 40% is a convention used to extract hazard rates; even the observed $R$ at settlement is a market price rather than a realized outcome. So there are three distinct objects routinely called "recovery"—the assumed recovery in a pricing model, the auction recovery that determines the payout, and the eventual distribution—and only the second is what a CDS holder actually receives. A holder who was right about the eventual recovery and wrong about the auction is paid on the auction.

*And the auction is now worth a great deal of money to influence.* A single number determines the payout on every contract, so participants with large positions have an interest in the clearing price, and the auction's design (caps on the price range, the mechanics of the net open interest)

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exists to constrain that. It is a smaller problem than the squeeze it replaced, but it is not none.

*What the question tests.* That the original mechanism failed because of the decoupling in question 1, and that the auction price is an expectation formed early rather than a realized outcome.

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**25.5** *Explain why an index CDS trades below the arithmetic average of its constituents' spreads, what the index skew is, and why it widens when the index is used as a macro hedge.*

*Why the index trades below the arithmetic average.* Because an index prices at the **annuity-weighted** average of its constituents, not the arithmetic one. Balancing the portfolio's two legs rather than each issuer's separately gives (25.7),

$$s_{\text{index}} = \frac{\sum_j V_j}{\sum_j A_j} = \frac{\sum_j A_j\, s_j}{\sum_j A_j}$$

and question 2 established that a wide-spread issuer has a *shorter* risky annuity, because its higher hazard rate discounts the later premium payments more heavily. So the widest issuers, the ones pulling the arithmetic mean up, carry **less** weight in the correct formula than in a simple mean, and the index prints below it.

Example 25.4 sizes the effect: a 125-issuer index of 115 issuers at 55 bp and 10 distressed issuers at 367.5 bp has an arithmetic average of 80.0 bp and a theoretical index spread of **77.3 bp**. The 2.7 bp difference is arithmetic, not a mispricing, and the wider the dispersion the larger it grows: an index of identical issuers would show no gap at all.

This is the same structure as Section 19.7's index-versus-constituents relation in the volatility market: a portfolio instrument does not price at the average of its components, and the wedge has an identifiable cause rather than being a market imperfection.

*What the index skew is.* The **residual** difference between the index's actual traded level and this theoretical annuity-weighted value. Having removed the arithmetic wedge, whatever remains is a genuine dislocation between the index and its constituents, and it is a real trading opportunity when it is large enough to cover the cost of trading 125 single issuers against one index. That cost is substantial: 125 bid–ask spreads, in issuers that question 13 notes are far less liquid than they were, against one liquid index trade.

*Why it widens when the index is used as a macro hedge.* Because that demand moves the index **without moving the constituents**. An investor wanting to reduce broad credit exposure ahead of a macro event does not buy protection on 125 issuers individually, which would be slow, expensive, and would move each issuer's market. They buy the index, which is the credit market's principal liquidity: bid–ask spreads on the index are a fraction of those on constituents, and a desk hedging a portfolio will usually hedge with the index and accept the basis.

So a wave of hedging demand lands entirely on one instrument. The index widens; the single issuers, which no one traded, do not. The skew is the accumulated pressure, and it is a measure of how much macro hedging is being done through the index rather than of anything about the underlying credits.

Two riders follow. The gap does not arbitrage away quickly, because closing it requires exactly the 125-issuer trade whose cost created the opportunity. This is Section 5.15's limits-of-arbitrage argument again, and the reason the skew can persist for weeks. And the direction is predictable: macro hedging is overwhelmingly *buying* protection, so the skew typically has the index trading *wide* of theoretical in stressed periods, which is precisely when a desk holding the constituent side is least able to fund the position.

This is Section 19.7's index protection demand appearing in a different market, and the parallel is exact: in equities, index put demand steepens the index skew relative to single stocks;

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in credit, index protection demand widens the index relative to its constituents. Both are the price of liquidity in the instrument everyone reaches for.

*What the question tests.* Separating the arithmetic wedge from the traded dislocation, and recognizing macro hedging demand as the cause of the second.

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**25.6** *Explain why restructuring creates a cheapest-to-deliver problem that bankruptcy does not, and why a risk system treating restructuring and no-restructuring contracts as substitutes has mismeasured the exposure.*

*Why restructuring creates a cheapest-to-deliver problem that bankruptcy does not.* Because after a restructuring the entity is **still alive**, and a living issuer's obligations trade at very different prices.

After a bankruptcy, all senior unsecured obligations of the same issuer are worth roughly the same thing: they become pari passu claims on one estate, and maturity ceases to matter because nothing is going to be paid on schedule anyway. Which bond the protection buyer delivers hardly matters, so the optionality embedded in "deliverable obligation" is nearly worthless.

After a restructuring the issuer continues to operate and continues to pay. Its obligations therefore retain a term structure, and a distressed but living issuer's **thirty-year bond is worth far less than its one-year bond**: the near maturity will probably be repaid, the distant one probably will not. A protection buyer holding a contract that pays $\mathcal{N}(1-R)$ where $R$ is the price of whatever they choose to deliver will deliver the long bond, collect a payout far exceeding their actual loss, and the contract will have overpaid.

That is a genuine free option handed to the protection buyer, and it is worth a material amount on a distressed issuer where the curve is steep.

*The market's responses.* Successive restrictions on what could be delivered (first limits on the maturity of deliverable obligations, then a system of **buckets** matching the deliverable's maturity to the contract's) and, in North America, largely **abandoning restructuring as a trigger** for corporate issuers. Europe kept it, mainly because bank regulators require it: a hedge that does not pay on restructuring does not obviously transfer the risk of the loan it is hedging, so a no-restructuring contract may not earn capital relief.

The result is that identical-looking contracts on the same issuer differ by jurisdiction and vintage: full or "old" restructuring (CR), modified restructuring (MR), modified-modified restructuring (MMR), and no restructuring (XR) are distinct instruments with distinct prices. The first three differ in how tightly they limit the deliverable maturities.

*Why treating them as substitutes mismeasures the exposure.* Because **they are different contracts**, and the difference has a price. Example 25.7 puts a number on it: an issuer with a 3% intensity of bankruptcy or failure to pay, recovering 40%, prices at 180.0 bp with no restructuring; restructuring events that raise that intensity by a further 15%, or 0.45 percentage points, and recover 70% take it to 193.5 bp. The clause is worth 13.5 bp, or 7% of the spread.

Three specific failures follow from netting them:

*The basis is real and one-directional.* A risk system offsetting a bought CR contract against a sold XR contract reports zero and holds a position that is long the restructuring clause, a genuine exposure worth 7% of the spread, with no capital and no limit against it.

*The hedge may not pay.* A bank hedging a loan with an XR contract is unprotected against precisely the outcome most likely for a distressed corporate borrower, which is a negotiated maturity extension rather than a filing. The hedge fails in the modal case.

*And the two cannot be compressed.* Question 13's compression relies on contracts being fungible; contracts differing in the restructuring clause are not, so they accumulate as gross notional that looks offsetting and is not.

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*The general lesson, which is this chapter's thesis.* A CDS insures against a *list of events*, not against loss. Change the list and you have changed the instrument, however similar the ticker looks—and question 7 is the case where the list and the loss came apart completely.

*What the question tests.* That a surviving issuer retains a price curve while a bankrupt one does not, and that a documentation difference worth 7% of the spread is an exposure rather than a detail.

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**25.7** *State what the Credit Suisse AT1 write-down established about the meaning of subordination, and explain why CDS on Credit Suisse did not pay.*

*What the write-down established.* That **subordination is a property of the documents, not of the word**—and that the ordering it implies is jurisdiction-specific.

In March 2023, as part of the arrangements accompanying Credit Suisse's acquisition, roughly **CHF 16 billion of Additional Tier 1 instruments were written down to zero** while shareholders received approximately **CHF 3 billion**. An instrument ranking *senior* to equity was extinguished while equity retained value. The capital structure was inverted, which is the opposite of the ordering the word "subordination" is supposed to guarantee.

Three observations follow, and the third is the one for this chapter.

*It was contested in law, not only in markets.* The Swiss AT1 terms allowed write-down on a viability event determined by the regulator, and holders had been told so. In October 2025, however, the Federal Administrative Court found that no such event had occurred and that the Swiss regulator's order lacked a legal basis; the appeal is pending, and the bonds remain written down. Whichever way it ends, almost no one had priced the possibility, because the instruments had been sold and analyzed as though the ordinary hierarchy applied.

*The repricing was immediate and general.* AT1 instruments across European banks fell sharply, and several regulators issued statements affirming that in *their* jurisdictions equity would absorb losses first. The market had been pricing a legal ordering that turned out to be jurisdiction-specific, so the repricing was of every AT1 document, not of Credit Suisse.

*And no credit event occurred.*

*Why CDS on Credit Suisse did not pay.* Because Credit Suisse **did not default**. It was acquired, and its senior obligations were performing throughout: every payment was made on time. None of the listed credit events (failure to pay, bankruptcy, restructuring, and, for a European bank, governmental intervention) occurred: the determinations committee ruled in May 2023 that the write-down fell on instruments junior to every obligation the contracts reference. A government-brokered acquisition of a solvent-but-failing bank is not on the list, and a CDS pays against the list.

This is Section 25.2's thesis in its cleanest form: a CDS insures against a list of events, not against loss. The asymmetry is stark. A protection buyer who had correctly identified Credit Suisse as the weakest large European bank, and had bought protection to express exactly that view, would have been **right about the institution and received nothing**. An AT1 holder with no view at all lost the entire position. The instrument designed to pay on credit deterioration did not pay; the instrument designed to absorb losses in going concern absorbed them completely, out of order.

*The general lesson extends past this instrument.* Wherever a payoff is defined by a *legal* event rather than by an *economic* outcome, the document is the instrument, and the residual risk is not credit risk but the risk that the document does not say what everyone assumed. Chapter 1's account of why the ISDA Master Agreement's netting clause matters was the same observation from the other side: the documentation is not an administrative wrapper around the trade, it is the trade.

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Chapter 24's question 12 made the related point for sovereigns: there a loss is imposed by a decision rather than forced by insolvency, so no balance-sheet model predicts it. Credit Suisse shows the same can happen to a corporate, once a resolution authority holds a switch the contract did not anticipate.

*What the question tests.* That seniority is contractual and jurisdictional rather than conceptual, and that a correct credit view can pay nothing when the trigger is a legal list.

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**25.8** *Explain why an equity tranche's expected loss falls as correlation rises while a senior tranche's rises, given that the portfolio's expected loss is unchanged, and use Example 25.5 to state the factor by which a 15–30% tranche's expected loss grows between $\rho = 0.10$ and $\rho = 0.50$.*

*The mechanism, which is Section 24.13's redistribution seen through the capital structure.* Correlation leaves the portfolio's expected loss **unchanged** (4.200% of notional at every $\rho$ in Example 25.5) and moves probability out of the middle of the loss distribution into both tails. The tranches sit at different points of that distribution, so they experience the redistribution in opposite directions.

*The equity tranche gains.* It absorbs the *first* losses, so it is destroyed by the ordinary case of a few scattered defaults and pays in full only if **almost nothing happens**. Raising correlation raises the probability of the "nothing happens" state enormously (Example 24.7 gives Pr(no defaults) rising from 0.59% at $\rho = 0$ to 53% at $\rho = 0.6$), so the one outcome under which the equity tranche survives becomes far more likely. Its expected loss falls from 95.76% at $\rho = 0$ to 26.59% at $\rho = 0.80$.

*The senior tranche loses.* It suffers only if losses exceed a large attachment point, which requires **many simultaneous** defaults. At low correlation that is essentially impossible: the issuers fail independently, the loss distribution is tightly concentrated near its mean, and a 15% portfolio loss, a quarter of the issuers defaulting, is a statistical impossibility rather than a remote risk. Raising correlation fattens exactly that tail. The 15–30% tranche's expected loss rises from 0.00% at $\rho = 0$ to 6.96% at $\rho = 0.80$.

*The two are consistent because the mean is fixed.* What the equity tranche stops losing, the senior tranche starts losing. Correlation does not create or destroy expected loss; it decides *which layer bears it*. Selling equity protection is buying correlation; selling senior protection is selling it.

*The factor for the 15–30% tranche.* From Example 25.5:

| $\rho$ | **Expected loss** |
|---|---|
| 0.10 | 0.098% |
| 0.50 | 4.272% |

$$\frac{4.272}{0.0983} = \mathbf{43.5}, \text{ a factor of about } \mathbf{43}$$

Read what has and has not changed. The same 125 issuers, the same 7% default probabilities, the same 40% recovery, the same 4.2% portfolio expected loss. Only a parameter that no market quoted has changed, and the tranche's expected loss has moved by a factor of forty-three.

*Two consequences follow.* The **rating** on such a tranche is a statement about $\rho$. Section 24.10 observed that the letter did not disclose the correlation assumption; this table says what that omission was worth: the difference between an instrument that almost never loses and one that loses 4.3% of its notional in expectation. And the market's own correlation number is not a correlation, which is question 9.

*What the question tests.* That the mean is invariant and the layers are not, and the size of a senior tranche's dependence on a parameter no one observes.

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**25.9** *Explain why implied correlation differing across tranches on the same portfolio is logically impossible, and what base correlation does about it.*

*Why it is logically impossible.* $\rho$ in the one-factor Gaussian copula is a property of **the portfolio**: the pairwise correlation of the latent variables driving 125 issuers. It describes how the constituents move together, and the constituents do not change when one slices their losses differently.

So a single number $\rho$ should reproduce *every* tranche on that portfolio simultaneously: pick the correlation, integrate the loss distribution, and read off each layer's expected loss. That is what the model asserts. Quoting each tranche in terms of the $\rho$ that reproduces *its own* price produces a different value for each—and if $\rho$ were what the model says it is, that cannot happen. One portfolio has one dependence structure. Getting five answers means the model is wrong, not that correlation varies by attachment point.

*What the pattern is really saying.* The Gaussian copula's loss distribution has the wrong shape. Its tail is too thin: Gaussian dependence has *zero tail dependence*, so in the limit extreme joint defaults are less likely than any credible account of contagion allows. The market, pricing senior tranches, demands more for that tail than the model at any single $\rho$ will produce, while simultaneously pricing the equity tranche at a level implying much less. Forcing one parameter to fit both is impossible, so the fitted parameter absorbs the misspecification and varies across the capital structure.

This is exactly Section 19.1's implied volatility smile, and the parallel is worth stating in full: a single-parameter model quoted across strikes, producing a different parameter at each strike, with everyone agreeing to use it as a *quoting convention* rather than a belief. Implied correlation is the credit market's smile, and it exists for the same reason: the model's distribution is the wrong shape in the tails.

*What base correlation does about it.* It removes the inconsistency by **re-parameterizing**, not by fixing the model. Instead of quoting each tranche $[a, b]$ with its own correlation, base correlation quotes a correlation for each *first-loss* tranche $[0, b]$, the equity tranche widened to detach at $b$, which are the "base" tranches. Any tranche is then the difference of two base tranches:

$$\mathrm{EL}[a, b] = \mathrm{EL}[0, b] - \mathrm{EL}[0, a]$$

each priced at its own base correlation. Because the base correlations are attached to a nested sequence rather than to disjoint slices, the construction reproduces every quoted tranche and the reconstructed tranches sum to the portfolio by design, although interpolating between quoted detachment points can still imply an arbitrage (a thin tranche with negative expected loss).

*But it removes the inconsistency without removing the fact that generated it.* The base correlation curve slopes, rising with detachment point on an equity-tranche basis, and that slope *is* the misspecification, relabeled. Base correlation is a change of coordinates, exactly as Black–Scholes implied volatility is a change of coordinates for an option price. It makes the numbers interpolate and compare sensibly; it says nothing about how firms actually fail.

*The practical dangers, and Chapter 26 is what happened when they were ignored.* Interpolating a base correlation curve to price a *bespoke* tranche with attachment points the market does not quote is extrapolating a fitted artifact. Hedging one tranche with another using base-correlation sensitivities assumes the curve moves in a particular way, which it need not. And a risk report carrying a base correlation as though it were an estimated dependence parameter has converted a quoting convention into a belief, which is the sentence that ends the chapter's example, and the subject of the next chapter.

*What the question tests.* That a portfolio has one dependence structure, that a strike-varying parameter is evidence of misspecification, and that base correlation is a consistent coordinate

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system rather than a repair.

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**25.10** *A convertible has a conversion ratio of 25 and the stock trades at 30. Compute the conversion value and the conversion price on a face of 1000, and explain why the instrument behaves like a bond at low share prices and like equity at high ones.*

*The two quantities.* With a conversion ratio of 25 and a face of 1000:

$$\text{Conversion value (parity)} = 25 \times \$30 = \$\mathbf{750}$$

$$\text{Conversion price} = \frac{\$1000}{25} = \$\mathbf{40}$$

So the bond is convertible into stock currently worth 75% of its face, and the share price would have to rise 33% to make conversion worth par.

The conversion price is the *effective strike* of the embedded option, and reading it that way is the point of computing it: the convertible is a bond plus a call on 25 shares struck at $40. The issuer has sold equity at $40 in exchange for a cheaper coupon.

*Why it behaves like a bond at low share prices.* The holder has two claims and takes the better one. At low prices the conversion option is deep out of the money, worth little, and the instrument's value rests on the **bond floor**: the present value of coupons and principal discounted at the issuer's credit spread. The equity delta approaches zero, and the security trades on credit and rates like any other corporate bond. Example 25.8 shows the pattern on a comparable structure: at a share price of 20 the convertible is worth 827.87 against a parity of only 400, with a delta of 6.00 shares out of a possible 20.

*Why it behaves like equity at high share prices.* The option moves deep into the money, its delta approaches the full conversion ratio, and the instrument tracks parity nearly one for one. At a share price of 70 the same example gives 1484.96 against a parity of 1400, a delta of 17.18 of 20 shares, and a premium over parity of only 6%. The bond floor has become irrelevant, because no holder would take 1000 of principal instead of 1400 of stock.

The transition is governed by the embedded call's delta, which is what makes the instrument interesting rather than merely a package. It is a bond whose sensitivity to the equity rises smoothly from zero to one as the equity rises—which is to say, the convertible is **long gamma**, and that is what a convertible arbitrage fund is buying (question 11).

*What each side gets.* The issuer obtains debt at a coupon far below its straight-debt cost (the example's 2% against a 7% discount rate) in exchange for equity upside above the conversion price. That is Section 17.9's warrant-attached-to-a-bond with the two components *fused* rather than separable, and Section 17.4's dilution analysis applies to the embedded call, since conversion creates new shares. The holder obtains downside protection they would not have holding the equity, at the cost of the coupon given up against the issuer's straight debt.

*One asymmetry stands out.* The bond floor is only as good as the issuer's credit, and it falls when the equity falls, because both respond to the firm's deterioration. So the "bond-like" region protects least in exactly the scenario that sends the instrument there, which is question 11's subject and the reason convertible arbitrage failed in 2008.

*What the question tests.* Two definitional computations, and that the instrument's character is the embedded call's delta rather than a switch between two regimes.

---

**25.11** *State the four exposures of a delta-hedged convertible arbitrage position, and explain why two of them go wrong together and why the timing of their recognition differs.*

*The four exposures.* The classic trade buys the convertible and shorts the delta in the

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underlying stock, which removes the first-order equity exposure and leaves:

1. **Long gamma.** The embedded call is unhedged in second order, so the position profits from movement in either direction. This is Chapter 18's dealer position with the sign reversed: the fund is long the convexity a dealer is usually short, and it earns that convexity by rehedging the delta as the stock moves.

2. **Long vega.** The same option is long volatility, and the convertible is typically bought because its embedded option looks cheap against listed options on the same stock.

3. **Short credit.** The bond floor is a *risky* bond, so the position loses if the issuer's spread widens. This is the exposure the equity hedge does not touch.

4. **Long carry.** The coupon on the convertible, less the cost of maintaining the short (stock borrow, less any short rebate).

*Which two go wrong together, and why.* The **credit** and **volatility** exposures, because a single event drives both: credit widens precisely when equity volatility spikes. A deteriorating issuer has a wider spread and a more volatile, more levered equity. Chapter 24's question 7 derived exactly this, since $\sigma_E/\sigma_V$ diverges as the firm approaches its default boundary.

The size is not trivial. In Example 25.8's structure, a 100 bp widening in the issuer's spread with the stock unchanged costs 36.66 on the bond floor, or 3.58% of the convertible's value.

Now the apparent paradox: on that description the two should *offset*. The position is long volatility and short credit, and the event raises volatility while widening credit, so the volatility gain and the credit loss arrive together and net out. That is the argument, and it is why the trade is described as hedged.

*Why the timing of their recognition differs, which is where the argument fails.*

*The credit loss is a mark-to-market and arrives* ***immediately***. The bond floor reprices the instant the spread widens. It appears in today's net asset value, in today's margin call, and in today's investor report. Nothing needs to happen for it to be realized; it is realized by the market's quote.

*The volatility gain is earned only by* ***gamma-hedging through the move***. Long gamma is not a payment—it is the right to rehedge repeatedly at favorable prices, collected in small increments over days and weeks as the stock oscillates, exactly as Section 18.4 describes. Realizing it requires the fund to **still be there**, holding the position, able to trade, and able to fund the short.

So the offset is real in the model and asymmetric in time: the loss is booked at once and the compensating gain is a claim on future trading. Under stress, three things then attack the second leg simultaneously. Prime brokers raise haircuts on convertible inventory. Investors redeem. And stock borrow becomes expensive or is recalled, which attacks the fourth exposure (carry) and can force the short to be covered, destroying the hedge itself.

*In 2008 many funds were not there.* Convertible arbitrage, a strategy that is genuinely hedged in the sense that its Greeks offset, produced one of the worst drawdowns of any hedge fund style. Convertibles traded well below their theoretical value for months, which is to say the arbitrage was available and no one could hold it.

*The general lesson.* Being hedged in the model and being able to hold the position are different things. A Greek-neutral position with a funding requirement is not risk-free; it has converted market risk into liquidity risk, and liquidity risk arrives on a timetable set by other people. This is Section 5.15's limits-of-arbitrage argument in yet another market, and it recurs in Chapter 27's Archegos and in Chapter 3's Metallgesellschaft stack-and-roll hedge.

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*What the question tests.* Decomposing the trade into its Greeks, recognizing that two of them share a driver, and that a theoretical offset fails when its two halves are realized on different clocks.

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**25.12** *Explain why a total return swap position does not appear in ownership disclosures, and who bears the risk if the receiver cannot meet a margin call.*

*Why it does not appear in ownership disclosures.* Because **the receiver does not own the asset**. One party pays the total return on a reference asset (coupons, dividends and price change) and receives a financing rate. The receiver has the full economics of ownership without the ownership, and without paying for it: the position is funded by the swap.

Disclosure regimes are written around *ownership* and *voting rights*. Beneficial-ownership thresholds, substantial-shareholding notifications, and short-position reporting all key off holding or controlling shares. A total return receiver holds neither. The dealer paying the total return buys the asset as a hedge and appears as the owner, so the position shows up in the disclosure record under the *dealer's* name, spread across several dealers, none of whom is taking any risk.

Three properties follow, and they recur whenever an instrument separates economics from ownership:

*Leverage is set by the margin, not by the asset.* A cash buyer of a bond posts its full price. A total return receiver posts *initial margin*, which may be a small fraction. The economics are identical and the balance sheet is not.

*The exposure is invisible where people look for it.* This is Section 7.12's phrase about leverage accumulating quietly. Because each dealer sees only its own book, no one—not the dealers, not the issuer, not the regulator—observes the aggregate position. Chapter 27's case study is what that enabled: a single family office accumulated economic exposure exceeding disclosure thresholds many times over across several prime brokers, and each broker believed it was facing a moderately levered client.

*And the risk is counterparty risk presented as market risk.*

*Who bears the risk if the receiver cannot meet a margin call: the* ***dealer paying total return***. The dealer is hedged in the market (it holds the asset), so it has no market exposure while the client performs. If the asset falls, the client owes variation margin. If the client cannot pay, the dealer is left holding the asset at a price above where it now trades, and must liquidate it into the market. The hedge is perfect against the market and worthless against the client, which is the configuration Chapters 8 and 9 spent their length on.

The loss is the shortfall between the collateral held and the cost of unwinding the hedge, and three things make it larger than the initial margin was sized for. The move that caused the default is correlated with the size of the position, since a large concentrated position *is* what moves the price. Several dealers unwind the *same* hedge at the same moment, so the liquidation is not into a normal market but into other dealers' liquidations. And the margin period of risk (the time between the missed call and completed liquidation) stretches exactly when it matters, which is Chapter 27's subject.

*The wider point.* A structure that is opaque to disclosure is opaque to *risk management*, and the two failures are the same failure. Each dealer priced its exposure to a client whose true leverage it could not see, and the initial margin each collected was appropriate for the position it knew about. Post-Archegos, the responses run in both directions: better client-level disclosure of aggregate positions, and higher initial margin on concentrated single-name total return swaps.

*What the question tests.* That disclosure keys off ownership while risk keys off economics,

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and that a perfectly market-hedged dealer is fully exposed to the counterparty.

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**25.13** *Distinguish the three causes of the CDS market's post-crisis shrinkage, and state which of them reduced risk rather than merely reducing gross notional.*

*The three causes.* Gross notional outstanding peaked before the crisis in the tens of trillions of dollars and fell by roughly an order of magnitude over the following decade. Three forces did it, and separating them matters because only one reduced risk.

*1. Compression removed offsetting trades.* A dealer with a bought and a sold contract on the same issuer, maturity and coupon holds two contracts and **no risk**. The Big Bang's standardization made those contracts fungible (identical coupons, identical maturities, upfront settlement of the difference, as in question 2), and multilateral compression services then tore them up in bulk. This eliminated enormous gross notional and *no net exposure at all*, which is precisely why gross notional was never a risk measure.

*2. Clearing concentrated and netted the rest.* Index CDS moved to central clearing under the mandates of Chapter 2. A CCP nets *multilaterally* rather than bilaterally (A's exposure to B and B's to C collapse against a single counterparty), which shrinks the reported figure again without changing anyone's economic position.

*3. Single-name activity genuinely declined.* Capital charges on dealer inventory rose, the instrument acquired a reputational cost after the crisis, and several large participants withdrew.

*Which of them reduced risk: only the third*—and it reduced the market's *usefulness* at the same time, which is why the answer is uncomfortable rather than reassuring.

The first two are **measurement**, not risk. Compression and clearing removed offsetting contracts and reorganized the netting; they did not change what anyone was economically exposed to. Indeed clearing arguably *improved* risk while reducing the reported number, by replacing a web of bilateral exposures with margined positions against a mutualized counterparty, a change Chapter 2 argues is a genuine improvement and Chapter 8 notes concentrates risk somewhere new.

The third is real and it supports both conclusions. *Against*: liquid single-name CDS now exists for only a few hundred reference entities, so a bank hedging a loan to a mid-sized borrower usually **cannot buy protection on it**. The risk-transfer function that justified the instrument works only for the largest issuers, and the market that remains serves index macro hedging and the CVA desks of Chapter 9 better than it serves the original purpose. Section 9.10 already flagged this as the reason most CVA is hedged by proxy, which introduces basis risk in place of the credit risk it removes. *For*: the market that remains is standardized, cleared, compressed, and settled by auction, and none of those was true in 2007. The instrument that failed in the crisis and the one that exists now differ in every respect that failed.

*The methodological point worth extracting.* A headline number falling by an order of magnitude invites a story about risk retreating from the system. Two of its three causes were bookkeeping. This is Section 1.6's argument in its most consequential form: gross notional counts contracts, and contracts are mostly intermediation. The measure that would have answered the question, net notional by reference entity, was not published before the crisis, and is now, because the trade repositories that the reforms created exist to produce exactly that number.

*What the question tests.* Separating a change in measurement from a change in exposure, and being even-handed about a decline that improved the market's safety and narrowed its purpose at once.

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**25.14** *Explain why a credit-linked note contains two credit exposures rather than one, and explain what the quanto spread between a dollar- and a euro-denominated CDS on a euro-area sovereign measures.*

*Why a credit-linked note contains two credit exposures.* A CDS is **unfunded**: no principal changes hands at inception, and the protection seller's obligation is backed only by their creditworthiness and whatever collateral the CSA of Chapter 8 requires. A credit-linked note converts the same exposure into a **funded** one. The investor buys a note, paying cash upfront; the issuer invests the proceeds and pays a coupon; if a credit event occurs on the reference entity, principal is reduced accordingly. Economically the investor has sold protection and prepaid the collateral in full.

The difference is entirely about counterparty risk, and it runs both ways.

*The protection buyer is fully secured.* They hold the investor's cash, so there is no exposure to the seller at all, which is why credit-linked notes are the format of choice when the seller is an entity no one wants unsecured exposure to: a hedge fund, a special-purpose vehicle, a counterparty in a jurisdiction whose netting is untested.

*The investor acquires exposure to the issuer.* Their principal now sits with the note's issuer and is invested in something. So the note carries:

1. the **reference entity's** credit risk, which is what the investor intended to take and what the note is named after; and

2. the **issuer's** (and the collateral's) credit risk, which is what happens to the prepaid principal.

The second is easy to overlook precisely because it is not what the instrument is called. Several structures in 2008 delivered losses through the *collateral* rather than the reference entity: investors who had formed a correct view on the named credit lost money because the cash backing the note was invested in something that failed.

*The same distinction organizes Chapter 26.* A cash securitization is funded: investors buy notes and the proceeds buy the assets. A synthetic one is unfunded: investors sell protection on a reference portfolio that stays where it is. The synthetic version can be created without anyone originating a loan, which is why the outstanding amount of synthetic exposure was not limited by the amount of debt: the same decoupling question 1 identified in single-name CDS, now at portfolio scale.

*What the quanto spread measures.* Sovereign CDS is usually denominated in a currency other than the sovereign's own, for an obvious reason: protection paying in a currency the defaulting sovereign controls is worth less than it appears.

A euro-denominated contract on a euro-area sovereign pays euros in a state of the world in which the euro may have depreciated sharply, precisely because that sovereign defaulted. The dollar contract pays the same notional in a currency that has not moved, so it is worth more. The **quanto spread** between the two is therefore the market's price for devaluation conditional on default, a Chapter 16 quanto effect, with the correlation running through a credit event rather than through a continuous co-movement.

That interpretation is unusually clean. It is a jump-at-default expectation. If the dollar contract trades meaningfully above the euro one, the market is pricing a substantial currency move *conditional* on the credit event, not a general currency view, since an unconditional expected move affects both contracts through the same discounting and cancels.

*And during the euro-area crisis it was a market-implied redenomination probability.* The quanto spread on peripheral sovereigns widened when the possibility of leaving the currency union was being discussed and narrowed when it receded. A market that had no instrument for

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pricing the breakup of a currency union priced it anyway, in the basis between two contracts that differed only in their settlement currency.

*The common thread between the two halves of this question*, and the chapter's thesis in small: in both cases a second exposure is hiding inside an instrument named after the first. The credit-linked note is named for the reference entity and carries the issuer; the sovereign CDS is named for the sovereign and carries the currency. Reading the contract is what finds them.

*What the question tests.* That funding a protection position substitutes one counterparty exposure for another rather than removing it, and that a currency basis between two otherwise identical contracts is a conditional expectation with a direct reading.

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**25.15** *State the four residual risks a protection buyer faces even when right about the credit, and explain which of them is least measured and why it is the one most likely to produce the largest loss.*

*The four residual risks* are Section 25.13's taxonomy, and each is a different way for a correct view on the credit to fail to be paid.

*1. Definitional risk.* The loss occurs and no listed credit event does. The protection references a list of events, not a loss, and if the borrower imposes its loss through a route the list does not name (a write-down of instruments junior to the reference obligation, a resolution, a voluntary exchange), the contract is silent. Question 7's Credit Suisse case is the instance: CHF 16 billion of AT1 written to zero, senior debt performing throughout, no credit event.

*2. Determination risk.* An event arguably occurs and the determinations committee rules that it did not, or takes long enough that the position must be marked, financed and margined in the meantime. Greece in 2012 and the 2022 sovereign cases are both instances; the committee had to rule on questions its drafters had not anticipated, and the answer was not knowable in advance.

*3. Settlement risk.* The event is determined, the contract pays, and the auction recovery differs from the loss on the asset actually held. Question 4's point: the auction price is the market's forecast of a defaulted claim weeks after the event, struck on the cheapest deliverable, and a hedger holding a long-dated bond is made whole on a different instrument.

*4. Basis risk.* The protection references one entity and the exposure is to another: a subsidiary, a guaranteed affiliate, or a proxy issuer entirely, as in Chapter 9's CVA hedging. The hedge and the exposure can be set side by side and the gap measured.

*Which is least measured: definitional risk.* The other three have a place in a risk system. Determination risk shows up as a marking and funding problem that can be provisioned for; settlement risk is a sensitivity to $R$ that can be stress-tested; basis risk is quantified, reported and hedged on every desk. Definitional risk is not a parameter that can be shifted. It is binary—the event is on the list or it is not—and no VaR engine generates a scenario for "the contract does not say what everyone assumed," because that is not a market variable.

*Why it produces the largest loss.* When it materializes the payout is not reduced; it is **zero**. The other three are matters of degree: a recovery a few points from the auction, a basis that widens, a determination that arrives late. Definitional risk is all or nothing, and it arrives in exactly the states the hedge was bought for: a resolution authority stepping in, a loss taken through a form other than default, a capital structure treated unconventionally. The protection buyer who identified Credit Suisse as the weakest large European bank lost the premium paid and the entire exposure the premium was supposed to cover.

*What can be done.* It cannot be modeled; it can only be *read*. Check the credit event list against the specific routes by which this borrower could impose a loss; check that the reference obligation's seniority matches the exposure; for a bank, ask how its instruments are treated under the resolution regime; and write down, as an explicit list, the circumstances in which the

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hedge does not pay, rather than assuming that it always does.

*What the question tests.* Sorting the residual risks by whether they can be parameterized, and recognizing that the one that cannot is the one that can produce a total loss.

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**25.16** *Five issuers each trade at 150 bp. Explain why the first-to-default basket costs the sum of the five spreads at $\rho = 0$ and falls as $\rho$ rises while the fifth-to-default rises, and why the five baskets' protection legs add up to the five single-name legs (Example 25.6).*

*Why the first-to-default costs the sum at zero correlation.* If defaults never coincide, protection on the first default is protection on each of the five issuers in turn, so its spread is $5 \times 150 = 750$ bp. As $\rho$ rises, defaults cluster: the first default is more likely to arrive together with others, which the contract has already stopped covering, so the basket pays on fewer separate occasions and its spread falls to 591 bp at $\rho = 0.30$ and 269 bp at $\rho = 0.90$.

*Why the fifth-to-default moves the other way.* It pays only if all five default, which at $\rho = 0$ is a joint event so remote that the spread rounds to zero. Correlation makes the joint event more likely, and the spread rises to 67 bp at $\rho = 0.90$. The first-to-default is the equity tranche of this pool and the fifth is its senior tranche, which is why the two want opposite things from the same number.

*Why the legs add up.* Since $\sum_{n=1}^{5} \mathbf{1}\{K \geq n\} = K$, every default is paid for exactly once, by the basket whose turn it is, so the five protection legs sum to the five single-name protection legs at every correlation. Correlation redistributes the premium among the baskets and cannot create or destroy it.

*What the question tests.* That a basket is a CDS on an order statistic of the default times, and that its price is a statement about correlation as much as about the spreads.

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# Chapter 26
# Securitization, Structured Credit, and the 2007–2009 Crisis

**26.1** *State three legitimate purposes of securitization, and explain why the failure of 2007–2009 does not condemn all of them.*

*Three legitimate purposes follow.*

*1. It converts illiquid assets into tradeable ones.* A single mortgage cannot be sold easily: it is a bespoke contract on one property with one borrower, and valuing it requires work no one will do for one loan. A security backed by ten thousand of them can be sold, because the pool's behavior is statistical rather than idiosyncratic and the instrument is standardized.

*2. It matches risk to appetite.* A pension fund that cannot hold individual mortgages (by mandate, by expertise, or by the operational burden of servicing them) can hold a senior claim on a pool of them. Someone with the appetite and the analytical capacity takes the first-loss piece. The pool has one risk profile; the tranches have several, and investors sort themselves across them.

*3. It transfers risk out of banks.* A bank that originates a loan and holds it ties up capital against it. Selling the risk frees that capital to support new lending, which is what the capital frameworks of Chapter 27 are designed to encourage when the transfer is genuine.

*Why 2007–2009 does not condemn all of them.* Because what failed was not securitization but a specific set of things built on top of it, and the distinction is not a rhetorical rescue—it is supported by the parts of the market that kept working.

*The plain instruments functioned throughout.* The agency mortgage-backed market is larger than everything that failed in 2008 put together; its central risk (prepayment, and the negative convexity of question 3) is genuinely harder to model than credit default; and it worked, through the crisis and since. **Risk that is understood can be borne at enormous scale.**

*And the failures were specific and nameable*, which is what makes the diagnosis useful: tranching applied to collateral exposed to *one* factor (question 6); re-securitization, which pooled the systematic residual; a correlation parameter that no one observed and everyone needed (question 7); a chain in which no one retained the risk they created (question 10); and a funding structure that forced sales (question 12). Chapter 26's own comparisons isolate each: covered bonds avoid the failures by transferring no risk at all (question 14), and CLOs share the tranching while avoiding four of the five mechanisms (question 13).

The honest form of the conclusion is the one the chapter insists on: securitization's benefits and its failure modes come from the *same* feature—the separation of origination from risk-bearing—so a reader must hold both halves at once. A system that securitizes nothing extends less credit for a given capital base (question 14); a system that securitizes without retaining anything loses the ability to see where the risk went.

*What the question tests.* That a general technique is not refuted by the failure of a particular application, and the discipline of naming which mechanisms failed.

**26.2** *Explain why a tranche is a call spread on the pool's loss, identify the strikes, and use Example 26.1 to state the factor by which the pool's and the senior tranche's expected losses each rise when the default probability triples.*

*Why a tranche is a call spread.* Write $L$ for the pool's loss as a fraction of notional. A tranche attaching at $a$ and detaching at $b$ absorbs losses only between those points. Measured as a fraction of the pool's notional, which is $(b-a)$ times the chapter's normalized $L_{[a,b]}$,

$$(b-a)\,L_{[a,b]} = \min\big(\max(L-a,\,0),\; b-a\big) = \max(L-a,0) - \max(L-b,0)$$

which is exactly a **call spread on** $L$, long the call struck at $a$ and short the call struck at $b$. The strikes are the attachment and detachment points: $a$ and $b$.

The tranche *investor* is short that call spread: they receive a spread and pay out the losses, so they have written a capped option on the pool's loss. An equity tranche is the low-strike piece, a senior tranche the high-strike one, and everything the chapter says about tranches follows from options theory rather than from credit theory.

*The consequence: convexity.* An option's value responds *nonlinearly* to its underlying, and a deeply out-of-the-money option responds most violently in relative terms. A senior tranche is a far out-of-the-money call spread on pool loss, so a parameter error that moves the pool's loss modestly moves the tranche's value enormously.

*The factors, from Example 26.1.* Tripling the default probability from 5% to 15%:

| True default probability | Pool expected loss | Senior 12–30% expected loss |
|---:|---:|---:|
| 5% | 3.00% | 0.266% |
| 15% | 9.00% | 7.374% |

$$\text{pool: } \frac{9.00}{3.00} = \mathbf{3}\times \qquad \text{senior tranche: } \frac{7.374}{0.266} = \mathbf{28}\times$$

Read what this means for an analyst. Someone who was wrong about subprime default rates by a factor of three was wrong by a factor that would have been embarrassing but *survivable* for a whole-pool investor: a 3% loss becoming 9%. The same error made a senior tranche holder wrong by a factor of nearly thirty, taking an instrument rated on a 0.27% expected loss to 7.4%.

*And this requires no story about fraud or stupidity.* That is the chapter's central point. A tranche is an option, options are convex, and convexity amplifies parameter error. The mechanism is arithmetic, and it operates on honest analysts using correct formulas with slightly wrong inputs.

*The same convexity operates on the second parameter.* Correlation enters a senior tranche's value with equally high leverage (Chapter 25's question 8 gives a factor of 43), and it is not observable either. Two unobservable inputs, each amplified by a factor of tens, is not a model that should be used to rate an instrument AAA, which is question 8's subject.

*What the question tests.* Recognizing the tranche payoff as a call spread, and that the resulting convexity is what converts a survivable pool error into a catastrophic tranche error.

**26.3** *Explain why an agency mortgage pass-through has negative convexity, identify the option and who holds it, and use Example 26.2 to give the effective convexity of the pass-through and of the same bond at a fixed prepayment speed.*

*The option and who holds it.* A US residential mortgage borrower may **prepay at any time without penalty**: by refinancing, by moving, or simply by paying down. That is a call option on the loan, struck at par, and *the borrower holds it*. The pass-through investor has therefore

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bought a bond and **written a call on it**, and the mortgage's yield includes the premium for that option.

*Why the convexity is negative.* A written call has negative gamma, and here the mechanism is visible in the cash flows. When rates *fall*, refinancing becomes attractive, prepayments accelerate, and the investor's principal is returned early, to be reinvested at the new, lower rates. The bond's price is therefore capped near par: the upside is given away exactly when it would have been worth having. When rates *rise*, prepayments slow, the investor is left holding a below-market coupon for longer, and the duration *extends*. Price appreciation is truncated and price depreciation is not, which is the definition of negative convexity.

*The measured figures, from Example 26.2:*

| | Effective duration | Effective convexity |
|---|---|---|
| Pass-through, prepayments respond | 3.89 | **−304** |
| Same bond, fixed prepayment speed | 5.07 | **+48.5** |
| Seven-year bullet | 5.78 | +39.9 |

*The comparison is the whole answer.* The *same bond*, with prepayments frozen at a constant speed, has convexity of +48.5: ordinary, positive, like any amortizing bond. Letting prepayments respond to rates takes it to −304. A swing of **some 350 points of convexity**, produced entirely by the option, on identical cash flow rules. The duration shortens at the same time, from 5.07 to 3.89, because expected principal now returns sooner in exactly the states where rates have fallen.

The price table shows it directly: a two-percent rally lifts the seven-year bullet 12.4% and the pass-through only 2.7%, while a two-percent sell-off costs the bullet 10.8% and the pass-through 11.9%. The investor receives roughly a fifth of the upside and all of the downside.

*Why this belongs in this chapter.* It is the counterexample that disciplines the rest of it. The agency mortgage market is larger than everything that failed in 2008 combined, its central risk is *harder* to model than credit default (prepayment depends on borrower behavior, on housing turnover, on refinancing frictions, none of which follows from a no-arbitrage argument), and it worked. Risk that is genuinely understood can be borne at enormous scale, and the chapter's indictment is therefore of specific structures rather than of securitization.

*What the question tests.* Locating the option and its holder, and that convexity is a property of the *response* of cash flows to rates rather than of the cash flows themselves.

---

**26.4** *Explain why a mortgage investor hedging to a duration target buys duration as rates fall, and relate this to Section 18.12.*

*The mechanism.* An investor targeting a fixed portfolio duration must trade whenever the portfolio's own duration drifts. Question 3 established that a mortgage's duration is not constant: it **shortens** as rates fall, because prepayments accelerate and principal comes back sooner, and it **extends** as rates rise. In Example 26.2's bond the effective duration is 3.89 with prepayments responding, against 5.07 with them frozen.

So when rates fall:

1. prepayment expectations rise;
2. the portfolio's duration *falls* below target;
3. restoring the target requires **buying** duration (receiving fixed in swaps, or buying Treasuries).

And when rates rise the portfolio *extends* past target, so the hedger must **sell** duration.

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*In price terms the hedger buys into a rally and sells into a sell-off.* A mortgage hedger buys duration as rates fall and sells it as rates rise, so it is always trading with the move. Expressed in *rate* terms it is the same trade: receiving fixed when rates fall and paying fixed when rates rise, in both cases pushing rates further in the direction they have already moved.

*The relation to Section 18.12.* This is precisely the short-gamma rebalancing rule, and it is short gamma for a genuine reason rather than by analogy: the investor is short an option (question 3), so they have negative gamma, so their delta hedge must be adjusted *with* the move rather than against it. Section 18.12's dealer buys the underlying as it rises because they wrote a call; the mortgage investor does the same thing in the rates market because they wrote the homeowner one.

*Why it matters more here than in most places.* **Scale.** The investor base is measured in trillions of dollars, and it is highly homogeneous: banks, agencies, money managers and insurers all hedging the same convexity in the same direction at the same time, because they all own versions of the same instrument. That makes it one of the largest identifiable feedback flows in any fixed-income market, and "convexity hedging" is a standard explanation offered for sharp, self-reinforcing moves in US swap and Treasury yields, particularly rallies, where the flow is concentrated because refinancing incentives are nonlinear in the rate.

*The general pattern, which recurs through the book.* A mechanical rule that forces trading in the direction of the move produces convexity in the market regardless of whether anyone holds an option: portfolio insurance (Chapter 18), inverse volatility products (Chapter 22), 0DTE dealer hedging (Chapter 23), and structured-note barrier hedging (Chapter 35). Here the option is real and the homeowner holds it, but the market effect would be identical if the rule came from a prospectus.

*What the question tests.* That a duration target plus a duration that moves with rates is a short-gamma rebalancing rule, and recognizing the same mechanism across two unrelated markets.

---

**26.5** *Name the three sources of credit support in a securitization, explain which is consumed first, and explain why the accumulating-support calculation of Example 26.3 overstates protection in exactly the scenario where protection is needed.*

*The three sources*, from Example 26.3's \$1,000 million pool with an 8.5% weighted-average coupon, bonds costing 5.5%, and 0.5% of servicing and fees:

*Excess spread*: what the pool earns above what the deal pays out: $8.5 - 5.5 - 0.5 = \mathbf{2.5\%}$ a year, or \$25 million.

*Overcollateralization*: the excess of assets over liabilities. Issuing \$980 million of bonds against \$1,000 million of loans starts the deal with **2%** of cushion.

*Subordination*: the tranching itself. For a senior bond attaching at **20%**, everything below it must be exhausted first.

*Which is consumed first: excess spread.* It is the deal's running income, and losses are charged against it before any tranche is touched. Only when a period's losses exceed that period's excess spread does the deal begin eating overcollateralization, and only when that is gone does subordination start to absorb.

With no defaults, the support at the end of the first year is therefore

$$20\% + 2.5\% + 2\% = \mathbf{24.5}\% \text{ of the pool}$$

and if excess spread is *retained* rather than released to the equity, it accumulates:

| | **Year 1** | **Year 2** | **Year 3** | **Year 4** | **Year 5** |
|---|---|---|---|---|---|
| Cumulative support | 24.5% | 27.0% | 29.5% | 32.0% | 34.5% |

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Which is why the structures looked so robust: a senior bond apparently protected by more than a third of the pool by year five.

*Why the calculation overstates protection in exactly the scenario that matters.* Because **the support grows only if the excess spread is actually earned, and it is earned only while the loans keep paying**.

A pool that starts defaulting loses its 2.5% a year at the same moment it begins consuming the 22% it already has. In the example's stressed scenario (8% of the original balance defaulting a year at a 40% recovery), the support is not 34.5% and rising; it is 19.6% after the first year and 8.0% after the fifth, and *falling*. The models that projected the accumulating column ran it under scenarios in which the accumulation happened—which is to say, under scenarios in which the protection was not needed.

*The failure has a name, and this book keeps meeting it.* It is a **correlation between the buffer and the loss**: a protection that shrinks in the state where it is required. Structurally it is identical to

- the wrong-way risk of Section 9.9, where exposure is largest when the counterparty is least able to pay;
- the recovery–default correlation of Section 24.4, where recoveries are lowest in the years defaults are highest;
- and the domestic bank selling sovereign protection in Chapter 24's question 12.

In each case a quantity relied upon as independent is in fact driven by the same event it is meant to protect against, and in each case the naive calculation multiplies two expectations where the covariance term is the whole story.

*The practical test* follows from the diagnosis: never quote a protection level computed under a base-case path. The number that matters is support conditional on the stress, and here that means projecting excess spread and losses *jointly* rather than layering an independently-forecast buffer on top of an independently-forecast loss.

*What the question tests.* The order of consumption, and recognizing the buffer–loss correlation as an instance of a failure mode the book has already named.

---

**26.6** *Explain why mezzanine tranches were pooled and re-tranched rather than sold directly, why doing so did not diversify, and use Example 26.4 to state the range of the "AAA" tranche's expected loss across inter-deal correlations.*

*Why mezzanine tranches were re-pooled rather than sold.* Because they were hard to sell. The senior pieces were absorbed by investors buying a rating, and the equity was taken by specialists who understood it and were paid for it. The **mezzanine** sat between: too risky for the rating-driven buyer, too thin and too complex for the equity buyer, and carrying a spread that looked unattractive relative to its perceived risk. An arranger with unsold mezzanine could either cut the price or find a way to make it look like something else.

Re-securitizing did the second. Pool a hundred mezzanine tranches, tranche the pool, and, by the same subordination arithmetic that produced a AAA from a pool of subprime loans, produce a new AAA from a pool of BBB. The output was rated on a diversification argument, and the diversification argument is what fails.

*Why it did not diversify.* Pooling removes idiosyncratic risk *only if the risk is idiosyncratic*, and the claims being pooled **had already had their idiosyncratic risk stripped out by the first securitization**, which is precisely what a mezzanine tranche is. The first tranching sold

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the idiosyncratic variation to the equity holder and left the mezzanine holding the systematic residual.

So re-securitizing pools the systematic residual, and the systematic residual does not diversify. The mezzanine tranches of a hundred subprime deals from the same vintage are exposed to *one thing*: national house prices. They are not a hundred independent bets on a hundred pools; they are one bet, made a hundred times, on whether house prices keep rising.

*The range, from Example 26.4.* Taking the 3–7% mezzanine piece (expected loss 20.19%) from a hundred different deals and tranching the result, everything depends on the correlation *between* the deals:

| Inter-deal $\rho$ | "AAA" 30–100% | "AA" 12–30% | 3–12% | Equity 0–3% |
|---|---|---|---|---|
| 0.10 | 1.08% | 44.13% | 94.36% | 99.96% |
| 0.30 | 3.83% | 40.89% | 80.19% | 97.71% |
| 0.60 | 7.55% | 36.64% | 63.34% | 86.84% |
| 0.90 | 11.25% | 32.18% | 49.18% | 69.72% |

The "AAA" tranche's expected loss runs from **1.08% to 11.25%**—a factor of **ten**, from a rounding error to a substantial loss, driven entirely by a parameter about which no market quoted anything.

*And the parameter was not 0.10.* Given what the collateral was, the honest inter-deal correlation was near the top of that range, not the bottom. The structures were rated near the bottom.

*The compounding is what makes re-securitization worse than tranching, not merely more of it.* Question 2's convexity is applied twice: an error in the pool's default rate is amplified once into the mezzanine tranche's loss, and that amplified error is then amplified again into the re-securitized senior tranche. A parameter error survivable at the pool level becomes catastrophic at one remove and unrecoverable at two.

*What the question tests.* That pooling requires idiosyncratic variation to work on, that a mezzanine tranche is by construction the piece from which it has been removed, and the size of the resulting dependence on an unobserved parameter.

---

**26.7** *Explain what tail dependence means, why the Gaussian copula's lack of it matters specifically for senior tranches, and why calibrating the correlation to pre-2006 data could not have captured the event that mattered.*

*What tail dependence means.* It is the tendency of extreme events to occur *together*. Formally, the limiting conditional probability that one variable is extreme given that another is:

$$\lambda = \lim_{u \to 1} \Pr\left(U_1 > u \,\middle|\, U_2 > u\right)$$

A distribution with $\lambda > 0$ says that conditional on one issuer failing catastrophically, another is *more* likely to fail than it would otherwise be, however far into the tail one looks.

*The Gaussian copula has $\lambda = 0$.* Under it, extreme events are **asymptotically independent**: conditional on one issuer defaulting catastrophically, the probability that another does converges to its *unconditional* value. This holds for *any* correlation short of one: a Gaussian copula at $\rho = 0.9$ still has zero tail dependence. Correlation controls behavior in the body of the distribution and dies in the tail.

*Why this matters specifically for senior tranches.* Because a senior tranche **only exists in the tail**. It is a far out-of-the-money call spread (question 2): it pays in full in every ordinary scenario and its entire risk is concentrated in the states where many issuers fail at once. So it is priced by exactly the region of the loss distribution where the copula's dependence has been switched off.

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The bias therefore runs one way and lands in one place. Equity and mezzanine tranches are priced by the body of the distribution, where the correlation parameter has its largest effect and the model is roughly calibrated. Senior tranches are priced by a tail the model has made too thin, so senior tranches look safe for a structural reason that no choice of $\rho$ corrects. Raising the correlation fattens the shoulder of the distribution without restoring tail dependence, which is one reason the implied correlation had to differ across attachment points (Chapter 25's question 9). The smile in base correlation is the missing tail dependence, relabeled.

Empirically the assumption is simply false for credit. Defaults cluster: they arrive in waves driven by a common economic state, and in a severe recession the conditional probability of a second default given a first is far above its unconditional value. The model treats as independent precisely the events that are most dependent.

*Why calibrating to pre-2006 data could not have captured the event that mattered.* Because the data contained no instance of it. Whatever correlation was estimated came from a sample in which **US house prices had not fallen nationally**. Regional declines had occurred and had diversified away across a national pool, which is exactly what the estimate recorded, and exactly why the estimate was low.

So the calibration was not wrong given its sample. The sample was wrong for the question. A national house price decline was not a rare draw from the estimated distribution; it was an event whose probability the estimator had no information about at all, and no amount of statistical care extracts it from data that does not contain it.

Note that this is not a criticism of the copula. It is the sampling problem of Section 19.9's pre-1987 discussion (where option markets showed no skew because no crash had happened), and no choice of copula would have repaired it. A t-copula with fat tails would still have had its degrees of freedom estimated from the same benign sample. The chapter's verdict separates the two failures precisely: the missing tail dependence is a criticism of the model, the calibration sample is not, and the third failure, treating the output as a measurement, is institutional rather than mathematical and carries the most weight.

*What the question tests.* That tail dependence is a limiting property distinct from correlation, that senior tranches live entirely where it operates, and the difference between a model that is wrong and a sample that is uninformative.

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**26.8** *Explain why a AAA tranche's rating is closer to binary than a AAA corporate bond's, and why rating triggers made downgrades destructive rather than merely informative.*

*Why a tranche's rating is closer to binary.* The two instruments were notionally calibrated to the same idealized default probability, but their loss distributions conditional on things going wrong are completely different.

*A corporate bond's rating can be wrong by one notch.* If the issuer deteriorates it becomes AA, then A, then BBB. Chapter 24's transition matrix describes exactly this: default is reached *through downgrade*, over years, and at every stage a holder can sell into a market that has repriced gradually. The rating is a point on a continuum and the instrument travels along it.

*A tranche has no such continuum.* Question 2's convexity is the reason. A senior tranche is fine—completely fine, untouched—until the pool's losses approach its attachment point, and then it is not. Example 26.1 shows the expected loss going from 0.27% to 15.2% on a default-rate move that raised the pool's own expected loss only fourfold, from 3% to 12%. There is no gradual path because the payoff itself has no gradual region: a call spread is flat, then steep, then flat.

So the rating is a statement about which side of the attachment point the pool will end up on, which is much closer to a binary proposition than to a position on a scale. And when

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downgrades came they came in **blocks of many notches at once** (AAA to junk in a single action), because the underlying quantity had crossed a threshold rather than drifted.

*Why rating triggers made downgrades destructive rather than merely informative.* A downgrade of a corporate bond conveys information; a downgrade of a structured tranche *caused transactions*, because the rating was embedded in contracts:

*Mandate-constrained holders were forced to sell.* Insurance companies, money market funds and many institutional mandates may hold only investment-grade paper. A multi-notch downgrade converts a permitted holding into a prohibited one overnight, and the holder must sell into a market where every similarly-constrained holder is selling the same instrument for the same reason on the same day.

*Collateral calls were triggered.* Downgrade provisions in derivative and financing documentation required additional collateral or termination, so a rating action became an immediate demand for cash.

This is Section 24.10's mechanism operating on an instrument whose downgrades arrive in blocks. A gradual mechanism applied to a discontinuous instrument produces a discontinuous outcome: forced selling concentrated in time, into no bid, which is question 12's point that the losses fell on those who *had* to sell rather than on those whose instruments defaulted.

*And the deeper problem behind both halves.* An investor holding a AAA tranche was, in practice, holding **an agency's correlation assumption**. The rating was not a summary of an analysis the investor could have performed themselves (as it broadly is for a corporate bond, whose financials are public) but was *the entire analysis*, resting on an unobservable parameter (question 7), and it was not disclosed as such. The letter was identical; what stood behind it was not.

*What the question tests.* That the convexity of question 2 implies a discontinuous rating, and that contractual triggers convert information into forced transactions.

---

**26.9** *Explain why the May 2005 episode caused losses on both legs of a trade that was supposed to be hedged, and why inferring a correlation from tranche prices and then using it to hedge tranches is circular.*

*The trade and the episode.* The standard position was long the equity tranche of an index and short the mezzanine, in a ratio making the package neutral to the index spread. Chapter 25's question 8 explains why: the equity tranche is strongly long correlation and the mezzanine much less so, so the package was a way to be **long correlation while hedged against the credit market's direction**. It carried positively, it had performed for years, and it was widely held.

In May 2005 two large US automakers were downgraded to junk. The shock was **idiosyncratic**, and the model had no room for idiosyncratic shocks: in the one-factor copula, an issuer defaults because the common factor moves or because its own draw is bad, and the second channel is a residual that diversifies away across 125 issuers. A downgrade large enough to move two issuers' spreads violently while leaving the other 123 alone is, in the model's terms, an event of negligible probability.

*Why both legs lost.*

*The equity tranche fell*, because idiosyncratic widening in a few issuers is exactly what an equity tranche is exposed to. It absorbs the first losses, and two issuers blowing out is a first loss.

*The mezzanine did not widen to compensate*, because the hedge ratio had been computed from a model in which spreads move *together*. In that model, whatever hurts the equity tranche must be a move in the common factor, which would also widen the mezzanine, so the short mezzanine leg was sized to gain from a co-movement that did not occur. As implied correlation fell, the mezzanine spread tightened instead, and the short leg lost too.

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So the trade lost money on **both legs simultaneously**, which the model said could not happen. Implied correlation then moved in ways no single parameter could reconcile, and the base-correlation apparatus of Chapter 25 was in part a response to the resulting inconsistencies.

*Why inferring a correlation from tranche prices and using it to hedge tranches is circular.* Because the parameter has no existence independent of the prices it is extracted from.

The procedure is: observe tranche prices → invert the model to obtain $\rho$ → use $\rho$ to compute sensitivities → hedge one tranche with another. But $\rho$ was *defined* as whatever number reproduces the observed prices under the assumed model. It is not a measurement of anything in the world; it is a restatement of the price in different units, exactly as an implied volatility is a restatement of an option price.

The circularity bites at the hedging step. A hedge ratio derived this way asserts that the two tranches will move as *the model* says they move when $\rho$ changes, and if the model's dependence structure is wrong, the ratio is wrong in a way no amount of recalibration reveals, because recalibrating simply reproduces the new prices too. The model can never be falsified by the prices it is fitted to. What falsifies it is a scenario the model excludes, which is precisely what May 2005 delivered.

*Three lessons were available in 2005 and were drawn in 2008 instead.*

- *A hedge ratio is model output, not market data.* A position hedged by a model is hedged against the model's world. Section 21.11's barrier measurement is the same observation with a different instrument.
- *A parameter that must be inferred from the prices it explains is not a parameter*; it is a quoting convention, and Chapter 25's question 9 says so in the market's own vocabulary.
- *And crowding turned a model error into a market event.* Everyone ran the same model and therefore held the same position, so the unwind moved the prices being unwound (the mechanism of Chapters 22 and 23).

*What the question tests.* That a one-factor model excludes idiosyncratic shocks by construction, and that a parameter inferred from prices cannot validate a hedge between those same prices.

---

**26.10** *Describe the originate-to-distribute chain, identify at which point the risk was supposed to stop, and explain how the risk that banks appeared to have distributed returned to their balance sheets.*

*The chain.* A mortgage was

1. *originated* by a broker paid a fee per loan;
2. *sold* to a bank that pooled it;
3. *tranched* by an arranger paid a fee per deal;
4. *rated* by an agency paid by the arranger;
5. *sold* to an investor who relied on the rating.

At every step the compensation was for **volume** and the risk moved on.

*Where the risk was supposed to stop: with the end investor.* That is the design, and it is not objectionable in itself. Originate-to-distribute is the same separation of origination from risk-bearing that makes any capital market work (a corporate bond underwriter does not hold

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the bond), and Section 24.16's account of banks distributing credit describes the identical structure functioning correctly. The problem was not the shape of the chain.

*What made it fail is that the distribution was not real.* The risk did a full circle and came back to the banks, by three routes that the capital framework of the time did not see:

*Banks retained the super-senior tranches.* These were the thickest pieces, they were unsellable at any spread investors would accept, and they appeared riskless, so banks kept them. The largest exposures in the system were the pieces no one had wanted to buy.

*They were held in off-balance-sheet vehicles.* Conduits and structured investment vehicles held the tranches and were not consolidated, so the exposure did not appear in the bank's own figures and attracted little or no capital.

And those vehicles were funded with short-term paper, backed by liquidity lines from the sponsoring bank. This is the binding one. The vehicle funded long-dated tranches with commercial paper that had to be rolled every few months. When it could not be rolled (late 2007), the liquidity backstop was drawn, and the assets came back onto the sponsoring bank's balance sheet at exactly the moment they were worth least.

So the bank had sold the credit risk and retained the liquidity risk, and in a crisis those are the same risk. The exposure was *contingent*, so it did not appear as an asset; it was in an *unconsolidated* vehicle, so it did not appear on the balance sheet; and it was a *commitment*, so it attracted almost no capital. Three accounting facts, each individually defensible, combined to make a large exposure invisible.

That is the specific mechanism Chapter 27 exists to address, and Section 26.13 records the responses: consolidation rules that bring such vehicles onto the balance sheet, capital against contingent liquidity commitments, risk retention requirements obliging the originator to keep a slice, and the leverage ratio of Section 27.10 as a backstop against exactly this kind of risk-weight arbitrage.

*The general lesson, and it is the chapter's closing one.* A system loses the ability to *see* a risk before it loses the ability to bear it. Every link in this chain was individually rational and individually disclosed; what no one could observe was that the chain was a loop.

*What the question tests.* That the incentive structure is not the flaw—incomplete transfer is—and the three specific accounting routes by which the transfer failed.

---

**26.11** *Explain why a synthetic CDO multiplies notional without multiplying the underlying default risk, and what this does to interconnection.*

*Why the notional multiplies but the default risk does not.* A cash CDO must *buy* the assets, so the amount of exposure that can exist is bounded by the amount of debt that was originated. A **synthetic** CDO sells protection on a reference portfolio through the CDS of Chapter 25, and Chapter 25's question 1 established that a protection buyer need not own the underlying obligation. The synthetic format therefore removed the constraint that there had to be a loan.

Demand for AAA-rated yield exceeded the supply of mortgages, and a synthetic structure could reference the same pool as many times as there were counterparties willing to take the other side.

But a hundred synthetic deals referencing the same mortgages **do not make those mortgages more likely to default**. The underlying event is unchanged; what has changed is the number of contracts whose value depends on it. So the two quantities separate cleanly:

- *The economic loss* is bounded by the mortgages: a fixed pool of borrowers, a fixed amount that can be lost.
- *The notional of claims* on that loss is unbounded, limited only by willingness to trade.

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*What this does to interconnection.* It converts a bounded shock into an unbounded *transmission* problem. The loss on the mortgages was bounded; the transfer of that loss through the system was not.

That distinction explains the shape of the crisis, and it is why question 12's sequence matters. The mortgage losses alone were survivable: large, but not in aggregate larger than the banking system's capital. What was not survivable was that those losses arrived as collateral calls on marks, propagating through a web of bilateral contracts among institutions that could not establish who was solvent. A bounded shock propagated through an unbounded web of contracts is a different problem from a large shock, and the crisis's transmission was through counterparty exposure rather than through mortgage losses.

The mechanism is Chapter 25's question 1 at portfolio scale: gross notional counts contracts, not exposures, and the same decoupling that made gross notional a meaningless risk measure in single-name CDS made synthetic CDO issuance unlimited.

*A fair counter-argument.* Synthetic structures also allowed the risk to be held by *whoever wanted it*, including investors who correctly thought subprime was overvalued and used the format to take the short side. That is a genuine improvement in market completeness: a view that could not otherwise be expressed became tradeable, and the people who were right made money. The instrument was symmetric; what was asymmetric was who understood it.

*And the consequence for what to regulate.* If the problem is transmission rather than the shock, then the remedies are the ones Chapters 2, 8, 9 and 27 describe (clearing, collateral, margin, capital against counterparty exposure) rather than restrictions on the reference asset. The post-crisis reform agenda is legible as a response to this diagnosis.

*What the question tests.* Separating the size of a loss from the number of claims on it, and that unbounded notional is an interconnection problem rather than a credit one.

---

**26.12** *State the sequence in which the crisis unfolded, identify the step at which mortgage losses stopped being the binding problem, and explain why senior tranches largely paid in the end and yet produced enormous losses.*

*The sequence.*

*2006–2007: the pool assumptions were wrong.* Subprime delinquencies rose above anything in the calibration sample (loosening underwriting, adjustable-rate resets, house prices flattening), for reasons visible in the data before they were visible in the ratings.

*Mid-2007: the tranches repriced nonlinearly.* Question 2's convexity operated. Pool losses that were bad became tranche losses that were catastrophic, and the mezzanine and re-securitized tranches went first.

*Late 2007: the funding withdrew.* The off-balance-sheet vehicles of question 10 funded long-dated tranches with short-dated paper. When the paper could not be rolled, the sponsoring banks took the assets back onto their balance sheets. This was Chapter 8's funding-fragility problem, arriving through a route the capital framework had not anticipated.

*2008: the counterparty channel opened.* Institutions that had sold protection on senior tranches faced collateral calls as the marks fell. The largest single case was an insurer that had written protection without holding capital against the *mark-to-market* exposure, on the reasoning that the tranches would never actually default—which was, in the narrow sense, true for many of them. They did not need to default; the collateral calls on the marks were sufficient.

*And then valuation itself failed.* With no trades, marks came from models, and Chapter 21's model risk arrived at the worst possible moment: institutions marking the same instrument differently, disputing collateral calls, and unable to establish who was solvent.

*Where mortgage losses stopped being the binding problem: at the third step, late 2007.* From

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the funding withdrawal onward the binding constraint was **liquidity and then counterparty solvency**, not credit losses. Everything after that point would have happened in substantially the same way had the ultimate credit losses been smaller, because what propagated was the *mark*, not the default. Total realized US subprime losses were large but not, in aggregate, larger than the banking system's capital. The crisis was in the transmission, not the shock.

*Why senior tranches largely paid and yet produced enormous losses.* These are two different questions about two different quantities, and the gap between them is the whole lesson.

*They paid because the instruments performed as written.* A senior tranche attaching at 20% suffers only if pool losses exceed 20%, and on most deals they did not. Holders who could **fund them to maturity** mostly recovered.

*The losses fell on those who had to sell.* Three mechanisms forced sales, and none required a default:

- *Funding.* The vehicles of question 10 could not roll their paper, so the assets had to be liquidated or absorbed.
- *Collateral.* Protection sellers faced calls on marks, and meeting them required selling something.
- *Ratings.* Question 8's triggers forced mandate-constrained holders out on downgrade.

Each is a demand for cash *today* against an instrument that pays in ten years. Section 5.15's limits-of-arbitrage argument names the general case: being right about an instrument and being able to hold it are different things, and the second is decided by one's funding rather than by one's analysis. Convertible arbitrage in 2008 (Chapter 25's question 11) failed the same way, on the same timetable, for the same reason.

*What the question tests.* That the binding constraint migrated from credit to funding to counterparty solvency, and the distinction between an instrument's performance and its holder's.

---

**26.13** *State four respects in which a CLO differs from a 2007 subprime CDO, say which of them is probably decisive, and use Example 26.6 to explain what is nonetheless worth watching in its collateral.*

*Four respects in which a CLO differs.*

*1. The collateral is not exposed to one factor.* A CLO holds loans to 150–300 companies across many industries, so there is genuine idiosyncratic variation for pooling to work on. Subprime mortgages from one vintage had none: they were a single bet on national house prices (question 6). The diversification a CLO claims actually exists, and this is the single largest difference.

*2. There is no re-securitization.* CLOs hold *loans*, not tranches of other deals, so question 6's compounding of convexity does not occur. The structure is applied once.

*3. The liabilities are term-funded and match the assets.* A CLO issues notes with maturities matched to the loan portfolio and **cannot be forced to sell when spreads widen**. Question 12's funding withdrawal has no analog.

*4. There is a manager with capital at risk.* A CLO is actively managed within indenture constraints and the manager typically holds part of the equity. Question 10's alignment problem is addressed by construction rather than by regulation.

To which should be added the structural feature the 2007 deals lacked: **coverage tests**. The senior overcollateralization test compares collateral par with the notes it supports, and on a breach diverts cash from the equity to amortize the senior notes until the test passes. The deal deleverages itself, automatically, in the direction of the senior holder, without anyone deciding to.

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*Which is probably decisive: the term funding.* The chapter says so and the evidence supports it. Question 12 established that the instruments which failed in 2008 were largely fine on a hold-to-maturity basis: senior tranches mostly paid. What killed their holders was having to sell: funding withdrawal, collateral calls, and rating triggers, each converting a long-dated asset into an immediate demand for cash.

A CLO's liabilities cannot be withdrawn. The AAA noteholder is locked in for the deal's life, so there is no run to be had, and the manager cannot be forced into a liquidation at distressed marks. That removes the mechanism that did the actual damage, as distinct from the mechanisms that produced the losses in the accounts.

*The empirical answer supports the structural one.* CLOs performed well through 2008 and through 2020 (two genuine tests, in different shocks) while sharing the tranching, the convexity of question 2, and the rating dependence. Four of the five mechanisms that produced the crisis are absent, and the market has been tested twice. That is about as good an answer as this subject admits.

*What is nonetheless worth watching*, since a fair assessment includes it. Loan documentation has weakened over the cycle: covenant-lite is now standard, so the early warning a covenant breach used to provide arrives later. Recoveries on leveraged loans have been trending below historical averages, which Section 24.4's identification problem makes easy to underweight. Example 26.6 puts numbers on the second: the enterprise outcome that pays a first lien 100% in a traditional structure pays 70% in a loan-only one, 63% when covenant-lite documentation delays the event, and 50.4% after a liability-management exercise. And the investor base for the AAA tranches is concentrated, which is a liquidity exposure rather than a credit one.

*What the question tests.* Diagnosing which of several differences addresses the mechanism that actually caused the losses, rather than the one that produced them in the accounts.

---

**26.14** *Explain what dual recourse means in a covered bond, why it removes the incentive problem, why covered bonds free up no bank capital, and what that implies about lending capacity.*

*What dual recourse means.* A covered bond is issued by a bank and secured on a pool of mortgages that **stays on the bank's balance sheet**. The holder has two claims:

1. an ordinary claim on *the bank*, as any senior creditor has; and
2. if the bank fails, a *preferential* claim on the ring-fenced pool.

The pool is **dynamic**: loans that deteriorate must be replaced by the issuer out of its own assets. So the holder is protected twice, and the bank's obligation does not end when a loan goes bad.

*Why this removes the incentive problem.* Because no risk is transferred at all—which is exactly the point. Question 10's chain has no one to pass the risk to. An issuer that writes bad loans must replace them; the credit risk stays where it was created, and stays there permanently.

Two further failures of this chapter are absent by construction. There is no tranching, so question 2's convexity does not arise: the holder's claim is on the whole pool and on the bank, not on a slice. And there is no re-securitization, because there are no tranches to re-pool. The result is a very long history (covered bonds predate almost everything else in this book) with a default record close to nil, through the crisis included.

*Why they free up no bank capital.* For the same reason they remove the incentive problem: the risk never leaves the bank. Regulatory capital is held against risk the bank retains, and a covered bond issuer retains all of it. A covered bond is a **funding** instrument (it raises cash against an asset pool at a good rate, because the dual claim makes it very safe), not a **risk-transfer** instrument. Securitization, done genuinely, is the reverse: it moves risk off the balance sheet and releases the capital held against it.

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*What that implies about lending capacity.* A banking system financed entirely this way would **extend less credit for a given capital base**. Each loan consumes capital for its whole life, and the bank can lend only until its capital is used up. A system that genuinely distributes risk recycles capital and supports more lending on the same equity.

So the comparison is not that Europe was wiser. It is that two systems made different choices about whether risk transfer was the objective, and the system that did not attempt it did not suffer the failures that attempting it produced. That is a real trade-off with a real cost on both sides, and it is the honest form of the comparison rather than the triumphal one.

*The sentence the chapter wants a reader to hold.* Securitization's benefits and its failure modes come from the same feature. Risk transfer is what frees capital and expands lending; risk transfer is also what breaks the incentive chain, creates the tranches whose convexity amplifies error, and makes the risk's location hard to observe. One cannot have the first without exposure to the second, and a reader deciding what to think about the instrument must hold both halves at once.

*What the question tests.* That capital relief and incentive alignment are two consequences of one property, and that the covered bond's safety is purchased with lending capacity.

---

**26.15** *Explain why the incentive problem in significant risk transfer is inverted relative to 2007, and state one respect in which private credit is safer than the pre-crisis structure and one that warrants monitoring.*

*Why the incentive problem is inverted.* In a significant risk transfer trade a bank buys protection on the **first-loss piece** of a portfolio of its own loans and **retains the senior exposure**, to reduce regulatory capital.

That is the mirror image of the 2007 chain. Then the originator was paid per loan and kept none of what it made, and the banks that pooled the loans kept only the *super-senior*, the piece that is safe unless everything fails, and so told its holder nothing about underwriting. Now the bank keeps the loans themselves and every loss beyond the first-loss piece it sold, so the originator keeps most of the exposure to its own underwriting: if it lends badly enough, the losses run through the sold slice into the senior tranche it retained, and the bank cannot escape its own portfolio.

Two further improvements come with the structure. The protection is usually **funded** (the buyer posts cash), so Chapter 25's counterparty problem is largely absent. And the portfolios are **the bank's own loans**, which it underwrote and can monitor, rather than loans acquired from brokers paid per origination.

*What is the same.* The capital relief depends on a *supervisory judgment* that the transfer is genuine, and the history of this chapter is a history of transfers that appeared genuine and were not. And the buyers are concentrated: a small number of specialist funds, often themselves levered, which is a different exposure from the one being measured.

*Private credit: one respect that is safer.* **The holders cannot be run on.** Private credit funds are closed-end with locked-up capital, so there is no maturity transformation of the kind that broke the vehicles in question 10 and no mechanism forcing a sale into a falling market. Question 12 identified forced selling as what actually caused the losses in 2008 (the instruments themselves largely paid), so removing the forced seller removes the mechanism that did the damage. Two related points reinforce it: the loans are held *whole* rather than tranched, so question 2's convexity does not apply at all, and the fund's liabilities match its assets by construction.

*And one that warrants monitoring: the marks.* Valuations are infrequent and produced by *the holder*. So the marks that would signal deterioration may simply not exist—which is not

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the same as the risk being absent, and question 12 records that the last time marks became unreliable was the moment the crisis became unmanageable. Two adjacent developments deserve the same attention: leverage has entered at the fund level, and the banks that no longer make these loans often finance the funds that do, which reconstructs a version of the exposure at one remove and puts it back on the balance sheets it was supposed to have left.

Neither is a repeat, and the chapter is careful about this. Both differ from the 2007 structure in ways that address its specific failures. What they share is the general feature that the risk's location is *harder to observe* than it was, and the lesson of the chapter is not that any particular structure is dangerous but that a system loses the ability to see a risk before it loses the ability to bear it.

*What the question tests.* That which slice the originator retains determines the incentive, and the discipline of assessing a new market by which specific past failure it does or does not reproduce.

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# Chapter 27
# Counterparty Credit Risk and Capital

**27.1** *Explain the difference between CVA as a price and CVA capital as a buffer, and why reducing one need not reduce the other.*

*CVA as a price* is an adjustment to the *value* of a derivative: the expected loss from the counterparty defaulting, computed under the risk-neutral measure from market-implied default probabilities, and charged to the client at inception. It is Chapter 9's quantity. It is a **number in the valuation**, it moves every day with spreads and exposures, and it can be hedged.

*CVA capital* is a **buffer against the possibility that the price moves**. It is equity the firm must hold because the CVA it has already marked may become larger if a counterparty's spread widens, marking down the derivative book, *without anyone defaulting*. It is not a valuation; it is a quantity of loss-absorbing capacity.

The distinction is the same one running through Chapter 9: a price is what you charge, capital is what you set aside in case the charge was too low. One is an expectation, the other a quantile.

*Why reducing one need not reduce the other.* Because the two are computed by different rules, and the capital rule is a *standardized formula* rather than a valuation model. There are two concrete asymmetries, and they run in opposite directions:

*A mitigation the price recognizes and the capital does not.* A firm can reduce its CVA price substantially by tightening a CSA: lowering a threshold, shortening the call frequency, adding initial margin. The valuation model sees the reduced exposure profile immediately. The standardized capital formula may not recognize the particular mitigation at all, so the price falls and the capital barely moves. This is Section 9.7's point, and it means a desk can genuinely reduce economic risk while the capital charge stands still.

*A hedge the capital recognizes and the accounts punish.* Conversely, hedging CVA with a single-name CDS reduces the capital charge under the standardized approach, while a hedge the framework does not recognize is capitalized as market risk instead, and either kind brings profit-and-loss volatility that appears in earnings every quarter. The firm has traded a capital charge for earnings volatility rather than removed the risk.

*The practical consequence* is that managing the two is a genuine optimization problem with no reason to have a common solution. A desk minimizing CVA price, one minimizing CVA capital, and one minimizing earnings volatility would hold three different portfolios. Question 9 develops the sharpest case: the hedge a desk can actually buy is the one the framework recognizes least.

*And the asymmetry is not a defect to be corrected.* The capital rule is deliberately coarse. A formula that tracked every valuation model's treatment of every CSA would be a model the bank could tune, and Chapter 26 is a chapter about what happens when a capital number inherits an institution's own assumptions. Standardization buys robustness at the cost of precision, and the

gap between price and capital is where that cost shows up.

*What the question tests.* That an expectation and a buffer answer different questions, and that the capital formula's coarseness is what decouples them.

---

**27.2** *Four trades with a counterparty mark at* $+30, -12, +8, -20$. *Compute the exposure with and without netting, and explain why a jurisdiction's netting opinion is described as worth more than any risk model.*

*Without netting.* The counterparty's administrator would enforce the contracts on which the firm owes and repudiate (or pay pennies on) those on which it is owed. That is Chapter 1's cherry-picking problem, and it leaves the exposure equal to the sum of the *positive* marks:

$$30 + 8 = \$\mathbf{38} \text{ million}$$

*With an enforceable netting agreement.* The whole portfolio collapses to a single claim:

$$\max(30 - 12 + 8 - 20,\ 0) = \max(6, 0) = \$\mathbf{6} \text{ million}$$

An **84% reduction**, obtained from a legal clause rather than from a hedge, a model, or a dollar of collateral.

And the benefit applies twice. Under SA-CCR the potential future exposure add-on is *also* reduced, because offsetting positions within a hedging set partly cancel before the add-on is computed. So netting reduces both components of EAD = $\alpha$(RC + PFE), not merely the replacement cost.

*Why a netting opinion is worth more than any risk model.* Because it is the assumption every number in the framework rests on, and it is binary.

*The magnitude.* No risk model produces an 84% reduction in exposure. Across a real dealer's book (tens of thousands of trades per major counterparty, most of them offsetting) the netting benefit is typically 80% or more. A model that improved exposure estimation by ten percent would be a significant achievement; netting is worth an order of magnitude more, and it is worth that *before* any modeling is done.

*And it is all-or-nothing.* A model that is somewhat wrong gives an answer that is somewhat wrong. Netting either is or is not enforceable in the relevant jurisdiction against the relevant counterparty type in insolvency. If it is not, because the jurisdiction has no close-out netting statute or because the counterparty type is carved out, then the exposure and the capital revert to the **gross** figure, and every model output computed on the net figure was answering a question about a world that does not apply. The $6 million becomes $38 million by operation of law, instantly, with no market move.

This is why Section 1.5 called the enforceability of close-out netting a precondition for a derivatives market existing at all. It is not a piece of administrative tidying attached to the trade; the enforceability of that clause is the difference between the framework working and the framework being arithmetic about the wrong number. Dealers commission and maintain legal opinions per jurisdiction and per counterparty type for exactly this reason, and a deterioration in one is a balance-sheet event.

*One rider is worth adding.* Netting is a mitigant whose failure is correlated with the event it mitigates: an insolvency in an untested jurisdiction is precisely when the clause gets litigated. That places it alongside the protections of question 12, and it is why the opinions are purchased rather than assumed.

*What the question tests.* The cherry-picking arithmetic, and that a legal precondition can dominate every quantitative refinement built on top of it.

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**27.3** *Compute the SA-CCR exposure at default for a portfolio with a replacement cost of $8 million and a PFE add-on of $3 million. Then compute the maturity factor for a ten-day margin period of risk, state what it does to the add-on, and state what assumption the margin period embodies.*

*The exposure at default.* By (27.1),

$$\text{EAD} = \alpha \times (\text{RC} + \text{PFE}) = 1.4 \times (8 + 3) = \$\mathbf{15.4}\text{ million}$$

*The maturity factor for a ten-day margin period of risk.* By (27.7),

$$\text{MF} = 1.5\sqrt{\frac{\text{MPOR}}{250}} = 1.5\sqrt{\frac{10}{250}} = 1.5 \times 0.2 = \mathbf{0.30}$$

*What it does to the add-on.* It multiplies it, so the $3 million add-on becomes

$$3 \times 0.30 = \$\mathbf{0.90}\text{ million}$$

a **70% reduction**, and the exposure at default falls to

$$1.4 \times (8 + 0.90) = \$\mathbf{12.46}\text{ million}$$

The replacement cost is unaffected (it is a loss already incurred, not a future move), so margining compresses the forward-looking component and leaves the current one alone.

*The logic of the scaling.* Exposure grows with the *square root* of time, because that is how far the market can have traveled. Margining does not make the market less volatile; it shortens the *window* over which an unhedged move can hurt. Variation margin resets the mark-to-market to zero daily, so the only exposure that can accumulate is what happens between the last call that was met and the completed close-out. Shorten that window and you shorten the exposure, by $\sqrt{t}$.

*The assumption the margin period embodies: that a defaulting counterparty's portfolio can be closed out in two weeks.* Ten days is a **supervisory figure**, not a measurement. It presumes that once the counterparty stops meeting calls, the surviving party can identify the failure, declare an event of default, and liquidate or replace the portfolio, all within ten business days. That requires the portfolio to be *liquid* and the position not to be *concentrated*.

Neither held in Section 27.14. A position representing many days of an underlying stock's trading volume cannot be closed out in ten days at anything like the marked price, and when six brokers attempt it simultaneously the assumption fails twice over. The maturity factor is where a liquidity assumption enters a capital calculation disguised as a constant, and question 16 lists it among the framework's gaps: SA-CCR's supervisory factors are by asset class, not by the liquidity of the specific position.

*Why the incentive is deliberate.* A 70% reduction in the add-on for signing a margin agreement is the framework's central lever. Chapters 2 and 8 described clearing and margin as regulatory mandates; this calculation is the economic reason a dealer would want them anyway. Capital, not virtue, is what moved the market—which is question 7's subject.

*What the question tests.* Substitution into SA-CCR, the $\sqrt{t}$ logic of the maturity factor, and that the ten days is a liquidity assumption rather than a fact.

---

**27.4** *Compute the margin period of risk at which the maturity factor reaches one, and state what this means for an illiquid portfolio.*

*The margin period at which the maturity factor reaches one.* Setting (27.7) equal to the unmargined value:

$$1.5\sqrt{\frac{\text{MPOR}}{250}} = 1 \quad \Longrightarrow \quad \text{MPOR} = \frac{250}{1.5^2} = \mathbf{111}\text{ business days}$$

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roughly five months.

*What it means for an illiquid portfolio: beyond about five months of close-out risk, margining provides no add-on relief at all.* The maturity factor is capped at one, the unmargined value, so a portfolio whose realistic close-out horizon exceeds 111 days is capitalized exactly as though no margin agreement existed.

*The logic is sound.* Margining helps because it shortens the window in which an unhedged move can hurt (question 3). Past 111 days the window is as long as the unmargined horizon, and there is nothing left to shorten. The formula is not being punitive; it has simply run out of benefit to confer.

*Three consequences follow.*

*The relief is concave, so the first days are worth most.* Going from 10 to 20 days raises the factor from 0.30 to 0.42, a 41% increase in the add-on. Going from 100 to 110 days raises it from 0.95 to 0.99, an increase of under 5%. Daily margining against weekly margining is a large difference; monthly against quarterly is almost none. Operational frequency matters at the short end and stops mattering at the long one.

*And the cap creates a threshold.* A supervisor extending a firm's assumed margin period (because the portfolio is concentrated, or because of past disputes, or because the counterparty is a fund with a history of late payment) moves the firm along a curve that flattens. Extending from 10 to 20 days raises the add-on by 41%; extending from 90 to 120, a step three times as long, raises it by 11%, because the relief was nearly exhausted already.

The deeper point is what happens when the assumption is wrong in the other direction. A firm holding a concentrated position in an illiquid issuer may be assuming ten days and facing sixty. It receives the full 70% relief while the economics justify relief of only about a quarter, since sixty days gives a maturity factor of 0.73. The capital understatement is largest exactly where the liquidity assumption is most heroic, and nothing in the formula detects this: SA-CCR's supervisory factors are by asset class, so a concentrated position in an illiquid issuer and a diversified one of the same notional carry the same add-on (question 16).

That is the gap Archegos ran through. The framework charged capital as though the portfolio could be closed out in days; it could not be closed out in weeks; and the shortfall was not a modeling error but a parameter the framework treats as a constant when it is a property of the position.

*What the question tests.* That the maturity factor is capped, where the cap binds, and that the relief is most overstated where the underlying assumption is least defensible.

---

**27.5** *A bank pays fixed on a three-year $100 million swap, unmargined, with no collateral. Compute its supervisory duration, effective notional and add-on. It then adds a seven-year receive-fixed swap of $47.2 million, chosen so that the two have equal and opposite dollar duration. Compute the hedging set's effective notional and add-on, and explain why a position with no net rate sensitivity still carries most of the add-on, and what would change if the second trade had a four-year maturity instead.*

*The single swap.* Paying fixed gains when rates rise, so $\delta = +1$; the trade runs from $t = 0$ to $T = 3$, so by (27.6)

$$d = \frac{1 - e^{-0.15}}{0.05} = \frac{1 - 0.8607}{0.05} = 2.7858$$

Unmargined with three years remaining, $\text{MF} = \sqrt{\min(3, 1)} = 1$, so by (27.5)

$$D = 1 \times 2.7858 \times 1 \times 100 = \$278.58 \text{ million}, \qquad \text{AddOn} = 0.50\% \times 278.58 = \$\mathbf{1.393} \text{ million}$$

*The hedged position.* The receiver has $\delta = -1$ and $d = (1 - e^{-0.35})/0.05 = 5.9062$, so

$$D_3 = -1 \times 5.9062 \times 1 \times 47.2 = -\$278.77 \text{ million}$$

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which cancels the first leg's dollar duration to within rounding. But the two sit in *different* maturity buckets (one to five years and over five years), so they do not sum. By (27.8),

$$\text{EN} = \sqrt{278.58^2 + 278.77^2 + 2 \times 0.7 \times 278.58 \times (-278.77)} = 215.86$$

and the add-on is $0.50\% \times 215.86 = \$\mathbf{1.079}$ million: the hedge earns 23% relief on a position that a trader would call flat.

*Why.* The supervisory correlation between adjacent buckets is 0.7, not 1. The formula does not believe that a three-year rate and a seven-year rate move together perfectly, and on that view the position retains curve risk. It does: a steepening leaves the payer profitable and the receiver losing. The framework is charging for a real exposure, at a size fixed by a supervisory number rather than by the portfolio's own history.

*And if the second trade were four-year.* Both legs would fall in the one-to-five-year bucket, where effective notionals are added before anything is squared. A duration-matched four-year receiver has $D = -278.58$, the bucket sums to zero, and the add-on with it. The same economic hedge earns complete relief or 23% of it depending only on which side of the five-year line the second trade falls, which is what a bucketed correlation structure costs in exchange for being uncontestable.

*What the question tests.* That the effective notional is a rate sensitivity in disguise, that offsets are complete within a bucket and partial across them, and that a supervisory correlation is an assumption a hedger can find expensive.

---

**27.6** *Using Example 27.3, state the ratio of capital across the three regimes and the two changes that produce it, then convert a capital requirement of $0.20 million on $100 million of notional into a running charge in basis points at a 12% hurdle rate.*

*The ratio.* From Example 27.3, on a five-year $100 million swap at zero mark-to-market against a 100% risk-weighted corporate, at a 10.5% capital requirement:

| | EAD | Risk wt. | RWA | Capital |
|---|---|---|---|---|
| Uncleared, unmargined | \$3.097m | 100% | \$3.097m | \$0.3252m |
| Uncleared, margined (10d) | \$0.929m | 100% | \$0.929m | \$0.0975m |
| Cleared at a qualifying CCP (5d) | \$0.657m | **2%** | \$0.013m | \$0.00138m |

**236** : **71** : **1**

Clearing the trade costs about 0.4% of the capital of leaving it uncleared and unmargined.

*Two changes produce it, and they compound.*

*1. Margining and clearing shorten the margin period of risk*, which cuts EAD by **79%** through the maturity factor of question 3, from \$3.097m to \$0.657m.

*2. Clearing replaces the counterparty's risk weight with the **2%** weight applied to trade exposures against a qualifying CCP*, a further factor of fifty on what remains.

Neither alone produces the result. The shorter margin period alone gives a factor of 4.7; the risk weight without the shortened period would give fifty. Multiplied, they give 236.

*The running charge.* Converting a capital requirement of \$0.20 million on \$100 million of notional at a 12% hurdle rate, by the method of Section 9.7's KVA calculation, gives what the client actually sees:

$$\frac{\$0.20\text{m} \times 12\%}{\$100\text{m}} = \frac{\$24{,}000}{\$100{,}000{,}000} = \mathbf{2.4} \text{ basis points per year}$$

Read that against the product. A charge of 2.4 basis points a year sounds negligible, and on a swap that trades at a fraction of a basis point of bid–ask it is **decisive**. A dealer quoting the

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same swap to a cleared and an uncleared client is quoting from different cost bases, and the gap is several times the spread. No amount of relationship or salesmanship closes it.

*Which answers a question Chapter 2 raised and did not settle.* The clearing mandate covered standardized products; it did not force the rest of the market to clear voluntarily, and much of it did anyway. The table is why. Capital rules did more to move the market than the mandate did. That is worth noticing because it is the general pattern: regulation that changes a price changes behavior more reliably than regulation that issues an instruction.

The table omits one qualification, in fairness. Clearing adds a capital charge for the member's default fund contribution, and the clearing *member* providing access (question 14) faces an economics far worse than the table suggests. The 2% weight accrues to the party whose trade is cleared, not to the intermediary who makes clearing possible.

*What the question tests.* Decomposing a capital ratio into its two multiplicative drivers, and converting a stock of capital into the running price a client is quoted.

---

**27.7** *Explain why the clearing mandate is described as less important than the capital rules in moving the market to clearing.*

*The claim.* The clearing mandate of Chapter 2 required standardized products to be cleared. It did *not* require the rest of the market to clear, and much of the rest cleared anyway. Something other than the instruction moved it.

*The reason is Example 27.3's ratio of 236 : 71 : 1.* A cleared trade consumes 0.4% of the capital of an uncleared, unmargined one. Converted to a running charge at the example's 10% hurdle, that is the difference between about 3.25 basis points a year and 0.01—on a product whose entire bid–ask spread is a fraction of a basis point.

So a dealer quoting an uncleared client must recover a cost that is several times the product's spread. The uncleared price is not slightly worse; it is outside the range in which the client will transact. Clients cleared because the alternative was unaffordable, and dealers pushed them to clear because the alternative was unprofitable. No instruction was required, because the price did the work.

*Why this is the more reliable mechanism.* Three reasons, and they generalize past this case.

*A mandate has a boundary and a price does not.* A rule specifying which products must clear invites classification: a product just outside the definition escapes entirely, and the incentive is to design products that sit outside it. A capital charge applies to whatever is left uncleared, so there is nothing to arbitrage into; the exemption carries its own cost.

*A price is enforced by the party bearing it.* A mandate requires supervision to detect non-compliance. A capital charge is deducted automatically by the firm's own finance function, and the desk feels it in its own profitability. No one has to be caught.

*And a price is proportionate by construction.* A mandate is binary: clear or do not. The capital framework charges more for exposures that are genuinely riskier, so a trade that is expensive to clear and cheap to margin bilaterally can rationally stay bilateral. That flexibility is a feature: the mandate would have forced trades to clear that had no business clearing.

*One qualification keeps the claim honest.* The capital rules did not act alone, and it would be too neat to say they did. The **uncleared margin rules** of Chapter 8 imposed initial margin bilaterally, removing much of the funding advantage of staying uncleared; and clearing delivers genuine multilateral netting and compression benefits that have nothing to do with capital. The fair statement is that the mandate set a floor and the price structure did the rest, and that the part of the market which moved *voluntarily* moved because of the price.

*The general lesson recurs.* Regulation that changes a price changes behavior more reliably than regulation that issues an instruction. Chapter 17's Sarbanes–Oxley two-day reporting

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rule is the same observation from a different angle: there, changing *when* a record is made succeeded where prohibition had failed.

*What the question tests.* That a capital differential larger than a product's bid–ask spread is a binding constraint, and why price-based regulation is harder to arbitrage than rule-based regulation.

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**27.8** *Explain what $\alpha = 1.4$ is calibrated to bridge, and why it is not a statement about any individual trade.*

*What $\alpha$ bridges.* Under the internal model method a bank computes an exposure profile by simulation and reduces it to **effective expected positive exposure** (the running maximum of expected exposure, averaged over the first year), then treats that number as a *loan-equivalent amount* and pushes it through the same capital formula a loan would use.

But a loan equivalent computed this way **understates** what a full portfolio credit model would require, for two reasons the reduction discards:

*It ignores the correlation between exposures and defaults across counterparties.* The loan equivalent is computed counterparty by counterparty, as though each firm's exposure and each firm's default were independent draws. In reality both are driven by common factors: the market moves that make exposures large are the same moves that make counterparties fail, and they happen to many counterparties at once.

*And it ignores granularity.* The capital formula behind a loan assumes an infinitely fine-grained portfolio in which idiosyncratic risk has diversified away. A derivatives book with a few large counterparties does not satisfy that, and the concentration is unpriced by the formula.

$\alpha$ is the **supervisory bridge over that gap**. Its value of 1.4 comes from studies of representative portfolios, with a floor of 1.2 for banks permitted to estimate their own.

*Why it is not a statement about any individual trade.* Because it is a **buffer for model error**, not a property of an exposure. The 40% uplift does not assert that this swap's exposure is 40% higher than the simulation says. It asserts that when a whole book's worth of loan-equivalent numbers are fed into a formula built on assumptions the book violates, the answer comes out roughly 40% too low on the portfolios the supervisors examined.

So it is a portfolio-level correction applied at the trade level for administrative convenience, and its size reflects the average of a study rather than anything about the trade in front of you. Two counterparties with quite different concentration and correlation properties receive the same 1.4.

*Why the distinction matters practically*, and it is worth saying plainly because the alternative reading does real damage. If a desk reads $\alpha$ as "exposures really are 40% higher than the model says," it will try to **hedge something that does not exist**. It will buy protection against a 40% larger exposure, pay for it, and find the hedge uncorrelated with anything its book ever experiences—because there is no such exposure to hedge. The uplift cannot be reduced by any trade; it can only be reduced by the bank being permitted to estimate its own $\alpha$, which is a supervisory conversation rather than a market transaction.

*The general pattern.* A capital framework contains numbers of two kinds: estimates of a quantity in the world, and buffers against the framework's own error. Confusing the second for the first leads to hedging the regulation instead of the risk. Question 10's DVA filter and question 13's leverage ratio are both instances of the second kind (rules justified by what capital is *for* rather than by measurement), and they resist the same category error.

*What the question tests.* That $\alpha$ corrects a modeling shortcut at portfolio level, and that a buffer for model error is not a hedgeable exposure.

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**27.9** *Explain what the CVA capital charge covers that the default charge does not, why this loss went uncapitalized before the charge existed, and why the hedge a desk can actually buy is the one the framework recognizes least.*

*What the CVA charge covers that the default charge does not.* The default charge capitalizes the loss from a counterparty **actually defaulting**. The CVA charge capitalizes the risk that **CVA itself moves**: that a counterparty's credit spread widens, marking down the value of the derivative book, without anyone defaulting at all.

These are genuinely different events. A counterparty whose spread goes from 100 to 500 basis points and then recovers has cost the firm a large mark-to-market loss and no credit loss. The pre-crisis framework capitalized only the second.

*Why this loss went uncapitalized before the charge existed.* Roughly **two-thirds** of counterparty-related losses in 2008–2009 came from CVA marks rather than from actual defaults, and the framework of the time held capital against none of it. Chapter 26's question 12 shows the mechanism operating: the largest single case was an insurer that had written protection on senior tranches without capital against the mark-to-market exposure, reasoning that the tranches would never default, which was, in the narrow sense, true for many of them. They did not need to default; the collateral calls on the marks were sufficient.

So the gap was not a mis-estimate. It was a whole category of loss the framework did not look at, and it was the category that did the damage.

*Why the hedge a desk can actually buy is the one the framework recognizes least.* The standardized CVA charge is sensitivity-based, so eligible hedges reduce it, but eligibility is graded:

- A **single-name CDS** on the counterparty is *fully* recognized. It is the exact instrument, and the capital relief matches the economics.
- An **index CDS** is recognized *only in part*. The framework applies a correlation assumption between index and issuer, and disallows a portion of the benefit.
- A **proxy** (a CDS on a different issuer in the same sector and region) is recognized further down still.

Now recall Chapter 25's question 13: the CDS market has contracted to a few hundred liquid reference issuers, while a dealer has *thousands* of counterparties. For most of them **no single-name CDS exists at any price**. The instrument the framework rewards is unavailable; the instruments available are the ones it discounts.

*The result is a genuine dilemma rather than a rule to memorize.* An index hedge removes most of the *economic* risk, earns only partial capital relief, and introduces a basis (the index-to-single-name gap of Chapter 25's question 5) that the CVA charge still capitalizes, because the index is recognized only in part, and that shows up in earnings every quarter. So:

*hedging the risk and hedging the capital charge are different trades,*

and the desk that attempts both ends up holding a position it wanted neither of. Section 9.10's observation that most CVA is hedged by proxy is the same fact seen from the pricing side.

*And the exemptions compound it.* The European framework exempts transactions with non-financial counterparties, sovereigns and pension funds from the CVA capital charge, on the argument that these clients cannot post collateral and would be priced out of hedging. The exposure is real and the capital is not held, so the charge's coverage is a policy choice rather than a measurement, which is worth knowing when comparing capital ratios across jurisdictions.

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*What the question tests.* That mark-to-market credit risk is a distinct category from default risk, its historical size, and that hedge eligibility and hedge availability point in opposite directions.

---

**27.10** *Explain why regulatory capital filters DVA out of common equity tier 1 while accounting requires it to be recognized.*

*The two rules.* Accounting standards require a firm to recognize a **debit valuation adjustment**, a gain when its *own* credit deteriorates, because its derivative liabilities are now less likely to be paid in full and are therefore worth less. Regulatory capital requires DVA gains and losses to be **filtered out of common equity tier 1**. A bank whose spread widens by 200 basis points may book a large accounting profit, and none of it counts as capital.

*The justification is a solvency argument, not an accounting one.* Ask how the gain could be *realized*. There are exactly two routes:

1. *Default.* The firm fails and does not pay the liability in full. The gain is realized by the event capital exists to prevent.

2. *Extinguishing the liability at its distressed value.* For debt this means buying it back, which requires cash, and a deteriorating firm does not have spare cash, which is precisely why its spread widened. For a derivative it means persuading the counterparty to tear up the trade at a price crediting one's own default risk, which none will.

So the gain is realizable only in default, or by an action a firm in that condition cannot take. A gain that can be monetized only by being closer to failure is not a buffer against failure, whatever its accounting treatment. And the perverse consequence of counting it is immediate: a bank's regulatory capital would *rise* as it approached insolvency, which inverts the entire purpose of the measure.

*Both rules are right about different questions.* Fair value asks *what is the liability worth?*—and it is genuinely worth less, since a rational buyer of that liability would pay less for it. Capital asks *what can absorb a loss?*—and this cannot. The apparent contradiction is a **category error about what the two systems are for**, not a defect in either.

*The same reasoning explains a second asymmetry.* The CVA capital charge of question 9 is computed on CVA *gross* of DVA, while the accounting number is *net*. A firm may not offset the deterioration of its counterparties against the deterioration of itself, for exactly the reason above: the second is not loss-absorbing capacity.

*One general principle is worth extracting.* An accounting standard and a capital rule that disagree are not necessarily in conflict; they may be measuring different things with the same inputs. Chapter 17's question 10 is the same lesson in equity compensation: diluted EPS is an accounting convention rather than a measure of value transfer, and reading it as the second produces nonsense. The discipline is to ask what a number is *for* before objecting that it differs from another number computed from the same data.

*One honest complication follows.* DVA is not merely an accounting artifact: a firm that *does* default genuinely pays less, and its creditors bear the loss. So the value is real; it simply accrues to the wrong party from capital's point of view: to the shareholders of a failed firm, at the creditors' expense, in the state where creditors were relying on capital to protect them.

*What the question tests.* That realizability, not existence, is what makes a gain loss-absorbing, and the discipline of distinguishing a valuation question from a solvency one.

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**27.11** *A counterparty has a 4% default probability, 60% loss given default, and an average exposure of $20 million. Compute CVA under independence, and again if the exposure conditional on default is twice the average. Distinguish specific from general wrong-way risk and state how each is treated.*

*CVA under independence.*

$$\text{CVA} = \text{LGD} \times \text{PD} \times \mathbb{E}[\text{exposure}] = 0.60 \times 0.04 \times \$20\text{m} = \$\mathbf{0.48}\text{ million}$$

*With conditional exposure twice the average.* CVA depends on the exposure *conditional on default*, not on the unconditional average:

$$0.60 \times 0.04 \times (2 \times \$20\text{m}) = \$\mathbf{0.96}\text{ million}$$

The charge **doubles**. The multiplier is exactly linear in the conditional exposure ratio, so a correlation strong enough to double the conditional exposure doubles the charge, and a ratio of five quintuples it.

Note what this calculation does not contain: a correlation parameter. The conditional exposure ratio is the only thing that matters, and it is *one number* rather than a joint distribution, which is convenient, and is also the reason the quantity is so hard to estimate. It is a conditional expectation on an event that has never occurred for this counterparty.

*Specific wrong-way risk* is **structural**: it follows from the trade's own economics and is identifiable in advance:

- buying protection on a sovereign from a bank domiciled in that country (Chapter 24's question 12);
- taking a counterparty's *own shares* as collateral;
- a commodity swap with a producer whose solvency depends on that commodity's price.

In each case the exposure and the default are driven by the same variable, by construction, and no estimation is needed to see it.

*The regulatory treatment is a rule, not a model:* where the counterparty is legally connected to the underlying or reference entity, the exposure must be computed as if that entity had already defaulted. That is deliberately crude, and deliberately so: modeling it would require a correlation no one can estimate, so the framework substitutes a conservative assumption for an unavailable number.

*General wrong-way risk* is the milder version: a statistical tendency for exposure to be larger when the counterparty is weaker, without a structural link. A firm whose derivative positions happen to correlate with the macroeconomic conditions that also drive its default probability has general wrong-way risk. It has no structural link to write a rule around, so it is handled inside the framework instead: through the alpha multiplier and the stressed calibration of the exposure model, backed by supervisory monitoring and stress testing.

*The distinction determines what can be done.* Specific wrong-way risk can be *avoided* by not doing the trade, or priced by the conservative rule. General wrong-way risk can only be monitored and buffered, because identifying it requires estimating a correlation on a rare event.

*And this pattern has appeared often.* Wrong-way risk is the general name for something the book keeps meeting: a protection whose value is negatively correlated with the need for it. Question 12 collects the instances.

*What the question tests.* That CVA scales linearly in conditional exposure, and that the specific/general split is a split between what can be identified in advance and what can only be estimated.

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**27.12** *Give three examples from this book of a protection whose value is negatively correlated with the need for it.*

Three examples follow, from different chapters, with the mechanism in each case:

*1. Excess spread in a securitization* (Chapter 26, Example 26.3). The first line of credit support is the pool's excess spread (2.5% a year in the example, accumulating to a projected 34.5% of support by year five). But excess spread is earned *only while the loans keep paying*. A pool that starts defaulting loses its 2.5% a year at the same moment it begins consuming the buffer it has already accumulated. In the bad scenario the support is not 34.5% and rising; it is 8% and falling. The models that projected the accumulating column ran it under scenarios in which the protection was not needed.

*2. Recovery in a credit portfolio* (Chapter 24, Section 24.4). A fixed 40% recovery assumption treats severity as independent of frequency. It is not: recoveries are lowest in exactly the years when defaults are highest, because a firm defaulting in a recession sells its assets into a market full of other distressed sellers, to buyers who are themselves impaired. So a portfolio loss model using an unconditional average understates the tail **twice over** (once because more issuers default, once because each recovers less), and capital is set by a tail quantile.

*3. Sovereign protection sold by domestic institutions* (Chapter 24, Section 24.12). The natural sellers of protection on a sovereign are banks and insurers domiciled in it. They hold its bonds, their loan book depends on its economy, their rating is capped by its rating, and they may be recapitalized by it. So the protection is worth most at exactly the moment the seller cannot pay. This is not a statistical correlation between exposure and counterparty quality but a *mechanical identity* between them.

Others follow briefly, because the pattern is the point.

- *Portfolio insurance* (Chapter 18): a synthetic put replicated by trading fails precisely when the market gaps, which is the scenario the put was bought for. October 1987.
- *The 0DTE stabilizing flow* (Chapter 23): offsetting customer buying and selling keeps net dealer gamma near zero, but the systematic sellers supplying that offset are free to withdraw, and would withdraw because conditions had deteriorated.
- *Netting enforceability* (question 2): the clause is litigated only in an insolvency, which is the one state in which it matters.
- *The margin period of risk* (question 4): ten days is assumed, and the assumption fails for concentrated illiquid positions, which are the positions that produce large losses.

The general name for it is **wrong-way risk**, and the general diagnostic is the same algebra every time. The naive calculation multiplies two expectations:

$$\mathbb{E}[\text{loss}] = \mathbb{E}[\text{frequency}] \times \mathbb{E}[\text{severity}]$$

when the correct one carries a covariance term that is *positive* in every case above. Chapter 20's relative-TSR award is the same error with the sign running in the holder's favor, which is why it is worth stating as an identity rather than as a warning: whenever the amount and the price share a driver, the covariance term is part of the answer.

*And a practical test follows.* Never quote a protection level computed under a base-case path. Ask what the protection is worth conditional on the state in which it is needed, and if that number is materially smaller, the difference is the real exposure.

*What the question tests.* Recognizing one structural pattern across four chapters, and the covariance term as its common algebraic form.

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**27.13** *Derive the risk weight below which the leverage ratio binds at a 10.5% capital requirement and a 5% leverage requirement, and explain why a risk-insensitive constraint makes a dealer's safest businesses expensive first.*

*The derivation.* Two constraints apply to the same trade. The risk-based requirement is capital $\geq$ EAD $\times$ RW $\times$ CR; the leverage requirement is capital $\geq$ EAD $\times$ LR, regardless of quality. The leverage constraint binds when it is the larger:

$$\text{EAD} \times \text{LR} \;>\; \text{EAD} \times \text{RW} \times \text{CR} \qquad \Longleftrightarrow \qquad \text{RW} \;<\; \frac{\text{LR}}{\text{CR}}$$

With LR = 5% and CR = 10.5%:

$$\text{RW} < \frac{5\%}{10.5\%} = \mathbf{47.6}\%$$

Below a 47.6% risk weight the leverage ratio is the binding constraint. At the 3% baseline the crossover is 29%; at 6% it is 57%.

*And essentially every high-quality exposure sits below it.* Sovereigns at 0%, qualifying CCPs at 2%, highly rated banks at 20%: all of them. For exactly those businesses, capital is set by a measure that does not care that the risk is low.

*Why a risk-insensitive constraint makes the safest businesses expensive first.* Because the leverage ratio charges the *same* capital per unit of exposure regardless of quality, while the *revenue* on an exposure scales with its risk. A safe, low-margin business earns a few basis points; a risky one earns much more. Charging both the same 5% of exposure means the safe business must cover its capital cost out of a far thinner margin, so it is the first to become unprofitable—and it becomes unprofitable *because* it is safe.

The businesses this hits are identifiable and they are the infrastructure: **Treasury market-making**, **repo intermediation**, and **client clearing** (question 14). All are low-margin, high-volume, and genuinely low-risk. A constraint insensitive to risk pushes a dealer out of them first.

This is intended, and the intention is defensible. A backstop that varied with risk weights would inherit whatever error is in them, and Chapter 26 is precisely a story about risk weights being wrong, about AAA ratings resting on an unobservable correlation, and about capital frameworks being talked into treating dangerous exposures as safe. The leverage ratio exists to bound what the risk-based framework can be persuaded of. Its insensitivity is the whole point: a measure that cannot be argued with cannot be gamed.

*But the cost is specific and has been observed.* In March 2020 Treasury market functioning deteriorated under a surge in volume while dealer balance sheets were constrained, and the supervisory response included temporarily excluding Treasuries and central bank reserves from the leverage exposure measure. The exemption expired in March 2021 and was not renewed; the 2025 recalibration of the US enhanced leverage ratio lowered the requirement on all exposures instead.

And the underlying question is genuinely hard, which is why it is unresolved: should a backstop be relaxed at exactly the moment the activity it constrains becomes most necessary? Relaxing it concedes that the constraint fails when tested. Not relaxing it means the constraint impairs market functioning in a crisis. There is no clean answer, and a reader should hold the tension rather than resolve it. It is the same shape as Chapter 38's procyclicality problem.

*What the question tests.* The crossover arithmetic, and that a deliberately crude constraint allocates activity as well as loss-absorbency.

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**27.14** *Explain why a clearing member's capital requirement is not eliminated by the fact that its client-facing and CCP-facing legs offset, and how the resulting concentration weakens the porting mechanism of Chapter 2.*

*Why the legs do not cancel.* A client does not face the clearing house directly. A **clearing member** stands between them, facing the CCP on one side and the client on the other. The position is *economically flat* (whatever the member owes the CCP it is owed by the client), but the two legs are with *different counterparties* and therefore cannot be netted. The member owes the CCP whatever the client fails to pay, and holds the client's margin against that.

Three capital consequences follow, and together they explain a market structure that surprises people.

*1. The client-facing leg is a derivative exposure and is capitalized as one.* It attracts the client's risk weight and an SA-CCR exposure, exactly as a bilateral trade would, notwithstanding the offsetting CCP leg. Netting requires a single counterparty; here there are two.

*2. Client margin is segregated, so the member cannot use it to fund.* Chapter 8's segregation requirement, which exists so the client's collateral survives the member's failure, means the member funds its own contribution to the CCP without the benefit of what it holds. That is Section 9.5's MVA arriving as an operating cost of the business rather than as a charge on a trade.

*3. And the leverage ratio counts the exposure.* This is the binding one. Question 13's crossover applies with full force: the risk weight against a qualifying CCP is 2%, so the risk-based charge is negligible and **the leverage requirement binds**. Whether segregated client initial margin may be recognized as reducing the leverage exposure measure has been changed more than once and remains among the most consequential technical questions in the framework, because it determines whether client clearing is a viable business at all.

*The observable result is concentration.* Client clearing in the major currencies is provided by a small number of banks, and several large institutions have exited the business. A framework designed to reduce the concentration of counterparty risk has concentrated the access to the mechanism that reduces it. That is not a contradiction so much as a reminder that capital rules allocate activity as well as loss-absorbency.

*How this weakens porting.* Chapter 2's default management relies on **porting**: when a clearing member fails, its clients' positions and margin are transferred to a surviving member, so clients are not liquidated for a failure that was not theirs. Porting is what makes clearing safe for a client rather than merely safe for the system.

But porting requires **a surviving member willing and able to take the positions**, at short notice, in a crisis. Concentration attacks that requirement from both sides:

- *There are few candidates.* With a small number of members in a currency, the failure of one leaves few destinations, and if the failure is systemic, the survivors are themselves stressed.
- *Taking on the positions consumes exactly the capital that is already binding.* A surviving member accepting ported clients adds leverage exposure at the moment its own ratio is under pressure. The economics that made it exit the business in calm conditions are worse in a crisis.

So the mechanism that protects clients depends on spare capacity in precisely the business the capital framework has made unattractive. If porting fails, clients are closed out—and the risk that clearing was supposed to remove reappears as forced liquidation. That places porting alongside question 12's list: a protection whose availability is negatively correlated with the need for it.

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*What the question tests.* Why an economically flat position is not a capitally flat one, and that a capital-driven exit from an intermediation business degrades a safety mechanism two chapters away.

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**27.15** *In the Archegos arithmetic, state the move at which the client's capital is exhausted and the shortfall per broker at a 30% move, taking margin as all a broker can collect, and explain why the relevant denominator for concentration is liquidity rather than the dealer's capital.*

*The arithmetic.* A fund with $10 billion of capital taking five times leverage through total return swaps, split across six prime brokers:

| | |
|---|---|
| Total gross exposure | $50bn |
| What each broker sees | $8.3bn |
| Initial margin held by each, at 15% | $1.25bn |

*The move at which the client's capital is exhausted:* −20%. A 20% fall on $50 billion of gross exposure is a $10 billion loss, which is the entire capital. Beyond that the client is insolvent and the loss falls on the brokers.

*And 20% is not a stress scenario.* For an equity portfolio concentrated in a few stocks that is an ordinary week. The leverage was sized so that a routine move destroyed the client.

*The shortfall per broker at a 30% move.* A 30% fall costs $15 billion on the gross exposure. Spread across six brokers, each faces

$$\frac{\$15\text{bn}}{6} = \$2.5 \text{ billion of loss against } \$1.25 \text{ billion of margin}$$

$$\text{shortfall} = \$\mathbf{1.25} \text{ billion each}$$

on the assumption that a broker can look only to the margin it holds: the client's other $2.5 billion of capital is not posted, and six brokers would claim it at once. If all of it were collected, the total shortfall would be $15 − $10 = $5 billion rather than $7.5 billion, or $0.83 billion each. Neither figure allows for what happens when six brokers liquidate the same concentrated positions simultaneously. Actual losses exceeded $10 billion in total, with a single institution losing more than $5 billion, one of the largest single-counterparty losses in the history of the derivatives market, involving no exotic instrument and no pricing error.

*Why the relevant denominator is liquidity, not the dealer's capital.* Because the loss is not determined by the mark at the moment of default; it is determined by **where the position can actually be sold**.

Every element of the framework is calibrated to a close-out that completes inside the margin period of risk. Question 3 showed what the ten-day assumption decides: it is what turns a $3 million add-on into $0.90 million. That assumption is a statement about liquidity, and here it was false. A position representing many days of a stock's trading volume cannot be closed out in ten days at anything near the marked price, and when *six* brokers attempt the same liquidation at once the price impact is not additive but compounding: each seller's execution worsens the others'.

So the right question is not "how large is this exposure against the dealer's capital?" but "how many days of trading volume is this position, and who else is holding the same one?" A $8.3 billion exposure that can be liquidated in a day is a manageable risk; the same $8.3 billion representing thirty days of volume, held identically by five competitors, is not—and the two are indistinguishable in every number the capital framework produces.

Four failures stand out, and only one is about the instrument. *Invisibility*: each broker's exposure was individually unremarkable, and the swap format kept the positions out of ownership disclosures, so the concentration was invisible to the market too (Chapter 25's

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question 12). *Margin that was not risk-based*: initial margin reflected competition for the client's business rather than the concentration and liquidity of the underlying. *The bilateral framework's blind spot*: nothing in it sees aggregate leverage across brokers. *And correlated liquidation*: the framework asks what a firm loses if one counterparty fails, not what it loses when that failure forces five other firms to sell the same assets.

*What the question tests.* That leverage sized to a routine move is the whole story, and that the binding constraint on a close-out is market depth rather than the surviving party's balance sheet.

---

**27.16** *Explain why initial margin is described as the only mitigant that works without information, and state four things the capital framework does not capture.*

*Why initial margin is the only mitigant that works without information.*

Begin with what the others require. *Netting* requires **legal certainty**: an enforceable close-out netting opinion in the relevant jurisdiction against the relevant counterparty type (question 2), and where it fails the exposure reverts to gross. *Wrong-way risk* requires **identifying the correlation**: a conditional expectation on an event that has never occurred for this counterparty (question 11). *Concentration limits* require **knowing the aggregate**, which in Archegos no one did (question 15).

Each is a mitigant conditional on knowing something. Initial margin protects against a move of a given size regardless of why it happened. It does not need to know whether the counterparty is levered elsewhere, whether the position is concentrated, whether the exposure correlates with the counterparty's solvency, or whether the netting opinion holds. It is a buffer sized in dollars against an outcome, not against a cause.

Which makes it the right instrument when the thing you cannot see is the thing that hurts you. That is the recurring shape of this chapter: the losses that mattered came from what the framework could not observe. A mitigant that is robust to *unknown* causes dominates one that must first identify them.

The distinction from variation margin is worth being precise about. Variation margin removes current exposure: it is settled daily and brings the mark back to zero, so it is a transfer of value that has *already happened*. Initial margin removes gap risk, the move between the last margin call and the close-out, which is exactly the margin period of risk of question 3. It is a buffer against value that has *not* happened. That is why initial margin must be segregated and cannot be rehypothecated: it belongs to the poster and exists only to be seized on default.

Its cost is Section 9.5's MVA and its benefit is the EAD reduction of question 3, so a dealer's decision about how much to demand is a trade between a funding cost it pays and a capital cost it avoids. Archegos is what happens when competition for a client pushes that trade toward the client.

*Four things the capital framework does not capture.*

*1. Aggregate leverage.* Nothing in a *bilateral* framework sees a client's total position across dealers. Each prime broker held an unremarkable exposure; in the chapter's arithmetic the six together held $50 billion.

*2. Liquidity of the underlying.* SA-CCR's supervisory factors are by *asset class*, not by the liquidity of the specific position. A concentrated position in an illiquid issuer and a diversified one of the same notional carry the same add-on, and question 4 showed that the maturity factor grants the same relief to both.

*3. Correlated liquidation.* The framework asks what a firm loses if one counterparty fails. It does not ask what it loses if that counterparty's failure forces five other firms to sell the same assets, which is the mechanism that pushed actual losses well past the arithmetic's $1.25 billion

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shortfall per broker.

*4. Model risk in the exposure itself.* The exposure profile is a simulation output, and Chapter 21 established that model choice moves such outputs materially. SA-CCR's standardization is partly a response (a supervisory formula cannot be gamed the way an internal model can), at the cost of being wrong in a known direction for portfolios it was not calibrated to.

*None of this is an argument against the framework.* It substantially improved on what preceded it, and the CVA charge in particular addressed a loss that had gone entirely uncapitalized (question 9). The argument is narrower and harder: a capital framework measures what a firm can see, and the losses that matter most have repeatedly come from what the whole system could not.

*What the question tests.* That robustness to unknown causes is what distinguishes initial margin, and that the framework's four gaps are all failures of visibility rather than of calibration.

---

**27.17** *In Example 27.5, the countercyclical buffer rises from 0.5% to 2.0%. Compute the new CET1, tier 1 and total capital requirements, and state whether the leverage ratio still binds and why the buffer change does not affect the leverage requirement.*

*The new requirements.* The combined buffer becomes 2.5% + 1.5% + 2.0% = 6.0%, added to every layer, against RWA of \$850 billion:

| | Ratio | Capital (\$bn) |
|---|---|---|
| CET1 | 4.5% + 6.0% = 10.5% | 89.25 |
| Tier 1 | 6.0% + 6.0% = 12.0% | 102.00 |
| Total capital | 8.0% + 6.0% = 14.0% | 119.00 |

*The leverage ratio no longer binds.* The leverage requirement is still 3.75%×2,500 = \$93.75 billion of tier 1, now below the risk-based tier 1 figure of \$102.00 billion. In crossover terms, the risk weight below which leverage binds has fallen to 3.75%/12.0% = 31.25%, and the bank's average of 34% is above it.

*Why the leverage requirement did not move.* The countercyclical buffer is a risk-based buffer, set against risk-weighted assets to make banks hold more capital when credit is growing fast. The leverage framework's only buffer is the G-SIB one, set at half the bank's surcharge, so a change in the countercyclical buffer passes through to the risk-based ratios alone. A supervisor tightening the cycle buffer therefore shifts which rule binds, and with it which lever a desk has: before the change, cutting exposure saved capital; after it, cutting risk weights does.

*What the question tests.* Assembling the stack, and seeing that the binding constraint is a property of the whole bank that a macroprudential decision can move.

---

**27.18** *A bank has a \$50 million loan with four years to run to a BBB-rated corporate and hedges it with a two-year CDS, covering restructuring, bought from a bank rated A. Using Table 27.1, compute the loan's RWA before and after the hedge, and explain why protection shorter than the loan is recognized only in part.*

*Before.* A BBB corporate is weighted 75%, so RWA are $50 \times 0.75 = \$37.50$ million.

*After.* The protection has $t = 2$ years left against a loan with $T = 4$, so the recognized amount is

$$P_a = 50 \times \frac{2 - 0.25}{4 - 0.25} = 50 \times \frac{1.75}{3.75} = \$23.33 \text{ million}$$

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That part takes the protection seller's 30% weight for an A-rated bank, and the remaining \$26.67 million keeps the borrower's 75%:

$$23.33 \times 0.30 + 26.67 \times 0.75 = 7.00 + 20.00 = \$27.00 \text{ million}$$

The hedge cuts RWA by \$10.50 million, 28%, where a four-year CDS would have cut them by $50 \times (0.75 - 0.30) = \$22.50$ million.

*Why only in part.* Protection that ends in two years leaves the loan unprotected in years three and four, and a borrower's default is if anything likelier late than early (Example 24.6). The formula credits the protection in proportion to the share of the loan's remaining life it covers, measured net of a three-month stub, and gives none at all to protection with three months or less to run, because the bank could find itself unhedged before it could replace it.

*What the question tests.* Applying the substitution rule with a maturity mismatch, and seeing that a hedge's capital value depends on its terms as much as on who sold it.

---

**27.19** *Explain why a securitization tranche below the pool's own capital requirement is weighted 1,250%, and why the floor on the senior tranche's weight is what makes a significant risk transfer trade worth doing.*

*Why 1,250%.* The framework first asks what capital the pool would need if the bank held it whole, $K$, as a fraction of the pool. A tranche whose detachment point is below $K$ absorbs losses the pool's own capital was meant to cover, so the framework treats it as capital: it is weighted 1,250%, which at the 8% minimum is $12.5 \times 8\% = 100\%$ of its amount. The holder of a first-loss tranche holds capital dollar for dollar against it, whoever that holder is.

*Why the senior floor matters.* Above $K$ the charge falls quickly with the attachment point, but it never falls below 15% (10% for a tranche meeting the simple, transparent and comparable criteria). A bank with a corporate loan book weighted around 100% that sells or insures the first-loss and mezzanine tranches and keeps the senior one replaces 100% weights on the whole book with roughly 15% on the part it keeps, plus whatever the first-loss buyer charges. The trade pays if that charge is less than the capital released times the bank's hurdle rate, which is the calculation behind the significant risk transfer market of Chapter 26. If the floor were higher, the senior tranche would consume most of the capital the trade was meant to release, and the market would not exist; if there were no floor, banks could retain almost all of the economic exposure at almost no capital, which is why supervisors test that the transferred risk is commensurate with the relief.

*What the question tests.* Reading the securitization weights as statements about who absorbs the pool's capital, and connecting them to the trade they make possible.

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# Chapter 28
# Martingales and Measures

**28.1** *Using Example 28.1, construct the riskless portfolio and compute its excess return, then explain why the market price of risk must be the same for two assets driven by the same source of uncertainty.*

*The portfolio.* With $r = 3\%$ and

$$\text{A} : \mu = 8\%,\ \sigma = 20\% \Rightarrow \lambda_A = \frac{8-3}{20} = 0.25,$$
$$\text{B} : \mu = 6\%,\ \sigma = 15\% \Rightarrow \lambda_B = \frac{6-3}{15} = 0.20$$

hold one unit of A and $-\sigma_A/\sigma_B = -20/15 = -1.3333$ units of B. The $dW$ terms carry $1.0000 \times 20\%$ and $-1.3333 \times 15\%$, which cancel *exactly*, so the portfolio is riskless.

*Its excess return.* The return is

$$1.0000 \times 8\% - 1.3333 \times 6\% = 8\% - 8\% = 0.00\%$$

and the cost of establishing it is

$$1.0000 - 1.3333 = -0.3333$$

A portfolio with a *negative* cost (a net borrowing of 0.3333) and a zero return. Financing that borrowing at 3% costs $0.03 \times (-0.3333) = -0.0100$, so the position earns

$$\mathbf{1.00}\% \text{ of the notional of A, riskless, per year}$$

Scale it and the profit is unbounded, which is what an arbitrage means. Nothing here depends on anyone's risk preferences or on either asset's expected return being "correct": the trade exists as soon as the two ratios differ.

*Why the market price of risk must be equal.* Because the construction above works for *any* pair of assets driven by the same $dW$, and it works in whichever direction the inequality runs. If $\lambda_A > \lambda_B$, long A and short B; if $\lambda_A < \lambda_B$, the reverse. The only configuration admitting no such trade is $\lambda_A = \lambda_B$.

The economics behind the algebra is that a single source of uncertainty can carry only one price. $\lambda = (\mu - r)/\sigma$ is the excess return per unit of risk, the compensation demanded for bearing one unit of $dW$. If two assets offered different compensation for the identical risk, one could buy the generous one, sell the stingy one in the proportion that cancels the risk, and collect the difference. Two prices for one thing is the definition of an arbitrage, and $\lambda$ is that price.

Note also what the equality *pins down*: given $\lambda = 0.25$ and $\sigma_B = 15\%$, B's expected return is forced to

$$\mu_B = 3\% + 0.25 \times 15\% = \mathbf{6.75}\%$$

so no-arbitrage determines a relation among expected returns without determining any of them, which is the shape of every result in this book.

*Why this is the whole of the chapter.* Everything that follows is a restatement of it. Chapter 13's risk-neutral valuation is the case $\lambda = 0$: a measure in which every asset earns $r$, reached by absorbing $\lambda$ into the probabilities. The change-of-numeraire theorem of Section 28.6 is the general case, in which $\lambda$ is shifted by the numeraire's own risk rather than set to zero. And Girsanov (question 8) is the statement that these shifts move drifts only.

*One condition is worth naming.* The argument requires the two assets to be driven by the *same* $dW$. With several sources of uncertainty there is a $\lambda$ for each, and the cancellation requires a portfolio spanning them: Section 28.3's loadings matrix must be invertible, which is exactly the completeness condition of question 13.

*What the question tests.* That the equality of market prices of risk is an arbitrage result rather than an equilibrium assumption, and constructing the trade that enforces it.

---

**28.2** *Using Example 28.2, state the three values of $\Lambda = dQ/dP$, explain why they average to one under $P$ rather than under $Q$, and explain why the reweighting is monotone in the terminal price.*

*The three values.* With $S_0 = 100$, $u = 1.2$, $d = 1/1.2$, $r = 4\%$, real-world $p = 0.65$ and risk-neutral $q = (e^r - d)/(u - d) = 0.565848$:

| **Path** | $S(2)$ | $P$ | $Q$ | $dQ/dP$ |
|---|---|---|---|---|
| $dd$ | 69.4444 | 0.122500 | 0.188488 | **1.538680** |
| $ud$ | 100.0000 | 0.455000 | 0.491328 | **1.079842** |
| $uu$ | 144.0000 | 0.422500 | 0.320183 | **0.757831** |

*Why they average to one under $P$ and not under $Q$.* Because $dQ/dP$ is a density with respect to $P$: it is the factor by which each $P$-probability must be multiplied to obtain the corresponding $Q$-probability. Averaging it *under $P$* simply re-sums the $Q$-probabilities:

$$E^P\left[\frac{dQ}{dP}\right] = \sum_i P_i \cdot \frac{Q_i}{P_i} = \sum_i Q_i = 1$$

This is the defining property, and it is what makes $dQ/dP$ a legitimate *change of probability* rather than an arbitrary reweighting: the new weights are nonnegative and they sum to one, so $Q$ is a probability measure.

Under $Q$ the same average is

$$E^Q\left[\frac{dQ}{dP}\right] = \sum_i Q_i \cdot \frac{Q_i}{P_i} = \sum_i \frac{Q_i^2}{P_i} = 1.063225 \;>\; 1$$

and it exceeds one for a reason worth naming: by Jensen's inequality applied to $E^P[(dQ/dP)^2] \geq (E^P[dQ/dP])^2$, the second moment of a non-constant density with unit mean is strictly greater than one, and $E^Q[dQ/dP]$ *is* that second moment. It equals one only if the two measures coincide. So "the Radon–Nikodym derivative averages to one" is a statement about the measure in the denominator, and reversing it is a genuine error rather than a matter of convention. The inverse density $dP/dQ$ is what averages to one under $Q$.

*Why the reweighting is monotone in the terminal price.* Good states are weighted **down** (0.7578 at $S = 144$) and bad states **up** (1.5387 at $S = 69.44$), monotonically.

That is exactly what the market price of risk means, expressed as probabilities. Question 1 gave $\lambda$ as the compensation demanded for bearing risk, an adjustment to *returns*. The same adjustment can be made to *probabilities* instead: rather than requiring assets to earn $r + \lambda\sigma$ under the true probabilities, require them to earn $r$ under distorted ones that overweight the states in which a dollar is worth most. A dollar delivered in a bad state is worth more than one

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delivered in a good state, so a valuation that discounts every state at $r$ must treat bad states as more likely than they are.

The monotonicity is not incidental. It follows from the density being a decreasing function of the terminal price, which in turn follows from $q < p$, and $q < p$ holds precisely because the asset's real-world expected return exceeds $r$. An asset with $\mu < r$ would produce a density increasing in $S_T$.

*And the check that the whole construction works.* $E^Q[S(2)] = 108.3287$, which equals $S_0 e^{2r} = 108.3287$ to every digit: the discounted stock price is a $Q$-martingale, as it must be. The reweighting was not chosen for elegance; it is the unique one making that true.

*What the question tests.* That a density averages to one under the measure it is a density with respect to, and that the monotone tilt is the market price of risk in probability form.

---

**28.3** *Using Example 28.2, state the size of the pricing error from using real-world probabilities and explain why it is positive for a call.*

*The error.* Pricing a two-year call struck at 100 on Example 28.2's tree:

| | |
|---|---|
| $e^{-2r}\,E^P[\text{payoff}]$ | 17.160733 |
| $e^{-2r}\,E^Q[\text{payoff}]$ | 13.004930 |
| Error from using the wrong measure | **+32.0**% |

Same tree, same volatility, same payoffs. Only the probabilities differ, and the price is wrong by a third.

*Why it is positive for a call.* A call pays only in the **upper** states, and question 2 established that the reweighting is monotone: $Q$ weights good states *down* relative to $P$ ($dQ/dP = 0.7578$ at $S = 144$). So the real-world measure assigns more probability to precisely the states in which the call pays, and pricing under $P$ therefore *overstates* the value.

Concretely, the only paying node is $uu$ at $S = 144$, and

$$P(uu) = 0.4225 \quad \text{against} \quad Q(uu) = 0.320183$$

The payoff is identical in both calculations; the probability attached to it is 32% larger under $P$, and that is the entire discrepancy: $0.4225/0.320183 = 1.3196$, and the price ratio $17.160733/13.004930 = 1.3196$ to four decimals.

*The general statement.* For *any* payoff that is increasing in $S_T$, pricing under the real-world measure overstates. For a decreasing payoff, such as a put, it understates, by the same mechanism running the other way. This is not a subtlety about calls; it is the market price of risk appearing as a systematic bias whose sign is the sign of the payoff's dependence on the underlying.

*Why the error is so large, since a third is a striking number.* Because the option is a *leveraged* claim on the tail. The stock's own mispricing under $P$ would be modest ($E^P[S(2)] = 114.85$ against $E^Q[S(2)] = 108.33$, about 6%), but the call pays only in the single state whose probability is most distorted, so it inherits the distortion undiluted. An option magnifies the measure error exactly as it magnifies everything else, which is Chapter 26's convexity argument arriving in a different setting.

*And the practical warning.* The temptation to use real-world probabilities is not hypothetical; it is what a forecast produces, and $p = 0.65$ is the kind of number an equity analyst would supply with conviction. The distinction Section 28.1 draws is that the measure that prices correctly is not the measure that forecasts correctly, and they are used for different tasks: $Q$ for pricing and hedging, $P$ for risk management, capital, and expected returns. Chapter 24's question 9 is the same distinction in the credit market, where the ratio of risk-neutral to historical default probabilities exceeds thirty for the safest issuers.

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*One thing the error is* not. It is not evidence that $p = 0.65$ was wrong. The real-world probability may be exactly right and the price still wrong by 32%, because the price does not depend on it. A correct forecast used in the wrong place is still an error.

*What the question tests.* That the sign follows from monotonicity of the density, and that options amplify a measure error rather than diluting it.

---

**28.4** *Explain what it means for a ratio to be a martingale, and why finding one is the whole of the valuation procedure.*

*What it means.* A process $X$ is a martingale under a measure $Q$ if its expected future value, conditional on everything known now, is its current value:

$$E^Q[X(T) \mid \mathcal{F}_t] = X(t)$$

No drift. Whatever information arrives between now and $T$, the best forecast of $X(T)$ is what $X$ is worth today. A martingale is not a process that does not move; it is one whose movement is unpredictable in the mean.

The relevant object here is a **ratio**. For a numeraire $N$ (any traded asset with a strictly positive price) the fundamental result is that there exists a measure $Q^N$ under which $V/N$ is a martingale for *every* traded asset $V$. So the martingale property is a statement about prices *measured in units of the numeraire*, not about prices themselves.

*Why finding one is the whole of the valuation procedure.* Because the martingale property *is* the pricing formula. Write it out at $t = 0$:

$$\frac{V(0)}{N(0)} = E^N\left[\frac{V(T)}{N(T)}\right] \qquad \Longrightarrow \qquad V(0) = N(0)\,E^N\left[\frac{V(T)}{N(T)}\right]$$

The left side is what we want and the right side is an expectation of a known payoff. There is nothing else to the procedure: choose a numeraire, find the measure that makes the ratio a martingale, take an expectation.

Every valuation in the book is an instance:

- $N = B$, the money-market account: $V(0) = E^Q[e^{-\int_0^T r}V(T)]$, risk-neutral valuation, which is Chapter 13.
- $N = P(\cdot, T)$, a zero-coupon bond: $V(0) = P(0,T)E^T[V(T)]$, and the discounting comes out of the expectation entirely because $N(T) = 1$. This is why Black's caplet formula is exact.
- $N = A$, the swap annuity: the annuity in a swaption's payoff cancels against the numeraire, and Black's swaption formula follows (question 6).
- $N = S_2$, an asset: Margrabe's formula (question 10), and $N(d_1)$ itself (question 9).

*And it explains what a "model" is for.* The martingale property says the drift is zero; it says nothing about the *volatility*. So the theory determines the drift and a model must supply the diffusion. That division is exactly right and worth internalizing: drifts are theory, volatilities are data. Chapter 32's HJM drift condition is the sharpest instance: the drift is not a modeling choice at all, it is forced by the volatility structure.

*The practical skill this converts into.* Valuing a payoff becomes a search for the numeraire that makes the awkward part of the payoff cancel. A swaption's payoff contains an annuity, so use the annuity; a payoff contains $S_T$ multiplied by an indicator, so use the stock; an exchange option has a moving strike, so use the asset that moves. The theorem guarantees the answer

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does not depend on which is chosen, but the difficulty depends on it enormously, which is question 5.

*What the question tests.* That $E[X(T)] = X(t)$ rearranges into a pricing formula, and that choosing the numeraire is the valuation step rather than a preliminary to it.

---

**28.5** *Name the four numeraires that cover practice, state the measure each induces, and explain why the choice changes no price but changes the difficulty enormously.*

*The four numeraires.*

| Numeraire | Measure | Martingale | Used for |
|---|---|---|---|
| Money-market account | Risk-neutral | $V/B$, $B_t = e^{\int_0^t r}$ | General simulation |
| Zero-coupon bond $P(t,T)$ | $T$-forward | Any $T$-maturity forward | Caplets, single payoffs |
| Swap annuity $A(t)$ | Swap | The forward swap rate | Swaptions, CMS |
| Foreign money market | Foreign risk-neutral | Foreign assets | Quantos, FX |

*Why the choice changes no price.* That is the content of the change-of-numeraire theorem, (28.8). Two numeraires $N$ and $M$ induce measures related by a density

$$\left.\frac{dQ^M}{dQ^N}\right|_{\mathcal{F}_t} = \frac{M(t)/M(0)}{N(t)/N(0)}$$

and substituting it into either valuation formula returns the other. The proof is three lines and shows the theorem is a restatement of the definition rather than a new fact: the defining property of $Q^M$ applied to any traded $V$, rewritten as an expectation under $Q^N$, gives back $V(0)/M(0)$.

The economic content is that a numeraire is a choice of units. Measuring a portfolio in dollars, in bonds, or in shares cannot change what it is worth, any more than quoting a distance in meters or feet changes the distance. Section 31.7 verifies this numerically rather than merely asserting it (the same caplet comes to 0.22478353 under the forward measure and 0.22498411 by simulation under the risk-neutral measure, agreeing to 0.85 standard errors), and the invariance is worth *seeing* because everything else in Part VI depends on it.

*Why the choice changes the difficulty enormously.* Because a quantity requiring a simulation under one numeraire may have a **closed form** under another. The same caplet illustrates the whole point: under the forward measure it is a closed form evaluated in microseconds; under the risk-neutral measure it took two million antithetic paths and two hundred steps to reach three significant figures.

The mechanism is always the same: pick the numeraire that cancels the awkward part of the payoff:

- A payoff paid once at $T$ calls for $P(\cdot,T)$, because $N(T) = 1$ removes the discounting from inside the expectation, and with it the covariance between payoff and discount factor (Chapter 31's question 15).
- A swaption's payoff *contains* an annuity, so the annuity numeraire cancels it and leaves a driftless lognormal (question 6).
- An exchange option has a moving strike, so using the second asset turns the strike into the number 1 (question 10).
- A payoff $S_T\mathbf{1}_{\{S_T>K\}}$ is a product, so the stock numeraire divides the $S_T$ out and turns an expectation of a product into a probability, which is $N(d_1)$ (question 9).

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*The practical rule this yields.* Look at the payoff, identify what makes it hard, and ask whether some traded asset in it can be promoted to numeraire. If the difficulty is discounting, use a bond; if it is a multiplicative factor, use the asset that appears; if it is a moving strike, use the thing it moves with. The theory guarantees you cannot get a wrong answer by choosing badly—only a laborious one.

*What the question tests.* That invariance is the theorem and tractability is the reason anyone cares, and knowing which numeraire each standard product calls for.

---

**28.6** *Explain why Black's swaption formula is exact rather than approximate, being specific about what cancels.*

*The claim.* Black's swaption formula is not an approximation under its own measure. It is an exact consequence of the martingale property, and the reason is a cancellation.

*The setup.* A swap that starts at $T_0$ pays its fixed leg on $T_1, \dots, T_N$, and the **annuity**

$$A(t) = \sum_{i=1}^{N} \tau_i \, P(t, T_i)$$

is the present value of the fixed leg per unit of fixed rate on unit notional, $\tau_i$ being the year fraction of the $i$th period. It is a portfolio of zero-coupon bonds, hence a traded asset, hence an admissible numeraire. The forward swap rate is

$$S(t) = \frac{P(t, T_0) - P(t, T_N)}{A(t)}$$

a traded asset divided by the annuity (the numerator is the value of the floating leg, which is a portfolio of bonds), so $S$ is a **martingale** under the annuity measure $Q^A$. No drift, by construction rather than by assumption.

*What cancels.* A payer swaption's payoff is

$$A(T_0) \max(S(T_0) - K,\ 0)$$

The annuity appears in the payoff. Valuing with $A$ as numeraire,

$$V_0 = A(0)\, E^A\left[\frac{A(T_0)\max(S(T_0) - K, 0)}{A(T_0)}\right] = A(0)\, E^A[\max(S(T_0) - K, 0)]$$

and **the annuity divides out exactly**. What remains is the expectation of a vanilla call payoff on a driftless variable. Assume $S$ is lognormal and Black's formula follows—exactly.

So two things canceled, and both were needed. The annuity in the payoff canceled against the annuity in the numeraire, removing the stochastic discounting; and the drift vanished because $S$ is a ratio of traded assets to that same numeraire. Nothing was approximated and no model was chosen; a numeraire was picked so that the awkward part of the payoff divided out.

*What remains an assumption.* Lognormality of $S$ under $Q^A$. That is a modeling choice, and it is the *only* one: the market's smile shows it to be false, which is why Section 29.9's volatility cube exists and why the shifted and normal conventions of Section 29.8 were adopted after 2014. But lognormality is a statement about the *distribution*, not about the drift or the discounting, and it is the same assumption the Black–Scholes–Merton model makes for equities. The exactness claimed here is that no additional error is introduced by the interest-rate structure, which is exactly what a reader should want to know, since a swaption's payoff and its discounting both depend on rates.

*Why this is the chapter's thesis in its cleanest form.* Most of Part VI consists of cases where the cancellation is *imperfect*, and what fails to cancel is precisely what has to be computed.

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Question 7's CMS coupon is the nearest such case, differing only in how the rate is paid. Chapter 30's three adjustments are all residues of failed cancellations. And Section 28.12's cash-settled swaption is the case where the numeraire is not a traded asset at all, so the cancellation fails for a different reason and costs about 15% of the premium.

*What the question tests.* Identifying that the annuity appears in *both* the payoff and the numeraire, and separating what the construction proves from what it still assumes.

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**28.7** *Explain why a CMS coupon breaks the cancellation that makes Black's swaption formula exact, identify the numeraire its payoff calls for, and say what quantity the resulting adjustment consists of.*

*What breaks.* A constant-maturity swap coupon pays the ten-year swap rate on a **single date** rather than over the annuity. So the payoff is $S(T_0)$ received at $T_0$, and question 6's cancellation has nothing to cancel against: there is no $A(T_0)$ in the payoff to divide out the annuity in the numeraire.

*The numeraire the payoff calls for: a zero-coupon bond* $P(\cdot, T_0)$, or the payment date's bond generally. A single cash flow at one date is valued under the forward measure for that date, which is what makes the discounting come out of the expectation (question 5).

*And under that measure S is not a martingale.* The swap rate is a martingale under $Q^A$, because it is a traded asset divided by $A$. It is *not* a ratio of traded assets to a zero-coupon bond, so under $Q^{T_0}$ it acquires a drift. The two measures are related by (28.8), and the discrepancy is what has to be computed.

*What the adjustment consists of: a covariance.* By (28.10), changing measure introduces

$$E^{T_0}[S(T_0)] = E^A[S(T_0)] + \mathrm{Cov}^A\left(S(T_0),\ \frac{dQ^{T_0}}{dQ^A}\right)$$

and since $E^A[S(T_0)] = S(0)$ is the forward swap rate, the **CMS convexity adjustment** is exactly that covariance: the covariance between the swap rate and the ratio of the two numeraires. Section 30.5 computes, sizes, and draws it.

*What belongs here is where it comes from.* It is *not* a correction for a modeling deficiency and *not* an empirical fudge. It is the price of **quoting a rate that is a martingale under one measure in a contract that settles under another**. A desk that regards the adjustment as an approximation to be minimized has misunderstood it: the adjustment is the difference between two exactly-defined quantities, and setting it to zero prices a different contract.

*Even its sign follows from the cancellation that failed.* A swap rate that rises raises the coupon linearly, but it *lowers* the value of the annuity the coupon would have been paid over, so the states in which the payoff is large are the states in which the numeraire that was supposed to accompany it is small. The covariance is therefore positive in the direction that makes the adjustment **add** to the forward rate: a CMS coupon is worth *more* than the forward swap rate suggests. Chapter 33's question 5 pursues what that implies for an investor in a CMS-linked note.

*The general shape, which recurs throughout Part VI.* Whenever a rate is observed under one convention and paid under another, a covariance appears:

- paid *late* relative to its natural date: Chapter 30's timing adjustment;
- paid *early*, at the start of its accrual period: the in-arrears adjustment;
- settled daily rather than at maturity: the futures–forward convexity adjustment;
- paid in a *different currency*: the quanto adjustment.

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*Four names, one calculation*, and Chapter 31's question 16 traces the same covariance through three results, one of them the futures–forward adjustment.

*What the question tests.* That the annuity's absence from the payoff is the specific failure, and that a convexity adjustment is a covariance produced by a measure mismatch rather than a model error.

---

**28.8** *Explain why Girsanov changes drifts and leaves volatilities alone, and what this makes possible throughout the book.*

*The statement.* Let $dW^N$ be a Wiener process under $Q^N$, with correlation $\rho$ to the numeraire ratio $M/N$ whose volatility is $\sigma_{M/N}$. Then

$$dW^M = dW^N - \rho\,\sigma_{M/N}\,dt$$

is a Wiener process under $Q^M$. The drift shifts by the covariance with the numeraire ratio; the volatility does not move at all.

Two features of that sentence matter. It is a *new process*, not the old one relabeled; the two differ by that drift term. And only the part of $dW^N$ that *moves with the numeraire* is shifted, which is where $\rho$ comes from: a process uncorrelated with the numeraire ratio keeps the drift it had.

*Why volatilities are untouched.* Because a change of measure reweights *probabilities*, and volatility is not a probability-weighted quantity in the way a mean is; it is a property of the *paths*. Formally, the quadratic variation $\langle X\rangle_t$ is a pathwise limit of sums of squared increments, computed path by path with no reference to how likely any path is. Equivalent measures agree on which paths are possible (that is what equivalence means: they share the same null sets), so they agree on every pathwise quantity, and quadratic variation is pathwise.

The intuition worth carrying: a change of measure can make a path more or less likely, but it cannot make it smoother or rougher. Volatility is a statement about roughness.

*The mechanism in the simplest case.* Two assets under the risk-neutral measure with correlation $\rho$; elect $S_2$ as numeraire. The numeraire ratio $S_2/B$ has volatility $\sigma_2$, so the rule gives

$$\frac{dS_1}{S_1} = (r + \rho\sigma_1\sigma_2)\,dt + \sigma_1\,dW_1^{S_2}, \qquad \frac{dS_2}{S_2} = \left(r + \sigma_2^2\right)dt + \sigma_2\,dW_2^{S_2}$$

Both drifts moved and neither volatility did. The shifts differ ($dW_1$ by $\rho\sigma_2$, $dW_2$ by $\sigma_2$) because $S_2$ moves with itself perfectly and with $S_1$ only through $\rho$. The check is that $X = S_1/S_2$ is now driftless, as a traded asset over the numeraire must be.

*What this makes possible throughout the book.* Three things, in increasing order of consequence.

*1. Calibration survives the measure change.* A volatility estimated or implied under one measure is the *same* volatility under any other. This is why a model calibrated to caplet volatilities under the forward measure can be simulated under the risk-neutral measure without recalibration, and why the entire practice of quoting instruments in volatility terms is coherent across conventions.

*2. Only the drift has to be recomputed.* Every change of numeraire reduces to *one* calculation: find the covariance between the process and the log numeraire ratio. That single sentence generates Chapter 30's three adjustments and Chapter 32's two drift conditions, each derived where it is used rather than postulated.

*3. And it is why drifts are theory while volatilities are data.* Girsanov says the drift is determined by the numeraire choice, so no model is needed for it, and HJM's drift condition (Chapter 32) makes this explicit by deriving the drift entirely from the volatility structure.

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The volatility itself is invariant to the theory, so it must come from the market. The theory determines what cannot be observed and leaves what can be to the data, which is a good division and not an accident.

*What the question tests.* That quadratic variation is a pathwise quantity and equivalent measures agree on paths, and that drift-only shifting is what makes calibration transferable across measures.

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**28.9** *Explain why $N(d_1)$ and $N(d_2)$ are probabilities of the same event, name the measure each is computed under, and say why they differ.*

*The event is the same:* $\{S_T > K\}$, the option finishing in the money. What differs is the **measure** under which its probability is computed.

*The decomposition.* Split the Black–Scholes call into its two obvious pieces:

$$c = e^{-rT} E^Q\left[S_T \mathbf{1}_{\{S_T>K\}}\right] \; - \; K e^{-rT}\, Q(S_T > K)$$

*$N(d_2)$ is the risk-neutral probability.* The second piece is straightforward: under $Q$, $\ln S_T$ is normal with mean $\ln S_0 + (r - \sigma^2/2)T$, and $Q(S_T > K)$ evaluates to $N(d_2)$. So $N(d_2) = Q(S_T > K)$, computed under the measure induced by the *money-market account*.

*$N(d_1)$ is the same probability under the stock measure.* The first piece is awkward because it is the expectation of a *product*: a terminal price multiplied by an indicator depending on that same price. Done directly it requires integrating $S_T$ against a lognormal density over a half-line and completing a square. Instead, choose the numeraire that makes the product disappear.

The payoff $S_T \mathbf{1}_{\{S_T>K\}}$ is the promise to deliver *one share* if the share finishes above $K$. Value it with the **stock itself** as numeraire (admissible, since a non-dividend-paying stock is traded and strictly positive):

$$\frac{V_0}{S_0} = E^S\left[\frac{S_T \mathbf{1}_{\{S_T>K\}}}{S_T}\right] = E^S\left[\mathbf{1}_{\{S_T>K\}}\right] = Q^S(S_T > K)$$

Dividing by $S_T$ is exactly what cancels the $S_T$ in the payoff, and an expectation of a product has become a probability. So $V_0 = S_0\, Q^S(S_T > K)$, and $N(d_1) = Q^S(S_T > K)$.

*Why they differ.* Girsanov (question 8) supplies it. Moving from the money-market numeraire to the stock shifts every drift by its covariance with $\ln S$, so under $Q^S$ the stock's log-drift becomes $r + \sigma^2/2$ rather than $r - \sigma^2/2$. The variance is untouched, and the same calculation that produced $N(d_2)$ now produces

$$N(d_1), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

The entire difference between $d_1$ and $d_2$ is $\sigma\sqrt{T}$ (the drift shift $\sigma^2 T$ measured in standard deviations $\sigma\sqrt{T}$), and that is the Girsanov shift, nothing else.

*The economic reading.* $Q^S$ weights states by the stock's own value, so it *overweights* the states in which the stock is high. The event $\{S_T > K\}$ is exactly a high-stock event, so it is more likely under $Q^S$ than under $Q$, and $N(d_1) > N(d_2)$ always. This is the same monotone tilt as question 2, applied with a different numeraire.

*What this says about the formula that the usual derivation hides.* $c = S_0 N(d_1) - K e^{-rT} N(d_2)$ is not a probability-weighted average of two loosely-related quantities. It is the same event priced twice, in two different units: once in shares (worth $S_0$ each) and once in cash (discounted at $e^{-rT}$). The delta being $N(d_1)$ is then not a coincidence: holding $N(d_1)$ shares is holding the share-denominated leg, which is why the hedge ratio and the first term of the formula are the same number.

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*What the question tests.* That $d_1$ and $d_2$ differ by exactly the Girsanov shift $\sigma\sqrt{T}$, and that the two terms of the Black–Scholes–Merton formula are one event valued in two units of account.

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**28.10** *Derive Margrabe's formula by taking asset 2 as the numeraire, and explain why no interest rate appears in the result.*

*The payoff.* An exchange option confers the right to give up asset 2 and receive asset 1:

$$\max(S_1(T) - S_2(T),\ 0)$$

Two assets, two volatilities, a correlation, and—the real difficulty—*a strike that moves*. Attacked directly it is a two-dimensional problem: a partial differential equation in two state variables, or a double integral over a bivariate lognormal.

*Use asset 2 as the numeraire and the second dimension disappears.* Asset 2 is traded and strictly positive, so it is admissible:

$$\frac{V_0}{S_2(0)} = E^{S_2}\left[\frac{\max(S_1(T) - S_2(T),\ 0)}{S_2(T)}\right] = E^{S_2}[\max(X_T - 1,\ 0)]\,, \qquad X = \frac{S_1}{S_2}$$

The moving strike has become the number 1, which is what electing an asset as the unit of account buys. And $X$ is a traded asset divided by the numeraire, so it is a **martingale** under $Q^{S_2}$, with no drift to estimate.

*The volatility of X* follows from Itô's lemma applied to a ratio:

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2}$$

A driftless lognormal struck at 1 is Black's formula, and unwinding the numeraire gives **Margrabe's formula**:

$$\boxed{V_0 = S_1(0)\,N(d_1) - S_2(0)\,N(d_2)}$$

with $d_1 = \left[\ln(S_1(0)/S_2(0)) + \sigma^2 T/2\right]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$.

*Why no interest rate appears—and this is exact, not approximate.* In units of asset 2 there is nothing to finance. The numeraire *is* the funding: acquiring the option's replicating portfolio means holding units of asset 1 against units of asset 2, and no cash is borrowed or lent at any point. A rate can only enter a valuation through a cash position, and there is none.

Equivalently: the rate would have to enter through the drift of $X$, and $X$ is a martingale, so its drift is zero regardless of the rate. Both assets' drifts shift under the measure change (question 8), but they shift *together* (each by $r$ plus a covariance term), and the $r$ cancels in the ratio.

A trader who asks which curve to discount an exchange option on has misunderstood the instrument. That is a useful diagnostic: the question is not merely hard to answer, it has no referent.

*The Black–Scholes–Merton formula is the special case.* Let asset 2 be $K$ zero-coupon bonds maturing at $T$, so $S_2(0) = Ke^{-rT}$ and $\sigma_2 = 0$. Then the volatility collapses to $\sigma_1$ and the formula reads $S_1(0)N(d_1) - Ke^{-rT}N(d_2)$. An ordinary call is an option to exchange cash for stock, and the rate appears there only because *cash* is the thing being exchanged, which is exactly the right explanation of why $r$ is in the Black–Scholes–Merton formula at all.

*And the practical consequence.* Margrabe prices real instruments: spark spread options (exchange gas for power, Chapter 36), crack spread options, outperformance options, and the best-of and worst-of structures of Chapter 34. All are two-asset problems that reduce to one dimension by this device, and the reduction is exact rather than an approximation, which is rare enough among multi-asset payoffs to be worth noting.

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*What the question tests.* That promoting one asset to numeraire converts a moving strike into a fixed one and a two-dimensional problem into a one-dimensional one, and that the absence of $r$ is structural.

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**28.11** *State the two conditions a numeraire must satisfy, and give an instance in which each has failed in a real market.*

*The two conditions.* A numeraire must be

1. **traded**: its price must be the value of a *self-financing* portfolio; and
2. **strictly positive**, so that dividing by it is always permitted.

*Where each has failed.*

*Failure of self-financing: a dividend-paying stock.* The holder receives cash along the way, so the price is *not* the value of a self-financing position: value leaks out. The repair is complete and costs nothing: use the **reinvested** stock instead, whose value is self-financing by construction. Chapter 5's dividend adjustments are that repair in a different notation.

*Failure of positivity: the risky annuity in a CDS option.* An option on a credit default swap takes as its numeraire the risky annuity of Section 25.3 (the present value of one basis point paid until default or maturity), under which the forward CDS spread is a martingale. But that annuity is worth nothing in default, so the measure it defines assigns probability zero to an event that $Q$ does not, and the two are therefore **not equivalent**. Single-name convention steps around this by making the option knock out at default. *Index options cannot*, because they are written not to cancel: the protection buyer is owed the losses accumulated before expiration (**front-end protection**), so the correction matters most in precisely the state that destroys the numeraire.

*A third failure, worth naming because it costs money daily.* The **cash-settled swaption convention** values the contract with $G(S) = \sum_{i=1}^{mn} m^{-1}(1 + S/m)^{-i}$ for an $n$-year swap paying $m$ times a year, a formula applied to the swap rate rather than a portfolio of bonds. It is not a traded asset at all, so it does not cancel against the numeraire the way question 6 required. The cash annuity falls as rates rise (on a ten-year annual-pay swap, 8.98 at a 2% swap rate to 7.36 at 6%), so it is smallest exactly where a payer swaption pays most, and that covariance survives. On an at-the-money five-year option on a ten-year swap, at a 3% forward rate and 25% volatility, it is worth about **9.9 basis points, or 15% of the premium**, widening to 28% at a volatility of 40% because it is a convexity term. A physically settled swaption has a price once the volatility is given; a cash-settled one has a price once the volatility *and* a model of the annuity are given, and a convention reading as a settlement detail has moved the contract between those two categories.

*And a fourth, which violates neither condition and is the most consequential.* The risk-free bank account is traded and strictly positive, and it is **not the numeraire for a collateralized trade**. Nothing in the theorem is violated; the *wrong admissible asset* has been chosen, which is why this case is the expensive one.

It closes a loop opened in Part I. Chapter 7 argued that a collateral agreement determines the discount curve, and Chapter 8 that the terms of the CSA change the value of a trade. In this chapter's language those are statements about *which numeraire is correct*, and question 12 pursues them.

*The general lesson.* The question "what is the numeraire" is not abstract; it is the question of what portfolio actually replicates the trade, and every time the market's answer changed, to the collateral account after 2008, to a choice among collateral currencies, to the firm's own funding where no collateral is posted, a valuation adjustment appeared.

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*What the question tests.* The two conditions, an instance of each failing in a traded market, and that choosing an admissible but wrong numeraire is a separate and larger error than violating a condition.

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**28.12** *Explain why the discount curve of Chapter 7, ColVA, and FVA are all statements about the choice of numeraire, and why that made the FVA debate unresolvable by data.*

*Why all three are statements about the numeraire.* A price is $V(0) = N(0)E^N[V(T)/N(T)]$, so specifying the numeraire specifies the discounting. The three debates are three arguments about which traded asset the denominator should be.

*The discount curve of Chapter 7.* A fully collateralized trade under a cash CSA is replicated by a portfolio funded at the collateral rate, so the numeraire is the **collateral account**, not the risk-free bank account. That is why OIS (and now SOFR) discounting replaced LIBOR discounting. The change was not a refinement of an estimate; it was the recognition that the previous numeraire did not correspond to the portfolio that actually replicates the trade.

*ColVA* is what remains when the CSA permits choices: multiple eligible currencies, a range of eligible securities, thresholds. The posting party holds an *option* over which asset to deliver, so the effective numeraire is the cheapest-to-deliver collateral rather than any single account. ColVA prices that optionality: a numeraire that is itself chosen dynamically.

*FVA* is the case where the trade is **not** collateralized. There is no collateral account to serve as numeraire, and the replicating portfolio must be funded at the dealer's own borrowing rate. FVA asks whether the dealer's funding cost belongs in the price, which is the question of whether the dealer's own funding account is the correct numeraire.

*Why the FVA debate was unresolvable by data.* Because the disagreement was never about a *measurement*. Both sides agreed on every observable: the trade's cash flows, the dealer's funding spread, the absence of collateral. They disagreed about *which portfolio replicates the trade*, and that is a question about the theoretical setup rather than about the world.

Two coherent positions, and neither is refutable by a price:

- *FVA is not a price.* Funding cost is a property of the *dealer*, not of the *trade*. Two dealers with different credit quality would value the same contract differently, which violates the law of one price; and the dealer's funding spread compensates its own creditors for its own default risk, so charging it to a client double-counts against DVA. On this view the correct numeraire is risk-free and FVA is an accounting transfer between the desk and the treasury.

- *FVA is a cost.* The desk must genuinely raise cash at its actual borrowing rate to run the hedge. A valuation ignoring that reports profits the firm does not have. On this view the funding account *is* the replicating portfolio and belongs in the numeraire.

*No experiment distinguishes them*, because the quantity in dispute is not observed anywhere. There is no traded instrument whose price is FVA, so no market quote adjudicates. The debate was settled *institutionally* (by accounting standards and by large firms taking the charge) rather than empirically, and the announcements were of policy rather than of discovery.

*The general point worth extracting.* Disputes about valuation adjustments are usually disputes about the numeraire, and disputes about the numeraire are disputes about *what portfolio replicates the trade*. Those are settled by argument about the setup, not by more data, which is why they persist for years among competent people, and why the resolution when it comes is a convention rather than a proof.

*What the question tests.* Recognizing three separately-named adjustments as one structural question, and why a disagreement about a modeling premise cannot be closed by observation.

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**28.13** *Using* [TN28]*, state both fundamental theorems, and explain why stochastic volatility, jumps, and credit each produce incompleteness for a different specific reason.*

*The two theorems*, as [TN28] states and proves them.

*First fundamental theorem.* A market is free of arbitrage if and only if there exists an **equivalent martingale measure**: a probability measure equivalent to the real-world one under which every traded asset, discounted by the numeraire, is a martingale. (In continuous time the precise statement replaces "no arbitrage" with no free lunch with vanishing risk and "martingale" with *local martingale*, for technical reasons the note sets out.)

*Second fundamental theorem.* An arbitrage-free market is **complete** (every contingent claim is replicable by a self-financing trading strategy) if and only if the equivalent martingale measure is **unique**.

*Why they are a diagnostic rather than a reassurance.* Together they say: a price depends on a model if and only if the market is incomplete. That is genuinely predictive. Before pricing anything, one can ask whether the payoff is spanned by traded instruments; if it is, the price is forced and any two correct models agree; if it is not, the price lies in an interval and a modeling choice picks a point in it. Chapter 21's barrier options differing by 49.5% under two models calibrated to the same vanillas is the second case made concrete.

*Why each of the three produces incompleteness, and for a different reason.*

*Stochastic volatility: the hedging instrument does not exist.* Volatility is not traded. The model has two sources of uncertainty, the price and the variance, and only one traded asset to hedge with, so the loadings matrix of Section 28.3 is not square and cannot be inverted. There is no portfolio of the stock and cash whose value tracks a change in $v$. The practical form of this is Section 21.10's parameter identification problem: along the flat directions of the calibration objective, several parameter sets fit the same surface and imply different prices for anything not in it.

*Jumps: the instruments could not span it even if they existed.* A jump of unknown size cannot be hedged by *any* finite set of instruments, because the number of states reachable in an instant exceeds the number of assets available to span them. A diffusion moves to one of two neighboring states in an instant and one asset suffices; a jump can land anywhere on a continuum. So this is a stronger failure than the first: with stochastic volatility one could complete the market by *trading* an option on volatility, and no finite addition completes a jump market.

*Credit: the parameters are not separately identified.* This is Section 24.4's problem: recovery and default probability enter a CDS spread only through the product $\lambda(1 - R)$, so spreads determine one number where two are needed. The market does not determine what the model must supply, and anything depending on $\lambda$ alone (a digital default swap, a first-to-default basket) is priced by an assumption rather than by a quote.

The three are worth distinguishing because the responses differ. For stochastic volatility, add a traded instrument: calibrate to the smile and the market completes itself in the vanilla dimension. For jumps, accept the residual and charge for it; the unhedgeable jump risk is what the volatility risk premium of Section 19.10 compensates. For credit, obtain the missing parameter from outside the price: recovery studies, seniority, jurisdiction.

*And the honest summary.* When a price depends on a model, the theorems say the dependence is structural rather than a failure of effort. No better calibration removes it. The right response is to report a range (Chapter 21's question 10) rather than to keep refining a number the market never determined.

*What the question tests.* Both theorems stated precisely, and that "incomplete" names three structurally different failures with three different remedies.

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# Chapter 29
# Standard Market Models for Rate Derivatives

**29.1** *Explain why Black's model can be applied to a caplet without any adjustment for the drift of the forward rate. Which discount factor must be used, and why that one?*

*Why no drift adjustment is needed.* Black's model assumes the underlying is lognormally distributed at expiration *with no drift*: the forward price is its own expectation. Applied naively to a forward interest rate that would be an assumption; here it is a theorem, provided the right measure is used.

Chapter 28 supplies the argument. A forward rate $F_i$ for the period $[T_i, T_{i+1}]$ satisfies

$$F_i(t) = \frac{1}{\tau_i}\left[\frac{P(t,T_i)}{P(t,T_{i+1})} - 1\right]$$

so $\tau_i F_i P(t,T_{i+1}) = P(t,T_i) - P(t,T_{i+1})$ is the value of a *traded* portfolio (long one bond, short another). Divided by $P(t,T_{i+1})$, it is a traded asset over a numeraire, hence a **martingale** under the $T_{i+1}$-forward measure. Its drift is zero by construction, not by approximation.

*Which discount factor: $P(0,T_{i+1})$, the **payment** date.* Formula (29.7) uses it, and it is not the expiration date $T_i$.

*Why that one.* Because it is exactly the numeraire under which $F_i$ is a martingale. The two choices are not independent: picking the discount factor *is* picking the measure, and picking the measure is what removes the drift. Choosing $P(0,T_{i+1})$ and assuming no drift is one coherent calculation; choosing $P(0,T_i)$ and assuming no drift is two incompatible halves.

The formal version is Section 28.6 and question 5 of Chapter 28. With $N = P(\cdot,T_{i+1})$ the terminal numeraire value is $N(T_{i+1}) = 1$, so

$$V_0 = P(0,T_{i+1})\,E^{T_{i+1}}[\text{payoff}]$$

and the discounting comes *out* of the expectation entirely. No covariance between the payoff and the discount factor can survive.

*And that covariance is not negligible.* Example 31.5 measures it: computing the same caplet's expected payoff under the risk-neutral measure and discounting at today's zero rate, treating payoff and discount factor as independent, overstates the value by **2.52%**. A caplet pays when rates are high, and when rates are high the discount factor is small, so the covariance is negative and ignoring it flatters the price.

So Black's caplet formula is exact rather than approximate, in the specific sense that no error is introduced by the interest-rate structure. What remains an assumption is the *lognormality* of $F_i$ under $Q^{T_{i+1}}$, which the smile shows to be false, and which Section 29.8's shifted and normal conventions address. That is a statement about the distribution, not about the drift or the discounting.

*A practical rider.* This is why a cap can be valued caplet by caplet and summed: each caplet lives under its *own* forward measure, with its own numeraire and its own martingale forward

rate. The caplets are independent options, so no consistency between the measures is required to add up their prices, which is a convenience the swaption of question 7 does not share.

*What the question tests.* That the drift vanishes because of the numeraire rather than by assumption, and that the discount factor and the measure are one choice made once.

---

**29.2** *A caplet covers a three-month period beginning in two years, on a notional of \$20 million. The forward rate is 3.60%, the strike is 3.75%, the volatility is 18%, and the discount factor to the payment date is 0.9280. Value the caplet, treating the rate as fixed in advance.*

*Setup.* $L = \$20$ million, $\tau = 0.25$, $F = 3.60\%$, $K = 3.75\%$, $\sigma = 18\%$, $P(0, T_2) = 0.9280$. The rate is fixed in advance, so the option expires at $T = 2$ and

$$\sigma\sqrt{T} = 0.18\sqrt{2} = 0.254558$$

*The arguments.*

$$d_1 = \frac{\ln(0.0360/0.0375) + \frac{1}{2} \times 0.18^2 \times 2}{0.254558} = \frac{-0.040822 + 0.032400}{0.254558} = \mathbf{-0.0331}$$

$$d_2 = d_1 - 0.254558 = \mathbf{-0.2876}$$

so $N(d_1) = 0.48680$ and $N(d_2) = 0.38681$.

*The value*, from (29.7):

$$\begin{aligned} \text{caplet} &= L\,\tau\,P(0,T_2)\,[F\,N(d_1) - K\,N(d_2)] \\ &= 20{,}000{,}000 \times 0.25 \times 0.9280\,[0.0360 \times 0.48680 - 0.0375 \times 0.38681] \\ &= 4{,}640{,}000 \times (0.017525 - 0.014505) \\ &= 4{,}640{,}000 \times 0.0030195 = \mathbf{\$14{,}011} \end{aligned}$$

Two checks are worth running on an answer of this shape.

*The option is out of the money and should be worth less than an at-the-money one.* The forward is 3.60% against a 3.75% strike (15 basis points out), and $N(d_1)$ and $N(d_2)$ both sit below 0.5, confirming it. An at-the-money caplet on the same terms would be worth about \$16,918, so the moneyness costs roughly 17%.

*And the magnitude is right for the notional.* The caplet is worth 7.0 basis points of notional (14,011/20,000,000), or 28 basis points of the quarterly accrued amount. That is the order of magnitude a three-month caplet two years out should carry at these levels, and a figure ten times larger or smaller would indicate a units error, most commonly forgetting $\tau$, which alone would quadruple the answer.

Note which time enters where. The volatility is scaled by $\sqrt{T} = \sqrt{2}$, the expiration; the discount factor is $P(0, 2.25) = 0.9280$, the *payment* date; and the accrual factor $\tau = 0.25$ scales the payoff. Three different roles for three different periods, and question 1 explains why the discounting must use the third rather than the first.

*What the question tests.* Correct substitution into Black's caplet formula, and keeping expiration, payment date and accrual period distinct.

---

**29.3** *Repeat Question 29.2 treating the rate as compounded in arrears. By what percentage does the value increase?*

*What changes.* Treating the rate as compounded in arrears makes it a **backward-looking** caplet. Uncertainty does not stop accumulating at $T_1 = 2$: the compounding runs across

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$[T_1, T_2]$ and each day's fixing is locked in as it is observed, so the remaining sensitivity declines, approximately linearly, to zero at $T_2$. By (29.10) the total variance is

$$\text{Var} = \sigma^2 \left(T_1 + \frac{\tau}{3}\right) = (0.18)^2 \left(2 + \frac{0.25}{3}\right) = 0.0324 \times 2.08333 = 0.067500$$

so the effective $\sigma\sqrt{T}$ rises from 0.254558 to

$$\sqrt{0.067500} = 0.259808$$

*Revaluing.*

$$d_1 = \frac{-0.040822 + \frac{1}{2} \times 0.067500}{0.259808} = \mathbf{-0.0272}, \qquad d_2 = -0.0272 - 0.259808 = \mathbf{-0.2870}$$

$$\text{caplet} = 4{,}640{,}000\,[0.0360\,N(-0.0272) - 0.0375\,N(-0.2870)] = \$\mathbf{14{,}360}$$

*The increase.*

$$\frac{14{,}360}{14{,}011} - 1 = \mathbf{+2.50}\%$$

*Why it is an increase, and why it is small.* The backward-looking caplet is worth *more* because it is exposed to rate movements for longer: the accrual period contributes a further $\tau/3$ years of variance that the forward-looking contract does not see. More variance is more value for an option, unambiguously.

The magnitude follows from the ratio inside the square root:

$$\sqrt{\frac{T_1 + \tau/3}{T_1}} = \sqrt{\frac{2.08333}{2}} = 1.0206$$

a 2.1% increase in effective volatility, which translates into 2.5% of premium because the option is slightly out of the money, and an out-of-the-money option's premium rises proportionally faster than its volatility. The extra variance is $\tau/3$ of a year against $T_1$ of two years (one part in twenty-four), so the effect is bound to be small at this maturity.

*And the $\tau/3$ is worth understanding rather than memorizing.* If the whole period's rate were determined at $T_2$, the extra variance would be $\tau$. If it were determined at $T_1$, it would be zero. Because the rate is an *average* whose sensitivity decays linearly from 1 to 0 across the period, the accumulated variance is $\int_0^\tau (s/\tau)^2 ds = \tau/3$. The same decaying delta is what Section 30.3 describes for averaging futures, and the same arithmetic gives the limiting variance ratio of $1/3$ in Chapter 33's question 9.

*Where it matters and where it does not.* At two years with a quarterly period the adjustment is 2.5%: real, but comfortably inside a bid–ask on a single caplet. It becomes material when $\tau$ is large relative to $T_1$: for the **first** caplet of a spot-starting cap, $T_1 = 0$ and the entire variance is $\sigma^2\tau/3$, so the contract has genuine optionality where its forward-looking predecessor had none. That is question 6.

*What the question tests.* Applying $\sigma^2(T_1 + \tau/3)$, and recognizing that the correction is second-order except when the accrual period is a large fraction of the option's life.

---

**29.4** *State cap–floor parity and prove it from the payoffs. Explain how you would use it to check a set of broker quotes.*

*The statement.* For a cap and a floor on the same schedule, notional and strike $K$,

$$\boxed{\text{cap} - \text{floor} = \text{value of a payer swap struck at } K}$$

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*The proof from the payoffs.* Period by period, the cap pays $L\tau_i \max(R_i - K, 0)$ and the floor pays $L\tau_i \max(K - R_i, 0)$. Their difference is

$$L\tau_i \left[\max(R_i - K, 0) - \max(K - R_i, 0)\right] = L\tau_i \left(R_i - K\right)$$

which holds for every $R_i$, above or below $K$, by the identity $\max(x, 0) - \max(-x, 0) = x$. And $L\tau_i(R_i - K)$ is exactly the cash flow of a swap that *receives* floating and *pays* fixed at $K$ over that period. Summing over periods gives a payer swap.

This is put–call parity in interest-rate form: long a call and short a put at the same strike is a forward. It requires no model, no distribution, and no volatility—only that the two instruments reference the same rate on the same dates.

*Two corollaries follow immediately.* At the money, where $K$ equals the par swap rate, the swap is worth zero and **a cap and floor at the par rate have equal value**. And below the par rate the cap is worth more, above it the floor is, so parity fixes the *difference* of the two quotes while the market only has to determine one of them.

*How to check broker quotes with it.*

*Step 1: build the swap value independently.* From the discount curve, compute the par swap rate for the schedule and hence

$$\text{payer swap value} = A(0)\left(S_{\text{par}} - K\right)$$

where $A(0)$ is the annuity. This uses only the curve: no volatility, no smile, no model.

*Step 2: difference the quotes.* Take the quoted cap and floor prices at the same strike and subtract.

*Step 3: compare.* The two must agree to within bid–ask. If they do not, something is wrong, and the discrepancy localizes the fault:

- A gap that is *constant across strikes* points at the curve: a discounting convention mismatch, or a stale fixing.
- A gap that *varies with strike* points at the option quotes: an inconsistent smile between the cap and floor surfaces, most often because they were marked by different desks or interpolated differently in the wings.
- A gap on *one* strike alone is likely a data error in that quote.

*Why this test is worth running rather than assuming.* It is one of very few checks in the rate option market that is **model-free**. Any comparison of two caps requires a view on volatility; this one does not, so a failure is unambiguously an error rather than a disagreement. It is the same discipline as Chapter 11's put–call parity check on equity chains and Chapter 19's arbitrage test on a smile: run the checks that cannot fail on good data before fitting anything.

*One practical caution follows.* The identity holds only if the two instruments genuinely share every term: schedule, day count, fixing convention, and in particular whether the first caplet is included (question 6). Two quotes that differ on that convention will fail the parity check for a reason that has nothing to do with pricing.

*What the question tests.* The $\max(x, 0) - \max(-x, 0) = x$ identity, and using a model-free relation as a data-quality test with a diagnostic reading.

---

**29.5** *Explain the difference between a flat volatility and a spot volatility. Why is it wrong to compare the flat volatilities quoted for a three-year and a seven-year cap?*

*The distinction.*

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A **flat volatility** is the single volatility which, applied to *every* caplet in a given cap, reproduces that cap's market price. It is a *quotation device*, a way of expressing one price as one number, and it is not a property of any individual caplet.

A **spot volatility** (also called a caplet or forward volatility) is the volatility of *one specific* caplet: a genuine property of one option on one forward rate.

A flat volatility is therefore a kind of average of the spot volatilities of the caplets making up that cap, and the two coincide only for the shortest cap, the one containing a single caplet.

*Why comparing a three-year and a seven-year flat volatility is wrong.* **They average over different sets of caplets.**

- The three-year flat volatility summarizes caplets 1 through 3.
- The seven-year flat volatility summarizes caplets 1 through 7, including the same first three.

The two numbers overlap in their inputs, so their difference is not the volatility of anything. It is not "the volatility of years four to seven"; it is the change in an average caused by extending its window, which mixes the new caplets with a re-weighting of the old ones.

The consequence is a shape that misleads. Because a flat volatility summarizes every caplet up to its maturity, its curve peaks later, peaks lower, and decays more slowly than the spot curve underneath it, exactly as a moving average lags and smooths a series. A trader reading the flat curve and concluding that volatility peaks at, say, four years is reading an artifact of the averaging: the underlying spot curve peaks earlier and higher.

*What to compare instead: spot volatilities*, obtained by **bootstrap**, exactly parallel to the zero-curve bootstrap of Chapter 4, and for the identical reason. There, par swap rates are averages over the curve and forward rates are what one actually models; here, flat volatilities are averages over caplets and spot volatilities are what one actually models.

The procedure: the price of a two-year cap less the price of a one-year cap is the value of the caplets falling in the second year, and those caplets' spot volatilities are whatever reproduces that difference, given the spot volatilities already extracted for year one. Work outward, one maturity at a time.

*Two practical points.* The bootstrap is done on *prices*, never on volatilities: subtracting flat volatilities is meaningless, subtracting cap prices is arithmetic. And the extracted spot volatilities can be noisy or even non-monotone where the flat quotes are illiquid, because differencing amplifies quote error; a smooth flat curve can imply a jagged spot curve, which is a data problem rather than a market signal.

*The general lesson, which recurs.* An average quoted as a single number is not a property of any component. A par swap rate is not a forward rate (Chapter 4); a five-year CDS spread is not the hazard rate for years three to five (Chapter 24's question 4); a flat volatility is not a caplet's volatility. In every case the repair is a bootstrap, and in every case the error from skipping it grows with the slope of the underlying curve.

*What the question tests.* That a flat volatility is an average over an interval determined by the cap's maturity, and that overlapping averages cannot be differenced to recover a component.

---

**29.6** *Explain why the first caplet of a spot-starting cap was conventionally excluded under LIBOR, and why that convention is no longer appropriate.*

*Why it was excluded under LIBOR.* Because it was not an option. Under LIBOR the reference rate for each accrual period is **set in advance**, fixed at the start of the period, two business days before. For a *spot-starting* cap the first period begins immediately, so its rate is already known at inception.

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A caplet on a rate that is already fixed has no optionality left. Its payoff $L\tau\max(R_1 - K, 0)$ is a **known cash flow**, not a contingent one: either the fixing exceeds the strike, in which case the caplet is worth a discounted certainty, or it does not, in which case it is worth nothing. There is no volatility to price and nothing to hedge.

So the market convention excluded it, and the convention was correct. Including a deterministic cash flow in an instrument quoted by its volatility would corrupt the quote—it would add value that responds to no volatility, so the implied flat volatility backed out of the price would be biased, and biased by an amount depending on where the fixing happened to land relative to the strike. A cap struck just below the current fixing and one struck just above would appear to have very different volatilities on the same day.

*Why the convention is no longer appropriate.* Because Chapter 4 replaced the reference rate with a **compounded risk-free rate**, which is not known until the period *ends*. The first period's rate now accrues day by day across $[0, \tau]$ and is determined only at $\tau$.

So the first caplet of a spot-starting cap now has **genuine optionality**. Question 3's formula gives its variance directly: with $T_1 = 0$,

$$\text{Var} = \sigma^2\left(T_1 + \frac{\tau}{3}\right) = \frac{\sigma^2\tau}{3}$$

This is not zero. For a quarterly period at a 20% volatility that is an effective $\sigma\sqrt{T} = 0.20\sqrt{0.25/3} = 0.0577$—small, but a real option with real vega, and one whose exclusion would now be an error rather than a convention.

*What follows in practice.*

*The convention had to be restated, not merely reinterpreted.* A cap quoted under the old convention and one quoted under the new contain different numbers of options, so their prices are not comparable and their flat volatilities are quoted on different instruments. Documentation must say which applies. This is the same species of problem as Chapter 25's restructuring clause: a term that reads as a detail and changes what is being priced.

*And the historical time series is not continuous.* Flat volatilities quoted before and after the transition summarize different portfolios of caplets, so a chart spanning the change contains a break that is a convention artifact rather than a market move. Anyone estimating volatility dynamics across the transition must adjust for it.

*The wider point.* A market convention encodes a fact about the underlying instrument, and when the instrument changes the convention becomes a bug. The first caplet's exclusion was a correct response to advance fixing; it survived the disappearance of advance fixing purely by inertia, and several such conventions did.

*What the question tests.* That the exclusion rested on the rate being known, and that backward-looking compounding restores optionality with variance $\sigma^2\tau/3$.

---

**29.7** *A five-year option on a three-year annual-pay swap has a forward swap rate of 3.80%, a strike of 3.80%, and a volatility of 22%. The annuity is 2.6540 per unit of notional. Value the payer swaption on a notional of \$50 million, and state the value of the corresponding receiver.*

*Setup.* Notional \$50 million, forward swap rate $F = 3.80\%$, strike $K = 3.80\%$ (at the money), $\sigma = 22\%$, expiration $T = 5$ years, annuity $A = 2.6540$ per unit of notional.

$$\sigma\sqrt{T} = 0.22\sqrt{5} = 0.491935$$

*The arguments.* At the money the two arguments are symmetric about zero:

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = \frac{\sigma\sqrt{T}}{2} = \mathbf{0.245967}, \qquad d_2 = -d_1 = \mathbf{-0.245967}$$

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so $N(d_1) = 0.597146$ and $N(d_2) = 0.402854$.

*The payer swaption.*

$$\begin{aligned} V &= L \cdot A \cdot [F\,N(d_1) - K\,N(d_2)] \\ &= 50{,}000{,}000 \times 2.6540 \times 0.0380 \times (0.597146 - 0.402854) \\ &= 5{,}042{,}600 \times 0.194293 = \$\mathbf{979{,}740} \end{aligned}$$

*The receiver swaption:* ***also $979,740***.

*Why.* Payer–receiver parity is the swap analog of put–call parity:

$$\text{payer} - \text{receiver} = L \cdot A \cdot (F - K)$$

because being long a payer and short a receiver at the same strike replicates a forward-starting payer swap, whose value is the annuity times the rate difference. At the money $F = K$, the right-hand side is zero, and the two options are worth exactly the same. No calculation is needed for the second number once the first is known.

That identity is worth having for the same reason as question 4's cap–floor parity: it is model-free, so a broker quoting an at-the-money payer and receiver at materially different prices has a data error rather than a view.

*Two checks confirm the magnitude.* The premium is 1.96% of notional, or, more usefully, 73.8 basis points of the annuity ($979{,}740/(50\text{m} \times 2.6540)$), which is the natural unit for comparing swaptions across tenors. And the at-the-money approximation of question 9 gives

$$L\,A\,F\,\sigma\sqrt{\frac{T}{2\pi}} = 5{,}042{,}600 \times 0.22\sqrt{5/(2\pi)} = \$989{,}629$$

within 1% of the exact figure, which is the accuracy that approximation should have at this $\sigma\sqrt{T}$.

*Where the annuity comes from and why it appears.* $A = 2.6540$ is $\sum_i \tau_i P(0, T_i)$ over the three annual payment dates of the underlying swap: the value of receiving one unit of fixed. Chapter 28's question 6 explains why it multiplies the whole formula: the swaption's payoff *contains* the annuity, using $A$ as numeraire cancels it, and $A(0)$ reappears when the numeraire is unwound. It is not a discount factor, and treating it as one (by discounting at $P(0, 5)$ instead) would be wrong by a factor of more than three.

*What the question tests.* Black's swaption formula with the annuity in its correct role, and payer–receiver parity as a shortcut rather than a second calculation.

---

**29.8** *Show that at the money the Bachelier formula reduces to $\sigma_N\sqrt{T/2\pi}$ per unit of numeraire.*

*The Bachelier (normal) formula.* With the underlying assumed normally distributed rather than lognormally, per unit of numeraire the value of a payer option is

$$V = (F - K)\,N(d) + \sigma_N\sqrt{T}\,\varphi(d), \qquad d = \frac{F - K}{\sigma_N\sqrt{T}}$$

where $\varphi(\cdot)$ is the standard normal density and $\sigma_N$ is the normal (absolute) volatility, quoted in basis points per year rather than as a percentage of the rate.

*At the money.* Set $F = K$. Then

$$d = \frac{0}{\sigma_N\sqrt{T}} = 0$$

and the two terms simplify:

- The first term vanishes outright, since $F - K = 0$, whatever $N(0) = 1/2$ multiplies it.

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- The second term becomes $\sigma_N \sqrt{T}\, \varphi(0)$, and $\varphi(0) = 1/\sqrt{2\pi}$.

Hence

$$V = \sigma_N \sqrt{T} \cdot \frac{1}{\sqrt{2\pi}} = \boxed{\sigma_N \sqrt{\frac{T}{2\pi}}}$$

*Three things this compact form is good for.*

*It is an exact result, not an approximation.* Unlike the lognormal at-the-money shortcut of question 9, no expansion was used—the normal model's at-the-money value *is* this expression. That makes it the natural reference point for the rate market, and one reason the normal convention survived the move to negative rates so comfortably.

*It gives an immediate mental price.* Since $1/\sqrt{2\pi} = 0.3989$, the at-the-money premium is almost exactly $\mathbf{0.4} \times \sigma_N \sqrt{T}$. A one-year at-the-money option at a normal volatility of 80 basis points is worth about 32 basis points of the numeraire; a four-year one about 64. Doubling the maturity multiplies the premium by $\sqrt{2}$, not by 2.

*And it inverts trivially.* Given an at-the-money price, the implied normal volatility is

$$\sigma_N = \frac{V}{\sqrt{T/2\pi}} = V\sqrt{\frac{2\pi}{T}}$$

with no root-finding at all. Implying a lognormal volatility requires a numerical solve; implying a normal one at the money requires a division. That is a genuine operational advantage on a surface with hundreds of at-the-money points to mark daily.

*One caution attaches to units.* Because $\sigma_N$ is absolute rather than proportional, the premium carries the units of the rate, so if $\sigma_N$ is 80 basis points, $V$ is in basis points *of the numeraire*, and for a swaption the numeraire is the annuity rather than a discount factor (question 7). Multiplying by a discount factor instead of the annuity is the usual error and is worth about a factor of three at question 7's three-year tenor.

*What the question tests.* That $F = K$ kills the first term and $\varphi(0) = 1/\sqrt{2\pi}$ delivers the rest, and that the resulting form is exact and directly invertible.

---

**29.9** *An at-the-money swaption is quoted at a lognormal volatility of 24% with a forward swap rate of 3.20%. Estimate the equivalent normal volatility, and explain why the rule of thumb works.*

*The estimate.* The rule of thumb is

$$\sigma_N \approx F \times \sigma_{\text{LN}}$$

so

$$\sigma_N \approx 0.0320 \times 0.24 = 0.00768 = \mathbf{76.8} \text{ basis points}$$

*Why the rule works.* Both models have simple at-the-money forms, and equating them gives the conversion.

*The normal side* is question 8's exact result: at the money the Bachelier value is $\sigma_N \sqrt{T/2\pi}$ per unit of numeraire.

*The lognormal side* needs one expansion. Black's at-the-money value is $F\,[N(d_1) - N(d_2)]$ with $d_1 = \sigma_{\text{LN}}\sqrt{T}/2$ and $d_2 = -d_1$. For small $x$ the symmetric difference is linear in $x$ with slope $2\varphi(0)$:

$$N(x) - N(-x) \approx 2\,\varphi(0)\,x = \frac{2x}{\sqrt{2\pi}}$$

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Putting $x = \sigma_{\text{LN}}\sqrt{T}/2$ gives $N(d_1) - N(d_2) \approx \sigma_{\text{LN}}\sqrt{T/2\pi}$, so

$$\text{Black}_{\text{ATM}} \approx F\,\sigma_{\text{LN}}\sqrt{\frac{T}{2\pi}}$$

*Equating the two prices* gives the conversion immediately:

$$\sigma_N\sqrt{\frac{T}{2\pi}} = F\,\sigma_{\text{LN}}\sqrt{\frac{T}{2\pi}} \quad\Longrightarrow\quad \sigma_N = F\,\sigma_{\text{LN}}$$

The $\sqrt{T/2\pi}$ cancels, so the conversion does not depend on maturity, which is why the rule can be applied across a whole surface without adjustment.

*The economic reading is simpler still.* A lognormal volatility is a *proportional* move and a normal volatility is an *absolute* one. A 24% proportional move in a 3.20% rate is $0.24 \times 3.20\% = 0.768\%$ in absolute terms. The formula is just that sentence.

*How good is it.* Very good at the money, and it degrades slowly with maturity because the expansion's error grows with $\sigma_{\text{LN}}\sqrt{T}$:

| Expiration | Exact equivalent $\sigma_N$ | Rule |
|---|---|---|
| 1 year | 76.6 bp | 76.8 bp |
| 5 years | 75.9 bp | 76.8 bp |
| 10 years | 75.0 bp | 76.8 bp |

Within 0.3% at one year and 2.4% at ten. The rule always overstates slightly, because the lognormal distribution's positive skew makes the exact at-the-money value a little below the linear expansion.

*Where it fails, and this matters.* The derivation used $F = K$ throughout, so the rule is an *at-the-money* conversion only. Away from the money the two models imply genuinely different distributions (lognormal cannot go below zero, normal can), so their smiles are not related by any constant, and converting a whole smile by multiplying through by $F$ is wrong. Section 29.8's shifted lognormal exists precisely to interpolate between the two families, and the volatility cube of Section 29.9 is quoted in whichever convention the market has settled on for that currency.

*And the reason the market moved to normal quoting at all.* A lognormal volatility is undefined when $F \le 0$, since $\ln(F/K)$ does not exist. Negative rates after 2014 broke the convention rather than the mathematics, and normal volatilities, which are indifferent to the sign of the rate, were already the natural quote for a market where absolute moves are what traders think in.

*What the question tests.* That both models share a $\sqrt{T/2\pi}$ at-the-money form so the maturity cancels, and that the conversion is exactly the proportional-to-absolute translation.

---

**29.10** *Explain why the volatility of a swap rate generally declines as the tenor of the underlying swap increases.*

*The observation.* On a volatility cube, at a fixed option expiration, implied swap-rate volatility falls as the tenor of the underlying swap lengthens: a one-year swap rate is more volatile than a ten-year swap rate, which is more volatile than a thirty-year one.

*The main reason: averaging under imperfect correlation.* A swap rate is approximately a weighted average of the forward rates spanning the swap. Question 12's relation makes the consequence precise: with $n$ forwards each of volatility $\sigma$ and uniform pairwise correlation $\rho$,

$$\sigma_{\text{swap}} = \sigma\sqrt{\rho + \frac{1-\rho}{n}}$$

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which is *decreasing in n* for every $\rho < 1$. A longer swap averages more forward rates, and the more rates are averaged the more the idiosyncratic component of each is diversified away.

The limit is instructive: as $n \to \infty$ the volatility tends to $\sigma\sqrt{\rho}$, not to zero. Averaging removes the *uncorrelated* part of each forward's movement and leaves the common part untouched, so the decline flattens out. That is why the tenor dimension of a volatility cube falls steeply from one to five years and then flattens, exactly the shape $\sqrt{\rho + (1-\rho)/n}$ predicts.

*Two reinforcing effects operate.*

*Forward-rate volatility itself declines with maturity.* The forwards entering a thirty-year swap include very long-dated rates, and Chapter 31's $\sigma e^{-aT}$ shows why those move less: mean reversion damps the response of distant forwards to a shock. So a long swap averages rates that are individually less volatile as well as imperfectly correlated.

*And correlation decays with separation.* The uniform-$\rho$ assumption is a simplification; in practice the correlation between the two-year and three-year forwards is much higher than between the two-year and twenty-five-year. A longer swap therefore spans a wider range of *decorrelated* rates, so the effective $\rho$ in the formula is itself lower for a longer tenor, compounding the decline rather than merely extending it.

*What this is not.* It is *not* that long rates are intrinsically calmer in some economic sense, and it is not a statement about the level of rates. It is a portfolio-variance effect: the swap rate is a basket, and baskets are less volatile than their components unless the components move in lockstep. Chapter 19's question 9 is the identical arithmetic in the equity market, where an index's implied volatility sits below its constituents' for exactly this reason.

*The practical consequences.*

A model calibrated to short-tenor swaptions will **overstate** long-tenor volatilities if it assumes perfect correlation, which is Chapter 31's one-factor defect, and question 12's warning about calibrating to caps alone.

And the tenor dimension of the cube is *information about correlation*, not merely a third axis to interpolate along. Reading $\sigma_{\text{swap}}(n)$ across tenors and inverting the formula recovers an implied average correlation, which is the natural sanity check on any multi-factor calibration (Chapter 32).

*What the question tests.* That a swap rate is a basket and its volatility obeys portfolio arithmetic, and that the decline flattens toward $\sigma\sqrt{\rho}$ rather than continuing to zero.

---

**29.11** *Explain, without algebra, why a cap must be worth at least as much as a swaption written on the same schedule and strike. Under what condition would the two be equal?*

*The two instruments.* On the same schedule and strike:

- A **cap** pays $\sum_i \tau_i \max(F_i - K, 0)$. It is a **portfolio of options**, one on each forward rate, each exercised independently.
- A **payer swaption** pays $A \max(S - K, 0)$, and since the swap rate is approximately a weighted average of those same forward rates, it is an **option on a portfolio**.

*Why the cap is worth at least as much.* For any weights $w_i$ summing to one,

$$\sum_i w_i \max(F_i - K, 0) \;\geq\; \max\left(\sum_i w_i F_i - K,\; 0\right)$$

because a sum of positive parts is at least the positive part of the sum. Every term on the left is nonnegative, so the left side is nonnegative; and wherever the right side is positive, the left side matches it term by term and adds whatever the negative terms would have subtracted. Taking

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expectations preserves the inequality, so on comparable terms **a cap is always worth at least as much as the corresponding swaption**. This is Jensen's inequality applied to the convex function $\max(\cdot - K, 0)$.

*The intuition: diversification running backward.* The cap holder collects on *whichever* rates happen to exceed the strike and is not penalized for those that do not; the losers are floored at zero individually. The swaption holder owns *one* option on the average, so a rate finishing far above the strike is partly canceled by another finishing below it. The cancellation happens *inside* the payoff, before the floor applies, and it destroys value for an option holder.

Put the other way: diversification reduces variance, and reduced variance is good for a bondholder and bad for an option holder. The swaption is the diversified position, which is precisely why it is worth less.

*When the two are equal: when the forward rates are **perfectly correlated**,* $\rho = 1$.

If all forward rates move in lockstep, they are all above the strike together or all below it together, so no cancellation can occur and the inequality holds with equality. Question 12's formula confirms it from the other direction: at $\rho = 1$,

$$\sigma_{\text{swap}} = \sigma\sqrt{1 + \frac{0}{n}} = \sigma$$

and the swap rate carries the same volatility as each forward. With equal volatilities and no cancellation, the two products are the same trade.

*Two riders that keep the statement honest.* The equality also requires the rates to be at the same *level* relative to the strike: a set of perfectly correlated but differently-positioned forwards still produces some cancellation in the average. And "comparable terms" is doing work: the comparison assumes the same strike, schedule and notional, and in practice a cap and a swaption on the same schedule are quoted at different volatilities precisely because the market prices the correlation between them.

*The point of the comparison.* The gap between a cap and a swaption is a price for correlation, and it is one of the few places correlation is directly quoted in the rate market. That makes it the natural calibration target for the multi-factor models of Chapter 32, and it is why question 12 warns that a model fitted to caps alone has no information about the parameter that governs the difference.

*What the question tests.* That a portfolio of options dominates an option on the portfolio, and that perfect correlation is the boundary case where the two coincide.

---

**29.12** *Eight forward rates each have a volatility of 24% and enter a swap rate with equal weight. Using (29.17), find the implied swap rate volatility when the uniform pairwise correlation is 1.0, 0.6, and 0.2. Comment on what the results imply about calibrating a model to caps alone.*

*The computation.* With $n = 8$ forward rates, each of volatility $\sigma = 24\%$, entering the swap rate with equal weight and uniform pairwise correlation $\rho$, equation (29.17) gives

$$\sigma_{\text{swap}} = \sigma\sqrt{\rho + \frac{1 - \rho}{n}}$$

| $\rho$ | Calculation | $\sigma_{\text{swap}}$ |
|---|---|---|
| 1.0 | $0.24\sqrt{1.000 + 0.000}$ | **24.00**% |
| 0.6 | $0.24\sqrt{0.600 + 0.050} = 0.24\sqrt{0.650}$ | **19.35**% |
| 0.2 | $0.24\sqrt{0.200 + 0.100} = 0.24\sqrt{0.300}$ | **13.15**% |

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Read the spread. The same eight forward rates, each with an identical and fully specified 24% volatility, generate a swap rate volatility anywhere between **13.15% and 24.00%**—a range of nearly two to one—depending on a parameter that the caplet market does not contain.

*What this implies about calibrating to caps alone.*

*Caps determine the forward-rate volatilities and nothing else.* Each caplet is an option on *one* forward rate (question 1), so a cap surface pins down $\sigma_1, \ldots, \sigma_n$ individually. It contains no information whatever about how those rates move *together*, because no caplet's payoff depends on two of them.

So a model calibrated to caps alone has a free parameter that moves swaption prices by nearly a factor of two. It will reprice every cap exactly and misprice swaptions by an amount determined by whatever correlation structure the model happened to assume, and if that assumption came from the model's functional form rather than from data, the mispricing is invisible to the calibration. The fit is perfect and the prices are wrong, which is the worst configuration because nothing signals it.

*The direction of the error is knowable.* Question 11 established that a cap is worth at least as much as the corresponding swaption, with equality only at $\rho = 1$. A one-factor model imposes $\rho = 1$ by construction, so it produces the *maximum* swap-rate volatility consistent with the caps—and therefore **overprices swaptions**, systematically, always in the same direction. Chapter 31's question 12 pursues why that sign is the dangerous one.

*What follows for practice.* A model intended to price both products must be calibrated to **both**, and the swaption surface is where the correlation information lives. That is the entire argument for the multi-factor models of Chapter 32: not that more factors fit better in some abstract sense, but that a single factor cannot represent a quantity the market quotes.

*And the general lesson, which is the one to carry.* An instrument tells you only about the quantities its payoff depends on. Caps price levels of volatility; the cap–swaption spread prices correlation; the tenor dimension of the cube (question 10) prices it again from another angle. A calibration is only as informative as the payoffs it is fitted to, and a parameter no traded payoff depends on is being asserted rather than measured, which is Chapter 26's correlation problem in a different market.

*What the question tests.* Substitution into the basket-variance formula, and that a perfect fit to one instrument class can coexist with a factor-of-two error in another.

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# Chapter 30
# Convexity, Timing, and Quanto Adjustments

**30.1** *Explain why a forward rate is a martingale under the $T_2$-forward measure but not under the risk-neutral measure. Which measure is relevant for valuing a payoff made at $T_2$?*

*Why it is a martingale under $Q^{T_2}$.* A forward rate for $[T_1, T_2]$ satisfies

$$\tau F(t)\, P(t, T_2) = P(t, T_1) - P(t, T_2)$$

The right-hand side is the value of a *traded* portfolio: long one zero-coupon bond, short another. Dividing by $P(t, T_2)$ expresses that traded value in units of the $T_2$ bond, and Chapter 28's central result says any traded asset divided by the numeraire is a martingale under the measure that numeraire induces. So

$$E^{T_2}[F(T_1)] = F(0)$$

*by construction*, with no drift to model and no assumption made.

*Why not under the risk-neutral measure.* There the numeraire is the money-market account $B_t = e^{\int_0^t r}$, and $F$ is *not* a traded asset divided by $B$. What is a martingale under $Q$ is $[P(t, T_1) - P(t, T_2)]\,/B_t$, and dividing that by $P(t, T_2)$ to recover $F$ reintroduces a stochastic factor. By Girsanov the change from $Q^{T_2}$ to $Q$ shifts $F$'s drift by its covariance with the log of the numeraire ratio, and that covariance is not zero, because the ratio $B_t/P(t, T_2)$ is driven by the same rates $F$ is.

So the forward rate acquires a drift under $Q$, and the size of that drift is exactly what Chapter 32's LIBOR market model must compute.

*The relevant measure for a payoff made at $T_2$: $Q^{T_2}$.* Two reasons, and they are the same reason twice.

*The discounting comes out.* With $N = P(\cdot, T_2)$ the terminal numeraire value is $P(T_2, T_2) = 1$, so

$$V_0 = P(0, T_2)\, E^{T_2}[\text{payoff}]$$

The discount factor leaves the expectation entirely, and no covariance between payoff and discounting can survive.

*And the underlying has no drift.* $F$ is a martingale under precisely this measure, so its expectation is today's forward, which is what Black's formula assumes.

*The two facts are one choice.* Picking the discount factor *is* picking the measure. That is why Chapter 29's caplet formula discounts at $P(0, T_2)$ and assumes no drift: both halves come from the same election, and mixing them (discounting at $P(0, T_1)$ while assuming no drift) is incoherent. Chapter 31's Example 31.5 measures what the incoherence costs on a two-year caplet: **2.52%**.

*And this is the template for the whole chapter.* Every adjustment that follows arises when the payoff is *not* paid at $T_2$, so the natural measure for the payoff differs from the one under which

the rate is a martingale, and the mismatch leaves a covariance behind. Timing adjustments, in-arrears adjustments, CMS adjustments and quanto adjustments are four instances of that one sentence.

*What the question tests.* That the martingale property is a consequence of the numeraire, and that the payment date selects the measure.

---

**30.2** *Without performing any calculation, state the sign of the required adjustment in each case, and justify your answer in one sentence:*

*(a) a variable uncorrelated with interest rates, paid three months after its natural date;*

*(b) a ten-year swap rate paid as a single cash flow in five years;*

*(c) a quanto forward on a foreign equity index, where the index and the foreign currency are positively correlated.*

*(a) A variable uncorrelated with interest rates, paid three months after its natural date:* ***no adjustment at all****.*

The timing adjustment (30.7) is proportional to $\rho$, the correlation between the variable and the forward rate covering the deferral period. Deferring a payment means discounting it over that period at a rate not known today, but if the payoff is independent of that rate, the expectation of the product is the product of the expectations and no covariance term appears. The deferral can be arbitrarily long and the adjustment remains zero. Length is not what creates the correction; correlation is.

*(b) A ten-year swap rate paid as a single cash flow in five years:* ***positive****.*

This is the CMS convexity adjustment. The swap rate is a martingale under the *annuity* measure, but the coupon pays once, at a single date, so the natural numeraire is a zero-coupon bond and the annuity does not cancel. What survives is a covariance with a definite sign: a swap rate that rises raises the coupon linearly but *lowers* the annuity the coupon would otherwise have been paid over, so the states with the largest payoff are the states in which the missing numeraire is smallest. The correction adds to the forward swap rate: a CMS coupon is worth more than the forward rate suggests, always.

*(c) A quanto forward on a foreign equity index, with index and foreign currency positively correlated:* ***negative****.*

By (30.20), $F^{\text{quanto}} = F^f e^{-\rho\sigma_V\sigma_S T}$, so $\rho > 0$ makes the exponent negative and the quanto forward sits *below* the ordinary foreign-currency forward. The hedging argument gives the economics: a dealer short the quanto must hold the foreign asset, whose gains arrive in foreign currency, and with positive correlation they arrive when that currency is dear, so in domestic terms the hedge systematically outruns the payoff it replicates. That recurring advantage is passed back to the buyer by lowering the forward.

*The common structure, which is the point of the question.* Each answer is the sign of a **covariance**, and none required a calculation:

| | **What is correlated with what** | **Sign** |
|---|---|---|
| (a) | nothing, by assumption | zero |
| (b) | swap rate with the annuity it is *not* paid over | positive |
| (c) | foreign asset with the currency its gains are paid in | negative |

*The general rule to carry.* Ask: in the states where the payoff is large, is the thing being multiplied by it large or small? If large, the naive calculation understates and the adjustment is positive; if small, it overstates and the adjustment is negative; if unrelated, there is no adjustment. That question is answerable without a calculation, and getting the sign right before computing the magnitude is the discipline this chapter is teaching.

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*What the question tests.* That every adjustment in the chapter is one covariance, and that its sign is determined by economics rather than by algebra.

---

**30.3** *A three-month interest rate futures contract has $T_1 = 3$ years and $T_2 = 3.25$ years. The short-rate volatility is 90 basis points per year. The futures rate is 3.85%. Calculate the forward rate using the Ho–Lee adjustment.*

*The Ho–Lee adjustment.* Under $dr = \theta(t)dt + \sigma dW$, equation (30.12) gives

$$\text{Forward rate} = \text{Futures rate} - \frac{1}{2}\sigma^2 T_1 T_2$$

with $\sigma$ the standard deviation of the *absolute* change in the short rate per year, a normal volatility, which is the market convention for this parameter.

*Substituting* $\sigma = 90$ bp $= 0.0090$, $T_1 = 3$, $T_2 = 3.25$:

$$\frac{1}{2}\sigma^2 T_1 T_2 = \frac{1}{2} \times 0.0090^2 \times 3 \times 3.25 = \frac{1}{2} \times 0.000081 \times 9.75 = 0.00039487$$

or **3.95 basis points**. Hence

$$\text{Forward rate} = 3.85\% - 0.0395\% = \mathbf{3.8105}\%$$

*The direction is the one to check first.* The forward rate is *below* the futures rate, always. Question 5 gives the economic reason; the algebraic one needs no model at all. Writing $Z$ for the density taking $Q$ to $Q^{T_2}$, the forward rate is $E^Q[RZ]$ and the futures rate is $E^Q[R]$, so their difference is $\text{Cov}^Q(R, Z)$. Both factors of $Z = P(T_1, T_2)/[B(T_1)P(0, T_2)]$ move *against* $R$ (a higher rate lowers the bond and raises the accumulated money-market account), so the covariance is negative and the futures rate exceeds the forward **unconditionally**. Only the monotonicity of bond prices in rates was used, which is why the *sign* needs no model while the *size* does.

*Two remarks attach to the parameter.* The $\sigma$ here is a **normal** volatility in absolute rate units, not a proportional one: 90 basis points a year of short-rate movement, not 90% of the rate. Substituting a lognormal volatility of, say, 20% would give an answer several hundred times too large. And Ho–Lee has no mean reversion, so it treats the short rate's volatility as constant across maturities; Hull–White's $\sigma e^{-aT}$ (Chapter 31) damps the long end and produces a smaller adjustment far out, which is why the Ho–Lee figure overstates the adjustment for distant contracts.

*Why 3.95 basis points matters.* On a three-year contract this is several times the typical bid–ask on the underlying swap, so it is not decorative, and question 4 shows what happens to it further along the strip.

*What the question tests.* Substitution into the Ho–Lee formula with the correct units for $\sigma$, and knowing the sign before computing the size.

---

**30.4** *Repeat Question 30.3 with $T_1 = 8$ and $T_2 = 8.25$. By what factor does the adjustment increase, and does that match what equation (30.12) predicts?*

*The adjustment at eight years.* With $T_1 = 8$, $T_2 = 8.25$ and the same $\sigma = 0.0090$:

$$\frac{1}{2}\sigma^2 T_1 T_2 = \frac{1}{2} \times 0.000081 \times 8 \times 8.25 = \frac{1}{2} \times 0.000081 \times 66 = 0.002673$$

or **26.73 basis points**, giving a forward rate of

$$3.85\% - 0.2673\% = \mathbf{3.5827}\%$$

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*The factor.*

$$\frac{26.73}{3.95} = \mathbf{6.77}$$

*And it matches* (30.12) *exactly.* The formula's only maturity dependence is the product $T_1T_2$, and $\sigma^2/2$ is common to both, so the ratio must be

$$\frac{T_1T_2\big|_8}{T_1T_2\big|_3} = \frac{8 \times 8.25}{3 \times 3.25} = \frac{66}{9.75} = 6.769$$

which is 6.77 to the precision quoted. The agreement is not a coincidence to be verified but an identity: the adjustment is exactly quadratic in maturity, so the ratio of two adjustments is the ratio of two products of maturities and nothing else survives.

*The structural lesson.* The adjustment grows with the square of maturity. Since $T_2 \approx T_1$ for a three-month contract, $T_1T_2 \approx T_1^2$, so moving from three years to eight years (a factor of 2.67 in maturity) multiplies the adjustment by roughly $2.67^2 = 7.1$, and exactly by 6.77 once the quarter-year offset is included.

That quadratic growth carries the practical rule: the adjustment is negligible for near-dated contracts and dominant for far-dated ones. On the front contract of a strip it is a fraction of a basis point and can be ignored; eight years out it is 27 basis points and cannot. A desk building a curve from a futures strip that applies no convexity adjustment will price the front correctly and be wrong by a quarter of a percent at the back, and the error is one-directional, always making forward rates look too high.

*Where the Ho–Lee figure overstates.* Because Ho–Lee has no mean reversion, its short-rate volatility does not decay with maturity, so the quadratic growth continues indefinitely. Under Hull–White the effective volatility of a distant forward is $\sigma e^{-aT}$ (Chapter 31's question 4), which damps the long end substantially: at $a = 0.05$ the eight-year factor is $e^{-0.4} = 0.67$, so the adjustment at eight years is materially smaller than 26.73 basis points (Figure 30.1 shows about a fifth smaller at eight years for $a = 3\%$). The quadratic law is the right shape and Ho–Lee is the wrong magnitude far out, which is why dealers' pricing systems use a mean-reverting model for the back of the strip.

*What the question tests.* That the adjustment is exactly quadratic in maturity, and reading a formula's structure to predict a ratio without recomputing it.

---

**30.5** *Explain in words why a trader who is long the rate through a futures position benefits from daily settlement, and why this requires the futures rate to exceed the forward rate.*

*The mechanics.* A forward contract settles once, at maturity. A futures contract is **marked to market daily**: gains and losses are paid in cash as they accrue. Consider a trader who is *long the rate*: short the futures price, positioned to profit when rates rise.

*When rates rise*, the trader **receives** variation margin, and can invest that cash at the new, *higher* rate.

*When rates fall*, the trader must **post** variation margin, and funds it at the new, *lower* rate.

*Both legs favor the trader.* Cash arrives exactly when reinvestment is attractive and must be paid exactly when funding is cheap. The timing of the cash flows is systematically favorable, and the favor is not a matter of luck; it follows from the sign of the position and applies on every move, in both directions, for the whole life of the contract.

*Why this forces the futures rate above the forward rate.* Because it is a genuine, repeatable advantage **with no offsetting cost**, and in an arbitrage-free market such an advantage must be priced away. The forward contract, settling once, confers no such benefit: its holder receives the same terminal payoff with none of the favorable interim financing.

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If the two contracts traded at the same rate, a trader could go long the rate through futures and short it through a forward, hold both to maturity, and collect the financing advantage with no terminal exposure at all—the payoffs cancel and the margin timing does not. That is an arbitrage, so the market removes it by making the futures position more expensive to enter, which for someone long the rate means

$$\boxed{\text{Futures rate} > \text{Forward rate}}$$

Equivalently, in price terms, the futures price sits *below* the forward price. The trader who is long the rate pays for the privilege by entering at a worse rate.

Note what the argument does and does not use. It uses only that margin flows are correlated with the rate at which they are financed, which is to say, once again, an adjustment is a covariance. It does not use any model of the short rate, and this is why question 3's sign statement holds unconditionally while its magnitude required Ho–Lee.

*Two riders are worth attaching.*

*The advantage is symmetric, not directional.* A trader *short* the rate faces the mirror image (receiving margin when rates fall and reinvesting at the lower rate), so their financing is systematically unfavorable, which is the same fact from the other side. The convexity adjustment compensates them, and the trade is fair.

*And it survives the move to SOFR.* The mechanism is about *settlement*, not about the reference rate, so SOFR futures carry the same adjustment as their predecessors. Question 8 notes which adjustments the transition removed; this is not one of them.

*What the question tests.* Articulating the financing asymmetry in both directions, and that no-arbitrage requires a free advantage to be priced into the entry level.

---

**30.6** *A payoff equal to the two-year swap rate is observed in three years and paid six months later. The forward two-year swap rate is 3.50% with 22% volatility; the forward rate for the deferral period is 3.20% with 25% volatility; the correlation is 0.8. Calculate the timing adjustment in basis points.*

*Setup.* A two-year swap rate observed at $T^* = 3$ and paid at $T = 3.5$, so the deferral is $\tau = 0.5$. The variable is $V$ with forward $V_0 = 3.50\%$ and volatility $\sigma_V = 22\%$; the deferral-period forward rate is $y_0 = 3.20\%$ with $\sigma_y = 25\%$; the correlation is $\rho = 0.8$.

*The formula*, from (30.8):

$$E^T[V(T^*)] = V_0 \exp\left(-\frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1 + y_0\tau}\right)$$

*The exponent.*

$$\frac{0.8 \times 0.22 \times 0.25 \times 0.0320 \times 0.5 \times 3}{1 + 0.0320 \times 0.5} = \frac{0.002112}{1.0160} = 0.00207874$$

*The adjusted rate.*

$$E^T[V] = 3.50\% \times e^{-0.00207874} = \mathbf{3.4927}\%$$

so the timing adjustment is

$$3.4927\% - 3.5000\% = \mathbf{-0.73} \text{ basis points}$$

The linear form (30.7) gives the same −0.73 bp, since the exponent is small enough that $e^{-x} \approx 1 - x$ to well within a hundredth of a basis point.

*Why it is negative.* The correlation is positive, so the swap rate tends to be large exactly when the deferral-period rate is high, and a high deferral rate means the payment is discounted more heavily over $[T^*, T]$. The large payoffs are the ones that get discounted most, so the

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deferred claim is worth less than the undeferred forward suggests. Had $\rho$ been negative the adjustment would be positive, and at $\rho = 0$ it would vanish however long the deferral.

*Why it is so small, which is the useful part of the answer.* Read the exponent's factors:

- $y_0\tau = 0.016$: the deferral is half a year at a 3.2% rate, so only 1.6% of interest is at stake at all;
- $\tau T^* = 1.5$: the adjustment scales with the *product* of the deferral length and the time to observation, and a six-month deferral keeps that product small;
- and the volatilities multiply to 0.055.

The result is a correction of *under one basis point* on a rate quoted to a fraction of a basis point of bid–ask—real, but not decisive.

*The scaling generalizes; the number does not.* The adjustment is proportional to $\tau T^*$, so it grows linearly in *both* the deferral and the observation date. Deferring the same rate by two years instead of six months quadruples it; observing it in twelve years instead of three quadruples it again. A payment deferred two years on a twelve-year observation would carry roughly $-11$ basis points, which is a different order of consequence.

*And contrast the two families.* A timing adjustment is proportional to $\tau$ and is small because deferrals are usually short. A CMS convexity adjustment (question 2b) does not depend on a deferral at all (it arises from the *annuity* failing to cancel) and is measured in tens of basis points. They are different mechanisms with different magnitudes, and a coupon can carry both.

*What the question tests.* Substitution into the timing formula with the right rate in each slot, and reading the $\tau T^*$ scaling to see why the answer is sub-basis-point.

---

**30.7** *A rate for a six-month period beginning in four years is paid at the start of the period rather than the end. The forward rate is 3.75% with volatility 28%. Calculate the in-arrears convexity adjustment using both the exact form (30.14) and the approximation in (30.15), and comment on the difference.*

*Setup.* A six-month rate ($\tau = 0.5$) for a period beginning at $T_1 = 4$, paid at the *start* of the period rather than the end. The forward is $F = 3.75\%$ with volatility $\sigma = 28\%$.

*The exact form*, from (30.15):

$$E^{T_1}[R(T_1)] = F + \frac{F^2\left(e^{\sigma^2 T_1} - 1\right)\tau}{1 + F\tau}$$

With $\sigma^2 T_1 = (0.28)^2 \times 4 = 0.3136$ and $e^{0.3136} - 1 = 0.368342$:

$$E^{T_1}[R] = 0.0375 + \frac{(0.0375)^2 \times 0.368342 \times 0.5}{1 + 0.0375 \times 0.5} = 0.0375 + \frac{0.000258991}{1.01875} = \mathbf{3.7754}\%$$

an adjustment of $+\mathbf{2.54}$ basis points.

*The approximation*, replacing $e^{\sigma^2 T_1} - 1$ by $\sigma^2 T_1$:

$$E^{T_1}[R] \approx F + \frac{F^2\sigma^2 T_1 \tau}{1 + F\tau} = 0.0375 + \frac{(0.0375)^2 \times 0.3136 \times 0.5}{1.01875} = \mathbf{3.7716}\%$$

an adjustment of $+\mathbf{2.16}$ basis points.

*The approximation understates by 14.9%.* That is a large relative error on a small absolute number, and its source is visible: $e^x - 1 = 0.3683$ against $x = 0.3136$, a 17% gap, because $\sigma^2 T_1$ is not small here. At a 28% volatility over four years the linearization is being applied outside the regime where it is accurate: the neglected term is $x^2/2 = 0.049$, a sixth of $x$ itself. For a one-year option at 15% the same comparison would differ by under 2%.

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*Why the adjustment is positive*, in contrast with question 6's timing adjustment. Here the payment is being **accelerated**, not deferred. Receiving a rate early is worth more when rates are high, so the scenarios that get upweighted are the *favorable* ones, the opposite of the deferral case, where the large payoffs were the ones most heavily discounted. The sign follows from the direction of the timing mismatch, and question 2's rule delivers it without algebra.

*Why this case admits an exact answer at all.* It is exact because the change of measure can be computed rather than approximated. Since $P(T_1, T_2) = 1/(1 + R(T_1)\tau)$, the Radon–Nikodym derivative is a simple function of $R$ itself:

$$\left.\frac{dQ^{T_1}}{dQ^{T_2}}\right|_{\mathcal{F}_{T_1}} = \frac{1 + R(T_1)\tau}{1 + F\tau}$$

which has unit mean under $Q^{T_2}$ precisely because $E^{T_2}[R] = F$. Taking the expectation produces $\mathrm{Var}^{T_2}(R)$ directly, so (30.14) is exact and the *only* modeling input is a variance. Most adjustments in this chapter require linearizing a density; this one does not.

*The practical reading.* The adjustment depends on the variance of $R$, not on any correlation, so it cannot be made to vanish by choosing a benign correlation; it is present whenever the rate is paid early and the rate is uncertain. And since it is a *variance* term, it carries a vega: an in-arrears swap has volatility exposure that a plain swap does not, which is Chapter 33's question 4 in a different instrument.

*What the question tests.* Both forms of the in-arrears adjustment, that the linearization's error grows with $\sigma^2 T_1$, and that acceleration reverses the sign relative to deferral.

---

**30.8** *Explain why the transition from LIBOR to compounded risk-free rates eliminated most of the in-arrears convexity adjustment. Identify two situations in which a convexity adjustment is still required under RFR conventions.*

*Why the transition eliminated most of it.* The classical in-arrears adjustment of question 7 existed because LIBOR conventions created a *mismatch* that a contract could break. Under LIBOR the rate for $[T_1, T_2]$ was **set at** $T_1$ and **paid at** $T_2$, the natural alignment, under which the rate is a martingale (question 1). An in-arrears swap broke that alignment by paying the rate at $T_1$, immediately on observation, and the adjustment was the price of the break.

Under compounded risk-free rates the rate for $[T_1, T_2]$ is **not known until** $T_2$, because it accrues day by day across the period. So a payment at $T_2$ is now the only natural arrangement, and it is also the arrangement under which the rate is a martingale. The structure that required the adjustment no longer arises, because paying a compounded rate "in arrears" is now simply paying it when it is known.

Put differently: LIBOR's advance fixing created a gap between *observation* and *payment* that a product could exploit; RFR compounding closes the gap by construction. The adjustment did not become smaller—the contract that needed it stopped being written.

*Two situations in which an adjustment is still required.*

*1. Constant maturity swaps.* Section 30.5 is explicit that this family was **entirely unaffected** by the transition, and the reason is structural rather than conventional. A CMS coupon pays a *ten-year swap rate* for a three-month or annual period. The swap rate is a martingale under the *annuity* measure; the coupon is paid on a single date, whose natural numeraire is a zero-coupon bond. The annuity fails to cancel, and the residual covariance is the CMS convexity adjustment. Nothing about which reference rate underlies the swap touches that argument: the mismatch is between a rate's own annuity and a single payment date, and it would exist in any rate regime. These adjustments run to tens of basis points, far larger than question 7's.

*2. Futures against forwards.* Questions 3 to 5 are about *settlement mechanics*, not about

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fixing conventions. A futures contract is margined daily, so its rate is a martingale under the risk-neutral measure while the forward rate is a martingale under $Q^{T_2}$; the gap is the financing covariance. SOFR futures are margined exactly as their predecessors were, so the adjustment survives unchanged: 26.73 basis points at eight years on question 4's numbers.

The RFR conventions themselves leave three residuals, which Section 30.4 and Section 33.4 point to. **Payment delays** pay the compounded rate a few business days after $T_2$ and need a question 6-style timing adjustment. **Lookbacks** without an observation shift misalign the weights on the daily fixings from those the discounting implies, leaving a small convexity. And **rate-set-in-advance** structures deliberately restore an in-advance fixing. Term SOFR does it the LIBOR way, fixing at the start of the period and paying at the end, which needs no adjustment beyond a small basis; last-reset conventions pay the previous period's rate one period late, a deferral whose timing adjustment is opposite in sign to question 7's. The first two are typically well under a basis point; all three are the residue of the transition rather than its abolition. *Quanto* adjustments (question 9) concern currency and are untouched.

*The general lesson.* An adjustment arises from a mismatch between the measure a rate is a martingale under and the measure its payment lives under. Changing the reference rate removed *one* such mismatch. It could not remove the others, because they come from the payment structure, from settlement mechanics, or from the currency—none of which the reform touched.

*What the question tests.* That the in-arrears adjustment was an artifact of advance fixing, and distinguishing convention-driven mismatches from structural ones.

---

**30.9** *A quanto forward on a foreign stock index is settled in domestic currency. The index has 25% volatility, the exchange rate (domestic per foreign) has 12% volatility, and the correlation is 0.4. The one-year foreign-currency forward level is 5,000. Calculate the quanto forward level.*

*The formula.* From (30.20),

$$F_0^{\text{quanto}} = F_0^{f}\, e^{-\rho\, \sigma_V\, \sigma_S\, T}$$

where $\sigma_V$ is the index's volatility, $\sigma_S$ the exchange rate's, and $\rho$ their correlation, with $S$ quoted as **units of domestic currency per unit of foreign currency**.

*The drift adjustment.*

$$\rho\, \sigma_V\, \sigma_S = 0.4 \times 0.25 \times 0.12 = 0.0120$$

*The quanto forward.*

$$F_0^{\text{quanto}} = 5{,}000 \times e^{-0.0120 \times 1} = 5{,}000 \times 0.988072 = \mathbf{4{,}940.4}$$

about **59.6 index points** below the ordinary foreign-currency forward, or $-1.19\%$.

*Why it is lower.* The correlation is positive, so the index and the foreign currency tend to rise together. Question 10 gives the hedging argument: a dealer short the quanto must hold the foreign asset, so the hedge's gains arrive in foreign currency, and here they arrive when that currency is strong; in domestic terms the hedge outruns the payoff it replicates. That recurring advantage is passed back by setting the quanto forward below the foreign forward.

*Getting the convention right is essential, and reversing it reverses the answer.* If $S$ were quoted the other way (*foreign* per *domestic*), then a positive correlation between the index and *that* rate means the index rises as the foreign currency *weakens*, and the adjustment flips sign to $+1.20\%$, giving 5,060.4. The two answers differ by 120 index points on a 5,000 index, so this is not a pedantic point. Before computing, state which currency is in the numerator and check that a rising $S$ means a stronger foreign currency.

*Two checks confirm the magnitude.* The exponent is $\rho\sigma_V\sigma_S T$, a product of two volatilities, so it is inherently second-order and 1.2% over a year is the right order for a 25% equity and a

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12% currency at moderate correlation. And it scales *linearly* in $T$, unlike the futures–forward adjustment of question 4 which is quadratic: a five-year quanto on the same parameters carries $e^{-0.06} - 1 = -5.8\%$, still modest, whereas the futures adjustment would have grown twenty-five-fold.

*Where the adjustment matters most.* It is largest when the index and the currency are strongly correlated, and that correlation is neither stable nor observable in the forward market; it must be estimated from history or implied from the handful of quanto structures that trade. The quanto adjustment therefore shares with the timing adjustment an input no market quotes, a correlation, which makes these two the members of this chapter's family most exposed to estimation error. A desk quoting quantos runs a correlation book whether it intends to or not.

*What the question tests.* Substitution into the quanto forward formula, and that the sign hangs entirely on the exchange-rate quoting convention.

---

**30.10** *Continuing Question 30.9, the domestic interest rate is 3%. Value a one-year quanto call on the index, struck at the quanto forward level and paying one unit of domestic currency per index point, and compare it with the same call if the correlation were zero. Why does the correlation matter when the contract's exchange rate is fixed?*

*The quanto forward.* By Question 9, $F_0^{\text{quanto}} = 5{,}000\, e^{-0.4\times 0.25\times 0.12} = 4{,}940.4$. A quanto call is valued with Black's model using this forward in place of the ordinary one and discounting at the domestic rate.

*At the quanto forward.* With the strike equal to the forward, $d_1 = \sigma_V\sqrt{T}/2 = 0.125$ and $d_2 = -0.125$, so $N(d_1) = 0.5497$ and $N(d_2) = 0.4503$ and

$$C = e^{-0.03} \times 4{,}940.4 \times (0.5497 - 0.4503) = 476.9$$

per index point.

*With zero correlation.* The forward is the ordinary 5,000 and the strike stays at 4,940.4, so $d_1 = 0.1730$, $d_2 = -0.0770$, $N(d_1) = 0.5687$, $N(d_2) = 0.4693$ and

$$C = e^{-0.03}\,(5{,}000 \times 0.5687 - 4{,}940.4 \times 0.4693) = 509.3$$

The positive correlation makes the call cheaper by 32.4, about 6.8%, though the exchange rate appears nowhere in the payoff.

*Why.* A dealer short the call hedges with the foreign index, and the hedge's gains arrive in foreign currency and are converted at the rate prevailing when they arrive. With $\rho > 0$ the gains arrive when the foreign currency is dear, so the hedge outruns the payoff it replicates and the price must be lower. The contract has no exposure to where the exchange rate ends up, only to how it moves with the index, which is why the adjustment contains both volatilities and the correlation and vanishes when any of them is zero. Question 30.14 develops the dealer's hedge.

*What the question tests.* That the quanto effect enters the price through the forward, and that its size can be read off a price, not just a drift.

---

**30.11** *Without using algebra, explain why the convexity adjustments on the two legs of a CMS spread do not cancel. Under what condition would they cancel?*

*Why they do not cancel.* It is tempting to assume the two adjustments largely offset, since the payoff is a *difference*. They do not, and the reason is that the ratio $-G''/G'$ driving the adjustment **depends strongly on the tenor of the underlying swap**, and the two legs of a CMS spread have *different* tenors by construction.

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State the mechanism in words. The CMS adjustment measures how curved a bond's price is in its own yield: convexity relative to duration. A ten-year bond carries far more convexity for its duration than a two-year bond, which has very little. So the long leg takes a much larger correction, and the difference leaves a substantial net.

Example 30.4 puts numbers on it. For a ten-year less two-year spread at $T_1 = 5$:

| | 2-year leg | 10-year leg |
|---|---|---|
| Forward swap rate $S_0$ | 3.90% | 4.40% |
| Volatility $\sigma$ | 24% | 18% |
| Annuity $A = -G'$ | 1.8888 | 7.9518 |
| $-G''/G'$ | 2.8690 | 9.8580 |
| Convexity adjustment | 6.28 bp | 15.46 bp |

The forward spread of 50.0 basis points widens to $4.5546\% - 3.9628\% = 59.18$ basis points, a net effect of **+9.2 basis points**.

Note where the example is pushed toward cancellation. It assumes the two-year leg is *more* volatile than the ten-year (24% against 18%), as the short end usually is, and that works against the effect, since the adjustment is proportional to $\sigma^2$. It is nowhere near enough: $-G''/G'$ is 3.4 times larger on the ten-year leg, and that tenor mismatch overwhelms everything else.

*When they would cancel: when both legs reference swaps of the* ***same tenor****.* Then $-G''/G'$ is the same for both, the two $S_0$ and the two $\sigma$ are nearly the same as well, and the adjustments are near enough term by term to vanish in the difference. A spread written on two swap rates of the same tenor but different *start dates* needs almost no correction; a spread on two different *tenors* needs a large one. It is the tenor mismatch that drives this, not the fact that the payoff is a spread.

*The direction is systematic, which is the practical point.* Because the long leg's adjustment is the larger unless the short leg's volatility is far higher than in the example, convexity *widens* the fair spread on a steepener. A dealer who omits the adjustment sells every such note too cheaply—persistently, not at random. This is the same defect as the one-directional futures error of Question 30.4: an omitted adjustment does not average out across trades, because its sign never changes.

One last caution follows, beyond this question but worth connecting. The above values a *linear* spread. A real steepener note pays $\max(\text{spread}, 0)$ or a leveraged multiple of it, and an option on the difference between two rates depends on the **correlation** between them, a quantity no convexity adjustment captures. The adjustment is therefore necessary and not sufficient: get it right and you have the forward spread; to value a floored payoff you additionally need a two-factor model or a replication argument, and a correlation that is neither observable nor stable.

*What the question tests.* That $-G''/G'$ varies strongly with tenor, so the two legs of a spread take materially different corrections; and that equal tenors, not equal levels or volatilities, are the condition that makes them cancel.

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# Chapter 31
# Short-Rate Models

**31.1** *Explain why Black's model cannot price a Bermudan swaption, and be specific about what it fails to produce.*

*What Black's model is and is not.* Black's swaption formula prices a *European* option by taking one expectation of a payoff at one date, under the measure in which the forward swap rate is a martingale (Chapter 28's question 6). It is exact for that product, and it needs no model of how rates evolve, only a distribution at expiration.

*Why that is not enough for a Bermudan.* A Bermudan swaption may be exercised on any of several dates. At each of them the holder must compare the value of exercising now with the value of **waiting**—and the value of waiting is the value of a Bermudan option with the remaining dates. So the problem is recursive, and solving it requires knowing, at every intermediate date and every rate level, what the surviving option is worth.

*What Black's model fails to produce: a **continuation value**.* It gives one number today. It does not give the option's value at a future date conditional on where rates have gone, because it has no notion of "where rates have gone": there is no state variable, no dynamics, and no intermediate time. Without a continuation value there is no exercise decision, and without exercise decisions there is no Bermudan.

Two related things are missing with it, and they are worth naming separately.

*The joint distribution across dates.* A Bermudan's value depends on the *relationship* between rates at different exercise dates: how much the curve at year three tells you about the curve at year four. Black's model prices each date in isolation, under its own measure, with no statement about how the dates connect. Question 8 shows this is exactly the information that determines the answer.

*And consistency between the marginals.* A strip of Black prices for the co-terminal Europeans is a set of marginal distributions with no guarantee that any single process generates all of them. The Bermudan requires one process.

*What the two bounds tell you, since they are worth having.* A Bermudan is worth at least the *largest* single European (the holder can always commit to one date) and at most the *sum* of them, since it cannot exercise more than once. Example 31.8 gives 1.1953 and 4.6484 against a true value of 1.6995, so the bounds bracket the answer by a factor of four. They are free and they are far too wide to trade on, which is precisely why a model is needed.

*And what the model must supply.* A short-rate model gives the continuation value *at every node* of a tree, so the early-exercise comparison costs nothing extra, which is why these models remain the standard tool for Bermudan and callable books (question 13) despite being less realistic than Chapter 32's alternatives.

*What the question tests.* That early exercise requires a conditional value at intermediate dates, and that a formula producing one unconditional number cannot supply it.

**31.2** *In Vasicek's model with $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, compute the limiting long rate and explain why it is below $b$. Then explain why an equilibrium model cannot match an observed term structure, and why this is arithmetic rather than calibration.*

*The limiting long rate.* Vasicek's model $dr = a(b - r)dt + \sigma dW$ has

$$R_\infty = b - \frac{\sigma^2}{2a^2} = 5\% - \frac{(0.01)^2}{2 \times 0.10^2} = 5\% - \frac{0.0001}{0.0200} = \mathbf{4.50}\%$$

*Why it is below $b$: a **convexity** effect.* The short rate reverts to $b = 5\%$, so a naive reading expects long yields to approach 5%. But a long bond's *price* is a convex function of the rate, so uncertainty about the path raises the bond's expected price (by Jensen's inequality), and a higher price means a lower yield. The 50 basis point gap is the accumulated value of that convexity, and it grows with $\sigma^2$ and shrinks with $a^2$ because more volatility and weaker mean reversion both mean more dispersion in the rate path.

This is the same effect as Chapter 30's, arriving automatically rather than as an adjustment added to it. There, a convexity correction had to be computed and added; here the model produces it as a consequence of its own dynamics. That is a genuine attraction of an equilibrium model, and it is worth noticing before the criticism.

*Why an equilibrium model cannot match an observed term structure.* Because it has too few degrees of freedom, and the shortfall is not close.

Vasicek generates the *entire* zero curve from $r_0$ and three constants $(a, b, \sigma)$, four numbers in total. An observed term structure is dozens of independent quotes: a two-year rate, a five-year rate, a ten-year rate, a thirty-year rate, and every point between, each set by supply and demand in its own segment. Four numbers cannot reproduce dozens of independent observations except by coincidence.

*And this is arithmetic rather than calibration.* No optimizer, no objective function and no amount of effort changes the count. The model spans a four-parameter *family* of curves, and the market's curve almost certainly does not lie in that family, so the best fit is a fit with residuals, and the residuals are mispricings against instruments that trade.

Example 31.1 shows what the family looks like: from the same parameters the model produces upward-sloping, flat and downward-sloping curves depending on where $r_0$ sits relative to $b$, which is a genuinely appealing economic story. But every curve in the family converges to the same $R_\infty$, and the shapes available are exactly those the model's three constants permit. A hump of realistic size, or a kink at seven years, is simply not in the set.

*The consequence, and it is the whole reason for the rest of the chapter.* A model that does not reprice the curve cannot be used to price anything off the curve: the errors in the underlying contaminate every derivative. So the no-arbitrage generation replaced the constant $b$ with a function $\theta(t)$ chosen to fit the curve exactly (question 3), converting the model from a description into an interpolation device with a stochastic component.

*What is lost in that trade.* It is precisely the economic story. Vasicek *explains* why the curve slopes as it does; Ho–Lee and Hull–White *assume* it and say nothing about why. The exchange is explanation for exactness, and for pricing purposes exactness is what matters.

*What the question tests.* That $R_\infty < b$ is Jensen's inequality on bond prices, and that a four-parameter model failing to span a many-point curve is a counting fact rather than a fitting failure.

---

**31.3** *Explain what $\theta(t)$ does in Ho–Lee, why it carries no information, and state the defect that rules the model out in practice together with the quantity that exposes it.*

*What $\theta(t)$ does.* In Ho–Lee, $dr = \theta(t)dt + \sigma dW$, the drift is a *function of time* chosen

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so that the model reproduces today's observed curve exactly. Given the initial forward curve $f(0,t)$,

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + \sigma^2 t$$

It is not a parameter to estimate; it is solved for. One function of time, computed mechanically from the curve, converts a model that cannot price into one that can, and it is worth pausing on how cheap that is.

*Why it carries no information.* Because $\theta(t)$ is *whatever it has to be*. It is determined entirely by the input curve, so it makes no assertion about the world and can never be wrong. Whatever curve is observed, $\theta$ absorbs it; a different curve tomorrow produces a different $\theta$ with no implication that anything has been learned.

That is the essential difference from Vasicek's $b$, which is a constant with an economic meaning: a long-run level the rate reverts to, estimable from history, and capable of being wrong. Replacing $b$ with $\theta(t)$ buys an exact fit and *surrenders the explanation*: a model in this class says nothing about why the curve has the shape it has.

The right way to read every model in the rest of the chapter follows from this. They are **interpolation devices with a stochastic component**, not descriptions of the economy. That is not a criticism—for pricing a derivative off a curve, exactness is what matters and explanation is not—but it disciplines what one may conclude from a calibration. A fitted $\theta$ is not evidence about anything.

*The defect that rules Ho–Lee out:* ***no mean reversion***.

And the quantity that exposes it is the volatility of the $T$-maturity forward rate, which under Ho–Lee is $\sigma$ **regardless of** $T$. The model asserts that a thirty-year forward rate is exactly as volatile as a one-year forward rate.

That is straightforwardly false. Question 4's $\sigma e^{-aT}$ gives the Hull–White answer and the data support a substantial decay: long-dated forwards move much less than short-dated ones, because a shock to the short rate is expected to revert rather than persist. Example 31.3's table makes the comparison concrete. At $a = 0.01$, which is nearly Ho–Lee, a thirty-year forward is three-quarters as volatile as a one-year forward, and no rate market behaves that way.

*Why this defect is fatal rather than cosmetic.* A model overstating long-forward volatility will overprice every long-dated option, and it will do so systematically and in one direction. It cannot be repaired by calibration, because $\sigma$ sets the *level* of the volatility term structure and Ho–Lee has nothing that sets its *shape*, so fitting the long end then breaks the short end and vice versa. The fix requires a second parameter, which is Hull–White.

*What the question tests.* That an exactly-fitted function is unfalsifiable and therefore uninformative, and that a flat forward-rate volatility term structure is the observable Ho–Lee gets wrong.

---

**31.4** *Using $\sigma e^{-aT}$, compute the ten-year forward-rate volatility at $\sigma = 100$ bp for $a = 0.05$ and $a = 0.20$, and explain which feature of the volatility term structure is set by $a$ and which by $\sigma$.*

*The computation.* Under Hull–White the volatility of the instantaneous forward rate maturing at $T$, seen from today, is $\sigma e^{-aT}$. At $\sigma = 100$ basis points and $T = 10$:

| | Calculation | Ten-year forward volatility |
|---|---|---|
| $a = 0.05$ | $100 \times e^{-0.50}$ | **60.7** bp |
| $a = 0.20$ | $100 \times e^{-2.00}$ | **13.5** bp |

A factor of **4.5** between them, from a parameter that leaves the short end almost untouched: at one year the same two settings give 95.1 and 81.9 basis points, a difference of only 16%.

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*Which feature each parameter sets.*

*$\sigma$ sets the **level** of the volatility term structure.* It multiplies the whole curve, so doubling it doubles the volatility at every maturity and changes no ratio between them. It is read off the overall level of quoted cap and swaption volatilities.

*$a$ sets the **shape**.* It controls how fast volatility decays with maturity and nothing else. At $T = 0$ the factor $e^{-aT}$ is 1 whatever $a$ is, so mean reversion has no effect at the very short end and progressively more further out.

The two extremes bracket the sensible range and both are wrong:

- At $a = 0.01$ the curve is nearly flat. That is Ho–Lee, and it says a thirty-year forward is three-quarters as volatile as a one-year forward.
- At $a = 0.20$ the thirty-year forward has essentially no volatility at all (0.2 bp), which is equally wrong in the other direction.

*Why the separation matters.* Because the two parameters act on *different features*, they are separately identified by the data. Section 29.9's volatility cube contains caps and swaptions at many expirations: the *term structure* of their volatilities determines $a$, while the *level* determines $\sigma$. Two observables, two parameters, and no degeneracy, which is a much healthier situation than Chapter 21's Heston calibration, where no parameter moved one feature alone.

*Confusing them is the most common error in using the model.* A desk that raises $\sigma$ to fit long-dated options, when the real problem is that $a$ is too high, will break the short end; and one that reads a fitted $a$ as an economic estimate of how fast the central bank mean-reverts has mistaken a shape parameter for a forecast.

*The table conceals one caution.* The clean identification holds for the *volatility surface*. Question 8 shows that a *single* European swaption contains no information about $a$ at all: one option observes the curve once and says nothing about how successive dates relate. The cube identifies $a$ because it spans expirations; a single instrument does not.

*What the question tests.* Evaluating $\sigma e^{-aT}$, and that level and shape are controlled by different parameters and read off different features of the market.

---

**31.5** *Describe the two stages of the Hull–White tree construction, state which quantities are solved rather than fitted, and explain why the tree reproduces the initial curve to machine precision while still mispricing options.*

*The two stages.*

*Stage one: build a tree with no drift level.* Let $x$ follow $dx = -ax\,dt + \sigma dW$ with $x(0) = 0$, the mean-reverting part alone, centered on zero. Set the spacing $\Delta x = \sigma\sqrt{3\Delta t}$ and, at each node, choose branching probabilities matching the *mean and variance* of the move. This stage uses only $a$ and $\sigma$; it knows nothing about the curve.

*Stage two: displace the whole tree.* Shift every node at time $t$ by an amount $\alpha(t)$, so that $r = x + \alpha(t)$ reprices today's zero curve. The displacements are computed *forward*, one time step at a time, from the Arrow–Debreu prices already accumulated.

*Which quantities are solved rather than fitted: the displacements $\alpha(t)$.* At each step there is exactly one unknown, the shift at that step, and exactly one condition, that the tree reproduce the market discount factor $P(0, t)$. One equation, one unknown, no optimization and no residual. They are solved, not fitted, and the distinction is the same one question 3 drew about $\theta(t)$: the displacement is whatever it must be, carries no information, and cannot be wrong.

The parameters $a$ and $\sigma$ are the opposite: they are *not* determined by the curve, they are calibrated to option prices, and they can be wrong. That is the division the construction enforces: curve-fitting is exact and separate, volatility-fitting is approximate and separate.

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*Why the tree reproduces the curve to machine precision.* Because stage two *solves* for the displacements rather than searching for them. Each $\alpha(t)$ is the unique number making the sum of the Arrow–Debreu prices at that level equal the market discount factor, and the equation is solved exactly. Section 31.8 reports the result: errors of at most $1.1 \times 10^{-16}$, one part in $10^{16}$, which is floating-point rounding and nothing else. There is no fit quality to assess because there is no fitting.

*Why it can still misprice options.* Because the curve and the options are different information, matched by different mechanisms.

The displacements guarantee that today's *bond prices* are right. They say nothing about how much rates *move*, which is governed entirely by $a$ and $\sigma$, and those two constants must reproduce an entire volatility surface of caps and swaptions across expirations, tenors and strikes. Two numbers cannot match hundreds of quotes, so there will be residuals, and every residual is an option mispriced.

Two further sources compound it. The model has no smile: Hull–White's $r$ is Gaussian, so its implied volatilities, quoted in normal terms, are close to flat in strike, and the market's skew cannot be matched at any parameter setting. And the tree is a discretization, so even a perfectly calibrated model carries lattice error, which question 7 shows is first-order and not monotone in the step count.

*The lesson worth extracting.* An exact fit to one set of instruments is no evidence about another. A reader shown $10^{-16}$ errors on the curve might reasonably conclude the model is extraordinarily accurate; it is extraordinarily accurate about the thing it was solved to match, and that thing is the input rather than an output. Chapter 21's local-volatility surface is the same trap: an exact fit to today's vanillas, and a badly wrong forward smile.

*What the question tests.* That the construction separates an exactly-solved curve fit from an approximately-calibrated volatility fit, and that machine precision on the first says nothing about the second.

---

**31.6** *Explain why the branching pattern must change at $\pm j_{\max}$, and what would go wrong if it did not.*

*Why the pattern must change.* Mean reversion makes the standard up/middle/down branching fail far from the center of the tree.

At a node high above the mean, the drift $-ax$ is strongly *negative*: the process is expected to fall sharply over the next step. Symmetric branching offers three destinations (up one level, stay, down one level), and the probabilities must match both the mean and the variance of that move. Once the expected drop per step, $u = a\,j\,\Delta t$ in units of $\Delta x$, exceeds $\sqrt{2/3} = 0.8165$, no admissible set exists: the outer probabilities stay positive, but the middle one, $p_m = 2/3 - u^2$, becomes a **negative probability**.

The remedy is to change the branching at the level $j_{\max}$, the smallest integer with

$$j_{\max} \geq \frac{0.1835}{a\,\Delta t}$$

At the top node the pattern becomes *down-down-middle*, with all three destinations at or below the current level, and at the bottom node *up-up-middle*. Below $j_{\max}$ the ordinary symmetric pattern is used.

*The consequence is that the tree stops widening.* At $\pm j_{\max}$ every branch points inward, so no path can travel further from the center. Section 31.8's production tree has $j_{\max} = 10$ over 44 steps, so it spans 21 levels rather than 89.

And that is not an approximation—it is what mean reversion means, expressed in the geometry. A mean-reverting process genuinely cannot wander arbitrarily far: the further it goes, the harder it is pulled back, so its stationary distribution has finite variance. A tree that

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kept widening would be representing a process without that property. The truncation is the model being right, not the lattice being crude.

*What would go wrong if the pattern did not change.*

*Negative probabilities.* This is the immediate failure. Probabilities outside $[0, 1]$ mean the scheme is no longer computing an expectation, and the consequences are those of Chapter 20's question 10: the recursion *amplifies* rather than averages, so rounding error grows geometrically and the calculation explodes. It is the same disease as an unstable explicit finite-difference scheme, and for the same reason: a weighted average with a negative weight is not an average.

*And before it explodes, it is wrong.* At settings where the probabilities are merely implausible rather than negative, the tree misrepresents the process: it lets paths reach rate levels the model assigns negligible probability to, so option values pick up contributions from states that should not exist. Long-dated and deeply out-of-the-money instruments, which depend most on the tails, are affected most.

Note what fixes the constant in $j_{\max} \geq 0.1835/(a\Delta t)$. It is $1 - \sqrt{2/3}$, the value of $u$ at which the switched, down-down-middle probabilities first become admissible. Symmetric branching stays admissible up to $u = 0.8165$, so the switch can be made anywhere between the two, and taking it at the first legal level keeps the tree narrowest. Stronger mean reversion or a coarser time step therefore means a *narrower* tree, and the narrowing discards little: in the production tree the edge $j_{\max}\Delta x = 10 \times 0.00866 = 0.0866$ sits about three and a half stationary standard deviations ($\sigma/\sqrt{2a} = 0.025$) from the center. The constant is not a tuning choice; it is where the algebra first allows the switch.

*What the question tests.* That the drift outruns the fixed spacing far from the center, and that the truncation encodes mean reversion rather than approximating it away.

---

**31.7** *In Example 31.8, explain why the Bermudan is worth more than any single European and much less than their sum, and why the tree's convergence to the Jamshidian value is not monotone.*

*The numbers*, from Example 31.8, at 24 steps per year:

| | |
|---|---|
| European, 1y only | 1.1586 |
| European, 2y only | 1.1953 |
| European, 3y only | 1.0559 |
| European, 4y only | 0.7974 |
| European, 5y only | 0.4412 |
| Sum of all five | 4.6484 |
| **Bermudan, any one of the five** | **1.6995** |

*Why it exceeds any single European: the right to **choose** has value.* A European holder must commit today to one exercise date; a Bermudan holder decides at each date, having *seen* where rates have gone. Optionality over the timing is itself an option, and it is worth 42.2% above the best single European (1.6995 against 1.1953).

The comparison against the *best* one is the right one, since the holder could always replicate the strongest European by pre-committing, so the Bermudan is bounded below by 1.1953, and the excess is the value of not having to commit.

*Why it is far below their sum: the rights are **mutually exclusive**.* Owning all five Europeans means exercising up to five times; the Bermudan permits *one* exercise, after which it is gone. So the sum is an upper bound and a loose one: 1.6995 is 37% of 4.6484.

The bound is loose because the five Europeans are highly overlapping claims. They are options on nearly the same underlying, driven by one factor, so they pay off in largely the same states. Adding them counts the same scenario five times, and the Bermudan can monetize it once.

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So the two comparisons together locate the product. A Bermudan is not "several Europeans" and not "one European with flexibility." It sits between two bounds separated by a factor of four, and where it sits within them is exactly what a model must determine, which is question 1's point and question 8's problem.

*Why the convergence to the Jamshidian value is not monotone.* The closed-form European value is **1.1465** by Jamshidian's decomposition (a European option on a coupon bond is a portfolio of options on the zeros), and the tree approaches it as

| Steps/year | Tree | Error |
|---|---|---|
| 4 | 1.1637 | +0.0172 |
| 8 | 1.1696 | **+0.0231** |
| 12 | 1.1667 | +0.0202 |
| 24 | 1.1586 | +0.0121 |
| 48 | 1.1502 | +0.0037 |

*Eight steps a year is worse than four.*

Section 20.2 established why, and the explanation transfers unchanged: a lattice's error depends on where the nodes fall relative to the payoff's kink, and refining the grid *moves* them. A finer tree is not uniformly better; it is a different grid, whose alignment with the strike is essentially arbitrary. The error therefore need not fall monotonically (here it grows from four steps to eight before it turns), and the convergence is roughly first-order in $1/n$ overall.

*The practical warning.* A desk that extrapolates two tree values without checking for this will get a worse answer than either. Richardson extrapolation assumes the error is a smooth function of $1/n$; here it is not, and Chapter 20's question 1 shows an extrapolation across a parity change producing an answer off by twice its own inputs' error. The remedies are the same as there: extrapolate only along a consistent branch, or average adjacent step counts, or—best—check convergence against a closed form, which is precisely what the Jamshidian value is doing in this example.

*What the question tests.* That timing optionality places the Bermudan strictly between two wide bounds, and that lattice error is governed by node placement rather than by step count alone.

---

**31.8** *Explain why a single European swaption contains no information about the mean-reversion rate, use Example 31.9 to state the range of Bermudan values consistent with an exactly matched European, and describe three defensible responses and one that is not.*

*Why a single European contains no information about $a$.* A European swaption observes the curve **once**, at its expiration. Its value depends on the *distribution* of the swap rate at that one date, essentially on the total variance accumulated by then, and many $(a, \sigma)$ pairs deliver the same total variance. The mean reversion rate governs how variance is *distributed across maturities* and how successive dates *relate to one another*, and a single observation date sees neither.

Example 31.9 demonstrates it by construction. Hold the one-year-into-five-year European receiver fixed at its tree value of 1.1586; for each $a$, solve for the $\sigma$ that reproduces it exactly; then reprice the Bermudan:

| $a$ | $\sigma$ | European | Bermudan | vs. $a = 0.08$ |
|---|---|---|---|---|
| 0.01 | 0.00823 | 1.1586 | 1.6474 | −3.1% |
| 0.08 | 0.01000 | 1.1586 | 1.6995 | — |
| 0.30 | 0.01724 | 1.1586 | 1.8705 | +10.1% |

*Every row prices the calibration instrument identically, to four decimals.*

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The range of Bermudan values: **1.6474 to 1.8705**, a span of 0.2231, or **13.1%** of the product's own value, across parameter choices the market does not distinguish.

*The mechanism, and it is the reason two-factor models exist.* A Bermudan's value depends on the *relationship* between rates at different exercise dates: how much the curve at year three tells you about the curve at year four. Strong mean reversion means rates at successive dates are less tightly linked, so waiting is more likely to bring a different outcome and the choice among dates is worth more, which is why the Bermudan rises with $a$ in the table. That relationship is governed by $a$, and a single European contains no information about it.

*Three defensible responses follow.*

*1. Calibrate to a co-terminal strip.* Use the swaptions expiring on each of the Bermudan's exercise dates and maturing on its final date. This is standard practice, and it constrains $a$ far better than a single option because the strip observes several dates, which is exactly the information the single instrument lacked.

*2. Fix a from history and calibrate only $\sigma$.* This is cruder, and defensible when the strip is not liquid. Section 29.9 reaches the same conclusion about SABR's $\beta$: when the data cannot support a parameter, choose it outside the calibration rather than letting the fit invent it.

*3. Quote a range.* Report the Bermudan at both ends of the plausible $a$ and treat the spread as a model reserve. Chapter 21's question 10 argues for exactly this on barrier options, and the governance rules are the same: take the reserve through P&L at inception, and report the range alongside the mark.

*And one that is not defensible: calibrating a to the single European.* The optimizer will return a number, and it will be determined by the starting guess and the tolerance rather than by the data, because every $a$ fits equally well. The fit is perfect, the parameter is arbitrary, and the Bermudan price inherits the arbitrariness while carrying the authority of a calibration. A parameter the data cannot identify is being asserted, not measured, and letting an optimizer assert it merely hides who did.

*What the question tests.* That one observation date cannot identify a parameter governing the relation between dates, the size of the resulting price range, and the discipline of sourcing an unidentified parameter from outside the fit.

---

**31.9** *Explain the difference between a Z-spread and an option-adjusted spread, and what the difference between them measures.*

*The Z-spread* is the constant addition to the discount curve that reproduces the bond's market price **while ignoring the call**, valuing the bond as though its scheduled cash flows were certain. It is computed from the price and the curve alone: no model, no volatility, no assumption about rates. It is a fact about the price.

*The option-adjusted spread* is the constant addition that reproduces the price **with the call modeled**. Computing it requires a term-structure model, a lattice, mean reversion, volatility, and an exercise rule at every node. It is a fact about the price and a model.

*What the difference measures: the embedded option, expressed as a running yield.*

Example 31.10 makes it concrete. A ten-year 5% bond, callable at par annually from year three, trading at 99.00:

| | |
|---|---|
| Z-spread | 94.6 bp |
| Option-adjusted spread | 52.1 bp |
| **Difference** | **42.5 bp** |

The same bond priced with the call switched off is worth 106.9013 against 101.6743 with it on, so the embedded call is worth **5.2270** points of price, and 42.5 basis points a year is that value

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amortized over the bond's life.

*Why the distinction matters practically.* A portfolio manager comparing this bond with a non-callable one on **Z-spread** would see 94.6 basis points and conclude it offered generous credit compensation. Roughly half of that is payment for an option the investor has written, not compensation for credit risk, and the two are not comparable.

The OAS exists precisely to strip that out. It answers: after paying for the option, what spread is the investor actually earning for the issuer's credit? That is the number comparable across callable and non-callable bonds, and it is what the acronym is for.

*Which side each number errs on.* The Z-spread **overstates** the credit compensation on any bond with a call, always, because the call's value is being counted as spread. So ranking callable bonds by Z-spread systematically favors the ones with the most valuable options, the ones the investor has given away the most on.

*The limitation, which is question 10's subject.* The OAS is the better number and it is not a clean one. It inherits every assumption in the model that produced it, and question 10 shows it moving from 20.9 to 71.5 basis points across parameter choices no market quotes. The word "adjusted" does not convey how much of it is the model, and a reader should hold both halves: the Z-spread is observable and answers the wrong question; the OAS answers the right question and is not observable.

*What the question tests.* That the two spreads differ by the option's running value, and that one is computed from data while the other requires a model.

---

**31.10** *Using Example 31.10, state the OAS at $a = 0.08$ with $\sigma = 1.0\%$ and with $\sigma = 1.5\%$, and explain what this implies about comparing OAS across dealers.*

*The two figures*, from Example 31.10 at $a = 0.08$:

| $a$ | $\sigma$ | **OAS** |
|---|---|---|
| 0.03 | 1.0% | 39.5 bp |
| 0.08 | 1.0% | **52.1 bp** |
| 0.20 | 1.0% | 71.5 bp |
| 0.08 | 1.5% | **20.9 bp** |

Raising volatility from 1.0% to 1.5% at the same mean reversion cuts the OAS from **52.1** to **20.9** basis points, a fall of **60%**, from one parameter change.

*Why volatility moves it so hard.* The OAS is what remains of the spread after the embedded call has been paid for. A higher volatility makes the call *more valuable*, so more of the bond's yield is attributed to the option and less to credit. The relationship is direct and steep, because a callable bond's price is far more sensitive to volatility than to anything else the model contains.

Across the whole table the OAS ranges from **20.9 to 71.5 basis points** (a factor of nearly three and a half) across parameter choices no market quotes.

*What this implies about comparing OAS across dealers.* That the comparison is close to meaningless unless the models agree.

Two houses quoting 52 and 21 basis points on the same bond at the same price are not disagreeing about the bond. They are reporting the same observable through two different volatility assumptions, and neither is wrong: there is no market in $a$ and $\sigma$ to adjudicate. A trader treating the gap as an opportunity has mistaken a modeling difference for a mispricing.

*The practical rule follows immediately.* OAS comparisons are meaningful within a book run on one model with one calibration, and close to meaningless across two houses that have not agreed on $a$ and $\sigma$. Within a book, the model is a common yardstick and relative value is real; across books, it is not a yardstick at all.

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*Two riders are worth attaching.*

*The Z-spread, for all its faults, is a fact about the price.* It answers the wrong question (question 9) but it answers it identically at every institution. When comparing across dealers, a number that is wrong in a known and common way can be more useful than one that is right in an unknown and idiosyncratic way.

*And the honest disclosure is the parameters.* A quoted OAS should travel with the $a$ and $\sigma$ that produced it, exactly as an implied correlation should travel with the copula (Chapter 26's question 9) and a base correlation with its interpolation scheme. The number alone is not enough information to use it.

*The general pattern, which this chapter keeps returning to.* A quantity derived through a model inherits the model's unidentified parameters, and question 8 showed the same effect on a Bermudan's price: 13.1% across values of $a$ that the calibration instrument could not distinguish. Here the range is larger because the OAS is a *difference* between two nearly-equal quantities, so a modest change in the option's value is a large change in the residual.

*What the question tests.* That the OAS is a residual and therefore amplifies model error, and that a model-dependent number is only comparable within a common calibration.

---

**31.11** *Explain why a one-factor model forces perfect correlation between all rates, and why this is an assertion rather than an approximation.*

*Why one factor forces perfect correlation.* In every model in this chapter there is a single state variable, $r(t)$. At any future date, *every* rate on the curve is a deterministic function of that one number: the two-year rate is $g_2(r(t))$, the ten-year rate is $g_{10}(r(t))$, and both functions are increasing.

Two deterministic increasing functions of the same random variable are perfectly correlated. So:

- a two-year rate and a ten-year rate **cannot move in opposite directions**;
- they **cannot move by unrelated amounts**, since knowing one determines the other exactly;
- and their correlation is **exactly one**, at every horizon, for every parameter setting.

*Why this is an assertion rather than an approximation.* An approximation is a statement that gets closer to the truth as some parameter is refined, or that is accurate within a stated tolerance. This is neither. The model does not approximate a correlation of one; it asserts it, as a structural consequence of having one state variable, and no choice of $a$, $\sigma$ or $\theta(t)$ changes it. There is no limit in which the model gets this right.

That distinction matters because it determines what can be done. An approximation can be improved by better calibration or finer discretization; an assertion can only be removed by changing the model.

*Where it is tolerable.* For a swaption or a callable bond the payoff depends on *one* rate, or on a monotone function of the state, so the model's inability to represent curve shape never enters. A Bermudan swaption is a *level* product, and one factor is defensible for it (question 13).

*Where it is fatal.* For anything whose payoff depends on the curve's **shape**. A CMS spread option pays on the *difference* between two rates, and under perfect correlation that difference is nearly deterministic. Question 12 prices how fatal.

*And the empirical fact it contradicts.* Rates at different maturities are demonstrably not perfectly correlated: Chapter 32's principal-component analysis finds a first factor explaining most but far from all of the variance, with slope and curvature factors accounting for the rest.

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Those second and third factors *are* the curve's shape, and a one-factor model has set them to zero.

*The general lesson, and it recurs across the book.* A model's structure determines which questions it can answer, independently of how well it is calibrated. Chapter 21's local-volatility model fits every vanilla and predicts the wrong forward smile; Chapter 24's Merton model has the right shape in the middle and zero short-dated spreads; here a one-factor model prices every swaption and cannot price a spread. In each case the failure is structural, the fit is perfect, and no recalibration helps, which is why diagnosing *which* feature a model asserts rather than estimates is the most useful thing to know about it.

*What the question tests.* That one state variable implies deterministic co-movement, and the difference between a claim that can be refined and one that can only be replaced.

---

**31.12** *Using Example 31.11, state the CMS spread option's value at $\rho = 1$ and at $\rho = 0.70$, and explain why the one-factor error always has the same sign and why that sign is the dangerous one.*

*The two values.* From Example 31.11, a five-year at-the-money option on the ten-year minus two-year swap spread, forward rates 3.40% and 2.80%, normal volatilities 90 and 75 basis points, annuity 4.20:

| $\rho$ | Spread vol | Option value | Ratio |
|---|---|---|---|
| 1.00 | 15.0 bp | **56.2 bp** | 1.00 |
| 0.70 | 65.4 bp | **245.0 bp** | 4.36 |

A one-factor model is forced to the first row: at $\rho = 1$ the option is worth **56.2 basis points**. At a market-plausible correlation of 0.70 it is worth **245.0—4.4 times as much**.

*And the 15.0 is not even a correlation assumption.* At $\rho = 1$ the spread volatility collapses to a difference,

$$\sqrt{\sigma_{10}^2 + \sigma_2^2 - 2\rho\,\sigma_{10}\sigma_2} \;\longrightarrow\; |\sigma_{10} - \sigma_2| = |90 - 75| = 15 \text{ bp}$$

the residual from two rates moving in lockstep at slightly different speeds. The one-factor model is not making a poor estimate of the spread's volatility; it is reporting the arithmetic leftover of having no spread risk at all.

*Why no calibration repairs it.* One could force a one-factor model to reproduce 245 basis points by inflating $\sigma$, but then every swaption in the book would be mispriced, because $\sigma$ sets the level of all rate volatility simultaneously. The error is in the model's **structure**, not its parameters, which is exactly the distinction Chapter 21 drew for equity models and question 11 states in general.

*Why the error always has the same sign.* Because $\rho = 1$ **minimizes** the spread's volatility. Inspect

$$\sigma_{\text{spread}} = \sqrt{\sigma_{10}^2 + \sigma_2^2 - 2\rho\,\sigma_{10}\sigma_2}$$

It is decreasing in $\rho$ throughout, reaching its smallest value at $\rho = 1$. So a one-factor model, forced to $\rho = 1$, produces the *lowest possible* spread volatility consistent with the two rates' own volatilities, and therefore the lowest possible option value. A one-factor model always undervalues a spread option, never overvalues it, for any parameters whatever.

*Why that sign is the dangerous one.* Because dealers are **short** these options. CMS spread options are among the most widely sold structured coupons (the building blocks of steepener notes, sold to investors seeking yield), so the desk's position is short the option and long the model.

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A desk short these options on a one-factor model is systematically short at a price it computed to be too low, which is the worst configuration a model error can take. It combines four bad properties:

- the position is *short*, so the loss is unbounded in the direction of the error;
- the error is *one-directional*, so it never averages out across trades;
- it is invisible to the calibration, since every swaption reprices correctly;
- and it grows with the size of the business, because the same model prices every ticket.

Had the sign run the other way, overvaluing the option, the desk would simply have failed to win business, which is a survivable outcome. Undervaluing means winning *all* the business, at a price that is wrong, in the same direction, every time.

The remedy is a second factor (Section 31.13), and question 13's rule states when it is required: use one factor when the payoff depends on the *level* of rates, two when it depends on their *shape*.

*What the question tests.* That $\rho = 1$ minimizes a spread's volatility so the bias is signed, and that a signed error on a systematically short book is the configuration to fear.

---

**31.13** *State the decision rule for choosing between a one-factor and a two-factor model, and give one product on each side of it.*

*The decision rule.*

*Use one factor when the payoff depends on the **level** of rates,*
*and two when it depends on their **shape**.*

*On the one-factor side: a **Bermudan swaption**.* Its payoff at each exercise date is $A\max(S-K,0)$, a function of a single swap rate. The exercise decision compares that one rate against a strike, so everything the product depends on is a monotone function of the state variable. A one-factor model represents it faithfully, and question 11's perfect-correlation assertion never enters because no two rates are being compared.

The same applies to callable bonds, caps, floors, and ordinary swaptions. These are among the largest users of interest-rate models by notional, and one factor is genuinely adequate for them.

*On the two-factor side: a **CMS spread option**.* Its payoff is $\max(S_{10}-S_2-K,0)$, a function of the *difference* between two rates. Under perfect correlation that difference is nearly deterministic, so a one-factor model prices the option at the residual $|\sigma_{10}-\sigma_2|$ and undervalues it by a factor of 4.4 (question 12). No calibration repairs it, and the error is signed.

*Why the rule is the right one.* A one-factor model has exactly one source of uncertainty, so it can represent exactly one degree of freedom in the curve's movement. That degree of freedom is the *level*. Any product whose value depends on a second degree of freedom (slope, curvature, the relation between two points) is asking the model for information it does not contain.

*The two-factor repair, and its costs.* The standard construction adds a stochastic mean-reversion level:

$$dr = [\theta(t) + u - ar]\,dt + \sigma_1 dW_1, \qquad du = -bu\,dt + \sigma_2 dW_2$$

with $dW_1 dW_2 = \rho\,dt$; equivalently, $r$ as the sum of two mean-reverting factors, the **G2++** form. Now short and long rates are driven by different combinations of two shocks, so they are

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imperfectly correlated and the curve can steepen, flatten and twist. The model remains affine, so bond prices stay closed-form.

The costs are real and should be weighed. Five parameters instead of two, with a correspondingly less stable calibration, the problem question 8 already identified with two, and Chapter 21's Heston account in another market. And the lattice becomes two-dimensional: a $3 \times 3$ branching structure whose node count grows as the square, which is why two-factor implementations usually price by Monte Carlo or by finite differences on a two-dimensional grid rather than on a tree, and that sacrifices the natural early-exercise handling which was the short-rate model's main advantage.

So the rule is not "more factors are better." It is that the number of factors should match the number of degrees of freedom the payoff depends on. A two-factor model priced on a Monte Carlo engine is a worse tool for a Bermudan than a one-factor tree, despite being more realistic: slower, harder to calibrate, and requiring regression for the exercise decision.

*What the question tests.* Matching model dimension to payoff dimension, and that the extra factor buys shape at the cost of the tractability that made the class attractive.

---

**31.14** *Explain why Gaussian short rates were regarded as defective before rates went negative and as necessary afterward, and name four properties that keep short-rate models in use together with the single property that rules them out.*

*Why Gaussian short rates were regarded as defective before rates went negative.* Hull–White's $r$ is **normally distributed**, so it goes negative with positive probability, and the probability is not small. Example 31.12, on a flat 3% curve:

| | 1y | 2y | 5y | 10y |
|---|---|---|---|---|
| $a = 0.05$, $\sigma = 1.0\%$ | 0.1% | 1.3% | 6.5% | 11.6% |
| $a = 0.05$, $\sigma = 1.5\%$ | 2.0% | 6.9% | 15.7% | **21.3%** |

At ten years the $\sigma = 1.5\%$ row assigns better than one chance in five to a negative short rate. This was cited as the model's principal flaw and as the reason to prefer CIR, whose square-root diffusion keeps $r$ positive, or a lognormal short rate.

*Why it became necessary afterward.* Rates went negative and stayed there for years. A Gaussian model priced the market that existed; a positive model could not be calibrated to it at all. CIR and lognormal models do not merely misprice negative rates—they cannot represent them, so the calibration has no solution. The property that had been the objection became the reason Hull–White survived while its competitors did not.

*The general lesson*, and it is the one Section 19.9 drew about 1987 and Chapter 26's question 7 drew about house prices: a constraint imposed because the unconstrained case "cannot happen" is a statement about the sample, and the sample is always shorter than the future. A model's implausible region is where its next test comes from.

*Where a floor is genuinely wanted*, the standard repair is a **shifted** model (apply a positive model such as CIR or a lognormal short rate to $r + s$ for a constant shift $s$, which moves the lower bound from zero to $-s$ rather than removing it), or the displaced and normal-versus-lognormal machinery of Section 29.8, which is the same idea applied to the quoting convention rather than to the process.

*Four properties that keep short-rate models in use.*

*1. They price American and Bermudan exercise naturally.* A tree gives the continuation value at every node, so an early-exercise decision costs nothing extra. A forward-rate model does not, and needs the regression machinery of Section 20.11. For the callable and Bermudan books, among the largest users of interest-rate models by notional, this is decisive.

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*2. They are fast.* A 44-step tree prices a Bermudan in milliseconds, which matters when a desk revalues thousands of positions and computes their sensitivities every night.

*3. They have few parameters*, which is a defect when the market is rich and a virtue when it is not. Question 8 showed that two parameters were already one too many for the available data.

*4. And they handle path dependence in the short rate directly*, which the compounded fixings of the post-transition market require.

*The single property that rules them out: the* ***one-factor defect****.* If the payoff depends on the *shape* of the curve rather than its level, a one-factor short-rate model is the wrong tool and no amount of calibration will make it the right one (questions 11 and 12). The case against is that and nothing else, which is worth stating precisely, because the class is often dismissed as outdated when its actual limitation is narrow, diagnosable in advance, and irrelevant to most of the notional it prices.

*What the question tests.* That an objection based on an unobserved region is an argument about the sample, and that this model class has one specific disqualifying limitation rather than a general obsolescence.

---

**31.15** *Using Example 31.5, state the caplet's value under the forward measure and under the risk-neutral measure and comment on their agreement, then explain what the 2.52% error in part (c) consists of and its sign.*

*The two values.* From Example 31.5, a caplet on the simple rate over $[2.0, 2.5]$ paid at 2.5, struck at 4%, notional 100, in a Hull–White economy with $a = 0.08$, $\sigma = 1\%$:

| | |
|---|---|
| (a) Under the 2.5-year forward measure, closed form | **0.22478353** |
| (b) Under the risk-neutral measure, 2 million antithetic paths | **0.22498411** |
| | (standard error 0.00023576) |

*The agreement.* The difference is 0.00020058, which is

$$\frac{0.00020058}{0.00023576} = \mathbf{0.85} \text{ standard errors}$$

Well inside sampling noise—this is the theorem confirmed rather than approximated. The numeraire is a choice of units and prices do not depend on it.

The comparison also shows what the choice buys. Route (a) is a closed form evaluated in microseconds, because under the forward measure the forward rate is a martingale (no drift to model) and the caplet is a European option on a zero-coupon bond, which is lognormal in this economy. Route (b) required simulating the short rate along two million paths and discounting each payoff pathwise. Same answer, six orders of magnitude apart in effort.

*What the 2.52% error in (c) consists of.* Part (c) computes the expected payoff under the risk-neutral measure and discounts it at today's zero rate, treating the payoff and the discount factor as independent. That gives 0.23044665, an error of +2.52%.

*That 2.52% is exactly the covariance between the payoff and the discount factor.* The correct risk-neutral valuation is

$$V_0 = E^{Q}\left[e^{-\int_0^T r}\,\text{payoff}\right] = E^{Q}\left[e^{-\int_0^T r}\right] E^{Q}[\text{payoff}] + \text{Cov}^{Q}\left(e^{-\int_0^T r},\ \text{payoff}\right)$$

and (c) drops the last term.

*Its sign is negative, so the error is positive.* A caplet pays when **rates are high**, and when rates are high the discount factor is **small**. The two move in opposite directions, the covariance is negative, and omitting it *overstates* the value. That is not a small effect on a two-year option in a low-volatility rate environment: 2.52% is many times the bid–ask on such a contract.

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*And this is precisely what the forward measure exists to prevent.* Using $P(\cdot, T_2)$ as numeraire folds the discounting *into* the numeraire, so $N(T_2) = 1$ and no covariance term can survive:

$$V_0 = P(0, T_2)\, E^{T_2}[\text{payoff}]$$

The expectation and the discount factor are separated by construction rather than by assumption.

Which answers a question a reader may have carried since Chapter 29: the reason Black's formula for a caplet is *exact* rather than approximate is that the market convention already made this choice: discounting at $P(0, T_2)$ and taking an expectation of the payoff alone is the forward-measure formula, arrived at by convention rather than by derivation.

*The general moral, in one sentence.* A measure change is not a mathematical formality; it is the difference between an exact formula and a systematic 2.52% error.

*What the question tests.* That invariance is checkable numerically, and that the quantity a measure change removes is a covariance with a determinable sign.

---

**31.16** *Using Example 31.6, state the range of $E^Q[S_T]/F(0,T)$ across the correlations shown, explain what quantity produces it, and name the two other results in this book that are the same quantity.*

*The range.* From Example 31.6, with $S_0 = 100$, equity volatility 25%, $T = 5$, and $P(0,T) = 0.824639$ so that the forward price is $F(0,T) = 100/0.824639 = 121.2652$:

| $\rho(S,r)$ | $E^Q[S_T]/F(0,T)$ | **Gap** |
|---|---|---|
| −0.50 | 0.989442 | −1.056% |
| 0.00 | 1.003125 | +0.313% |
| +0.50 | 1.016997 | +1.700% |

The ratio runs from **0.9894 to 1.0170**, a spread of 2.76% across the correlations shown.

*What produces it: the covariance between the asset and the discount factor.* The forward price is a *no-arbitrage fact* (buy the asset, finance it, deliver it), and it does not depend on any correlation. $E^Q[S_T]$ is an *expectation*, and under the risk-neutral measure the numeraire is the stochastic money-market account, so

$$S_0 = E^Q\left[e^{-\int_0^T r}\, S_T\right] = E^Q\left[e^{-\int_0^T r}\right] E^Q[S_T] + \mathrm{Cov}^Q\left(e^{-\int_0^T r},\, S_T\right)$$

The forward is $S_0/P(0,T)$, so the two coincide only when the covariance vanishes, and in general that requires deterministic rates. Zero correlation is not enough: the asset's risk-neutral drift is the short rate itself, so $S_T$ and the discount factor still covary at $\rho = 0$ through the discount factor's own convexity, and that row sits 0.31% above one.

*The two other results in this book that are the same quantity.*

*1. The 2.52% error of Example 31.5* (question 15), where computing the caplet's expected payoff under $Q$ and discounting at today's zero rate ignored the covariance between the payoff and the discount factor.

*2. Section 30.3's futures–forward convexity adjustment*, where daily settlement means gains are reinvested and losses financed at the prevailing rate, so a futures price differs from a forward price by a term set by the covariance between the underlying and the short rate, above it when that covariance is positive.

Three results that look unrelated are one covariance term appearing under three names, and that is the clearest single argument for learning the measure machinery rather than the formulas. Under the $T$-forward measure all three vanish: with $P(\cdot, T)$ as numeraire, $E^T[S_T] = F(0,T)$ exactly, the caplet formula is exact, and there is no convexity adjustment because the quantity being adjusted for has been absorbed into the unit of account.

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*A practical warning follows.* A Monte Carlo engine that simulates under the risk-neutral measure and reports "the expected terminal price" is not reporting the forward. At $\rho = 0.5$ over five years the difference is 1.7%, larger than the bid–ask on the forward it is being compared with, and large enough that a reconciliation break will be blamed on the simulation rather than on the question it was asked.

And note which of the two quantities is model-independent. The forward price is fixed by arbitrage; $E^{Q}[S_T]$ moves with a correlation no one observes directly. A desk quoting a forward is quoting a fact; a desk quoting an expected terminal price is quoting a model output, and the two should never appear in the same column without a label.

*What the question tests.* That the forward and the risk-neutral expectation differ by a covariance, and recognizing one quantity behind three separately-named results.

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# Chapter 32
# HJM, LMM, and Multi-Curve Term Structure Models

**32.1** *Explain what the state variable is in an HJM model, why today's curve poses no fitting problem, and state the HJM drift condition in words together with what it says about the model's degrees of freedom.*

*The state variable: the* ***entire forward curve***. HJM does not model a short rate and derive the curve from it. It takes $f(t,T)$, the instantaneous forward rate for every maturity $T$, as the object that evolves. The state is therefore a *function*, not a number, which is both the framework's power and the reason question 2 says it cannot be put on a tree.

*Why today's curve poses no fitting problem.* Because it is the **initial condition**. The model evolves $f(t,T)$ forward from $f(0,T)$, and $f(0,T)$ is whatever the market says it is. There is nothing to fit: no $\theta(t)$ to solve for as in Ho–Lee, no displacement to compute as in the Hull–White tree, no residual to minimize. The curve is an input.

That is a genuine simplification over Chapter 31, where fitting the curve required a separate construction, and it is worth noticing that the simplification comes from choosing a richer state rather than from a cleverer technique.

*The drift condition, in words.* Once the volatility functions are specified, the drift is determined. No-arbitrage leaves no freedom in it:

$$m(t,T) = \sum_{k=1}^{K} s_k(t,T) \int_t^T s_k(t,\tau)\, d\tau$$

Each forward rate's drift is its own volatility multiplied by the integral of that volatility over the maturities between now and $T$, a quantity built entirely from the volatility structure.

*What it says about the model's degrees of freedom: there is exactly* ***one*** *free input.* Choose the volatility structure and everything else follows: the drift, the bond prices, every derivative price. Nothing else may be specified, and nothing else needs to be.

*Two things are worth extracting from the derivation.*

*It used almost nothing.* Only that bond prices are traded, and that $f(t,t) = r(t)$. No equilibrium argument, no utility function, no assumption about preferences. That is why the condition binds *every* arbitrage-free term-structure model rather than a particular family of them.

*And it is Chapter 13's result in a new setting.* There, a derivative's price does not depend on the underlying's expected return; here, drift is not an independent degree of freedom once no arbitrage is imposed. Drifts are theory and volatilities are data. Chapter 28's question 8 states the general principle and this is its sharpest instance.

*The consequence that organizes the rest of the chapter.* If volatility is the only input, then modeling is entirely about choosing volatility structures, and calibration is entirely about identifying them from prices. Questions 6 through 9 are that program, and question 2 explains why the choice is constrained by tractability as well as by fit.

*What the question tests.* That the state is the whole curve so the initial curve is an input rather than a fit, and that no-arbitrage collapses the model's freedom to a single function.

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**32.2** *Explain in what sense Ho–Lee and Hull–White are HJM models, why a general HJM model cannot be put on a recombining tree, and identify the volatility structure that makes one Markov.*

*In what sense they are HJM models.* They are HJM models with particular volatility structures. Question 1 established that specifying $s(t,T)$ specifies everything, so any arbitrage-free term-structure model *is* an HJM model with some volatility function, and Chapter 31's two are the simplest cases:

| Model | HJM volatility structure |
|---|---|
| Ho–Lee | $s(t,T) = \sigma$ |
| Hull–White | $s(t,T) = \sigma e^{-a(T-t)}$ |

Ho–Lee's flat structure is exactly Chapter 31's question 3 complaint (every forward equally volatile), and Hull–White's exponential decay is the $\sigma e^{-aT}$ of that chapter's question 4, now visible as a *choice* rather than a property of the model.

So Chapter 31 was HJM all along, with a volatility structure chosen for tractability rather than for fit. The two chapters are not competing frameworks but one framework and a family within it.

*Why a general HJM model cannot be put on a recombining tree.* Because it is generally **non-Markovian** in any finite-dimensional state.

The drift condition makes each forward rate's evolution depend on the volatility structure integrated over maturities, and once the volatility is allowed to depend on the current curve or on time in a general way, the future evolution depends on the *entire path* the curve has taken, not on any finite summary of where it is now. A tree node represents a state; if the state is the whole history, an up-then-down move and a down-then-up move reach genuinely different states, so the tree cannot recombine. Node count grows like $2^n$ and the method is unusable beyond a small number of steps.

This is the same failure as Chapter 20's cash-dividend tree and Chapter 21's rough volatility, and it has the same shape: a process whose drift depends on its history has no state variable to put on a lattice.

*The volatility structure that makes one Markov:* ***separable exponential***, of the form

$$s(t,T) = \sigma(t)\, e^{-a(T-t)}$$

Under this structure the entire curve at any date is a deterministic function of a *single* number, the short rate, so the state collapses from a function to a scalar, the process is Markov, and a recombining tree exists. Hull–White is precisely this case with $\sigma$ constant, which is why Chapter 31 could build a tree at all.

The generalization is the multi-factor version: a sum of $K$ separable exponential terms gives a $K$-dimensional Markov state, which is the G2++ of Chapter 31's question 13.

*The trade-off this exposes, and it is the chapter's organizing tension.* Markovian structures buy lattices, speed, and natural early exercise; they cost realism, because the exponential form is chosen for its mathematical convenience and not because forward-rate volatilities have that shape. Non-Markovian structures buy fit and cost the lattice, which is why the LIBOR market model of the next section is simulated rather than lattice-priced, and why Chapter 31's models survive for callable books (question 16).

*What the question tests.* That Chapter 31's models are HJM special cases, that non-Markovian dynamics is what breaks recombination, and that separable exponential volatility is

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exactly the condition restoring it.

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**32.3** *Explain why the LIBOR market model reproduces Black's caplet formula by construction rather than by calibration, and why the forward rates cannot all be martingales under one measure.*

*Why Black's caplet formula is reproduced by construction.* The LMM *posits* that each forward rate $F_k$ is lognormal under its own forward measure $Q^{k+1}$:

$$dF_k = \sigma_k(t) F_k \, dW^{k+1}$$

with zero drift, which Chapter 29's question 1 established is not an assumption but a consequence of choosing $P(\cdot, T_{k+1})$ as numeraire.

A caplet on $F_k$ pays at $T_{k+1}$, so it is valued under exactly that measure, where $F_k$ is a driftless lognormal. Black's formula *is* the value of a call on a driftless lognormal. So the model reproduces the market's caplet formula because it assumes the distribution the formula assumes, under the measure the formula uses.

*Why that is "by construction" rather than "by calibration."* A calibrated model reproduces prices because parameters were chosen to make it do so, with a residual. Here there is no optimization and no residual: equation (32.8) determines each forward's total variance directly from its quoted caplet volatility,

$$\sigma_k^{\text{Black},2} \, T_k = \int_0^{T_k} \sigma_k(t)^2 \, dt$$

and any instantaneous volatility function satisfying that integral reprices the caplet *exactly*.

This is the model's chief practical attraction, and it is worth being precise about what it buys: the desk's most liquid quotes are matched perfectly, with no fitting error to explain, before any harder calibration begins. Question 9 makes it the first of three stages, and it is the only one that is exact.

*Why the forward rates cannot all be martingales under one measure.* Because each is a martingale under a *different* numeraire, and there is only one numeraire in a simulation.

$F_k$ is a martingale under $Q^{k+1}$; $F_j$ is a martingale under $Q^{j+1}$; and these are different measures whenever $j \neq k$. A Monte Carlo path must evolve *all* the forwards together, under *one* measure, so at most one of them can retain a zero drift, and the rest acquire drifts equal to their covariance with the log of the numeraire ratio.

Under the terminal measure, with $P(\cdot, T_N)$ as numeraire, the drifts are

$$\mu_k(t) = -\sigma_k(t) \sum_{j=k+1}^{N-1} \frac{\rho_{kj} \, \tau_j \, F_j(t) \, \sigma_j(t)}{1 + \tau_j F_j(t)}$$

Two checks confirm it. At $k = N - 1$ the sum is *empty* and the drift vanishes: the longest forward is a martingale under its own measure, as it must be. And the drift is negative, because moving each forward to a longer numeraire discounts it over an additional stochastic period.

*Three consequences follow.* The drift is **state-dependent**, since $F_j$ appears on the right, which is what makes the simulation non-trivial and is the source of question 10's discretization error. It is a no-arbitrage consequence, not a parameter, the same statement as question 1's HJM condition, in discretized language. And simulating under the *spot* measure instead gives sums over $j \leq k$ with the opposite sign: neither choice is more correct, and implementations pick whichever puts the zero drift where the payoff is most sensitive.

*What the question tests.* That the LMM's assumption is exactly Black's, and that a single simulation measure forces drifts onto every forward but one.

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**32.4** *Compute $\tau F/(1+\tau F)$ at $F = 3\%$ and at $F = 8\%$ with annual accrual, and explain the relevance to the drift.*

*The computation*, with annual accrual so $\tau = 1$:

| $F$ | Calculation | $\tau F/(1+\tau F)$ |
|---|---|---|
| 3% | 0.0300/1.0300 | **0.0291** |
| 8% | 0.0800/1.0800 | **0.0741** |

*The relevance.* This factor appears in *every summand* of the LMM drift,

$$\mu_k(t) = -\sigma_k(t) \sum_{j=k+1}^{N-1} \frac{\rho_{kj}\,\tau_j F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)}$$

so it scales the entire drift. Its smallness is the point: at a 3% rate each term in the sum is roughly **three percent** of a product of two volatilities. The drift is two orders of magnitude below the diffusion.

*Three consequences follow, in increasing order of usefulness.*

*1. The drift is a small correction, not a driver.* A simulation that got the drift badly wrong would still produce approximately the right answer at low rates, because the diffusion dominates. That is reassuring about robustness and is also a warning: an implementation error in the drift may not show up in a low-rate test case.

*2. It explains question 10's finding.* The predictor–corrector scheme exists to correct the *frozen-drift* approximation, re-evaluating the state-dependent drift at the predicted point rather than at the start of the step. Correcting a term that is 3% of the dynamics cannot move the answer measurably, and Example 32.4 confirms it: Euler and predictor–corrector agree to a fraction of a basis point at every step count. The residual bias comes from the frozen volatility, not from the frozen drift.

*3. And it makes the rule conditional rather than universal.* The factor at 8% is **2.5 times** its value at 3%, and it keeps rising: at 15% it is 0.130, at 25% it is 0.200. So the drift correction earns its cost when the drift is large (high-rate currencies, emerging markets, and many forwards between the simulated rate and the numeraire) and not otherwise.

Note also that $\tau$ does not work the same way. A semiannual accrual roughly halves the factor, but over the same stretch of curve it also doubles the number of terms in the sum, so the total drift is nearly unchanged: the sum approximates an integral over the maturities out to the numeraire, which is the HJM drift again. Annual and semiannual conventions at the same rate carry about the same drift; what moves it is the rate level and the length of curve.

*Where the factor comes from.* It is the diffusion coefficient of $\ln(1+\tau_k F_k)$, which is ln of the ratio $P(t,T_k)/P(t,T_{k+1})$, the numeraire ratio between adjacent forward measures. Itô gives $\tau F\sigma/(1+\tau F)$, and walking from one measure to the next accumulates one such term per step (question 3). The factor is the price of one change of numeraire, and its smallness is the statement that adjacent forward measures are close together at low rates.

*What the question tests.* Evaluating the drift's scaling factor, and that its size determines whether a numerical refinement is worth implementing.

**32.5** *Using Example 32.1, state the share of variance explained by one and by three factors at $\beta = 0.05$, and explain why the often-quoted figure of 99% for three factors is misleading in this context.*

*The shares.* From Example 32.1, modeling correlation as $\rho_{ij} = e^{-\beta|T_i - T_j|}$ over thirty annual forwards, at $\beta = 0.05$:

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| $\beta$ | **1 factor** | **2** | **3** | **4** | **5** |
|---|---|---|---|---|---|
| 0.05 | **64.8**% | 82.2% | **88.6**% | 91.7% | 93.5% |

One factor explains **64.8%** and three explain **88.6%**.

The eigenvectors have the familiar interpretation: the first is a roughly parallel shift (**level**), the second changes sign once (**slope**), the third twice (**curvature**). Empirical $\beta$ for major currencies falls around 0.05 to 0.10, putting the first factor at 47–65% and the first three at 78–89%.

*Why the often-quoted figure of 99% is misleading here.* Because it is computed on a **different object**. The familiar "three factors explain 99% of the curve" comes from principal components of **yield levels**, which are dominated by a common trend: rates move up and down together, so a single level factor absorbs almost everything and the residual is small.

The table above is computed on the **correlation structure**, which is what actually governs a spread option. And on that object three factors explain closer to 80%, not 99%.

The difference matters because the residual is precisely what the product is written on. A CMS spread option pays on $S_{10} - S_2$. The level factor, the one that dominates the 99% figure, contributes almost nothing to that difference, because a parallel shift moves both rates equally and leaves the spread unchanged. The spread lives entirely in the variance the level factor does not explain, so quoting a statistic dominated by the level factor is quoting the one part of the decomposition irrelevant to the payoff.

Chapter 31's question 12 prices the consequence: a one-factor model values a CMS spread option at 56.2 basis points against 245.0 at a plausible correlation, a factor of 4.4, and 64.8% of variance explained sounds comfortable while producing that.

*The working answer.* Three factors for most books and two for many, with the caveat that the marginal factor is cheap in an LMM (adding one costs a little simulation time and no structural change), so the argument for three rather than two is easier to make than the argument for two rather than one.

*The general lesson, and it recurs.* A summary statistic is only as relevant as the object it summarizes. A variance share computed on levels answers a question about levels. The same trap appears in Chapter 29's flat-versus-spot volatilities, where an average over caplets is not a statement about any caplet, and in Chapter 23's contract counts. Before quoting a percentage, ask what it is a percentage of.

*What the question tests.* Reading a variance-decomposition table, and that a statistic computed on levels does not describe the correlation structure a spread option depends on.

---

**32.6** *Distinguish flat, spot, and instantaneous volatility, state which the market quotes, and explain why the market determines the integral of instantaneous variance but not its shape.*

*The three objects.*

*Flat volatility* (Section 29.5): the single number which, applied to *every* caplet in a cap, reproduces that cap's market price. A quotation device, not a property of any caplet.

*Spot volatility*: the Black volatility of *one* caplet, the piecewise-constant number that prices that individual option.

*Instantaneous volatility* $\sigma_k(t)$: the volatility of forward rate $k$ *at each moment of calendar time*. This is the LMM's own object, and it is a function rather than a number.

*Which the market quotes:* ***flat volatilities***, one per cap. Spot volatilities are extracted from them by the bootstrap of Chapter 29's question 5. Neither is the instantaneous volatility.

*Why the market determines the integral but not the shape.* Because a caplet's price depends

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only on the *total variance* accumulated to its expiration:

$$\sigma_k^{\text{Black},2}\, T_k = \int_0^{T_k} \sigma_k(t)^2\, dt$$

The option cannot distinguish *when* the variance arrived. A forward rate that is highly volatile for one year and quiet thereafter, and one that is uniformly volatile throughout, produce the *same* caplet price if their integrals match, so no caplet quote discriminates between them.

*The market determines the integral and not the integrand.* The shape of $\sigma_k(t)$ is therefore a modeling choice with **no caplet price to discipline it**.

*What does discipline it:* ***swaptions***. A swaption depends on how forward rates move *together over time*, so it is sensitive to when the variance arrives and how the rates are correlated while it does. That is why question 9's calibration fits the shape to swaptions after fixing the total variances from caplets; the two instrument classes carry different information and are used in that order.

*The standard parametrization* is Rebonato's, in which volatility depends only on time to maturity:

$$\sigma_k(t) = [a_R + b_R(T_k - t)]\, e^{-c_R(T_k - t)} + d_R$$

Four parameters produce a humped shape, which is what the data show: a forward rate two years out is more volatile than one thirty years out *and also* more volatile than one settling next week. The hump is a real empirical feature (uncertainty about near-term policy is low because policy is nearly known, and uncertainty about the distant future is damped by mean reversion), and a monotone parametrization cannot represent it.

*Why the freedom matters rather than being a technicality.* Two calibrations reproducing every caplet exactly can imply quite different dynamics, and therefore different prices for anything path-dependent or multi-date. This is Chapter 21's local-volatility problem in the rate market: an exact fit to today's quotes constrains the model far less than it appears to, and what is unconstrained is precisely the forward evolution that exotics depend on.

*What the question tests.* That three distinct volatility objects are in play, that only the integral of the third is observable from caplets, and that swaptions supply the missing shape information.

---

**32.7** *Explain why a time-homogeneous volatility parametrization is preferred, and relate this to Section 21.3.*

*What time-homogeneous means.* The instantaneous volatility depends only on **time to maturity** $T_k - t$, not on calendar time $t$ and $T_k$ separately. Rebonato's form

$$\sigma_k(t) = [a_R + b_R(T_k - t)]\, e^{-c_R(T_k - t)} + d_R$$

is time-homogeneous by construction: every forward traces the *same* volatility curve as it approaches its own fixing, merely displaced in time.

*Why it is preferred.* Three reasons, in increasing order of importance.

*1. It is what the data show.* The volatility of a forward rate depends on how far it is from fixing, not on which calendar year it is. A two-year forward behaves like a two-year forward whether observed today or in five years' time.

*2. It is parsimonious and identifiable.* Four parameters describe the whole surface, so they are determined by many instruments rather than one each. A non-homogeneous specification, with a separate volatility for each $(k, t)$ pair, has enough freedom to fit anything and is therefore constrained by nothing, which is question 6's freedom exploited rather than disciplined.

*3. And it is the only form under which the model's own forecasts are self-consistent.* This is the decisive reason. A time-homogeneous model predicts that the volatility *term structure*

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seen in five years will look like today's, which is what makes the model's statements about the future usable. A non-homogeneous fit implies a future term structure that differs from today's in whatever arbitrary way the calibration happened to require, and the desk is then holding a model that predicts its own inputs will change for no stated reason.

*The relation to Section 21.3.* This is exactly the local-volatility failure, in the rate market.

Chapter 21's local-volatility model fits today's vanilla surface *exactly* and, because $\sigma_{\text{loc}}(S,t)$ is a fixed function with no freedom left, makes a definite and testable prediction about the *forward smile*. The prediction is that the skew decays (by a factor of more than four between three months and three years), so a forward-starting option lives in a nearly flat world. *The market disagrees*: forward smiles are roughly stationary in moneyness, and a local-volatility model therefore marks forward-starting products too cheap and hedges them with the wrong vega profile.

The structural parallel is exact:

| | **Chapter 21** | **Here** |
|---|---|---|
| Exact fit to | today's vanilla surface | today's caplet volatilities |
| Unconstrained by the fit | the forward smile | the volatility shape |
| Non-stationary choice implies | a decaying forward smile | a changing volatility term structure |
| Product that pays for it | cliquets, forward-starts | swaptions, exotics on several dates |

In both cases the error is invisible to the calibration, because the calibration instrument prices correctly at every setting. And in both cases the repair is the same: impose *stationarity* as a modeling principle rather than letting the fit choose. Time-homogeneity in the LMM is the counterpart of the stationary forward smile that local volatility fails to produce, and Chapter 21's local-stochastic construction is the analogous repair there.

*One honest caveat follows.* Perfect time-homogeneity cannot generally fit the whole swaption surface, so implementations allow a small non-homogeneous correction on top of a homogeneous core, accepting a slightly worse claim about the future in exchange for a materially better fit today. The discipline is to keep the correction small and to know its size, which is the same governance rule as Chapter 21's mixing parameter.

*What the question tests.* That time-homogeneity is what makes a model's forward statements consistent with its present ones, and recognizing the local-volatility failure as the same defect.

---

**32.8** *Distinguish instantaneous from terminal correlation, state which one a spread option pays on, use Example 32.3 to give the maximum decorrelation produced by volatility shape alone, and explain why that result strengthens rather than weakens Chapter 31's criticism.*

*The distinction.*

*Instantaneous correlation* $\rho_{ij}$ is the correlation between the Brownian increments driving two forward rates *at a single moment*, a parameter of the model's dynamics.

*Terminal correlation* is the correlation between the two forward rates' *levels at a future horizon* $H$, accumulated over the whole path:

$$\frac{\rho_{ij}\int_0^H \sigma_i\sigma_j\,dt}{\sqrt{\int_0^H \sigma_i^2\,dt\int_0^H \sigma_j^2\,dt}}$$

These differ in general, because two rates can be instantaneously correlated while their volatility *shapes* differ (one front-loaded, one back-loaded), so their accumulated moves need not line up.

*Which a spread option pays on:* ***terminal*** *correlation.* A CMS spread option pays $\max(S_{10}-S_2-K,0)$ at one date. It depends on the joint distribution of the two rates *at that date*, which is terminal, not on how they co-moved instant by instant along the way.

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*The maximum decorrelation from volatility shape alone.* Example 32.3 isolates the channel by setting the instantaneous correlation to 1 and asking what the Rebonato shapes produce on their own:

| $T_i$ | $T_j$ | **Horizon** | **Terminal corr. at $\rho = 1$** |
|---|---|---|---|
| 2 | 3 | 2y | 0.9968 |
| 5 | 10 | 5y | **0.9936** |
| 10 | 20 | 5y | 0.9994 |

*At most **0.6%** of decorrelation*: the worst case, the five- and ten-year forwards over a five-year horizon, gives 0.9936 rather than 1.

This is a genuinely useful negative result, and not the one the literature emphasizes. The distinction between the two correlations receives extensive theoretical attention; the measurement says the gap is negligible for realistic volatility shapes.

*Two things follow.*

*In the LMM, the correlation you calibrate is the correlation you get.* Terminal correlation is instantaneous correlation multiplied by a factor very close to one. That makes the *correlation matrix* the object to spend calibration effort on (question 9's third and hardest stage), and it makes the model's spread-option prices directly interpretable: a price implies a correlation, without a shape-dependent translation in between.

*And it strengthens rather than weakens Chapter 31's criticism.* One might hope that a one-factor short-rate model, though it sets $\rho_{ij} = 1$ by construction, could recover some effective decorrelation through volatility shape, so that terminal correlations might be materially below one even with instantaneous correlation pinned at one. The table says no. With $\rho = 1$ the shape channel delivers at most 0.6%, so a one-factor model's terminal correlations are essentially 1 as well.

So Chapter 31's defect **cannot be repaired by clever volatility structures**. It requires more factors, which is what this chapter provides. Had the shape channel been large, the criticism would have been softened; because it is small, the criticism is confirmed and the remedy is narrowed to exactly one option.

*What the question tests.* That the two correlations differ in principle and barely in practice, and that the negative result closes off a possible defense of one-factor models.

---

**32.9** *Describe the three stages of LMM calibration, state which is exact and which is underdetermined, and explain what is lost when a full correlation matrix is reduced to K factors.*

*The three stages, in the order that matters.*

*Stage 1: caplet volatilities. **Exact**.* Equation (32.8) determines the total variance of each forward directly from the market's caplet quotes, and the LMM reproduces them by construction (question 3). No optimization is involved, which is the model's chief practical attraction: the most liquid quotes are matched perfectly with no residual to explain.

*Stage 2: the volatility shape. A **well-posed fit**.* The four Rebonato parameters are chosen to fit *swaption* volatilities (which, unlike caplets, depend on the shape and on correlation), with the total variances held fixed at their stage-1 values. This is a genuine optimization with a genuine residual, but it is well-posed: four parameters against many swaption quotes, so the parameters are identified.

*Stage 3: correlation. **Genuinely underdetermined**, and the hard part.* It can be estimated historically, parametrized (the $e^{-\beta|T_i - T_j|}$ form of Example 32.1 is standard), or implied from spread options where those are liquid, which is rarely. Historical and implied correlations differ *systematically*, and by more than the volatility analogs of Section 19.10, which should be

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expected: a correlation risk premium has the same logic as a volatility risk premium and *less to discipline it*, since far fewer instruments quote correlation directly.

So LMM calibration is one exact step, one well-posed fit, and one genuinely underdetermined estimate. A desk that reports the three separately is doing better than one that reports a single fitting error, because a combined residual conceals which of the three is failing, and they fail for different reasons and demand different responses.

*What is lost when a full correlation matrix is reduced to K factors.*

*A rank-K approximation does not reproduce the original entries.* A full $n \times n$ correlation matrix has $n(n-1)/2$ free entries; a $K$-factor representation has roughly $nK$. For thirty forwards that is 435 numbers compressed into perhaps 90, so most entries must change.

Example 32.1's table is exactly the statement of how much is lost. At $\beta = 0.05$, three factors capture 88.6% of the variance, so **11.4% of the correlation structure is discarded**, and question 5 established that the discarded part is not randomly distributed. The retained factors are level, slope and curvature, in that order; what is lost is the higher-order relative movement between nearby rates.

*Three specific consequences follow.*

*Correlations between* adjacent *maturities are distorted most.* A low-rank approximation reproduces the broad structure (distant rates are less correlated than near ones) and smooths away the fine detail, which is precisely the detail a short-dated spread option depends on.

*The reduction always* raises *correlations among nearby rates*, because dropping factors removes sources of independent variation. So factor reduction biases spread-option prices *downward*, in the same direction as Chapter 31's one-factor error and for the same reason, merely by less.

*And the loss is invisible in the calibration.* Swaption prices depend on averages of correlations and are relatively insensitive to the rank reduction; spread options depend on differences and are not. A calibration fitted to swaptions will look fine at any $K$.

*The practical response* is to choose $K$ by reference to the *product* rather than to a variance threshold (three factors for a book containing spread options, fewer if not) and to price the most correlation-sensitive positions at two ranks to measure the gap, which is Chapter 21's model-reserve discipline applied to a different parameter.

*What the question tests.* That the three stages differ in how determined they are, and that rank reduction discards exactly the structure the correlation-sensitive products depend on.

---

**32.10** *Using Example 32.4, state the change in implied volatility between 4 and 200 time steps, explain why the predictor–corrector makes no difference there, and state the conditions under which it would.*

*The change.* From Example 32.4, a one-year option on a five-year swap, five annual forwards flat at 3%, instantaneous correlation 0.85, four hundred thousand antithetic pairs:

| Steps | Scheme | Implied vol |
|---|---|---|
| 4 | Euler | 12.5200% |
| 4 | Predictor–corrector | 12.5185% |
| 200 | Euler | 12.5812% |
| 200 | Predictor–corrector | 12.5811% |

From 4 to 200 steps the answer moves by

$$12.58 - 12.52 = \mathbf{6} \text{ basis points of volatility}$$

against a Monte Carlo standard error of 2.2 basis points on each level. The step counts do not share draws, so the difference carries about 3.1, and 6 is only **two standard errors**. It is real all the same, because it can be computed without simulation: holding $\sigma_k$ at its value at the start

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of each step, across the hump, gives a shortfall of 7.7 basis points at four steps and 0.6 at fifty. It is also finished quickly: the increment from 50 to 200 steps is smaller than the error, so fifty steps a year is where this converges.

*Discretization bias is real and it is cheap to remove.* That is the first finding: fifty steps a year buys all of it.

*Why the predictor–corrector makes no difference.* The two schemes agree to a fraction of a basis point at every step count: 0.15 bp at 4 steps, 0.003 bp at 200. The reason is question 4's factor.

The predictor–corrector exists to correct the **frozen-drift** approximation: a naive Euler step holds the state-dependent drift fixed over the interval, and the corrector re-evaluates it at the predicted point. But every term of the drift carries a product $\sigma_k \sigma_j$, so the drift is second order in volatility where the diffusion is first order, and at a 3% rate the factor $\tau F/(1+\tau F) \approx 0.029$ damps it further. Over the option's one-year life that leaves a drift of 0.0019 in $\ln F$ against a diffusion standard deviation of 0.153. Correcting a term that small cannot move the answer.

So the residual bias is coming from the frozen volatility, not from the frozen drift. It comes from holding the humped $\sigma_k(t)$ at its starting value across each step, which no drift correction addresses and which integrating the variance over the step would remove. The refinement is being applied to the wrong part of the error, which is why more steps help and a better drift scheme does not.

*The conditions under which it would matter.* The correction earns its cost when **the drift is large**, so that it is a material part of the dynamics:

- *High rates.* At 8% the factor is 0.0741, two and a half times its value at 3%; in a high-rate or emerging-market currency at 15–20% it is 0.13–0.20, and the drift becomes a genuine contributor.
- *Not, on its own, a long accrual period.* $\tau$ scales each term, but halving it doubles the number of terms over the same stretch of curve, so annual and semiannual conventions carry about the same total drift at the same rate.
- *Long horizons and many forwards.* The drift is a *sum* over the forwards between $k$ and the terminal date, so a thirty-year simulation accumulates far more terms than a five-year one.
- *Coarse time steps*, since the frozen-drift error is what a corrector removes and it grows with step size.

The textbook advice "use a predictor–corrector" is right in some regimes and pointless in others, and the way to tell is to compute $\tau F/(1+\tau F)$ before deciding.

*The general lesson.* Measure which term dominates before refining any of them. A numerical improvement targeted at a term contributing 3% of the dynamics cannot repay its cost, however theoretically correct it is, and the effort is better spent on the step count, which addresses the error that is actually there.

*What the question tests.* Reading a convergence table against its standard error, and diagnosing why a theoretically-motivated refinement has no measurable effect in a particular regime.

---

**32.11** *Explain why forward rates and swap rates cannot both be exactly lognormal, explain the significance of Rebonato's weights summing to one, and use Example 32.5 to state the size of the inconsistency in each of the two cases and compare it with a typical bid–ask.*

*Why both cannot be exactly lognormal.* A forward swap rate is a *weighted average* of

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forward rates, with weights that are themselves functions of those rates. A weighted sum of lognormal variables is not lognormal: the lognormal family is closed under multiplication, not addition. So if the forwards are exactly lognormal (the LMM's assumption), the swap rate cannot be; and if the swap rate is exactly lognormal (the swap market model's assumption, which makes Black's swaption formula exact), the forwards cannot be.

The two market conventions are therefore **mutually inconsistent**, and both are used daily.

*The significance of Rebonato's weights summing to one.* His approximation expresses the swaption's Black volatility as a weighted average of forward volatilities, with

$$w_k = \frac{\partial S}{\partial F_k} \cdot \frac{F_k}{S}$$

For Example 32.5's swap these are 0.2120, 0.2058, 0.1998, 0.1940, 0.1884, summing to **1.0000**.

That is an identity, not a normalization, and the distinction matters. The forward swap rate is **homogeneous of degree one** in the forwards: scaling every $F_k$ by $\lambda$ scales $S$ by $\lambda$. Euler's theorem for homogeneous functions then gives

$$\sum_k F_k \frac{\partial S}{\partial F_k} = S \qquad \Longrightarrow \qquad \sum_k w_k = 1$$

So the weights sum to one *necessarily*, for any curve and any swap, and no one chose them to. That is the check that the approximation is a *genuine weighted average* rather than a fitted formula: a fitted formula would have no reason to satisfy the identity, and one that failed it would be misspecified.

*The size of the inconsistency, in two cases.*

| | **1y into 5y at 3%** | **5y into 10y at 6%** |
|---|---|---|
| Rebonato approximation | 12.5818% | 9.0048% |
| Simulation, 800 steps | 12.58% | 9.00% |
| **Error** | under 1 bp | under 1 bp |

*Against a typical bid–ask of half a volatility point (50 bp):*

*Both cases are far inside it.* Under a basis point is about a fiftieth of the spread, and both errors are smaller than the simulation's own standard error, so neither is measurable let alone tradeable. The effort spent on this in the literature looks out of proportion to its size, and the long trade does not rescue it: lengthening the option and adding forwards leaves the approximation as good.

*Why it holds up under all three changes.* The second trade raises the rate (6% against 3%, which takes question 4's drift factor from 0.029 to 0.057), lowers the correlation (0.70 against 0.85), and lengthens the option while adding forwards (ten over five years rather than five over one). Each makes the frozen weights move more, and the error still does not appear, because the weights must keep summing to one, so a weight that drifts up is paid for by others drifting down, and what survives is second order in the displacements rather than first.

*The practical conclusion.* The incompatibility is not a reason to reject either model, and on the evidence here it is not a reason to do anything at all: use the approximation for quoting and for calibration. What the exercise is worth is the habit rather than the answer: the quantity was measurable, so it got measured instead of asserted, which is the same discipline as question 10.

*What the question tests.* That lognormality is not preserved under addition, that Euler's theorem forces the weights to sum to one, and that this constraint is what keeps the resulting error second order and therefore negligible, a theorem with no price attached.

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**32.12** *Compute the number of simulations a bump-and-revalue risk run requires for a book with forty curve points and twenty-five volatility points, and explain why common random numbers are essential rather than merely helpful.*

*The count.* With $M$ curve points and $V$ volatility points, a full bump-and-revalue sensitivity run requires $M + V + 1$ simulations: one base valuation and one per bumped input:

$$40 + 25 + 1 = \mathbf{66} \text{ simulations}$$

(Central differences would double the bumped runs to 130, at the cost of a second full pass, so dealers' pricing systems generally use one-sided bumps.)

*Why this, and not the path count, is the binding constraint.* A price is one number; a book needs its derivatives with respect to *every* calibration input. Adding a currency to the book does not make each simulation slower; it makes the number of simulations larger. At twelve minutes per pricing run, 66 runs is over thirteen hours and does not fit in an overnight window; the chapter's larger example of 81 runs is 16.2 hours. An LMM book's overnight risk run, not its pricing, is the binding computational constraint.

*Why common random numbers are essential rather than merely helpful.* Because without them the difference of two noisy estimates carries the noise of both, which can exceed the bump's whole effect—so the run produces no information at all.

Example 32.6 makes it concrete. A swaption prices at 100.0000 with a standard error of 0.0500 at 100,000 paths, and a one basis point curve bump moves it by 0.0480.

*With independent draws*, the difference of two runs has standard error $0.0500\sqrt{2} = 0.0707$:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0707} = \mathbf{0.68}$$

The delta is smaller than the uncertainty in the delta. A risk report built this way is reporting sampling noise with a sign attached.

*With common random numbers*, the two runs differ *only* through the bump, most of the sampling error cancels, and a standard error near 0.0004 is routine:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0004} = \mathbf{120}$$

*The cost of not sharing the draws is not a factor of two.* Matching that precision with independent draws requires

$$\left(\frac{0.0707}{0.0004}\right)^2 \times 100{,}000 \approx \mathbf{3.1 \text{ billion paths}}$$

*per bump*, a factor of 31,250 in paths, and there are 65 bumps. One line of implementation (seed the bumped run identically) buys what no amount of hardware can.

This is Chapter 20's question 8 in a setting where the stakes are higher: there, common random numbers improved a delta; here they are the difference between a usable number and no number.

*And the structural fix, since 66 runs is the real problem.* Adjoint algorithmic differentiation computes *all* $M + V$ sensitivities in a fixed multiple (typically three to five times) of the cost of *one* price, independently of how many inputs there are:

| Curve points | Vol points | Bump-and-revalue | AAD |
|---:|---:|---:|---:|
| 50 | 30 | 81 runs | ~4 runs |
| 120 | 70 | 191 runs | ~4 runs |

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The right column does not move, which is the whole of the argument for adjoint methods (question 13). Sixty-six runs at twelve minutes is 13.2 hours; four runs is 48 minutes.

*What the question tests.* The $M + V + 1$ scaling, and that common random numbers change the signal-to-noise ratio by a factor that no additional computation can substitute for.

---

**32.13** *State the condition under which pathwise differentiation fails and name one repair, then explain what adjoint algorithmic differentiation delivers and why its cost does not grow with the number of inputs.*

*When pathwise differentiation fails:* ***discontinuous payoffs***.

The method differentiates the simulated payoff with respect to the input *along each path*, which requires the payoff to be differentiable in the state. Where it is, the estimator is unbiased, has far lower variance than bumping, and needs only *one* simulation.

Where the payoff jumps (a **digital**, which pays a fixed amount above a threshold and nothing below, or a **barrier**, whose value drops discontinuously on touching), the derivative of the payoff does not exist at the discontinuity. Differentiating pathwise gives zero almost everywhere, since almost no path lands exactly on the boundary, and the estimator misses the entire sensitivity: it reports that moving the underlying does not change the payoff, which is true on each individual path and false in expectation.

*Two standard repairs follow.* **Smoothing the payoff** replaces the step with a steep but differentiable ramp, which reintroduces a small bias controllable by the ramp width. The **likelihood-ratio method** differentiates the *density* rather than the payoff and therefore tolerates any payoff at all, at the cost of higher variance.

*What adjoint algorithmic differentiation delivers.* All $M + V$ sensitivities in a **fixed multiple (typically three to five times) of the cost of a single price**, independently of how many inputs there are.

*Why the cost does not grow with the number of inputs.* Because AAD traverses the computation **backward**, once.

A pricing calculation is a composition of elementary operations from inputs to output. Forward-mode differentiation, which is what bumping approximates, propagates the derivative with respect to *one* input from the start to the end, so it must be repeated once per input: cost $\propto$ number of inputs.

The adjoint reverses the direction. It starts from the *single output* and propagates sensitivities backward through the same computation graph, accumulating $\partial(\text{price})/\partial(\text{intermediate})$ at every node. Because there is one output and many inputs, one backward pass reaches *all* the inputs at once. The cost is set by the size of the computation graph (the price calculation itself) and not by how many leaves that graph has.

So the scaling is the reverse of bumping's, and question 12's table shows it: 81 runs against ~4 at 50 curve and 30 volatility points, 191 against ~4 at 120 and 70. The right column does not move.

*The costs, since nothing is free.* AAD requires the computation graph to be *stored* during the forward pass, so memory grows with the number of steps and paths, and checkpointing trades computation for memory to manage it. It requires either an operator-overloading library or source transformation, so it is an implementation commitment rather than a change of settings. And it inherits pathwise differentiation's limitation exactly: a discontinuous payoff breaks AAD for the same reason it breaks pathwise, and needs the same repairs.

*Why this belongs in a chapter on the LMM.* Because question 12 established that the input count, not the path count, is what makes an LMM risk run infeasible, and AAD is the technique that removes the input count from the cost. It is what makes a large LMM book viable overnight, and the online note [TN39] returns to it as the industrial form of the same idea.

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*What the question tests.* That pathwise differentiation needs a differentiable payoff, and that reverse-mode accumulation costs one pass regardless of input count because there is one output.

---

**32.14** *Name four products that require a multi-factor model, state for each what makes a single state variable insufficient, and explain why dealers' implied correlations sit systematically below historical ones.*

*Four products that require a multi-factor model.*

*1. CMS spread options*, and the notes built on them. A coupon paying $\max(S_{10} - S_2 - K, 0)$, sold to investors who want yield and will be long the curve's slope to get it: the coupon pays only while the ten-year rate exceeds the two-year by more than $K$. A single state variable makes the spread nearly deterministic: with one factor both rates are increasing functions of the same number, so their difference carries only the residual $|\sigma_{10} - \sigma_2|$. Chapter 31's question 12 prices it: 56.2 basis points against 245.0, a factor of 4.4. These are the largest single use of the model.

*2. Range accruals on two rates.* A coupon accruing only on days when a rate, or the spread between two, sits inside a band. The payoff depends on the joint distribution at many dates, not on one date's marginal, so a model that gets each date's distribution right and the dates' relationship wrong will misprice it.

*3. Ratchets and target redemption notes.* A ratchet's coupon is capped relative to the *previous* coupon, making it path-dependent in the forward-rate *vector*; a target redemption note (TARN) terminates once cumulative coupons reach a threshold, making the termination date itself a functional of the path. Neither is a function of a single state variable at a single date, which is precisely the condition under which Chapter 31's tree stops being applicable.

*4. Callable structured notes*, which combine the above with the exercise problem of Section 32.12. They need both the joint distribution and a continuation value, and are the hardest products in the rate market for exactly that reason.

*Why dealers' implied correlations sit systematically below historical ones.*

*Because the flow is one-directional.* The buyers of these structures are yield-seeking investors: private banks, insurers, regional banks. In a callable note or a range accrual they are short optionality and are paid a coupon for it, but the product that sets the correlation market is the steepener, and its buyer is **long** a CMS spread option: the coupon pays only while the curve is steep enough. The dealer takes the other side and is therefore **short** spread optionality, which it must hedge by trading swaptions and CMS in the correlation.

A spread option is long *decorrelation*: it is worth more when the two rates move independently, because the spread is then more volatile. A dealer short it is short decorrelation, and must be paid for warehousing that position, and the payment appears as marking spread-option volatility above what historical correlation would imply, which is to say marking the implied correlation *below* the historical one. The correlation exposure created by structured-note issuance is a large and persistent one-way flow, and a persistent one-way flow moves the price away from the statistical estimate.

This is the same asymmetry Section 19.10 identified in equity volatility, arising the same way. There, institutional put demand is one-directional, dealers are structurally short, and implied volatility sits systematically above realized—a volatility risk premium. Here investors buy spread optionality, dealers are short it, and spread volatility sits above its historical estimate: a **correlation risk premium**, with implied correlation below historical.

*Two consequences are worth drawing.* A desk calibrating to *historical* correlation will find its prices systematically off the market, and the gap is a premium rather than an error (question 9's warning that historical and implied correlations differ systematically, by more than the volatility analogs, because there is less to discipline them). And the premium is a *position*

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as well as a price: a dealer accumulating the same one-way exposure across every ticket is running a concentrated correlation book whose risk is not diversified by doing more business.

*What the question tests.* That each product needs a second degree of freedom for a specific identifiable reason, and that a systematic implied-versus-historical gap is a risk premium created by one-directional flow.

---

**32.15** *Explain why using the same Monte Carlo paths to fit and to apply an exercise rule biases the price upward, and why a suboptimal rule produces a lower bound rather than an error of unknown sign.*

*Why using the same paths biases the price upward.* Because the exercise rule then has **foresight**.

Longstaff–Schwartz estimates the continuation value by regressing realized continuation values on state variables at each exercise date, then uses the fitted function as the exercise rule. If the *same* paths are used to fit the regression and to price, the fitted rule has been tuned to the particular realizations it will be applied to. It encodes not only the general relationship between state and continuation value but also the *sampling noise* of those specific paths—and applying it to them exploits that noise.

Concretely: the regression will have overfitted the paths where continuation happened to do well, so the rule holds on exactly those, and exercised on exactly the ones where exercising happened to do well. That is a decision made with information the holder would not have had, and it produces a value above the true one.

*The standard discipline is to fit on one set of paths and price on another.* That removes the foresight bias entirely, at the cost of a second simulation, and leaves only the second bias, below.

*Why a suboptimal rule produces a lower bound rather than an error of unknown sign.* Because the true American or Bermudan value is a **supremum** over exercise policies:

$$V = \sup_{\tau} \mathbb{E}[\text{discounted payoff at } \tau]$$

Any particular rule, including a bad one, attains at most this supremum. Applying a specific, implementable policy and averaging its realized payoffs estimates the value *under that policy*, which by definition cannot exceed the value under the best policy.

So the error has a **known sign**. It is not that the estimate is approximately right with unpredictable deviations; it is that the estimate is *at most* the true value, always, for every imperfect rule.

Two things follow, and they are why the sign matters.

*The result is a genuine lower bound.* One can state with confidence that the option is worth *at least* the computed figure, a defensible statement to a risk committee, and one that no two-sided error bar supports. Pairing it with an **upper bound** from the dual (Andersen–Broadie) formulation brackets the answer, and the width of the bracket measures how good the exercise rule is. Those upper bounds exist and are expensive.

*And a poor regression basis looks like conservatism and is not.* With a high-dimensional forward-rate vector, the natural basis functions are the swap rate and the annuity, sometimes with a second swap rate to capture the curve's shape. Choosing too small a basis produces a suboptimal exercise rule and therefore a downward-biased price, which a desk may mistake for prudence. It is not prudence: it is an unquantified error in a known direction, and a desk *buying* the option at that price is overpaying relative to what a better rule would extract.

*The two biases are separable, which is the practical point.* Foresight bias is upward and is removed by out-of-sample pricing; suboptimality bias is downward and is reduced by a better basis. Reporting one number without stating which discipline was applied conflates them, and a

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naive implementation that does both (in-sample pricing with a poor basis) produces two errors of opposite sign whose net is genuinely unknown.

*And the whole apparatus is slow.* Chapter 31's Bermudan takes milliseconds on a trinomial tree; the same product in a three-factor LMM with regression takes seconds to minutes. For a book of thousands of Bermudans revalued nightly with sensitivities, that difference decides the model, which is question 16's subject and the reason Chapter 31 survives.

*What the question tests.* That in-sample fitting lets the rule use information it should not have, and that an optimal-stopping value being a supremum is what signs the suboptimality error.

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**32.16** *State which model you would use for a callable bond, for a CMS spread option, and for quoting a single swaption, with reasons.*

*A callable bond: a **one-factor short-rate model** (Hull–White).*

The payoff depends on the *level* of rates (the issuer calls when refinancing is cheaper), so Chapter 31's question 11 defect never bites: no two rates are being compared, and perfect correlation among them is harmless. What the product does need is **early exercise at many dates**, and a tree supplies the continuation value at every node for free. A 44-step tree prices it in milliseconds, which matters when a book of thousands is revalued nightly with sensitivities.

*The price of this choice* is Chapter 31's question 8: mean reversion is not identified by a single European swaption, so the value spans 13.1% across values of $a$ the calibration cannot distinguish. The remedy is to calibrate to a co-terminal strip and to quote a range—not to change model.

*A CMS spread option: an **LMM**.*

The payoff depends on two rates and their joint distribution, which is exactly what a one-factor model cannot represent: it prices the option at 56.2 basis points against 245.0 at a plausible correlation, and no calibration repairs it because the error is structural (Chapter 31's question 12). An LMM carries an explicit correlation matrix, question 8 shows that the calibrated correlation is essentially the correlation delivered, and question 5's factor analysis says three factors capture enough of the structure to price it.

*The price of this choice* is speed and, if the structure is also callable, the regression machinery of question 15. Accept both—there is no alternative that prices the product at all.

*Quoting a single swaption: **SABR**.*

Here the question is interpolation across strikes at one expiration, and nothing more. No dynamics are needed, no joint distribution, no exercise decision: just a smooth arbitrage-free smile through the quoted points, fitted in microseconds and refittable on every tick. That is most of what a swaption desk does most days. Using an LMM for it would be absurd: minutes of computation to answer a question about one date's marginal distribution, with a calibration that is underdetermined in exactly the dimension the question does not ask about.

*The general principle, which outlasts all three.*

*A model should be chosen by which of its errors you can afford,*
*not by which is most realistic.*

Chapter 31's one-factor model is unrealistic in a way that costs a *factor of four* on a spread option and *nothing* on a callable bond. The LMM is realistic in a way that costs seconds instead of milliseconds—which is free for one trade and prohibitive for ten thousand overnight. Neither fact is a defect; both are the specification.

The discipline this implies is to identify, before choosing, which dimension the payoff actually depends on: level or shape, one date or several, marginal or joint, exercise or not. That question has an answer for every product, and the answer selects the model. A desk that instead

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asks “which model is best” will reach for the most elaborate one and pay for realism it cannot use, while a desk that asks “which error can I afford” will sometimes correctly choose the model it knows to be wrong.

*What the question tests.* Matching model to payoff dimension across three deliberately different products, and articulating the selection criterion as affordable error rather than realism.

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# Chapter 33
# Swaps Revisited

**33.1** *Using Example 33.1, state the amortizing par rate and the error from the weighted-average-life shortcut, and explain why the error grows with the curve's curvature.*

*The par rate.* For a ten-year annual swap amortizing linearly from 100 to zero, the notional enters every term of both legs:

$$\frac{\sum_k N_k \,[P(0,t_k) - P(0,t_{k+1})]}{\sum_k N_k \,P(0,t_{k+1})} = \mathbf{3.8386}\%$$

*The shortcut and its error.* The tempting simplification is to price the amortizing swap as a *bullet* swap at its weighted average life, weighting each payment date by the notional outstanding over the period ending on it. That life is exactly 4.0 years here, and the four-year bullet par rate is 3.6576%:

$$3.6576\% - 3.8386\% = \mathbf{-18.1} \text{ basis points}$$

For context, the bullet ten-year rate is 4.0187% and the bullet five-year 3.7432%, so the amortizing rate sits between them, sensibly, while the shortcut lands below both.

*Why the shortcut fails.* Weighted average life summarizes the schedule by its first moment, and the par rate is a nonlinear function of the discount factors. Replacing a schedule by its mean is exact only when the function being applied is **linear**, and the par rate (a ratio of two weighted sums of discount factors) is not.

The amortizing swap's rate is an annuity-weighted blend of the whole curve out to ten years, with weights $N_k P(0,t_{k+1})$ that decline as the notional amortizes. Collapsing that blend to a single point at 4.0 years discards the entire back half of the schedule, which on an upward-sloping curve is the part carrying the higher rates. Hence the error is *negative*, and systematically so on an upward-sloping curve.

*Why the error grows with the curve's curvature.* Jensen's inequality supplies the mechanism. Applying a nonlinear function to a mean differs from the mean of the function by an amount proportional to the function's *second* derivative times the spread of the distribution. Here the distribution is the schedule of payment dates and the function is the par rate's dependence on maturity. So:

- a *straight* curve (linear in maturity) makes the second derivative zero and the shortcut exact;
- a *curved* one makes it wrong, by more the more curved it is.

The shortcut is therefore at its worst precisely when the curve is most interesting (steep, humped, or inverting), which is exactly when someone reaches for a simplification.

*The pattern worth recognizing.* This is the same class of error as Chapter 20's naive expectation: replacing a distribution by its mean inside a nonlinear function. It recurs throughout the book: the weighted-average-life shortcut here, the credit triangle applied to

a five-year spread in Chapter 24's question 4, a flat volatility read as a caplet volatility in Chapter 29's question 5, and $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ in Chapter 20's question 11. In every case a summary statistic is being pushed through a nonlinear map, and in every case the fix is to do the sum rather than the shortcut.

*And the sum is not hard.* The correct calculation is one line of arithmetic over the schedule, using discount factors the desk already has. The shortcut saves nothing and costs 18 basis points.

*What the question tests.* That the par rate is nonlinear in the schedule so its first moment is not sufficient, and recognizing the mean-inside-a-nonlinear-function error as a recurring pattern.

---

**33.2** *Explain why a forward-starting swap amplifies curve errors, and why compounding at a spread produces an adjustment while flat compounding does not.*

*Why a forward-starting swap amplifies curve errors.* Because it is a **difference** of two spot swaps, and differencing amplifies relative error.

A swap starting in five years and running to fifteen is, in value terms, a fifteen-year swap less a five-year swap. Its par rate is therefore determined by the *forward* segment of the curve (the part between years five and fifteen), which is extracted by subtracting one nearly-equal quantity from another.

An error of one basis point in the fifteen-year rate does not produce a one basis point error in the forward rate; it produces a larger one, because the forward must absorb the whole discrepancy over a shorter span. Concretely, if the forward covers ten of the fifteen years, an error $\epsilon$ in the long rate is magnified by roughly $15/10 = 1.5$; a five-year-into-two-year forward magnifies by $7/2 = 3.5$. The shorter the forward window relative to its start date, the worse it gets.

Two aggravating factors follow. The two spot rates being differenced are *highly correlated*, so the error in the difference is dominated by whatever is *not* common between them—which is precisely the interpolation scheme in the region where quotes are sparse. And the far end of the curve is where quotes are sparsest, so the forward-starting swap depends most on the part of the curve that is least observed.

This is the same structure as Chapter 24's question 4, where a forward hazard rate extracted from two spot spreads exceeded the naive figure by 41%, and Chapter 29's question 5, where flat volatilities had to be bootstrapped rather than differenced. Whenever a quantity is a forward, its error is a magnified version of the spot errors it was built from.

*Why compounding at a spread produces an adjustment while flat compounding does not.*

*Flat compounding* accrues the floating rate on the accumulated balance and pays it at the end. Each period's accrual is $(1 + \tau_k R_k)$ applied to the running balance, so the terminal amount is a *product* of terms each of which is a martingale under its own measure, and the telescoping is exact. The value of receiving the compounded amount equals the value of receiving the individual payments and reinvesting them, which is what the floating leg is worth anyway. Nothing is left over.

*Compounding at a spread* breaks the telescoping. The accrual becomes $(1 + \tau_k(R_k + s))$, and multiplying out produces cross terms in which the spread multiplies the *stochastic* rate:

$$\prod_k (1 + \tau_k R_k + \tau_k s) = \prod_k (1 + \tau_k R_k) + s \sum_k \tau_k \prod_{j \neq k} \left(1 + \tau_j R_j\right) + O(s^2)$$

The second term is the spread multiplied by a *random* accumulation factor, so its expectation is not the spread times the expected factor: there is a covariance, and that covariance is the adjustment.

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The economics is that the spread is earning interest at a rate that is itself uncertain, and correlated with the balance it is applied to. A deterministic spread compounded at a stochastic rate is not a deterministic quantity, and the difference is a convexity term of exactly the kind Chapter 30 catalogs.

*The magnitude is small* for ordinary spreads and tenors (second-order in $s$ and in the rate's volatility), but it is not zero, and it is the reason a spread-compounded leg is not simply a flat leg plus a spread annuity.

*What the question tests.* That differencing two spot rates magnifies their errors, and that compounding is exact for the rate alone and generates a covariance once a spread is added.

---

**33.3** *Using Example 33.2, state the classic in-arrears adjustment at ten years and the post-transition adjustment at ten years, and explain why the ratio between them says nothing about rate volatility.*

*The two adjustments at ten years.* From Example 33.2, with $F = 3.5\%$, $\tau = 0.5$, $\sigma = 25\%$:

| | |
|---|---|
| **Classic in-arrears adjustment, 10y** | **3.76** bp |
| **Post-transition, 2-day payment lag, 10y** | **0.0607** bp |
| Ratio | about **62** |

*Why the second is so much smaller.* A compounded-in-arrears SOFR rate is *determined* at the end of its accrual period and *paid* then. **The payment is not early**—so the mismatch generating the classical adjustment does not arise at all. What survives is only the small delay between the end of the observation period and the payment date, typically two business days, and the adjustment scales with that delay.

*Why the ratio says nothing about rate volatility.* Because *both* figures were computed at the *same* volatility, 25%, on the same forward rate, in the same currency. Nothing about the market changed between the two rows of the calculation.

*A convexity adjustment is a property of a convention, not of a market.* What changed was the *timing convention*: the rate stopped being paid six months before its natural date and started being paid two days after it. The adjustment followed the convention, mechanically, because a convexity adjustment measures a mismatch between when a rate is observed and when it is paid—and that mismatch is a contract term.

So a reader who saw the adjustment fall from 3.76 to 0.06 basis points and concluded that rate volatility had collapsed by a factor of sixty would be badly wrong. Volatility enters the formula identically in both cases; the deferral length is what moved, from half a year to two days.

*The general lesson, and it is the sharper of the two.* Twenty years of literature on in-arrears convexity was solving a problem the market later removed by changing the contract. That is not an argument against the analysis (it was correct, and the instruments existed), but it is a caution about where to invest effort. An adjustment arising from a *convention* can be legislated away; one arising from *structure* cannot.

Chapter 30's question 8 draws the line. The CMS adjustment survives the transition untouched, because it comes from a rate's annuity failing to cancel against a single payment date, a structural mismatch present under any reference rate. The futures–forward adjustment survives, because it comes from settlement mechanics. The in-arrears adjustment did not survive, because it came from a fixing convention, and the fixing convention was replaced.

*And the new convention has its own problem, in a different place.* Chapter 29's backward-looking caplet (that chapter's question 3) exists precisely because compounding-in-arrears creates optionality where the old convention had none: the variance $\sigma^2(T_1 + \tau/3)$ rather than $\sigma^2 T_1$. The reform removed one adjustment and created another, which is the usual outcome

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and worth expecting.

*What the question tests.* That the ratio measures a change in contract timing rather than in volatility, and distinguishing convention-driven adjustments from structural ones.

---

**33.4** *Using Example 33.3, state the fair fixed rate with and without convexity adjustments and the difference, and explain why a product with a linear-looking payoff carries a vega.*

*The two rates.* From Example 33.3, paying the ten-year swap rate annually for ten years, forward ten-year rate flat at 3.40%, swap-rate volatility 25%, discounting at 3.2%:

| | |
|---|---|
| Fair fixed rate, with convexity adjustments | **3.5426**% |
| Fair fixed rate, ignoring them | **3.4000**% |
| **Difference** | **14.26** bp |

The adjustment grows linearly in the payment date (2.72 bp on the first payment, 27.23 bp on the tenth), and the fair fixed rate is the annuity-weighted average of the adjusted rates.

Read that against the market. A ten-year swap trades at a bid–ask of well under a basis point. So fourteen basis points on the fixed leg, every year for ten years, is not a refinement: a desk quoting a CMS swap without the adjustment is not slightly wrong, it is not in the market. It would win every trade it quoted and lose on all of them.

*Why a linear-looking payoff carries a vega.* Because the adjustment depends on a volatility, and the adjustment is part of the price.

The payoff itself is linear: the coupon *is* the swap rate, with no option, no cap, no floor. Everything about the contract's cash flows suggests it should be priced off the curve alone, as an ordinary swap is. But question 7 of Chapter 28 explains why it cannot be: the swap rate is a martingale under the *annuity* measure, while a CMS coupon is paid on a *single date* whose natural numeraire is a zero-coupon bond. The annuity fails to cancel, and what is left behind is a covariance between the swap rate and the ratio of the two numeraires.

*A covariance is a second moment, and second moments are volatility.* So the price depends on $\sigma$ even though the payoff does not. The vega is not in the contract; it is in the measure change.

*Three consequences follow.*

*The CMS desk is a volatility desk.* Its hedges are **swaptions**, not swaps. A trader hedging a CMS book with swaps alone would be delta-hedged and unhedged in the exposure that actually moves the mark.

*The product cannot be priced off a curve system.* It requires a swaption volatility surface, which means the CMS book must be marked and risk-managed alongside the options book rather than the linear rates book, an organizational consequence of a pricing fact.

*And the vega is one-directional across the market.* Question 5 shows why: the adjustment always has the same sign, so every CMS payer's exposure points the same way, and a dealer intermediating the flow accumulates a concentrated position.

*The general lesson.* A linear payoff does not imply a linear price. The test is not what the cash flows look like but whether the quantity being paid is a martingale under the measure the payment lives in—and Chapter 30's five adjustments are five instances where it is not.

*What the question tests.* The size of the adjustment against the bid–ask, and that a measure-change covariance introduces volatility dependence into a payoff that has none.

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**33.5** *Explain why a CMS convexity adjustment always has the same sign, and what that implies for the investor in a CMS-linked note.*

*Why the sign is always the same.* The CMS convexity adjustment is a covariance produced by a failed cancellation, and the direction of that failure is fixed by the arithmetic rather than by market conditions.

A swap rate that *rises* does two things at once:

- it raises the coupon **linearly**, since the CMS payment is the rate itself;
- and it *lowers* the value of the **annuity** the coupon would ordinarily have been paid over, because a higher rate discounts every future payment more heavily.

The swap rate is a martingale under the annuity measure, but the coupon is paid on a single date. So the states with the *largest* payoff are exactly the states in which the missing numeraire is *smallest*. The covariance therefore runs one way, and the correction **adds** to the forward swap rate.

*The sign follows from the cancellation that failed*, not from any parameter. No choice of volatility, correlation or curve shape reverses it. Volatility changes the *size* of the adjustment and never its direction, since a variance is nonnegative. Example 33.3's adjustments are positive at every payment date, growing from 2.72 to 27.23 basis points.

*What this implies for the investor in a CMS-linked note.*

*The coupon is worth more than the forward curve suggests.* A stream paying the ten-year swap rate is worth 3.5426% a year where the forward ten-year rate is 3.4000%; the 14 basis points are the value of the convexity attached to receiving the rate on a single date.

*And the investor pays for it.* The investor receiving the CMS coupon is **long** that convexity, and the issuer prices the note with the adjustment included, in the issue price or in a margin below CMS. The investor is therefore *buying* convexity, as the chapter puts it, and whether at a fair price is something the headline coupon does not reveal: the term sheet shows the rate paid, not the adjustment charged.

*And the position is a volatility position.* By question 4, the adjustment depends on swaption volatility, so an investor in a CMS note is *long* volatility, whether or not the term sheet mentions volatility anywhere. If the volatility used to price the note was high, the investor overpaid for the convexity, and the note's mark falls when implied volatility falls.

The contrast with question 6 is the instructive part. The range accrual investor sells volatility and is paid for it through an enhanced coupon; the CMS note investor buys volatility and pays for it through the pricing. In both cases the term sheet describes the coupon rather than the option, so the investor cannot see which side of a volatility trade they are on.

The honest way to present such a note is therefore to state the forward rate, the adjustment, and the volatility assumption that produced it, so the buyer can see that 14 of the 354 basis points are the price of a specific exposure they are buying. Chapter 35's question 15 makes that a general evaluation step.

*What the question tests.* That the sign is forced by the annuity's inverse relation to the rate, and that a systematically positive adjustment is a price the investor pays for convexity it holds.

---

**33.6** *Using Example 33.4, state the accrual fraction, the enhanced coupon, and the coupon's value if realized volatility is 50%, then explain what the enhancement actually is.*

*The three quantities.* From Example 33.4 (a coupon accruing only when the three-month rate is inside $[2.50\%, 4.50\%]$, forward rate 3.50%, volatility 25%, observation at two years):

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*The accrual fraction.* Under the forward measure the probability of being inside the band is a difference of two digitals:

$$P(\text{inside}) = 0.8126 - 0.2192 = \mathbf{0.5934}$$

*The enhanced coupon.* A coupon paid only on in-range days must be scaled up by the reciprocal to match a vanilla coupon's value:

$$\frac{3.50\%}{0.5934} = \mathbf{5.8980}\%$$

The headline enhancement is **240 basis points, 69% above the vanilla coupon**, which is what sells the note.

*The value at 50% realized volatility.* A wider distribution spends less time inside the band:

| Realized volatility | Accrual fraction | Value of the 5.8980% coupon |
|---|---|---|
| 15% | 0.8322 | 4.9082% |
| 25% | 0.5934 | **3.5000**% |
| 50% | 0.3095 | **1.8254**% |

At 50% realized volatility the coupon is worth **1.8254%**—*just over half* the vanilla 3.50%, and less than a third of its headline rate.

*What the enhancement actually is:* ***the short vega, priced.***

At the pricing volatility of 25% the coupon is worth exactly the vanilla 3.50%, *by construction*; that is how the 5.8980% was computed. Above it the investor receives less; below it, more. So the 240 basis points is not a spread, not a credit premium, and not compensation for complexity. It is the premium for selling volatility, spread over the coupon rather than paid up front.

*Nothing is being given away, and nothing is being received for free.* The note is a vanilla coupon plus a short volatility position, repackaged so that the short position appears as a higher number on the front page and the risk appears nowhere.

*Why the packaging matters.* A term sheet quoting "5.90% versus 3.50% on the vanilla" is describing the premium and not the sale. An investor comparing it against a deposit rate is comparing a risk-adjusted quantity against a risk-free one, and the 240 basis points is exactly the size of the adjustment they are not making.

The mechanism is the mirror image of question 5's CMS note, whose investor buys volatility rather than selling it, and identical to Chapter 35's autocallables, where the same short volatility appears through a barrier. The range accrual and the autocallable differ as instruments; the trade does not.

*One structural feature is worth adding.* The exposure is to **realized** volatility over the observation period, not to implied volatility at inception, so the investor cannot exit the position by unwinding a vega hedge: they are short the actual path. And question 7 shows that the path in which they lose most is the one in which they most need the income.

*What the question tests.* Computing an accrual probability as a difference of digitals, and identifying an enhanced coupon as the premium on a short volatility position rather than as a spread.

---

**33.7** *Explain why a range accrual's coupon fails in the states where its holder most needs income, and name the earlier section whose structure this repeats.*

*The failure.* A range accrual pays a full coupon when rates sit quietly inside the band and pays *nothing* on days when they do not. So the coupon is largest in calm markets and collapses

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in volatile ones—and volatile rate markets are precisely the environments in which the holder's other income is under pressure.

Question 6 measures it: at 50% realized volatility the accrual fraction falls to 0.3095 and the 5.8980% coupon is worth 1.8254%, below the 3.50% vanilla the investor gave up, and less than a third of the headline rate.

*Why the states coincide, rather than merely being correlated.* The investor is a yield-seeking holder (a private bank client, an insurer, a regional bank) whose reason for buying the note is income. The scenarios that widen rate distributions are macroeconomic dislocations: a policy shock, an inflation surprise, a credit event. In those scenarios the holder's other assets are falling, their funding is tightening, and their need for reliable income is at its highest. The coupon disappears exactly then.

*And the timing compounds it.* The accrual is measured day by day over the coupon period, so a single volatile month can destroy most of a year's coupon, and the holder discovers this at the payment date, after the period has passed, with no opportunity to have acted.

The earlier section whose structure this repeats: Section 27.9's wrong-way risk, in which a counterparty's exposure is largest in the states where it is most likely to default. Here the coupon is smallest in the states where income is most needed.

Section 26.4's excess spread, the first line of credit support in a securitization, has the same shape. Excess spread accumulates to a projected 34.5% of support by year five—but it is earned *only on loans that keep paying*. A pool that starts defaulting earns less spread at the same moment its losses begin consuming the buffer, so in that section's stress case the support does not rise from 22% to 34.5% but falls, to 19.6% after one year and 8.0% after five. A protection that shrinks in the state where it is needed.

In all three the mechanism is the same: a quantity relied upon as independent is in fact driven by the same event it is meant to offset.

*The general failure mode, which the book names repeatedly.* Chapter 27's question 12 collects the instances: wrong-way risk in counterparty exposure, recovery correlated with default frequency, domestic banks selling sovereign protection, portfolio insurance failing in a gap. In every case the naive calculation multiplies two expectations where the covariance term is the whole story, and in every case the diagnostic question is the same: in the state where this matters most, what is it actually worth?

*The practical test for an investor*, and Chapter 35's question 15 formalizes it: do not evaluate a structured coupon at its expected value. Evaluate it conditional on the scenario that motivated buying it. A range accrual bought for income should be valued in a market disruption; if it pays nothing there, it is not an income instrument.

*What the question tests.* That the coupon's failure state and the holder's need state are the same state, and recognizing the buffer–loss correlation pattern across two unrelated markets.

---

**33.8** *Using Example 33.5, state the ambiguity in the quoted cancellable swap rate implied by Example 31.9 and compare it with a typical bid–ask, then explain why this makes model risk a pricing problem.*

*The quoted rate.* From Example 33.5, a six-year swap with annual cancellation rights from year one, whose Bermudan value is 1.6995 per 100 of notional:

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| | |
|---|---|
| Annuity of the six-year swap | 5.272406 |
| Bermudan option value | 1.6995 |
| **Spread on the fixed leg** | **32.23** bp |
| Vanilla six-year par rate | 3.9862% |
| **Cancellable swap fixed rate** | **4.3085**% |

A party who wants the right to cancel pays 32 basis points a year for it, and that is the whole economics of the instrument.

*A first-order estimate.* Example 31.8's option is struck at 4%, not at the cancellable rate of 4.3085% that this table produces, so the spread slightly overstates the option's value, and the ambiguity below inherits that; a full solution would iterate the strike to a fixed point.

*The ambiguity.* Chapter 31's question 8 showed the Bermudan's value spanning **13.1%** across mean-reversion assumptions that all reproduce the calibrating European *exactly* (1.6474 to 1.8705 against a base of 1.6995). Annuitized, that becomes

$$0.131 \times 32.23 = \mathbf{4.2} \text{ basis points}$$

of ambiguity in a quoted swap rate.

*Against the bid–ask.* A six-year swap trades at a spread of a fraction of a basis point. So the model ambiguity is **several times the market it is being quoted into**—not a rounding error absorbed by the spread, but a number several times the spread.

*Why this makes model risk a pricing problem rather than only a risk-management one.*

The usual framing treats model risk as a control matter: a validation group measures it, a reserve is held against it, a committee reviews it. That framing presumes the model's uncertainty is small relative to the price and shows up as a valuation adjustment.

Here it does not. Two dealers quoting the same cancellable swap from the same European volatilities can differ by more than the market they are quoting into. That is a competitive fact before it is a risk fact:

- The dealer with the *lowest a* quotes the tightest rate and wins the trade, not because it has better information or a better hedge, but because its unidentified parameter happened to sit at one end of a range the data cannot distinguish.
- So the business systematically accrues to whoever assumes the most favorable value of an unobservable parameter, which is a selection mechanism with no relation to skill.
- And the winner is not obviously mispricing, because *every* value in the range reprices the calibration instrument to four decimals.

This is the clearest statement in the book of why model risk is a pricing problem. Chapter 21's barrier options differ by 49.5% across models, which is a striking number, but barrier options trade at wide spreads and are marked with reserves. A cancellable swap trades in a market measured in tenths of a basis point, and carries four basis points of model ambiguity in its quoted rate.

*The remedies are Chapter 31's.* Calibrate to a *co-terminal strip* rather than a single European, which constrains $a$ far better because the strip observes several dates. Fix $a$ from history if the strip is illiquid. And quote a range internally even while quoting a number externally, because a desk that does not know its own ambiguity cannot know whether it is winning business or losing money.

*What the question tests.* Converting an option value into a running spread, and that model ambiguity exceeding bid–ask changes model risk from a control question into a competitive one.

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**33.9** *Derive the variance ratio for an average of n observations, evaluate it at n = 21, and state its limit. Explain why averaging changes an option's value and not the swap's.*

*The derivation.* Let $S$ be lognormal with $\ln S_t$ having variance $\sigma^2 t$, and take the average of $\ln S$ over $n$ equally spaced dates $t_i = iT/n$. Since $\mathrm{Cov}(\ln S_{t_i}, \ln S_{t_j}) = \sigma^2 \min(t_i, t_j)$,

$$\mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} \ln S_{t_i}\right) = \frac{\sigma^2}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} \min(t_i, t_j) = \frac{\sigma^2 T}{n^3}\sum_{i=1}^{n}\sum_{j=1}^{n} \min(i, j)$$

The double sum is the standard identity $\sum_i \sum_j \min(i, j) = n(n+1)(2n+1)/6$, so

$$\mathrm{Var} = \frac{\sigma^2 T}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \boxed{\frac{(n+1)(2n+1)}{6n^2}\,\sigma^2 T}$$

*At* $n = 21$, one month of business days:

$$\frac{22 \times 43}{6 \times 441} = \frac{946}{2646} = \mathbf{0.3575}$$

so the effective volatility multiplier is $\sqrt{0.3575} = \mathbf{0.5979}$.

*The limit.* As $n \to \infty$ the ratio tends to $2n^2/6n^2 = \mathbf{1/3}$, giving a volatility multiplier of $1/\sqrt{3} = 0.5774$.

Note how fast it converges. Twenty-one observations already reach within four percent of the continuous limit:

| $n$ | **Variance ratio** | **Vol multiplier** |
|---|---|---|
| 1 | 1.0000 | 1.0000 |
| 5 | 0.4400 | 0.6633 |
| 21 | 0.3575 | **0.5979** |
| $\infty$ | 0.3333 | 0.5774 |

Twenty-one fixings do essentially all the work continuous averaging would, so the distinction between daily and continuous averaging is immaterial while the distinction between a single fixing and any average is enormous.

*Why averaging changes an option's value and not the swap's.*

*The swap is linear.* A commodity swap settles against the average price, and its fair fixed price is the *average of the forwards*. Averaging is a linear operation and expectation is linear, so

$$\mathbb{E}\left[\frac{1}{n}\sum_i S_{t_i}\right] = \frac{1}{n}\sum_i \mathbb{E}[S_{t_i}] = \frac{1}{n}\sum_i F_i$$

The variance of the average is irrelevant, because the swap's value depends on the mean alone.

*The option is not.* An option's payoff $\max(\cdot - K, 0)$ is a *convex* function, so its value depends on the whole distribution and in particular on its dispersion. Averaging cuts that dispersion by a factor of $\sqrt{3}$, and the option's value falls accordingly. Example 33.6 prices it: an at-the-money one-year option on an \$80 commodity at 35% volatility is worth 11.1136 on a single fixing at expiration and **6.6669** on the average of 21 fixings spread across the year. Averaging alone removes 40% of the premium, with no change to the underlying's volatility.

*The trap that follows.* A commodity option quoted on an average is **not comparable** with one quoted on a fixing, and comparing their implied volatilities without adjusting is a common and expensive error, the subject of question 10.

*What the question tests.* Deriving the variance of an average from the covariance structure, and that linearity is exactly what makes the swap indifferent and the option not.

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**33.10** *Explain why an implied volatility quoted on an average is not comparable with one quoted on a single fixing.*

*The core reason.* An implied volatility is the number that, inserted into a particular formula, reproduces a price. Change the underlying quantity from a single fixing to an average, and the formula's meaning changes with it, so the two numbers are answers to different questions.

Question 9 gives the size. An average over $n$ dates has variance $(n+1)(2n+1)/(6n^2)$ times a single fixing's, so its effective volatility is about **0.598** times as large at $n = 21$ and **0.577** in the limit. The averaging alone accounts for a factor of $1/\sqrt{3}$ before any market view enters.

*What goes wrong in practice.* Suppose an option on an average over its life quotes at an implied volatility of 21% and one on a single fixing at 35%. A naive comparison concludes that the average option is far cheaper and that there is relative value in buying it. There is not: Example 33.6's numbers are exactly these: 35% on the fixing corresponds to 20.93% on the average of the *same* underlying with the *same* volatility. The two quotes are the same market.

The error is symmetric and equally expensive in the other direction. A desk marking an average-based book against fixing-based implied volatilities will overstate its vega by a factor of nearly two, and hedge accordingly.

*Three specific hazards follow.*

*The averaging window must be known, not assumed.* A quote on a single fixing, a calendar-month average of daily prices, and a five-day settlement window imply different multipliers (1.000, 0.598 and 0.663 in the table), so "the average" is not a sufficient specification.

*The multiplier depends on when the averaging occurs relative to expiration.* The formula above assumes the averaging runs from inception to expiration. An option whose averaging period is a single month *one year out* has almost the full year's variance accumulated before averaging begins, so the reduction is far smaller. The window's *position* matters as much as its length.

*And the distribution is not lognormal.* An arithmetic average of lognormals is not lognormal (Chapter 32's question 11 makes the general point), so quoting it through a lognormal formula is a convention, not an identity. Chapter 20's geometric-average control variate exists precisely because the geometric average *is* lognormal and the arithmetic one is not.

*The general principle.* An implied volatility is a price in disguise, and prices are only comparable across identical contracts. The same trap appears throughout the book: Chapter 29's flat volatility is not a caplet volatility; Chapter 23's 0DTE implied volatility prints 10% below the 30-day for a purely mechanical reason; Chapter 22's variance swap strike is not a volatility forecast. In every case the number is a coordinate, and coordinates from different systems cannot be subtracted.

*What the question tests.* That the averaging multiplier is a mechanical factor of about $1/\sqrt{3}$, and the discipline of checking that two implied volatilities describe the same contract before comparing them.

**33.11** *Explain why the mark-to-market of a fixed-notional cross-currency swap grows with the exchange rate, what that does to collateral flows and exposure, and what the resettable structure does about it.*

*The mark.* A fixed-notional cross-currency swap exchanges principal at inception and at maturity **at the same rate**. So the two legs' notionals are matched in value on day one and never again: if the exchange rate moves, the final re-exchange is worth more to the party owed the appreciated currency, by the move times the notional, and the swap's mark-to-market moves with it, one-for-one.

*What a CSA does with it.* The final principal exchange is part of the swap's value, so

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variation margin covers it: each day the party in the money receives collateral equal to the swap's current value, and the terminal exchange is not uncollateralized. The difficulty is the size of the mark being margined. It follows the exchange rate and can reach a large share of the notional, on a swap whose principal is paid once, at the end.

Three consequences follow from a mark that large. The collateral flows are large and growing, each a real cash transfer and a funding demand. The exposure over the margin period of risk is the change in a very large mark since the last margin call, and the SA-CCR add-on, driven by the large FX delta, scales with it. And the CVA charge and the capital held against the swap follow the add-on.

*Chapter 9's exposure profile is where this shows.* A single-currency swap's exposure peaks about a third of the way through and declines to zero (Section 9.1), because a swap with one payment left cannot be worth much. A fixed-notional cross-currency swap does *not* decline to zero: the principal exchange sits at the very end, so the potential mark, and with it the potential exposure, grows with the accumulated FX move right up to maturity—exactly the wrong shape, since it is largest when the counterparty has had the longest to deteriorate.

*What the resettable structure does.* A **mark-to-market** or **resettable** cross-currency swap resets one leg's notional at each payment date to the current spot equivalent, with the difference **paid in cash** at that date.

The effect is to convert one large terminal exchange into a series of small periodic settlements. The mark is reset to near zero at each payment date rather than accumulating, so between resets it can grow only by one period's move rather than by the whole life's. The potential exposure profile flattens dramatically, and since an FX move grows with the square root of time, the peak falls by roughly the square root of the ratio of one reset period to the swap's maturity.

*Three consequences are worth naming.* The margin flows, the SA-CCR add-on and the CVA charge of Chapter 27 all fall sharply, which is why resettable structures dominate the interbank market. The reset payments are *real cash flows*, so the structure has a funding profile a fixed-notional swap does not, and a treasury must plan for them. And the swap now contains a series of short-dated FX exposures rather than one long-dated one, which *prices differently* (question 12's subject).

*What the question tests.* That a fully collateralized mark can still be a very large mark, and that periodic resetting keeps the mark small instead of letting it accumulate.

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**33.12** *Explain why a resettable cross-currency swap prices differently from a fixed-notional one, and identify the configuration from Chapter 28 that produces the correction.*

*Why the prices differ.* The two structures have different cash flows, so they are different instruments. A fixed-notional swap exchanges one pair of principals at maturity; a resettable swap exchanges a *series* of smaller amounts at every reset date, each determined by the spot rate prevailing then.

Those interim payments are not a redistribution of the same value. Each reset payment is a *stochastic* amount, proportional to the FX move over that period, settled at a date when interest rates are also uncertain. A stochastic cash flow paid at a stochastic discount factor is not worth the discounted expectation of the amount, and the difference is a covariance.

*The configuration from Chapter 28 that produces the correction: the* ***quanto***, or more precisely, the change from the foreign to the domestic risk-neutral measure, whose price is the covariance between the asset and the exchange rate.

The mechanism is Chapter 30's question 10 in a different instrument. The resettable swap's periodic payments depend on the product of an FX move and a rate-dependent discount factor,

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so their expectation under the domestic measure picks up a term

$$-F\rho\,\sigma_S\,\sigma_F\,T_i$$

on the forward rate $F$, exactly as a quanto forward does (Example 33.7's adjustment), with $\sigma_S$ the exchange-rate volatility and $\sigma_F$ the rate volatility. Each reset carries a small correction of this form, and they accumulate over the swap's life.

*The economics, stated without the measure change.* A dealer hedging a resettable swap must adjust its FX hedge at every reset, and the size of the adjustment depends on where spot has gone. If the FX rate and the interest rate differential move together, the dealer is systematically rebalancing at unfavorable levels. This is the same recurring cost as the quanto's rebalancing argument, and it must appear in the price.

*Three practical points.*

*The correction is small but not negligible*, being second-order in two volatilities, and it grows with the number of resets and with the correlation between rates and FX. On a long-dated swap in a currency pair where the two are strongly linked, it is basis points rather than fractions.

*It requires a correlation no one observes directly.* Like every quanto adjustment (Chapter 30's question 9), the input is a correlation that must be estimated from history or implied from a thin market, so a resettable swap carries a correlation exposure that a fixed-notional one does not.

*And the two structures are therefore not interchangeable quotes.* A desk comparing a resettable quote against a fixed-notional one must adjust for both the correction and the different exposure profile of question 11. The resettable is cheaper in capital and more expensive in model dependence, which is a trade rather than a free improvement.

*The general pattern.* A structural change that improves the exposure profile usually introduces a valuation adjustment, because it replaces a simple cash flow with a state-dependent one. Resettable notionals do it here; capped variance swaps do it in Chapter 22's question 5, where the cap converts a model-free instrument into a model-dependent one. Risk reduction is rarely free of model risk.

*What the question tests.* That interim stochastic settlements introduce a covariance the terminal exchange did not, and identifying it as the quanto configuration of Chapter 28.

---

**33.13** *Using Example 33.8, state the year-on-year adjustment at ten years and its average, and explain why the zero-coupon market needs no adjustment.*

*The adjustment.* From Example 33.8, with breakeven inflation flat at 2.50% and inflation volatility 1.2% in normal terms:

| Year | Year-on-year convexity adjustment |
|---|---|
| 1 | 0.00 bp |
| 2 | 1.44 bp |
| 5 | 5.76 bp |
| 10 | **12.96** bp |

*Averaged over ten years:* ***6.48 basis points***.

So a ten-year year-on-year swap does *not* strike at the ten-year zero-coupon breakeven, and the gap is not a quoting error—it is the accumulated convexity of ten ratio payments. The adjustment grows linearly in the payment date, from zero on the first payment (which has no preceding reset to be uncertain about) to 12.96 basis points on the tenth.

*Why the zero-coupon market needs no adjustment.* Because the configuration that produces one does not arise.

*A zero-coupon inflation swap* exchanges a fixed compounded rate for the realized index ratio $I_T/I_0 - 1$, settled **once** at $T$. There is one payment, one observation ratio, and *no timing*

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*mismatch*: the quantity referenced is observed at $T$ and paid at $T$. The quoted rate is therefore the breakeven inflation rate directly, with nothing between the observation and the settlement to generate a covariance.

*A year-on-year swap* pays $I_k/I_{k-1} - 1$ each year. Each payment is a ratio of index levels observed at two different dates and paid at the later one, and that is exactly the configuration Chapter 28 says produces an adjustment. The denominator $I_{k-1}$ is itself random when viewed from today, so the payment is a ratio of two stochastic quantities settled under a measure in which neither is a martingale, and the residual is a covariance.

Note the structure: the adjustment is zero for the first payment, where $I_0$ is known, and grows with each subsequent one as more of the denominator becomes uncertain. That linear growth is visible in the table and is the signature of the mechanism.

*Why this explains the market's shape.* The two inflation markets are *quoted independently* rather than one being derived from the other, and the reason is exactly this six and a half basis point wedge. A desk that quoted year-on-year swaps off the zero-coupon curve would be systematically wrong, in one direction, by several times a typical bid–ask.

*The general rule, and it is worth carrying beyond inflation.* Whenever a market quotes two instruments that look like transformations of each other, the difference is usually a convexity adjustment and is usually worth locating. Caps against swaptions (Chapter 29's question 11), futures against forwards (Chapter 30's question 5), CMS against the forward swap rate (question 4 here): in each case the apparent redundancy conceals a measure mismatch, and the market's insistence on quoting both separately is evidence that practitioners have located it.

*Two features have no rate-swap analog. Indexation lag*: the index used is typically the one published two or three months earlier, so the swap references a rate already partly known, which reduces volatility near each reset and complicates interpolation between monthly points. And *seasonality*: consumer price indices have a pronounced, stable annual pattern, so a year-on-year swap spanning exactly twelve months is nearly seasonality-neutral while one that does not is emphatically not.

*What the question tests.* That a single terminal ratio generates no mismatch while a sequence of ratios does, and reading the linear growth of the adjustment as evidence of its mechanism.

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**33.14** *Explain why, where pension demand for inflation protection is concentrated, inflation swap breakevens can sit above bond breakevens, and why this is a demand imbalance rather than a forecast disagreement.*

*The observation.* In markets where pension demand for inflation protection is concentrated, such as the sterling market, inflation swap breakevens (the fixed rate on a zero-coupon inflation swap) have tended to sit *above* bond breakevens, the difference between a nominal government bond's yield and an inflation-linked bond's real yield. The two should measure the same quantity, expected inflation plus a risk premium, and there they persistently do not.

*Why it is a demand imbalance rather than a forecast disagreement.*

Because the two markets have different, one-directional client bases, and each is priced by whom it must accommodate.

*In the swap market the structural payer of fixed (the receiver of inflation) is a pension fund or an insurer.* Their liabilities are explicitly inflation-linked: index-linked pensions, annuities with cost-of-living escalators. They must hedge that exposure and they are large, so there is a persistent one-way demand to *receive* inflation in swap form. A dealer taking the other side is short inflation and must be paid for it, and the payment appears as a higher quoted breakeven.

*In the bond market the flow runs the other way and is weaker.* An inflation-linked bond requires capital to buy (it is a funded instrument), so the buyer base is limited to those with

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cash to deploy, and the issuance is controlled by the sovereign rather than by demand. The swap is *unfunded*, so a hedger can obtain the exposure without balance sheet, which is why the marginal hedging demand goes there.

So the wedge is a price for absorbing a one-way flow, not two groups of forecasters disagreeing about future inflation. Three pieces of evidence support the reading:

- the gap is *persistent* and one-signed, whereas a forecast disagreement would fluctuate in direction;
- it is largest at the long maturities where pension demand is concentrated, and small at the front where it is not;
- and it *widens* when regulatory or accounting changes force pension hedging, which is a demand event rather than an information event.

*The same structure appears repeatedly in this book.* Chapter 19's index put demand steepens the equity skew; Chapter 32's question 14 shows structured-note issuance pushing dealers' implied correlations below historical ones; Chapter 22's roll cost is the volatility risk premium collected by whoever will take the other side of a one-way hedging need. In each case a persistent one-directional flow moves a price away from its statistical estimate, and the gap is a premium rather than an error.

*Why the distinction matters practically.* A forecast disagreement invites a trade: buy the cheap one, sell the dear one, wait for convergence. A demand imbalance does not converge, because the flow creating it does not stop. An investor selling swap breakevens against bond breakevens is collecting a premium for warehousing pension hedging demand, and should size the position as an insurance sale rather than as an arbitrage. The trade is a carry position with a tail, and the tail arrives when the demand intensifies.

Two frictions reinforce it, worth naming because they explain why the gap persists rather than being arbitraged: closing it requires simultaneously shorting an inflation-linked bond (often difficult to borrow) and receiving in a swap, which consumes balance sheet and counterparty limits. Chapter 5's limits-of-arbitrage argument again.

*What the question tests.* That two markets measuring the same quantity can differ persistently because of who must trade in each, and distinguishing a premium from a mispricing by its persistence and direction.

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**33.15** *Distinguish unwind, novation, and offsetting, state what each does to market risk and to counterparty risk, and explain why headline notional statistics describe compression more than activity.*

*The three routes out of a swap, and what each does.*

*Unwind.* The two parties agree on a termination payment equal to the mark-to-market and the trade ceases to exist. **Market risk: eliminated. Counterparty risk: eliminated.** Simple and complete— and it requires the original counterparty's *cooperation and pricing*, which is exactly what may be unavailable when it is most wanted.

*Novation.* A third party steps into one side. The trade is preserved and the counterparty changes. **Market risk: eliminated for the exiting party. Counterparty risk: transferred, not removed.** It requires consent, because the remaining party is acquiring a *new credit exposure*, Chapter 27's subject appearing as a documentation clause rather than as a capital formula.

*Offsetting.* Enter an opposite trade with someone else. **Market risk: flat. Counterparty risk: *doubled*.** The original trade and its mirror both remain live, with two counterparties, two sets of collateral flows, and two exposures that do not net because they are with different parties.

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Offsetting is exactly the wrong trade under Chapter 27's framework—it leaves the economics unchanged and doubles the capital—and it is done anyway, whenever an unwind cannot be agreed or a novation cannot be consented to. That is the point of listing the three together: the operationally easiest route is the one that consumes the most capital.

*Compression exists because of the third.* Multilateral compression services tear up offsetting trades *across many participants* while preserving each one's net market risk. A dealer holding a receiver with A and a payer with B, where a third party holds the mirror pair, can have all four torn up and replaced by nothing, removing four contracts and no risk.

*Why headline notional statistics describe compression more than activity.*

Because gross notional counts *contracts outstanding*, and compression removes contracts without removing exposure. Chapter 25's question 13 separates the three forces behind the roughly tenfold fall in the CDS market's gross notional (compression, clearing, and a genuine decline in single-name activity), and only the last was a fall in activity; the same pattern has held in rates.

So a fall in headline notional is, in the main, a measurement event:

- *compression* eliminates offsetting contracts: pure bookkeeping, zero risk change;
- *clearing* nets multilaterally rather than bilaterally, shrinking the reported figure again with no change in anyone's economic position;
- and only a genuine decline in *new business* would represent reduced activity.

The first two dominate, so the notional statistics that headline the derivatives market are largely a story about compression rather than about activity—and a reader inferring that the market has shrunk by an order of magnitude has read an infrastructure improvement as a retreat.

*The point for this chapter.* A swap's life does not end when its risk does. Between the moment a position is economically flat and the moment the contracts are actually gone, the firm carries a capital cost under Chapter 27 and an operational cost always (settlement, collateral, reconciliation, lifecycle events). Compression is the industry's answer to that gap, and its scale is the measure of how large the gap had become.

*What the question tests.* That the three exits differ in what they do to counterparty risk rather than to market risk, and that a notional statistic measures contracts rather than exposure.

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# Chapter 34
# Exotic Options

**34.1** *State the replication test that organizes this chapter, and use it to explain why a compound option carries less model risk than a forward-start.*

*The test.*

*Can the payoff be replicated by traded instruments—and if so,* ***statically*** *or only* ***dynamically****?*

*If it can be replicated statically* (a package, a chooser, a digital as a limit of call spreads), then the vanilla market **determines** the price, model risk is small, and the residual is a *hedging cost* rather than a *valuation uncertainty*.

*If it can only be replicated dynamically*, the price depends on the assumed dynamics, and calibration to vanillas constrains but does not determine it. Barriers, cliquets, lookbacks and everything multi-asset live here.

*Why a compound option carries less model risk than a forward-start.*

*A compound option is an option on an option, and both are terminal-value payoffs.* Its value depends on the distribution of the underlying option's *price* at the compound's expiration, and that price is itself a function of the underlying's level at that date. So the whole payoff is a function of $S_{t_1}$, a single terminal value. Pricing needs one root search for the critical price at which the inner option is worth the strike, and then the terminal distribution the vanilla market already reveals. Model risk is **low**.

*A forward-start option's strike is set at a future date*, so its payoff is $\max(S_T - \alpha S_{t_1}, 0)$, a function of a *ratio* of prices at two dates. Its value depends on the distribution of the return over $[t_1, T]$ *conditional on arriving at* $t_1$, which is exactly the **forward smile**. Chapter 21's question 3 established that the vanilla surface does not determine the forward smile: a local-volatility model fitted exactly to today's vanillas predicts a forward skew decaying by a factor of more than four, and the market's is roughly stationary. So a forward-start's price is a statement about dynamics that no vanilla constrains, and its model risk is as large as anything in the book.

*Why the test is more useful than the taxonomy.* Because it predicts the right thing, and intuition does not. A compound option *sounds* harder than a barrier (an option on an option, two layers, a root search) and is easier. A forward-start *sounds* trivial (it is just a call whose strike is set later) and carries severe model risk. Sorting instruments by apparent complexity gets both backward; sorting by replicability gets both right.

And it is a different question from whether a closed form exists, which is question 17's point and is worth separating now. The arithmetic Asian has no formula and very little model risk; the continuous barrier has eight exact formulas and severe model risk. A formula tells you how quickly you can compute a number; replicability tells you whether the number means anything.

*What the question tests.* Stating the organizing test as a test, and applying it to a pair of instruments whose difficulty is the reverse of their apparent complexity.

**34.2** *Using Example 34.1, state the fraction of the vanilla premium the barrier removes, and explain why knock-out deltas can exceed one.*

*The fraction.* From Example 34.1, with $S_0 = K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, continuously monitored:

| | |
|---|---|
| Vanilla call | 12.335999 |
| Down-and-out call | 9.111221 |
| Down-and-in call (the difference) | 3.224778 |

$$\frac{3.224778}{12.335999} = \mathbf{26.1}\%$$

The barrier removes a quarter of the premium—a discount large enough to be the entire reason the instrument is bought, and large enough that the pricing had better be right. A buyer who believes the underlying will not fall 10% is paying 9.11 instead of 12.34 for the same upside, and the 3.22 saved is the price of the belief.

*Why knock-out deltas can exceed one.* Because the option's value responds to spot through **two** channels that point in the same direction near the barrier.

For a vanilla call, delta is bounded by one: the option can never be worth more than the stock, so its sensitivity cannot exceed the stock's. For a down-and-out call there is a second effect:

- *approaching the strike* from below makes the option *more* valuable, the ordinary call effect;
- *approaching the barrier* makes the option *less* valuable, because knock-out becomes more likely, and at the barrier the option is worth zero.

Moving *up* therefore both increases intrinsic value *and* moves away from the barrier, so the two contributions add. Near the barrier the second dominates: a small rise in spot sharply reduces the knock-out probability, and the option's value rises by more than the stock does. Delta above one is not a numerical artifact; it is the survival probability moving.

The same mechanism makes *gamma change sign* near the barrier, which is why these instruments are difficult to hedge: the required position is large, direction-changing, and most extreme exactly where the payoff is most uncertain.

*And the hardest case is worse still.* A **reverse knock-out** (barrier above the strike, so the option is knocked out when it is *deep in the money*) has a *discontinuous payoff* at the barrier: at expiration a price just below $H$ pays almost $H - K$, and a touch of $H$ at any moment pays nothing. It is the most difficult vanilla-like instrument in the market, and Section 21.11 measures the model disagreement on one: 49.5% between two models calibrated to the same vanillas.

*The structural point, which organizes the rest of the chapter.* The vanilla depends only on the terminal distribution, which Section 19.2 showed the option chain reveals. The barrier depends on the probability of touching, which it does not. Everything difficult about barriers follows from that sentence.

*What the question tests.* That the discount is large enough to be the instrument's purpose, and that a survival probability moving with spot adds a second delta channel.

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**34.3** *Using Example 34.2, state the tree prices at 50 and 51 steps, explain the difference, and state the general lesson in a form that does not mention barriers.*

*The two prices.* From Example 34.2, pricing the down-and-out call on a Cox–Ross–Rubinstein tree that knocks out at any node at or below $H = 90$. The analytic answer is **9.111221**:

| Steps | Tree | Error | Effective barrier |
|---|---|---|---|
| 50 | 9.142046 | +0.031 | 89.94 |
| 51 | 10.494157 | **+1.383** | 86.93 |

*Fifty-one steps is forty-five times worse than fifty.*

*The explanation is entirely in the last column.* The tree can only knock out at a node, so the barrier it actually enforces is the highest node level at or below $H$. At $n = 50$ that level is 89.94, within six cents of the true barrier. At $n = 51$ the node spacing shifts and the nearest level below 90 is **86.93**, more than three points too low.

So the tree is not approximating the contract badly; it is pricing a different contract at every step count. A barrier at 86.93 is genuinely harder to breach than one at 90, so the option genuinely survives more often and is genuinely worth more. The 1.383 is not discretization error—it is the correct price of the wrong instrument.

*The error is also one-signed*, which makes it worse than random. The effective barrier is *always* at or below the true one, so the option is *always* over-priced. Averaging two grids does not help, because both are biased the same way; and Richardson extrapolation, which Chapter 20's question 1 already warned assumes a smooth error in $1/n$, makes it worse.

*The general lesson, in a form that does not mention barriers:*

> *A numerical method that resolves the payoff's discontinuities converges; one that does not, does not—and refining a grid that misses the discontinuity refines the wrong problem.*

That statement covers a great deal of this book. Chapter 20's binomial oscillation is node placement relative to the *strike's* kink, and its parity effect is this same phenomenon in a milder form. Chapter 20's control variate works because it removes the part of the error the grid handles badly. Chapter 20's trinomial tree exists precisely to supply the free parameter that lets a node be placed on a feature for *any* $n$. And Chapter 23's 0DTE gamma band is the same geometry in a market rather than a lattice.

The practical rule follows immediately, and it is a rule about software rather than about mathematics. Never let a user choose the step count on a barrier product. The pricing library should choose $n$ from the barrier (question 5's alignment condition), because a user who doubles the steps expecting a better answer will sometimes get one a thousand times worse.

*What the question tests.* That the effective barrier is what the tree prices, and stating the lesson at the level of discontinuity resolution rather than of barriers.

---

**34.4** *Explain why grid-averaging and Richardson extrapolation both make the barrier tree error worse rather than better.*

*Why grid-averaging fails.* Averaging the prices from $n$ and $n + 1$ steps works for a vanilla because Chapter 20 established that the binomial error *oscillates in sign* with the parity of $n$, so the two errors are roughly equal and opposite, and averaging cancels them.

That premise is false here. Question 3 showed the barrier error is **one-signed**: the effective barrier is *always* at or below the true one, so the option is *always* over-priced. Averaging two

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positive errors gives a positive error. At $n = 50$ and $n = 51$ the errors are +0.031 and +1.383, and their average is +0.707: worse than the better of the two, and no cancellation whatever.

The mechanism is that the two errors have *different causes*. In a vanilla the error comes from where the terminal nodes fall relative to the *strike*, and that alternates. In a barrier it comes from where the node *levels* fall relative to the *barrier*, and the resulting displacement is always inward. There is no configuration in which the tree enforces a barrier *above* $H$, because it must knock out at the highest node at or below it.

*Why Richardson extrapolation fails, and fails harder.* Richardson assumes the error behaves like $C/n$, a smooth, decaying function of the step count. Then $2V(2n) - V(n)$ removes the leading term.

The barrier error is not a function of $n$ in any smooth sense. It is a function of where the node lattice happens to land, which jumps discontinuously as $n$ changes: 89.94 at $n = 50$, 86.93 at $n = 51$, 88.25 at $n = 100$, 89.94 again at $n = 200$. The sequence is not decaying and not monotone (1600 steps is worse than 200), so there is no leading term to remove.

Extrapolating amplifies the mismatch. The formula $2V(2n) - V(n)$ subtracts one price with weight $-1$ and adds another with weight +2, so an error that does not shrink between the two step counts is *magnified* by the weights rather than canceled. Chapter 20's question 1 warns of the same failure on a vanilla when parity is crossed (an extrapolated answer worse than either of its inputs), and here the condition fails at essentially every pair.

*The common diagnosis.* Both techniques are **error-cancellation** methods, and both assume something about the error's structure: averaging assumes it alternates, Richardson assumes it is smooth in $1/n$. The barrier error is neither. Applying a cancellation technique to an error whose structure it misdescribes does not merely fail to help—it combines two wrong numbers with weights chosen for a different problem.

What works instead is to remove the cause rather than to cancel the symptom: place a node layer *on* the barrier, which is question 5, and which reduces the error by three orders of magnitude with *fewer* steps.

*The general principle worth extracting.* Before applying an acceleration technique, verify its assumption about the error. Both methods here are standard, correct in their own settings, and actively harmful in this one, and nothing in their output signals the misapplication, because they return a number either way.

*What the question tests.* That averaging needs alternating error and Richardson needs smooth error, and that the barrier's node-placement error has neither property.

---

**34.5** *Derive the step count that places a node layer on the barrier, and state the error it achieves against an unaligned grid with more steps.*

*The derivation.* In a Cox–Ross–Rubinstein tree the node levels are $S_0 u^{\pm j}$ with $u = e^{\sigma\sqrt{T/n}}$. A layer of nodes falls exactly on the barrier $H$ when

$$S_0 u^{-m} = H \quad \Longleftrightarrow \quad \frac{\ln(S_0/H)}{\sigma\sqrt{T/n}} = m \quad \text{for an integer } m$$

Solving for $n$:

$$\boxed{n = \left\lfloor \frac{m^2\sigma^2 T}{\ln^2(S_0/H)} \right\rfloor}$$

With $S_0 = 100$, $H = 90$, $\sigma = 25\%$, $T = 1$, we have $\ln^2(100/90) = 0.011101$, so $m = 4$ gives $n = 90$ and $m = 9$ gives $n = 456$.

*The error it achieves.*

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| $m$ | Steps | Tree | Error |
|---|---|---|---|
| 4 | 90 | 9.111965 | **+0.00074** |
| 9 | 456 | 9.111153 | −0.00007 |
| 16 | 1441 | 9.112680 | +0.00146 |

*Against an unaligned grid with more steps:* at 100 steps without alignment the error is **0.833**.

$$\frac{0.833}{0.00074} \approx \textbf{1,100} \text{ times larger}$$

with more steps. Ninety aligned steps beat one hundred unaligned ones by three orders of magnitude.

*Why the alignment matters so much.* Question 3 established that the tree prices a *different contract* whenever the effective barrier differs from $H$. Alignment makes the effective barrier *exactly* $H$, so the tree is finally pricing the right instrument. Once it is, the remaining error is ordinary discretization error, which is small and behaves smoothly.

So the improvement is not a better approximation to the same problem; it is a switch to the correct problem. That distinction explains why the residual errors in the table are erratic in the fourth decimal but uniformly tiny: they are the ordinary lattice error, which no longer competes with a contract-specification error a thousand times larger.

*Three practical consequences follow.*

*The step count is not a free parameter.* It is determined by the barrier, and only certain values are admissible. A library must compute $n$ from $H$, not accept it from a user (question 3's rule).

*Multiple barriers over-determine it.* A double-barrier option needs node layers on *two* levels, and a single $n$ generally cannot deliver both. This is exactly why the **trinomial** tree of Chapter 20's question 3 exists: its free parameter $\Delta x$ can be chosen independently of $\Delta t$, so node placement is achievable for any $n$ and any number of barriers.

And a discretely-monitored barrier needs the time grid aligned too: the monitoring dates must fall on tree steps, or the same error reappears in the time dimension. Question 11 shows why the monitoring correction and the node placement must both be made.

*What the question tests.* Deriving the alignment condition from the CRR node geometry, and that resolving the discontinuity beats refining the grid by three orders of magnitude.

---

**34.6** *Using Example 34.3, state the static hedge error when $r = 5\%$ and $q = 0$, and explain why the condition for exactness is zero drift rather than zero rates.*

*The static hedge.* When the drift is zero ($r = q$), a **reflection** argument gives an exact static replication: a down-and-in call with strike $K$ and barrier $H < K$ has the same value as $K/H$ puts struck at $H^2/K$.

The argument has two cases. If the barrier is never touched, the call never comes alive and the puts, struck at $H^2/K$ below $H$, expire worthless too. If it is touched, at that moment the puts are worth exactly one call struck at $K$, so they are sold, the call is bought, and the hedge has become the option.

Example 34.3 confirms it without computing anything: at $r = q$ the block $C$ of (34.11) and the value of $K/H$ puts struck at $H^2/K$ are term for term the same expression, and that expression contains no $r$.

*The error when the condition breaks.* At $r = 5\%$, $q = 0$:

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| | |
|---|---|
| Down-and-in call | 3.224778 |
| Replicating puts | 1.861532 |
| **Static hedge error** | **−42.3%** |

*The replication is not robust to drift*, and 42% is not a rounding error: the hedge is worth little more than half the instrument it is supposed to replicate.

*Why the condition is zero **drift** rather than zero **rates**.* The reflection argument works by mapping each path that touches $H$ to its mirror image reflected about $H$. That mapping, with the weight $K/H$ attached, preserves value only if the distribution is *symmetric* about the barrier in the sense of put–call symmetry, which requires the price itself to have no drift under the pricing measure, so that the forward from the moment of touching equals $H$. The $-\sigma^2/2$ in the log-price's drift does not break this; the weight $K/H$ is what absorbs it.

The risk-neutral price drift is $r - q$, so the symmetry condition is $r = q$, *not* $r = 0$. The example's algebra shows it directly: the bracket contains no $r$, so at $r = q = 5\%$ the replication is exact, even though rates are far from zero; raising the common carry only discounts the same number. What matters is that the two carry terms cancel, not that either is small.

The distinction is worth keeping because it determines where the technique is usable. It works for **currency** options where the two interest rates are close, for **futures** options where the drift is zero by construction, and for **equity** options only where the dividend yield approximately matches the funding rate. It fails on a non-dividend-paying stock at 5% rates, which is the most common textbook case.

*And the refinements are what desks actually use.* Replicating with a strip of vanillas at *several maturities* rather than one restores accuracy under drift, at the cost of exactness; it trades a clean identity for a robust approximation. That is the standard construction, and question 7 explains why a desk prefers it to a dynamic hedge even knowing it is inexact.

*What the question tests.* That symmetry about the barrier requires the carry terms to cancel rather than to vanish, and the size of the failure when they do not.

---

**34.7** *Explain why a static hedge that is wrong by a known amount can be preferable to a dynamic hedge that is right in a model.*

*The comparison, stated precisely.* A **static hedge** is a portfolio of vanillas bought once and left alone, traded again at most once, when the barrier is touched. A **dynamic hedge** is a position in the underlying, rebalanced continuously according to a model's delta.

*Why the static one can be preferable even when it is wrong by a known amount.*

*1. A known error is a manageable error.* Question 6's 42.3% is a number: it can be computed, reserved against, priced into the quote, and reported to a risk committee. A dynamic hedge that is "right in the model" has an error that is *unknown* because it depends on how the world differs from the model, and Section 21.11 shows the disagreement between two calibrated models reaching 49.5% on a reverse knock-out. An error you can size is worth more than a correctness claim you cannot test.

*2. It requires no rebalancing, so it survives the conditions that break dynamic hedging.* A dynamic hedge must be traded, continuously, at the prices the market offers. Question 2 established that a knock-out's delta can exceed one and its gamma changes sign near the barrier, so the required trades are largest, most frequent, and most direction-changing exactly where the underlying is moving fastest. Chapter 18's question 9 prices the resulting transaction costs even in a controlled simulation; in a gap, the rebalancing is not merely expensive but impossible.

*A static hedge is indifferent to all of this.* It is bought once and does not care whether the market is liquid on the day the barrier is approached.

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*3. It removes the model from the hedge, not merely from the price.* A dynamic hedge is a *model output*; Chapter 26's question 9 makes the general point that a position hedged by a model is hedged against the model's world. If the model's dynamics are wrong, both the price and the hedge are wrong, and the errors compound. A static hedge is a portfolio of instruments whose prices are *observed*; whatever the true dynamics, the hedge is worth what those vanillas are worth.

*4. And it is robust to jumps.* A delta hedge assumes a continuous path. A gap through the barrier leaves the dynamic hedger with a position sized for a level the market has already passed. The static hedger holds vanillas whose values move with the jump, so the replication degrades gradually rather than failing at once.

*The general principle.* Prefer a hedge whose failure mode you can characterize. A 42% error that is constant, computable, and independent of market conditions is a business decision: charge for it or decline the trade. An error that appears only under stress, whose size depends on an unobservable, and which is largest when execution is hardest, is not.

This is why Section 34.7 lists static hedging as the middle of three honest responses, above pricing with a model and reserving, below refusing the trade. It converts a *valuation uncertainty* into a *hedging cost*, which is question 1's distinction and the whole point of the chapter's organizing test.

*What the question tests.* That characterizable error beats untestable correctness, and that a static hedge's advantages are concentrated exactly where a dynamic hedge's difficulties are.

---

**34.8** *Explain why two models calibrated to the same vanilla surface can price a barrier 49.5% apart, and state the three defensible responses.*

*How it is possible.* Because a vanilla depends on the terminal distribution and a barrier depends on the path, and two models can agree on every terminal distribution while disagreeing about how the underlying gets there.

The vanilla surface identifies the marginal distribution of $S_T$ at each maturity (Chapter 19's Breeden–Litzenberger result) and nothing more. Infinitely many path laws share a given set of marginals. A barrier's payoff depends on the probability of *touching* a level, which is a statement about the joint behavior across time, and no vanilla isolates it.

So neither model commits an arbitrage: both reproduce every price the market quotes, and there is no trade that profits from their disagreement, because the barrier is not quoted.

*The experiment, and its design matters.* Section 21.11 calibrated local volatility and a stochastic-volatility model to the *same* vanilla surface, *verified* that each reproduces it, and only then priced barriers. They differed by **49.5%** on an up-and-out call with the barrier at 110. The design is Hull and Suo's (2002), and the verification step is what makes the number mean anything: without it, the spread could be a calibration failure rather than a genuine indeterminacy.

*What the spread measures* is therefore not an error in either model—both are right about every quoted vanilla—but the size of the question the market has left unanswered.

This is the sharpest statement in the book of what "the surface does not determine the price" means. Every vanilla was matched; the barrier was not.

Three defensible responses follow, in ascending order of what they give up.

*1. Price with the model, reserve against the range.* Compute under two or three models and hold the spread as a reserve. Chapter 21's question 10 gives the governance rules: take the reserve through P&L at inception rather than carrying it as a memo item, and report the range *alongside* the mark, since a 1% range and a 49.5% range demand different oversight.

*2. Hedge statically where possible.* This converts the model question into a residual and a

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known error (question 7's argument). A static hedge's error is computable and independent of dynamics, so the unanswerable question becomes an answerable one.

*3. Or refuse the trade.* A reverse knock-out at a barrier where the models disagree by half the price is an instrument whose value the market has not determined. Declining is not timidity; it is the recognition that the desk would be taking a position in a modeling assumption rather than in a market.

And the response that is not defensible is the one most often taken: reporting a barrier price to four decimals from a calibrated model. The precision is real (the model computes it exactly), and it describes the model rather than the market. A number carrying more precision than information is worse than a range, because it invites reliance the calculation cannot support.

*What the question tests.* That marginals do not determine path law, that the experiment's verification step is what licenses the conclusion, and that the honest outputs are a range, a static hedge, or a refusal.

---

**34.9** *Using Example 34.5, state the cost of the* $[95, 100]$ *overhedge relative to the digital, and explain why the replicating call spread's notional diverges while its value converges.*

*The cost of the overhedge.* From Example 34.5, a cash-or-nothing call paying \$1 if $S_T > 100$, with $S_0 = 100$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, is worth **0.504049**. Buying a one-sided call spread struck entirely below the strike:

| One-sided spread | Cost | Premium over the digital |
|---|---|---|
| $[99, 100]$ | 0.511635 | +1.5% |
| $[98, 100]$ | 0.519249 | +3.0% |
| $[95, 100]$ | **0.542210** | **+7.6%** |

The $[95, 100]$ overhedge costs **7.6% more** than the digital it replaces.

*Why the replicating call spread's notional diverges while its value converges.* A digital is the limit of a symmetric call spread of width $\varepsilon$ straddling the strike, held at notional $1/\varepsilon$:

| $\varepsilon$ | Notional | Value |
|---|---|---|
| 5.00 | 0.2 | 0.504159 |
| 0.10 | 10.0 | 0.504050 |
| 0.01 | 100.0 | 0.504049 |

*The value converges and the notional diverges*, which is the whole difficulty in two columns.

The reason is that the two columns measure different things. The *value* converges because the call spread's payoff converges pointwise to the digital's: narrowing the ramp changes the payoff only on a shrinking interval, whose probability weight vanishes. The *notional* diverges because the payoff must reach \$1 over a width of $\varepsilon$, so the slope, and hence the position size needed to produce it, is $1/\varepsilon$.

*What that means for whoever has to trade it.* At $\varepsilon = 0.10$ the dealer holds ten call spreads per unit of payoff, and in the last moments before expiration the delta approaches $1/\varepsilon = 10$ per unit: ten shares must be traded per dollar of payoff, with no time to trade them. That is what Section 18.13's "gamma diverges" means when someone has to do it.

So dealers overhedge. They sell the digital and buy a call spread struck **entirely below** the strike, which *dominates* the digital's payoff everywhere: it pays at least what the digital pays, at every terminal price. The position is then safe by construction rather than by rebalancing: there is nothing to trade near expiration, and no configuration in which the hedge falls short.

*And the overhedge premium sets the floor under the bid–ask on a digital.* The 7.6% is not a markup or a profit margin; it is the cost of a dominating portfolio. At a one-point window

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the dominating spread and its mirror image, which the digital dominates, bracket the digital at 0.4965 and 0.5116, so no two-way price narrower than 3.0% is hedged on both sides; skew, jumps, buying every spread at the offer and the capital the position ties up widen the quote beyond that. A tighter window is a better price and a larger position to trade; a wider one is a worse price and a smaller one. Where a desk settles that trade-off sets the width, and it depends on how much of the position the desk expects to hold to expiration rather than on the option's complexity.

*The pattern generalizes.* A discontinuous payoff cannot be dynamically hedged near its discontinuity at any finite cost, so the market prices the *dominating static hedge* instead. Chapter 35's barrier-linked notes inherit the same problem (question 12 of that chapter shows the delta jump at a knock-in barrier) and the same response.

*What the question tests.* That pointwise payoff convergence and position-size divergence are compatible, and that the overhedge premium sets the floor under the quoted spread rather than being a markup on it.

---

**34.10** *Explain why an Asian option is worth about half a vanilla and a lookback about 1.67 times it, and why Asians dominate commercial usage while lookbacks do not.*

*The two ratios.* From Example 34.7, all at $S_0 = K = 100$, $\sigma = 25\%$, $r = 5\%$, $T = 1$:

| | Value | Relative to vanilla |
|---|---|---|
| Vanilla at-the-money call | 12.3360 | 1.00 |
| Floating-strike lookback call | 20.5522 | **1.67** |
| Geometric-average Asian call | 6.5298 | **0.53** |

Same underlying, same volatility, and a factor of three between the two exotics.

*Why the Asian is worth about half.* An average has less variance than an endpoint. Chapter 33's question 9 computed the ratio exactly: $(n+1)(2n+1)/(6n^2)$, tending to $1/3$, so the effective volatility is about $1/\sqrt{3} = 0.577$ of the true one.

So an Asian option is a vanilla on a less volatile quantity, and that explains most of the discount: at an effective 14.4% rather than 25%, an at-the-money call is worth 8.38, about two-thirds of the vanilla. The rest comes from the average's lower forward (the average of a rising path sits below its end, which is the drift $b_G$ in the formula), which takes the ratio to 0.53 in the table.

*Why the lookback is worth about 1.67 times.* It removes the timing decision. A floating-strike lookback call pays $S_T - m_T$: the holder buys at the best price achieved over the whole path. There is no strike to get wrong and no entry point to mistime. Nothing is left to get wrong, and perfect hindsight is worth 67% more than a vanilla.

Note that a floating lookback can never finish out of the money ($S_T \geq m_T$ always), which is most of the reason it costs what it costs.

*Why Asians dominate commercial usage and lookbacks barely exist.*

*Because Asians match a real exposure and lookbacks do not.* A corporate hedging a year of fuel purchases, or a treasurer converting revenue monthly, has an *average* exposure: they buy every day, so their realized cost *is* an average. An Asian option is the right shape, and the reduced premium is not a discount but the correct price of the correct instrument. Section 33.8 makes the same point from the swap side: the averaging convention exists because the physical exposure is itself an average.

*No one has an economic exposure to the minimum of a price path.* A lookback pays for hindsight, which no business activity generates and no hedge requires. It is bought as a speculative or marketing instrument, and it is taught more than it is traded.

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*And their model risk runs the same way. The Asian's is low*: an average smooths the path, so an Asian is *less* sensitive to the path distribution than the vanilla is to the terminal one. *The lookback's is high*: it is a bet on the extremes of the path, which is precisely what a calibrated vanilla surface says least about. This is question 8's argument, applied to a running minimum instead of a barrier.

*One elegant consequence follows.* The geometric Asian has an exact closed form (it is lognormal, with $\sigma_G = \sigma/\sqrt{3}$), while the arithmetic Asian actually traded has none, and it barely matters, because the geometric version is an outstanding control variate: on this contract the two payoffs correlate at 0.9994, and subtracting the geometric one cuts the variance by a factor of 330, or about 850 with the regression-optimal coefficient. Example 20.6 finds the same on a monthly-fixed contract, at 0.99914 and 583. The instrument that is traded is the one without a formula, and it does not matter—which is the reverse of the situation for every barrier in this chapter.

*What the question tests.* That the Asian's discount is the $1/\sqrt{3}$ variance ratio and the lookback's premium is removed timing risk, and that commercial usage follows exposure shape rather than mathematical interest.

---

**34.11** *Using Example 34.4, state the value of daily and of monthly monitoring relative to continuous, and explain why an unaligned tree that omits the continuity correction can land near the right price for the wrong reason.*

*The values.* From Example 34.4, applying the Broadie–Glasserman–Kou correction

$$H_{\text{eff}} = H\, e^{\pm\beta\sigma\sqrt{T/m}}, \qquad \beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} = 0.5826$$

to the down-and-out call whose continuous value is 9.111221:

| Monitoring | $m$ | Effective $H$ | Price | vs. continuous |
|---|---|---|---|---|
| Daily | 252 | 89.18 | 9.527218 | **+4.57**% |
| Monthly | 12 | 86.29 | 10.675121 | **+17.16**% |

*Even daily monitoring is worth 4.6% of the option, and monthly monitoring 17%.*

The direction is intuitive once stated: a knock-out priced as continuous is **undervalued**, because a path that dips below the barrier *between* closes is not caught. Discrete monitoring makes the option harder to knock out, so it survives more often and is worth more. Equivalently, the effective barrier sits *further from spot*: 89.18 rather than 90 for daily, 86.29 for monthly.

*Why an unaligned tree without the correction can look right.*

The two adjustments act in **two different dimensions** of the same lattice, and both move the barrier the *same way*, away from spot.

*Node placement* (questions 3 and 5) is an error in the *price* dimension: the tree can only knock out at a node, so the barrier it enforces is the highest node level at or below $H$. That level is always *at or below* the stated barrier, so against the continuous contract the option is always over-priced.

*Discrete monitoring* is a feature of the *contract*, in the time dimension: it checks the barrier only at specified dates, so paths that breach between them survive. The correct barrier for the discrete contract is $H_{\text{eff}}$, also below $H$, and the correct price is higher than the continuous one.

So the misalignment pushes the enforced barrier down, and the omitted correction leaves the target too high; the two errors have *opposite* signs against the discrete price and partly offset, by an amount that depends on where the nodes happen to fall. Against the daily price of 9.527218, the unaligned tree that ignores monitoring is **0.42 high** at 100 steps (9.944477) and

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**0.39 low** at 50 steps (9.142046). Neither number tells the user anything, and a tree that is close at one step count is far off at the next.

*The practical rule.* Handle the two separately and in order: *first* compute the effective barrier from the monitoring frequency, *then* align the tree to that effective barrier using question 5's step count. Both corrections are cheap, and neither substitutes for the other; an accidental offset between two omitted ones is not a hedge.

And the result is worth verifying independently, which the example does: simulating the daily-monitored contract directly with 400,000 antithetic paths gives 9.523044 against the correction's 9.527218, agreeing well within the simulation's standard error of 0.028437.

*What the question tests.* The size of the discrete-monitoring premium, and that two errors of opposite sign can offset by accident, which makes a price right for the wrong reason.

---

**34.12** *Using Example 34.9, state the worst-of put's value relative to a single-asset put at $\rho = 0.20$, and explain why the correlation dependence runs opposite to a basket's.*

*The value.* From Example 34.9 (three assets each at 100 with 28% volatility, $T = 1$, $r = 5\%$, a worst-of put struck at 100 paying on the *minimum*):

| $\rho$ | $E[\min S_T]$ | **Worst-of put** | **÷ single-asset put** |
|---|---|---|---|
| 0.20 | 83.87 | 17.64 | **2.05** |
| 0.60 | 89.86 | 14.71 | 1.71 |
| 0.95 | 99.62 | 10.64 | 1.24 |

At $\rho = 0.20$ the worst-of put is worth **2.05 times** a single-asset put—*twice as much*, and that factor of two is what pays the coupon on a worst-of note.

*Why the correlation dependence runs opposite to a basket's.*

*A basket averages.* An option on $(S_1 + S_2)/2$ depends on the dispersion of the *average*, and low correlation *reduces* that dispersion, since the assets' idiosyncratic moves partly cancel. Less dispersion means a cheaper option, so a basket option is worth *more* as correlation *rises* (question 13).

*A worst-of takes a minimum.* There is no cancellation: low correlation *increases* the chance that *at least one* asset falls a long way, and the minimum records whichever did. So the worst-of put is worth more as correlation *falls*.

The arithmetic is striking. With three assets at $\rho = 0.20$, the expected minimum at one year is **83.87** against a starting level of 100. The drift is upward and the minimum still falls sixteen percent, purely from picking the loser three times. Averaging would have produced something near 105.

So an averaging operator and a minimum operator respond to dependence in opposite directions, and the same correlation parameter that cheapens one instrument enriches the other. A dealer holding both is not diversified: they hold two positions in the same parameter with opposite signs, which is a correlation book.

*What this means for the investor in a worst-of note.* They are **long correlation**: having sold the put, they lose as correlation falls, and the direction of the exposure is the one that should worry them.

*Correlation rises in crises*, which on its own would lower the put's value. But it rises because everything is falling together, and with all three assets below the strike the put is deep in the money whatever the correlation.

So the investor carries two losses for one coupon. The low-correlation premium pays for one of them, a single asset collapsing while the others hold up; the other is the market-wide

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fall, against which the rise in correlation offers no protection. Notes are issued when markets are calm and correlation is low, so the coupon is typically set on the first risk alone.

*The general shape.* This is Chapter 26's question 7 in an equity setting: a correlation estimated in benign conditions, applied to an instrument whose payoff depends on the crisis, priced by whoever had the calm-period data. Chapter 35's question 14 pursues what the doubled coupon actually measures.

*What the question tests.* That min and mean respond oppositely to dependence, and that the investor's long-correlation position is mispriced by the conditions in which notes are issued.

---

**34.13** *Using Example 34.8, state the basket call's value relative to the basket of calls at $\rho = -0.5$ and at $\rho = 1$, explain what the gap represents, and explain why fifty assets cannot be handled with a single correlation.*

*The two ratios.* From Example 34.8, a basket call on $(S_1 + S_2)/2$ struck at 100, with the assets at 100 and volatilities 25% and 30%:

| $\rho$ | Basket call | Basket of calls | Ratio |
|---|---|---|---|
| −0.50 | 8.3481 | 13.2836 | **0.628** |
| 0.00 | 10.3260 | 13.2836 | 0.777 |
| +1.00 | 13.2792 | 13.2836 | **1.000** |

At $\rho = -0.5$ the basket call is worth **62.8%** of the basket of calls; at $\rho = 1$ they are *equal*, though the last row still shows a gap of 0.0044, because the two volatilities differ.

*What the gap represents: the **diversification the basket option does not let the holder keep**.*

An option on *each* asset pays on whichever rises, and the loser's decline is floored at zero separately. An option on the *average* pays only if the average rises, so a rise in one asset is partly canceled by a fall in the other *inside* the payoff, before the floor applies.

That is Chapter 29's question 11 in a different market: a portfolio of options dominates an option on the portfolio, by Jensen's inequality applied to the convex function $\max(\cdot - K, 0)$. There the comparison was a cap against a swaption and the gap priced rate correlation; here it is a basket of calls against a basket call and the gap prices equity correlation. One inequality, two markets.

Equality holds only at perfect correlation *and* equal volatilities, which is why the $\rho = 1$ row is not exactly equal: with volatilities of 25% and 30% the assets still move by different amounts even when perfectly correlated, so a little cancellation survives.

*Why fifty assets cannot be handled with a single correlation.*

Because a single $\rho$ asserts that every pair is equally related, and no real basket is like that. A fifty-asset basket has $50 \times 49/2 = 1{,}225$ pairwise correlations. Replacing them with one number imposes a structure the market contradicts in identifiable ways:

- *Sector blocks.* Two banks are far more correlated than a bank and a miner, and a basket concentrated in one sector behaves quite differently from a diversified one with the same average correlation.
- *The average is not sufficient.* The basket's variance is $\sum_i \sum_j w_i w_j \rho_{ij} \sigma_i \sigma_j$, a *weighted* sum, so correlations among the *large* weights matter more, and a uniform $\rho$ weights them all alike.
- *And the tails differ from the body.* Chapter 19's question 9 established that correlation rises in declines. A single $\rho$ fitted to the body understates the basket's downside dispersion.

*The practical response* is the factor structure of Chapter 32's question 5: represent the correlation matrix by a small number of common factors plus idiosyncratic terms, which

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captures sector and market structure with $O(nK)$ parameters rather than $O(n^2)$, and reduces to a single $\rho$ only in the one-factor equicorrelated case.

And note what the basket lacks that the exchange option has. Margrabe's formula is *exact* because the ratio of two lognormals is lognormal; the basket has *no closed form*, for the same reason the arithmetic Asian does not: a weighted sum of lognormals is not lognormal. The standard workaround matches the basket's first two moments to a lognormal, which is question 17's point about formulas and model risk being nearly unrelated.

*What the question tests.* That the option-on-portfolio gap is Jensen's inequality pricing correlation, and that a single correlation parameter cannot represent a large basket's structure.

---

**34.14** *Explain why a capped cliquet is short volatility, why its value sits entirely in the forward smile, and why that makes calibration to today's surface actively misleading rather than merely insufficient.*

*Why a capped cliquet is short volatility.* An uncapped cliquet pays $\sum_i \max(r_i, 0)$, a strip of forward-starting at-the-money calls, unambiguously *long* volatility. Capping each period at $C$ changes the period payoff to $\min\big(\max(r_i, 0),\ C\big)$, which by (34.21) is a forward-start *call spread*:

$$V = c(1,\ 1,\ \tau) - c(1,\ 1 + C,\ \tau)$$

long the at-the-money call and **short** the call struck at $1 + C$.

With $C = 8\%$ the short strike sits close to the one-year forward, so higher volatility helps the short leg more than the long one: the call struck at the cap sits nearer the forward than the at-the-money call does, so its vega is the larger of the two. The net position is short vega.

Example 34.10 measures it on a three-year annual-reset cliquet capped at 8%:

| Forward vol | Per period | Three-year cliquet |
|---|---|---|
| 15% | 0.03800 | 0.11401 |
| 25% | 0.03561 | 0.10682 |
| 35% | 0.03372 | 0.10115 |

*The value falls as volatility rises.* That sign reversal is the whole design: the cap is what converts a long-volatility instrument into a short-volatility one, and it is what lets the structure be marketed as a high participation rate.

*Why the value sits entirely in the forward smile.* Each period's payoff is a *forward-starting* call spread, struck at a multiple of the level prevailing at that period's reset date. By the scaling in (34.9) the period value is a number, not a function of the level: it depends only on the **forward volatility** for that period, and not at all on where the underlying happens to be when the period begins.

So the entire contract is a bet on the volatilities of three future one-year windows, and nothing else. The spot smile enters only insofar as it constrains those forward volatilities, which, as question 1 established, it barely does.

*Why calibrating to today's surface is actively misleading rather than merely insufficient.*

*Because it produces a confident number that is wrong in a known direction.*

An exact fit to today's vanillas is *achievable*, and a local-volatility model achieves it. Having achieved it, the model then makes a definite prediction about the forward smile, and Chapter 21's question 3 shows the prediction is wrong: the local-volatility skew decays by a factor of more than four between three months and three years, while the market's forward smile is roughly *stationary* in moneyness.

So the calibration:

- *succeeds* on every instrument it is tested against, leaving no residual to warn anyone;

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- *determines* the forward smile, so the model produces a single, apparently model-independent number;
- and *gets it wrong*, systematically, in the same direction every time.

*Insufficient* would mean the calibration leaves the price undetermined, so a range must be reported. *Misleading* is worse: it determines the price, precisely, from an assumption the market contradicts—and the exactness of the vanilla fit is what makes the output persuasive.

*The remedy* is Chapter 21's local-stochastic construction with an explicit mixing parameter, which reproduces the vanillas at every setting while sweeping the forward-smile assumption: the model risk made into a dial. A capped cliquet priced at both extremes gives the range, and the range is the honest output.

*What the question tests.* That a cap converts a call strip into a spread strip and reverses the vega sign, and the distinction between a calibration that under-determines a price and one that determines it wrongly.

---

**34.15** *Explain why dealers run a structural long-exotic, short-vanilla book, and why exotics with static hedges trade at narrower spreads than others.*

*Why the book is structurally long-exotic and short-vanilla.* Because the client is almost always short the exotic optionality and the dealer long it—and the reason is economic rather than accidental.

*The exotic feature is what cheapens the note or the hedge.* A barrier makes a call cheaper by 26% (question 2); a cap on a cliquet raises the participation rate; a worst-of underlying doubles the coupon (question 12); a knock-in makes a note's protection conditional. In every case the client accepts a feature that reduces the instrument's value *to them*, in exchange for a better headline number. So the client has **written** the exotic optionality and the dealer has **bought** it.

The dealer then hedges in the *vanilla* market, selling the vanillas whose risks approximate the exotic's, so the aggregate position is long exotics against short vanillas, in the same direction across every client and every ticket. The flow is one-directional, so it does not net down as the business grows.

*And the flow shows up in observable prices.* The surface marks of Section 19.8 and Section 22.11's implied correlation both reflect it: dealers hedging long exotic optionality by selling vanillas supply volatility at the strikes the products embed, and dealers left short correlation by buying worst-of puts from note investors bid implied correlation above realized. Chapter 32's question 14 finds the same mechanism in rates, where the flow runs the other way and implied correlation sits below historical. The dealer's hedging need is visible in the market's prices, which is one of the more useful facts in the book.

*Why the spreads differ so much.* Because the two groups have different *residual risks*, and the spread prices the residual rather than the complexity.

*Exotics with static hedges trade at narrow spreads.* A digital, a chooser, a package: each can be replicated by a portfolio of vanillas bought once and held (question 1). The dealer's residual is a *known hedging cost*: question 9's overhedge premium sets the floor under a digital's bid–ask, and it is a computable number that does not depend on any model of dynamics. Once the hedge is on, the dealer's remaining exposure is the vanillas' own market risk, which is priced and liquid.

*Exotics without them trade in percent.* A reverse knock-out, a cliquet, a lookback, anything multi-asset. Here question 8's finding governs: two models calibrated to the same vanilla surface differ by **49.5%** on a barrier at 110. The dealer's residual is a *valuation uncertainty*,

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not a hedging cost, and a spread must cover it, because the dealer cannot know which model is right and must be protected either way.

So the spread is a direct reading of question 1's test. Statically replicable means the vanilla market determines the price and the dealer charges for execution; dynamically replicable only means the price depends on assumed dynamics and the dealer charges for the range. A quoted spread of 5% rather than 5 basis points is the market telling you which side of that line the instrument sits on.

*Two riders follow.* The relationship holds across instruments and not within one: a barrier far from spot is cheap to be wrong about and trades tighter than one near it, though both are dynamically hedged. And a spread wide enough to cover model risk is also wide enough to be uncompetitive, which is why the third of question 8's honest responses, *refuse the trade*, is exercised more often than the literature suggests.

*What the question tests.* That the one-directional flow follows from the exotic feature being the concession, and that the quoted spread measures residual valuation uncertainty rather than complexity.

---

**34.16** *Using (34.11), price the down-and-in call of Example 34.1 from the single expression C, and explain why the up-and-in call with $H < K$ is worth the vanilla.*

*Pricing the down-and-in call from C.* For a down-and-in call with $H < K$, the value is the single block $C$, with $\phi = +1$ (call) and $\eta = +1$ (down-barrier). With $S_0 = K = 100$, $H = 90$, $r = 5\%$, $q = 0$, $\sigma = 25\%$, $T = 1$:

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2} = \frac{0.05 + 0.03125}{0.0625} = 1.3$$

so the two power factors are

$$\left(\frac{H}{S_0}\right)^{2\lambda} = 0.9^{2.6} = 0.760380, \qquad \left(\frac{H}{S_0}\right)^{2\lambda-2} = 0.9^{0.6} = 0.938740$$

and

$$y_1 = \frac{\ln\left(H^2/(S_0K)\right)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T} = -0.517884$$

Substituting into $C$:

$$C = 100 \times 0.760380N(-0.517884) - 100e^{-0.05} \times 0.938740N(-0.767884) = \mathbf{3.224778}$$

which matches Example 34.1's down-and-in value exactly; and, by in–out parity, $12.335999 - 3.224778 = 9.111221$ recovers the knock-out.

Note how little the eight barrier types cost. All eight have closed forms under Black–Scholes with continuous monitoring, assembled from four expressions $A, B, C, D$ with two sign switches $\phi$ and $\eta$. The formulas are not the difficulty, which is question 17's point.

*Why an up-and-in call with $H < K$ is worth the vanilla.*

Because the knock-in is certain whenever the option would pay anything.

An up-and-in call pays $(S_T - K)^+$ only if the running maximum $M_T$ has reached $H$. For the option to be worth anything at expiration we need $S_T > K$. But $H < K$, so

$$S_T > K > H \quad \Longrightarrow \quad M_T \geq S_T > H$$

The path has *necessarily* crossed the barrier; indeed it is above it at expiration. So the indicator $\mathbf{1}_{\{M_T \geq H\}}$ equals one on every path where the payoff is nonzero, and the barrier does nothing:

$$(S_T - K)^+ \mathbf{1}_{\{M_T \geq H\}} = (S_T - K)^+$$

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The two payoffs are *identical*, path by path, so the two options have the same value. No calculation is required.

*The corollary follows immediately from in–out parity.* Since in + out = vanilla, an *up-and-out* call with $H < K$ is worth **zero**, which is also obvious directly: to pay, the option must finish above $K > H$, and to be above $H$ at expiration it must have knocked out.

*The general discipline.* Check the barrier's position relative to the strike before computing anything. Several of the eight cases degenerate (to the vanilla, to zero, or to a simpler instrument), and a formula applied blindly will return the right answer while concealing that the instrument was not really a barrier option. A quoted price for an up-and-out call struck above the barrier is a quote for nothing, and someone should notice before it trades.

*What the question tests.* Substituting into the barrier block formula, and reasoning about degenerate cases from the payoff's indicator rather than from the formula.

---

**34.17** *Name three instruments in this chapter with closed forms and severe model risk, and one with no closed form and very little. Explain why the two properties are nearly unrelated.*

*Three have closed forms and severe model risk.*

*1. The continuous barrier option.* All eight types have exact Black–Scholes–Merton formulas (question 16): four expressions and two sign switches. And two models calibrated to the same vanilla surface price one 49.5% apart (question 8). The formula is exact and the answer is not determined.

*2. The lookback.* A floating-strike lookback has a closed form, and its value depends entirely on the distribution of the path's *extremes*, which is what a vanilla surface says least about (question 10). Model risk is high.

*3. The cliquet*, in its capped form. Each period is a forward-start call spread with a clean scaling formula, and its whole value sits in the *forward smile*, which the spot surface does not determine and which a local-volatility model gets systematically wrong (question 14).

*One with no closed form and very little model risk: the* ***arithmetic-average Asian***. A sum of lognormals is not lognormal, so no formula exists; the market prices it by moment-matching or by simulation. And its model risk is *low*: an average *smooths* the path, so the Asian is less sensitive to the path distribution than a vanilla is to the terminal one. The instrument that is traded is the one without a formula, and it barely matters, which is the reverse of the situation for every barrier in this chapter.

*Why the two properties are nearly unrelated.* Because they answer different questions.

*A closed form is a statement about* ***tractability under an assumed model***. It exists when the payoff happens to be a function of quantities that are analytically convenient under geometric Brownian motion: when a reflection principle applies, when a ratio of lognormals is lognormal, when a geometric average preserves the family. It says nothing about whether geometric Brownian motion is the right model.

*Model risk is a statement about* ***what the market determines***. By question 1's test, it is low when the payoff can be replicated *statically* from traded vanillas, and high when it can only be replicated dynamically. That depends on whether the payoff is a function of the terminal distribution, which the option chain reveals, or of the path, which it does not.

*Nothing links the two.* Analytic convenience under one model and determination by the market are independent properties, and the chapter's summary table sets them side by side precisely because they are so easily confused:

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| Instrument | Closed form | Model risk |
|---|---|---|
| Package, chooser | Exact | None |
| Compound | Exact, plus a root search | Low |
| Digital | Exact | Low; the cost is hedging |
| Barrier, continuous | Exact, all eight | **Severe** |
| Lookback | Exact | High |
| Asian, geometric | Exact | Low |
| Asian, arithmetic | **None** | Low |

*The practical warning.* A closed form is the most misleading reassurance in derivatives. It produces a number instantly, to arbitrary precision, with no simulation error and no convergence diagnostics to inspect, so it *looks* more reliable than a Monte Carlo estimate with a standard error attached. The Monte Carlo number is honest about its uncertainty; the formula's precision is entirely conditional on a model no one believes.

The right pairing of question and technique is therefore: use the formula for speed, and use question 1's replication test—not the formula's existence—to decide how much to trust the answer and how wide to quote.

*What the question tests.* That analytic tractability and market determination are orthogonal, and that a closed form's precision is conditional on the model rather than evidence for it.

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# Chapter 35
# Structured Products and Autocallables

**35.1** *Using Example 35.1, compute the participation rate before and after a 2% fee, and explain why the same note's participation would be 59.4% at 2% issuer funding and 107.3% at 6%.*

*The participation rate before fees.* From Example 35.1 (a five-year principal-protected note, \$1,000 face, issuer funding at 4.5%, index yielding 2.0% with 20% volatility):

| | |
|---|---|
| Zero-coupon bond to guarantee the \$1,000 | 798.52 |
| **Left to buy optionality** | **201.48** |
| Cost of a 100% participation at-the-money call | 209.60 |
| **Participation rate** | **96.1**% |

*After a 2% distribution fee.* The fee comes out of the optionality budget, not out of the guarantee:

$$\frac{201.48 - 20.00}{209.60} = \mathbf{86.6}\%$$

Ten points of participation, gone, and it appears nowhere in the term sheet. The face amount is unchanged, the maturity is unchanged, the guarantee is unchanged—the fee is paid entirely out of the one term the buyer is being sold on.

*Why funding drives the participation rate.* The bond leg must be discounted at *the issuer's own* curve, because the guarantee is the issuer's promise. A higher funding rate means a *cheaper* zero-coupon bond, which leaves more of the \$1,000 available to buy options:

| Issuer funding rate | Participation (no fee) | (2% fee) |
|---|---|---|
| 2.0% | **59.4**% | 46.9% |
| 4.5% | 96.1% | 86.6% |
| 6.0% | **107.3**% | 99.0% |

The rate *nearly doubles* (a factor of 1.80) as funding goes from 2% to 6%, *with the index unchanged*. Nothing about the equity market, its volatility, or its prospects has moved.

So the participation rate is mostly a statement about the issuer's funding cost, not about the index. Two consequences follow, and the second is uncomfortable.

*A note looks better when rates are high.* This is why principal-protected structures nearly vanished in the zero-rate years and returned after 2022. The product's attractiveness is a term-structure phenomenon, not an investment insight, and a salesperson explaining that participation has risen because the outlook has improved has the causation backward.

*And it explains where the money comes from.* A 107% participation rate at 6% funding is not generosity; the issuer has borrowed at 6% and spent the saving on options. The investor has made an unsecured five-year loan to the issuer and been paid in optionality rather than in coupon.

Which sets up question 2. If participation tracks funding, then comparing two issuers' notes on their headline rates is comparing their credit spreads—and the better-looking note is the weaker credit.

*What the question tests.* That the fee is charged against the option budget rather than the principal, and that the headline term is largely a function of the issuer's borrowing cost.

---

**35.2** *Explain why the best-looking note in a comparison across issuers at one moment is systematically the weakest credit.*

*The argument, in one line.* The issuer paying more to fund can buy more optionality.

Question 1 established the mechanism. A note's optionality budget is the face amount less the present value of the guaranteed principal, discounted at *the issuer's own* curve. A weaker credit discounts at a higher rate, so its zero-coupon bond is cheaper, so more of the $1,000 is left over:

| Issuer funding rate | Participation (no fee) |
|---:|---:|
| 2.0% | 59.4% |
| 4.5% | 96.1% |
| 6.0% | **107.3%** |

Now hold the moment fixed and compare across issuers. At a single point in time the risk-free curve, the index level, and the index's volatility are the *same for everyone*. The only other inputs that vary across issuers are the funding spread and the fee, which question 1 showed also cuts participation. So across a set of otherwise identical notes carrying the same fee, the *entire* dispersion in headline participation rates is credit spread. Nothing else can produce it.

Hence, fees equal, the ranking is exact and inverted: sort a set of comparable notes by participation rate and you have sorted them by issuer credit quality, worst first. The best-looking note is systematically the weakest credit.

*Why this is not a mispricing.* Each note is fairly priced against its own issuer's curve. The weak issuer is offering more optionality because it is offering more credit risk, and the extra participation is exactly compensation for it. Nothing is being given away, and nothing is being stolen: the investor is being paid to lend to a riskier borrower, in a currency (participation) that does not look like credit compensation.

*Why it nevertheless does damage.* Because of *how the products are chosen*. A buyer comparing term sheets sees participation rates, coupon levels and barrier positions, terms that look like product features, and does not see funding spreads. Selecting the most attractive terms is therefore selecting the weakest credit, without the buyer having made a credit decision at all.

Three features aggravate it:

- The note is *unsecured*, so the buyer is a senior creditor with no collateral, so the credit exposure is real and total.
- Distribution channels rank notes by headline terms, so the selection pressure runs the wrong way across an entire market rather than in one transaction.
- And the credit exposure is *correlated* with the payoff: an issuer under stress is more likely to fail in exactly the market conditions that trigger the note's knock-in.

The remedy is question 15's second step: price the bond leg at the issuer's own curve. A note from a AA issuer and one from a BBB issuer are not comparable on their headline terms, and the difference between them is credit compensation rather than investment skill. Doing that step converts an apparent product comparison into the credit comparison it actually is.

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*What the question tests.* That headline terms are a monotone function of funding spread when everything else is common, and that selecting on terms is unknowingly selecting on credit.

---

**35.3** *Decompose a reverse convertible paying 10% when deposits pay 4%, and state what the 6% actually buys.*

*The decomposition.* A reverse convertible pays a high fixed coupon and repays principal in full *unless* the underlying has fallen below a barrier, in which case it repays in shares. That is

**a bond + a short put that pays only below the barrier**

The investor is *short* an option, and the coupon is the premium.

*What the 6% buys.* A note paying 10% when deposits pay 4% is **not offering 6% of extra yield**. It is paying 6% for a put the investor has written. The investor has sold downside protection on the underlying and is being paid for it in installments, labeled as coupon.

To be precise about the exchange: the buyer has given up the right to their principal in states where the underlying falls below the barrier, and receives 600 basis points a year for it. Whether that is a good sale depends entirely on whether 600 basis points is a fair price for that put, which is a question about volatility, about the barrier's distance, and about the underlying's skew, none of which the term sheet mentions.

*Why the mis-selling risk here is structural rather than incidental.*

The product is described in the vocabulary of fixed income (coupon, maturity, principal, issuer), and its risk is entirely equity. Every term on the front page belongs to the bond leg; the option leg appears only as a barrier condition in the small print, phrased as a contingency rather than as a position.

So the buyer's natural comparison is wrong by construction. A buyer who compares it with a bond is comparing the wrong thing, and the comparison flatters the note enormously: 10% against a 4% deposit looks like a 6% pickup for accepting an issuer's credit, when it is a 6% premium for writing an equity put.

*The correct comparison is with selling puts*—and the same buyer would very likely decline that trade. A retail investor asked "would you like to write a one-year put on this index, struck near the money, for 6% of notional?" would recognize the question as an equity derivatives decision requiring a view. The same investor asked "would you like a 10% coupon instead of 4%?" does not.

That gap between the two framings is the mis-selling risk, and it is structural because it is created by the product's *name and format* rather than by anything a salesperson says. No misrepresentation is required; the vocabulary suffices.

*The general pattern, which the chapter returns to repeatedly.* If the headline term is attractive, the buyer is short something; question 15's first evaluation step states it as a rule. A range accrual's enhanced coupon is a short volatility position (Chapter 33's question 6); an autocallable's 18.42% is a short put. In every case the premium is real, the risk is real, and only the premium appears on the front page.

*What the question tests.* Decomposing the note into a bond and a written option, and that the fixed-income vocabulary is what makes the equity risk invisible.

---

**35.4** *Using Example 35.2, state the fair coupon and the three probabilities, and explain why a payoff sold on its hit rate is negatively skewed.*

*The fair coupon and the three probabilities.* From Example 35.2 (a two-year note on an

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index at 100, autocall at 100% observed monthly after a two-month lockout, knock-in at 75% observed daily, volatility 25%, rate 2.5%, 80 million antithetic paths):

| | |
|---|---|
| $P$(autocalls at some point) | **0.8311** |
| $P$(knocks in at some point) | **0.4132** |
| $P$(knocked in and never autocalled) | **0.1645** |
| **Fair coupon** | **18.42% per annum** |

Note that the second probability is much larger than the third: 41% of paths touch the barrier at some point, but most of those recover and autocall later, so only 17% end badly. The residual 0.44% is the paths that never autocall and never knock in, returning principal plus the accrued coupon at maturity.

*An 18.4% coupon, and the note prices at par.* Nothing is being given away: the coupon is exactly what the embedded short put is worth. A reader who finds 18.4% implausibly generous should read it as a measurement of the option rather than as an offer.

*Why a payoff sold on its hit rate is negatively skewed.*

*The note produces its advertised return about 83% of the time.* That is a genuinely high win rate. It is why these products are easy to sell, and why holders report satisfaction for years at a stretch, rolling the proceeds of each autocall into the next note.

The remaining 17% is where all the loss is, and it is not a small loss. A knocked-in note pays $\min(S_T/100, 1)$, so the investor absorbs **the full downside from par**: there is no partial protection once the barrier is breached, and the loss is whatever the index has done.

That is the definition of negative skew: many small gains, few large losses, and a fair price obtained by multiplying them out. Question 7 gives the magnitudes: wins average 8.22% and losses average 32.05%, a nearly four-to-one asymmetry at a five-to-one win rate, and the two ratios multiply to par.

*Why this shape recurs rather than being particular to autocallables.* It is what sells. A product is chosen on the two statistics a buyer looks at, the headline coupon and the probability of success, and both are maximized by pushing the loss into a small, distant, high-severity tail. The mean is unchanged, because the price is fair; only the shape is engineered.

The chapter names the family: this is the same structure as Chapter 22's inverse volatility products, whose losses compounded into a 96% single-day destruction, and Chapter 33's range accrual, which pays a 69% enhancement and fails exactly when income is needed. Different underlyings, one payoff shape, one reason for it.

And the exposure the buyer does not see is volatility (question 5's subject), because the note was priced at 25% and its value falls as realized volatility rises.

*What the question tests.* Reading the three probabilities as a description of the payoff's shape, and that a high hit rate and a fair price together imply negative skew.

---

**35.5** *Explain why an autocallable is short volatility in the worst possible way, being specific about when volatility rises.*

*The exposure.* From Example 35.2, revaluing the note at its fair 18.42% coupon across realized volatilities:

| **Realized volatility** | **Note value at $c$ = 18.42%** |
|---|---|
| 15% | 1.0520 |
| 20% | 1.0204 |
| 25% | **1.0000** |
| 30% | 0.9847 |
| 40% | 0.9574 |

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*The note is short volatility.* It was priced at 25% and prices at par there; above that the investor loses, below it they gain.

*Why the exposure exists.* The embedded short put is a short volatility position, and the barrier sharpens it: higher volatility raises the probability of touching 75%, which is the only route to a loss. It also *reduces* the probability of autocalling early, so the investor is left holding the risk for longer.

*Why this is the worst possible way to be short volatility.* Because volatility rises exactly when the underlying falls, so the two exposures are not independent risks that might offset across scenarios—they are one risk, arriving together.

Chapter 19's leverage effect and skew are the general statement: equity volatility is negatively correlated with the index level, structurally and reliably. So the state in which the note's short volatility position loses is the *same* state in which its short put position loses:

- the index falls, so the knock-in approaches;
- volatility rises, so the probability of breaching rises further;
- autocalls stop occurring, so the position cannot terminate early and escape;
- and the barrier, once touched, exposes the full downside.

*Four effects, one direction, one trigger.*

*The compounding is what distinguishes this from an ordinary short vega position.* A trader short vega on a delta-hedged book loses when volatility rises and is indifferent to direction. The note's holder is short vega *and* short the underlying's downside, with the vega loss concentrated in the downside states. The two exposures do not diversify; they multiply.

*And the pricing was set in the wrong conditions.* Notes are issued when markets are calm and volatility is low, because that is when investors are comfortable buying them, although low implied volatility pays a smaller coupon than a stressed market would. So the coupon reflects calm-period volatility and the risk is realized in stressed-period volatility. Chapter 34's question 12 makes the parallel point about the correlation on a worst-of note: the compensation is set at the point in the cycle least representative of the state that matters.

*One further consequence follows for the holder.* They cannot hedge it. The exposure is to *realized* volatility over the note's life, not to implied volatility at any date, so there is no vega position to unwind, and question 10 shows the exit is worst precisely when they would want it.

*What the question tests.* That the note's vega and delta exposures share a trigger through the leverage effect, and that issuance timing systematically prices the risk in the wrong regime.

---

**35.6** *State the note's and the index's skewness from the simulated distribution, and explain what a buyer switching between them is exchanging when both earn the risk-free rate.*

*The two skewness figures.* From Example 35.3, simulating the note at its fair 18.42% coupon against simply holding the index for two years, on the same 80 million paths:

| | **The note** | **The index** |
|---|---|---|
| Mean total return | +1.60% | +5.13% |
| Standard deviation | 17.25% | 38.36% |
| **Skewness** | **−1.59** | **+1.14** |
| $P$(positive) | 83.55% | 48.59% |

Both instruments earn the risk-free rate by construction, since everything is computed under the pricing measure. The note's lower mean is not a worse deal: it terminates early, after 0.70

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years on average, so it earns that rate over a shorter life. The check is $2.5\% \times 0.70 = 1.75\%$ against the simulated 1.60%; the small shortfall is there because the longest-lived paths, which carry the largest growth factor, are the knocked-in ones that lose, and it confirms the pricing is consistent.

So with the mean neutralized, the comparison is purely about shape—and the shape is the product.

*What the buyer is exchanging: a **positively** skewed payoff for a **negatively** skewed one.*

*The index is positively skewed* (+1.14): most two-year holding periods deliver a modest result (the probability of a positive return is only 48.6%), and the distribution's long tail lies on the right, in large gains. An equity holder is paid mainly by infrequent large upside.

*The note is negatively skewed* (−1.59): 83.55% of outcomes are positive and modest, and the distribution's long tail lies on the *left*. The holder is paid frequently and small, and loses rarely and large.

A buyer who moves from the index into the note has not reduced risk; they have exchanged a positively skewed payoff for a negatively skewed one of the same expected value. That is a genuine choice and it may be the right one for some purposes, but it is not the choice the marketing describes, which is "equity-like returns with more certainty."

*Why the exchange is easy to make unknowingly.* Every statistic a retail buyer naturally consults improves:

- the probability of a positive return rises from 48.59% to 83.55%;
- the standard deviation falls from 38.36% to 17.25% (question 9's subject);
- and the 5th percentile improves slightly, from −44.79% to −40.03%.

*Only the third moment worsens*, and it is the one no one reports.

*The deeper point about what skewness is worth.* Positive skew is valuable: it is why lottery tickets are overpriced. Negative skew is the thing investors must be *paid* to hold (it is why out-of-the-money index puts trade rich), and here the payment is the coupon. So the note's holder is collecting a skewness premium, and the question is whether they know they are selling insurance. Chapter 19's question 7 makes the identical argument about straddle sellers, and Chapter 22's question 9 about dispersion.

*What the question tests.* That equal means make the comparison purely one of shape, and that a skewness reversal is invisible in every statistic a buyer normally consults.

---

**35.7** *State the average win, the average loss, and the worst one percent, and reconcile them with the 83.55% probability of a positive return.*

*The three figures.* From Example 35.3:

| | |
|---|---|
| $E$[return \| positive] | **+8.22**% |
| $E$[return \| negative] | **−32.05**% |
| Average of the worst 1% | **−60.23**% |
| Median return | +4.61% |
| $P$(positive) | 83.55% |

*The reconciliation.* The three numbers are not in tension with the 83.55% win rate; they are what makes it consistent with a fair price:

$$0.8355 \times 8.22\% + 0.1645 \times (-32.05\%) = 6.87\% - 5.27\% = \mathbf{+1.60}\%$$

which is exactly the simulated mean total return. The arithmetic closes to the last decimal.

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Read the two ratios. Wins average 8% and losses average 32%, *a nearly four-to-one asymmetry*, at *a five-to-one win rate*. Those two ratios are the product, and they multiply out to a fair price. Neither on its own describes the note; their product is what the price reflects and the marketing reports only the second.

*Why an 83.55% win rate is compatible with a poor outcome.* Because frequency and severity are separate quantities, and a fair price constrains only their product. Engineering the frequency up requires pushing the severity up too, and there is no configuration in which both improve: the price enforces the trade-off.

*The worst one percent is where the risk sits.* At −60.23%, the tail is not merely worse than the average loss; it is nearly twice as bad. A knocked-in note pays $\min(S_T/100, 1)$, so once the barrier is breached the investor holds the index outright, with no floor and no participation in any recovery beyond par. The note's downside is the index's downside, undiminished, while its upside was capped at the coupon.

That asymmetry is the whole instrument: the buyer has exchanged all of the index's upside beyond the coupon for none of the index's downside protection.

This makes two things clear about how such products should be described.

The pair of statistics the term sheet reports, the coupon and the hit rate, are the two the product was engineered to optimize. A buyer who sees only $E[\text{return}]$ and $P(\text{positive})$ has seen exactly the statistics that are maximized subject to the price being fair.

*And the statistic that would inform them is the conditional loss.* "When this loses, it loses 32% on average and 60% in the worst one percent" is a sentence that conveys the position; "83.55% of the time you receive 18.42% per annum" does not. Question 15's fourth evaluation step is exactly the instruction to compute the first sentence.

*What the question tests.* That the three conditional statistics reconcile to the fair mean, and that a high hit rate is purchased with severity rather than being evidence of safety.

---

**35.8** *Explain why the 25th and 50th percentiles of the note's return coincide, and why that makes the product unusually persuasive.*

*Why the two percentiles coincide.* Because the note's return distribution has a large **atom** (a single outcome occurring with substantial probability), and both percentiles fall inside it.

The mechanism is the autocall. The note is observed monthly after a two-month lockout, so the *first* observation date is at three months. If the index is at or above 100% then, the note terminates and pays the coupon for a three-month holding:

$$18.42\% \times 0.25 = \mathbf{4.61}\%$$

which is exactly the median return reported in Example 35.3.

Because the index starts *at* the autocall level, the probability of being above it three months later is close to a half: 49.5% of simulated paths deliver precisely this return. The distribution therefore has a point mass at +4.61% spanning every percentile from the 17th to the 65th, and the quantile function is *flat* across that whole range.

*Why that makes the product unusually persuasive.*

*Every quantile a cautious buyer checks returns the same reassuring number.* Someone who asks "what is the median outcome?" and someone who asks "what happens in the bad quartile?" (a deliberately conservative question) receive *identical* answers. The distribution appears not merely favorable but *robust*: it seems that even a poor outcome delivers the advertised return.

That impression is precisely wrong. The distribution is not flat because outcomes are stable; it is flat because half the probability mass has been concentrated onto one point, and the

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variation has been pushed entirely into the tail below. Question 7 shows what is down there: an average loss of 32% and a worst-one-percent of −60%.

So the usual diagnostic fails silently. Checking a lower quantile is a standard way to probe a distribution's downside, and it works because most distributions vary smoothly: the 25th percentile of an equity return genuinely tells you about a bad-but-not-extreme outcome. Here the same check returns the median, and the buyer concludes the product is safe on evidence that contains no information about its risk.

*The threshold is sharp and unmarked.* The distribution stays flat down to roughly the 17th percentile and then falls off a cliff. A buyer sampling at the 25th percentile sees nothing; one sampling at the 15th sees the whole risk. Nothing in the term sheet indicates where the cliff is, and the natural round numbers a person would check are all above it.

*The remedy* is question 15's fourth step (simulate the distribution, not the scenarios), and specifically to report the *conditional* loss rather than more quantiles. "When this loses, it loses 32%" cannot be defeated by choosing a percentile above the cliff.

*What the question tests.* That the autocall atom flattens the lower half of the quantile function, and that a standard downside check returns no information when the mass is concentrated.

---

**35.9** *Explain why a variance-based risk measure reports this note as safer than the index, and what that says about variance as a risk measure.*

*What variance reports.* From Example 35.3:

| | **The note** | **The index** |
|---|---|---|
| Standard deviation | **17.25**% | **38.36**% |
| Skewness | −1.59 | +1.14 |

The note's standard deviation is **55% lower**. On any variance-based measure (volatility, a Sharpe ratio, a mean–variance optimizer, a risk rating derived from historical standard deviation), the note is emphatically the safer instrument.

*Why variance reports it that way.* Because variance is symmetric and squares deviations, so it treats a gain and a loss of the same magnitude identically and is dominated by the *body* of the distribution rather than by the tail.

The note's body is extremely tight: question 8 showed a point mass at +4.61% covering the 17th through 65th percentiles, and 83.55% of outcomes are modest positives clustered near the coupon. That concentration produces a small variance almost by itself. The 17% of outcomes carrying the entire loss, averaging −32% and reaching −60%, contribute to the variance but do not dominate it, because they are few.

Meanwhile the index's variance is inflated by its *upside*: a positively skewed distribution with a long right tail of large gains scores badly on a measure that penalizes deviation in either direction. Variance counts the index's best outcomes as risk.

*What this says about variance as a risk measure.* That it is a measure of **dispersion**, not of **danger**, and the two coincide only for symmetric distributions.

Three specific failures are visible here:

*It is blind to sign.* A distribution can be made to score well by removing upside, which is exactly what the note's coupon cap does. Any instrument that truncates gains and retains losses will look safer on variance, and that is a design principle rather than an accident—it is how the product is engineered.

*It is dominated by the body.* Risk lives in the tail, and squaring deviations weights the tail by its probability, which for a rare severe loss is small. Chapter 38's expected shortfall exists for this reason: it conditions *on* the tail rather than averaging over the whole distribution.

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*And it can be gamed by construction.* The note was not built to have low variance, but low variance is an automatic consequence of concentrating outcomes, so a product designer optimizing for apparent safety and one optimizing for saleability produce the same instrument.

*The general lesson.* A risk measure that a payoff can be shaped to satisfy is not measuring risk. Chapter 38 develops the formal version (subadditivity and coherence), and this note is the concrete case: a negatively skewed instrument with a 60% tail loss scoring as half as risky as the index it references, on the measure most widely used.

And note that no statistic reported here is wrong. The standard deviation genuinely is 17.25%. The failure is in the inference from it, which is why question 6's skewness and question 7's conditional loss belong on the same page.

*What the question tests.* That symmetric squared deviation rewards truncated upside, and that dispersion and danger are different quantities.

---

**35.10** *Explain why the exit from a structured note is worst precisely when it is most wanted.*

*Why the exit is poor at all times.* A structured note is quoted daily and is not liquid, and the distinction matters more than it sounds.

*The issuer is usually the only bid.* There is no exchange, no competing market maker, and no obligation to quote. A buyer who wants out sells back to the party who sold it to them, at a price that party computes.

Two consequences follow immediately. The bid embeds a second spread: having paid the issue premium of Section 35.10, an early seller pays again on the way out, and round-tripping a note in the first year can cost several percent of principal before any market move. And the quoted price is a model output: these are Level 2 or Level 3 valuations, and Chapter 26's observation about the crisis applies in miniature: when the only price comes from the party on the other side, the holder cannot distinguish a mark from a quote.

*Why it is worst precisely when it is most wanted.* Because the conditions that create the desire to sell are the same conditions that move the bid against the seller. An investor most wants to exit after a sharp fall, when the knock-in is close and the outlook has changed, and at that moment:

*The embedded option has moved most against them.* The short put is now near the money, so it is worth far more, and the note is worth correspondingly less. This is not the dealer being opportunistic—it is the correct valuation.

*Volatility has risen, compounding it.* Question 5 established the note is short volatility, and volatility rises when the index falls. So the value falls through two channels at once.

*The issuer's own funding spread has probably widened*, which lowers the bond leg, and in a market-wide stress the issuer is under pressure for the same reasons the underlying is.

*And the issuer's willingness to quote deteriorates*, because its own hedging book is under strain at exactly that moment (question 12).

So four effects push the same way, triggered by one event. The exit is worst when it is most wanted, which is this book's recurring wrong-way structure appearing one last time; Chapter 27's question 12 collects the family: excess spread, recovery rates, sovereign protection, portfolio insurance, netting enforceability. A facility whose availability is negatively correlated with the need for it.

*The practical conclusion, and it is question 15's fifth step.* A note should be evaluated as a hold-to-maturity instrument, because that is what it is. A buyer who is not certain they can hold it for the full term, and through the scenario in which they would most want not to, should not consider the liquidity feature as mitigation, because it will not be there.

*The Korean episode is the case study.* Notes referencing an offshore China index worked as

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designed for years, with holders repaid early and rolling into new notes. When the index fell a long way and stayed there, holders who wanted out found the bid reflecting a near-certain knock-in, and the loss was crystallized either way.

*What the question tests.* That the sole bidder is also the counterparty whose valuation moves with the same trigger, and identifying this as an instance of the book's wrong-way pattern.

---

**35.11** *Explain why a product with a 17% loss probability sold to a hundred thousand people does not produce 17% of them losing independently.*

*The error.* The 17% is a probability over states of the world, not a fraction of holders. It answers "how likely is this note to lose?" and not "what proportion of buyers will lose?"

*Why they are different.* The hundred thousand holders own notes referencing the *same* underlying, with barriers set at the *same* level, observed over overlapping periods. Their outcomes are therefore **almost perfectly correlated**, not independent:

- if the index never breaches 75%, *essentially none* of them loses;
- if it does breach and stays down, *essentially all* of them lose.

There is no mechanism producing 17,000 losers and 83,000 winners. The 17% is the probability of the bad state, and in that state the losses are collective.

An independence assumption would predict a binomial distribution of losers with mean 17,000 and a standard deviation of $\sqrt{100{,}000 \times 0.17 \times 0.83} \approx 119$, so between 16,600 and 17,400 losers, essentially every time. The true distribution is bimodal: near zero with probability 83%, near a hundred thousand with probability 17%. Same mean, entirely different distribution, and the second is the one a supervisor cares about.

*Why this matters at three levels.*

*For the individual buyer*, nothing changes: their own probability is 17% either way. The correlation is irrelevant to a single decision.

*For a distributor*, it changes everything. A bank selling a hundred thousand notes is not running a diversified book of small risks; it is running *one* risk at a hundred thousand times the size. Its conduct exposure, its reputational exposure, and its complaint volume all arrive at once or not at all. A distributor reasoning "some will lose, that is the nature of the product" has assumed a diversification that does not exist.

*And for a supervisor*, it is the whole of the concern. A hundred thousand simultaneous retail losses is a political and social event, not a statistical one. It generates redress schemes, parliamentary attention and regulatory intervention, which is why the Korean episode and China's snowball market both produced supervisory responses rather than merely losses.

*The mechanism compounds through question 13's clustering.* Barriers are set at round fractions of the level prevailing at issue, so notes issued during any period of stable prices share nearly the same *absolute* barrier. The correlation therefore extends beyond one issuer's book to the entire cohort of notes sold across a market over months, and a single sustained decline resolves all of them together.

*The general statistical point.* A marginal probability says nothing about a joint outcome, and the whole of Chapter 24's correlation discussion is this observation in a different market: correlation leaves every marginal untouched and moves probability to both tails. Here the marginal is 17% at every correlation, and the distribution of *aggregate* losses is entirely determined by the correlation—which is one.

*What the question tests.* That a probability over states is not a proportion of a population when the population shares an underlying, and that the distinction is invisible to the individual and decisive to everyone else.

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**35.12** *Using Example 35.6, state the delta just above and just below the barrier and the jump, then explain why the withdrawal of the issuer's bid matters more than the forced sale at the barrier.*

*The two deltas and the jump.* From Example 35.6 (the embedded down-and-in put struck at 100 with a barrier at 75, one year, 25% volatility, a 2.5% rate and no dividends, valued exactly with the barrier formulas of Chapter 34):

| Index | Delta |
|---|---|
| 130 | −0.055 |
| 100 | −0.399 |
| 80 | −0.921 |
| **75.33** (just above) | **−1.017** |
| **74.67** (just below) | **−0.827** |
| 70 | −0.885 |

*The jump is* +0.190: the delta moves from −1.017 to −0.827 on crossing the barrier.

Above the barrier the delta steepens *monotonically* from −0.055 to −1.017 as the market falls, and it exceeds one in magnitude, the barrier feature of Chapter 34's question 2. The issuer, *long* this option, must therefore hold a growing *long* position in the index as the market falls: from 5.5% of notional at 130 to **101.7%** at the barrier.

So the issuer is buying into a falling market, continuously, and that flow is stabilizing.

*Why the withdrawal of the bid matters more than the forced sale.*

At the barrier, two things happen at once.

First, the issuer must sell about 19% of notional immediately: the delta jump, executed as a discrete order. This is what the popular account of these episodes emphasizes.

*Second, and larger: the buying stops.* Below the barrier the option is an *ordinary* put, whose delta steepens far more slowly, from −0.827 at 74.67 to only −0.885 at 70. So the persistent bid that had been supporting the market on the way down *simply disappears*.

Compare the magnitudes. The one-off sale is 19% of notional. The flow that vanishes had been running from 5.5% to 101.7% of notional over the descent, a cumulative purchase of some 96% of notional, arriving steadily as a bid underneath the market. Losing that is a far larger change in the market's demand than a single 19% sale.

*And the arithmetic changes what should be monitored.* The popular account says: watch for the forced selling at the barrier. The arithmetic says: watch the size of the stabilizing flow that is about to stop. Those are different measurements. The first is a one-off order size, the second is the accumulated hedge position that will cease to grow, and only the second is proportional to the risk.

Which sets up question 13. If the relevant quantity is the aggregate stabilizing bid, then it must be measured across *all* issuers, and no single issuer can see it.

*One structural note.* This is Chapter 34's question 2 in a market rather than in a pricing exercise: a knock-in delta exceeding one is what makes the hedge large enough to matter, and the discontinuity at the barrier is what makes its disappearance abrupt.

*What the question tests.* Reading the delta profile as a hedging flow, and that the loss of a persistent bid is a larger market event than the discrete trade that accompanies it.

**35.13** *Explain why barriers cluster across issuers, why no issuer can observe the clustering, and what that implies about what should be monitored.*

*Why barriers cluster.* The clustering is not coincidence but arithmetic. Barriers are set at **round fractions** (75%, 80%) of the level prevailing *at issue*. Notes are issued continuously

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against a rolling reference level, so any period of stable prices produces a cohort of notes sharing nearly the *same absolute barrier*.

A sustained decline of a given size therefore brings a large cohort into range *at once*. Nothing coordinates the issuers; the convention of round percentages and the fact that they are all referencing the same index does it automatically.

*Why no issuer can observe it.* Because each dealer sees only its own book. There is no central register of outstanding structured notes by barrier level, no reporting requirement that aggregates them, and no commercial incentive to share: a dealer's book of barriers is competitively sensitive information.

So each issuer can compute, precisely, the stabilizing flow *it* will provide and the point at which *its* bid will withdraw. None can compute the aggregate. The stabilizing flow is larger than any one issuer knows, and this is Chapter 27's question 15 aggregation problem in a different market: in its Archegos arithmetic six prime brokers each see an unremarkable $8.3 billion and none sees the $50 billion.

*The three consequences, in increasing seriousness.*

*The flow is larger than anyone knows*, so its stabilizing effect is under-appreciated on the way down and its withdrawal is under-anticipated.

*The withdrawal is synchronized.* When the index passes the cluster, many issuers' bids disappear within the same few sessions. Question 12 established that the vanishing bid is the larger effect for one issuer; here it happens to all of them at once.

*And the feedback is self-reinforcing while it lasts.* The buying on the way down slows the decline, which keeps the index above the barriers longer, which sustains the buying—until it does not. A stabilizer that works until it stops is not a stabilizer, and Chapter 23's saturating pin risk made the identical point about a different product: a market that damps small moves and cannot damp large ones is arguably worse than one that damps neither, because the calm is manufactured and the warning signs are suppressed.

*What should be monitored.* Not any single issuer's position, and not the size of the sell order at the barrier. The aggregate outstanding notional by barrier level: the accumulated stabilizing bid that will cease when a given index level is passed.

That is a measurement no market participant can make and a supervisor could: it requires only the outstanding notes' barriers and notionals, which every issuer knows for its own book and a regulator could collect. Neither the Korean nor the Chinese response went this far: Korea's targeted sales practice, and China's suitability rules and its limits on private funds' snowball holdings and on dealers' issuance, in place before 2024, still did not measure outstanding notional by barrier level.

*The general principle, and it is the chapter's closing one.* A risk that is individually measured and collectively invisible is the shape that produces incidents. Chapter 26's securitization chain, Chapter 27's Archegos, Chapter 23's dealer gamma and this market are four instances, and in each the remedy is aggregation rather than a limit on any participant.

*What the question tests.* That round-number conventions plus a common underlying generate clustering mechanically, and that the monitorable quantity is aggregate notional by barrier rather than anyone's individual position.

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**35.14** *Explain why a note linked to the worst of three indices pays roughly double the coupon, and why that is a measure of risk rather than of value.*

*Why the coupon roughly doubles.* Because the embedded short put is worth roughly twice as much.

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Chapter 34's question 12 prices it. With three assets each at 100 with 28% volatility, a **worst-of** put struck at 100 (paying on the *minimum* of the three) is worth

| $\rho$ | Worst-of put | Relative to single-asset put |
|---|---|---|
| 0.20 | 17.64 | **2.05** |
| 0.60 | 14.71 | 1.71 |
| 0.95 | 10.64 | 1.24 |

At a realistic correlation the worst-of put is worth about twice a single-asset put, and the coupon is the premium on that put. Double the option, double the coupon.

The mechanism is that a minimum has no cancellation. A *basket* averages, so idiosyncratic moves partly offset; a *worst-of* records whichever asset fell furthest, so more assets and lower correlation both make the outcome worse. With three assets at $\rho = 0.20$ the expected minimum at one year is **83.87** against a starting level of 100: the drift is upward and the minimum still falls sixteen percent, purely from picking the loser three times.

*Why that is a measure of risk rather than of value.*

*Because the note prices at par either way.* The single-underlying note and the worst-of note are both fairly priced; the coupon differs because the risk differs, not because one offers a better deal. A doubled coupon on a doubled risk is not an improvement—it is the same trade at twice the size.

So the coupon is a *thermometer*, not a bargain. Reading across a set of notes, the one with the highest coupon is the one whose embedded option is most valuable, which is to say the one most likely to cost the investor money. The coupon ranks the products by risk, in descending order of attractiveness.

That is the same inversion as question 2, where the best-looking participation rate identified the weakest credit. Here the best-looking coupon identifies the worst-of structure with the most underlyings and the lowest correlation.

*The higher coupon is paying for three specific things.*

*More ways to lose.* A single-underlying note fails if one index breaches; a worst-of note fails if *any* of three does. The failure probability rises with the number of underlyings, and issuers respond to demand for higher coupons by adding underlyings, which is the cheapest way to raise a coupon and the least visible way to raise the risk.

*A correlation position that offers no protection.* The investor is long correlation (Chapter 34's question 12): a low correlation makes the worst-of put dear, and the investor loses when one asset falls alone. Correlation *rises* in crises, which on its own would help, but it rises because everything is falling together, and with every asset below the strike the put is deep in the money whatever the correlation. The coupon pays for two losses, and the crisis delivers the second.

*And a selection effect in which underlyings are chosen.* Issuers pick volatile indices precisely because volatility pays the coupon: Korea's notes referenced an offshore China index and China's snowballs reference the CSI 500 and CSI 1000 for exactly this reason. The underlying is selected for the property that makes the note dangerous.

*The practical rule.* Treat a coupon as a risk disclosure. Two notes at 9% and 18% are not a poor deal and a good one; they are a smaller position and a larger one in the same trade, and the correct question is whether the investor wants twice as much of it.

*What the question tests.* That the coupon equals the option premium so it scales with risk, and that ranking products by coupon ranks them by danger.

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**35.15** *List the five evaluation steps of Section 35.11, state what each is designed to reveal, and apply them to Example 35.2's note in one sentence.*

*The five steps, and what each reveals.*

*One. Decompose it.* Every note is a bond plus one or more options. Identify the bond, identify each option, and, critically, identify which side of each option the buyer is on. *Reveals:* whether the buyer is long or short optionality. The rule is blunt and reliable: if the headline term is attractive, the buyer is short something.

*Two. Price the bond leg at the issuer's own curve. Reveals:* how much of an attractive participation rate or coupon comes from the issuer's funding spread rather than from the underlying. Question 1 showed participation running from 59.4% to 107.3% on funding alone, and question 2 that the best-looking note is the weakest credit. A note from a AA issuer and one from a BBB issuer are not comparable on their headline terms.

*Three. Value the option leg and compare with par. Reveals:* the fee. The gap between the sum of the parts and 100 is what the buyer paid for distribution and structuring. Where an estimated initial value is disclosed this step is done for the reader; where it is not, the rule that the fee widens with complexity is reliable.

*Four. Simulate the distribution, not the scenarios.* A term sheet presents three or four illustrative outcomes. *Reveals:* the skewness (question 6), the conditional loss given a loss (question 7), and the fact that the median and the 25th percentile can be the same number (question 8). A buyer who sees only $E[\text{return}]$ and $P(\text{positive})$ has seen the two statistics the product was engineered to optimize.

*Five. Ask what the exit is worth. Reveals:* that the issuer is the only bid, that the spread applies again, and that the bid is lowest when the need to sell is greatest (question 10). A note should be evaluated as a hold-to-maturity instrument, because that is what it is.

*Applied to Example 35.2's note, in one sentence:*

> It is a two-year unsecured loan to the issuer, bundled with a short down-and-in put on the index struck at par with a 75% barrier and a short position in realized volatility, whose coupon is the premium on those written options rather than a yield, fair at 18.42% and about 2.1% above fair value when sold at Example 35.7's 12.65%; it returns a modest positive result 83.55% of the time and loses an average of 32% the rest, with no liquid exit and a bid that is lowest exactly when the holder would want it.

That is a description no marketing document contains, and every clause in it came from one of the five steps.

*Why the procedure is worth having as a procedure.* It requires no model the reader does not now have (a bond price, an option price, a Monte Carlo simulation), and it answers the questions a term sheet is designed not to raise. The term sheet is not dishonest; it is an accurate description of the cash flows, organized so that the coupon and the hit rate are prominent and the written option, the funding spread, the fee, the skewness and the exit are not.

*And the steps are ordered deliberately.* Decomposition must come first because everything else depends on knowing which options exist and which way round they are. The distribution comes fourth because it requires the decomposition to simulate. And the exit comes last because it is the step buyers skip, on the assumption that a daily quote means a market.

*What the question tests.* The procedure as a whole, what each step is diagnostic of, and the ability to state a product's economics in language its documentation avoids.

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# Chapter 36

# Energy, Commodity, and Environmental Derivatives

**36.1** *State the Samuelson relation and compute the twelve-month volatility at $\sigma_S = 50\%$ and $a = 1.0$. Using Example 36.2, state the one-month to twelve-month ratio at $a = 1.5$ and explain what a cross-maturity hedge ratio of one would cost.*

*The relation.* If the spot mean-reverts at rate $a$, a shock decays before a distant contract expires, so by (36.2)

$$\sigma_F(T) = \sigma_S\, e^{-a(T-t)}.$$

The decay is the one Hull–White gives forward rates in Section 31.6, but here it is a percentage volatility of a price, not a basis-point volatility of a rate.

*The twelve-month volatility* at $\sigma_S = 50\%$ and $a = 1.0$:

$$50\% \times e^{-1.0\times 1} = 50\% \times 0.3679 = \mathbf{18.4}\%$$

*The ratio at $a = 1.5$.* From Example 36.2, at a 60% spot volatility the one-month contract carries **52.9%** and the twelve-month **13.4%**:

$$\frac{52.9}{13.4} = \mathbf{3.96}$$

Nearly a factor of four, and it is independent of $\sigma_S$: the ratio is $e^{a(T_2-T_1)}$ and the level cancels.

*What a cross-maturity hedge ratio of one would do.* It would **over-hedge the volatility by that ratio**. Hedging a twelve-month exposure with front-month futures on a one-for-one basis puts on roughly four times the volatility exposure the position carries, so the "hedge" introduces more risk than it removes on the far leg.

This is the commodity version of the duration mismatch of Chapter 6: matching notionals rather than sensitivities. There, hedging a ten-year bond with a two-year futures contract one-for-one leaves an enormous residual; here, hedging a twelve-month barrel with a one-month barrel does the same. The correct ratio is $e^{-a(T_2-T_1)}$, and question 9 shows what happens when a firm ignores it at scale.

Two further consequences follow, and the second is the one desks get wrong.

*Options on different contract months are not comparable at the same strike.* A 40% implied volatility on the front month and on the one-year are describing very different markets, so a commodity volatility surface needs a maturity dimension that means something quite different from an equity surface's. On an equity surface the underlying is the same asset at every expiration; here it is not.

*And the effect competes with a second one. Storage* links adjacent contracts, which *raises* correlation between them; *mean reversion lowers* the far contract's volatility. So a commodity with cheap, ample storage has a flat volatility term structure and a highly correlated curve; one with costly or full storage has a steep one and a curve whose near and far ends can move independently.

The volatility term structure is therefore a direct read on storage economics, which is what makes it worth quoting, and which is why the same $\sigma_S$ and different $a$ describe genuinely different physical markets rather than different calibrations.

*What the question tests.* Substituting into the Samuelson relation, and that a notional-matched cross-maturity hedge is mis-sized by the volatility ratio.

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**36.2** *Using Example 36.3, state the ratio of spike-model to lognormal cap values at \$200 and at \$400, and explain why calibrating the lognormal to at-the-money options does not close the gap.*

*The two ratios.* From Example 36.3, comparing the spike model (an ordinary day lognormal with median \$43 and log standard deviation 30%, multiplied on 2.4% of days, about nine a year, by a lognormal spike factor with mean 5 and log standard deviation 60%) against a lognormal fitted to the *same mean and the same standard deviation*:

| Strike | Spike model | Lognormal | Ratio |
|---|---|---|---|
| \$60 | 5.3300 | 10.4989 | 0.51 |
| \$200 | 1.6578 | 0.6518 | **2.54** |
| \$400 | 0.5056 | 0.0483 | **10.48** |
| \$800 | 0.0739 | 0.0016 | 46.70 |

At \$200 the spike model is worth **2.54 times** the lognormal; at \$400, **10.48 times**.

And note the disagreement runs in opposite directions at the two ends: 0.51 at \$60 and 46.70 at \$800.

*Why calibrating to at-the-money options does not close the gap.* Because the lognormal already *over*-prices the cap near the money, so fitting it there moves the error the wrong way.

The lognormal was fitted on the **mean and the standard deviation** (\$49.30 and \$40.42 in both models), and it reproduces those two moments exactly. It does not reproduce the at-the-money cap: like the \$60 cap, that one is worth less in the spike model than in the lognormal. Calibrating the lognormal to it therefore *lowers* its volatility, and a lower volatility makes the far caps cheaper still, widening the gap the question asks about.

*The mechanism is where the variance is spent.* The lognormal spends its variance on everyday fluctuation and the real distribution spends it on nine or so days a year. The two have the same total variance and completely different allocations of it:

- the lognormal makes *moderate* moves too likely, so it **over-prices** near-the-money caps (0.51 at \$60);
- and it makes *extreme* moves far too unlikely, so it **under-prices** the wings (46.70 at \$800).

The medians tell the same story: \$43.39 against \$38.12, and the 99.9th percentiles \$573.80 against \$349.60.

So no re-scaling of a single volatility parameter can fix it. A lognormal has *one* shape parameter, and matching one point on the strike axis merely relocates the error rather than removing it. Raising $\sigma$ to price the \$400 cap correctly would make the \$60 cap absurdly expensive.

*The mechanism generalizes beyond power*, which is why it belongs in a chapter rather than a footnote. Any distribution whose variance is dominated by rare large events is misdescribed by a variance-matched diffusion, and matching moments is not matching the distribution. Chapter 21's jumps make the same point in the equity market, Chapter 24's Merton short-end failure makes it in credit, and question 14's bimodal carbon density makes it again later in this chapter.

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*The remedy* is to model the spikes explicitly (a jump component, a regime, or a non-parametric density) and to treat implied volatility as a quoting convention rather than a parameter, which is question 16's conclusion in the carbon market.

*What the question tests.* That moment-matching leaves the shape free, and that a one-parameter lognormal calibrated at the money moves the error in the wings the wrong way.

---

**36.3** *Using Example 36.5, state the value of the option on one megawatt-hour at $\rho = 0.9$ and at $\rho = 0$, and explain why high power–gas correlation makes the spread option cheap rather than expensive.*

*The two values.* From Example 36.5, a peaking plant with power at \$42.00/MWh, gas at \$3.20/MMBtu, heat rate 12.5, six months, power volatility 55%, gas volatility 45%, so fuel cost is \$40.00/MWh and the intrinsic spread \$2.00:

| $\rho$ | Effective vol | Option value | Time value |
|---|---|---|---|
| 0.9 | 24.4% | \$**3.929** | \$1.929 |
| 0.0 | 71.1% | \$**9.171** | \$7.171 |

Modeling power and gas as independent values the option on each megawatt-hour at \$9.171 instead of \$3.929—**133% too high**.

*Why high correlation makes the spread option cheap.* Because the option is on the *spread*, and the volatility that enters Margrabe's formula is that of the ratio of power to the cost of HR units of gas,

$$\hat{\sigma} = \sqrt{\sigma_e^2 + \sigma_g^2 - 2\rho\,\sigma_e\,\sigma_g}$$

which is *decreasing* in $\rho$ throughout. The heat rate sets the quantity of gas, not its percentage volatility, so it does not appear here. At $\rho = 0.9$ the effective volatility is **24.4%**; at $\rho = 0$ it is **71.1%**, nearly three times as large, from two legs whose own volatilities never changed.

The economics is that power and gas are strongly correlated because gas frequently sets the marginal price of power. When gas rises, the generator's fuel cost rises *and* the price it receives rises, so the margin between them barely moves. The plant is long a spread, not long power, and the spread is far less volatile than either leg.

This is the opposite of the intuition that two volatile inputs must make a volatile output, and the intuition is what misleads. A modeler who reasons "power is 55% volatile and gas is 45%, so the spread must be very volatile" has added variances where the formula subtracts a covariance.

Note the direction of the error. Assuming independence (or simply omitting the correlation term) *overstates* the plant's value, always, because $\rho = 0$ is far from the strongly positive correlation the market shows and the value is monotone in $\rho$. The error flatters the asset, which is the dangerous direction for a valuation used to justify building or buying one.

*Why this is the single most consequential correlation assumption in energy finance.* Because it values the *entire generation fleet*. A gas-fired generator earns $\max(P_e - \text{HR} \cdot P_g, 0)$ per MWh, running when the spread is positive and shutting when it is not, so the plant *is* a strip of call options on the spark spread, and by Chapter 34 each is an **exchange option**: power against HR units of gas.

Every gas plant in a market is valued by the same correlation, so an error in it is not diversified across a portfolio—it is one error, repeated, in the same direction, across an asset class.

*And Margrabe's formula prices it exactly*, which is worth noting: an exchange option has a closed form (Chapter 28's question 10), no interest rate appears, and correlation enters *only*

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through $\hat{\sigma}$. So the whole table is a table of Black–Scholes–Merton prices at five volatilities, and the modeling question reduces entirely to one number.

*What the question tests.* That a spread's volatility falls with correlation, and that the assumption prices an entire asset class in one direction.

---

**36.4** *Compute the 3-2-1 crack spread at crude \$80/bbl, gasoline \$2.50/gal, heating oil \$2.55/gal, and state what changes if the refinery's actual yield is 2-1-1.*

*The 3-2-1 crack spread.* The convention takes three barrels of crude in and two barrels of gasoline plus one of heating oil out. Products are quoted per gallon and crude per barrel, so the products must be converted at 42 gallons to the barrel:

$$\begin{aligned} \text{Gasoline} &: 2 \times \$2.50 \times 42 = \$210.00 \\ \text{Heating oil} &: 1 \times \$2.55 \times 42 = \$107.10 \\ \hline \text{Product revenue} &= \$317.10 \text{ per three barrels} \\ \text{Crude cost} &= 3 \times \$80 = \$240.00 \\ \hline \text{Gross margin} &= \$77.10 \text{ per three barrels} \end{aligned}$$

Expressed per barrel of crude, which is the market convention:

$$\frac{\$77.10}{3} = \$\mathbf{25.70}/\text{bbl}$$

*What changes at a 2-1-1 yield.* The 3-2-1 ratio approximates a *typical* US refinery's yield; it is not a description of any particular refinery. At 2-1-1 (two barrels in, one gasoline and one heating oil out):

$$\frac{(1 \times 2.50 + 1 \times 2.55) \times 42 - 2 \times 80}{2} = \frac{212.10 - 160.00}{2} = \$\mathbf{26.05}/\text{bbl}$$

a different number, and more importantly *a different hedge*.

*The practical consequences matter more than the 35 cents.*

*The hedge ratio is wrong.* A refiner with a 2-1-1 yield who hedges with 3-2-1 crack spreads is short the wrong proportion of gasoline against heating oil. On six barrels of crude the 3-2-1 hedge sells four gasoline and two heating oil; the 2-1-1 output is three and three. The residual is short one barrel of gasoline and long one of heating oil per six barrels of crude, which amounts to a short position in the gasoline–heating oil spread, taken unintentionally.

*And the residual is seasonal and volatile.* The gasoline–heating oil differential swings with the driving season against the heating season, so a mismatched crack hedge leaves a position that gains in one season and loses in the other.

*The general point.* The 3-2-1 exists because it approximates a *typical* US refinery's yield, and a standardized contract must pick one ratio. A refiner's actual yield is set by its configuration (a cracking refinery, a coker, a simple topping unit all differ) and by the crude slate it runs, which changes with relative crude prices. So the standard spread is a liquid proxy carrying a *basis* against every real refinery, which is question 7's subject: the hedge removes most of the risk and leaves a residual that is different in kind.

*One further caution attaches to units.* The factor of 42 is where most errors in this calculation occur. Products in dollars per gallon and crude in dollars per barrel differ by a factor of 42, so omitting it understates product revenue by that factor and produces a spread that is deeply negative, which should be caught by inspection, since a refinery with a negative gross margin would not be running.

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*What the question tests.* The unit conversion and the standard ratio, and that a standard crack ratio is a proxy whose mismatch against a real yield is itself a position.

---

**36.5** *Using Example 36.6, state the extrinsic share at 60% volatility and $\rho = 0.75$, and explain why the facility is usually leased by a trader rather than operated by the utility that built it.*

*The extrinsic share.* From Example 36.6, at 60% monthly volatility and $\rho = 0.75$ between adjacent months:

| **Monthly vol** | $\rho = 0.90$ | $\rho = 0.75$ | $\rho = 0.50$ |
|---|---|---|---|
| 30% | 0.2% | 2.2% | 7.1% |
| 60% | 5.1% | **15.7**% | 27.0% |
| 80% | 11.0% | 25.0% | 37.4% |

**15.7%** of the facility's total value is extrinsic. At plausible parameters the share runs 10–30%, and it is the part a spreadsheet does not capture.

*The two components.* The *intrinsic* value is the spread a facility can lock in today (inject in the five cheapest months at an average \$2.95, withdraw in the five most expensive at \$3.67, for \$0.72 per MMBtu per cycle), and it can be hedged immediately with calendar spreads. The *extrinsic* value is *the right to change one's mind*: to withdraw early if winter arrives early and cold, to re-inject if the spread widens. It is the time value of a strip of calendar-spread options.

Note what drives it. The extrinsic share rises with volatility (more reason to re-optimize) and falls with the correlation between adjacent months (if months move together, there is nothing to arbitrage between them). At $\rho = 0.90$ and 30% volatility it is 0.2%, essentially nothing, and at $\rho = 0.50$ and 80% it is 37.4%. The same physical facility is worth quite different amounts depending on parameters that describe the market rather than the asset.

*Why a trader leases it rather than the utility operating it.* The reason is not that traders are better at running compressors.

A utility that books the intrinsic spread has sold the extrinsic value to whoever leases the capacity, because capturing the extrinsic value requires something a utility is not organized to do:

- *Continuous re-optimization.* The option must be exercised (injection and withdrawal schedules changed) as the curve moves, which requires a trading desk watching the forward curve daily rather than an operations team executing a plan.
- *A view on the option's value*, and therefore a model of it. The intrinsic spread is arithmetic; the extrinsic value requires pricing a strip of calendar-spread options and knowing what they are worth.
- *And a tolerance for mark-to-market variation*, since the position is a long option and its value moves. A regulated utility's incentives reward a locked-in margin over a volatile one worth more in expectation.

So the lease price is the intrinsic value and the trader keeps the extrinsic. That is a market structure fact explained by a valuation fact, which is why the example is in the chapter—the 15.7% is not an abstraction but the trader's business.

*The general pattern.* An asset with embedded optionality migrates to whoever can exercise the option. Chapter 26's CLO manager, Chapter 35's structured-note issuer, and the storage lessee here are the same phenomenon: the party who values the option most and can act on it acquires it, and the transfer looks like an ordinary commercial arrangement.

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*What the question tests.* That extrinsic value requires active re-optimization to realize, and that the lease structure is a transfer of optionality rather than of operating skill.

---

**36.6** *Invert the carry relation to give the implied convenience yield, and compute it for $S = 80$, $F = 77$ at six months, $r = 4\%$, $u = 5\%$. State what a negative value would indicate and why it cannot be arbitraged away.*

*Inverting the carry relation.* For a storable commodity,

$$F = S\,e^{(r+u-y)T}$$

where $u$ is the storage cost rate and $y$ the convenience yield, the benefit of holding the physical commodity rather than a claim on it. Solving,

$$\boxed{y = r + u - \frac{1}{T}\ln\left(\frac{F}{S}\right)}$$

*The computation.* With $S = 80$, $F = 77$ at $T = 0.5$, $r = 4\%$, $u = 5\%$:

$$\frac{1}{0.5}\ln\left(\frac{77}{80}\right) = 2 \times (-0.038221) = -0.076443$$

$$y = 0.04 + 0.05 - (-0.076443) = \mathbf{16.64}\%$$

A large positive convenience yield, which is what a market in **backwardation** implies: the forward sits below spot despite the cost of carrying, so holders of the physical must be receiving a benefit exceeding the full carry.

*What a negative value would indicate.* $y < 0$ requires $F > Se^{(r+u)T}$, with the forward above full carry. That is normally an arbitrage: buy the physical, borrow to finance it, pay storage, sell the forward, and lock in a riskless profit. So a negative implied convenience yield is a statement that the arbitrage is unavailable.

*Why it cannot be arbitraged away: **storage is full**.* The cash-and-carry trade requires somewhere to put the commodity. When tanks, caverns or warehouses are at capacity, the marginal storage cost is not $u$; it is effectively infinite, because no additional barrel can be stored at any price. The relation $F \le Se^{(r+u)T}$ is enforced by a trade that no longer exists, and the forward is free to rise above it.

This happened, and spectacularly. In April 2020 Cushing storage approached capacity, the curve steepened beyond full carry, and the expiring WTI contract settled *below zero* because holders of expiring longs had nowhere to take delivery. The carry relation itself cannot produce a negative futures price (it makes $F$ a positive multiple of $S$), so what went negative was the spot value of a barrel at the delivery point. That is the same constraint at the other extreme: the trade that normally stops a price falling too far, buying the cheap barrel and storing it, also requires storage, and its absence removed the floor.

*The general asymmetry, as a rule.* The carry relation is a one-sided bound for a physical commodity. Its upper bound, $F \le Se^{(r+u)T}$, is enforced by cash and carry and holds only while storage is available. Its lower bound is not enforced at all, because the *reverse* arbitrage requires borrowing the physical commodity, and there is no reliable lending market for barrels of crude the way there is for shares. So a commodity forward can trade arbitrarily far below full carry, and the shortfall is what the convenience yield measures.

Which means the convenience yield is a residual, not a measurement. It is defined as whatever makes the equation hold, so it absorbs storage scarcity, the option value of holding inventory, and any error in $u$ or in the curve. Question 11 shows what happens when the same

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calculation is applied to an allowance, where nothing is stored and the residual cannot be physical at all.

*What the question tests.* Inverting the carry relation, and that its enforcement is one-sided and conditional on storage capacity.

---

**36.7** *Using Example 36.8, state the risk reduction achieved by hedging the flat price, and explain why that figure overstates the protection in exactly the episodes the hedge was bought for.*

*The risk reduction.* From Example 36.8, a producer hedging with a light sweet crude contract at a 32% flat-price volatility and a 28% volatility *of the basis*:

| | |
|---|---|
| Unhedged standard deviation of the realized price | \$24.96/bbl |
| Hedged standard deviation (basis only) | \$1.67/bbl |
| **Risk reduction** | **93.3**% |

*The hedge works: it removes 93% of the risk.* That is the headline, and it is why producers hedge.

The basis itself is substantial (grade −\$3.20, location −\$2.40, timing −\$0.35, totaling −\$5.95 or **7.6%** of the contract price), but its *variability* is what survives the hedge, and \$1.67 against \$24.96 is a genuine and large improvement.

*Why 93.3% overstates the protection in exactly the episodes the hedge was bought for.*

Because the residual is different in kind, and it is correlated with the events that motivate hedging.

*Flat price is liquid, continuously quoted, and hedgeable. Basis is none of those.* That asymmetry is invisible in a variance decomposition, which treats the two components as interchangeable contributors to a standard deviation.

And the basis components move for reasons that arrive precisely in a dislocation:

- *Grade differentials* move with refinery configuration and with regulation on sulfur content, and they gap when a refinery outage removes the natural buyer for a heavy sour crude.
- *Location differentials* move with pipeline capacity and can jump when a line is constrained. A pipeline outage or an apportionment does not move the basis by a standard deviation; it moves it by a multiple.
- *Timing basis* moves with local storage, and question 6 established that storage constraints are exactly what break the carry relation.

So the 28% basis volatility is an unconditional average, dominated by ordinary quarters in which the basis drifts. In a constrained market it is not the relevant number, and the residual the hedge leaves behind becomes a multiple of \$1.67 at the same moment the producer most needed protection.

This is the book's recurring wrong-way structure, and Chapter 27's question 12 collects the family: a protection whose effectiveness deteriorates in the state it was bought for. Here it operates through the *residual* rather than through the hedge itself: the hedge continues to perform exactly as designed on flat price, and what fails is the assumption that the remainder is small.

*Three practical consequences follow.* Basis is hedgeable in *some* markets and not others. Where basis swaps trade, the exposure can be removed at a price; where they do not, it cannot. A hedge program should be sized and reported on *basis-conditional* scenarios rather than on an unconditional variance reduction. And a producer quoting "93% hedged" to a board has reported a number that is true on average and misleading about the tail.

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*What the question tests.* That a variance decomposition treats hedgeable and unhedgeable components alike, and that basis volatility is conditional on exactly the events that make hedging valuable.

---

**36.8** *Using Example 36.9, state the annual roll cost at 1% monthly contango on a \$780m stack, then state the cash loss and the present value of the offsetting gain at a 10% price fall and explain why the difference can end a firm.*

*The annual roll cost.* From Example 36.9, a supplier selling ten years of fixed-price delivery at a million barrels a year, stacking the whole ten-million-barrel exposure in the front month:

| Monthly contango | Annualized | Cost per year on a \$780m stack |
|---|---|---|
| 0.5% | 6.2% | \$48.1m |
| 1.0% | 12.7% | **\$98.9m** |
| 2.0% | 26.8% | \$209.2m |

At 1% monthly contango the roll costs **\$98.9 million a year**, which is Chapter 5's roll yield, applied ten times over because the stack is ten times the annual exposure.

*The cash loss and the offsetting gain at a 10% price fall.*

| Price fall | Futures loss (cash, now) | PV of contract gain |
|---|---|---|
| −10% | **\$78.0m** | **\$61.4m** |
| −20% | \$156.0m | \$122.7m |
| −40% | \$312.0m | \$245.5m |

*Why the difference can end a firm.* The two numbers differ in **timing**, not merely in size, and the timing is the whole problem.

*The futures loss is cash, now.* Variation margin is settled daily, in cash, and must be funded on the day it is called.

*The contract gain accrues over a decade.* The supplier will sell oil at above-market prices for ten years, which is worth \$61.4 million in present value—and *none* of it is available today. It arrives a barrel at a time over ten years, and the table discounts each year's deliveries from mid-year at 5%, continuously compounded.

So a firm whose hedge is economically correct can be unable to fund it. In present-value terms the position loses only \$16.6 million at −10%, the amount by which a one-for-one stack over-hedges (question 9's discounting point)—but the firm must find \$78 million in cash while holding an asset that will pay it back slowly. If it cannot, it must liquidate the hedge, at which point it is unhedged into whatever the market does next.

*Metallgesellschaft is the case.* The firm's short-dated futures needed cash while its long-dated obligations were merely valuable, and the positions were liquidated near the bottom, crystallizing a loss that the unliquidated hedge would not have produced.

*And it recurred in 2022 at national scale.* European generators that had sold their output forward owed variation margin daily as gas and power prices rose several-fold, while the offsetting gain would arrive only on delivery. Example 8.9 works the arithmetic and Section 40.12 tells the episode.

*Two conclusions are worth carrying.* A hedge sized on price risk alone is not sized, because the funding leg scales with the same move the hedge was bought to neutralize. And the more complete the hedge, the larger the funding requirement, so the prudent firm has the biggest problem, which is the uncomfortable part.

*What the question tests.* That the mismatch is one of timing rather than of value, and that a correctly hedged firm can fail on liquidity.

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**36.9** *Give the two independent reasons a one-for-one stack is the wrong size, state the remedy, and explain what the remedy requires that cannot be reliably estimated.*

*There are two reasons, and they compound.*

*1. Discounting.* A one-for-one stack matches *nominal* barrels, but ten years of future gains are worth less than an immediate loss of the same nominal size. Question 8's table makes it concrete: at −10% the present value of the gain is \$61.4m against a \$78.0m cash loss, or **79%**. So a one-for-one stack over-hedges in present-value terms by construction, and the over-hedge is roughly one minus the average discount factor over the exposure's life.

*2. Samuelson.* Question 1 established that the front contract is roughly **four times** as volatile as the twelve-month at $a = 1.5$. Hedging a ten-year exposure with front-month futures therefore over-hedges the *volatility* as well: the hedge instrument moves far more per barrel than the exposure does.

*The two are independent and multiply.* Discounting says the far cash flows are worth less; Samuelson says the far cash flows *move* less. Either alone would make a one-for-one stack too large; together they make it substantially too large, and the excess is a directional position taken unintentionally.

*The remedy: a* ***maturity-weighted stack***. It holds fewer front contracts, with each year's exposure weighted by *both* corrections:

$$\text{weight for year } T \;\propto\; e^{-aT} \times P(0,T)$$

the Samuelson factor times the discount factor. That is the hedge ratio that equates present-value sensitivities rather than notionals, and it is straightforwardly the right construction.

*What it requires that cannot be reliably estimated: the mean-reversion parameter $a$.*

The discount factors are observable: they come from the curve. The mean-reversion rate is not. It is identified only from the volatility term structure, which is short, noisy, contaminated by storage effects working in the opposite direction (question 1), and unstable across regimes, and Example 36.7 shows that no single $a$ fits both ends of the curve, so the estimate depends on which end it is fitted to. Estimates differ by factors rather than by percentages, and Example 36.2's table puts the ratio of twelve-month to one-month volatility at 0.63 at $a = 0.5$, 0.25 at $a = 1.5$ and 0.06 at $a = 3.0$.

So the choice is between a hedge that is knowably wrong and one that is wrong by an unknown amount. A one-for-one stack is wrong in a direction and by an amount that can be computed and disclosed (79% on discounting alone, worse with Samuelson). A maturity-weighted stack is right in principle and depends on a parameter whose plausible range spans a factor of six.

*And the first is preferable only if its error is reported.* That qualification is the point. An unweighted stack is defensible as a deliberate, documented approximation whose bias is stated; it is indefensible as an unexamined default, which is what it was at Metallgesellschaft. The same logic governs Chapter 31's question 8: when a parameter cannot be identified, choose it outside the calibration and *say that you have*.

*What the question tests.* That discounting and Samuelson are separate, compounding over-hedges, and that the correct remedy depends on an unestimable parameter, so the honest choice is about which error to disclose.

---

**36.10** *Compute the two-year allowance forward at a spot of €65 and a 4% rate, and state which side of that value is enforced by arbitrage.*

*The forward.* An emissions allowance has no storage cost and no convenience yield: it is a registry entry, costless to hold, and delivers no physical benefit. Allowances are also **bankable**:

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a compliance entity may hold one indefinitely and surrender it in a later year. So the carry relation reduces to pure financing:

$$F = S\,e^{rT} = €65 \times e^{0.04\times 2} = 65 \times 1.083287 = \mathbf{€70.41}$$

*Which side is enforced by arbitrage: the **upper** side only.*

*If* $F > Se^{rT}$, the arbitrage is available and easy. Buy the allowance spot, borrow to finance it, sell the forward, hold the registry entry for two years (which costs nothing), and deliver. The profit is locked, the trade requires no storage capacity, and unlike question 6's commodity there is no physical constraint that can prevent it. So the forward cannot exceed full carry.

*If* $F < Se^{rT}$, the reverse trade requires *borrowing* an allowance: sell spot, invest the proceeds, buy the forward, and return the allowance in two years. That requires a lending market in allowances, which is thin, and it requires a holder willing to give up their compliance position, which a compliance entity facing an uncertain future obligation may decline at any price. So the lower bound is not reliably enforced.

This makes allowances an unusually clean case. Because banking is permitted and storage is costless, the *upper* bound is tighter than in any physical commodity: there is no "tanks full" scenario, no location basis, no deliverable-grade question. The forward curve of a bankable allowance should be a pure interest-rate curve, sloping upward at $r$ and nothing else.

*And that is what makes any deviation informative.* If the observed two-year forward trades *below* €70.41, the implied convenience yield is positive, and question 11 asks what a positive convenience yield can possibly mean for an asset that costs nothing to store and confers no physical benefit. The residual cannot be physical, so it must be something else, and identifying what is the point of the calculation.

*Two riders attach to the arbitrage.* It requires the allowance to remain valid, so it embeds *regulatory* risk: a scheme redesign that cancels or devalues banked allowances would break the trade, and this is not hypothetical in schemes that have been revised. And it requires financing at $r$, so a participant funding above the risk-free rate faces a wider no-arbitrage band, which is Chapter 5's limits-of-arbitrage argument again.

*What the question tests.* That zero storage cost and banking reduce the carry relation to pure financing, and that the enforceable side is the one requiring no borrowing of the asset.

---

**36.11** *Explain why a positive implied convenience yield in allowances cannot be physical, what its decay with maturity indicates, and how dealer financing of compliance inventory contaminates the reading.*

*Why a positive implied convenience yield cannot be physical.* Question 10 established that an allowance is a registry entry: costless to store, conferring no physical benefit, and bankable. A convenience yield measures the value of holding the physical commodity rather than a claim on it: avoiding a stock-out, keeping a refinery running, meeting an unexpected order. None of those exist for a number in a database. There is no operational advantage to holding an allowance today rather than a forward on one.

So if the inversion of question 6's relation returns a positive $y$ on an allowance curve, the number is **not measuring convenience**. It is a residual, and it is absorbing something else.

*What its decay with maturity indicates.* In Example 36.10 the forward sits a constant 3% below carry at every horizon, and the implied yield falls as $1/T$: 6.09% at six months, 1.02% at three years. That is the signature of a **one-off loss of value** spread over a longer horizon, not of a running yield, since a constant $y$ would not decay at all. The chapter reads it as event risk: a probability that a banked allowance will not be usable as expected, because the scheme is redesigned at a phase boundary or banked allowances are devalued.

A one-off discount has other possible sources, and they produce the same shape:

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- *A compliance deadline.* Entities must surrender allowances by a fixed date, so demand concentrates ahead of it and the spot market tightens relative to forwards that mature after the deadline has passed.
- *A supply schedule.* If auction volumes are known to increase, the near term is tight relative to a better-supplied future, and the curve reflects it.
- *Or a positioning squeeze*, with short compliance entities buying into a thin spot market.

A yield that stayed constant across maturities would indicate something quite different: a continuing cost or benefit of holding, which an allowance does not have. So the decay identifies a one-off event rather than a running yield, and telling the regulatory event from a near-term squeeze needs more than the curve's shape.

*How dealer financing of compliance inventory contaminates the reading.*

The calculation infers $y$ from $F$, $S$ and $r$, and it assumes $r$ is the relevant financing rate. For a compliance entity it usually is not.

A common arrangement is that a **dealer holds the allowances** on its own balance sheet and sells them forward to the compliance entity, financing the inventory. The entity gets its allowances when it needs them without tying up cash; the dealer earns a financing spread. The forward price then embeds the dealer's funding cost and balance-sheet charge, not the risk-free rate.

So the observed $F$ reflects $r_{\text{dealer}} > r$, which makes $F$ *higher* than the risk-free carry would imply, and inverting with the risk-free $r$ produces a *negative* implied convenience yield that is really a financing spread. Conversely, a market in which dealers are unwilling to carry inventory (because of capital charges, or because Chapter 27's leverage ratio binds on a low-risk inventory position) shows the opposite.

*The reading is therefore contaminated in a way that is invisible in the number.* A single figure absorbs scarcity, regulatory expectation, and the dealer sector's balance-sheet capacity, and nothing in the curve separates them.

*The general lesson is question 6's in a purer setting.* A convenience yield is defined as a residual, so it absorbs every misspecification in the relation that defines it. On a physical commodity that residual has at least a plausible physical interpretation; on an allowance it has none, so the same number is unambiguously a measurement of something the model does not contain. The cleaner asset makes the residual's emptiness visible.

*What the question tests.* That a convenience yield requires a physical benefit to be meaningful, reading the term structure of a residual as a diagnostic, and that the financing rate assumption is what the calculation is most sensitive to.

---

**36.12** *Using Example 36.11, state the price change from two different 20 Mt shifts in required abatement, and explain what that implies for a value-at-risk estimated on a historical window.*

*The two price changes.* From Example 36.11, with abatement following a merit order and supply fixed by the cap:

| Required abatement | Clearing price | Change |
|---|---|---|
| 190 → 210 Mt | €42.50 → €49.17 | **+15.7**% |
| 290 → 310 Mt | €100.00 → €150.00 | **+50.0**% |

The same 20 Mt shift moves the price 15.7% in one place and 50.0% in another (a factor of 3.2), with nothing changed but the position on the curve.

*The mechanism.* Supply is *vertical*: the cap fixes the number of allowances, so a demand shift cannot be absorbed by producing more. It must be absorbed entirely by price, moving

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along a marginal abatement cost curve that is steeply **convex**: cheap measures first, then progressively more expensive ones. The curve runs from €8/t at 50 Mt to €190/t at 320 Mt.

So the price elasticity is not a property of the market; it is a property of where on the curve the market currently sits. Deep in the cheap region a shock is absorbed at modest cost; near the expensive end the same shock produces a violent move.

*What this implies for a value-at-risk estimated on a historical window.* It understates the risk, systematically.

The reason is precise: the elasticity that generated the history is not the elasticity now in force. A VaR computed from the last two years of allowance returns measures the volatility produced by the curve's slope *over those two years*. If the cap has tightened since (which a declining cap does by design, as the EU's does every year), the market has moved up the curve and the current slope is steeper. The historical sample was drawn from a flatter regime and cannot contain the current one.

Three consequences follow, and each inverts an ordinary intuition.

*Volatility rises as the market matures.* A tightening cap moves the market up a steepening curve, so an allowance market becomes *more* volatile over time, the reverse of what happens as a financial market deepens and liquidity improves. A risk model extrapolating the usual maturation pattern will be wrong in direction.

*The error is one-signed.* Because the cap only tightens, the elasticity only rises, so historical VaR only understates. There is no compensating period of over-estimation to average against.

*And the correct state variable is not the price.* It is the position on the abatement curve (cumulative required abatement), because that is what determines the elasticity. A risk model that conditions on the price level is fitting the wrong thing: two identical prices at different points in the scheme's life imply quite different volatilities.

*The general pattern.* This is Chapter 26's question 7 in a different market: a parameter estimated from a sample that structurally cannot contain the regime being forecast. There it was correlation estimated before a national house-price decline; here it is elasticity estimated before the cap tightened. In both cases the estimator is unbiased for the sample and uninformative about the question.

*What the question tests.* That a convex supply curve makes elasticity state-dependent, and that a historical VaR inherits the elasticity of a regime that has been deliberately superseded.

---

**36.13** *Explain how a buy-out price caps a renewable certificate and what that does to the vega of an option written on it.*

*How a buy-out price caps a certificate.* A renewable certificate carries an obligation proportional to sales rather than a cap, so supply grows with generation and question 12's vertical supply does not apply. Where the obligation can be discharged at a fixed **buy-out** price (paying a stated amount per unit instead of surrendering a certificate), that price is a hard ceiling.

The argument is one line. No obligated party would ever pay more than the buy-out price for a certificate, because the buy-out discharges the same obligation. So the certificate's price cannot exceed it, and the payoff is **capped by construction** rather than by market convention.

*What that does to the vega of an option written on it.*

*It changes the sign of the vega as the price approaches the cap.* An option on a capped underlying is not an ordinary call; it is a *call spread* (long the strike, short the buy-out level), because the underlying can never settle above the cap:

$$\text{option on capped underlying} = c(K) - c(B)$$

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where $B$ is the buy-out price. And the vega of a call spread is the difference of two vegas:

$$\mathcal{V} = \mathcal{V}(K) - \mathcal{V}(B)$$

- Far below the cap, $\mathcal{V}(K) \gg \mathcal{V}(B)$ and the option is **long volatility**, behaving like an ordinary call.
- As the price rises toward $B$, the short leg's vega grows faster (an out-of-the-money option's vega rises as it approaches the money), and the net vega *shrinks*.
- Roughly midway between $K$ and $B$ the two vegas are equal and the net vega is **zero**; once the price is above that point, the short leg's vega exceeds the long leg's and the net vega is *negative*.

So past a threshold the vega changes sign with no change in the contract. A desk that is long volatility on the position at one price level is short it at another, and nothing was traded.

This is Chapter 34's capped cliquet in another market. There, an 8% cap on each period's return converted a strip of forward-start calls into a strip of call spreads and reversed the sign of the whole contract's vega: the value falls as volatility rises. The mechanism is identical; only the source of the cap differs, contractual there and regulatory here.

*Two practical consequences follow.*

The hedge must be re-signed as the market moves, and the crossover point is not marked on any term sheet. A vega hedge established at €30 against a €50 buy-out is the wrong sign at €45, and the transition is smooth enough that a desk monitoring vega monthly can be on the wrong side of it for weeks.

*And the cap is a policy variable.* Unlike a contractual cap, the buy-out price is set by a regulator and can be changed, so the option's terminal payoff depends on a level that is not a market quantity, and the position carries a regulatory exposure that no volatility surface prices.

*What the question tests.* That a hard ceiling converts a call into a call spread, and that the resulting vega reversal is a level effect requiring the hedge to change sign.

---

**36.14** *Using Example 36.12, state the at-the-money implied volatility and the two wing values, explain the shape, and state what a standard skew parametrization fitted to these quotes would get wrong.*

*The three values.* From Example 36.12, modeling the one-year allowance price as a mixture (with probability $p = 0.35$ a tightening takes the forward to €110, otherwise it stays at €45), each regime lognormal with 25% volatility, giving a mixture forward of $F =$ €67.75:

| Strike | Call | Implied vol |
|---|---|---|
| €40 | 28.167 | **45.5%** |
| €67.75 | 14.535 | **56.4%** |
| €120 | 2.407 | **48.4%** |

The at-the-money implied volatility is **56.4%**, falling to **45.5%** and **48.4%** in the wings.

*The shape: it **humps** instead of smiling, the opposite of an equity index.*

*Why.* The density has two modes and a valley between them. An at-the-money option straddles the valley (it is struck where the distribution has *least* mass, between the €45 and €110 regimes). The mass a lognormal would put near the strike, where the payoff is small, the mixture puts out in the two modes, far from the strike; for the same variance that raises the expected distance from the strike, which is what an at-the-money option pays. So relative to a lognormal it is expensive, and inverting its price returns a high implied volatility. The wings, by contrast, have *less* mass than a lognormal of the same variance, because the mixture's

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probability is concentrated in two lumps rather than spread through the tails. So the wings are cheap and their implied volatilities are low.

An equity index has the opposite structure: a single mode with a fat left tail, which produces a downward-sloping skew. Here the underlying's uncertainty is dominated by a scheduled binary event (a regulatory tightening), and that produces a fundamentally different density.

*What a standard skew parametrization would get wrong.* It would fit the wrong sign and extrapolate in the wrong direction.

Standard parametrizations such as SABR and SVI are built to represent a smile or a skew: a curve with one minimum, sloping up in at least one wing. A humped surface has a *maximum* at the money and falls away on both sides, which those families cannot represent at all. Fitting one to these quotes produces:

- a fitted skew of the wrong sign, since the parametrization will interpolate a slope through points that are symmetric about a peak;
- and, worse, extrapolation in the wrong direction, so strikes beyond the quoted range are priced with volatilities that rise where they should fall.

The second is the damaging one, because the far wings are exactly where a desk relies on the fit rather than on quotes.

*The treatment generalizes beyond carbon.* It applies to any underlying whose uncertainty is dominated by a scheduled binary event (an election, a court ruling, a drug trial readout), and carbon is simply the case where the structure is **permanent** rather than resolving on a date. A biotech stock before a trial result and an allowance before a policy review have the same density shape, and both are misdescribed by a smile.

*The remedy* is question 16's: hedge with the *mixture's* delta and use implied volatility only as a quoting convention. The right diagnostic is the density, not the volatility, and Chapter 19's Breeden–Litzenberger construction recovers it directly from the same quotes.

*What the question tests.* That a bimodal density produces a humped rather than smiling surface, and that standard parametrizations fail on sign and on extrapolation rather than merely on fit quality.

---

**36.15** *Explain why the variance-matched lognormal volatility of 50.3% differs from the 56.4% at-the-money implied volatility, and what each measures.*

*The two numbers.* The mixture's standard deviation is €36.32, which a lognormal matched *on variance* would report as **50.3%**. The at-the-money implied volatility is **56.4%**. They differ by six volatility points.

*What each measures.*

*The variance-matched 50.3% measures total dispersion.* It answers: what single lognormal has the same standard deviation as this distribution? It is a summary of the *second moment* and nothing else, and it says nothing about shape.

*The at-the-money implied 56.4% measures the price of one option.* It answers: what single lognormal reproduces the price of the at-the-money call? That price depends on the density *near the money*, which for a bimodal distribution is the **valley between the two modes** (question 14).

So they differ because the at-the-money option is not a function of the variance. It is struck in the region where the mixture has *least* mass relative to a lognormal of the same variance. That mass sits instead in the two modes, far from the strike, so the expected distance from the strike, which is what the option pays, is larger than the lognormal's; the option is expensive and

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its implied volatility is high. Move to the wings and the same comparison reverses: the €40 and €120 options imply 45.5% and 48.4%, both *below* the variance-matched 50.3%.

*And **neither number is the volatility of anything**.* That is the chapter's phrase and it is the point. The mixture has no single volatility; it has a regime uncertainty and a within-regime diffusion, and no scalar summarizes both. Each of the two numbers is a projection of the distribution onto a one-parameter family, and different projections give different answers because the distribution is not in that family.

*The decomposition makes the emptiness explicit.* Splitting the mixture's variance:

| | |
|---|---|
| Regime uncertainty (which mode) | **73**% |
| Within-regime diffusion | 27% |

*Nearly three-quarters of what a model calls volatility is not diffusion at all.* It is uncertainty about a discrete outcome, and it will not be realized as day-to-day price variation—it will be realized as a single jump when the policy decision arrives. A desk delta-hedging on 56.4% expecting gamma P&L from continuous movement will collect a fraction of it and then be hit by a gap.

*Two practical consequences follow.*

*Realized volatility will look far below implied*, right up until the event, and a desk short options will appear to be winning. Chapter 19's volatility risk premium is a different phenomenon producing the same appearance, and distinguishing them requires knowing which regime the density is in.

*And the two numbers should be quoted separately.* Reporting a single "volatility" for such an underlying compresses a mixture into a scalar and loses the information a hedger needs, which is why question 14's remedy is to work with the density rather than with any implied volatility.

*What the question tests.* That variance-matching and at-the-money inversion are different projections of a non-lognormal density, and that a regime-dominated variance is not diffusion.

---

**36.16** *Using Example 36.12, state the largest delta error, explain why a book spread across strikes does not reliably net it out, and state what the residual will be mistaken for.*

*The largest delta error.* From Example 36.12, where delta error is the mixture's true delta minus the Black delta computed at the fitted implied volatility:

| Strike | Implied vol | Delta error |
|---|---|---|
| €40 | 45.5% | −0.036 |
| €50 | 51.1% | **−0.069** |
| €60 | 55.0% | −0.059 |
| €67.75 | 56.4% | −0.026 |
| €80 | 56.3% | +0.034 |
| €90 | 54.7% | +0.067 |
| €100 | 52.6% | **+0.078** |
| €120 | 48.4% | +0.060 |

The largest is **0.078 at €100**—eight delta points on a position the trader believes is hedged.

*Why a book spread across strikes does not reliably net it out.* Because the sign flips across the strikes.

The errors are *negative* below about €73 (reaching −0.069 at €50) and *positive* above it (reaching +0.078 at €100). A trader holding a mixture of strikes does not average two errors of the same sign toward zero; the book's net error depends on the *composition* of the position, and it can be as large as the individual errors if the book is concentrated on one side.

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Worse, a book that *is* spread evenly across strikes has a net error near zero only by coincidence, and any change in the book's strike distribution (one large trade, one expiration rolling off) moves the net error without anyone trading the underlying. The hedge error is a function of the book's shape rather than of its size, which no risk report tracks.

*And the errors are not random.* A random hedging error averages toward zero as trades accumulate. These are **systematic**: the same strike always carries the same error, in the same direction, because it comes from a fixed feature of the density rather than from noise. So the residual accumulates with every rehedge rather than diversifying.

*What the residual will be mistaken for:* ***gamma P&L.***

The signature is identical. A delta-hedged position generates a residual profit and loss that correlates with movement in the underlying, which is exactly what gamma P&L looks like. A trader reviewing the book sees a position that is flat on delta, produces P&L when the market moves, and concludes that the gamma is behaving as expected.

*It is model error.* The delta was computed from the wrong density, so the hedge is mis-sized, so the position has a residual directional exposure that reveals itself when the market moves. The magnitude is comparable to genuine gamma P&L and the sign depends on the book's strike composition, so there is no diagnostic in the P&L itself that separates them.

That is what makes it dangerous rather than merely inaccurate. An error disguised as an expected phenomenon is not investigated, and a desk can run it for years, attributing the variation to gamma and adjusting nothing.

*The remedy*, and it is stated as an instruction rather than an observation: hedge with the mixture's delta and use the implied volatility only as a quoting convention. The two roles are separable (the market quotes in Black volatilities and the hedge is computed from the density), and conflating them is what produces the error. The right diagnostic is the density, not the volatility.

*What the question tests.* That sign-flipping systematic errors do not net across a book, and that model error masquerading as gamma P&L escapes investigation because it has the expected signature.

---

**36.17** *Explain why the Bachelier form rather than Black's is appropriate for a degree-day index, and use* (36.6) *to verify the put–call parity relation in Example 36.13.*

*Why Bachelier rather than Black.* Three reasons apply, and the first is decisive.

*A degree-day index is not an asset and has no natural scale.* Black's model assumes the underlying is *lognormal*: proportional moves, strictly positive, and a distribution whose width scales with the level. A heating-degree-day index is a *count* of temperature deficits: it is a physical quantity with an additive structure, and a season with 1200 HDD is not "20% more volatile" than one with 1000 in any meaningful sense. Its variation is naturally expressed in *HDD*, not in percent.

*The distribution is approximately normal by construction.* The seasonal index is a *sum* of daily deficits over five months, so the central limit theorem applies directly. Example 36.13's index has mean 1200 and standard deviation 140, and a normal is the right description of a sum of a hundred and fifty imperfectly correlated daily quantities.

*And there is no carry, because the index cannot be held.* Black's model prices a forward on a traded asset; no one can buy a degree day, so there is no cost-of-carry relation, no discounting of a replicating portfolio, and no reason for the lognormal machinery. The price is an actuarial expectation, not a replication cost.

*Verifying parity.* Example 36.13 prices an HDD put struck at 1100 with a \$5,000 tick, on an index with $\mu = 1200$ and $\sigma = 140$, so $d = 100/140 = 0.7143$:

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| | |
|---|---|
| $P(\text{HDD} < 1100) = \Phi(-d)$ | 23.8% |
| $E[\text{shortfall} \mid \text{below}]$ | 82.2 HDD |
| **Value of the put** | **\$97,618** |
| Value of the matching call | \$597,618 |

The parity relation is

$$C - P = \tau(\mu - K) = \$5{,}000 \times (1200 - 1100) = \$\mathbf{500{,}000}$$

and directly from the table,

$$\$597{,}618 - \$97{,}618 = \$500{,}000 \quad \checkmark$$

exactly.

Two features of this parity are worth noting, because they differ from every other parity relation in the book.

*It holds with no discounting.* Ordinary put–call parity carries a factor $e^{-rT}$ on the strike, because the strike is cash paid at expiration and cash has a time value relative to a *held asset*. The index is not an asset, so there is nothing to carry and nothing to discount: the relation is between two expected payoffs at the same date, in the same units.

*And it is the Bachelier analog of forward–strike parity.* $C - P = \tau(\mu - K)$ says the difference between a call and a put is the expected value of the linear payoff $\tau(\text{HDD} - K)$, which follows from $\max(x, 0) - \max(-x, 0) = x$ applied under the expectation, the same identity as Chapter 29's cap–floor parity, with an expectation in place of a swap value.

Which makes it a useful data check, exactly as cap–floor parity is: it is model-free given $\mu$, so a broker quoting a call and a put that fail it has an error rather than a view.

*What the question tests.* That the underlying's additive, non-tradeable, sum-of-daily-quantities structure selects the normal model, and verifying a parity relation that carries no discount factor.

---

**36.18** *Explain the difference between burn analysis and index modeling, state what each assumes, and quantify the effect of a 1 HDD-per-year trend on Example 36.13's contract.*

*The two conventions.*

*Burn analysis* takes the empirical distribution of the last 20 to 40 years of the index and averages the payoff directly: no model, no simulation, just the historical record run through the contract. It is **transparent** and it *assumes stationarity*: that the last thirty years are draws from the distribution now in force.

*Index modeling* fits a daily temperature process (seasonality, mean reversion, autocorrelation, a stochastic component) and simulates the index forward. It **imposes structure** and can be wrong in ways burn analysis cannot: a misspecified daily process produces a seasonal distribution that no amount of data would have suggested.

The trade-off is the usual one between a non-parametric estimator with high variance and a parametric one with a bias you cannot see. Burn analysis uses 30 observations to estimate a distribution, which is very few; index modeling uses ten thousand daily observations but only through the lens of an assumed process.

*And a warming trend makes stationarity false*, which breaks burn analysis at its foundation. So detrending is a pricing input rather than a refinement: a choice that changes the answer, not a tidying step.

*Quantifying it on Example 36.13's contract.* A 30-year window with a 1 HDD per year trend misstates $\mu$ by roughly **15 HDD**, the trend accumulated between the window's midpoint

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and the present. At the \$5,000 tick that is

$$15 \times \$5{,}000 = \$\mathbf{75{,}000}$$

on a contract linear in the index, such as a swap on the seasonal total. The put moves by less, because it pays only below the strike: repricing it with $\mu = 1185$ in (36.6) gives about \$116,700 against \$97,618, a change of about **\$19,000**—a fifth of the premium from one modeling choice, which settles the question of whether detrending is a refinement.

*And the direction is systematic.* A warming trend means the historical average *overstates* expected HDD, so an un-detrended burn analysis overstates $\mu$, which *understates* the value of a put struck below the mean and *overstates* the value of a call. The error has a sign, it is the same sign every year, and it grows with the length of the window.

*The uncomfortable consequence.* A *longer* history makes burn analysis *worse*, not better. More data reduces sampling error and increases trend bias, and past thirty years the second dominates. The estimator's accuracy is not monotone in the sample size, which inverts the ordinary intuition and is why the window length is itself a pricing decision.

*Why this matters more here than elsewhere.* Question 17 established that a weather derivative cannot be delta-hedged (no one can buy a degree day), so the price is the expected payout plus a risk load, and every disagreement about value in this market is a disagreement about the distribution rather than about the model. There is no replication to appeal to and no market price to calibrate against, so $\mu$ and $\sigma$ *are* the entire pricing problem.

Which is why the two conventions coexist rather than one winning. Burn analysis is auditable and defensible to a counterparty; index modeling is more efficient and less transparent. Desks typically run both and treat the gap as a model reserve, which is Chapter 21's discipline arriving in an actuarial market.

*What the question tests.* That stationarity is burn analysis's binding assumption, and that a modest trend moves the premium by a fifth, so detrending is a pricing decision with a signed consequence.

---

**36.19** *Using Example 36.13, state the minimum-variance hedge ratio and the residual standard deviation at $\rho = 0.85$, and explain why the remainder cannot be hedged.*

*The minimum-variance hedge ratio.* Hedging a load exposure $L$ with a station index $S$, the ratio that minimizes the variance of the hedged position is

$$h^* = \rho \, \frac{\sigma_L}{\sigma_S}$$

which is Chapter 3's minimum-variance hedge ratio, unchanged. At $\rho = 0.85$ that is $0.85\,\sigma_L/\sigma_S$, and if the two have comparable volatilities, roughly 0.85 contracts per unit of exposure.

*The residual standard deviation.*

$$\sigma_L\sqrt{1-\rho^2} = \sigma_L\sqrt{1-0.7225} = 0.527\,\sigma_L$$

so **53%** of the original standard deviation survives, and the variance reduction is **72%**:

| Station–load $\rho$ | Variance reduction | Residual s.d. |
|---|---|---|
| 0.95 | 90% | 31% |
| 0.90 | 81% | 44% |
| 0.85 | **72%** | **53%** |
| 0.70 | 49% | 71% |

Read the table's shape. The relationship is severely nonlinear: a correlation of 0.85, which sounds high, leaves *more than half* the standard deviation in place. Getting the residual below a

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third requires a $\rho$ of about 0.95. Correlations that sound adequate are not, because the residual depends on $\sqrt{1-\rho^2}$ rather than on $\rho$, and that function is flat near $\rho = 0$ and steep near $\rho = 1$.

*Why the remainder cannot be hedged.* Because there is nothing to hedge it with.

The problem is **geographic basis**: the index is measured at one station; the utility's load is spread over a service area. The residual is the difference between the weather at the reference station and the weather over the utility's territory, a real, physical quantity that:

- *has no traded instrument*. Contracts exist on a limited set of reference stations, chosen for the quality and continuity of their records, and not on arbitrary service areas;
- *cannot be replicated*. Question 17 established that no one can buy a degree day at the reference station, let alone a weighted average over a region;
- and could only be removed by a bespoke contract written on the utility's own weighted index, which a counterparty will price with a wide load, because it cannot hedge it either and cannot verify the index.

So the residual is not a modeling failure or a sizing error. It is the irreducible difference between the exposure and the only instrument that exists, and it is Chapter 3's basis risk in its purest form—with the aggravation that the underlying is not tradeable at all.

Two consequences follow for how such a hedge should be presented. A utility reporting "72% hedged" has reported a *variance* reduction, and question 7's warning applies: the residual is concentrated in exactly the unusual weather that motivated hedging, since a regional anomaly is more likely in an extreme season than in an average one. And the residual's size should be quoted in the units that matter (53% of $\sigma_L$, or in dollars) rather than as a variance percentage, because 72% sounds like far more protection than 53% of the standard deviation surviving.

*What the question tests.* The minimum-variance ratio and its residual, that $\sqrt{1-\rho^2}$ makes a high correlation less protective than it sounds, and that the remainder is unhedgeable for want of an instrument rather than for want of a model.

---

**36.20** *Explain why a catastrophe bond multiple falls as the expected loss rises, and why multiples widen after an unanticipated event with no change in the physical hazard.*

*The multiple.* A catastrophe bond pays a coupon and returns principal unless a defined event occurs, in which case principal is written down to the sponsor. Pricing is an expected loss plus a load, and the load is most of it. The market quotes the ratio of spread to modeled expected loss:

| Expected loss | Spread | Multiple | Risk load |
|---|---|---|---|
| 1.00% | 4.50% | **4.50**× | 3.50% |
| 2.00% | 6.50% | 3.25× | 4.50% |
| 4.00% | 10.00% | 2.50× | 6.00% |
| 8.00% | 16.00% | **2.00**× | 8.00% |

*Why the multiple falls as the expected loss rises.* Read the last column: the risk load grows, but far more slowly than the expected loss. From 1% to 8% of expected loss (a factor of eight), the load rises only from 3.50% to 8.00%, a factor of 2.3. The multiple is 1 + load/EL, so a load growing more slowly than EL produces a falling multiple arithmetically.

The economics behind it: remote risks carry proportionally more compensation because the investor is being paid for the **uncertainty in the loss estimate** as well as for the loss.

A 1% expected loss is estimated from a vendor catastrophe model extrapolating into a region of the distribution with very little historical support. If the true expected loss were 2%,

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the investor's compensation would be halved, so the *relative* parameter uncertainty is enormous at the remote end and modest at the frequent end, where events are observed often enough to calibrate against.

This is the same pattern as credit, and the parallel is exact. Chapter 24's question 9 found the ratio of risk-neutral to historical default probability reaching **32.8** for AAA issuers and falling to **1.6** for single-B—whatever the excess compensates for, it does not scale with the risk. The mechanisms are identical: a systematic-risk premium that does not shrink with probability, non-credit components that are roughly fixed in absolute terms, and parameter uncertainty that is proportionally worst where the estimate is thinnest.

*Why multiples widen after an unanticipated event with no change in the physical hazard.*

Because the loss estimate comes from a vendor catastrophe model rather than from a market, so the multiple is partly a discount for model risk—and an event the models did not anticipate is direct evidence that the models are wrong.

The physical hazard need not have changed at all. A hurricane making landfall in an unmodeled location, a wildfire loss an order of magnitude above the modeled distribution, or a flood in a region the model treated as low-risk each reveal that the model's tail was mis-specified, which affects the credibility of every expected-loss figure the model produces, including those for perils it got right.

Three effects compound:

- *Parameter uncertainty is repriced* across the whole market, because one model failure implicates the methodology.
- *Capital withdraws.* Investors who took losses reduce allocations, and the remaining capital demands a higher price, which is a supply effect independent of the hazard.
- *And model revisions follow*, so the expected-loss denominators themselves change, often upward.

*The general lesson.* When a price is quoted as a multiple of a model output, the multiple prices the model. A widening multiple after a surprise is not investors becoming irrational about hurricanes; it is investors correctly reducing the weight they place on an estimate that has just been shown to be capable of being badly wrong. Chapter 26's question 8 makes the same point about a AAA tranche's rating: the letter was an agency's correlation assumption, and repricing it was repricing the assumption rather than the collateral.

*What the question tests.* That a slowly growing load over a fast-growing expected loss produces a falling multiple, and that the multiple contains a model-credibility discount that a surprise revises.

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# Chapter 37
# Crypto Derivatives

**37.1** *Using Example 37.1, state the annualized basis and the return on capital at 20% initial margin, and give three reasons the spread is wider than Chapter 6's Treasury basis.*

*The annualized basis.* From Example 37.1, spot at \$60,000 and the three-month futures price at \$61,800:

$$\frac{61{,}800}{60{,}000} - 1 = 3.00\% \text{ over the quarter} \quad \Longrightarrow \quad (1.03)^4 - 1 = \mathbf{12.55}\% \text{ annualized}$$

Against a dollar financing cost of 5%, buying spot, selling the futures contract and waiting earns **7.55%** a year over the cost of carry.

*The return on capital at 20% initial margin.* Leverage of 5.0× multiplies that excess over carry:

| Initial margin | Leverage | Return on capital |
|---|---|---|
| 50% | 2.0× | 15.1% |
| 35% | 2.9× | 21.6% |
| 20% | 5.0× | **37.8**% |

This is Chapter 6's Treasury basis trade with the numbers moved up by two orders of magnitude, and the reason the spread is 12.55% rather than a few basis points is not that the arbitrage is harder. It is that three things Chapter 6 took for granted are absent.

*1. There is no CCP.* A Treasury basis trade faces a clearing house with a default waterfall, mutualized loss-sharing and a regulated capital structure (Chapter 2). Here the position is an **unsecured exposure to a venue**: a private company, often offshore, with no resolution regime and no supervisor of the kind Chapter 2 describes.

*2. Margin is set by the venue rather than by a model, and can be raised without notice.* A CCP's margin follows a published methodology with a governance process around changes. A crypto venue can change its schedule intraday, and question 2 shows what that does to a position already sized against the old one.

*3. The collateral is held by the counterparty.* There is no segregation of the kind Chapter 8 requires, so the trade's return is conditional on the counterparty surviving the quarter. The posted margin is an unsecured claim on the venue, and question 19's FTX case is what happens when that claim fails.

So the basis is not an inefficiency but a price. It compensates for venue credit, for margin uncertainty, and for the absence of segregation, three exposures a Treasury basis trader does not carry. Chapter 5's observation that arbitrage requires a balance sheet applies here with the addition that it also requires trusting the venue with the balance sheet.

This reframes the 37.8%. It is not an unexploited opportunity; it is the market's price for bearing three risks simultaneously, levered five times. Question 3 pursues why quoting the leverage without the margin schedule is the specific error.

*What the question tests.* Annualizing a quarterly basis and levering a carry, and identifying which of Chapter 6's institutional preconditions are missing rather than treating the spread as mispricing.

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**37.2** *Using Example 37.2, state the rally that exhausts the posted margin on a fully hedged basis trade, and explain why a position that has lost nothing can be closed.*

*The rally that exhausts the margin.* From Example 37.2, at \$60,000 spot the futures price is \$61,800, and 20% initial margin on the contract is \$12,360 per unit; a rally of $x$ raises the futures price by $x \times \$61{,}800$:

| Rally | Cash called | Share of margin | Value of the position |
|---|---|---|---|
| +10% | \$6,180 | 50% | unchanged |
| +20% | \$12,360 | 100% | unchanged |
| +33% | \$20,394 | 165% | unchanged |

$$\frac{\$12{,}360}{\$61{,}800} = \mathbf{20}\%$$

A 20% rally, the margin rate itself, exhausts the margin on a position that has not lost a cent.

*Why a position that has lost nothing can be closed.* Because the trade is flat in value and its cash flows are not.

The basis trade is genuinely hedged: the spot leg and the short future move together, so the profit was locked at the moment of entry and the position's value column reads "unchanged" at every rally. But the futures position is margined daily and the spot is not. A rally calls cash on the short futures leg against an *unrealized* gain on the spot leg, an asset that will not produce cash until it is sold.

So the trader must fund the entire rally in cash while holding an equal and offsetting gain that delivers none. When the posted margin runs out, the venue liquidates. The trade is liquidated for want of cash while the thing it is hedged against is working exactly as intended, and the liquidation forfeits the 7.55%, which was locked in only for a position held to expiration. The futures leg is closed at the top of the rally with its loss realized in cash; the coin is either sold alongside, which recovers nearly all of that loss but none of the carry, or kept as an unhedged long.

This is Chapter 8's margin call as a liquidity risk, and the reader has met it twice already:

- *Metallgesellschaft* (Chapter 36's question 8), whose short-dated futures needed cash while its long-dated supply obligations were merely valuable;
- *the 2022 European utilities*, whose hedges required variation margin measured in tens of billions while the offsetting power was worth more and would arrive on delivery.

*A hedge which is correct in value and wrong in timing is not a hedge.*

*What this market adds* is that *the venue sets the margin*, so the schedule in force at entry is not a contractual commitment. A venue that raises initial margin from 20% to 35% during a rally converts a position that was surviving into one that is not, and it can do so without notice and without any change in the trade's economics.

*The remedy is to size on the funding leg rather than on the value leg.* The relevant question is not "what can this position lose?"—the answer is nothing—but "what move can I fund?" At 20% margin the answer is 20%, and on an asset that, at 70% volatility, rises that far at some point within a quarter about 55% of the time, the answer is inadequate.

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*What the question tests.* That daily margining on one leg against an unmargined other creates a funding exposure independent of value, and recognizing the shape from two earlier chapters.

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**37.3** *Explain why the leverage in Example 37.1 and the rally in Example 37.2 must be quoted together, and which earlier chapter's case has the same shape.*

*Why they must be quoted together.* Because they are the two halves of one statement, and either alone is misleading in a predictable direction.

*The leverage measures the reward.* Question 1's 37.8% return on capital is achieved *only* at 20% initial margin. At 50% margin the same trade returns 15.1%. So the headline return is a statement about the margin schedule, not about the basis.

*The margin schedule measures the survivable move.* Question 2 showed that the same 20% margin that produces the 37.8% return is exhausted by a **20% rally**, and the two numbers come from the identical input. Halving the leverage halves the return and doubles the survivable move.

So the two are mechanically linked and inversely related:

| Initial margin | Return on capital | Rally that closes the trade |
|---|---|---|
| 50% | 15.1% | 50% |
| 35% | 21.6% | 35% |
| 20% | **37.8**% | **20**% |

Quoting the left column without the right describes a trade that returns 37.8% and omits that it does not survive a move the underlying makes routinely. The return is not a property of the arbitrage; it is a property of how much liquidation risk has been accepted to obtain it.

*And the omission is not symmetric.* The pressure runs one way, toward quoting the return and omitting the survivable move, because the first sells the trade and the second qualifies it.

*The earlier case with the same shape: Chapter 27's Archegos.*

The parallel is exact in structure. There, a fund with $10 billion of capital took five times leverage through total return swaps, and Chapter 27's question 15 established that the client's capital was exhausted at a 20% move, which, for a concentrated portfolio of a few stocks, is an ordinary week. The leverage was sized so that a routine move destroyed the position.

Here the arithmetic is the same and the direction is reversed: a fully *hedged* position, levered five times, closed by a 20% rally that costs it nothing. In both cases:

- the leverage was quoted as a return enhancement;
- the move that exhausts the capital was within the underlying's normal range;
- and the liquidation forfeited a return that existed only for a position held to expiration.

*The general rule.* A leverage figure is meaningless without the move it survives, and the two should be quoted as a pair (**"5×, closed at** +20%**"**), exactly as an option is quoted with its strike rather than merely its premium.

*What the question tests.* That return on capital and survivable move are two readings of one margin number, and recognizing the Archegos configuration in a hedged rather than a directional trade.

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**37.4** *Using Example 37.3, state the forward price and basis at a 3.5% staking yield, and the yield at which the future trades at spot.*

*The forward and basis at a 3.5% staking yield.* A staked asset pays a yield, so the carry relation acquires a dividend term: $F = Se^{(r-q)T}$. From Example 37.3, with spot \$3,000, $r = 5\%$ and $T = 0.25$:

$$F = 3{,}000\, e^{(0.05-0.035)\times 0.25} = 3{,}000 \times 1.003757 = \$\mathbf{3{,}011.27}$$

| Staking yield $q$ | Forward | Quarterly basis | Annualized |
|---|---|---|---|
| 0.0% | \$3,037.74 | +1.258% | +5.13% |
| 3.5% | **\$3,011.27** | **+0.376%** | **+1.51%** |
| 5.0% | \$3,000.00 | 0.000% | 0.00% |
| 7.0% | \$2,985.04 | −0.499% | −1.98% |

*The yield at which the future trades at spot:* $q = r = \mathbf{5.0}\%$.

At that point the exponent vanishes and $F = S$ exactly. The benefit of holding the asset equals the cost of financing it, so there is nothing to carry and the forward is flat. This is the same condition that makes Chapter 34's reflection argument exact for a barrier option, and Chapter 5's result that a stock yielding exactly the risk-free rate has a flat forward curve.

*Above $q = r$ the curve inverts.* At $q = 7\%$ the three-month future trades *below* spot, at a $-1.98\%$ annualized basis.

Read what this does and does not mean. Nothing about the asset changed; it acquired a dividend. An asset that pays a yield exceeding its financing cost trades backwardated, and this is an arithmetic consequence of the carry relation rather than a signal about supply, demand or positioning.

That distinction is question 5's subject, and it matters because a persistent backwardation that would signal scarcity in Chapter 36 signals only that the staking yield exceeds the funding rate.

*The practical consequence for reading the basis.* Question 1's 12.55% basis is often used as a positioning indicator, a proxy for leveraged long demand. For a staked asset that reading is contaminated: the observed premium mixes leveraged demand with the yield, and a desk reading it as sentiment will misread it whenever $q$ moves.

*And the correction is available.* The two components are separable if the staking yield is quoted independently, *which it is*, since staking yields are published and observable. So the adjustment is a subtraction, requires no model, and is *routinely not made*. That is the useful observation: the data exist and the convention has not caught up.

*What the question tests.* Substituting a yield into the carry relation, and locating the point at which the curve flattens as the condition $q = r$ rather than as a market phenomenon.

**37.5** *Explain why a persistent backwardation in a staked asset does not mean what the same shape means in Chapter 36, and what must be quoted to separate the two.*

*The contrast.* In Chapter 36, a persistent **backwardation** (forward below spot) implies a large positive *convenience yield*. Chapter 36's question 6 inverts the relation: with $S = 80$, $F = 77$ at six months, $r = 4\%$ and $u = 5\%$, the implied convenience yield is **16.64%**. That number means something physical: holders of the commodity are receiving a benefit (avoiding a stock-out, keeping a refinery supplied) worth more than the full cost of carry. It is a *scarcity* signal.

Here it means nothing of the kind. Question 4 showed that a staked asset trades backwardated whenever $q > r$, purely arithmetically. At $q = 7\%$ against $r = 5\%$ the curve inverts by $-1.98\%$

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annualized, and *nothing is scarce*. There is no storage constraint, no physical benefit, and no inventory. The asset simply pays a dividend larger than its financing cost, exactly as a high-dividend-yield equity does.

So the same observable shape carries entirely different information in the two markets. A trader importing Chapter 36's reading ("backwardation means the physical is tight") would conclude that a staked asset in backwardation is in short supply, when the correct conclusion is that its protocol pays well.

And the ambiguity is genuine rather than pedantic, because both effects can be present. The observed basis is

$$\text{basis} \approx \underbrace{(r-q)T}_{\text{carry}} + \underbrace{\text{leveraged demand}}_{\text{positioning}}$$

and a single number cannot separate them. A basis of zero could be $q = r$ with no positioning, or $q > r$ offset by heavy long demand, or the reverse. The two components have opposite policy implications (one is a yield, the other is a crowded trade), and both move.

*What must be quoted to separate them: the **staking yield**, independently.*

Given $q$, the carry component is computable exactly and the residual is the positioning signal:

$$\text{positioning} = \text{observed basis} - (r-q)T$$

No model is required. And the staking yield is observable (protocols publish it, and it is reported by every major data provider), so the correction is available and is routinely not made, which is the chapter's point. The market quotes the raw basis as a sentiment indicator, and the indicator is contaminated by a quantity everyone can look up.

*Two further separations would help.* The *funding rate* $r$ should be the trader's actual dollar funding rather than a risk-free proxy, since the trader's actual funding may exceed it. And for a proof-of-stake asset the yield itself varies with network participation, so $q$ is not constant, meaning the carry component moves even when neither the price nor positioning does.

*The general lesson.* A curve shape is a residual, and residuals absorb everything the model omits. Chapter 36's question 11 makes the identical point about allowances, where a positive implied convenience yield cannot be physical and must be absorbing a financing spread. Here it is a dividend. In both cases the fix is to model the omitted term rather than to interpret the residual.

*What the question tests.* That backwardation is a carry consequence rather than a scarcity signal for a yielding asset, and that separating carry from positioning requires only an observable quantity the convention ignores.

---

**37.6** *Explain what holds a perpetual futures contract to spot in the absence of delivery, and why the mechanism is an adaptation to missing infrastructure.*

*What holds a perpetual to spot: the **funding rate**.*

A perpetual futures contract never expires, so the convergence mechanism every other futures contract relies on, *delivery*, is unavailable. Chapter 2's argument that a futures price must converge to spot at expiration has no force here, because there is no expiration.

The substitute is a periodic cash payment between longs and shorts, typically every eight hours (some venues hourly), proportional to the gap between the perpetual's price and a spot index:

- when the perpetual trades *above* spot, **longs pay shorts**;
- when it trades *below*, **shorts pay longs**.

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So a persistent premium becomes expensive to hold and a persistent discount becomes profitable to hold, and the flow of capital toward the profitable side closes the gap. The contract is tethered to spot by a running cost rather than by a terminal obligation.

*Why the mechanism is an adaptation to missing infrastructure.*

*Delivery requires infrastructure this market did not have.* A deliverable futures contract needs a defined delivery location, a custody arrangement, a settlement process, an inspection or verification standard and, crucially, a clearing house standing between the parties to guarantee that delivery occurs. Chapter 2 describes all of it, and question 1 established that none of it exists here: no CCP, venue-set margin, and collateral held by the counterparty.

Rather than build that infrastructure, the market engineered around its absence. A perpetual requires no delivery, no custody transfer at expiration, no settlement standard, and no roll, so it sidesteps every institutional gap at once.

*And the adaptation has genuine advantages, which is why it dominates.* There is no roll, so no roll cost and no expiration-week liquidity fragmentation (contrast Chapter 22's VIX products, whose entire character is set by rolling). Liquidity concentrates in a single contract rather than fragmenting across months. And the price of a levered long is *observable*: question 7's point, and an unexpected improvement over the classical market.

*What it gives up.* The tether is *economic* rather than *contractual*. Delivery convergence is enforced by arbitrage that cannot fail; funding convergence is enforced by the willingness of capital to take the profitable side, which can be overwhelmed in a dislocation. A perpetual can trade meaningfully away from spot for as long as no one is able to fund the offsetting position, which is precisely when it matters, and which is Chapter 5's limits-of-arbitrage argument appearing in a contract designed to avoid needing it.

*The pattern worth naming.* A market lacking an institution will invent a contract that does not require it, and the invention usually trades a hard guarantee for a soft incentive. Question 8 shows the same trade in the liquidation mechanism, where a margin call has been replaced by automatic close-out.

*What the question tests.* That funding replaces delivery as the convergence mechanism, and that the design is a response to absent clearing infrastructure rather than a financial innovation for its own sake.

---

**37.7** *Using Example 37.4, convert a funding rate of 0.03% per eight hours into an annualized rate and into the equivalent three-month futures premium.*

*The annualized rate.* Funding is paid every eight hours, so there are $3 \times 365 = 1{,}095$ periods a year:

$$0.03\% \times 1{,}095 = \mathbf{32.85}\% \text{ annualized}$$

*The equivalent three-month futures premium.* A quarter is one-quarter of the annual funding:

$$\frac{32.85\%}{4} = \mathbf{8.21}\%$$

From Example 37.4:

| Funding per 8 hours | Annualized | Equivalent 3-month premium |
|---:|---:|---:|
| +0.0100% | +10.95% | +2.74% |
| +0.0300% | **+32.85%** | **+8.21%** |
| −0.0200% | −21.90% | −5.47% |

*The two are the same statement about the price of a levered long.* A quoted funding rate of 0.01% per eight hours and a three-month futures contract at a 2.74% premium describe an identical cost of holding a levered long, and a desk running both should see them agree. When

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they do not, the difference is the cost of the two venues' credit and margin terms (question 1's three missing institutions) rather than anything about the underlying.

That makes the pair a cross-venue arbitrage check, and a useful one: it is model-free, requires only two quoted numbers, and any persistent gap localizes to a specific pair of venues.

*One more point is worth noting.* In one respect this market reports more than the regulated one does.

Chapter 5 had to *infer* a convenience yield as the residual that made an equation hold, and Chapter 36's questions 6 and 11 showed how much that residual absorbs (storage scarcity, financing spreads, regulatory expectation) with no way to separate the components. The quantity was never observed; it was always backed out.

Here that price is printed every eight hours. The cost of holding a levered long is quoted directly, by the venue, in real time. So a quantity of the kind Chapter 36 spent a section extracting from a curve is here simply *observable*, and it is observable at eight-hour resolution rather than at the frequency at which forward curves are marked.

That is an unexpected place to find better data than the regulated market provides, and it is worth stating plainly because the market's other properties are so much worse. Question 9's liquidation profile is a second instance: the venues publish open interest and leverage tiers in real time, which Chapter 35's structured-note market could not observe at all.

*One caution attaches to the conversion.* The arithmetic above uses simple scaling, which is right to the precision the market quotes at. Compounding 1,095 periods of 0.03% gives $e^{0.3285} - 1 = 38.9\%$ rather than 32.85%, so at high funding rates the two conventions diverge, and the market's convention is the simple one, which should be checked before comparing against a compounded futures basis.

*What the question tests.* The unit conversion between a per-period funding rate and a term premium, and that the perpetual's design makes an unobservable classical quantity directly quoted.

---

**37.8** *Explain why liquidation on a crypto venue is not a margin call, and what the client gives up when it replaces one.*

*Why liquidation is not a margin call.* A margin call, as Chapter 8 describes it, is a **demand**: the counterparty is notified that additional collateral is required, given a period to deliver it (usually a business day), and only on failure to deliver does the surviving party have the right to close out. The client has *time* and *choice*: post more, reduce the position, or let it be closed.

*Liquidation on a crypto venue is automatic and immediate.* When account equity falls below a maintenance threshold, the venue's engine closes the position at market, without notice, without a cure period, and without the client's participation. There is no call, no negotiation, and typically no opportunity to post: the close-out happens in the same instant the threshold is crossed.

*What the client gives up.*

Not the ability to protect the position. A client on a crypto venue can add collateral at any moment, unprompted, so a position that falls below the threshold has been allowed to fall there. What is lost is the *timing* of that decision. A cure period lets the client decide *after* the trigger, having seen the market, with a business day to raise cash and to learn whether the move reverses. Without it the same decision has to be made *before* the trigger, without knowing whether it will be needed.

That costs the client money in two ways:

*Protection must be funded ahead of time.* To survive a move of a given size the client must post the collateral for it now and leave it idle, or watch the position continuously and move cash

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at any hour—and bank wires do not move at every hour.

*And a fast market outruns the client.* The threshold is crossed precisely when the market has moved against the position, the engine closes at market into whatever liquidity exists, and question 9 shows that many accounts cross their thresholds together, so the executions are simultaneous and the price impact is shared among exactly the people being liquidated. A move that a margin call would have allowed a day to answer is over in minutes.

Question 2's case is where this bites hardest: a fully hedged basis trade, losing nothing, is closed for want of cash. Its owner could have avoided that only by holding, in advance, collateral against a rally nobody knew would come.

*The general reading.* Removing a cure period is not an operational simplification; it shifts a cost to the client, replacing the option to wait with the cost of funding protection in advance. The client gives up the value of deferring the decision and should want compensation for it (in lower fees or lower margin) but rarely gets it, because the cost appears nowhere in the documentation.

And question 20's auto-deleveraging goes one step further: there, even a *correctly positioned and fully margined* trader can have the position removed, which no margin regime in Part I permits.

*What the question tests.* That the missing cure period is what distinguishes the two mechanisms, and that what it removes is the right to decide after the trigger, not the ability to protect the position.

---

**37.9** *Using Example 37.5, state the forced selling triggered by a 10% decline, and explain why publishing the liquidation profile does not make the cascade smaller.*

*The forced selling.* From Example 37.5, taking \$12 billion of long-biased perpetual open interest distributed across the leverage the venues actually offer, where a position at $\ell$ times leverage liquidates on a fall of roughly $1/\ell$:

| Leverage | Liquidates on | Notional in tier | Cumulative |
|---|---|---|---|
| 25× | −4% | \$1.2bn | \$1.2bn |
| 20× | −5% | \$1.8bn | \$3.0bn |
| 10× | −10% | \$4.2bn | **\$7.2bn** |
| 5× | −20% | \$3.6bn | \$10.8bn |

A 10% decline mechanically forces **\$7.2 billion** of selling (**60%** of the open interest), and the selling arrives into a market already falling.

*The second-round effect is the point.* That flow pushes the price further, which triggers the next bucket, which pushes further still. It is Chapter 35's issuer hedging with the feedback running the same way, and with no participant choosing the timing, because the triggers are mechanical.

*Why publishing the profile does not make the cascade smaller.*

Here the disclosure is *the opposite* of Chapter 35's, and the contrast is instructive. Auto-callable barriers cluster because issuers independently choose round fractions, and no issuer can see the aggregate. That was the whole difficulty of Chapter 35's question 13, and the remedy proposed there was for a supervisor to collect what no participant could observe.

Here open interest and leverage tiers are published by the venues in real time, so the liquidation profile is computable by anyone who wants it. The information problem is solved.

*And the cascade happens anyway*, for three reasons.

*No one can act on it individually.* A leveraged long who knows the profile still faces a choice between closing, which forgoes the position, and holding through a decline they now

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expect. Knowing that \$7.2 billion sells at −10% does not give any individual the ability to prevent it, and the rational response for a levered holder is to *sell first*, which accelerates the cascade rather than damping it.

*The triggers are mechanical, not discretionary.* A liquidation is executed by the venue's engine on a price condition (question 8), so no amount of foreknowledge changes whether it fires. Compare a discretionary margin call, where a counterparty aware of a cluster might grant forbearance.

*And transparency invites the opposite trade.* A visible cluster of stops is a target: a participant who can push the price down the required 4% triggers \$1.2 billion of forced selling, into which a short can be covered. Publication shows the flow in advance to those best placed to trade ahead of it.

So the two chapters together make the sharper point. Chapter 35's problem was that no one could see the aggregate; here everyone can, and the outcome is the same. Transparency is necessary and not sufficient: what removes a mechanical cascade is changing the mechanism (lower maximum leverage, partial rather than full liquidation, a cure period), not illuminating it.

*What the question tests.* Reading a liquidation ladder as cumulative forced flow, and that disclosure addresses an information failure while the cascade is a mechanism failure.

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**37.10** *Explain why crypto skew changes sign with the market while equity index skew does not, and name the chapter whose pattern it resembles.*

*The observation.* Equity index skew is *persistently* negative: out-of-the-money puts trade above out-of-the-money calls, in every market, in nearly every period, for the structural reasons Chapter 19's question 1 sets out: leverage, crash asymmetry, and one-directional protective demand.

*Crypto skew changes sign with the market.* In a rally, calls trade above puts and the skew is *positive*; in a decline it inverts to the equity shape. The surface's tilt is a *state variable* rather than a fixture.

*Why.*

*There is no leverage effect.* Chapter 19's first structural reason is that equity is a residual claim on a levered balance sheet, so a falling price mechanically raises the debt-to-equity ratio and hence equity volatility. A crypto asset has no balance sheet, no debt, and no residual-claim structure, so the mechanism that anchors equity skew simply does not exist.

*The demand is two-sided and regime-dependent.* Equity index option demand is dominated by institutions buying downside protection on portfolios they hold, a permanent, one-directional flow that dealers must absorb. Crypto option demand is dominated by *leveraged directional speculation*, which switches sides: in a rally, call buying for upside participation; in a decline, put buying for protection. The skew follows the flow, and the flow follows the market.

*And upside is not bounded in the way equity upside is.* An equity index that doubles is extraordinary; a crypto asset that doubles is a normal year. So the right tail carries genuine probability mass, and a positively-skewed density is not implausible in the way it would be for an index.

*The chapter whose pattern it resembles: Chapter 19's* ***currency*** *smile.*

Chapter 19's question 1 explains why a currency pair shows a roughly symmetric smile rather than a skew: a currency is a ratio, so a fall in one direction is a rise seen from the other desk, and neither side is privileged. There is no leverage effect and no asymmetry of catastrophe, so there is no structural reason for a persistent tilt, and what tilt exists is transient and reflects positioning.

Crypto resembles this closely: no leverage effect, no privileged direction, and a tilt driven

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by flow rather than by structure. The **risk reversal** (the quoted difference between 25-delta call and put volatilities) is the instrument that prices the residual asymmetry in both markets, and in both it is quoted *because* it moves rather than because it is stable.

*Two practical consequences follow.*

A skew parametrization calibrated on one regime will not fit the next, so the fitted parameters carry no forecasting content and a desk marking off a stale skew is marking off a positioning snapshot.

*And skew is a positioning indicator here in a way it is not in equities.* An equity skew tells you almost nothing about current positioning because it is always steep; a crypto risk reversal tells you which way the leveraged flow is leaning, which makes it genuinely informative. Question 5's warning applies, that reading a market-structure quantity as sentiment requires first removing the mechanical component.

*What the question tests.* That the absence of a leverage effect removes equity skew's structural anchor, and matching the resulting flow-driven symmetry to the currency case rather than the equity one.

---

**37.11** *Using Example 37.8, state the coin profit and loss of a long inverse position at a 50% fall and at a 50% rise, and explain why the two are not symmetric.*

*The two coin profits and losses.* From Example 37.8, a long inverse position of \$100,000 notional entered at \$60,000. An inverse contract settles in *coin*, so the position's coin value is $Q/F_0 - Q/S$:

| Price | Move | P&L in coin | P&L in dollars |
|---|---|---|---|
| \$30,000 | −50% | **−1.6667** | −\$50,000 |
| \$90,000 | +50% | **+0.5556** | +\$50,000 |
| \$120,000 | +100% | +0.8333 | +\$100,000 |

*Why the two are not symmetric.* The dollar column is a straight line and the coin column is not.

The dollar P&L is linear by construction (the contract pays \$100,000 per unit move in the price), so a 50% fall and a 50% rise are ∓\$50,000, symmetric. But converting that to coin divides by a price that is *itself* the variable:

$$\Pi_{\text{coin}} = \frac{N}{F_0} - \frac{N}{S}$$

which is linear in $1/S$ and therefore concave in $S$. Differentiating twice,

$$\frac{\partial^2 \Pi_{\text{coin}}}{\partial S^2} = -\frac{2Q}{S^3} < 0$$

so a long inverse position is short gamma measured in coin.

The asymmetry is dramatic and it runs the wrong way for the holder:

- a 50% fall costs **1.6667 coin**;
- a 50% rise earns only **0.5556 coin**;
- and the coin gain is *capped* at $Q/F_0 = 1.6667$ however far the price rises (the second term $Q/S$ goes to zero and no more can be earned), while the coin *loss is unbounded*, since $Q/S$ diverges as $S \to 0$.

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*The upside is bounded and the downside unbounded, in the unit that matters.*

And the practical consequence is that margin is computed in the unit where the position is curved. Starting with 0.5 coin of margin:

| Price | Equity (coin) | Equity (dollars) |
|---|---|---|
| \$54,000 | +0.3148 | \$17,000 |
| \$48,000 | +0.0833 | \$4,000 |
| \$42,000 | −0.2143 | −\$9,000 |

Equity in coin falls faster than the price does, so liquidation (question 8) arrives sooner than a dollar-denominated calculation suggests. A trader sizing the position on the linear dollar column and margined on the curved coin column has mis-measured exactly the quantity the venue will act on.

*The general point.* A payoff's convexity depends on the numeraire it is measured in. A position that is linear in dollars is short gamma in coin, and neither statement is more correct; they are the same position seen in two units of account. Question 12 makes that a formal observation about change of numeraire.

*What the question tests.* That dividing a linear dollar payoff by a stochastic price introduces convexity, and that margin being computed in the curved unit is what makes it operationally consequential.

---

**37.12** *Explain why an inverse contract's valuation is a change of numeraire, and which adjustment a desk drops by discounting the dollar payoff instead.*

*Why it is a change of numeraire.* An inverse contract pays in **coin**, so the natural unit of account is the coin rather than the dollar. Chapter 28's framework says exactly what to do: the valuation is

$$V_0 = S_0 \, E^S\left[\frac{\text{payoff in coin}}{1}\right]$$

using the *coin* as numeraire, an admissible choice, since it is a traded asset with a strictly positive price (Chapter 28's question 11).

That is precisely Chapter 28's question 9 construction, where using the *stock* as numeraire turns $E[S_T\mathbf{1}]$ into a probability and produces $N(d_1)$. Here the same device turns a coin-settled payoff into an expectation under the coin measure, and Girsanov supplies the drift shift: moving from the dollar measure to the coin measure shifts every drift by its covariance with $\ln S$, which for the coin itself is $\sigma^2$.

So the valuation is not a dollar valuation converted at the end. It is an expectation taken under a different measure, and the two differ by exactly the covariance between the payoff and the exchange rate. That is question 11's convexity, stated formally.

*Which adjustment a desk drops by discounting the dollar payoff instead: the **quanto** adjustment.*

Computing the dollar P&L, discounting it at the dollar rate, and converting to coin at spot treats the payoff and the coin/dollar rate as *independent*. They are not—the payoff *is* a function of that rate. The omitted term is

$$-\rho \, \sigma_V \, \sigma_S$$

which is Chapter 30's quanto drift adjustment, with the "foreign asset" being the contract's dollar value and the "exchange rate" being the coin price itself.

And here the correlation is not an estimate—*it is one*. The payoff and the numeraire are driven by the same variable, so $\rho = \pm 1$ and the adjustment is at its maximum possible

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size. In an ordinary quanto the correlation must be estimated and is usually well below one (Chapter 30's question 9); here it is structural.

*The adjustment shows in the sensitivity rather than in the level.* At any single price the coin profit can be recovered by converting the dollar profit at that price (a 50% fall costs 50,000/30,000 = 1.6667 coin either way), but the coin profit's *sensitivity* to the price is not the dollar sensitivity converted, and it is the sensitivity a hedger uses.

*Two practical consequences follow.*

*The delta is wrong.* A desk hedging an inverse position with a linear dollar delta is under-hedged on the downside and over-hedged on the upside, systematically, because the true coin delta carries the $-2Q/S^3$ curvature.

*And the margin is computed correctly by the venue and incorrectly by the trader.* The venue marks in coin, which is the curved unit; a trader running a dollar book sees a linear position. The discrepancy is not a modeling refinement; it is the difference between the position the trader thinks they hold and the one that will be liquidated.

*The general lesson, which is Chapter 28's thesis in an unexpected market.* The question "what is the numeraire" is not abstract; it is the question of what portfolio actually replicates the trade. A coin-settled contract is replicated in coin, and pricing it in dollars is choosing an admissible but wrong numeraire—Chapter 28's fourth failure case, and the expensive one.

*What the question tests.* That a coin-settled payoff calls for the coin as numeraire, and identifying the dropped term as a quanto adjustment with structurally perfect correlation.

---

**37.13** *Using Example 37.9, state the break-even coin price, and the revenue change from a 30% price fall before and after the difficulty adjustment.*

*The break-even.* From Example 37.9, a miner producing 500 coins a year against $14 million of electricity and $4 million of other costs:

$$\frac{\$18\text{m}}{500} = \$\mathbf{36{,}000} \text{ a coin}$$

and the margin moves much more than the price, because costs are fixed while revenue is proportional to price:

| Coin price | Revenue | Costs | Margin | Margin/revenue |
|---|---|---|---|---|
| $30,000 | $15.0m | $18m | −$3.0m | −20.0% |
| $60,000 | $30.0m | $18m | +$12.0m | +40.0% |
| $90,000 | $45.0m | $18m | +$27.0m | +60.0% |

*The revenue change from a 30% price fall.* Suppose the price falls from $60,000 to $42,000 and 20% of the network's hash rate leaves:

| | Revenue | Change |
|---|---|---|
| Before | $30.0m | — |
| After, at constant share | $21.0m | **−30**% |
| After the difficulty adjustment | $26.3m | **−12.5**% |

*The protocol converts a 30% revenue shock into a 12.5% one.*

*The mechanism is what separates this from a spark spread.* A gas generator's output is a fixed quantity: it earns $\max(P_e - \text{HR} \cdot P_g, 0)$ per MWh on however many MWh it produces, and nothing about the power market changes its production. A miner earns a *share* of a fixed issuance, and the protocol's difficulty adjustment keeps blocks arriving on schedule regardless of how much hardware is running.

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So if a price fall drives competitors off, the survivor's share rises. The quantity produced is not fixed; it is the residual after the marginal miner exits. Twenty percent of the hash rate leaving raises the survivor's share by $1/0.8 = 1.25$, and $0.70 \times 1.25 = 0.875$, a 12.5% decline rather than 30%.

So the protocol supplies a partial hedge that no counterparty wrote. It is an automatic stabilizer built into the issuance rule, and it operates without a contract, a premium, or a counterparty. That is a genuinely unusual feature and worth naming: the industry's principal hedge is a property of the asset rather than an instrument written on it.

*Two riders attach to the arithmetic.*

The exit is *endogenous*: the 20% assumed here is not given but determined by which miners' costs exceed the new revenue, so the hedge's strength depends on the cost distribution across the network, which the miner does not observe.

And the stabilizer works in *both* directions. A price rise attracts hash rate, the difficulty is adjusted upward, and the survivor's share *falls*, so the upside is damped too. The table's +60% margin at \$90,000 assumes constant share and overstates what a miner would actually earn.

Question 14 pursues why the hedge is only partial, and why miners nonetheless prefer options to forwards despite having a natural short.

*What the question tests.* Computing a break-even and the margin's sensitivity to price when costs are fixed, and that a variable production share converts a price shock into a smaller revenue shock through an endogenous supply response.

---

**37.14** *Explain why the difficulty adjustment is only a partial hedge, giving the three qualifications, and why miners prefer puts to forwards.*

*Why the difficulty adjustment is only a partial hedge. Three qualifications apply.*

*1. It is slow.* The difficulty is adjusted on a schedule (roughly every two weeks for bitcoin) while the price moves continuously. So between adjustments the miner has *no* hedge at all: revenue falls one-for-one with price, and the stabilizer arrives only after the interval closes. A sharp decline is fully borne for up to two weeks, which is precisely the window in which a levered miner fails.

*2. It depends on who leaves, and whether they do.* The protection comes from the miners who exit, and those are the high-cost ones, so a high-cost miner is near the exit itself when it needs protection most, while a low-cost miner gains the most. Nor is the exit guaranteed. Question 13's arithmetic assumed 20% of the hash rate exits. Whether it does is not determined by the price but by the cost distribution across the network, by whether competitors have hedged, by power contracts that may be take-or-pay, and by the sunk cost of hardware that has no alternative use. A miner with a fixed-price power contract will continue running at a cash loss rather than default on the contract, so the exit can be far smaller and slower than the model assumes, and the miner cannot observe the distribution.

*3. It works through quantity, which a forward of fixed size cannot match.* After a fall the miner's share rises and it produces more than it sold; after a rise the share falls and it may produce less, and must buy coin at the high price to deliver. The puts discussion below develops this.

Two further points might be added. It damps the upside too. The stabilizer is symmetric: a price rise attracts hash rate, the difficulty is adjusted *up*, and the survivor's share falls. So the mechanism that limits the downside also caps participation in a rally, and a miner valuing their business on constant-share revenue overstates both tails.

*And it hedges the wrong quantity.* Difficulty responds to the *hash rate*, which responds to profitability, which depends on price *and* on power costs. A power price shock with an

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unchanged coin price produces no difficulty response at all for the affected miner, because the network-wide profitability that drives difficulty has not moved.

*Why miners prefer puts to forwards.*

Because the quantity is uncertain and the difficulty adjustment is only a partial offset.

A **forward** sale fixes the price on a fixed quantity. But question 13 established that a miner's production is *not* fixed: it is a share of issuance that rises when others leave and falls when they arrive. So a miner who sells forward the quantity they expect to produce is short a fixed amount against an uncertain output:

- if the price *falls*, competitors leave, the miner's share rises, and they produce *more* than they sold, so the surplus is unhedged exactly when the price is low;
- if the price *rises*, competitors arrive, the miner's share falls, and they may produce *less* than they sold, leaving them short coin into a rally, and forced to buy at the high price to deliver, which turns part of the gain into a loss on the shortfall.

A **put** has no delivery obligation. It protects the downside without creating an exposure to producing less than expected, and it leaves the upside intact, which matters because question 13's leverage means the upside is where the business's value is.

Three supporting reasons follow. The difficulty adjustment already provides *some* downside protection (12.5% rather than 30%), so what the miner needs is protection against the *residual* tail rather than against the whole move—which is what an out-of-the-money put is. Miners are typically levered against hardware, so the binding constraint is *survival* rather than average revenue, and an option protects survival while a forward protects the average. And a forward requires posting variation margin, which question 2 and Chapter 36's question 8 both show can close a correct hedge for want of cash; a purchased put requires only the premium, paid once.

*What the question tests.* The three limits of an automatic stabilizer, and that quantity uncertainty plus an upside amplified by fixed costs makes an option the appropriate instrument where a forward would be natural for a fixed-output producer.

---

**37.15** *Using Example 37.11, compute the fair value at 50% recovery on the exposed reserve, and explain what the observed 12.3% discount priced that the disclosure did not.*

*The fair value at 50% recovery.* From Example 37.11, where a large stablecoin disclosed in March 2023 that $3.3 billion of its roughly $40 billion reserve, about 8%, sat at a bank that had just failed:

$$1 - 0.50 \times 0.08 = 1 - 0.04 = \mathbf{0.9600}$$

a discount of **4.00%**. The full table:

| Recovery on the exposed reserve | Fair value | Discount |
|---|---|---|
| 100% | 1.0000 | 0.00% |
| 50% | **0.9600** | **4.00%** |
| 0% | 0.9200 | 8.00% |

*What the observed price did not fit.* The token traded to about **0.8774**, a **12.3%** discount, against an *arithmetic worst case* of 8.00% at zero recovery.

*No recovery assumption produces that price.* Even writing the entire exposed reserve to nothing leaves the token worth 0.9200, so the extra **4.3 points** were not pricing the disclosed exposure at all.

*What they were pricing: the **redemption mechanism** itself.*

The peg is held by an arbitrage: if the token trades below one dollar, redeem it with the issuer for a dollar and pocket the difference. That arbitrage needs an open window. It requires

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the issuer to be processing redemptions, at par, in a reasonable time, for the holder in question. A holder who fears the window closes before their turn will sell at any price above what they expect a queue to yield.

So the 12.3% decomposes into two quite different things:

- a *credit loss* on a known, disclosed exposure, at most 8.00%, and probably less;
- and an option-like charge for access to the redemption facility, worth about 4.3 points.

*Only the first is in the disclosure.* The second is a claim about the mechanism, and no reserve report addresses it.

Which is the useful general observation. A stablecoin's disclosure documents the *assets*; the market prices the assets *plus* the reliability of the process for converting a token into them. The second is a Chapter 5 limits-of-arbitrage question (the peg is enforced by a trade, and a trade that may not be executable does not enforce anything), and it is invisible in every reserve attestation.

The link to the rest of the book is direct. The failure of Silicon Valley Bank (Section 40.14) reached the derivatives market of this chapter *through the settlement asset*—not through any position, model, or counterparty in it. A trader perfectly hedged in coin terms lost 12.3% on the units the hedge was denominated in, which is a category of risk that appears nowhere in the position's Greeks.

Question 16 pursues the consequence for collateral, and it is the sharpest form of the point: the settlement asset is not a numeraire that can be assumed constant.

*What the question tests.* Pricing a token as a claim on a reserve, and that a price outside the arithmetic bounds is evidence of a second exposure the disclosure does not describe.

---

**37.16** *Explain why margin posted in a stablecoin is wrong-way risk in the collateral, and what remedy Section 37.9 says is almost never applied.*

*Why stablecoin margin is wrong-way risk in the collateral.* Because the collateral's value falls in the same states that make the exposure it secures most valuable.

Chapter 27's question 12 collects the family: a protection whose value is negatively correlated with the need for it. The usual instances concern the *counterparty*: a domestic bank selling sovereign protection, an insurer writing tranche protection. Here the correlation sits in the collateral itself, which is a purer form of the problem.

The mechanism is specific and it runs in two directions:

*A stablecoin depegs in a stress*, as in question 15's case, where a bank failure took 12.3% off a token in days. That is exactly when a counterparty is most likely to fail and when the surviving party most needs the collateral to be worth what it says. The collateral fails at the moment of close-out, and Chapter 27's question 3 established that the margin period of risk is the only window in which an unhedged move can hurt, so a collateral impairment inside that window is unhedged by construction.

*And crypto market stress and stablecoin stress share causes.* A venue failure, a large liquidation cascade (question 9), or a broad decline all pressure the redemption mechanism at the same time they pressure counterparties. The two are not merely correlated in the data; they have common drivers.

So a position that is fully margined in dollar terms is not fully margined. A trader holding $10 million of stablecoin collateral against a $10 million exposure is protected against the exposure and not against the collateral, and the residual appears in no Greek and no margin calculation, because the calculation treats the settlement asset as a unit rather than as a claim.

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The remedy that Section 37.9 names and that is almost never applied: **a haircut on the stablecoin**.

The technique is entirely standard. Chapter 8 describes haircuts as the mechanism for exactly this problem: collateral whose value is uncertain is accepted at less than face, with the discount sized to the plausible impairment over the margin period of risk. Even a 5% haircut, applied to a token with an 8% worst-case reserve exposure and a documented 12.3% observed discount, would still have been too small.

Why it is almost never applied is worth stating because the reasons are institutional rather than analytical:

- the token is *named* a dollar, and the naming substitutes for an assumption;
- haircutting it would make the venue's margin terms less competitive than a rival's;
- and the depegs have been *temporary*, so a haircut looks like an overreaction between episodes—which is when it would have to be imposed.

*The general lesson.* The settlement asset is an assumption, not a constant. Every calculation in this book that discounts a payoff, computes a margin requirement, or nets an exposure presumes a unit of account whose value is known. Chapter 7's discount curve, Chapter 8's eligible collateral, and Chapter 28's numeraire are all statements about that unit—and this chapter is where the assumption fails visibly.

*What the question tests.* That collateral impairment correlated with close-out is wrong-way risk in its purest form, and that the standard remedy is unapplied for competitive rather than technical reasons.

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**37.17** *Using Example 37.10, state the shortfall at a price ratio of 2 and of 4, and explain why the function is symmetric in $k$ and $1/k$.*

*The shortfalls.* From Example 37.10, a constant-product liquidity position's shortfall against simply holding the initial basket depends only on the price ratio $k$:

$$\mathrm{IL}(k) = \frac{2\sqrt{k}}{1+k} - 1$$

| **Price ratio** | 0.25× | 0.5× | 0.8× | 1.25× | 2× | 4× |
|---|---|---|---|---|---|---|
| **Shortfall** | −20.00% | −5.72% | −0.62% | −0.62% | **−5.72%** | **−20.00%** |

At $k = 2$ the shortfall is **5.72%**; at $k = 4$ it is **20.00%**.

*Why the function is symmetric in $k$ and $1/k$.* Substitute $1/k$ directly:

$$\mathrm{IL}(1/k) = \frac{2\sqrt{1/k}}{1+1/k} - 1 = \frac{2/\sqrt{k}}{(k+1)/k} - 1 = \frac{2\sqrt{k}}{1+k} - 1 = \mathrm{IL}(k)$$

The algebra is a two-line identity, and the table confirms it: 0.5× and 2× both give −5.72%; 0.25× and 4× both give −20.00%.

*The economic reason is that the pool has no privileged direction.* A constant-product pool holds two tokens and rebalances between them; which one is called the numeraire is a labeling choice. Doubling the price of token A against token B is the same event as halving the price of B against A, so the position's shortfall cannot depend on which way round the ratio is quoted. Symmetry in $k$ and $1/k$ is the mathematical statement that a pool is agnostic between its two assets.

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*And the symmetry is the signature of a short straddle.* A payoff that loses the same amount for a move up as for the equivalent move down, with the loss growing with the size of the move in either direction, is a written straddle. The resemblance is not loose. The position's value is

$$V(p) = 2L\sqrt{p}$$

a *concave* payoff in the underlying, which is what a written option is. Differentiating twice gives $V''(p) = -L/(2p^{3/2}) < 0$ *everywhere*, so a liquidity provider is short gamma, and Chapter 18 established what a short gamma position earns and what it loses.

*Two features of the delta go with it.* $V'(p) = L/\sqrt{p} = x$, *exactly the token inventory*, so the pool is automatically delta-hedged to its own holdings, with no rebalancing decision to make. And that automatic hedging is precisely what generates the short gamma: the pool sells the token as its price rises and buys it as its price falls, mechanically, so its delta falls as the price rises, which is what being short gamma means.

*On the name.* The market calls the shortfall *impermanent loss*, a name for a loss that is real unless the price returns. It is "impermanent" only in the sense that it reverses if the price ratio returns to its starting point, which is true of any short option position before expiration, and is not a reason to describe the loss as unreal.

*What the question tests.* Evaluating the shortfall function and verifying its symmetry algebraically, and recognizing that symmetry plus growth in both directions identifies a written straddle.

---

**37.18** *Derive the $\sigma^2/8$ bleed of (37.4) from Chapter 18's gamma cost, and explain why it does not depend on the pool size.*

*The derivation.* Chapter 18's gamma cost per unit time for a position with second derivative $V''$ is

$$\frac{1}{2}|V''|\,\sigma^2 p^2$$

which is the running cost of being short gamma: the rebalancing loss a short-gamma position pays as the underlying moves.

From question 17, a constant-product position has $V(p) = 2L\sqrt{p}$ and

$$V''(p) = -\frac{L}{2p^{3/2}}$$

Substituting:

$$\frac{1}{2} \cdot \frac{L}{2p^{3/2}} \cdot \sigma^2 p^2 = \frac{L\,\sigma^2\sqrt{p}}{4}$$

And dividing by the position's own value $V(p) = 2L\sqrt{p}$ to express it as a rate:

$$\frac{L\sigma^2\sqrt{p}/4}{2L\sqrt{p}} = \boxed{\frac{\sigma^2}{8}}$$

*Why it does not depend on the pool size.* Because $L$ *cancels*. The liquidity parameter appears linearly in *both* the gamma cost and the position value (the cost scales with the position and so does the base it is measured against), so the *rate* is independent of how much capital is committed.

And it does not depend on the price either, for the same reason: $\sqrt{p}$ appears in both and cancels. The bleed is $\sigma^2/8$ of the position, whatever the price and whatever the pool size. That is an unusually clean result, and one that makes the economics immediately legible:

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| Volatility | 40% | 60% | 80% | 100% | 120% |
|---|---|---|---|---|---|
| **Fee yield needed to break even** | 2.00% | 4.50% | **8.00%** | 12.50% | 18.00% |

Read that as a business. A liquidity provider is running a short-volatility position whose running cost is a known function of realized volatility, and the fee income must exceed it. At an 80% volatility, ordinary for a crypto pair, the pool must earn **8% a year in fees** simply to break even against holding the tokens. That is a demanding threshold, and it converts a question about whether providing liquidity is profitable into an arithmetic comparison between two observable numbers.

*The general point about scale-invariance.* The result is a rate rather than an amount, so it answers the question a provider actually has ("is this worth doing?") without reference to size. Contrast Chapter 35's structured note, where the risk is naturally quoted as an amount and the comparison requires normalizing.

*And it identifies exactly what is being sold.* The provider is short realized volatility and long a fee stream, which is Chapter 19's straddle-seller position with the premium paid continuously rather than up front. Chapter 19's question 7 established that a short volatility position earns a premium because it is insurance; here the premium is the trading fee, and whether it is adequate is determined by the same comparison of implied against realized.

*One limit applies.* The derivation is a continuous-time, small-move result: Chapter 18's gamma cost is a local approximation. The exact expected shortfall, $1 - e^{-\sigma^2 T/8}$, confirms the formula and shows where it stops applying: $\sigma^2 T/8$ is the first term of that exponential's expansion, so it always *overstates* the loss, modestly at ordinary horizons (8.000% against an exact 7.688% over a year at 80% volatility) and badly at extreme ones (54.0% against 41.7% at 120% volatility over three years). The reason is that the quadratic approximation keeps growing for large price ratios while the shortfall is bounded and flattens: at $k = 4$ question 17's table gives $-20.00\%$ where $(\ln k)^2/8$ gives $-24.0\%$.

*What the question tests.* Deriving the bleed from Chapter 18's gamma cost, and that $L$ and $p$ canceling is what makes the answer a scale-free rate.

---

**37.19** *Explain why the FTX failure is described as a custody failure rather than a market failure, and what the underlying did throughout.*

*Why it is a custody failure.* The underlying did nothing unusual throughout.

Bitcoin, ether and the major crypto assets continued to trade, on other venues, at prices that were quoted continuously; bitcoin settled every block throughout. There was no pricing error, no model breakdown, no contract that failed to perform as written, and no market in which positions could not be valued.

*What failed was the safekeeping of client assets.* Customer funds held for safekeeping were not segregated and were not there. That is a failure of the arrangement by which a client's property is held separate from the firm's, an arrangement Chapter 8 treats as a precondition rather than as a risk.

*The distinction matters because it locates the exposure.* A market failure is addressed by better models, wider spreads, more capital against price risk, and better hedging. None of those would have helped here. A trader with a perfectly hedged, fully margined, correctly valued book lost everything held at the venue—because the loss had nothing to do with the book.

*And it is question 1's third missing institution, realized.* That question identified three things Chapter 6's Treasury basis trade takes for granted and this market lacks: no CCP, venue-set margin, and the collateral held by the counterparty. The third is the one that failed. The trade's return is conditional on the counterparty surviving the quarter, and the counterparty did not.

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*Three structural comparisons make the gap concrete.*

*Segregation.* Chapter 8 requires client collateral to be held separately so that it survives the intermediary's failure, and Chapter 27's question 14 shows the cost: a clearing member's client-facing and CCP-facing legs offset, yet both attract capital. That cost is the price of the protection, and a venue that does not pay it offers better terms and no protection.

*Porting.* Chapter 2's default management transfers a failed member's client positions to a surviving member, so clients are not liquidated for a failure that was not theirs. No analog existed.

*And a resolution regime.* A regulated intermediary's failure runs through an insolvency process designed for the purpose, with client assets ring-fenced. Here it ran through an ordinary bankruptcy in which clients were unsecured creditors.

So the correct summary is that the derivatives worked and the infrastructure did not. That is the reverse of the usual account, and it locates the reform agenda precisely. What this market needs is not better pricing but the institutional layer Part I describes: segregation, a central counterparty, and a resolution regime.

*The general lesson, and it recurs.* An instrument's risk is not exhausted by its payoff. Chapter 25's Credit Suisse case made the same point from the documentation side (a correct credit view paid nothing because the trigger was a legal list), and question 15 made it from the settlement-asset side. In each case the position was right and something outside the position determined the outcome.

*What the question tests.* Separating the performance of contracts and prices from the safekeeping of assets, and identifying which of Part I's institutions was the missing one.

---

**37.20** *Explain what auto-deleveraging is, why it has no analog in Part I, and why a correctly positioned and fully margined trader can still lose the position.*

*What auto-deleveraging is.* When a liquidated position cannot be closed in the market (the price has gapped through the liquidation level, or no one will buy), the venue faces a shortfall it must allocate. *Auto-deleveraging* closes the shortfall by **forcibly closing profitable positions on the opposite side**, at the bankruptcy price, selected by a ranking that typically targets the most profitable and most levered accounts first.

So a trader on the winning side of a move can have their position *canceled*—not margined, not called, but terminated—to fund a loss on the losing side.

*Why it has no analog in Part I.*

Because Part I's institutions handle the same problem without touching the winners' positions. When a clearing member defaults and its portfolio cannot be liquidated at the marked price, Chapter 2's **default waterfall** absorbs the shortfall in a defined order: the defaulter's margin, then its default fund contribution, then the CCP's own capital, then the mutualized default fund of surviving members, and finally assessments on those members and the CCP's other recovery tools.

*Every layer is capital that was posted in advance for exactly this purpose.* Surviving members contribute money; their positions are untouched. A member who was correctly positioned keeps the position and pays a share of the loss from a fund it knowingly funded.

Auto-deleveraging inverts that. It takes the *position* rather than the capital, from the party whose only distinguishing feature is that they were right, and it does so without any pre-funded arrangement. There is no waterfall because there is no disclosed, pre-funded capital behind the venue, so once its undisclosed insurance fund is gone the only resource available is the winners' profits.

*Why a correctly positioned and fully margined trader can still lose the position.*

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Because nothing about their own account triggers it. The trader has:

- posted the required margin, and more;
- taken a position that is profitable;
- and done nothing that any rule prohibits.

The trigger is entirely on the *other side of the market*: someone else's leverage, someone else's liquidation, and a gap the venue's engine could not close. The trader's own risk management is irrelevant to whether it happens.

And the loss is real even though the position is closed at a defined price: the trader loses the *future* profit on a correct view, and is left flat in a market that has just moved their way, with no ability to re-enter at the price they were closed at.

So the exposure is to the venue's other clients, and it is unhedgeable. A trader cannot size it, diversify it away within a venue, or contract out of it; it is a term of the venue's rulebook. The only mitigations are to spread across venues and to prefer venues with genuine insurance funds, which is a counterparty-selection decision rather than a market one.

*The general point, and it completes the chapter's argument.* Every mechanism in this chapter substitutes an economic incentive for an institutional guarantee: funding replaces delivery (question 6), automatic liquidation replaces a margin call (question 8), and auto-deleveraging replaces a default waterfall. Each substitution works in ordinary conditions and reallocates the failure in extreme ones—and in this last case it reallocates it onto the participants who were right.

*What the question tests.* That the waterfall protects winners' positions by using pre-funded capital, and that auto-deleveraging's exposure is to other clients' leverage rather than to anything the trader controls.

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# Chapter 38

# Value at Risk and Expected Shortfall

**38.1** *Using Example 38.1, state the 99% one-day VaR and ES for a \$100m book at 2% daily volatility, and explain why the ratio between them falls as confidence rises.*

*The two measures.* A \$100 million position at 2.0% daily volatility has a daily standard deviation of \$2.0 million. At 99% confidence, with $z = 2.3263$ and normal returns:

$$\text{VaR}_{99\%} = 2.3263 \times \$2.0\text{m} = \$\mathbf{4.65}\text{ million}$$

$$\text{ES}_{99\%} = \frac{\phi(z)}{1-c}\,\sigma = \frac{0.02665}{0.01} \times \$2.0\text{m} = \$\mathbf{5.33}\text{ million}$$

From Example 38.1:

| Confidence | $z$ | VaR | ES | ES/VaR |
|---|---|---|---|---|
| 95.0% | 1.6449 | \$3.29m | \$4.13m | 1.254 |
| 99.0% | 2.3263 | \$**4.65m** | \$**5.33m** | 1.146 |
| 99.9% | 3.0902 | \$6.18m | \$6.73m | 1.090 |

*Why the ratio falls as confidence rises.* Because the normal tail thins so fast that conditioning on being past a distant point adds little.

Expected shortfall is the *average* loss given that the VaR threshold is exceeded. How much larger that average is than the threshold depends on how much probability mass lies beyond it and how far out it extends. For a normal distribution the density decays as $e^{-x^2/2}$, faster than any exponential, so once past $2.5\sigma$ there is very little left, and the conditional mean sits close to the threshold. At 95% the ES exceeds the VaR by 25%; at 99.9%, by only 9%.

And that ratio is a property of the normal distribution rather than of the portfolio, which is exactly why it cannot be trusted. A desk that observes ES running 15% above VaR and treats the relationship as stable has learned a fact about a modeling assumption. On a fatter tail the ratio *widens* rather than narrowing, and question 8's table shows by how much: at 99.9% the fat-tailed ES is \$13.70m against a fat-tailed VaR of \$10.14m, a ratio of 1.35 where the normal gives 1.09.

So the ratio is a diagnostic. A book whose ES/VaR ratio behaves like the normal column is a book whose tail the model believes is normal; a ratio persistently above it is evidence of fat tails. The signature is deep in the tail: the normal's ratio falls toward one as confidence rises, while a fat tail's falls only slowly and settles well above it: question 8's book gives 1.35 at 99.9% against 1.09.

*What the question tests.* Computing both measures from the normal, and that their ratio measures the assumed tail shape rather than anything about the position.

**38.2** *Show that 99% VaR and 97.5% ES nearly coincide for a normal loss, and explain what the regulator was trying to achieve by choosing those two levels.*

*The near-coincidence.* For a normal loss distribution,

$$\mathrm{VaR}_{99\%} = 2.3263\,\sigma, \qquad \mathrm{ES}_{97.5\%} = \frac{\phi(1.9600)}{0.025}\,\sigma = 2.3378\,\sigma$$

$$\frac{2.3378}{2.3263} = \mathbf{1.0049}$$

The two are half a percent apart. Under normality the two requirements are, for practical purposes, identical.

*What the regulator was trying to achieve.* The confidence level was chosen to make the change roughly capital-neutral for a normal book, so that what moved was not the level of the requirement but its sensitivity to the shape of the tail.

That is a carefully engineered piece of policy and it is worth reading as such. Chapter 39's framework replaced 99% VaR with 97.5% ES, and had the regulator kept the confidence level at 99%, the switch would have raised requirements for everyone, which would have made the reform a capital increase presented as a methodological improvement, and would have been resisted as such.

By choosing 97.5%, the level was pinned to the old one for a book whose losses are normal. The result is:

- banks with normal books pay the same;
- banks with genuinely fat-tailed books pay more, because ES integrates the tail and VaR samples one point of it.

The redistribution is the entire content of the reform. No aggregate capital was raised or lowered by construction; capital was *reallocated* toward the portfolios whose tails are heavy. That is a much cleaner intervention than a level change, because it targets exactly the property the old measure could not see.

Three motivations lay behind the switch, all served by this calibration.

*Subadditivity.* VaR is not a coherent risk measure and ES is (questions 3 to 5), so ES can be used for desk limits and aggregated across a firm without the failure Example 38.2 exhibits.

*Tail sensitivity.* Question 8 shows VaR can *fall* when tails fatten, which is the wrong sign and would be reported with a straight face. ES rises at every level.

*And it removes an incentive.* Question 6: a desk measured on VaR is rewarded for selling far-out-of-the-money options, because the loss lies beyond the quantile. ES prices it.

*One cost remains, and question 23 addresses it.* ES cannot be backtested by counting exceptions the way VaR can, so the framework had to work around the loss of its most straightforward validation test.

*What the question tests.* The arithmetic of the coincidence, and reading a confidence-level choice as a deliberate capital-neutrality constraint that isolates the intended change.

**38.3** *State the subadditivity condition and explain why a measure that violates it cannot be used for desk limits.*

*The condition.* A risk measure $\rho$ is **subadditive** if for any two portfolios

$$\rho(A + B) \;\le\; \rho(A) + \rho(B)$$

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Combining two books cannot produce more risk than holding them separately. It is one of the four axioms of a coherent risk measure (alongside monotonicity, positive homogeneity, and translation invariance), and it is the one VaR fails.

*Why a measure that violates it cannot be used for desk limits.*

*1. The limits do not aggregate.* A firm sets a total limit and allocates it across desks, relying on the sum of the parts bounding the whole. If the measure is not subadditive, a set of desks each inside its limit can produce a firm-wide risk *exceeding* the total, so the allocation controls nothing. Question 4 gives the case: two positions each measuring zero combine to $10 million.

*2. It penalizes diversification.* Under a subadditive measure, merging books can only reduce measured risk, so the measure never argues against combining. Under VaR it can *rise*, which means the measure sometimes reports that diversifying made things worse, and a limit framework built on it will occasionally prohibit a genuinely risk-reducing trade.

*3. And it creates an arbitrage against the limit system.* This is the decisive one. If splitting a position across two desks reduces total measured risk, then the firm can manufacture limit capacity by reorganizing, with no change to what it owns. A trader facing a binding limit books half the position with a colleague and both are compliant. A control that can be defeated by an internal transfer is not a control.

*4. Risk allocation becomes incoherent.* Question 19 allocates total VaR across positions, and the allocation sums exactly—but only because VaR is *homogeneous*, not because it is subadditive. Where subadditivity fails, a position's allocated risk can exceed its standalone risk, which is not interpretable.

*Why the failure is not a corner case.* VaR fails whenever the loss distribution has a discrete jump near the quantile, which describes credit portfolios, insurance, digital payoffs, and anything with default risk. Question 4's example uses two ordinary bonds with a default probability *smaller* than the tail being measured, and nothing pathological is constructed.

*The remedy.* Expected shortfall is subadditive, always, for any distribution, because it is an average over a tail rather than a quantile, and averages combine linearly in a way quantiles do not. Question 5 verifies it on the failing example.

*What the question tests.* The axiom, and that its practical content is whether a limit framework can be defeated by relabeling positions.

---

**38.4** *Using Example 38.2, compute the 99% VaR of each bond alone and of the two together, and explain what feature of the loss distribution produces the failure.*

*Each bond alone.* Two independent corporate bonds, $10 million each, 0.8% default probability, zero recovery, measured at 99%.

A loss occurs with probability 0.008, which is *less* than the 1% tail being measured. So the cumulative probability of no loss is $1 - 0.008 = 0.992 \geq 0.99$, and the 99th percentile falls *inside* the no-default branch:

$$\text{VaR}_{99\%}(\text{one bond}) = \$\mathbf{0}$$

Measured at 99%, a bond that defaults with probability 0.8% carries no risk at all, because the quantile is reached before the default outcomes are encountered.

The condition is worth stating generally, because it determines when the failure occurs: $\text{VaR}_c$ of a single bond is zero exactly when $p \leq 1 - c$, the default probability at most the tail being measured.

*The two together.* With independence:

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| Loss | Probability | Cumulative |
|---|---|---|
| \$0m | 0.984064 | 0.984064 |
| \$10m | 0.015872 | 0.999936 |
| \$20m | 0.000064 | 1.000000 |

Since $0.984064 < 0.99$, the 99th percentile falls in the \$10m outcome:

$$\text{VaR}_{99\%}(\text{both}) = \$\mathbf{10}\text{m} \;>\; \$0 + \$0$$

Diversifying made the measured risk go from nothing to \$10 million.

And nothing pathological was constructed: two ordinary bonds, independent, with a default probability *smaller* than the tail being measured. This is not a contrived counterexample; it is what happens to any credit portfolio whose individual default probabilities sit below the tail probability, which is the ordinary case for investment-grade issuers measured at 99% or beyond.

*The feature that produces the failure: a discrete jump in the loss distribution near the quantile.*

VaR reads one point off the cumulative distribution. If the distribution is *continuous* near that point, small changes in the portfolio move the quantile smoothly. If it has an *atom* (a lump of probability at a single loss level), then the quantile can jump discontinuously as the portfolio changes, because adding a position shifts the cumulative distribution across the lump.

Here the atoms are at \$0, \$10m and \$20m. For one bond, 99.2% of the mass sits at \$0 and the 99th percentile is inside it. Adding the second bond redistributes mass so that only 98.41% sits at zero, and the 99th percentile falls through into the next atom. The measure moved because the quantile crossed a boundary, not because the risk changed.

So VaR fails whenever the loss distribution has a discrete jump near the quantile, which describes credit portfolios (default is a discrete event), insurance, digital options, and every structured product with a barrier. Question 6 shows the incentive this creates.

*What the question tests.* Constructing the joint loss distribution and locating the quantile in it, and identifying atoms near the threshold as the structural cause.

---

**38.5** *Compute the 99% expected shortfall for the same two bonds, and verify that it satisfies subadditivity.*

*Each bond alone.* Expected shortfall averages the worst $1 - c$ of outcomes rather than reading its boundary. For one bond the worst 1% consists of the 0.008 default outcomes at \$10m and, filling out the remaining $0.01 - 0.008 = 0.002$, part of the no-loss outcome:

$$\text{ES}_{99\%}(\text{one bond}) = \frac{0.008 \times 10 + 0.002 \times 0}{0.01} = \$\mathbf{8}\text{m}$$

Note the contrast with question 4 immediately. VaR reported \$0 and ES reports \$8m *on the identical position*. VaR read the quantile and found it inside the no-loss region; ES averaged the tail and found the default branch sitting in it at full weight.

*The pair.* The worst 1% runs from cumulative probability 0.99 to 1. It consists of the 0.000064 probability of both defaulting (\$20m) and the part of the \$10m atom above 0.99, $0.999936 - 0.99 = 0.009936$:

$$\begin{aligned}\text{ES}_{99\%}(\text{both}) &= \frac{(1 - 0.999936) \times 20 + (0.999936 - 0.99) \times 10}{0.01} \\ &= \frac{0.000064 \times 20 + 0.009936 \times 10}{0.01} = \frac{0.00128 + 0.09936}{0.01} = \$\mathbf{10.06}\text{m}\end{aligned}$$

*Verifying subadditivity.*

$$\text{ES}_{99\%}(A + B) = \$10.06\text{m} \;\le\; \text{ES}_{99\%}(A) + \text{ES}_{99\%}(B) = \$8\text{m} + \$8\text{m} = \$16\text{m} \quad \checkmark$$

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Comfortably satisfied, with a \$5.94m diversification benefit, which is the sensible answer, since two independent bonds *are* less risky together than the sum of their standalone risks.

*Why ES behaves where VaR does not.* Because it integrates the tail rather than sampling one point of it.

A quantile is a *boundary*, and boundaries can jump when a distribution has atoms (question 4). An average over a region is *continuous* in the distribution: shifting probability mass changes the average smoothly, because the average uses all the mass in the region rather than the location of its edge.

That is the whole of the mathematical difference, and it is why ES is subadditive *for every distribution* while VaR is guaranteed to be subadditive only for elliptical ones, a class that excludes essentially every credit or option portfolio.

*And the two answers on one bond are the useful comparison.* \$0 against \$8m on the same position is not a refinement; it is the difference between a measure that reports a possible total loss as no risk at all and one that carries it at its probability-weighted size. Question 6 shows that the gap is exploitable and that desks exploit it.

*What the question tests.* Computing ES as a probability-weighted tail average including a partial atom, and that continuity in the distribution is the property delivering subadditivity.

---

**38.6** *Explain why a desk measured on VaR has an incentive to sell far-out-of-the-money options, and what expected shortfall does to that incentive.*

*The incentive.* A desk measured on VaR is rewarded for selling options whose strikes lie beyond the confidence level, because VaR does not see losses past the quantile at all.

Consider a far-out-of-the-money option with a 0.5% chance of finishing in the money, measured at 99%. The loss occurs only in the worst 0.5% of outcomes, *strictly beyond* the 99th percentile, so:

- the premium is collected and appears as revenue;
- the loss scenario lies outside the region VaR examines;
- and the position's contribution to measured VaR is zero, or near it.

The desk books revenue against no measured risk, which is the definition of an exploitable metric.

Question 4's bonds are the same phenomenon in credit form: a 0.8% default probability measured at 99% reports zero, because the losses live past the quantile.

*And the incentive is strong precisely because it is invisible.* The desk is not concealing anything: the position is booked, marked, and reported. It simply does not register in the measure the desk is judged on. A risk manager reviewing VaR sees a book that has become more profitable without becoming riskier, which is the appearance of skill.

The exploitation shows up in three ways, all documented:

- selling deep out-of-the-money puts (small premiums collected against a large loss), and Chapter 22's short variance position is the institutional version;
- writing credit protection on remote tranches, which is Chapter 26's senior tranche and Chapter 24's question 9's ratio of risk-neutral to historical default probability at the safe end;
- and taking positions whose loss distribution is deliberately shaped to place the mass just beyond the threshold, which requires no more than knowing where the threshold is.

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*What expected shortfall does to the incentive: it prices the tail.*

ES averages *all* losses beyond the threshold, so a loss at the 99.5th percentile enters the 99%-ES calculation at its full magnitude. The far-out-of-the-money option now contributes to the measure roughly in proportion to how bad its loss is, not according to whether the loss happens to fall on one side of an arbitrary line.

So the arbitrage closes. There is no strike at which the position becomes invisible, because ES has no boundary past which it stops looking. There is only the confidence level, beyond which *everything* counts.

*Two qualifications keep the answer honest.* ES removes the *cliff* but not all incentive: a loss in the far tail still enters with small probability weight, so extremely remote risks are still underweighted relative to their severity. And ES is only as good as the assumed tail: question 8 shows a normal assumption reporting less than half the true ES at 99.9%, so a desk selling tails against a normal model is exploiting the *distribution* rather than the measure.

*The general principle, and Chapter 35's question 9 is its other instance.* A risk measure that a payoff can be shaped to satisfy is not measuring risk. There it was variance rewarding truncated upside; here it is a quantile ignoring everything beyond itself.

*What the question tests.* That a quantile creates a threshold beyond which losses are unpriced, and that averaging the tail removes the boundary rather than merely raising the number.

---

**38.7** *Explain what historical simulation assumes, and state the largest loss it is capable of reporting.*

*What historical simulation assumes.* It takes the actual returns of the last $N$ days (typically 250 or 500), applies each of them to today's portfolio, and reads the required quantile off the resulting distribution of hypothetical profits and losses.

The assumptions are three, and they are strong:

*1. Stationarity.* The last $N$ days are draws from the distribution now in force. If volatility, correlation or the market's structure has changed, the sample describes a regime that has passed. Chapter 36's question 12 makes the sharpest version of this point about allowances, where the elasticity generating the history is not the elasticity now in force.

*2. That the sample contains the relevant scenarios.* The method can only produce losses that *some historical day* would have caused. A risk that has not been realized in the window is assigned probability zero.

*3. And independence across days*, so that the empirical distribution is a valid estimate of the one-day distribution rather than of a path.

*What it buys, and why it remains the most used method.* It requires *no distributional assumption*: no normality, no covariance matrix, no copula. Fat tails, skewness, and the actual dependence structure between assets are all carried automatically, because they are in the data. And it handles nonlinear instruments correctly, since each historical scenario is applied to the full portfolio and revalued.

*The largest loss it is capable of reporting: the worst single day in the window.*

That is a hard ceiling, and it is stark. A 500-day window contains 500 outcomes, so:

- the measure cannot report a loss larger than the worst of them, *at any confidence level*;
- the 99.9% quantile of a 500-point sample is somewhere between the worst and second-worst observation, so it is estimated from *one or two* data points;
- and a scenario worse than anything in the window is, to the method, impossible.

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Volatility-scaled historical simulation (Section 38.5) relaxes the ceiling by rescaling each day to today's volatility, but plain historical simulation, equal-weighted or age-weighted, cannot exceed it.

The simulation after Example 38.7 quantifies how binding this is. Each day stays below the true 99.9% VaR of \$10.14m with probability 0.999, so a 500-day window's largest loss does with probability $0.999^{500}$: 61% of windows have a maximum below the number they are being asked to estimate, and 218 of the example's 400 simulated windows (55%) do. So in the majority of windows, historical simulation *cannot* produce the right answer, whatever quantile is requested, because the answer exceeds the largest number in the sample.

Which is the essential limitation. At 99% over 500 days the method is estimating the fifth-worst day, which is thin but defensible. At 99.9% it is estimating something beyond the sample's reach, and question 11 pursues what that means. The method is a good estimator of the body and cannot see the tail, and its apparent freedom from assumptions is exactly what prevents it from extrapolating past the data.

*What the question tests.* That the assumption is stationarity plus sample adequacy, and that the sample maximum is an absolute ceiling on the reportable loss.

---

**38.8** *Using Example 38.6, explain why the fat-tailed VaR at 95% is* lower *than the normal VaR, and why the expected shortfall is not.*

*The comparison.* From Example 38.6, taking the same \$100 million book and changing *nothing except the distribution*, which becomes Student's $t$ with four degrees of freedom, rescaled to the *identical* 2.0% daily standard deviation:

| | VaR | | ES | | |
|---|---|---|---|---|---|
| **Conf.** | **Normal** | **Fat** | **Normal** | **Fat** | **ES error** |
| 95.0% | \$3.29m | **\$3.01m** | \$4.13m | \$4.53m | +10% |
| 99.0% | \$4.65m | \$5.30m | \$5.33m | \$7.38m | +39% |
| 99.9% | \$6.18m | \$10.14m | \$6.73m | \$13.70m | +103% |

*Why the fat-tailed VaR is* lower *at 95%.* \$3.01m against \$3.29m, 8% less risk by that measure.

The reason is a constraint that is easy to overlook: the two distributions have the same variance. Variance is a fixed budget, so a distribution with the same variance and a heavier tail must be thinner somewhere: mass moved into the extremes has to come from somewhere, and it comes from the shoulders.

At 95% the quantile sits in exactly that thinner region. The $t_4$ density lies *below* the normal between about $0.7\sigma$ and $2.6\sigma$ and above it beyond $2.6\sigma$, and $z_{95\%} = 1.645$ falls in that stretch. So the fat-tailed 95% quantile is closer to the mean.

Fat tails do not raise VaR at every confidence level, and the crossover is somewhere between 95% and 99% for this pair. So a bank that fits a fat-tailed model and measures at 95% reports a risk reduction, which is the wrong sign, and would be reported with a straight face, because the model is better and the number is lower.

*Why expected shortfall does this far less.* It rises at every level in the table (10%, 39%, 103%) because it integrates the tail rather than sampling one point of it.

ES at confidence $c$ is the average loss over the worst $1 - c$ of outcomes, so it depends on *all* the mass beyond the threshold, and once the threshold is far enough out, the heavier tail has more of that mass further out. It is not immune to the shoulder effect: equal variance forces the fat tail to be thinner somewhere, and below about 89% the $t$ ES also falls under the normal

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one. But that crossover sits far below VaR's, near 97.5%, and below any level risk is actually measured at.

So at the levels that matter ES moves with tail weight and VaR need not. That is one of the three reasons for the regulatory switch in question 2.

*The practical instruction.* A VaR that falls when a model is improved is not evidence of lower risk, and a risk system reporting one should be interrogated rather than believed. The diagnostic is question 1's ES/VaR ratio: for a normal tail it *falls toward one* as confidence rises, while for a fat tail it stays well above one (1.35 against 1.09 at 99.9%), so comparing the two columns identifies the tail shape directly.

*What the question tests.* That equal variance forces a fatter tail to be thinner in the shoulders, and that integrating the tail makes ES track tail weight at high confidence where a quantile need not.

---

**38.9** *Compute, from Example 38.6, the ratio of the fat-tailed to the normal expected shortfall at 99.9% and the same ratio for VaR, and explain why the expected shortfall error matters more than the VaR error.*

*The understatement.* From Example 38.6, at 99.9%:

| | Normal | Fat-tailed |
|---|---|---|
| $ES_{99.9\%}$ | \$6.73m | \$13.70m |

$$\frac{13.698}{6.734} - 1 = \mathbf{+103}\%$$

The true expected shortfall is 103% above the normal figure; the normal assumption reports less than half of it.

The ratio is $13.70/6.73 = 2.03$. For VaR it is $10.14/6.18 = 1.64$, so the normal VaR falls 39% short of the true one, against 51% short for ES. Both are large, and the ES error is the larger.

*Why the ES error matters more than the VaR error.* Four reasons follow, in ascending order.

*1. It is the number in the capital framework.* Chapter 39 replaced 99% VaR with 97.5% ES (question 2), so the ES error *is* the capital error. A firm using a normal model holds capital against a loss it has understated by half.

*2. It is the number that describes what actually happens.* VaR answers "how bad is the threshold of a bad day?"; ES answers "how bad is a bad day, on average, when one arrives?" The second is the quantity a firm's survival depends on: a firm does not fail because a threshold was crossed but because of what was lost after it was.

*3. The error compounds with the severity.* Reading down question 8's ES error column (10%, 39%, 103%), the understatement *grows* with the confidence level. So the worse the scenario being described, the more wrong the normal assumption is. The error is largest exactly where the measure is being relied upon, which is the same structural complaint as Chapter 26's question 7 about tail dependence.

*4. And the VaR error is partly self-correcting while the ES error is not.* Question 8 showed VaR can fall when tails fatten, so at some confidence levels a normal VaR is *conservative*. A firm measuring across several levels will encounter both signs and may notice the inconsistency. The ES error is one-signed at every level above about 89%, which covers every level risk is measured at (always an understatement there), so it produces no internal contradiction to alert anyone.

*The general point.* A measure that integrates the tail inherits the full error in the tail's assumed shape. That is the price of ES's virtues: it is subadditive and tail-sensitive (questions 5

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and 8) precisely because it uses all the mass beyond the threshold, and using all the mass means depending on all of it being right.

Which makes the distributional assumption, rather than the measure, the binding problem. Questions 10 to 12 pursue it: the tail cannot be estimated from a short sample at all, extreme value theory converts an impossible estimate into a difficult one, and the most precise estimator is the most wrong.

*What the question tests.* The size of the understatement, and that ES's error is one-signed at the levels that matter, growing in severity, and directly the capital number.

---

**38.10** *Using the simulation that follows Example 38.7, compare the bias and the variability of the three estimators, and explain why the most precise one is the most wrong.*

*The three estimators.* From the simulation after Example 38.7, estimating the 99.9% VaR of the fat-tailed book, whose exact value is \$10.14m, from 500 days, repeated over 400 independent windows:

| Method | Mean VaR estimate | Bias | Spread across windows |
|---|---|---|---|
| Empirical quantile | \$9.22m | −9.1% | \$2.85m |
| Normal fit | \$6.19m | **−39.0%** | **\$0.70m** |
| Peaks over threshold | \$10.11m | −0.3% | \$3.32m |

Read the normal row against the others. It is by far the most precise, with estimates that vary by \$0.70m across windows against \$2.8m or more for the alternatives—and it is wrong by 39%.

Precision and accuracy point in opposite directions here, and the reason is structural. A normal fit estimates two parameters (mean and variance) from 500 observations, so both are estimated well and the resulting quantile is stable. But the quantile is computed by extrapolating a shape that is wrong, and no amount of data improves a misspecified shape. The estimator converges precisely on the wrong answer.

*The uncomfortable practical consequence.* A risk report showing a stable number month after month is displaying the property that a normal assumption is good at producing. Stability is exactly what a reviewer looks for as evidence of a well-behaved model, and here it is evidence of the opposite: the two estimators that move around are the two that are closer to right.

So the usual diagnostic is inverted. A jumpy tail estimate is a sign that the estimator is responding to the tail, which is thin and therefore noisy; a stable one is a sign that it is not looking at the tail at all.

The peaks-over-threshold estimator cuts the bias to 0.3%, inside the \$0.17m sampling error of a 400-window average, and pays for it in variability (\$3.32m against the normal's \$0.70m). On expected shortfall the median window gives \$11.48m against a true \$13.70m, 16% low, while the average over the windows is \$15.19m, 11% high, pulled up by the few windows whose fitted shape $\hat{\xi}$ comes out high enough for the divisor $1 - \hat{\xi}$ to matter. That is the honest trade: a nearly unbiased estimate with wide confidence bounds, against a tight estimate of the wrong quantity.

And the empirical quantile sits between them, biased down 9.1% because a 500-point sample's 99.9th percentile is estimated from its worst one or two observations, which is question 11's subject.

*The general lesson, and it generalizes past risk measurement.* A model's stability is not evidence of its validity, and can be evidence against it. A model with few parameters and a wrong functional form produces consistent answers; the consistency comes from the constraint, not from the data. Chapter 31's question 10 makes the same point about OAS: the Z-spread is

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stable and answers the wrong question, the OAS is unstable and answers the right one.

*What the question tests.* Reading a bias–variance table, and that the estimator a reviewer would prefer on stability grounds is the one that is most wrong.

---

**38.11** *Compute the probability that a 500-day window's largest loss falls below the true 99.9% VaR, and explain what this means for historical simulation at that confidence.*

*The probability.* Against a true 99.9% VaR of \$10.14m, each day's loss stays below it with probability 0.999. The days are independent, so a 500-day window's largest loss stays below it with probability

$$0.999^{500} = e^{500 \ln 0.999} = e^{-0.5003} = \mathbf{0.606},$$

so **61% of windows have a maximum below the number they are being asked to estimate.**

*What this means for historical simulation at that confidence.* The method cannot produce the right answer in the majority of windows, because the answer exceeds the largest number in the sample.

Question 7 established that historical simulation's ceiling is the worst single day in the window. So in 61% of 500-day windows the correct 99.9% VaR is larger than any loss the method is capable of reporting, at any quantile, with any interpolation, however the percentile is computed. The constraint is not statistical imprecision; it is arithmetic impossibility.

*And the remaining 39% are no better in a useful sense.* There, the estimate is driven entirely by the single worst observation, so the reported number is a function of one data point. It will move sharply when that observation rolls out of the window, producing a step change in reported risk with no change in the portfolio or the market.

Three consequences follow.

*The estimate is biased downward, systematically.* Question 10's empirical-quantile row shows the bias: −9.1% on average. Since the sample maximum is a lower bound on any quantile beyond it, the error can only run one way.

*The reported number is unstable in a way that looks like risk changing.* A large loss entering the window raises reported VaR; the same loss leaving it 500 days later lowers reported VaR by the same amount, with no economic event. The risk report moves on the calendar, and Chapter 39's stressed measures exist partly to address this.

*And the confidence level is misrepresented.* A firm reporting "99.9% VaR" from a 500-day window is reporting a quantity the sample does not identify. The honest description is "approximately the worst day in the window," which is a different statement with a different meaning.

*What the number of observations would have to be.* Estimating a 99.9% quantile with reasonable precision requires the tail region to contain enough observations to average over, so roughly $10/0.001 = 10{,}000$ days, or forty years, over which stationarity (question 7's first assumption) is indefensible. The sample cannot be lengthened out of the problem.

Which is why extreme value theory exists, and question 12 explains what it does and does not solve. It extrapolates a fitted tail shape beyond the sample, converting a quantity that cannot be estimated into one that can be estimated badly.

*What the question tests.* That the sample maximum binds in a majority of windows, and that the resulting failure is one of identification rather than of precision.

---

**38.12** *Give the two reasons extreme value theory converts an impossible estimate into a difficult one rather than into a known answer.*

*What extreme value theory does.* It fits a *parametric tail* to the observations exceeding a

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high threshold (a generalized Pareto distribution, fitted by the peaks-over-threshold method) and then *extrapolates* that fitted shape beyond the data. Because the fit uses the shape of the whole tail region rather than one or two order statistics, it can produce estimates *past* the sample maximum, which question 11 showed historical simulation cannot.

Question 10 measures the improvement: bias falls from −9.1% (empirical) to −0.3%, and on expected shortfall the median window is 16% low and the average 11% high. That is a real gain and it is not a solution.

It remains difficult for two reasons.

*1. The tail index must be estimated from the tail, which is where the data are scarcest.* The whole method rests on a shape parameter governing how fast the tail decays, and that parameter is identified only by the observations *above* the threshold. In a 500-day window with a threshold at the 95th percentile, that is 25 observations—and the estimate must then be extrapolated to a quantile implied by one observation in a thousand.

The consequence is visible in question 10's spread column: \$3.32m across windows, *wider* than the empirical quantile's \$2.85m. The method trades bias for variance, and the variance is substantial. A single estimate carries wide confidence bounds that are rarely reported alongside it.

*2. The threshold choice is a free parameter with no clean answer.* Set it too high and there are too few exceedances to fit; too low and the observations included are not from the tail, so the asymptotic theory justifying the Pareto form does not apply. The results are sensitive to the choice, and the standard diagnostics (mean excess plots, stability of the shape estimate across thresholds) are themselves noisy on a few hundred points.

So the analyst chooses a number that materially affects the answer, using diagnostics that do not clearly determine it. Chapter 31's question 8 names this pattern precisely: a parameter the data cannot identify is being asserted rather than measured, and letting a procedure assert it merely hides who did.

*A third limitation is worth adding.* Extreme value theory's asymptotic results describe the tail of a *fixed* distribution. A market whose volatility and dependence structure change, which is question 7's stationarity problem and question 15's correlation problem, does not have one tail to estimate. The theory is exactly right about a situation that does not obtain.

So the honest summary is this. Extreme value theory converts an impossible estimate into a difficult one. Before it, the 99.9% quantile of a 500-day sample is not identified at all (question 11's 61%). After it, the quantity is estimable: averaged over windows its VaR is nearly unbiased (expected shortfall, at 11% to 16%, much less so), but a single window's estimate has a standard error of 33%, conditional on a threshold choice the analyst makes and a stationarity assumption that is false.

Which is a genuine improvement and should be reported as what it is: an estimate with wide bounds, not a number. Question 10's lesson applies: the stable estimate was the wrong one, and a method that honestly reports its own variability is preferable to one that conceals it.

*What the question tests.* That the shape parameter and the threshold are both estimated from the scarcest data, and that the improvement is in identifiability rather than in precision.

---

**38.13** *Using Example 38.8, compute the long-run volatility implied by* $\omega = 0.000002$, $\alpha = 0.06$, $\beta = 0.93$, *and state the one-year forecast under each estimator.*

*The long-run volatility.* A GARCH(1,1) model $\sigma_n^2 = \omega + \alpha u_{n-1}^2 + \beta\sigma_{n-1}^2$ has unconditional variance

$$V_L = \frac{\omega}{1-\alpha-\beta} = \frac{0.000002}{1-0.06-0.93} = \frac{0.000002}{0.01} = 0.0002$$

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$\sigma_L = \sqrt{0.0002} = \mathbf{1.4142}\%$ daily or $1.4142\% \times \sqrt{252} = \mathbf{22.4}\%$ annualized

*The one-year forecast under each estimator.* From Example 38.8, a market at 1.00% daily volatility takes a −3% day. Both estimators react almost identically: EWMA at $\lambda = 0.94$ moves to 1.2166%, GARCH to 1.2207%, a rise of about 22% from a single day. They disagree completely about what happens next:

| Horizon | GARCH average to horizon | EWMA |
|---|---|---|
| 1 day | 19.4% | 19.3% |
| 10 days | 19.5% | 19.3% |
| 3 months | 20.2% | 19.3% |
| 1 year | **21.4**% | **19.3**% |

GARCH forecasts 21.4% and EWMA 19.3%, both annualized.

*Why they diverge.* EWMA's forecast is flat at every horizon: its best guess for volatility in a year is today's estimate. That is a structural property: EWMA is the special case of GARCH with $\omega = 0$ and $\alpha + \beta = 1$, so it has *no* long-run level to revert to and its forecast is a random walk.

GARCH has $\alpha + \beta = 0.99 < 1$, so it *mean-reverts* toward $V_L$. The current estimate of 1.2207% daily (19.4% annualized) sits below the long-run 1.4142% (22.4%), so the forecast *climbs* toward it, though slowly, since the reversion rate $1 - \alpha - \beta = 0.01$ implies a half-life of about 69 days.

Note that $\alpha + \beta = 0.99$ is typical and it is very close to one. Volatility is extremely persistent, so the reversion is slow enough that the two estimators agree at horizons out to a month, and separate only over quarters.

Which is exactly question 14's point. The two are fit for different jobs: a one-day margin calculation cannot distinguish them, and anything with a term structure requires the mean reversion.

*One caution applies to the long-run figure.* $V_L$ is extremely sensitive to $\alpha + \beta$ because the denominator is small: at $\alpha + \beta = 0.99$ it is $\omega/0.01$, and at 0.995 it would be $\omega/0.005$—*double*. So a small estimation error in the persistence produces a large error in the long-run level, and the annualized 22.4% should be treated as an order of magnitude rather than a figure. That is the standard fragility of GARCH's most-quoted output.

*What the question tests.* The unconditional-variance formula, and that the estimators' agreement at one day and disagreement at one year is the presence or absence of a reversion level.

---

**38.14** *Explain why EWMA is adequate for a one-day margin calculation and inadequate for a volatility term structure.*

*Why EWMA is adequate for a one-day margin calculation.* Because at a one-day horizon the two estimators are indistinguishable, and EWMA is simpler.

Question 13's example makes it concrete: after a −3% day, EWMA moves to 1.2166% and GARCH to 1.2207%, a difference of four thousandths of a percentage point, far inside any margin calculation's tolerance. Both react at essentially the same speed, because the reaction is governed by $\alpha$ (or by $1 - \lambda$), and both are calibrated to react similarly.

The advantages that then decide it:

- *One parameter.* $\lambda$, conventionally 0.94, against GARCH's three, so there is nothing to estimate, nothing to re-estimate, and no risk of a fitted parameter drifting.
- *A trivial update rule.* $\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1 - \lambda)u_{n-1}^2$ requires only the previous estimate and the latest return, so a whole correlation matrix updates in one pass with no optimization.

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- *And it is transparent.* A margin methodology must be explicable to a counterparty and stable under audit, and an exponentially weighted average is both.

For a quantity that will be recomputed tomorrow anyway, sophistication about the long run buys nothing.

*Why it is inadequate for a volatility term structure.* Because EWMA has no long run.

It is GARCH with $\omega = 0$ and $\alpha + \beta = 1$, so its unconditional variance is undefined and its multi-period forecast is *flat*: today's estimate, at every horizon. Question 13's table shows it: 19.3% at one day, ten days, three months and one year alike.

That is a statement no market believes. After a volatile day, EWMA forecasts that elevated volatility persists forever; after a calm one, that calm persists forever. The forecast is a random walk, so the model asserts that volatility has no tendency to return anywhere.

*And the market's own prices contradict it directly.* Chapter 19's volatility term structure slopes upward in calm markets and inverts in stress because volatility mean-reverts, and Chapter 19's question 10 gives the mechanism. A model with no reversion cannot produce either shape: it predicts a flat term structure always, which is the one shape never observed.

Three specific failures follow. It cannot price options at different maturities consistently, since every expiration gets the same volatility. It cannot produce Chapter 22's VIX futures curve, whose entire character is mean reversion. And it cannot support a stress projection over any horizon longer than a few days, since it will simply extend today's conditions.

*The general principle.* Choose a model by the horizon of the question. A one-day question needs responsiveness and nothing else; a term-structure question needs a reversion level, which is the one thing EWMA has thrown away in exchange for its simplicity. Chapter 32's question 16 states the same rule for interest-rate models (a model should be chosen by which of its errors you can afford), and EWMA's error is entirely at long horizons.

*What the question tests.* That the two estimators differ only in the reversion term, and that the term is irrelevant at one day and decisive across maturities.

---

**38.15** *Using (38.4), recompute the VaR of Example 38.9's book at correlations of 0.30 and 0.90, and explain why the estimation error has a consistent sign.*

*The effect on VaR.* In Example 38.9, position A is \$60m at 1.0% daily volatility and position B \$40m at 2.5%, so their daily dollar standard deviations are \$0.6m and \$1.0m. The square root in (38.4) is the portfolio's standard deviation:

$$\rho = 0.30: \quad \sigma_P = \sqrt{0.6^2 + 1.0^2 + 2 \times 0.30 \times 0.6 \times 1.0} = \sqrt{1.72} = \$1.3115\text{m},$$
$$\text{VaR} = 2.3263 \times 1.3115 = \$3.051\text{m};$$

$$\rho = 0.90: \quad \sigma_P = \sqrt{0.6^2 + 1.0^2 + 2 \times 0.90 \times 0.6 \times 1.0} = \sqrt{2.44} = \$1.5620\text{m},$$
$$\text{VaR} = 2.3263 \times 1.5620 = \$3.634\text{m}.$$

The example's table carries the same calculation across the range:

| Correlation | VaR | Change |
|---|---|---|
| −0.50 | \$2.028m | −34% |
| 0.30 | \$3.051m | — |
| 0.90 | **\$3.634m** | **+19%** |
| 1.00 | \$3.722m | +22% |

Moving from 0.30 to 0.90 raises reported risk by **19%**—with no change in any position and no change in any volatility.

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*Why the estimation error has a consistent sign.* Because correlations move most in exactly the episodes the measure exists to describe, and they move in one direction.

The mechanism is worth stating precisely, because it is not merely that correlations are unstable. Assets that are unrelated in normal conditions become one asset in a liquidation, and the reason is not economic: the common factor in a crisis is that the same leveraged holders are selling all of it. A forced seller does not sell the assets that are economically linked to their problem; they sell what they can, which is everything.

The consequence follows in three steps:

- the input is *estimated* in calm periods, when the sample is dominated by ordinary days on which idiosyncratic factors dominate;
- it is *applied* in violent ones, when a single deleveraging factor dominates;
- and the calm-period estimate is therefore *too low*.

For a book long both assets, as this one is, the error has a consistent sign: measured risk is too low. There is no compensating regime in which correlations fall below their estimate at exactly the moment losses arrive. (A hedged book is the exception: there a rise in correlation lowers measured risk, and the danger runs the other way, through a hedge correlation that breaks down.)

*And the diversification benefit is the quantity destroyed.* A portfolio VaR sits below the sum of its standalone VaRs by an amount that is entirely a function of correlation; question 19's table shows an 18% benefit at $\rho = 0.30$. That benefit is what evaporates in a crisis, so a firm that has sized positions against a diversified VaR discovers that the diversification was the part that failed.

This is not a call for conservative correlations. Simply raising every correlation would overstate risk in ordinary conditions, misallocate capital, and penalize genuine diversification. The problem is not the level of the estimate but that one number is being asked to describe two regimes.

*It is a reason the framework does not let a bank choose them freely.* Chapter 39's constrained diversification rules (prescribed correlation scenarios, floors on aggregation, and limits on the recognized benefit) exist precisely because the input is estimated by the firm, has a signed error, and directly reduces the firm's own capital requirement.

*And question 16 shows the deeper version.* Correlation is only the second moment; what matters in a crisis is *tail dependence*, and a Gaussian model has none at any correlation. So even a correctly estimated correlation understates joint extremes if the dependence structure is wrong.

*What the question tests.* That correlation moves VaR materially with nothing else changing, and that the estimation error is signed because the estimating regime and the applying regime differ systematically.

---

**38.16** *Using Example 38.10, compute the probability that the second exposure is beyond its 99.9th percentile given that the first is, under each copula, and compare the two with the coefficient of tail dependence.*

*The conditional probabilities.* The first exposure is beyond its 99.9th percentile with probability 0.001, so the conditional probability is the joint probability divided by 0.001:

$$\text{Gaussian: } \frac{0.00005426}{0.001} = \mathbf{5.4}\%, \qquad t_4\text{: } \frac{0.0002635}{0.001} = \mathbf{26.3}\%.$$

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The coefficient of upper tail dependence is the limit of this conditional probability as the percentile goes to 100. Under the $t_4$ copula the 26.3% is already close to its limit of 25.3%; under the Gaussian copula the 5.4% is still on its way to a limit of zero, and it falls further at every more extreme percentile.

*The comparison.* From Example 38.10, two exposures each normally distributed with correlation 0.50, differing *only* in the copula (Gaussian against Student's $t$ with four degrees of freedom):

| Both beyond | Gaussian | $t_4$ | Ratio |
|---|---|---|---|
| 95th percentile | 0.01219 | 0.01694 | 1.39× |
| 99th percentile | 0.001294 | 0.002877 | 2.22× |
| 99.9th percentile | 0.00005426 | 0.0002635 | **4.86**× |

The ratio of the two conditional probabilities is the ratio of the joint ones: a joint 99.9th-percentile event is **4.86 times** more likely under the $t_4$ copula—with *identical margins and identical correlation*. (The tabulated probabilities are rounded; the ratio is computed from the unrounded values.)

The discrepancy grows as the event gets worse (1.39, 2.22, 4.86), which is the opposite of what a modeler wants from an approximation. A good approximation degrades gracefully in the region where it is used least; this one degrades fastest exactly where the risk measure is looking.

*The limiting statement is the cleanest.* The coefficient of upper tail dependence, the limiting probability that one exposure is extreme *given* that the other is, takes the values

$$\lambda_{\text{Gaussian}} = 0 \quad \text{at every correlation below one,} \qquad \lambda_{t_4} = 0.2532$$

Under a Gaussian copula, joint catastrophes have probability zero in the limit no matter how correlated the assets are. Correlation controls the body of the distribution and dies in the tail.

So correlation and tail dependence are different quantities, and a risk system that specifies only the first has said nothing about the second. Question 15's warning, that correlations are estimated in calm periods and applied in violent ones, is about getting a number wrong; this is about the number being insufficient however well it is estimated.

Chapter 26's question 7 made this point about CDO tranches, where it destroyed a market. There a senior tranche exists entirely in the region where Gaussian dependence has been switched off, so the model made senior tranches look safe for a structural reason no choice of $\rho$ corrected.

Here it decides something more basic: whether a portfolio risk measure can see simultaneous extremes at all. A firm's VaR or ES aggregates exposures across desks, currencies and asset classes, and the aggregation is a dependence assumption. If that assumption has zero tail dependence, the measure cannot report the scenario in which everything goes wrong together—which is the only scenario that threatens a firm.

The practical response is to use a copula with tail dependence (a $t$ copula is the standard choice and costs one extra parameter) and to test the sensitivity, since the degrees-of-freedom parameter is as hard to estimate as question 12's tail index and for the same reason.

*What the question tests.* That margins and correlation do not determine joint extremes, and that the Gaussian copula's zero tail dependence is a structural property rather than a calibration choice.

---

**38.17** *Using Example 38.11, explain why three factors keep 99% of the directional book's variance and 10% of the butterfly's, and state what a risk system must report as a result.*

*The factor structure.* From Example 38.11, nine tenors from three months to thirty years:

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| Factor | Variance | Cumulative | Shape |
|---|---|---|---|
| Level | 90.98% | 90.98% | all tenors move together |
| Slope | 3.94% | 94.92% | short end against long end |
| Curvature | 1.96% | **96.88**% | wings against the belly |

Three factors explain 96.88% of the variance *of the curve*, which is the number usually quoted and the reason three-factor models are standard.

*Why the two books give different verdicts.* Because a factor decomposition explains variance of the *curve*, and a portfolio's risk depends on which part of that variance its sensitivities touch.

*The directional book* has sensitivities scattered across the curve and is net long, so it is exposed principally to a parallel shift, which is the level factor, carrying 90.98% of the variance on its own. The level factor alone keeps 98.7% of the book's variance and three factors keep **99.2%**, because the factors retained are precisely the ones the book is exposed to.

*The butterfly* (long \$20,000 per basis point at two and five years, short \$40,000 at three) is constructed to be *nearly immune* to the level factor. A parallel shift moves the long and short legs by offsetting amounts and the position does not move at all; the level factor is close to a parallel shift but not exactly one, which is why one factor still keeps 3.0% of the butterfly's variance. It is nearly immune to slope as well, which adds under one percentage point of its variance.

So the butterfly's entire risk lives in the residual: the higher-order, tenor-specific movements that the first three factors discard. Three factors keep only **10.5%** of its variance, which in VaR terms is \$38,194 of \$117,925, and the discarded 3.12% of curve variance is essentially all of this book's exposure.

*The general statement.* A factor model's adequacy is a property of the portfolio, not of the market. The same decomposition, applied to the same curve, on the same day, is excellent for one book and useless for another. Any position constructed to be neutral to the retained factors has its risk concentrated in the discarded ones, and spread positions, relative-value trades and basis books are constructed that way deliberately.

And that is the uncomfortable part: factor reduction is *most* inaccurate for exactly the books that are designed to be factor-neutral, which are the books whose risk is hardest to see by inspection.

What a risk system must report as a result: the share of each book's risk that the factor model captures, alongside the VaR itself.

A single firm-wide statement ("we use three factors, explaining 97% of curve variance") is inadequate, because it describes the market rather than the positions. The system must report, per book:

- the VaR under the reduced model *and* under the full covariance matrix;
- the ratio between them, which is the coverage;
- and a flag where the coverage is low.

A book like the butterfly, at 10% coverage, should be measured on the full matrix, and the reduction should be understood as a computational convenience that is available for some portfolios and not others.

This is Chapter 32's question 9 in a risk-management setting: reducing a correlation matrix to $K$ factors discards structure, and what is discarded is precisely what the correlation-sensitive products depend on.

*What the question tests.* That factor coverage depends on the alignment between retained factors and portfolio sensitivities, and that the required disclosure is per-book coverage rather than an aggregate variance share.

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**38.18** *State the square-root-of-time rule, the assumptions it requires, and the direction of its error in a trending market.*

*The rule.* A $T$-day VaR is obtained from a one-day VaR by scaling with the square root of the horizon:

$$\mathrm{VaR}_T = \mathrm{VaR}_1 \times \sqrt{T}$$

so a ten-day requirement is $\sqrt{10} = 3.16$ times the one-day figure.

*The assumptions it requires: that returns are independent and identically distributed across days, with mean zero.*

Independence is what makes variances add. Over $T$ days,

$$\mathrm{Var}\left(\sum_{t=1}^{T} r_t\right) = \sum_{t} \mathrm{Var}(r_t) + 2\sum_{t<s} \mathrm{Cov}(r_t, r_s)$$

and the rule drops the covariance terms entirely. A nonzero mean would also break it, since the mean grows with $T$ while the standard deviation grows with $\sqrt{T}$. It also requires the distributional shape to be preserved under aggregation, so that the same quantile multiple applies, which is true for the normal and not true in general.

*The direction of the error in a trending market.* It understates the risk.

A trending market has *positively autocorrelated* returns: a down day is more likely to be followed by another down day. The covariance terms are then positive, so the true $T$-day variance exceeds the sum of the daily variances, and $\sqrt{T}$ scaling is too small.

The economics is straightforward and it is why trends occur in the relevant episodes: a liquidation takes days. A forced seller does not complete in one session, and the selling that pushes the price down on Monday continues on Tuesday. So the periods in which the ten-day figure matters are precisely the periods in which returns are most positively autocorrelated.

*And the error compounds with question 15's.* Correlations rise in the same episodes, so the cross-sectional aggregation understates *and* the temporal aggregation understates, in the same scenario, for related reasons: the same leveraged sellers driving both.

*The opposite case, for completeness.* A *mean-reverting* market has negatively autocorrelated returns, so $\sqrt{T}$ scaling *overstates* the risk. That describes ordinary conditions in many markets, which is why the rule appears adequate in backtests: it is conservative in the sample and optimistic in the crisis.

Two further reasons make the rule fail at longer horizons. The portfolio does not stay constant: over ten days a desk trades, so a ten-day VaR on today's position describes a position that will not be held. And the distributional shape changes with aggregation: by the central limit theorem a ten-day return is closer to normal than a one-day return, so a fat-tailed one-day distribution scaled by $\sqrt{10}$ overstates the ten-day tail, an error in the *opposite* direction to the autocorrelation effect, and not generally of the same size.

*The practical consequence.* The rule is a convenience that avoids estimating the $T$-day distribution directly, and its error is small in calm markets and signed against the firm in stressed ones. Chapter 39's liquidity horizons, which scale different risk factors by different periods rather than applying one $\sqrt{T}$, are partly a response to this.

*What the question tests.* That $\sqrt{T}$ scaling drops the autocovariance terms, and that positive autocorrelation in a liquidation makes the omission signed against the firm.

**38.19** *Using Example 38.12, state each position's share of the money and share of the risk, and explain what a total-VaR limit fails to notice.*

*The two shares.* From Example 38.12, on the book of question 15 (A is \$60m at 1.0%

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volatility, B is \$40m at 2.5%, correlation 0.30):

| | Position | Standalone VaR | Component VaR | Share of risk |
|---|---|---|---|---|
| A | \$60m (**60%**) | \$1.396m | \$0.958m | **31.4%** |
| B | \$40m (**40%**) | \$2.326m | \$2.093m | **68.6%** |
| Total | \$100m | \$3.722m | \$3.051m | 100% |

B is 40% of the money and 69% of the risk.

*And the standalone numbers do not show this.* They sum to \$3.722m against a portfolio VaR of \$3.051m, an 18.0% diversification benefit that has to be allocated somehow, and component VaR allocates it in the only way that adds up.

*What a total-VaR limit fails to notice: the composition.*

A limit on total VaR constrains one number. It says nothing about how that number is distributed across positions, and the marginal figures show why that matters. At the margin, a dollar moved into B adds 0.0523 of VaR against A's 0.0160, so B is more than three times as risk-intensive per dollar.

So consider shifting \$10m from A to B:

- the book's *size* is unchanged at \$100m;
- the total VaR rises by **12.8%**;
- and if the limit is not currently binding, nothing prevents it.

A limit on total VaR permits that trade and a limit on component VaR notices it.

Three consequences follow.

*Concentration is invisible.* A book can drift toward its most risk-intensive position while staying inside a total limit, and the drift is only detected when the limit binds, at which point the concentration is already established.

*Capital and revenue are attributed wrongly.* A desk earning a return on \$60m of position A and one earning on \$40m of B look like a 60/40 split of the firm's balance sheet; they are a 31/69 split of its risk. Return-on-risk comparisons made on notional are meaningless.

*And hedging decisions are misdirected.* A manager told to reduce VaR by 20% will act on the larger position if shown notionals, and on the correct one if shown components.

*One caution, and it is the same one as everywhere in this chapter.* Component VaR sums exactly only because VaR is homogeneous (scaling the whole book by $\lambda$ scales VaR by $\lambda$), and Euler's theorem then guarantees the components add to the total. It does not require VaR to be subadditive, and question 4 showed VaR is not. So the allocation is exact and the quantity being allocated may be unreliable, which is question 20's subject.

*What the question tests.* Reading component against standalone contributions, and that a total limit constrains a scalar while risk concentration is a property of the vector.

---

**38.20** *Explain how an allocation can be exact while the total it allocates is unreliable.*

*The apparent paradox.* Question 19's component VaRs sum *exactly* to the portfolio VaR: \$0.958m + \$2.093m = \$3.051m, to the last digit. That exactness looks like evidence of a well-founded calculation. It is not evidence about the total at all.

*Why the allocation is exact.* Because VaR is **positively homogeneous** of degree one: scaling every position by $\lambda$ scales VaR by $\lambda$. Euler's theorem for homogeneous functions then gives

$$\mathrm{VaR}(w) = \sum_i w_i \frac{\partial \mathrm{VaR}}{\partial w_i}$$

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and the component VaRs are exactly those terms. The identity holds for any homogeneous function whatever, regardless of whether that function is a good measure of anything.

So the exactness is a property of the *form* of the measure, not of its *content*. It would hold equally if VaR were computed from a wrong covariance matrix, a wrong distributional assumption, or a wrong confidence level; the parts would still sum to the whole.

*And the total is unreliable for reasons the allocation cannot detect.* This chapter has supplied several, and each leaves the allocation untouched:

- *the correlation input* (question 15), estimated in calm periods and applied in violent ones, with a signed error;
- *the tail assumption* (questions 8 and 9), where a normal model reports less than half the true ES at 99.9%;
- *the dependence structure* (question 16), where a Gaussian copula assigns zero probability to joint catastrophes;
- *the estimation of the quantile itself* (question 11), which a 500-day window cannot identify at 99.9%;
- *and subadditivity* (question 4), which VaR does not satisfy, so a position's allocated risk can in principle exceed its standalone risk, and the sum still adds up.

The right reading of an exact allocation, therefore: it answers "given this total, how is it distributed?" and it answers that question correctly. It does not answer "is this total right?"—and no property of the allocation bears on it.

Which makes component VaR genuinely useful for the purpose it serves. Question 19's concentration finding (40% of the money and 69% of the risk) is a *relative* statement, and relative statements are far more robust to a common error in the total than absolute ones are. If the covariance matrix understates everything by a third, the shares are roughly unaffected while the total is badly wrong.

*The general principle worth extracting.* An identity that holds by construction carries no information about the inputs. Chapter 19's question 8 is the sharpest instance in this book: an option chain's butterflies and digitals sum to exactly 100% for *any* six numbers, arbitrage-free or not, because the middle terms telescope. A check that cannot fail on bad input is a check on the implementation, not on the data, and the discipline is to know which kind you are looking at before relying on it.

*What the question tests.* That Euler's theorem delivers the exactness from homogeneity alone, and distinguishing a structural identity from evidence about the quantity it decomposes.

---

**38.21** *Using Example 38.13, state the range of exception counts consistent with a correct model at 99% over 250 days, and explain why zero exceptions is also a rejection.*

*The range.* From Example 38.13, at 99% over 250 days the expected number of exceptions is 2.5, and the Kupiec likelihood-ratio test rejects when the statistic exceeds 3.841:

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| Exceptions | Probability | Kupiec LR | Reject? | Basel zone |
|---|---|---|---|---|
| 0 | 0.0811 | 5.03 | **yes** | green |
| 1 | 0.2047 | 1.18 | no | green |
| 2 | 0.2574 | 0.11 | no | green |
| 3 | 0.2149 | 0.09 | no | green |
| 4 | 0.1341 | 0.77 | no | green |
| 5 | 0.0666 | 1.96 | no | yellow |
| 7 | 0.0097 | 5.50 | **yes** | yellow |
| 10 | 0.0002 | 12.96 | **yes** | red |

Anything from **one to six** exceptions is statistically indistinguishable from a correct model.

Read what that means for the test's power. The accepted counts are exception rates from 0.4% to 2.4%: a year whose exceptions run at 97.6% coverage and one at 99.6% both pass as 99%. The acceptance region spans a factor of six in the exception count, so a backtest at this sample size can detect only gross misspecification.

*Why zero exceptions is also a rejection.* It is the row people forget, and it follows directly from the arithmetic: the probability of zero exceptions under a correct model is 0.0811, and the Kupiec statistic is 5.03, above the 3.841 threshold.

A model that never breaches is not conservative but miscalibrated, in the direction that wastes capital. A firm reporting a 99% VaR that has not been exceeded in a year is reporting a number that is too large: it is holding capital against losses that do not occur, at the cost of the business it could otherwise do.

*The test is two-sided because a risk measure is a forecast, not a promise.* That sentence is the whole justification. A VaR is a statement that losses will exceed the threshold on 1% of days, not a statement that they will never exceed it. A forecast that is systematically too high is as much a failure of forecasting as one that is too low, and only the *consequences* differ: too low threatens solvency, too high wastes capital.

Two practical consequences follow.

*Backtesting cannot validate a model at high confidence.* At 99.9%, the expected exceptions over 250 days is 0.25, so observing none is the overwhelmingly likely outcome under both a correct and a badly wrong model. The test has essentially no power there, which is why regulatory backtesting is done at 99% even where capital is set at another level.

*And the sample cannot be enlarged out of the problem.* Distinguishing 99% from 98.5% with reasonable power requires several years of data, over which the portfolio and the market have both changed, so the longer sample tests a model that no longer exists. This is question 7's stationarity problem arriving in the validation step.

*What the question tests.* Reading an acceptance region off a likelihood-ratio table, and that a two-sided test follows from a risk measure being a forecast with costs on both sides.

---

**38.22** *Explain where the Basel zones and the Kupiec test disagree, and what the supervisor gains by setting the zones where they are.*

*Where they disagree.* Question 21's table sets the two side by side:

| Exceptions | Kupiec | Basel zone | |
|---|---|---|---|
| 0 | **reject** | green | Basel is lenient |
| 5 | do not reject | **yellow** | Basel is strict |
| 6 | do not reject | **yellow** | Basel is strict |
| 7 | **reject** | yellow | Basel stops short of red |

Five exceptions puts a bank in the yellow zone and raises its capital multiplier, while the Kupiec

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statistic of 1.96 is comfortably inside the acceptance region. At seven the test rejects, yet the bank is still yellow; red, which treats the model as failed, begins only at ten. The zones are a graded schedule rather than a single cut, so they disagree with the test in both directions.

*What the supervisor gains by setting them there.*

*1. Power.* At this sample size a test that waits for statistical significance has little of it: the Kupiec test rejects a model whose true coverage is only 97% in 63% of years. The yellow zone, starting at five, flags that model in 87% of years. The price is false alarms (a correct model lands in yellow or worse in 10.8% of years), and the 1996 framework accepted that trade deliberately.

*2. The asymmetry of the two errors.* A statistical test is designed to control the probability of falsely rejecting a correct model. A supervisor faces *asymmetric consequences*: a bank with too little capital may fail, and a bank with too much merely earns less. A threshold chosen to balance statistical errors is not the threshold that balances supervisory ones.

*3. The incentive.* The bank chooses its own model. A tolerance calibrated to the edge of statistical significance would let a firm undermeasure risk, knowing that five or six exceptions is defensible, and keep the capital saving. Whatever the stated rationale, a multiplier that starts rising at five attaches a cost well before the point of statistical rejection, and that removes the option.

The zero-exception row shows the asymmetry plainly. Kupiec rejects it and Basel calls it green, because the supervisor is not worried about a bank holding too much capital.

*The general principle.* A regulatory threshold is a price, not a test. Chapter 27's question 7 makes the same observation about the clearing mandate: capital rules moved the market more than the instruction did, because regulation that changes a price changes behavior more reliably than regulation that issues a rule. Here the price is the capital multiplier, set to buy power against bad models at an accepted rate of false alarms rather than at a statistician's critical value.

*What the question tests.* That the zones and the test disagree in both directions, that the zones trade false alarms for power, and that they price the bank's choice of model rather than testing a hypothesis.

---

**38.23** *Explain why expected shortfall cannot be backtested by counting exceptions, and how the capital framework works around this.*

*Why exceptions cannot backtest expected shortfall.* Because ES is not a threshold, so there is nothing to count.

VaR backtesting works because VaR makes a *binary, observable* prediction: on each day, the loss either exceeds the threshold or it does not, and the frequency of exceedances is a testable statistic (question 21's Kupiec test). The prediction is falsifiable one day at a time.

ES predicts the *average* of the losses in the tail. To test it one would need to compare the realized average of tail losses against the forecast, and that requires:

- observing enough tail events to average over. At 97.5% over 250 days there are about six expected exceedances, so the realized average is computed from roughly six observations;
- and those observations are the most variable ones in the sample, since they are draws from the tail. Averaging six draws from a heavy tail gives an estimate with enormous standard error.

So a direct test has almost no power, and the sample cannot be extended without violating stationarity (question 7).

A separate obstacle concerns comparing forecasts rather than testing one: ES is not **elicitable**. There is no scoring function whose expected value is minimized by reporting the

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true ES for every loss distribution (whereas VaR is elicitable, via the pinball loss), so two competing ES forecasts cannot be ranked by their average score alone. The pair of VaR and ES is jointly elicitable, which is what the comparative tests exploit. Elicitability is not needed to test a single model, and the tests of Acerbi and Székely, which compare each tail loss with the forecast ES on the days VaR was breached, do without it.

*How the capital framework works around it.* It backtests VaR and capitalizes ES.

Chapter 39's framework sets capital on 97.5% expected shortfall, and then validates the model by backtesting *VaR*, at 99% and 97.5%, using exception counts exactly as before. The logic is that a model producing the right *quantiles* is likely to produce the right *tail average*, since both come from the same simulated distribution.

Two supporting devices complete the arrangement:

*The P&L attribution test.* Rather than testing the tail directly, the framework compares the desk's *risk-theoretical* P&L (produced by the risk model) against its *hypothetical* P&L (produced by revaluing the actual book). A desk whose risk model tracks its actual P&L day by day is presumed to have the distribution roughly right, including its tail. This tests the model rather than the measure, which sidesteps the elicitability problem entirely.

*And the consequences of failure are coarse.* A desk that fails its own tests is moved to the standardized approach, and the bank-wide exception count raises the capital multiplier in steps through the yellow zone. Neither needs a precise statistical statement about how wrong the ES was: both respond to the VaR exceptions and the attribution test.

*The honest summary.* The framework chose the better measure and kept the older test, and the gap between them is bridged by an assumption: that a model calibrated to two quantiles has the tail right. Question 9 shows what that assumption is worth when the distributional shape is wrong: a normal model can match quantiles reasonably at 97.5% and report less than half the true ES at 99.9%.

So validation is weaker than the measure. That is a real limitation, acknowledged in the design, and it is the price of moving to a coherent measure whose principal virtue, integrating the tail, is exactly what makes it hard to observe.

*What the question tests.* That a tail average is not falsifiable by counting, that elicitability is the formal obstacle, and that the framework validates the model rather than the measure.

---

**38.24** *Using Examples 38.14 and 38.15, explain what a stress test reports that VaR does not, why a one-factor-at-a-time sensitivity report understated the scenario loss, and what the reverse stress test found about where the book's risk lies.*

*What a stress test reports.* A loss under a named scenario, with no probability attached. VaR answers how large a loss is reached at 99%, a question about the estimated distribution; a stress test answers what happens if a particular set of moves occurs, so it can describe events the estimation window never contained. In Example 38.14 the liquidation scenario loses \$6.4m, 2.10 times the 99% VaR of \$3.051m. The model puts its daily probability at about 5 in 10 million, and the stress test does not rely on that figure, because it is computed from the 0.30 correlation the scenario assumes has broken.

*Why the sensitivity report understated it.* Moving each asset alone by four of its own standard deviations loses \$2.4m and \$4.0m. The worst of these, \$4.0m, is 37.5% short of \$6.4m, because the two losses add only when both assets move at once. A report that moves one factor at a time cannot see a joint move, and a joint move is exactly what a liquidation produces.

*What the reverse stress test found.* Fixing the loss at $l$ = \$6.4m and asking for the most plausible move gives $\mathbf{r}^* = -l\,\Sigma\mathbf{w}/(\mathbf{w}^\top\Sigma\,\mathbf{w})$: A falls 3.35% and B 10.98%. That move lies 4.88 standard deviations out, against 4.96 for the hand-written scenario, so the joint density there is

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about one and a half times as high. On it, B falls 4.39 of its own standard deviations and A only 3.35: the risk is concentrated in B, and the symmetric scenario a risk manager would write by hand is not the one that matters most.

*What the question tests.* That stress tests and probability-based measures answer different questions, that joint moves are the point of a scenario, and that a reverse stress test locates risk rather than only sizing it.

---

**38.25** *A book's daily loss is normal with a standard deviation of \$2 million. Using the standard-error formula of Section 38.5, compute the standard error of a 99% VaR estimated from 250, 500 and 1,000 days, each as a share of the true VaR, and state how much longer a window must be to halve it.*

*The ingredients.* The true VaR is $2.3263 \times \$2\text{m} = \$4.653\text{m}$. The loss density at the VaR is the standard normal density at 2.3263 divided by the \$2m scale:

$$f(\text{VaR}_{99\%}) = \frac{N'(2.3263)}{2} = \frac{0.026652}{2} = 0.013326 \text{ per million dollars.}$$

*The standard errors.* Dividing $\sqrt{0.99 \times 0.01/N}$ by that density:

| Days | $\sqrt{c(1-c)/N}$ | Standard error | Share of VaR |
|---|---|---|---|
| 250 | 0.006293 | \$0.472m | 10.1% |
| 500 | 0.004450 | \$0.334m | 7.2% |
| 1,000 | 0.003146 | \$0.236m | 5.1% |

*Halving it.* The standard error falls as $1/\sqrt{N}$, so halving it takes **four times** as many days: 250 to 1,000 halves it. A 99% VaR as precise as 2.5% of its value would need about 4,100 days, some sixteen years, over which the assumption that every day is drawn from the same distribution is not credible. This normal book is also the easy case; the fat-tailed book of Example 38.3 has a lower density in the tail and a standard error of 14% at 500 days.

*What the question tests.* Applying the quantile standard error, and that precision bought by lengthening the window costs data at the rate of the square.

---

**38.26** *Repeat Example 38.5 with a dollar gamma of −\$200 million. Compute the 99% one-day loss under the delta–gamma expansion, and state by how much it exceeds the delta-only figure.*

*The expansion.* With $\Delta_\$ = \$10\text{m}$ and $\Gamma_\$ = -\$200\text{m}$,

$$\Delta V \approx 10\,r + \frac{1}{2} \times (-200)\,r^2 = 10\,r - 100\,r^2 \qquad (\$ \text{ millions}).$$

*The 99% loss.* The stock's 99% fall is still $r = -2.3263 \times 2.0\% = -4.653\%$:

$$\text{loss} = -10 \times (-0.04653) + 100 \times 0.04653^2 = 0.4653 + 0.2165 = \$0.6817\text{m}.$$

The loss still rises monotonically as the stock falls, and a rise produces a loss only beyond 10%, five standard deviations away, so this is the 99% VaR under the expansion.

*Against the delta-only figure.* The linear term alone gives \$0.4653m, so the expansion's figure is **46.5% higher**, twice the 23% of the example, because the quadratic term doubles with the gamma.

The doubling is also a warning. A gamma term that adds nearly half the linear loss means the position's delta changes by a large fraction over the move being measured, and that is the condition under which the next term, which the expansion drops, starts to matter. A book this short gamma is where full revaluation becomes worth its cost.

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*What the question tests.* Carrying the delta–gamma expansion through, and that the quadratic term scales with gamma while the linear VaR does not see it at all.

---

**38.27** *Using the GARCH parameters of Example 38.8, show that a deviation of the variance from its long-run level halves in about 69 trading days, and compute the annualized volatility forecast for the single day 100 trading days after the variance just estimated.*

*The half-life.* The forecast $k$ days ahead is $V_L + (\alpha+\beta)^k(\sigma^2_{n+1} - V_L)$, so the deviation from $V_L$ is multiplied by $\alpha+\beta = 0.99$ each day. It halves when $0.99^k = 0.5$:

$$k = \frac{\ln 0.5}{\ln 0.99} = \frac{-0.6931}{-0.01005} = \mathbf{69.0} \text{ trading days.}$$

*The forecast at 100 days.* The variance just estimated is $\sigma^2_{n+1} = 0.000002 + 0.06 \times 0.03^2 + 0.93 \times 0.01^2 = 0.000149$, and $V_L = 0.0002$. Then

$$\begin{aligned}\mathbb{E}[\sigma^2_{n+101}] &= 0.0002 + 0.99^{100} \times (0.000149 - 0.0002) \\ &= 0.0002 - 0.3660 \times 0.000051 = 0.0001813,\end{aligned}$$

a daily volatility of 1.3466% and an annualized one of $1.3466\% \times \sqrt{252} = \mathbf{21.4}\%$.

The figure is two-thirds of the way from today's 19.4% to the 22.4% long-run level, as a deviation that has gone through about one and a half half-lives should be. EWMA's forecast for the same day is still 19.3%.

*What the question tests.* Reading mean reversion off the GARCH forecast formula, and converting a daily variance to an annualized volatility.

---

**38.28** *Using (38.8), compute the ten-day 99% VaR of the one-day \$4.65m book of Section 38.14 at a daily autocorrelation of −0.10, and state the size and direction of the square-root rule's error.*

*The multiplier.* With $h = 10$ and $\rho = -0.10$,

$$\begin{aligned}h + 2\sum_{k=1}^{9}(10-k)\,\rho^k &= 10 + 2\left[9\times(-0.1) + 8\times 0.01 + 7\times(-0.001) + 6\times 0.0001 + \cdots\right] \\ &= 10 + 2\times(-0.8264) = 8.347.\end{aligned}$$

*The ten-day VaR.* $\$4.6526\text{m} \times \sqrt{8.347} = \$4.6526\text{m} \times 2.889 = \mathbf{\$13.44m}$, against the square-root rule's $\$4.6526\text{m} \times \sqrt{10} = \$14.71\text{m}$.

*The error.* The rule overstates the true figure by $(14.71 - 13.44)/13.44 = \mathbf{9.5}\%$. In a mean-reverting market a down day tends to be followed by a partial recovery, so ten days' losses partly cancel and the ten-day distribution is narrower than independence implies. Set beside the section's table, where an autocorrelation of +0.10 leaves the rule 9% too low, the same rule errs by about the same amount in opposite directions, which is why it is not conservative.

*What the question tests.* Using the autocorrelation-adjusted scaling formula, and that the direction of the square-root rule's error depends on the sign of the autocorrelation.

---

**38.29** *Redo Example 38.12 at a correlation of 0.60: compute the portfolio VaR, each position's marginal, component and incremental VaR, and state which of the three sets sums to the total.*

*The portfolio.* In return units the covariance with the portfolio is

$$\begin{aligned}(\Sigma\mathbf{w})_A &= 0.01^2 \times 60 + 0.6 \times 0.01 \times 0.025 \times 40 = 0.0120, \\ (\Sigma\mathbf{w})_B &= 0.6 \times 0.01 \times 0.025 \times 60 + 0.025^2 \times 40 = 0.0340,\end{aligned}$$

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so $\mathbf{w}^\top\Sigma\mathbf{w} = 60\times0.0120 + 40\times0.0340 = 2.08$, $\sigma_P = \$1.4422$m, and the VaR is $2.3263\times1.4422 = \$\mathbf{3.355m}$, the 0.60 row of Example 38.9.

*The three sets.* By (38.9), and with standalone VaRs of \$1.396m for A and \$2.326m for B:

| | **Marginal VaR** | **Component VaR** | **Incremental VaR** |
|---|---|---|---|
| A | $2.3263\times0.0120/1.4422 = 0.0194$ | $60\times0.0194 = \$1.161$m | $3.355 - 2.326 = \$1.029$m |
| B | $2.3263\times0.0340/1.4422 = 0.0548$ | $40\times0.0548 = \$2.194$m | $3.355 - 1.396 = \$1.959$m |
| Sum | — | \$3.355m | \$2.988m |

*Which sums.* Only the **components** sum to the total, by Euler's theorem. The incremental figures sum to \$2.988m, short of the total by the \$0.367m diversification benefit, which each of them credits in full to the position being removed. The marginal figures are rates per dollar and are not meant to sum to anything.

*What changed from 0.30.* B's share of the risk falls from 68.6% to 65.4% and A's rises from 31.4% to 34.6%: at a higher correlation A diversifies B less, so A carries more of the joint risk. The diversification benefit itself shrinks from \$0.671m to \$0.367m.

*What the question tests.* Computing the three allocation figures from (38.9), and that only the Euler components add up.

---

**38.30** *For the two-stock book of Example 38.4, find the most plausible pair of returns that loses \$1.0 million, state how many standard deviations out it lies, and compare it with the scenario in which both stocks fall by four of their own standard deviations.*

*The ingredients.* Positions $\mathbf{w} = (10, 5)$ in \$ millions, volatilities 2.0% and 1.0%, correlation 0.3. In return units

$$\Sigma\mathbf{w} = \big(0.02^2\times10+0.3\times0.02\times0.01\times5,\ 0.3\times0.02\times0.01\times10+0.01^2\times5\big) = (0.0043,\ 0.0011),$$

and $\mathbf{w}^\top\Sigma\mathbf{w} = 0.0485$, so $\sigma_P = \$0.2202$m, as in the example.

*The most plausible move.* By (38.10) with $l = 1.0$:

$$r_1^* = -1.0\times\frac{0.0043}{0.0485} = -\mathbf{8.87}\%, \qquad r_2^* = -1.0\times\frac{0.0011}{0.0485} = -\mathbf{2.27}\%.$$

The loss is $10\times8.87\% + 5\times2.27\% = \$1.0$m, and the move lies $1.0/0.2202 = \mathbf{4.54}$ standard deviations out. Stock 1 falls 4.43 of its own standard deviations and stock 2 only 2.27.

*The symmetric scenario.* Both stocks falling by $k$ of their own standard deviations loses $10\times0.02k + 5\times0.01k = 0.25k$, so $k = 4$: stock 1 falls 8% and stock 2 falls 4%. Its Mahalanobis distance is 4.96, against 4.54, so the joint density at the reverse-stress scenario is $e^{(4.96^2-4.54^2)/2} \approx 7.4$ times as high.

The gap is far wider than the one and a half of Example 38.15, because the two positions are far from balanced: stock 1 carries most of the risk, so the plausible route to the loss runs almost entirely through it, while the symmetric scenario asks a quiet stock to move four standard deviations for little contribution to the loss.

*What the question tests.* Applying the reverse-stress formula, and that the scenario loads on the position with the largest covariance with the book.

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# Chapter 39
# Market Risk Capital and Model Risk

**39.1** *Explain why the trading-book boundary was made prescriptive, compute the capital at stake in Section 39.2's bond portfolio, and explain why removing the capital benefit of reassignment matters more than tightening the definition.*

*Why the boundary was made prescriptive.* Under the earlier rule an instrument sat in the trading book if the bank intended to trade it, and intent is not observable, not verifiable, and revisable after the fact. Trading-book and banking-book positions attract different capital, so a bank holding a position could reclassify it into whichever book charged less. The reclassification required no transaction, no change in the risk and no counterparty, only a change of stated intention, so capital could be reduced by an act of description.

The current rule replaces intent with a presumptive list (instruments held for short-term resale, correlation trading, net short credit or equity positions, and underwriting and market-making inventory in the trading book; unlisted equities, real estate, retail and small-business credit, and funds that cannot be looked through in the banking book), and a bank may depart from it only with supervisory approval and documented reasons.

*The capital at stake.* The bank of Section 39.2 holds $500 million of corporate bonds that attract 3% capital in one book and 8% in the other:

$$\$500\text{m} \times (8\% - 3\%) = \$25 \text{ million}.$$

That is the value of the classification option on a single portfolio, and under the old rule the bank held it for free.

*Why removing the benefit matters more.* A definition can be argued with and a price cannot. However carefully the boundary is drawn, instruments sit near it, new instruments are created, and a bank with $25 million riding on the answer will construct a case for the classification it prefers. Tightening the definition raises the cost of making that case; it does not remove the case.

The decisive rule is different in kind. Reassignment requires supervisory approval, is permitted only in extraordinary circumstances, and may not produce a capital benefit: if moving a position lowers its charge, the bank holds a disclosed surcharge equal to the difference. The surcharge is fixed at the reassignment, and it may run off only with supervisory approval. A bank may still argue about classification, and winning the argument gains it nothing, so the incentive is removed rather than the opportunity, and the question stops being contested.

This is the pattern Chapter 27's question 7 identifies: regulation that changes a price changes behavior more reliably than regulation that issues an instruction. There the clearing mandate covered standardized products and the capital differential moved the rest of the market; here the presumptive list covers the clear cases and removing the benefit handles the rest. The rule also does not require the supervisor to be right about every instrument. A list that is wrong at the margin still produces the right outcome, because the margin is not worth exploiting.

*What the question tests.* That an intent-based boundary is unverifiable, computing the value of the classification option, and that removing the payoff from arbitraging a rule dominates tightening the rule.

---

**39.2** *Explain why the standardized approach supplies correlations rather than estimating them, and relate this to Section 38.11.*

*Why the correlations are supplied.* A bank estimating its own correlations is estimating a quantity that directly reduces its own capital, from data that systematically understates it.

Chapter 38's question 15 establishes both halves. Moving the correlation of a two-position book from 0.30 to 0.90 raises its VaR by 19% with no change in any position or volatility, so the whole diversification benefit hangs on this one input. And the input's estimation error has a consistent sign. Correlations are estimated in calm periods and applied in violent ones, and assets that are unrelated in normal conditions move together in a liquidation, because the common factor in a crisis is that the same leveraged holders are selling all of them. The calm-period estimate is too low, and the measured risk with it.

Combine the two and the supervisory problem is plain. The bank chooses a number, the estimation procedure biases it downward, and the bias reduces the bank's own capital. A bank acting in complete good faith, using the standard estimator on the standard window, still produces an understatement, so this is not mainly a question about incentives to cheat. The framework therefore prescribes the correlations, which makes requirements comparable across banks, takes the sample-period bias out of the bank's hands, and turns the diversification benefit into a policy parameter. That settles by fiat the instability of estimated correlations that Section 38.11 documented: the supervisor would rather be wrong in a known direction than depend on a bank's estimate.

*The framework goes further than one prescribed number.* Each bucket charge is computed three times: at the prescribed correlations, at 1.25 times them capped at one, and at the larger of $2\rho_{kl} - 1$ and $0.75\,\rho_{kl}$. The worst of the three is the capital. So the framework does not merely prescribe a correlation; it declines to commit to one and charges for the uncertainty.

That is a sharper response than uniform conservatism. Raising every correlation would overcharge directional books and undercharge hedged ones, since question 4 shows the two are hurt by opposite scenarios. Taking the worst of three charges each book for the scenario that hurts *it*.

The cost is real. A genuinely diversified book pays for a diversification it will probably realize, and the charge is insensitive to evidence: a bank with a well-founded estimate cannot use it. The framework has traded accuracy for robustness, which is defensible given the direction of the error it guards against.

*What the question tests.* That a self-estimated correlation is a downward-biased input to the bank's own requirement, and that the three-scenario device prices the uncertainty rather than assuming a level.

---

**39.3** *Using the weighted sensitivities and correlations of Example 39.1, compute the directional book's bucket charge under the high-correlation scenario, and explain why it equals the sum of the weighted sensitivities.*

*The high scenario.* Each prescribed correlation is multiplied by 1.25 and capped at one. The smallest of the three is the 2-to-10-year correlation, and even it exceeds the cap:

$$1.25 \times 0.8869 = 1.1086 > 1,$$

so all three correlations become exactly one.

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*The bucket charge.* With every $\rho_{kl} = 1$ the expression under the square root is a perfect square:

$$K_b^2 = \sum_k WS_k^2 + \sum_{k \neq l} WS_k\, WS_l = \Big(\sum_k WS_k\Big)^2,$$

so

$$K_b = \Big|\sum_k WS_k\Big| = 260{,}000 + 330{,}000 + 440{,}000 = \$1{,}030{,}000.$$

*Why it equals the sum.* A correlation of one says the three points of the curve move as one factor. The bucket is then a single exposure whose weighted sensitivity is the sum of the three, and a single exposure's charge is the absolute value of its weighted sensitivity. All three of the directional book's sensitivities have the same sign, so nothing offsets and the absolute value of the sum is the sum itself. This is the largest the charge can be for these weighted sensitivities: no correlation at or below one can make the cross terms larger.

The same identity gives the butterfly's high-scenario charge at once. Its weighted sensitivities are \$286,000, −\$506,000 and \$220,000, which sum to zero, so its charge is $|0| = 0$. Question 4 takes up what that implies.

*What the question tests.* Applying the cap, and seeing that perfect correlation collapses a bucket to one factor whose charge is the absolute value of the summed weighted sensitivities.

---

**39.4** *Using Example 39.1, explain why the directional book is worst under high correlation and the butterfly under low, why the butterfly's charge falls to zero under high correlation, and why the hedged book nonetheless pays the larger proportional penalty.*

*The two books.* Example 39.1 uses one currency and three points on the curve, at 2, 5 and 10 years. The directional book receives fixed at all three. The butterfly receives fixed at 2 and 10 years and pays fixed at 5, sized so that its weighted sensitivities sum to zero:

| Scenario | Directional book | Butterfly |
|---|---|---|
| Low correlation | \$988,376 | **\$100,854** |
| Medium (as prescribed) | \$1,009,403 | \$71,315 |
| High correlation | **\$1,030,000** | \$0 |
| **Charge (worst)** | **\$1,030,000** | **\$100,854** |
| Excess over medium | +2.0% | +41.4% |

In the low scenario each correlation becomes the larger of $2\rho_{kl} - 1$ and $0.75\,\rho_{kl}$. The first is larger whenever $\rho_{kl} > 0.8$, which all three prescribed correlations are, so the low correlations are 0.9120 (2–5 years), 0.9409 (5–10) and 0.7738 (2–10).

*Why the directional book is worst under high correlation.* Its weighted sensitivities all have the same sign, so in

$$K_b = \sqrt{\max\Big(0,\ \sum_k WS_k^2 + \sum_{k \neq l} \rho_{kl}\, WS_k\, WS_l\Big)}$$

every cross term $WS_k\, WS_l$ is positive. Raising the correlations raises every cross term and the charge with it, and the worst case is the perfect co-movement of question 3.

*Why the butterfly is worst under low correlation.* Its sensitivities have mixed signs (long the wings, short the belly), so the two cross terms that pair a wing with the belly are negative and large. High correlation makes the legs offset, which is what the butterfly was built to exploit. Low correlation breaks the hedge: if the three points move less closely together, the wings and the belly no longer cancel and the position has real risk.

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*Why the butterfly's charge falls to zero.* Question 3 showed that under the high scenario every correlation is one and the charge is the absolute value of the summed weighted sensitivities. The butterfly was sized so that the sum is zero, so a perfectly correlated curve makes its hedge perfect and the charge is exactly zero. The formula reports that correctly: if the three points move together by their prescribed shocks, 130 basis points at 2 years and 110 at 5 and 10, the position does not change in value.

*Why the hedged book pays the larger penalty.* The framework takes the worst of three scenarios, and the butterfly's worst lies far from its medium: 41.4% above it, against 2.0% for the directional book. Figure 39.1 shows why. The directional book's charge varies by about 4% across the scenarios, because a directional book is nearly as risky under any correlation; the butterfly's runs from zero to $100,854. A book whose risk depends heavily on a parameter the framework refuses to fix pays heavily for that dependence.

So the penalty is a charge for correlation exposure, and the butterfly has more of it. A hedged position is hedged conditional on a correlation, and Chapter 38's question 15 gives the reason to take that seriously: the diversification benefit is what evaporates in a crisis. Three consequences follow. Hedging does not always reduce capital under this rule, and a relative-value book can pay a larger penalty than a directional one. The penalty falls on books built to be factor-neutral, which is Chapter 38's question 17 arriving as a capital charge: a butterfly's risk lives in the part of the curve that a coarse factor structure discards. And the outcome is defensible rather than perverse, since charging the butterfly its medium-scenario $71,315 would assert a correlation on behalf of a position whose value depends on it.

A single prescribed correlation could not do this. A high one would overcharge the directional book and undercharge the butterfly, a low one the reverse, and either could be arbitraged by building positions that are riskless under it. Conservatism has no fixed direction, so the framework charges each book for the scenario that hurts it. A book whose risk does not depend on correlation at all (a single point on the curve) pays the same under all three and is charged nothing extra.

*What the question tests.* That the sign of the cross terms decides which scenario binds, that a perfectly correlated curve makes a hedge neutral to the prescribed shocks riskless, and that the hedged book is charged for its sensitivity to correlation rather than for its risk at the prescribed level.

---

**39.5** *Explain why a long option attracts no curvature charge while the same option sold attracts the whole of it, using Figure 39.2.*

*The calculation.* The curvature charge shocks the factor up and down by its delta risk weight, revalues in full, and asks how much worse the result is than the delta prediction:

$$CVR = \max\Big(0,\ -\min_{\text{up, down}}\big[V(\text{shocked}) - V(0) - \delta \times \text{shock}\big]\Big).$$

*A long option.* Figure 39.2 draws the long $10 million at-the-money call of Section 39.3 with its delta line. At a 55% rise the option is worth $5.505m against a delta prediction of $3.648m; at a 55% fall it is worth almost nothing against a prediction of −$2.239m. Revaluation beats the delta line by $1.86m on the upside and $2.24m on the downside. Both gaps lie above the line because the value curve is convex, so the bracket is positive in both directions, its minimum is positive, the negative of that minimum is negative, and the floor sets the charge to zero. Capital is not held against the possibility of making money.

*The same option sold.* Every value changes sign, and so does each bracket: −$1.86m up and −$2.24m down. Both directions are adverse, the worse is taken, and the charge is $2.24m, the whole of the larger gap. The floor at zero is what produces the asymmetry. Without it the

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long call would carry a negative charge of −$1.86m, which the framework does not allow; with it, all the curvature capital falls on the short side.

*Why this is the payoff's asymmetry.* For moderate moves the gap is approximately

$$V(\text{shocked}) - V(0) - \delta \times \text{shock} \approx \frac{1}{2}\,\Gamma\,(\Delta S)^2,$$

whose sign is the sign of gamma whatever the direction of the move. Chapter 18's question 3 made the same point about gamma P&L: a short-gamma book loses on movement in either direction. The rule revalues in full rather than using this approximation because a 55% shock is far outside the range where the expansion is accurate, which is why the two gaps differ. A rule that charged a long option for curvature would be capitalizing a convexity that pays.

Two consequences follow. The treatment is consistent under netting: the revaluation is of the desk's net position in the factor, so a long and a short in the same option cancel, and a book long some options and short others pays only on its net short convexity. And it makes the option-selling business expensive, deliberately. Chapter 38's question 6 showed that a desk measured on VaR is rewarded for selling far-out-of-the-money options, because the loss lives beyond the quantile; a shock-and-revalue calculation does not care where the quantile sits. That is why the delta-hedged short call of Example 39.2 pays so heavily, with 80% of its charge in the curvature term.

One caveat applies. Because of the floor, the charge measures adverse convexity only, not risk in the ordinary sense. A book with large positive gamma has a genuine cost—it pays the theta that finances the convexity—and its curvature charge is zero. The framework is measuring loss potential rather than variability, consistent with the expected-shortfall philosophy of Chapter 38.

*What the question tests.* That the floor at zero transmits the sign of gamma, so the asymmetry belongs to the payoff rather than to the formula.

---

**39.6** *Compute the curvature charge on a short $10 million at-the-money six-month put on the stock of Section 39.3 at 25% volatility and zero interest rates, and explain why it equals the short call's.*

*The put.* With interest rates at zero, put–call parity gives put = call − (stock − strike), so the at-the-money put is worth the same $0.704m as the call. Its delta is the call's less one, 0.5352 − 1 = −0.4648, a dollar delta of −$4.648m. The stock's delta risk weight, and so the curvature shock, is 55%.

*Up 55%.* The put is worth $0.0047m. The delta prediction is

$$\$0.704\text{m} + 0.55 \times (-\$4.648\text{m}) = -\$1.852\text{m},$$

so revaluation beats it by $1.857m.

*Down 55%.* The put is worth $5.500m, against a delta prediction of

$$\$0.704\text{m} + 0.55 \times \$4.648\text{m} = \$3.261\text{m},$$

so revaluation beats it by $2.239m.

*The charge.* For the long put both brackets are positive and the floor makes the charge zero. For the short put they are −$1.857m and −$2.239m, the worse is taken, and

$$CVR = \$2{,}239{,}369,$$

identical to the short call's charge to the dollar.

*Why it equals the short call's.* The two curvature terms are equal one by one: $1.857m up and $2.239m down for both options. Parity writes the put as the call minus a forward, and a forward is linear in the stock. A linear position's revaluation under any shock equals its delta

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prediction exactly, so it contributes nothing to the bracket; and the forward's delta of one is exactly the difference between the call's delta and the put's, so subtracting it changes the delta term by the same amount it changes the value. What remains in the bracket is the convexity, and a put and a call at the same strike and expiration have the same convexity, the same gamma.

So the curvature charge prices convexity and nothing else, and it cannot tell a short put from a short call. A desk that converts one into the other by trading the stock or a forward changes its delta charge and leaves its curvature charge exactly where it was.

*What the question tests.* Running the curvature calculation on a second payoff, and seeing through put–call parity that a linear position contributes nothing once delta is removed.

---

**39.7** *Recompute Example 39.2 for a small-cap stock, whose volatility has a 60-day liquidity horizon, keeping the curvature shock at 55% so that only the vega term changes, and state the total charge as a share of notional and the curvature term's share of it.*

*What changes.* Only the vega risk weight, which is $\min\left(55\% \times \sqrt{LH/10},\ 100\%\right)$. For a 60-day horizon

$$55\% \times \sqrt{60/10} = 55\% \times 2.4495 = 134.7\%,$$

which the cap reduces to 100%. The cap already binds at about 33 days, the horizon at which $55\% \times \sqrt{LH/10}$ reaches one, so 40-day and 60-day volatilities carry the same weight. The curvature shock is held at the example's 55% so that the comparison isolates the vega horizon, and the reported vega sensitivity is unchanged at \$702,488.

*The charge.*

| Component | Large-cap | Small-cap |
|---|---|---|
| Delta | ≈ \$0 | ≈ \$0 |
| Vega | \$546,407 | \$702,488 |
| Curvature | \$2,239,369 | \$2,239,369 |
| **Total** | **\$2,785,776** | **\$2,941,857** |
| Share of notional | 27.9% | 29.4% |
| Curvature share of total | 80.4% | 76.1% |

The small-cap book carries \$2,941,856, or 29.4% of its \$10 million notional, and the curvature term is 76.1% of it. The longer horizon raises the vega charge by 28.6% and the total by 5.6%.

*Why hedging still removes only delta.* Delta neutrality is a statement about the first derivative at a point, and the other two charges concern quantities the hedge does not touch. The stock has no vega and no gamma, since the underlying is linear and its second derivatives vanish (Chapter 18's question 5), so buying or selling it leaves the short option's volatility exposure and its short convexity exactly as they were. A book flat by the measure a trader watches minute to minute attracts capital close to 30% of its notional.

The comparison also shows where the liquidity horizon bites. Holding a less liquid volatility costs capital through the vega weight, but the effect is bounded by the cap, while the curvature term, which depends on the stock's delta risk weight rather than on the volatility's horizon, stays the dominant cost.

One qualification applies. The charge is on a single position with no offsets. In a real book vega and curvature net across positions in the same bucket, so a desk that both buys and sells options pays far less than the sum of its positions' stand-alone charges. The example measures the cost of a one-directional short option book, which is the configuration the framework is aimed at.

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*What the question tests.* Applying the capped vega weight, recomputing the total, and seeing that the curvature term delta-hedging cannot remove still dominates.

---

**39.8** *A desk holds $50 million notional of variance swaps and $200 million of products carrying gap risk. Compute its residual risk add-on, and explain why the framework charges a flat percentage rather than modeling these payoffs.*

*The add-on.* The residual risk add-on is 1% of the notional of instruments with an exotic underlying (future realized volatility, a variance swap's underlying, is one) and 0.1% of the notional of other instruments bearing residual risks such as gap risk:

$$1\% \times \$50\text{m} + 0.1\% \times \$200\text{m} = \$0.5\text{m} + \$0.2\text{m} = \$0.7 \text{ million}.$$

*Why a flat percentage.* The add-on exists for payoffs the sensitivities do not describe. Delta, vega and curvature capture how a price responds to the prescribed risk factors; a gap, the path of realized volatility, or a dependence on correlation or behavior is a risk those factors do not span. Any model of it would be the bank's own model of a payoff with little market evidence, which is the thing the framework has decided it cannot verify. Section 39.5 handles unverifiable factors in the internal model by charging a stress add-on outside the diversification; the standardized approach handles unverifiable payoffs by declining to model them at all.

The flat charge is crude and knowingly so. It falls on notional rather than risk, so it does not distinguish a well-hedged position from an unhedged one or a deep out-of-the-money payoff from an at-the-money one, and no hedge reduces it. That crudeness is what makes it honest: a charge that pretended to measure gap risk precisely would carry a false precision the supervisor could not check, while a flat percentage states openly that the risk is being priced as a category.

The behavioral effect is a tax on complexity. A desk choosing between a payoff the sensitivities describe and one they do not pays a fixed toll for the second, which is a modest but steady push toward simpler products. For this desk the $0.7 million is a small figure beside the curvature charge of a single short option position.

*What the question tests.* Applying the two rates, and that a flat charge on notional is the framework's deliberate alternative to modeling payoffs it cannot verify.

---

**39.9** *In Example 39.3, a factor class is reclassified from a 40-day to a 60-day horizon, raising* $\text{ES}_4$ *from 45 to 60. Compute the new aggregate, and explain why the desk's capital changed although nothing it holds did.*

*The new aggregate.* The weights are unchanged (1.0000, 1.0000, 1.4142, 1.4142 and 2.4495), and only $\text{ES}_4$ moves:

$$\sqrt{100^2 + 90^2 + (70 \times 1.4142)^2 + (60 \times 1.4142)^2 + (20 \times 2.4495)^2}$$

$$= \sqrt{10{,}000 + 8{,}100 + 9{,}800 + 7{,}200 + 2{,}400} = \sqrt{37{,}500} = 193.65,$$

against 185.34 before, a rise of 4.5%, and now 1.94 times the flat ten-day figure of 100.

*Why* $\text{ES}_4$ *rises.* $\text{ES}_4$ shocks only the factors whose horizon is at least 60 days. A factor class reassigned from 40 to 60 days now survives into that bucket, so it is shocked in four of the five terms instead of three. Its stand-alone contribution is scaled by the square root of the sum of the squared weights of the buckets it appears in:

$$\sqrt{1 + 1 + 2} = 2.00 \text{ at 40 days}, \qquad \sqrt{1 + 1 + 2 + 2} = 2.45 \text{ at 60 days},$$

and a 120-day factor, which appears in all five terms, is scaled by $\sqrt{12} = 3.46$ in total. The reclassification raised the weight on that class from 2.00 to 2.45 times its ten-day figure.

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*Why the capital changed when nothing held did.* The schedule assigns the horizon, not the bank. No position, volatility or correlation moved; a supervisory classification did, and the classification enters the formula as a multiplier. The judgment about how long a position takes to exit has moved from the bank's model to the regulator's table rather than disappearing.

That is the price of fixing Chapter 38's horizon problem. Chapter 38's question 18 showed the square-root-of-time rule failing when returns are autocorrelated, and a single ten-day horizon is wrong because the horizon is a property of the position: a liquid rate factor can be exited in days, a concentrated illiquid credit cannot. Chapter 27's question 4 shows the same assumption in the counterparty framework, where the margin period of risk is a liquidity assumption entering a capital calculation as a constant. The framework replaces one horizon with five, and the cost is that a bank's capital now depends on which bucket the regulator puts each factor in—a dependence the bank cannot trade away.

*What the question tests.* Substituting into the horizon-weighted aggregation, and that a reclassification moves capital through the schedule rather than through any change in risk.

---

**39.10** *State the risk factor eligibility test, and redo Example 39.4 with a correlation of 0.50 between the two parts, separating the cost of lost diversification from the cost of the stress calibration.*

*The eligibility test.* A risk factor is modellable only if it has at least 24 real price observations over the past twelve months with no 90-day period containing fewer than four, or at least 100 real price observations over the twelve months. A real price is an actual transaction or a committed quote. A factor that fails is non-modellable and is capitalized by a stressed expected shortfall, added to the modeled charge rather than diversified with it.

The test is about data rather than modeling ability. A factor can be perfectly well understood and still fail, because the question is whether the bank's estimate can be checked against prices: a long-dated volatility point that trades twice a year, a thin emerging-market tenor, a correlation between two illiquid issuers.

*Example 39.4 at a correlation of 0.50.* If every factor were modellable the desk would hold

$$\sqrt{80^2 + 30^2 + 2 \times 0.50 \times 80 \times 30} = \sqrt{9{,}700} = 98.5.$$

Under the non-modellable treatment it holds the modellable 80 plus the add-on $1.6 \times 30 = 48$, which is 128 as before, so the penalty is $128 - 98.5 = 29.5$, or 30.0% more for identical positions. It splits into two parts that add:

| | $\rho = 0.25$ | $\rho = 0.50$ |
|---|---|---|
| Fully modellable, diversified | 92.2 | 98.5 |
| Non-modellable treatment | 128.0 | 128.0 |
| Lost diversification, 80 + 30 − diversified | 17.8 | 11.5 |
| Stress calibration, 48 − 30 | 18.0 | 18.0 |
| **Increase** | +38.8% | +30.0% |

The higher correlation leaves less diversification to lose, so that part falls from 17.8 to 11.5; at a correlation of one it would vanish. The stress-calibration part does not depend on the correlation at all. The two effects are additive: the total is the modellable charge plus the stand-alone factor plus the stress uplift on it. Only the lost diversification is a property of the rule. The factor of 1.6 is the example's assumption; a real factor's stress scenario is calibrated on its own data and could be larger or smaller.

*Why the add-on is not diversified.* Diversification would require the correlation between the non-modellable factor and the rest of the book, and the factor failed the test precisely because there are not enough prices to estimate anything about it. Allowing diversification would

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ask the bank to estimate the one quantity the classification says it cannot, and Chapter 38's question 15 shows that such an estimate errs in a consistent direction, understating risk, worst for illiquid factors.

What the framework is pricing is opacity rather than risk. The desk may understand these factors well; what it cannot do is demonstrate its estimates against observable prices. A desk facing the add-on has a priced reason to prefer instruments whose factors are quoted, and the market a reason to quote factors many banks hold. One criticism recurs: the observation count is mechanical, so a factor can flip status on the arrival or absence of a few trades in a market the bank does not control, and data-pooling services exist largely to soften that.

*What the question tests.* Stating the current eligibility test, recomputing the example, and separating a penalty that depends on correlation from one that does not.

---

**39.11** *Using the closed form of Example 39.5, compute the share of unexplained variance at which the rank correlation equals 0.75, and state the zone the desk is in.*

*The share.* Invert the closed form $\rho_S = \frac{6}{\pi}\arcsin(\rho/2)$:

$$\rho = 2\sin\left(\frac{\pi\rho_S}{6}\right) = 2\sin\left(\frac{\pi}{8}\right) = 0.7654,$$

and since $\rho = \sqrt{1-f}$,

$$f = 1 - \rho^2 = 1 - 0.7654^2 = 41.4\%.$$

(Exactly, $f = 1 - 4\sin^2(\pi/8) = \sqrt{2} - 1$.)

*The zone.* A rank correlation of 0.75 lies between the red threshold of 0.70 and the green threshold of 0.80, so the desk is amber; equivalently, 41.4% lies between the 33.8% at which amber begins and the 48.6% at which red begins. The mean KS statistic in Example 39.5 is about 0.10 at that share, also between its 0.09 and 0.12 thresholds, so the two statistics agree.

An amber desk keeps its model and pays the surcharge of Section 39.8. So a desk whose risk model leaves more than two-fifths of its P&L variance unexplained is still capitalized on that model. The permissiveness is a property of the map from correlation to variance: the correlation falls slowly while the unexplained share grows, so a threshold that looks strict in correlation terms is generous in variance terms.

*What the question tests.* Inverting the arcsine relation, converting a correlation into a share of unexplained variance, and reading the zone from both statistics.

---

**39.12** *Explain why the rank correlation and the Kolmogorov–Smirnov statistic leave the green zone at nearly the same share of unexplained variance, and construct a model error that only one of them would detect.*

*Why they leave green together.* In Example 39.5 the rank correlation leaves green at 33.8% unexplained variance and the mean KS statistic at about 35%, and both reach red near half. That is because a missing factor does two things at once. It removes the part of the P&L that moves on days the model does not see, which weakens the day-by-day correlation, and the rank correlation reads that. And it shrinks the risk-theoretical series' variance to $1 - f$ of the hypothetical one's, which narrows its distribution, and KS reads that. The single error drives both statistics, so they cross their thresholds at nearly the same point.

*An error only KS detects: right ordering, wrong scale.* Let the risk-theoretical P&L be exactly half the hypothetical P&L every day. The ordering of the days is perfect, and a rank correlation is unchanged by multiplying a series by a positive constant, so the Spearman statistic is exactly 1: green by the widest margin. But the two distributions differ in scale by a factor of

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two. Measuring the hypothetical series in its own standard deviations, its distribution function is $\Phi(x)$ and the risk-theoretical series' is $\Phi(2x)$, so

$$KS = \max_x \left|\Phi(2x) - \Phi(x)\right| = 0.161,$$

reached at $x = 0.68$, well above the red threshold of 0.12. The KS statistic disqualifies this model on its own. A model that halves the risk would also understate VaR and show in backtesting, so this is not an error the framework misses—but within the attribution test only KS sees it.

*An error only the rank correlation detects: right variance, wrong ordering.* Take a model that omits a factor but scales up the volatilities of the factors it contains until its risk-theoretical P&L has the hypothetical series' variance. The distributions now match, so KS registers only sampling noise and stays green on average; but the omitted factor still moves the actual P&L on days the model does not see, so the pairing degrades exactly as in Example 39.5 and the rank correlation falls into amber and then red. This is the correctly sized but structurally wrong model of Section 39.7.

*What this implies.* The two statistics are complements. The rank correlation catches errors in which days the model thinks are good and bad, KS catches errors in how large the model thinks the moves are, and the ordinary failure, a missing factor, trips both. A model has to be wrong in a narrow way to pass both: right in ordering, right in scale, and wrong only in what neither statistic measures, such as the shape of the tail beyond the 250 days observed.

*What the question tests.* That a missing factor both decorrelates and shrinks variance, and constructing errors that separate a rank statistic from a distributional one.

---

**39.13** *Explain what kind of model failure the attribution test catches and what kind it does not, and relate the second to a learned pricing or hedging model.*

*What the attribution test catches: a missing risk factor.* The test compares the desk's risk-theoretical P&L, produced by the risk model, against its hypothetical P&L, produced by full front-office revaluation of the same positions on the same day's market moves. If the two diverge, the risk model does not contain something the book is exposed to. Question 12 established that a missing factor both mis-orders the days and narrows the distribution, so the test catches an incomplete specification: a curve tenor not represented, a basis not modeled, a smile dimension collapsed to a single volatility. That is what the framework wanted to test: a desk cannot use a model that does not know what the desk trades.

*What it does not catch: an error in the dynamics.* Both series are computed from the same historical market moves, so the test says nothing about whether the distribution the model assumes for future moves is right. A model containing every relevant factor, correctly specified, but assuming those factors are normally distributed will track its own hypothetical P&L closely—and, as Chapter 38's question 9 shows, understate the true expected shortfall at 99.9% by half. The test validates composition, not distribution, and the framework's cover for the second is backtesting (question 15) and the stress calibration.

*The connection to learned models.* This is the limitation of held-out testing for a learned model. In Example 10.1 of [TN39] a deep hedging policy, evaluated on fresh paths from the simulator it was trained on, performed as trained. Changing only the shape of the daily moves (a jump diffusion matched to the same 20% total volatility) cut its advantage over a tuned no-trade band from 19.2% to 6.4%, and the policy degraded by 40.1% against the classical rules' 19.7% and 20.9%. The learned pricer of Section 3 of [TN39] fails the same way: tested on points inside its training range it is accurate, and outside that range it returns a plausible number rather than an error.

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Fresh data from the same generating process tests estimation error; it does not test the generating process. The attribution test is the regulatory instance of that: replaying history through two calculations checks that they contain the same factors and cannot check whether the assumed dynamics describe tomorrow. That is why [TN39] insists that the simulator be delivered alongside the policy, and why question 24 argues that model risk is not a subset of market risk. A test conducted inside a model's own world cannot evaluate that world.

*What the question tests.* That attribution validates factor coverage on historical moves, and recognizing the same blind spot in a learned model validated on data from its own training process.

---

**39.14** *A desk records 9 exceptions at 99% and 34 at 97.5% over 250 days. State the consequence, and explain why VaR is backtested when capital is held against expected shortfall.*

*The consequence.* The desk backtests its one-day VaR at both levels, and the limits are 12 exceptions at 99% and 30 at 97.5% over 250 days. Nine exceptions at 99% is within its limit; 34 at 97.5% exceeds 30. Failing either test is enough, so the desk loses its model approval and its whole requirement moves to the standardized approach.

Both counts are far above what a correct model would produce: 2.5 exceptions are expected at 99% and 6.25 at 97.5%. Nine at 99% is already well outside the one to six that Chapter 38's question 21 found consistent with a correct model, and the 99% limit of 12 tolerates it; the 97.5% test, with more exceptions to count and so less noise, is the one that catches the miscalibration. Running two levels is what gives the desk test its power: a model understating the body of the loss distribution can survive the 99% count and still fail at 97.5%.

*Why VaR is backtested.* Capital is held against 97.5% expected shortfall, but expected shortfall offers no event to count. A VaR exception is a day on which the loss exceeded a number, and the count of such days has a known distribution under a correct model; an expected shortfall is an average over the tail, and no single day either exceeds it or does not in a way that tests it. Section 38.16 made the point, Chapter 38's question 23 asked how the framework works around it, and [TN38] treats the statistics. So the framework backtests the quantiles and trusts the tail average to come from the same model.

The choice of levels is not arbitrary. For a normal loss, 97.5% expected shortfall and 99% VaR nearly coincide (Chapter 38's question 2), so testing at both 97.5% and 99% checks two points of the tail that the capital measure averages over. It is still a check on quantiles rather than on the tail beyond them, which is the gap the stress calibration and supervisory review are there to cover.

*What the question tests.* Applying the desk-level limits, and why the framework backtests a quantile while capitalizing an average.

---

**39.15** *Explain what backtesting tests that attribution does not, and construct a desk that passes each while failing the other.*

*What backtesting tests that attribution does not: the size of the tail.* Backtesting counts days on which the actual loss exceeded the reported VaR, so it tests magnitude and frequency together: does a loss of the stated size occur at the stated rate? A model whose distribution is too narrow produces too many exceptions, and Chapter 38's question 21 shows that too few is a rejection as well. Attribution compares two daily P&L series over ordinary days; question 12 showed that KS can see a gross scale error in the body of the distribution, but nothing in the test asks how large the losses beyond the 97.5th or 99th percentile are.

*What attribution tests that backtesting does not: the composition of the model.* A missing

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factor mis-orders the daily predictions and is caught, whereas backtesting sees only whether the aggregate loss crossed a line, and can be satisfied by a model that is wrong in offsetting ways.

*A desk that passes backtesting and fails attribution.* Its risk model omits a factor (say a curve basis), but the volatilities of the factors it does contain have been calibrated up, so its VaR is the right size. Backtesting passes: exceptions occur at about the stated rate. Attribution fails: on days the basis moves, the predicted and actual P&L diverge, and the rank correlation degrades even though the two distributions match, as in question 12's second construction. The model is right on average and wrong about what it measures, so it will mis-size any hedge and fail when the compensating calibration stops compensating.

*A desk that passes attribution and fails backtesting.* Its risk model contains every relevant factor, correctly specified, but assumes the factors are normally distributed when the book's exposures are fat-tailed. Attribution passes: the risk-theoretical P&L revalues the positions on the realized moves through the right factors, with no distributional assumption, so it tracks the hypothetical P&L closely. Backtesting fails: the VaR is computed from the normal assumption and understates the tail (in Chapter 38's question 9 the true expected shortfall at 99.9% is 103% above the normal model's), so exceptions arrive far more often than the stated rate. A stale calibration does the same: right factors, wrong level.

So the two tests are complements, and the framework runs both because neither suffices: attribution asks whether the model contains the right things, and backtesting whether it sizes them right. Question 13 adds the limitation they share. Both replay history, so neither evaluates whether the assumed dynamics describe the future.

*What the question tests.* That the two tests probe composition and magnitude respectively, and constructing a desk that separates them in each direction.

---

**39.16** *Explain why desk-level approval is a cliff rather than a slope, what the amber surcharge changes, and what behavior the cliff induces.*

*Why approval is a cliff.* Approval is binary at the desk level. A red desk, or one that fails its backtest, moves its entire requirement onto the standardized approach, not a portion of it and not a graduated penalty, and for a book with genuine diversification the standardized charge typically runs 30% to 100% above the model number. Example 39.5 puts the red line at 48.6% of unexplained variance: a desk just below it keeps its model, and one just above loses it, on a 250-day sample that Section 38.16 showed is short.

The bank-wide backtesting multiplier is a slope by comparison. Exception counts map to multipliers of 1.50, 1.70 and 2.00, which question 18 shows act on the modeled charge rather than by a change of method: they raise it by 11.6% at amber and 28.9% at red, both measured from green. The cliff is the price of making approval mean something: "partly fit" is not a coherent supervisory statement about a model.

*What the amber surcharge changes.* An amber desk keeps its model but pays a surcharge: half the excess of the standardized charge over the modeled charge for the green and amber desks together, scaled by the amber desks' share of the standardized charge,

$$\text{surcharge} = \frac{1}{2} \times \max\left(0,\ SA_{G+A} - IMA_{G+A}\right) \times \frac{SA_A}{SA_{G+A}}.$$

That turns the first step down into a slope: the surcharge grows with the share of the bank's business in amber, and a single amber desk pays a fraction of the gap rather than all of it. It does not remove the cliff at red, which is still a switch to the standardized approach for the whole desk. Above both sits a cap: the bank's internal-model total may not exceed the standardized charge for all its desks.

*What behavior the cliff induces.* Four patterns follow.

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*Desk boundaries become a capital variable.* Approval is per desk, so a bank chooses how to draw its desks, and a desk close to failing can be reorganized. Splitting off the positions whose factors are poorly modeled leaves a desk that passes and a small one that fails, which is far cheaper than one large failing desk. The documentation requirements around desk definition exist to constrain this.

*Effort concentrates at the threshold.* A desk safely green gains nothing from improving its model; a desk near amber or red gains a great deal. Modeling resources flow to the desks nearest the line, which is not necessarily where the largest or least understood exposures are.

*Risk-taking is shaped around what the model can attribute.* A trade whose risk factors sit outside the model degrades the attribution score, so a desk near the threshold has a reason to decline it even when it is a good trade. The constraint is on measurability rather than on risk, which is question 10's opacity charge appearing as behavior.

*Risk models include every factor the front office prices with.* The chapter's conclusion is the rational response: a model conservative in the specific sense of containing every factor, whether or not it matters, is the cheapest insurance against a red result.

The cliff matters because of what sits on the other side: question 19 shows desks without model approval contributing 27.0% of a bank's total requirement. And question 20 shows the whole structure can be switched off for a bank at the output floor.

*What the question tests.* That binary desk-level approval creates a discontinuity, how the amber surcharge softens its first step, and the behaviors the discontinuity induces.

---

**39.17** *Using Example 39.6, recompute the modeled charge when yesterday's internally modeled charge is 650, state which term binds, and explain why the averaged term usually does.*

*The recomputation.* With $IMCC_{t-1} = 650$ and everything else as in Example 39.6:

$$\max(650 + 95,\ 1.5 \times 380 + 88) = \max(745,\ 658) = 745.$$

Yesterday's term binds, and the modeled charge is 745 rather than 658.

*Why the averaged term usually binds.* For yesterday's term to bind, $IMCC_{t-1} + SES_{t-1}$ must exceed $m_c \times \overline{IMCC} + \overline{SES}$. With the example's add-ons that needs $IMCC_{t-1} > 658 - 95 = 563$, which is 48% above its sixty-day average of 380. Two features make that rare. The multiplier is at least 1.5, so the averaged term starts half again above a typical day's charge. And an average over sixty days moves slowly, so it stays close to the typical level while a single day can jump. Yesterday's figure exceeds the averaged term only after a sharp one-day rise of the kind this question assumes (a 71% jump from 380 to 650), and not in the ordinary course.

That is the design. The averaged term is an anti-procyclicality device: a requirement keyed to the latest day's measure would spike in a crisis exactly when capital is scarcest, and averaging damps the spike at the cost of responding slowly. Yesterday's term is the backstop for the case averaging cannot handle, a sudden large increase in risk that the bank should not be able to carry for sixty days on the strength of a calm average.

*What the question tests.* Substituting into the two-term maximum, and why the multiplier and the averaging together make the averaged term the normal constraint.

---

**39.18** *State how the bank-wide backtesting exception count enters the capital formula, and compute the increase from a green to a red multiplier.*

*How the exception count enters.* The bank's model as a whole is backtested at 99% across all its approved desks, and the count of exceptions over 250 days places it in a zone (green,

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amber or red, as in Section 38.16) that sets the multiplier $m_c$:

$$\max\Big(IMCC_{t-1} + SES_{t-1},\ \ m_c \times \overline{IMCC} + \overline{SES}\Big).$$

So the exception count multiplies the sixty-day average of the internally modeled capital charge.

*The increase from green to red.* From Example 39.6, with $IMCC_{t-1} = 420$, $\overline{IMCC} = 380$, $SES_{t-1} = 95$ and $\overline{SES} = 88$:

| Multiplier | Modeled charge | Increase |
|---|---|---|
| 1.50 (green) | 658 | — |
| 1.70 (amber, five exceptions) | 734 | +11.6% |
| 2.00 (red) | 848 | +28.9% |

$$\frac{2.00 \times 380 + 88}{1.50 \times 380 + 88} - 1 = \frac{848}{658} - 1 = +28.9\%.$$

At every multiplier the averaged term exceeds $420 + 95 = 515$, so it is the constraint throughout, as question 17 explains. The increase is less than the 33% rise in the multiplier because the non-modellable add-on is not multiplied.

*Two observations about the design follow.* It is a slope, not a cliff. Unlike question 16's desk approval, a deteriorating bank-wide backtest raises the charge by 11.6% at amber and 28.9% at red, both measured from green, a graduated warning appropriate to a noisy test: Chapter 38's question 21 showed that anything from one to six exceptions is statistically consistent with a correct model. And the zones are tighter than statistics alone would set them, which Chapter 38's question 22 explains: five exceptions puts a bank in the amber zone while the Kupiec statistic is comfortably inside its acceptance region. The zones are a price rather than a test, set where they change behavior, because the bank chooses its own model and the error it would prefer has a sign.

*What the question tests.* Substituting the multipliers into the assembly formula, and that the multiplier acts on the averaged term that normally binds.

---

**39.19** *Using Example 39.6, state the share of the total requirement contributed by desks without model approval, and explain why this makes the attribution test the focus of capital management.*

*The share.* From Example 39.6:

| Component | Amount | Share |
|---|---|---|
| Modeled measure (ES and NMRF) | 658 | 60.2% |
| Default risk charge, approved desks | 140 | 12.8% |
| Desks on the standardized approach | 295 | 27.0% |
| of which residual risk add-on | 35 | 3.2% |
| **Total** | **1,093** | 100% |

Desks without model approval contribute 295 of 1,093, or 27.0%: more than a quarter of the bank's market risk requirement. The residual risk add-on of 35 belongs to the standardized charge, so it sits inside the 295 rather than beside it.

*Why this makes the attribution test the focus.* The attribution test is what decides which desks are in that 27.0%, and question 16 established that the decision is a cliff. Three features make it *the* focus rather than one concern among several.

*The size of the step.* A desk's standardized charge typically runs 30% to 100% above its modeled one, because the standardized approach is calibrated to be conservative: prescribed correlations (question 2), the worst of three correlation scenarios (question 4), and a curvature

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charge (question 5) that hits option books hard. Example 39.2's 27.9% of notional on a single delta-hedged short option position shows what that means.

*Its controllability.* The modeled measure responds slowly to modeling work and the default risk charge is largely mechanical. The 27.0% responds to a pass or a fail, and a pass is something a desk can work toward. A bank does not reduce it by hedging; it reduces it by passing the test on more desks.

*The leverage of the effort.* Improving a model that already passes lowers the requirement marginally; moving a desk from fail to pass removes a discontinuity. The return on modeling investment is concentrated at the threshold.

This explains why banks devote disproportionate resources to attribution performance relative to the test's difficulty: question 11 showed a desk can leave more than 40% of its variance unexplained and still keep its model. The test is not hard and failing it is expensive, and the second fact drives the behavior.

One qualification remains, and question 20 supplies it. All of this presumes the modeled number is the binding constraint. For a bank at the output floor it is not, and the incentive structure is switched off.

*What the question tests.* Reading the assembly table, placing the residual risk add-on inside the standardized charge, and that a large controllable share behind a binary threshold concentrates management attention.

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**39.20** *Using Example 39.7, find the modeled non-trading RWA at which failing the desk's test costs exactly 20, and explain what the floor does to the incentives of Section 39.6.*

*The break-even.* Let $x$ be the bank's modeled non-trading RWA. The floor is $0.725 \times 1{,}000 = 725$ whatever $x$ is, and the bank holds the larger of its modeled total and the floor:

$$\text{cost of failing} = \max(x + 100,\ 725) - \max(x + 60,\ 725).$$

There are three regimes. For $x \le 625$ the floor binds whether the desk passes or fails and the cost is zero. For $x \ge 665$ the floor binds in neither case and the cost is the desk's full 40. In between, the floor binds only when the desk keeps approval, and the cost is $x + 100 - 725 = x - 625$. Setting that to 20:

$$x - 625 = 20 \quad \Longrightarrow \quad x = 645.$$

Check: keeping approval gives $645 + 60 = 705$, below the floor, so the bank holds 725; losing it gives $645 + 100 = 745$, above the floor, so it holds 745; the cost is 20. Bank C of Example 39.7, at 640, sits in the same middle regime and pays 15.

*What the floor does to the incentives.* The difference between Bank A's 40, Bank C's 15 and Bank B's nothing has nothing to do with their trading books: the desk, the test and the failure are identical. It is a property of the rest of each balance sheet. The framework of Sections 39.6 and 39.8 makes attribution failure a cliff, puts more than a quarter of the requirement on desks without approval (question 19), and so concentrates capital management on the test. For a bank at the floor none of that applies: the test has no capital consequence, the cliff has no drop, and modeling effort earns no return. For a bank in the middle regime the incentive is partial, and it weakens the closer the bank sits to the floor.

So the framework's behavioral design is conditional on the bank not being at the floor, and whether it is depends on the non-trading book, which the market risk framework does not govern. A binding constraint switches off every constraint beneath it. Chapter 27's question 13 makes the same point about the leverage ratio, which binds below a risk weight of about 48% and makes a dealer's safest businesses expensive first. In both cases a deliberately crude

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backstop, introduced to bound what the risk-based framework can be talked into, also nullifies the risk-based framework's incentives for the banks it binds on.

*What the question tests.* Solving the floor arithmetic piecewise, and that the floor's bindingness is decided outside the trading book.

---

**39.21** *Explain why a bank near the output floor may rationally decline model approval, and why this is not evidence that its modeling is inadequate.*

*Why declining approval can be rational.* Question 20 establishes the condition: for a bank whose modeled total sits below the output floor, the modeled number is not the constraint. Bank B holds 725 whether its desk keeps approval or loses it, so the capital benefit of approval is zero. Against that sits a large, recurring cost: building and maintaining the expected-shortfall engine, the liquidity-horizon classification of question 9 and the eligibility testing of question 10; daily attribution and backtesting at desk level, with the data infrastructure to produce comparable risk-theoretical and hypothetical P&L; continuous validation and governance; supervisory engagement; and the operational disruption of failing. Positive cost and zero benefit give the answer. A bank that moves its trading book to the standardized approach eliminates the cost and holds the capital it was going to hold anyway.

*Why this is not evidence of inadequate modeling.* The decision is about the constraint, not the model. Bank B may have a better risk model than Bank A. Its lower modeled total, 660 against Bank A's 760, is if anything evidence of a less risky trading book or a more refined model—and it is that lower number that puts the bank under the floor and removes the reward.

That produces a perversity. A bank whose models produce a low number is more likely to be at the floor, and a bank at the floor gains nothing from having models, so the better the modeling, the weaker the incentive to keep it. An observer who sees a bank on the standardized approach and concludes that it could not obtain approval draws the wrong inference: the correct one is that approval was not worth having, which is a statement about the composition of the balance sheet rather than about risk management. Section 39.13 records that banks have made exactly this choice on some desks.

Two riders follow. A bank may still run internal models for management purposes while declining regulatory approval; the models remain useful for limits, pricing and hedging. And the floor phases in over time (to 2028 under the Basel timetable and 2030 in the European Union), so a bank near it faces a moving calculation and may rationally keep approval as an option on the floor ceasing to bind.

*What the question tests.* That a zero marginal benefit against a positive cost dictates the decision, and that being under the floor reflects the whole balance sheet rather than modeling quality.

---

**39.22** *Using Example 39.9, compute the deduction if the three overlapping adjustments were added in full rather than at 50%, explain why model risk cannot be reduced by trading better, and compute the business the actual deduction displaces at a 12% capital requirement.*

*The actual deduction.* From Example 39.9, market price uncertainty, close-out cost and model risk overlap in what they measure, so the core approach adds them and counts half:

$$0.5 \times (40 + 35 + 60) = 67.5 \text{ bp},$$

and the other six adjustments are added in full:

$$25 + 10 + 8 + 6 + 5 + 4 = 58 \text{ bp}.$$

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The total is 67.5 + 58 = 125.5 basis points, or $1,255,000 on the $100 million book.

*Added in full.* Without the 50% aggregation the deduction would be

$$135 + 58 = 193 \text{ bp} = \$1{,}930{,}000,$$

54% more. The aggregation exists because the three adjustments are three views of one uncertainty (how far from the mark the position could actually be realized), and adding them in full would count that uncertainty more than once.

*Why model risk cannot be reduced by trading better.* Model risk, at 60 basis points, is the largest single component, and it measures a valuation uncertainty rather than an execution cost. The other components respond to trading: a tighter market narrows the close-out cost, a smaller position reduces the concentration charge, better controls reduce the operational component. Model risk is the uncertainty in the price of a payoff nothing quotes. Chapter 21's question 10 measures it directly: two models calibrated to the same vanilla surface price a barrier 49.5% apart. A tighter market in the hedging instruments does not narrow that spread, because the spread comes from the market's failure to determine the price at all. Only a better-identified model, a market that starts quoting the instrument, or not holding the position reduces it.

*The business displaced.* The deduction comes straight out of common equity tier 1, and at a 12% capital requirement that equity would have supported

$$\frac{\$1.255\text{m}}{0.12} = \$10.5 \text{ million}$$

of risk-weighted assets, about a tenth of the book's own fair value. Added in full, the figure would be $16.1 million.

This is where Section 21.11's model spread becomes a number on a balance sheet. Hull and Suo's (2002) procedure measures it (calibrate competing models to a common surface, then price what the surface does not determine), and prudent valuation turns the measurement into a deduction. As a rule with prescribed adjustments it applies in the European Union and the United Kingdom; elsewhere supervisors expect valuation reserves of the same kind without prescribing them. The incentive points the right way: a desk facing a 60 basis point deduction on model-uncertain positions has a priced reason to prefer instruments the market quotes.

*What the question tests.* Applying the core-approach aggregation, distinguishing a valuation uncertainty from an execution cost, and converting a common-equity deduction into displaced risk-weighted assets.

---

**39.23** *Explain how a supervisory stress test becomes trading-book capital in the United States, and why reverse stress testing is the exercise most likely to find what the formula misses.*

*How the stress test becomes capital.* In the United States the largest banks' trading books take an instantaneous global market shock as part of the supervisory stress test. The resulting losses enter the projected decline in the bank's capital under the stress scenario, and that decline sets the **stress capital buffer**, a requirement on top of the minimum that rises with the bank's projected stress losses. So a trading book that loses heavily under the supervisory shock raises the bank's capital requirement directly, through a channel that never passes through the expected-shortfall formula. For those banks a supervisory scenario, not the formula of Section 39.9, often decides how much the trading book costs.

Its value lies in the comparison as much as the number. The scenario is common across banks, so a bank whose losses are unusual under a shared shock is showing a position, and the exercise finds it without anyone specifying in advance what to look for.

*Why reverse stress testing finds what the formula misses.* Every other exercise in the framework starts from a model or a scenario and computes a loss. Reverse stress testing starts

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from the outcome, a loss that would make the bank fail, and searches for the scenarios that produce it. That inversion matters for two reasons. The search is not confined to the risk factors the model contains, so it can reach the blind spots Section 38.19 listed: concentration, liquidity, and exposures to factors no measure includes. And it asks the question in the form management needs answered—what would have to happen—rather than the form a model can answer—how much would this scenario cost. Section 38.17 works one through for a two-asset book and finds that the most plausible route to a given loss runs through the position with the largest marginal risk, which is often not the largest position.

The formula handles what can be specified in advance. A supervisory stress test checks it against a common shock, and reverse stress testing looks for the shock nobody specified, which is where the losses of Chapter 40 come from.

*What the question tests.* The US route from a supervisory scenario to a capital requirement, and why starting from the outcome escapes the model's own factor set.

---

**39.24** *Explain why model risk is not a subset of market risk, and why the error and the measure move together.*

*Why model risk is not a subset of market risk.* The two are risks about different things. Market risk is the risk that prices move, and it is measured within a model: the model supplies a distribution of market outcomes, and VaR or expected shortfall is a functional of that distribution. Model risk is the risk that the model is wrong. It cannot be a functional of the model's own distribution, because a model assigns no probability to the proposition that it is misspecified; no amount of running it produces an estimate of its own error.

The distinction shows in what each responds to. Market risk falls if positions are reduced or hedged; model risk does not. Chapter 21's question 10 has two models calibrated to the same vanilla surface pricing a barrier 49.5% apart, and no trade in vanillas closes that gap. Question 22 makes the same point in capital terms: model risk is the largest component of the valuation adjustment and the one a desk cannot reduce by trading better.

*Why the error and the measure move together.* The same model produces the position's value and its risk measure. If the model is wrong, the mark is wrong, so the reported P&L is wrong; the sensitivities are wrong, so the hedge is mis-sized; and the risk measure is wrong, so the error is not reported as risk. There is no independent instrument to detect it: a market risk system asked whether the model is wrong answers from inside the model, and the answer is no.

The co-movement is worse than correlation. The circumstances that make a model wrong—an unprecedented move, a regime the calibration sample does not contain, a correlation breaking down—are the circumstances in which the position is losing money. Chapter 38's question 15 gives the mechanism for correlation, Chapter 26's question 7 for tail dependence, and Example 10.1 of [TN39] for a hedging policy under jumps. In each case the model's error is largest in the state where the loss is largest, and the risk measure is computed by the model that is failing, so the system reports least error when the error is largest. Model risk belongs to the family of Chapter 27's question 12: a protection whose reliability is negatively correlated with the need for it.

That is why it is treated outside the market risk measure. Prudent valuation deducts it from common equity directly, a calculation that does not depend on the model in question, and the governance of Section 39.14 (inventory, ownership, independent validation, documented limitations) makes the model's assumptions explicit without claiming to verify them. The only defensible treatment of an error the model cannot see is one the model does not compute.

*What the question tests.* That a model cannot represent its own misspecification, and that value, hedge and measure fail together because one model produces all three.

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**39.25** *Using Section 39.15, explain why reproducibility is an inadequate validation standard for a learned hedging policy, and what should be required instead.*

*The standard.* Reproducibility (re-running the code on the same data returns the same result) is a natural validation requirement and an adequate one for most models. A closed-form pricer, a lattice, a calibration routine: each is deterministic given its inputs, and a validator who reproduces the number has confirmed the implementation.

*Why it is inadequate for a learned hedging policy.* Section 39.15 cites the result, which is Example 12.2 in Section 12 of [TN39]. The same policy was trained three times, changing only the random seed:

| | | Holding at spot | | |
|---|---|---|---|---|
| **Seed** | **CVaR$_{95}$** | **95** | **100** | **105** |
| 7 | 1.9852 | 0.2047 | 0.5401 | 0.8485 |
| 17 | 2.0013 | 0.2007 | 0.5478 | 0.8641 |
| 27 | 2.0003 | 0.2062 | 0.5441 | 0.8652 |
| Black–Scholes delta | | 0.1082 | 0.5081 | 0.8879 |

The performance reproduces and the strategy does not. Expected shortfall spans 0.8% of its mean, so as far as the objective is concerned the three runs are the same answer. But at a spot of 105 the policies hold 0.8485, 0.8641 and 0.8652, a spread of 1.7 delta points on positions that are supposed to be identical: on a $100 million book, a $1.7 million difference in stock held between two desks running the same code on the same data.

So reproducibility can be satisfied trivially and prove nothing. Fixing the seed makes the training run reproducible, and a validator re-running it obtains the same weights, while a different and equally defensible seed would have produced a materially different position. The requirement is met and the question it was meant to answer is untouched. That question is whether the model gives a determinate answer. For a deterministic model, reproducibility is that question; for a learned policy the two come apart, because the procedure is reproducible and the object it produces is not determined by the specification.

*What should be required instead.* The properties that do reproduce, and the inputs that define the model:

- the structural features: all three runs hold well above the Black–Scholes delta of 0.1082 at 95 and below 0.8879 at 105, so every run learns to flatten and they differ on how much; a validator should test for flattening, a no-trade region, and recovery of the classical delta when costs are zero;
- the spread across seeds, reported as a number rather than eliminated by fixing one seed;
- the objective, since the policy optimizes whatever it is given (Section 9 of [TN39]);
- the simulator and the training range, with performance under alternative dynamics: Example 10.1 of [TN39] shows the advantage shrinking from 19.2% to 6.4% under jumps, and Section 3 shows a learned pricer failing silently outside its training box;
- and the labels or reference model the network learned from, since the case study in Section 4 of [TN39] is one where the pricer was wrong and the network smoothed the evidence.

The operational response is to treat the trained policy as a released artifact (trained once, weights recorded and versioned) and to treat retraining as a model change requiring revalidation, not a refresh.

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The general point is that a validation standard must test a property that can fail. Reproducibility tests implementation, and a learned policy's characteristic failure is not an implementation failure. The same reasoning is behind Example 12.1 of [TN39], where a smooth surrogate passes an arbitrage check that could not have failed.

*What the question tests.* That seed-invariant performance with seed-dependent positions defeats reproducibility as a standard, and substituting structural, specification and data checks for it.

---

**39.26** *A senior unsecured loan to a nonfinancial corporate has a one-year PD of 1% and a maturity of 2.5 years, and the bank uses the foundation IRB approach. Compute the prescribed correlation, the worst-case default rate, the maturity adjustment, the capital requirement and the risk weight, and state whether the output floor binds on a book of such loans if the standardized weight is 100%.*

*The correlation.* With PD $= 0.01$, $w = (1 - e^{-0.5})/(1 - e^{-50}) = 0.3935$, so

$$R = 0.12 \times 0.3935 + 0.24 \times 0.6065 = \mathbf{0.1928}.$$

*The worst-case default rate.* With $N^{-1}(0.01) = -2.3263$ and $\sqrt{0.1928} = 0.4391$,

$$\text{WCDR} = N\left(\frac{-2.3263 + 0.4391 \times 3.0902}{\sqrt{0.8072}}\right) = N(-1.0791) = \mathbf{14.03}\%.$$

*The maturity adjustment.* $b = (0.11852 - 0.05478 \ln 0.01)^2 = (0.11852 + 0.25227)^2 = 0.1375$, and at $M = 2.5$ the numerator is one, so the adjustment is $1/(1-1.5\times0.1375) = \mathbf{1.2598}$.

*Capital and risk weight.* The foundation LGD is 40%, so by (39.1)

$$K = 0.40 \times (0.1403 - 0.0100) \times 1.2598 = \mathbf{6.57}\%, \qquad \text{RW} = 12.5 \times 6.57\% = \mathbf{82.1}\%.$$

*The floor.* The modeled weight is 82% of the standardized 100%, above 72.5%, so on a book of such loans alone the floor does not bind. The comparison the floor makes is for the bank as a whole, though, and a bank with BBB loans beside these, modeled at 62% of standardized, could still be on it.

*What the question tests.* Running the formula end to end, and seeing that a 1% PD loan sits near the crossing point, where the floor's effect depends on the rest of the book.

---

**39.27** *Using Example 39.8, explain why the output floor binds on a bank that lends mainly to investment-grade companies and not on one that lends mainly to speculative-grade companies, and what this implies for the value of trading-book model approval at each.*

*Why the ratio rises with PD.* The standardized weights step up with the rating, but slowly: 50% at A, 75% at BBB, 100% at BB, 150% below. The IRB weight rises much faster, because it is a tail quantile and the worst-case default rate of a low-PD borrower is a small number. At A the model gives 17.5% against 50%, a ratio of 0.35, and at BBB 46.7% against 75%, a ratio of 0.62; at BB and B the ratios are 0.94 and 0.97. The floor compares the modeled total with 72.5% of the standardized total, so a book of mostly A and BBB loans is modeled below the floor and holds the floor, and a book of mostly BB and B loans is modeled above it and holds its modeled number.

*What it implies for the trading book.* For the investment-grade lender the floor already binds through its lending, so the trading book's modeled capital does not change the requirement: a desk that loses model approval adds RWA beneath a floor that is already above the total, as Bank B of Example 39.7 showed, and model approval is worth nothing until the desk's standardized charge is large enough to lift the bank off the floor. For the speculative-grade

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lender every unit of modeled trading capital saved is a unit of requirement saved, and the attribution test and the backtests keep the full force Sections 39.6 and 39.8 describe.

*The irony.* The floor exists because supervisors distrust model outputs, and it bites hardest on the banks whose loan books are safest by their own models. That is why the European Union added a transitional 65% weight for unrated corporates whose PD the bank estimates at no more than 0.5% inside the floor calculation, and why the floor's effect on lending to large companies was among the most contested parts of the 2017 reforms.

*What the question tests.* That a rule calibrated on the whole bank transmits the composition of the loan book into the incentives of the trading desks.

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# Chapter 40
# Derivatives Mishaps and What We Can Learn

**40.1** *Using Example 40.1, explain why the stack hedge is flat in value and catastrophic in cash, and identify which leg produces the cash demand.*

*Why the hedge is flat in value.* From Example 40.1, 160 million barrels of forward commitment hedged one-for-one with front futures:

| Price fall | Futures margin (cash out) | Supply contract (unrealized) |
|---|---|---|
| \$2.00/bbl | −\$0.32bn | +\$0.32bn |
| \$5.00/bbl | −\$0.80bn | +\$0.80bn |
| \$8.00/bbl | −\$1.28bn | +\$1.28bn |

*Every row is flat in undiscounted value.* The two columns are equal and opposite at every price, which is what a hedge is supposed to look like: the firm has fixed the price at which it will deliver, and the fixing works.

*Why it is catastrophic in cash.* The two legs settle on entirely different timetables.

*The leg producing the cash demand is the **futures**.* It is margined daily, so the loss is paid in **variation margin now**: \$1.28 billion of actual cash, paid into the clearing house as the price falls.

*The supply contract's offsetting gain arrives over ten years and cannot be pledged.* The firm will sell oil above market for a decade, which is worth \$1.28 billion undiscounted and delivers *nothing today*. It is not a receivable, not a security, and not collateral—it is a stream of future favorable margins on deliveries that have not happened.

So a firm whose hedge is working perfectly must fund the entire adverse move in cash while holding an asset it cannot monetize. If it cannot, it liquidates the hedge, at which point it is unhedged into whatever the market does next, and the unrealized gain it was holding is exposed to the next move without the position that offset it.

*The general form of the failure.* A hedge sized on price risk alone is not sized, because the funding leg scales with the same move the hedge was bought to neutralize. Chapter 36's question 8 makes the identical point on the same case, and the shape recurs:

- *European utilities in 2022* (question 14), where several of the hedged generators needed state liquidity support;
- *UK LDI in 2022* (question 9), where funding improved at every step while the schemes could not pay;
- *a crypto basis trade* (Chapter 37's question 2), closed by a 20% rally that cost it nothing.

*And the uncomfortable corollary*: the more complete the hedge, the larger the funding requirement. The firm that hedged nothing had no margin calls; the cash demand was created

by the hedge and scaled with its size, and question 3 adds that a one-for-one stack made that size larger than the exposure required.

*What the question tests.* That the columns net to zero in value and not in time, and identifying the margined leg as the source of the demand.

---

**40.2** *Using Figure 40.1, state the peak funding gap and when it occurs, and explain why the strategy can fail between the two points even though both curves end at the same total.*

*The peak funding gap.* Figure 40.1 plots two cumulative curves after an \$8/bbl fall spread over six months, on Example 40.1's position:

- the cash paid on the futures leg climbs evenly to **\$1.28bn** over the six months of the fall and stays there, because it is paid as the price falls;
- the gain realized on the supply contract accrues evenly at about \$0.128bn a year and reaches \$1.28bn only at the ten-year horizon.

The gap is the first curve less the second. It widens for as long as margin is still being paid, because cash goes out twenty times faster than the gain comes in, and it narrows from the moment the margin stops. So it peaks at six months, when the margin has all been paid and the supply contract has realized half a year's gain:

$$\$1.28\text{bn} - \$0.064\text{bn} = \$\mathbf{1.22}\text{ billion}$$

It closes only slowly after that (a year after the fall it is still \$1.15 billion), and not until the ten-year horizon does it reach zero.

*Why the strategy can fail between the two points.* Because the end points are what a valuation model computes, and the vertical distance between the curves is a loan the strategy never arranged.

A valuation model asks what the two legs are worth together once the contract has run off. The answer is zero: both curves reach \$1.28bn, so the hedge is exact in undiscounted terms, and question 1's table is flat at every price. That calculation is correct, and it answers a question about the end point.

The firm has to survive the middle. Throughout it, it is a net payer of cash, and the offsetting asset it holds can be neither sold (it is the firm's own supply business) nor pledged (it is not a receivable). So the vertical distance has to be financed from somewhere else (a credit line, the parent's capital, or the sale of other assets), and none of those is part of the hedge's design.

Two properties of the gap make it more dangerous than its size.

*It is largest at the worst moment.* The gap is proportional to the price fall, so the amount the firm must borrow grows with the very event that makes it need to borrow. A credit line arranged in calm conditions is sized to a calm-conditions gap.

*And it appears while the firm looks fully hedged.* No value-based risk report shows it: question 1's table is zero in every row. The gap is visible only when the position is described in cash, and most risk systems do not describe positions that way.

*The figure is the chapter's mechanism.* Every case in the first category of Section 40.15 is that gap: the LDI schemes (question 9), AIG's margin calls (question 10), and Example 37.2's basis trade all draw the same two curves with different labels. A valuation model computes where the curves end; the failure happens in between.

*The implication.* A hedge that is right in value and wrong in timing is a financing commitment attached to a correct opinion, and the commitment has to be arranged when the hedge is put on. The lesson of Section 40.1 sizes it: Section 8.10's stress test counts gross cash

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outflow rather than net loss, and on this position it reports $0.16 billion of variation margin for every dollar the oil price falls, where a value measure reports nothing. The funding line should be committed at that scale before the trade.

*What the question tests.* Reading the gap off the two cumulative curves, and that equal end points are a statement about value while the distance between the curves is an unarranged financing need.

---

**40.3** *Explain how a stack-and-roll converts a flat-price hedge into a calendar spread, and state the first-year cost in Example 40.1's contango case.*

*How the conversion happens.* A one-for-one stack holds the *entire* multi-year exposure in the **front** contract. That contract expires monthly, so the position must be rolled: closing the expiring month and opening the next.

*Each roll transacts the calendar spread.* The firm buys back a contract at one price and sells another at a different one, and the difference is the spread between adjacent months. Repeated monthly for the life of the commitment, the position's profit and loss is therefore driven by *the shape of the curve* as much as by its level.

So a hedge intended to fix the flat price has acquired a second, unintended exposure to the term structure, and that exposure is not offset by anything on the supply contract, which references the flat price and not the spread.

*The first-year cost.* From Example 40.1, the stack averages about 153 million barrels over the first year and pays $0.35 a barrel on each of twelve rolls:

$$153{,}000{,}000 \times \$0.35 \times 12 \approx \$0.64 \text{ billion}$$

*Contango costs $0.64 billion in the first year*, on a hedge whose purpose was to be neutral.

*And the number is large because the stack is large.* The roll is paid every month on the *whole* remaining commitment, about 153 million barrels on average in the first year, not on the fraction of the exposure arising that year, which is Chapter 36's question 9's point that the stack ratio is wrong even in principle, over-hedging both in present-value terms and in volatility.

*Why the exposure was not chosen.* No one decided to take a view on the calendar spread. The position arose *mechanically* from a decision about *how* to hedge, taken because the far-dated futures were illiquid and the front contract was the only place a position of that size could be established. An operational constraint became a market position, and it was not on any risk report as one.

*The general lesson.* A hedge instrument that does not match the exposure's maturity introduces a term-structure position. Chapter 33's question 2 makes the same point about forward-starting swaps, Chapter 36's question 1 about Samuelson and cross-maturity hedge ratios, and Chapter 5's roll yield is the general form. The residual is not noise—it has a sign, a size, and a name.

And it is the second of the case's three separately sufficient problems: question 1 gives the cash timing, this gives the roll, and Chapter 36's question 9 gives the sizing. Any one of them was enough.

*What the question tests.* That rolling a stack transacts the calendar spread monthly on the full notional, and reading the contango figure as a cost of the hedging method rather than of the hedge.

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**40.4** *Explain what Barings and Société Générale have in common, and why the same failure destroyed one firm and not the other.*

*What they have in common.* Neither is a derivatives failure; both are failures of the risk control system. The checks that should have caught an unauthorized position (the **segregation of duties** that keeps trading apart from its settlement, independent reconciliation, and verification of what the trader reported) were in the hands of the person they were meant to check. In both cases positions reported as low-risk arbitrage were outright directional. At Barings one person controlled trading **and the settlement of that trading**, and performed the reconciliation that would have exposed the position. At Société Générale the trader had come from the middle office and knew its checks well enough to hide the positions behind fictitious offsetting trades.

The instruments were exchange-traded, cleared, and margined daily: index futures in both, and at Barings also Nikkei options and Japanese government bond and Euroyen futures. Nothing about the products was hard to value. The positions were not mispriced, not exotic, and not model-dependent. What failed was the control that separates the person who takes a position from the person who settles and reconciles it, an operational control that would be just as necessary if the instruments were physical commodities or equities.

So the derivative was the *vehicle*, not the cause. Futures did let a concealed position reach a given size with less funding than a cash position would have needed, but that funding was paid daily and was observable, which is what the second lesson below turns on.

*Why the same failure destroyed one firm and not the other.* The size of the hidden position relative to the firm's equity. From the chapter's table:

| | Position | Firm equity | Ratio |
|---|---|---|---|
| Barings, 1995 | \$27bn | \$0.6bn | **45**× |
| Société Générale, 2008 | EUR 50bn | EUR 30bn | 1.7× |

$$\frac{\$27\text{bn}}{\$0.6\text{bn}} = 45\times \qquad \text{and} \qquad \frac{\text{EUR } 50\text{bn}}{\text{EUR } 30\text{bn}} \approx 1.7\times$$

The same failure, and the only difference is the ratio. Barings held about forty-five times its equity in directional futures and option positions, so a modest adverse move exceeded everything the firm had. Société Générale held a *larger* notional against a balance sheet that could absorb the EUR 4.9 billion the unwinding cost. The losses themselves tell the same story in secondary form: Barings lost £827 million, more than its equity, and failed; the French bank lost several times as much and survived.

Three lessons follow, and the third is the one usually missed.

*The severity of a control failure is set by a number the control system is not looking at.* Control failures are not survivable or unsurvivable on their own; their outcome is set by the ratio of the largest position that can be hidden to the equity behind it. No amount of modeling would have helped either firm.

*So bound the ratio, since the failure cannot be ruled out.* This is the lesson of Section 40.3, and its mechanism is Section 2.3's daily settlement: every exchange-traded loss is paid in variation margin the next morning, so a cap on the margin funding a desk can draw is a cap on the position it can carry, whether or not the position has been reported. At Barings the signal was there: the head office kept remitting margin to Singapore for a position it believed was hedged. The reform that followed was operational rather than quantitative (segregation of trading from settlement, independent reconciliation, mandatory leave), which is question 16's general finding.

*And including these cases in a derivatives book is itself the point.* They are routinely cited as evidence that derivatives are dangerous, and they are evidence of nothing of the kind—a firm

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that cannot reconcile its own positions will fail in whatever instruments it trades. Chapter 26's question 1 makes the parallel argument about securitization: a general technique is not refuted by a failure whose cause lies elsewhere.

*What the question tests.* Identifying the shared cause as a control failure rather than a derivatives one, and that survival was determined by the ratio of the position to equity.

---

**40.5** *Using Example 40.3, compute the move that exhausts LTCM's equity, and explain why this makes the strategy a view about the path rather than about value.*

*The wipeout move.* From Example 40.3, equity of \$4.7bn against a \$125bn balance sheet is **26.6**× leverage, before derivatives notionals far larger than either:

$$\frac{\$4.7\text{bn}}{\$125\text{bn}} = \mathbf{3.76}\%$$

| Adverse move on assets | Loss | Share of equity |
|---:|---:|---:|
| 0.5% | \$0.63bn | 13% |
| 1.0% | \$1.25bn | 27% |
| 2.0% | \$2.50bn | 53% |
| 4.0% | \$5.00bn | **106**% |

*A 3.76% adverse move on the assets exhausts the equity.*

*Why this makes the strategy a view about the path rather than about value.*

*The strategy did not require the convergence trades to be wrong.* They could be entirely correct (the spreads genuinely mispriced, the convergence genuinely coming), and the fund would still fail if the balance sheet *first* lost 3.8% of its value while the spreads widened.

And convergence trades widen before they converge. That is the entire reason they are available. A spread that never widened would be arbitraged away by someone with less capital; it exists and pays because holding it through the widening is uncomfortable and requires capital. The strategy's return is compensation for bearing exactly the risk that destroyed it.

So at 26.6× leverage the fund was not expressing a view about *value*, where the spread should be, but a view about the *path*: that the spreads would get there without first costing the balance sheet 3.8% of its value. It had priced the first and not the second.

Two features made the path view worse than it looks.

*The positions were correlated in a way the model did not contain.* Many separate convergence trades across markets shared one factor: a flight to liquidity. When it arrived they widened together, so the diversification the leverage was justified by was not there. Chapter 38's question 15 gives the mechanism: the common factor in a crisis is that the same leveraged holders are selling all of it.

*And the leverage itself generated the move.* A fund forced to reduce sells the positions it holds, which widens the spreads it is short, which forces more reduction. At sufficient size the path is not exogenous. And others held the same trades: LTCM's positions had been copied across the dealer community, so when the fund had to sell, so did they, and in September 1998 the copies were unwound together. That is Section 40.2's shared rule in another form, and it makes the adverse move larger than any one holder's own analysis would put it.

*The general statement, and it is the section's title.* Leverage prices the path, not the view. A position at low leverage can be right eventually; the same position at 26.6× must be right *soon and monotonically*, which is a far stronger claim and one the analysis supporting the trade did not make. The chapter's lesson turns this into a sizing rule: Section 38.17's reverse stress test finds the adverse move the position must survive on the way to converging (3.76% of the

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balance sheet here), and leverage should keep that move well beyond the widening such trades have shown.

Question 6 is the same argument with size in place of leverage, and Chapter 37's question 2 with a crypto basis trade: a position that has lost nothing being closed for want of capacity to hold it.

*What the question tests.* Converting leverage into a wipeout move, and that a small buffer converts a valuation view into a path view the strategy's own logic says will be tested.

---

**40.6** *Using the Amaranth figures, compute the days required to exit at a 10% participation rate, and explain why both of the measure's assumptions fail simultaneously.*

*The exit schedule.* From Section 40.5, the fund held positions approaching **40% of open interest** in the contracts that mattered. At that share the exit is not a market price—it is a schedule. Taking daily volume, for illustration, as equal to the open interest of those contracts, the days to exit are 40% divided by the participation rate:

| Maximum participation in daily volume | Days to exit |
|---|---|
| **10%** | **4.0 days** |
| 20% | 2.0 days |
| 30% | 1.3 days |

At a 10% participation rate the exit takes four days—and every one of them is a day the market knows a forced seller is present.

*Why both of the measure's assumptions fail simultaneously.* A one-day VaR assumes:

1. *a one-day horizon*: the position can be exited within the measurement period;
2. *observable prices*: the exit happens at the prices used to compute the measure.

The position's own size makes both false, and makes them false together.

*The horizon fails because of the size.* Four days is not one day, so the exposure runs four times longer than measured, and Chapter 38's question 18 shows that $\sqrt{T}$ scaling would understate even that, because a liquidation produces positively autocorrelated returns.

*And the prices fail for the* same *reason.* The four days are not four independent draws at market prices; they are four days during which the market can see a large seller who must complete. The price the position is marked at is not the price it can be sold at, and the gap widens as the liquidation proceeds.

So the two failures are not independent risks that might offset. They have one cause, and reducing participation to reduce the price impact lengthens the horizon, while raising it to shorten the horizon worsens the impact. There is no participation rate at which both assumptions hold.

*Why a distribution-based measure cannot see this.* Section 38.19 lists concentration among the things VaR does not capture, and here is why: two books with identical loss distributions have identical VaR whether the position is 1% of the market or 40% of it. The distribution is a statement about price moves; concentration is a statement about the relationship between the position and the market, and nothing in the distribution encodes it.

The distribution is the same; the exit is not.

*And the framework's response does not reach it.* Chapter 39's **liquidity horizons** (Section 39.4, that chapter's question 9) replace one horizon with five, and the **non-modellable risk factor** treatment (Section 39.5, its question 10) charges for factors with no observable prices. Both address horizon and observability, but by risk-factor class: the horizon is set by what the factor is, and neither responds to the holder's own size. A position at 40% of open

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interest gets the same horizon as one at 1%. Concentration is left to the supervisor's own review (Section 39.12), outside the formula, which is why it remains unmeasured, as question 20 discusses.

*What a risk manager should do instead.* The chapter's lesson sets the horizon by the holder's exit, not by the factor's class. Held whole until sold, the four-day exit scales the one-day VaR by $\sqrt{4} = 2$ under Section 38.14's square-root rule. That still assumes prices unmoved by the seller. So limit the share of open interest directly, and stress the exit with an impact cost.

*What the question tests.* Computing days-to-exit from a participation rate, and that horizon and price-observability fail together because they share one cause.

---

**40.7** *Using Example 40.4, explain why every prime broker was adequately margined and the system was not.*

*Why every broker was adequately margined.* From Example 40.4 and the per-broker arithmetic of Section 27.14 that it cites, a family office held concentrated single-stock exposure through **total return swaps** at six prime brokers:

| | |
|---|---|
| Gross position across all brokers | \$50bn |
| Held at each broker | \$8.3bn |
| Margin at each broker (15%) | \$1.25bn |
| **Total margin across the system** | **\$7.5bn** |

*Each broker saw its own slice and margined it at a level it judged adequate for that slice.* An \$8.3 billion single-stock exposure at 15% initial margin is a defensible arrangement, and any one broker examining its own book would find nothing wrong.

*Why the system was not.* The liquidation was not six independent liquidations. All six sold the same stocks at the same time, and with the client's capital gone the margin was all that stood against the loss:

| Impact of simultaneous exit | Loss | Against \$7.5bn margin |
|---|---|---|
| 10% | \$5.0bn | covered |
| 25% | \$12.5bn | short by \$**5.0**bn |
| 40% | \$20.0bn | short by \$**12.5**bn |

The margin was set for the exposure each broker measured and was wrong for the exposure that existed, and no broker's own book showed the difference. Section 27.14 adds that the margin levels themselves were set by competition for the client's business rather than by the concentration of the positions. The outcome was not shared out evenly either: the fund's capital was wiped out, and the dealers lost more than \$10 billion between them.

*Why no one could see it.* Two features of the instrument, and they are the reason the swap was chosen:

- *The client does not own the shares*, so the position appears in no ownership disclosure (Chapter 25's question 12). The concentration was invisible to the *market*, not merely to the brokers.
- *Each broker sees only its own book.* Nothing in that book shows what the client holds elsewhere, and the client has every reason not to say.

So the failure is one of *measurement*, not of modeling. Every model was right about the question it was asked. The question ("how much margin does an \$8.3 billion exposure require?") was the wrong question, and the right one was unanswerable from the broker's own book. It was not unanswerable altogether, and the chapter's lesson says what a broker can do:

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ask for the positions held elsewhere, treat a refusal to disclose them as a risk in itself, and set the margin rate by Section 40.5's exit arithmetic rather than by competition for the business. Measured on the whole position, the gap is visible at once: a joint exit at 25% impact loses $12.5 billion against $7.5 billion collected.

Chapter 27's questions 15 and 16 develop the same case: the relevant denominator for concentration is *liquidity* rather than the dealer's capital, and initial margin is the only mitigant that works without information, which is why it is the right answer here. Netting requires legal certainty, wrong-way risk requires identifying a correlation, and concentration limits require knowing the aggregate; margin protects against a move of a given size regardless of why it happened.

*And the pattern recurs.* Chapter 35's question 13 has structured-note barriers clustering across issuers with no issuer able to see the aggregate; Chapter 26's securitization chain has the same shape. A risk that is individually measured and collectively invisible is the configuration that produces incidents.

*What the question tests.* That per-broker adequacy and system adequacy are different questions, and that the swap format is what made the aggregate unobservable.

---

**40.8** *Explain what assumption of Chapter 2 the nickel case tested, and what the risk that materialized actually was.*

*The assumption tested.* That the clearing house does not fail.

Every calculation in Parts I and V treats the CCP as the thing that does not fail, the counterparty whose performance is certain. That assumption is what makes central clearing an improvement over bilateral trading at all: Chapter 2's default waterfall, Chapter 27's 2% risk weight for a qualifying CCP, and the 236:71:1 capital ratio of that chapter's question 6 all rest on it. Section 2.10 had flagged it as a judgment about institutional resilience rather than a fact.

*What happened.* In March 2022 the nickel price, which had closed near $29,000 a metric ton the previous Friday, more than tripled in two trading days to above $100,000, against a very large short position held by a producer as a hedge. On an illustrative 100,000-metric-ton short (the producer's actual position was larger, and much of it was held over the counter, out of the exchange's view):

| Price | Loss on a 100,000-metric-ton short from $29,000 |
|---|---|
| $40,000 | $1.10bn |
| $50,000 | $2.10bn |
| $80,000 | $5.10bn |
| $100,000 | $7.10bn |

*The exchange then canceled several hours of trades.* The CCP faced the prospect of member defaults it judged could exceed the resources standing behind it, and its resolution was to **void executed trades**. No member defaulted, and the default waterfall was never drawn on.

*What the risk that materialized actually was.* Not the price.

*A trader who was long nickel, correct, and fully margined had the profit taken away.* That is not market risk, not counterparty risk, and not liquidity risk. It is the risk that the venue changes the rules: that a completed transaction at an agreed price is undone by the institution whose function was to guarantee it.

The reader has met it before, in a market with no regulation at all. Chapter 37's question 20 is **auto-deleveraging**: when a crypto venue cannot close a liquidated position, it forcibly closes *profitable* positions on the other side. A correctly positioned and fully margined trader loses the position because of someone else's leverage, and there is no waterfall because there is no

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pre-funded capital.

The nickel case is that mechanism appearing in a regulated exchange, and the parallel is close enough to be uncomfortable: in both, the cost of protecting the venue fell on the winners. The one difference makes the nickel case the more striking: there the cost was imposed before any member had defaulted or any layer of the waterfall had been touched.

*The general lesson.* A CCP's guarantee is not an axiom; it is a balance sheet plus a rulebook, and both have limits. Section 40.7's lesson draws the consequences for each side: a member or end user should measure each layer of Section 2.8's waterfall against a move like March 2022's and read the venue's rules on canceling trades with it, and the venue should know the full size of a large position, over-the-counter holdings included (the part of the producer's short it could not see). Chapter 2 is right that clearing improves on bilateral trading, and Chapter 27's question 14 already noted that clearing concentrates risk somewhere new—this case shows what the concentration is worth in the tail.

*And the practical consequence for a hedger.* A firm using an exchange to hedge must now regard *the venue's rules* as a term of the hedge, and the relevant question is not only "will my counterparty pay?" but "will this trade stand?" That is a legal and governance exposure that no Greek measures.

*What the question tests.* That the tested assumption is the CCP's certainty of performance, and identifying the realized risk as trade cancellation rather than price movement.

---

**40.9** *Using Example 40.5, explain how a scheme's funding position could improve at every step while it was unable to meet collateral calls.*

*The apparent contradiction.* Example 40.5 describes a system with roughly £1.5tn of liabilities at a duration near twenty, hedged with leveraged gilt and swap positions requiring variation margin. The schemes' reported hedge ratios were higher than 40%, but part of each hedge was gilts held outright, which call for no collateral; the margined part is taken as covering 40% of the liabilities' rate sensitivity, so each margin call is 40% of the fall in the liabilities' value:

| Yield rise | Liability falls by | Margin call |
|---|---|---|
| 50 bp | 10% | £60bn |
| 100 bp | 20% | £120bn |
| 150 bp | 30% | £180bn |
| 250 bp | 50% | £300bn |

*The funding position improved at every row.* Because the hedge covered less than the whole of the liabilities' rate sensitivity, the liabilities fell faster than the assets, so the schemes became **better funded** as the crisis progressed—and were simultaneously unable to meet the margin calls.

*How both are true at once.* They measure different things:

- *Funding* is a **present-value** comparison: assets against the discounted value of future pension payments. A rising yield discounts those payments more heavily, so the liability shrinks and the ratio improves. This is a balance-sheet fact and it was genuinely favorable.
- *The collateral call* is a **cash** demand: the leveraged hedging positions were marked against the scheme daily, and variation margin had to be paid on the day.

The improvement was in a quantity measured over decades; the demand was in cash, that afternoon.

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So this is question 1's structure exactly (a hedge that is right in value and wrong in timing), and it is worth noting that the shape recurred in a completely different market twenty-nine years after Metallgesellschaft.

But the LDI case adds a mechanism the earlier one lacked, and it is the more alarming half.

*Selling gilts to raise the collateral pushed yields higher, which raised the next margin call.* The schemes' most liquid assets were the very instruments whose price was falling, so meeting a margin call required selling into the move that had generated it. That selling raised yields further, which generated a larger margin call, which required more selling.

*The hedge's own success generated the demand that forced its unwinding.* A hedge that works (the liability falls, the funding improves) produces the mark-to-market loss on the hedging leg that consumes the collateral, and the collateral is raised by selling the asset whose price move is the source of the gain.

Three general points follow.

*Leverage converts a timing mismatch into a feedback loop.* An unlevered hedge would have produced the same directional outcome without the forced selling; the leverage is what made the demand exceed the liquid assets.

*The participants who were most prudent were most exposed*, since a scheme that had not hedged faced no margin calls at all: question 1's uncomfortable corollary, again.

*And the aggregate was invisible.* No individual scheme could see how much gilt selling the system would produce at a given yield, which is question 7's Archegos structure and Chapter 35's question 13 in another market.

The book's tools measure all of this, as the chapter's lesson shows. The margined hedge is a DV01 of about £1.2 billion per basis point ($0.40 \times £1.5\text{tn} \times 20 \times 0.0001$), so Example 8.10's arithmetic makes a 100 basis point move a £120 billion margin call; and Section 38.17's reverse stress test, run on the buffer a scheme actually holds, finds the yield rise that exhausts it. That move is the number to report beside the funding ratio, and the plan should be for a larger one than any scheme's own stress, because the selling feeds back on yields (Example 8.11's loop).

*What the question tests.* That present-value funding and daily cash margin are different measures of the same position, and that the sale required to meet the margin call is what closed the feedback loop.

---

**40.10** *Explain why AIG's collateral demands required no defaults at all, and what distinction of Chapter 25 the case turns on.*

*Why no defaults were required.* By the end of 2007 AIG had sold protection on about $527 billion of super-senior tranches, most of it for European banks seeking regulatory capital relief. Its contracts required it to post **collateral against the mark-to-market** of those positions as spreads widened, and additional collateral on a downgrade of its own rating.

*Those are two triggers, and neither is a default.* Collateral is owed when the position's *value* moves against the seller, and a position's value moves long before, and often without, anything defaulting. The margin calls came mainly from about $78 billion written on multi-sector CDOs backed largely by subprime mortgages. At a duration of 5:

| Spread widening | Collateral demanded on $78bn |
|---|---|
| 100 bp | $3.9bn |
| 200 bp | $7.8bn |
| 400 bp | $15.6bn |
| 800 bp | $31.2bn |

$\$78\text{bn} \times 5 \times 0.0400 = \$15.6\text{bn}$

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*Zero defaults are required for any row.* The protection seller's view had been that the super-senior tranches would never take losses, and on realized defaults that view was defensible for much of the book: Chapter 26's question 12 records that senior tranches largely did pay in the end. They did not need to default. The collateral calls on the marks were sufficient.

*The distinction of Chapter 25 the case turns on:* the difference between **realized losses** and **marks**, Section 25.4's distinction between default risk and mark-to-market credit risk. A CDS insures against a list of events, not against loss, and its *value* moves for reasons that have nothing to do with that list. The position was a view on realized losses; the contract was a claim on marks.

Chapter 27's question 9 gives the regulatory form of the same distinction: the default charge capitalizes the loss from a counterparty actually defaulting, and the CVA charge capitalizes the risk that CVA itself moves (a spread widens, marking down a book, without anyone defaulting at all). Roughly two-thirds of counterparty-related losses in 2008–2009 came from CVA marks rather than from actual defaults, and the pre-crisis framework capitalized only the defaults.

*AIG is the largest single instance of that gap.* An institution had written protection without holding capital against the mark-to-market exposure, on a view about defaults that was largely defensible and irrelevant to the collateral it owed.

Three features made it worse.

*The rating trigger.* A downgrade required additional collateral, so the firm's own deterioration *generated* a cash demand. That is Chapter 26's question 8 mechanism, a rating change causing a transaction rather than conveying information. It is Section 27.9's wrong-way risk written into the contractual terms: the contract becomes most expensive exactly when the seller is least able to pay.

*The positions were illiquid and the marks were disputed.* With no trades, marks came from models, and Chapter 26's question 12 records the consequence: institutions marking the same instrument differently and unable to establish who was solvent. Question 15 pursues what that does.

*And the exposure was one-directional.* A protection *seller* posts as spreads widen and never receives as they tighten beyond the initial level, so in the state that was arriving the cash flowed only out.

The reforms that followed are legible as responses to exactly this. The CVA capital charge (Chapter 27's question 9), the uncleared margin rules of Chapter 8, and the clearing mandate all address the same gap: the exposure that is real, priced daily, and settled in cash before any credit event occurs.

*What the question tests.* That collateral responds to marks and ratings rather than to defaults, and locating the case in Chapter 25's separation of realized loss from the mark.

---

**40.11** *Explain what changed and what did not change when the credit index desk adopted a new VaR model, and state the loss as a multiple of each reported figure.*

*What changed:* ***the measure***. The credit index desk adopted a new VaR model, hastily approved and containing a spreadsheet error, and the reported number halved (from $132 million to $66 million), so a position that had been near its limit was suddenly comfortably inside it. No position changed on the day of the switch.

*What did not change:* ***the risk***. The same instruments, the same notional, the same counterparties, the same market. A model change is a change in the measuring instrument and not in the thing measured, and the desk's exposure to a widening in credit indices was exactly what it had been the day before.

So the new model did not reduce risk; it created room to add more—and the desk did,

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roughly tripling the book's notional over the following months, because the limit was expressed in the units the model produced.

*The loss as a multiple of each reported figure.* The desk's realized loss ran to about $6.2 billion. Against the $132 million the old model reported and the $66 million the new one reported,

$$\frac{\$6{,}200\text{m}}{\$132\text{m}} \approx \mathbf{47}\times \qquad \text{and} \qquad \frac{\$6{,}200\text{m}}{\$66\text{m}} \approx \mathbf{94}\times.$$

The comparison sets a one-day figure against a cumulative loss, so the multiples are not a backtest; but the scale is the point. The loss was nearly two orders of magnitude beyond either reported number, and neither number was measuring the exposure that materialized.

Three lessons follow, and the third is the general one.

*A limit expressed in model units is a limit on the model.* A desk constrained by reported VaR can relieve the constraint by changing the model, and the change requires no trade. The control was defeated without anything being concealed.

*A model change should be treated as a position change.* The correct governance response is to run old and new in parallel, to require the limit to be re-set rather than inherited, and to treat a large fall in reported risk from a methodology change as a reason for scrutiny rather than relief. The chapter's lesson adds two specifics: ask for Section 38.16's backtest on both models before the new number reaches a limit, and set a limit the model cannot relax (on gross notional) beside the VaR limit, so that a book tripling in size is visible whatever the measure says. Chapter 39's question 16 shows the framework attempting this through desk-level approval, and question 21 there shows where the incentive can nonetheless be switched off.

*And the measure could not see the exposure in any case.* The position was concentrated (large relative to the traded market in those indices), and Section 38.19 establishes that a distribution-based measure cannot see concentration: two books with identical loss distributions have identical VaR whether the position is 1% of the market or 40% of it. Question 6's Amaranth case is the same blind spot.

The section's title states it exactly. The measure changed and the risk did not—and a firm that responds to a change in its measuring instrument as though it were a change in its exposure has let the instrument set the position.

*What the question tests.* That a model change alters reported risk without altering exposure, and that the loss's scale relative to either figure shows neither was measuring the binding risk.

---

**40.12** *Using Example 40.6, explain how portfolio leverage and instrument leverage multiply, and compute the loss on a 300 bp rise at the pool's effective duration of 7.3 years.*

*The two leverages.* From Example 40.6, a $7.5 billion municipal investment pool levered to about $20.5 billion through repo (**2.73**×) and invested a large share of it in **inverse floaters**, whose coupon *falls* as rates rise.

*Why they multiply.* The instrument carries its own leverage. An inverse floater with coupon leverage $L$ pays something like $C - L \times r$, so a rise in rates hits it twice: through the discounting, as with any bond, and through the coupon. Its duration is roughly $(1 + L)$ times its underlying's.

*And portfolio leverage multiplies that, because it applies to the whole holding:*

$$\text{effective duration on the pool's capital} = \underbrace{(1 + L) \times D_{\text{underlying}}}_{\text{instrument}} \times \underbrace{2.73}_{\text{portfolio}}$$

Were the whole book in such notes, with the underlying's duration taken as 4 years:

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| Coupon leverage | Instrument duration | Portfolio effective duration |
|---|---|---|
| 1 | 8.0 yr | 21.9 yr |
| 2 | 12.0 yr | 32.8 yr |
| 3 | 16.0 yr | 43.7 yr |

Bonds of four-year duration become a thirty-three-year position from two ordinary multiplications, neither of which is hidden: $(1 + 2) \times 4 = 12$ years in the instrument, then $\times 2.73$ in the portfolio.

*The pool's actual duration.* The actual book mixed inverse floaters with other structured notes and plain agency paper, so its duration was lower than any row above. Its 1994 loss of about $1.64 billion, on a rise of roughly 300 basis points in short and intermediate yields, corresponds to an asset duration of about 2.67 years on the $20.5 billion, which the repo leverage turns into an effective duration of about **7.3 years** on the depositors' $7.5 billion.

*The loss on a 300 bp rise at 7.3 years.*

$$0.0300 \times 7.3 \times \$7.5\text{bn} = \$\mathbf{1.64}\text{ billion}$$

which is **21.9%** of the pool:

| Rate rise | Loss | Share of the $7.5bn |
|---|---|---|
| 100 bp | $0.55bn | 7.3% |
| 200 bp | $1.10bn | 14.6% |
| **300 bp** | **$1.64bn** | **21.9%** |

The multiplication shows in the comparison with an unlevered holding. A $7.5 billion pool of 2.67-year paper would have lost about 8% on the same move; the repo made the depositors' loss 2.73 times that. And had the whole book been in $L = 2$ notes, the same 300 basis points at 32.8 years would have cost $7.38 billion, 98.4% of the pool.

Two observations attach to the arithmetic.

*Neither leverage is exotic.* Repo financing is ordinary and an inverse floater's coupon formula is arithmetic. What was unusual was the product, and the product was never computed, which is question 13's subject.

*And the effective duration is the only number that matters.* A pool reporting its holdings, its repo, and its coupon formulas has disclosed everything needed to compute 7.3 years, and has disclosed nothing at all if no one multiplies.

*The pattern recurs*, and Chapter 37's question 11 is the closest analog: an inverse crypto contract is linear in dollars and curved in coin, so a position sized on the linear column is margined on the curved one. In both cases a transformation no one performed converted a comprehensible position into a different one.

*What the question tests.* That instrument and portfolio leverage compose multiplicatively into an effective duration, and computing the loss at the pool's calibrated duration.

---

**40.13** *Explain why nothing was concealed in the Orange County case and the buyer still lost, and what this implies about the limits of disclosure.*

*Why nothing was concealed.* The coupon formula was in the term sheet and the repo was on the balance sheet. Every input to question 12's calculation was disclosed:

- the inverse floaters' coupon structure, from which the instrument's duration follows;
- the repo borrowing, from which the portfolio leverage follows;
- the holdings themselves, in the pool's published reports.

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No one misrepresented anything, no document was false, and no exotic instrument was involved. The pool reported its yield advantage rather than its duration, and warnings during 1994 that further rate rises threatened it were dismissed.

*Why the buyer still lost.* Because the risk was in a quantity no one calculated.

The disclosed facts were *inputs*; the exposure was the *product* of them: question 12's effective duration of 7.3 years, an asset duration multiplied by 2.73 times leverage. That multiplication was available to anyone and was performed by no one, and until it is performed the disclosure conveys the components of a risk without conveying the risk.

*And the reported quantity was the wrong one.* The pool reported *yield*, which the strategy genuinely improved. Inverse floaters pay above-market coupons in a falling-rate environment, and for several years the pool's performance was exactly what was advertised: an income enhancement, accurately described in a sense. The metric being tracked was the one the position was good at, and the metric that would have revealed the exposure was not tracked at all.

*What this implies about the limits of disclosure.*

*Disclosure transfers information, not understanding*, and the two come apart precisely where a risk requires a computation to see. Three specific limits follow:

*Disclosure is not aggregation.* Facts disclosed separately may combine into an exposure disclosed nowhere. No single document said "effective duration 7.3 years," and nothing required one to.

*Disclosure does not choose the metric.* A buyer who reads everything and tracks yield has complied fully with every disclosure regime and measured the wrong thing. The regime specifies what must be said and not what must be computed.

*And a sophisticated-buyer presumption can be wrong about the buyer.* A municipal treasurer is a professional investor by every regulatory test and was not equipped to compute a levered duration, which is the argument of Chapter 35's question 22 about suitability, and the reason the chapter calls for disclosure and suitability rather than risk measurement as the response to this category.

*The general lesson, and it is question 16's.* The failure is rarely detection (the information was there), and the reform that helps is rarely quantitative. What would have prevented this is not a better model but a requirement that someone responsible compute and report an effective duration, which is a governance rule.

*What the question tests.* That the exposure was a product of disclosed inputs, and that disclosure delivers components while risk requires an aggregation no one was obliged to perform.

---

**40.14** *Explain why a European generator that had sold power forward faced margin calls while its economics improved, and compute the demand on 50 TWh for a move from 50 to 400 EUR/MWh.*

*Why a hedged generator faced margin calls while its economics improved.* A generator that has sold its future output forward is correctly hedged and short futures: it has fixed the price at which its power will be sold, which is the textbook position.

As European power prices rose several-fold through 2022:

- the short futures leg lost, and required **variation margin, in cash, every day**;
- the offsetting gain (electricity worth far more than the hedge assumed) would arrive only on delivery, quarters or years later.

*The hedged sales' economics were locked; the plant's value rose—and neither offset the cash.* The generator owns a plant selling into a far higher market, so its economics improved

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throughout; but the hedged output earns the price it was sold at, and the unhedged output's higher value arrives only as the power is generated. Its liquidity requirement meanwhile multiplied with a price it had deliberately insulated itself from.

*The demand on 50 TWh for a move from 50 to 400 EUR/MWh.*

$$50\,\text{TWh} = 50{,}000{,}000\,\text{MWh}$$

$$50{,}000{,}000 \times (\text{EUR } 400 - \text{EUR } 50) = 50{,}000{,}000 \times \text{EUR } 350 = \text{EUR } \mathbf{17.5}\text{ billion}$$

Seventeen and a half billion euros of cash, owed on a hedge that was working, by a firm whose underlying business had become far more valuable.

*The consequences were national.* Liquidity lines measured in tens of billions of euros were extended to power and gas companies across Europe, several such firms required state liquidity support to avoid failing, and Nordic governments opened backstop facilities for utilities facing margin on exchange-traded power. Nor was this a product of mandatory clearing: most of the utilities hedged on exchanges by choice.

And the striking feature is that it fell hardest on the participants who had hedged. A generator that had left its output unsold faced no margin calls at all and simply enjoyed the higher prices. The cash demand was created by the hedge, and it scaled with how much output had been sold forward.

This is question 1's mechanism, one generation on. Metallgesellschaft's short-dated futures needed cash while its long-dated obligations were merely valuable; a 2022 utility's futures needed cash while its power stations were merely valuable. The instrument, the market and the decade differ; the mismatch does not.

Two conclusions are worth carrying.

A hedge sized on price risk alone is not sized, because the funding leg scales with the same move the hedge was bought to neutralize—and here the move was a multiple rather than a few percent.

*And the more complete the hedge, the larger the funding requirement.* That is the uncomfortable part, and it means a firm cannot manage the exposure by hedging more carefully. The remedies are liquidity facilities sized to the hedge, non-cash-collateralized bilateral structures, or accepting less hedging, each of which trades one risk for another.

*What the question tests.* That daily margin on the short leg meets a delivery-date gain on the other, and computing a cash demand from a volume and a price move.

---

**40.15** *Explain why a valuation that cannot be independently checked responds to pressure, and why the pressure is asymmetric.*

*Why an uncheckable valuation responds to pressure.* Because there is nothing for it to be anchored to.

A position with a market has a price somebody will trade at, and the mark is disciplined by the possibility of testing it. A position with no market is marked by a model, and a model's output depends on inputs (a correlation, a volatility, a prepayment assumption, a recovery rate), each of which lies within a defensible range rather than at a point.

So the mark is a range and somebody must choose a point in it. Every choice inside the range is defensible, no choice is verifiable, and the person choosing has an interest in the outcome.

Chapter 21's question 10 sizes the range in one case: two models calibrated to the *same* vanilla surface pricing a barrier **49.5%** apart, with both reproducing every quoted price. Chapter 31's question 10 gives another: an option-adjusted spread running from 20.9 to 71.5

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basis points across parameter choices no market quotes. In neither case does a wrong mark announce itself.

*Why the pressure is asymmetric.* Because the consequences of marking too high and too low are not symmetric for the person marking.

*Marking optimistically* produces reported profit, which supports compensation, capital ratios, and the continuation of the business. It is defensible (the number is inside the range), and it is not falsified until the position is sold or matures, which may be years away.

*Marking conservatively* produces a loss today that the position may never realize. It invites the question of why the position is held, reduces reported performance, and is equally unverifiable.

So the two errors carry very different costs to the person making them, and the pressure runs one way at every level: the trader who marks the book, the desk head whose P&L it is, and the firm whose capital ratio depends on it. No one objects to a mark that produces a profit. No one has to act in bad faith; a series of individually defensible choices drifts in one direction because the range is wide and the incentives are not centered in it.

*And the drift compounds.* A mark that has been optimistic must stay optimistic (correcting it realizes the accumulated difference at once), so the position becomes progressively harder to mark down as the gap grows. That is why such episodes end abruptly rather than gradually.

*The structural responses*, and they are the right ones because they attack the asymmetry rather than the estimate:

- *independent price verification*, moving the choice to someone whose compensation does not depend on it;
- *prudent valuation*, which Chapter 39's question 22 works through: in Example 39.9 the model risk adjustment is the largest single component at 60 basis points as assessed, and as one of three overlapping adjustments counted at 50% it contributes 30 basis points of a 125.5 basis point deduction from capital, taken whether or not anyone suspects a problem;
- and Chapter 21's discipline of reporting a range alongside the mark and taking the reserve through profit and loss at inception.

*The general statement.* Where a number cannot be checked, the number that gets reported is a function of who benefits from it, which is a governance fact rather than a modeling one, and is why the remedy is organizational.

*What the question tests.* That an unverifiable mark is a choice within a defensible range, and that the asymmetry of consequences rather than dishonesty is what produces the drift.

---

**40.16** *Explain the case study's finding that the failure is rarely detection, and why so little of the resulting reform is quantitative.*

*The finding.* The case study's finding is that in almost every case the facts needed to prevent the loss existed inside the institution before it happened:

- *Barings*: the reconciliation discrepancies were reported and unresolved for months;
- *LME nickel*: the exchange-traded part of the short was visible to the exchange, though much of the position was held over the counter, out of its view;
- *the London Whale*: the credit index desk's own valuations were disputed internally before the loss was recognized;

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- *Silicon Valley Bank*: the unhedged securities portfolio had been the subject of supervisory findings;
- *Archegos*: the post-mortem commissioned by Credit Suisse found that the bank's risk function had escalated the concentration, and the escalation did not change the position.

Elsewhere in the chapter the number that mattered was at least computable from what was known: Orange County's effective duration from its term sheets and its repo (question 13), LTCM's wipeout move from its leverage in one line (question 5).

*So the failure is rarely detection and usually escalation.* Someone saw it; the observation did not reach a person with both the authority and the incentive to act, or reached them in a form that competed with a revenue argument and lost. What failed was not measurement but **attention, incentive, and responsibility**.

*Why so little of the resulting reform is quantitative.* Because a better measurement does not change what happens to an observation that was already made.

The reforms the case study names are all of that kind: segregating trading from settlement, requiring mandatory leave, making risk report outside the business line, and giving a desk-level test a capital consequence. Each is an attempt to change what happens to an observation, not to generate a better one. Run through what actually changed after these cases, as Section 40.16 records it:

| Case | What changed |
|---|---|
| Barings, Société Générale | segregation; independent reconciliation; mandatory leave |
| London Whale | desk-level P&L attribution test with a capital consequence |
| AIG | CVA capital charge; uncleared margin rules; clearing |
| Orange County | disclosure and suitability: who may sell what to whom |
| Amaranth, Archegos | illiquidity priced; concentration still unmeasured |
| Silicon Valley Bank | held-to-maturity treatment unchanged |

The entries that produced durable change are rules about who must do what, who must see what, or who must post what—not formulas. The entries where measurement was the problem changed least.

Three reasons make that the right response rather than a failure of ambition.

*The models were mostly not wrong.* Chapter 39's question 24 makes the point that model risk is not a subset of market risk; here the complementary point holds: most of these losses did not require a model error, and improving the models would have changed nothing.

*The binding constraints were institutional.* Cash timing (questions 1, 9 and 14), aggregate visibility (question 7), venue rules (question 8), accounting treatment (question 18). None of these is a quantity to be estimated better.

*And improving a measure has no effect if the existing measure was already being overridden.* The case study draws this corollary itself: Chapter 39's machinery is an attempt to make overriding expensive rather than to make measurement better, and that is a design rather than an accident.

One exception confirms the pattern. The one substantially quantitative reform on the list, the CVA capital charge, addressed a category of loss the framework had *not measured at all* (question 10), which is a gap in coverage rather than an error in estimation.

*What the question tests.* That the information was inside the firm and the failure lay in escalation, and that reforms addressing what happens to an observation are structural because the failure was.

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**40.17** *Using Example 40.7, compute the economic loss on a $120bn portfolio at duration 5.6 for a 300 bp rise, and state it as a share of a $16bn equity base.*

*The economic loss.* From Example 40.7, a $120bn portfolio at duration 5.6:

$$\$120\text{bn} \times 5.6 \times 0.0300 = \$\mathbf{20.2}\text{ billion}$$

| Yield rise | Economic loss | Share of a $16bn equity base |
|---|---|---|
| 100 bp | $6.7bn | 42% |
| 200 bp | $13.4bn | 84% |
| **300 bp** | **$20.2bn** | **126%** |
| 400 bp | $26.9bn | 168% |

$$\frac{\$20.2\text{bn}}{\$16\text{bn}} = \mathbf{126}\%\text{ of equity}$$

A 300 basis point rise produces an economic loss exceeding the firm's entire equity.

*And the hedge was available.* The bank had held pay-fixed swaps against part of the book and unwound most of them during 2022. A pay-fixed swap on about half the duration would have cost negative carry (in 2021, paying a fixed rate near 1% against an overnight rate near zero, on the order of 100 basis points a year on the swapped notional) and would have removed about half of that column, taking the 300 basis point loss from $20.2 billion to about $10 billion. The instrument was standard, liquid, cleared, and priced by Chapter 7's arithmetic. Nothing about the hedge was difficult.

That is what makes this case the right one to end on. Every other case involves a derivative position; this one involves its absence. The instrument that would have prevented the loss is the plainest in the book: a vanilla interest rate swap, the subject of Chapter 7.

Three observations follow.

*The exposure required no modeling to see.* Duration times portfolio times rate move is one multiplication. Question 16's finding applies exactly: the failure was not detection, since the portfolio had been the subject of supervisory findings.

*The bank had chosen the exposure.* A long-duration securities portfolio funded by demand deposits is a deliberate carry position (earning the term premium), and it is a position a bank may reasonably take. What it may not reasonably do is take it unhedged at 126% of equity, and no analysis was required to know the size.

*And the accounting concealed it*, which is question 18's subject and the second of three necessary ingredients.

The sequence the example identifies is worth stating as the chapter's closing structure: an unhedged exposure, an accounting treatment that concealed it, and a funding base that could leave in a day. Each is necessary; none alone is sufficient. A hedged bank with the same accounting and the same depositors would have survived; so would an unhedged one whose losses were visible early enough to raise capital.

*What the question tests.* The duration arithmetic, and that a loss from declining to hedge belongs in a chapter about derivatives mishaps as much as any loss from trading them.

**40.18** *Explain the role of held-to-maturity accounting in the same case, and why a loss disclosed in the footnotes did not reach capital until a sale and a capital raise put it in front of depositors.*

*The role of held-to-maturity accounting.* Securities classified as held-to-maturity are not marked through capital. The classification asserts that the bank intends and is able to hold them until they repay at par, so interim price movements are treated as irrelevant and the securities are carried at amortized cost.

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So the reported number stayed intact while the economic number did not. Question 17's stylized $20.2 billion of economic loss (126% of equity) would appear in *no* regulatory capital measure and no headline balance-sheet figure; the actual unrealized loss, about $15 billion at the end of 2022, did not either. The bank remained, on its reported numbers, adequately capitalized throughout.

*Why a loss disclosed in the footnotes did not reach capital.* Because the treatment makes the disclosure and the recognition two different events.

*The disclosure was complete and inert.* The unrealized loss was in the footnotes, where anyone could read it. But the footnote does not enter capital, so it triggers nothing: no capital ratio falls, no supervisory threshold is crossed, and the headline numbers depositors watch do not move.

*Recognition waits for a sale.* The treatment holds *until securities are sold*, at which point the loss crystallizes and must be recognized—and a sale is precisely what a bank does when it needs liquidity. In March 2023 a loss-making sale and a capital raise did both at once: the sale realized the loss, and the raise announced that capital was needed to cover it.

The sequence therefore produces the crisis in the act of revealing the loss:

1. The bank needs to reposition, and its liquid assets are the securities, so it sells some.
2. The sale forces recognition of a loss that had been in the footnotes only.
3. The capital raise that accompanies it is the first headline evidence of the exposure, and it arrives as news that the bank is short of capital.
4. Depositors, concentrated, uninsured, and now aware, leave in a day.

So the window in which the bank could have acted quietly (hedging, shrinking the book, or raising capital while the loss was only a footnote) went unused, because nothing in the reported numbers forced it. Here the recognition and the crisis are the same event, because the mechanism that brings the loss into capital is the mechanism that signals distress.

*And the concealment was structural rather than intentional.* The classification was permitted, applied correctly, and disclosed. This is question 13's limit of disclosure again: the information was available and the metric being watched was a different one.

Three observations follow.

*The accounting did not cause the loss; it removed the warning.* The economic exposure of question 17 existed regardless of classification. What held-to-maturity treatment removed was the *signal*: the steadily worsening mark through capital that would have prompted action while action was still possible. It also tilted the hedging decision: as Section 3.8 shows, a pay-fixed swap would have been marked through earnings while the held-to-maturity securities it hedged were not, so on the reported numbers the hedge would have added volatility rather than removed it. That is why the chapter's lesson says to measure the DV01 with the held-to-maturity securities included.

*The intent test is unverifiable in the relevant state.* A bank intends to hold to maturity until it cannot, and *whether it can* depends on its funding, so the classification's premise fails exactly when it matters. Chapter 39's question 1 makes the same criticism of an intent-based trading-book boundary, and the framework's answer there was to remove the benefit of reclassification.

*And the third ingredient was the depositor base.* A funding base that could leave in a day is what converted a slow problem into a two-day failure, and Example 40.7 concludes that each of the three is necessary and none alone is sufficient. The accounting problem itself, Section 40.16 notes, is unresolved: marking a held-to-maturity book through capital would import volatility into the accounts of institutions that genuinely intend to hold.

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*What the question tests.* That amortized-cost treatment keeps a disclosed loss out of capital until a sale forces recognition, and that the sale and the capital raise are simultaneously the recognition and the distress signal.

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**40.19** *Name the four mechanisms that recur across these cases, and give one case for each.*

*The four mechanisms, as Section 40.15 names them, with a case for each.*

*1.* ***Cash and value are different quantities.*** A position correct in value can be closed for want of cash, and no valuation model contains the closing.

*Case: Metallgesellschaft* (questions 1 to 3). Every row of Example 40.1 is flat in value and catastrophic in cash: $1.28 billion of variation margin against a supply-contract gain arriving over a decade, with a peak funding gap of $1.22 billion. The chapter counts the mechanism five times: Metallgesellschaft itself, the LDI schemes (question 9), the 2022 European generators (question 14), AIG (question 10), and Example 37.2's basis trade.

*2.* ***Size and leverage change the exit.*** The measure describes the position's distribution; the loss comes from the position's share of the market, or from leverage that made the path decisive, which are different variables.

*Case: Amaranth* (question 6). At 40% of open interest the exit took days rather than a day, and the market knew a forced seller was present for every one of them. LTCM (question 5), Archegos and the nickel case also failed at the point of liquidation rather than at the point of valuation; for LTCM it was leverage rather than share that made a 3.76% move fatal.

*3.* ***Someone must see the whole.*** The aggregate existed and no one held it.

*Case: Archegos* (question 7). Each of six brokers margined its $8.3 billion slice at a level it judged adequate; the $50 billion existed on no one's report, because a total return swap gives exposure without the disclosure that ownership would carry. Barings and Société Générale (question 4) are the same failure inside a firm rather than across firms.

*4.* ***The guarantee has a bottom.*** A counterparty's promise is worth its resources and its governance, not its label.

*Case: LME nickel* (question 8). Facing member defaults it judged could exceed its resources, the clearing house voided executed trades before the waterfall was touched. Section 37.10's crypto venue shows the same mechanism without regulation.

*And a fifth sits apart:* ***the buyer cannot price what it buys***. Orange County (questions 12 and 13) lost on a levered duration that was disclosed in every input and computed by no one. The chapter keeps it separate because it is not a failure of measurement but of *who is equipped to measure*, and the response to it is disclosure and suitability rather than risk measurement.

*What the four have in common.* None of them is a pricing error. In almost none of these losses did a derivative behave in a way its model did not predict; the cases turned on *cash*, *size*, *information* and the *venue*, which are four properties of the arrangement around the position rather than of the position itself.

*Which is why a better model addresses none of them, and why the reforms of question 20 addressed them unequally.* Segregation and reconciliation addressed the inside-the-firm version of the third mechanism durably; the clearing reforms widened the first, by converting credit risk into liquidity risk; the second, concentration, is still not measured; and the fourth remains a matter of each clearing house's resources and governance.

*What the question tests.* Abstracting the chapter's four recurring mechanisms from a heterogeneous set of cases, and that none of them is a valuation failure.

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**40.20** *Explain which category of failure produced durable reform and which produced the least, with an example of each.*

*Which category produced durable reform: the* ***control failures****.*

*The example is Barings* (question 4). The mechanism was specific and procedural: one person controlled trading, settlement, and the reconciliation that would have exposed the position. The response was equally specific: segregation of trading from settlement, independent reconciliation, and mandatory leave became standard after Barings, and losses of that type have become rarer. This is the profession's clearest success, and it required no modeling at all.

It is durable but not self-enforcing. Société Générale in 2008 and UBS in 2011 showed that a control exists only as long as someone operates it, which is why question 4's second lesson matters: bound the largest position that can be hidden, since hiding cannot be ruled out.

*Which produced the least: the* ***measurement failures****.*

*The example is Amaranth* (question 6), *repeated by Archegos* (question 7). Amaranth's lesson about concentration was available in 2006: a position at 40% of open interest has an exit that no distribution-based measure describes. Archegos repeated it in 2021 with a different instrument, and the aggregate was again invisible. Chapter 39's framework prices illiquidity (liquidity horizons, the non-modellable treatment) and still does not measure concentration, which Section 38.19 states as a known gap rather than a solved problem.

*Between the two sit the clearing reforms.* After 2008 they were substantial, and AIG (question 10) is the case they answer: the CVA capital charge, the uncleared margin rules and the clearing mandate all attach a cost to the mark-to-market exposure that had gone uncapitalized. But they also widened the exposure of the LDI and energy cases: pushing more of the market onto margin, cleared and bilateral, converts credit risk into liquidity risk, and the 2022 episodes were the bill for that conversion arriving.

*And one problem remains unresolved.* The accounting treatment behind Silicon Valley Bank (question 18) is still available, and the arguments for it are not frivolous. The gap is real and the fix is not obvious, which is why the case will recur.

*Why the two categories differ.*

*A control failure has a procedural fix inside one firm.* Separating two functions or requiring mandatory leave needs no model and no data the firm does not already hold, and compliance can be checked.

*A measurement failure needs information no single party holds.* Concentration is a relationship between a position and the whole market, so measuring it requires aggregating across counterparties and venues, which is exactly what each party's own measure cannot see. Pricing the symptom (illiquidity) is possible within the existing framework; measuring the cause is not.

*The uncomfortable summary*, and it is the honest one for a chapter of this kind: the failures that were easiest to fix inside one firm have been fixed, and the failure that recurs is the one whose mechanism lies across firms. Question 19's four mechanisms are not equally tractable.

*What the question tests.* Matching the reform record of Section 40.16 to its causes: control failures fixed durably by procedure, concentration left unmeasured because no one holds the aggregate, and clearing as a substantial reform with a liquidity cost.

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**40.21** *Using Example 40.2 and Figure 40.2, compute the futures the programs must sell as the index falls from 100 to 90, and explain why the selling per index point peaks near the floor and slows once the delta approaches* $-1$.

*The futures sold from 100 to 90.* The programs replicate a put with a floor at 95, three months to expiration, 20% volatility and a zero interest rate, on \$70 billion of insured portfolio. Because the portfolio itself falls with the index, the futures short at index level $S$ is $|\Delta| \times \$70\text{bn} \times S/100$. From Example 40.2:

| Index | Put delta | Futures short | Incremental selling |
|---|---|---|---|
| 100 | −0.287 | \$20.07bn | — |
| 95 | −0.480 | \$31.92bn | +\$11.85bn |
| 90 | −0.688 | \$43.35bn | +\$11.43bn |

$\$43.35\text{bn} - \$20.07\text{bn} = \$23.28\text{bn} \approx \$\mathbf{23.3}$ billion

A 10% decline obliges the programs to sell \$23.3 billion of index futures, and the rule leaves them no discretion over any of it.

*Why the selling per point peaks near the floor.* Differentiate the short with respect to a fall in the index. It has two terms:

$$\text{futures sold per index point} = \underbrace{\Gamma \times \$0.70\text{bn} \times S}_{\text{the delta steepens}} - \underbrace{|\Delta| \times \$0.70\text{bn}}_{\text{the portfolio shrinks}}$$

The first is gamma on the insured notional: each point of decline makes the delta more negative by $\Gamma$, and that is applied to the whole portfolio. The second is the delta on the portfolio's own fall: a smaller portfolio needs a smaller short for the same delta. At the floor the delta changes by about 0.042 per index point, so

$$0.042 \times \$0.70\text{bn} \times 95 \approx \$2.79\text{bn}, \qquad 0.480 \times \$0.70\text{bn} \approx \$0.34\text{bn},$$

and the net is about **\$2.45 billion of futures per index point**, the peak of Figure 40.2's right panel. Gamma is largest for an option near its strike, and the floor is the strike, so that is where the delta steepens fastest and the selling runs fastest. The table shows it in averages: \$11.85 billion over the five points from 100 to 95 is \$2.37 billion a point, and \$11.43 billion over the five from 95 to 90 is \$2.29 billion.

*Why it slows once the delta approaches* $-1$. As the index falls below the floor the put goes deep into the money. Its delta cannot fall below $-1$, so it flattens and gamma falls toward zero, and the first term shrinks: at 85 the delta is about $-0.86$ and the gamma term is about \$1.59 billion, at 80 the delta is $-0.95$ and it is about \$0.69 billion. The second term meanwhile grows toward $1 \times \$0.70\text{bn}$ and comes to dominate: \$0.60 billion at 85, \$0.67 billion at 80. The net selling is about \$0.99 billion a point at 85 and almost nothing at 80, which is why the table's last row covers ten points for \$9.98 billion, about \$1 billion a point. Once the delta is close to $-1$ the programs are already short almost the whole portfolio, and each further point of decline adds little to sell while the shrinking portfolio needs slightly less.

*Why the number matters.* Section 40.2's lesson is that this schedule is published in advance by the mathematics. A program should set its peak, summed over everyone running the same rule, against what the futures market can absorb, and where the market cannot absorb it, buy the protection as a listed put instead. On October 19, 1987 it could not, and the programs fell behind their schedules.

*What the question tests.* Reading the selling off the futures-short schedule, and decomposing the selling per point into gamma on the insured notional less the delta on the portfolio's own

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fall, so that its peak at the floor and its decline as the delta reaches −1 both follow from the shape of gamma.

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